Fleet Management

Use **VCF Operations** as a VI administrator to manage identity and access, certificates, passwords, tags, configuration profiles, and availability of the VCF fleet.

For VCF, you use VCF Operations and in certain cases individual components for these capabilities. For vSphere Foundation, you perform the supported operations on the individual management components.

In This Chapter

- Managing Identity and Access With VCF Single Sign-On
- Managing Certificates in VMware Cloud Foundation
- Managing Passwords for VMware Cloud Foundation Components
- Tags and Categories Overview
- Managing the Configuration of your Environment
- Managing Infrastructure Services for VMware Cloud Foundation
- Adding or Removing VCF Components Post Deployment
- Linking vCenter instances in VCF Operations
- Collecting Data with Cloud Proxy in VCF Operations
- Shutdown and Startup of VMware Cloud Foundation
- Component Backup and Restore of VMware Cloud Foundation
- VCF Instance Backup and Restore
- VCF Fleet Disaster Recovery
- FIPS Configuration for VCF Components
- Configuring Management Components

Getting to Know VCF

To apply this documentation, you must be acquainted with the VCF Product Overview, VCF Design, and VCF Release Notes.

Deploying or Converging to VCF and Setting Up Cloud Infrastructure

For information about deploying or converging to a VCF fleet or VCF Instance, see Deployment, Convergence, and Upgrade. For information about adding infrastructure and configuring storage and networking for workloads in VCF Operations and the vSphere Client as a VI administrator, see Building Cloud Infrastructure.

Managing the Lifecycle of Your VCF Platform

You can manage software binaries and update your VCF fleet and VCF Instance centrally in VCF Operations. See Lifecycle Management.

SDKs, APIs, and CLI for VCF Administration

You can build, operate, and manage your VCF private cloud by using the VCF SDK and APIs, and VCF PowerCLI. See Administration SDKs, APIs, and CLI.

Managing Identity and Access With VCF Single Sign-On

Streamline identity and access management across your environment by configuring VCF single sign-on (SSO). This centralized approach allows you to assign granular permissions for every component using global roles, all from a single pane of glass. Furthermore, you can manage custom roles across multiple vCenter instances simultaneously, while API access provides a reliable foundation for your automation workflows.

- To configure and set up VCF single sign-on, see Configuring VCF Single Sign-On.

- To create custom VCF roles and for more information about built-in VCF roles, see Managing VCF Roles.
- To provision a vCenter custom role to the vCenter instances in a VCF Instance, see Provisioning vCenter Custom Roles.
- To manage API clients, API tokens, and access tokens, see Managing API Clients and Tokens .
- To migrate VMware Identity Manager to identity broker, see Migrating VMware Identity Manager to Identity Broker.

Configuring VCF Single Sign-On

You can set up VCF single sign-on for VCF to streamline user access with centralized authentication. VCF single sign-on offers Enterprise admin users the ability to utilize single sign-on and access VCF components such as vCenter, VCF Operations, VCF Automation, VCF Operations for networks, VCF Operations orchestrator, VCF Operations HCX, Salt RaaS, and NSX, with one set of credentials. VCF single sign-on provides a secure login experience with a single authentication source that you configure from VCF Operations across all VCF components except SDDC Manager and ESX.

Benefits of VCF Single Sign-On

VCF single sign-on provides multiple benefits:

- Improved user experience as users can log in once and can gain access to the VCF components they have been granted access to.
- Enhanced security as you reduce password fatigue and strengthen protection against unauthorized access.
- Centralized management as you manage the identity provider configuration and user/group provisioning across VCF Instances at a central location in VCF Operations.
- Support for multiple identity providers protocols such as OIDC 2.0, SAML 2.0, and directory services such as OpenLDAP and Active Directory.

Identity Providers and Protocols Supported for VCF Single Sign-On

After you deploy VCF, you can configure VCF single sign-on by using an external identity provider and then sign in to VCF components by using VCF single sign-on. The identity broker component handles the single sign-on functionality.

> **Note:**
>
> You can configure a single identity provider per identity broker instance.

Identity Providers Supported

You can configure the following identity providers:

- Okta
- Ping Identity
- Microsoft Entra ID
- Microsoft Active Directory Federation Services (ADFS)
- Symantec Identity Security Platform
- Any SAML 2.0 Identity Providers
- Any OIDC 1.0 Identity Providers

You can configure the following directory services:

- Active Directory over LDAP (AD/LDAP)
- OpenLDAP

Authentication Protocols Supported

- Security Assertion Markup Language (SAML 2.0).
  An XML-based protocol where the identity provider issues digitally signed XML-based assertions to the service provider for user authentication.
- OpenID Connect protocol (OIDC).

An authentication protocol built on OAuth 2.0, where the identity provider issues an ID token in JSON format for user authentication.

User/Group Provisioning

As part of configuring VCF Single Sign-On, you sync users and groups in the identity broker component provisioned from your identity provider. The following protocols are supported:

1. **SCIM 2.0** (System for cross-domain identity management)
2. **JIT (Just-In-Time)**
   a. Just-In-Time Group Provisioning
      - Groups are created dynamically based on the SAML assertion or OIDC authentication response when the user logs in for the first time. You can use group pre-provisioning if you know the group names from the identity provider that requires entitlements.

   b. JIT pre-provisioning of groups
      - You create the required groups in advance to perform role assignments. Users will be assigned to these groups dynamically on their first log in based on SAML assertion or OIDC authentication response. Only the groups pre-provisioned are available for role assignments before the first log in.

      Note:

      When Just-In-Time (JIT) user provisioning is enabled, a user object is automatically created upon successful login, based on the SAML or OIDC authentication response. Group memberships are assigned according to the group claims in the response. However, based on how JIT provisioning works, if the user later leaves the organization and is removed from the external identity provider, the JIT-provisioned user is not automatically deleted from the system.
3. **AD/LDAP (Active Directory over LDAP) and on-demand provisioning for AD/LDAP**
   Users and groups are synced at periodic intervals. The entire directory is not synced, only users and groups that are specified. Passwords are never synced.

   Note:

   You can use SAML 2.0 or OIDC as an authentication source and SCIM, JIT, and AD/LDAP as user-group provisioning.

For operational guidance and prerequisites, see Identity and Access Management, Choosing an Identity Provider Type, and VCF SSO User and Group Provisioning Options.

VCF Single Sign-On Architecture

VCF allows organizations to configure VCF single sign-on across a fleet of VCF Instances based on your preference for manageability and availability.

There are one or more potential designs for VCF single sign-on configuration which depend on the deployment of the identity broker instances across your VCF Instances. For more information, see VCF Single Sign-On Detailed Design.

Deployment Modes of the Identity Broker

When you configure VCF single sign-on, you can implement the identity broker that manages it, as embedded in the management domain vCenter or you can deploy it as a VCF management services component.

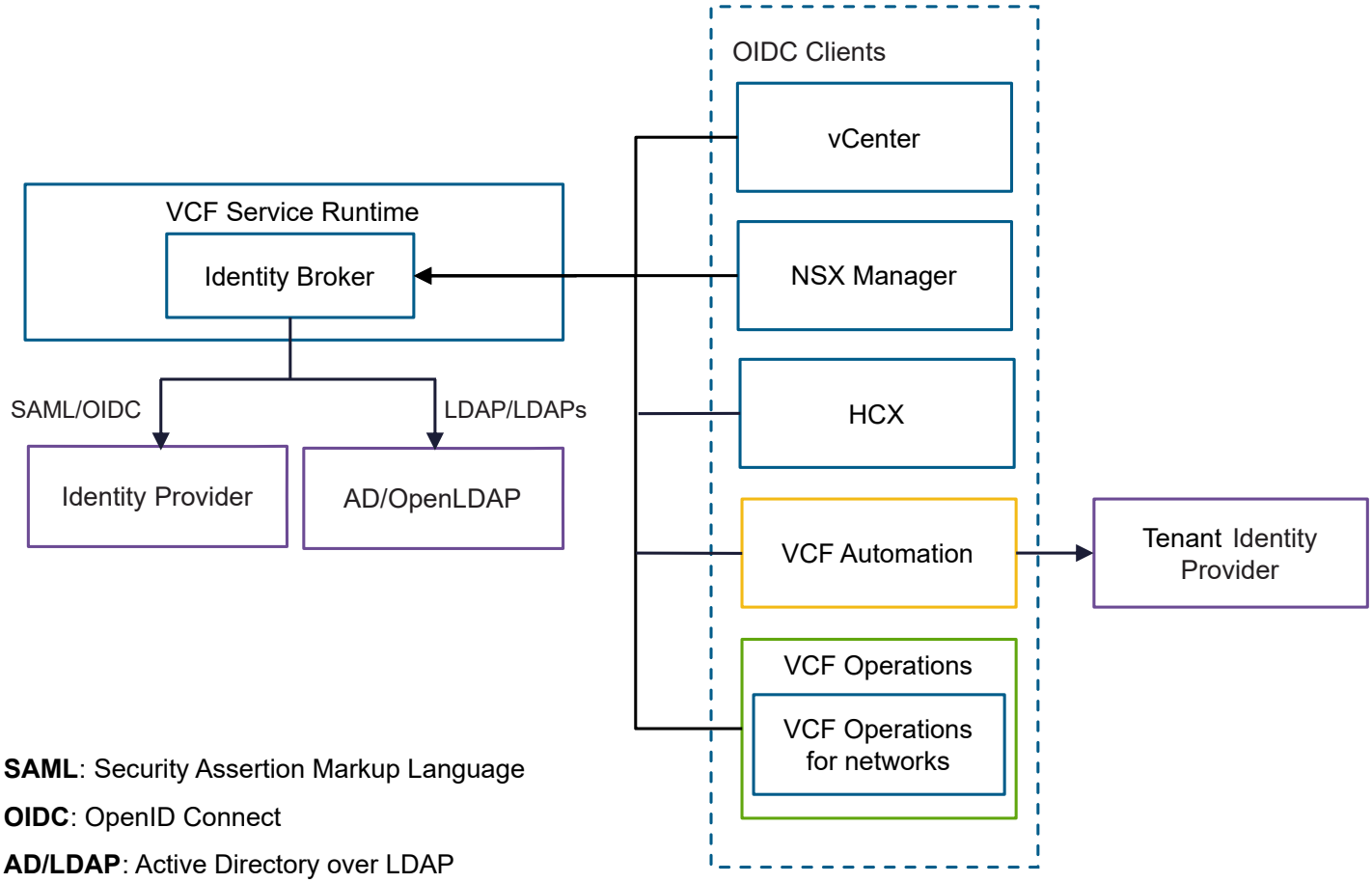
Instance Deployment Mode

The identity broker component is deployed as a VCF management services component within the management domain of the chosen VCF Instance. Your environment must meet the minimum requirements for the identity broker instance.

Figure 434: Instance Deployment Mode

The following diagram shows an identity broker instance that is deployed as part of VCF Management Services. The first identity broker instance is deployed in the primary VCF Instance. You can optionally deploy identity broker in other VCF Instances.

Figure 435:



SAML: Security Assertion Markup Language

OIDC: OpenID Connect

AD/LDAP: Active Directory over LDAP

Embedded Deployment Mode

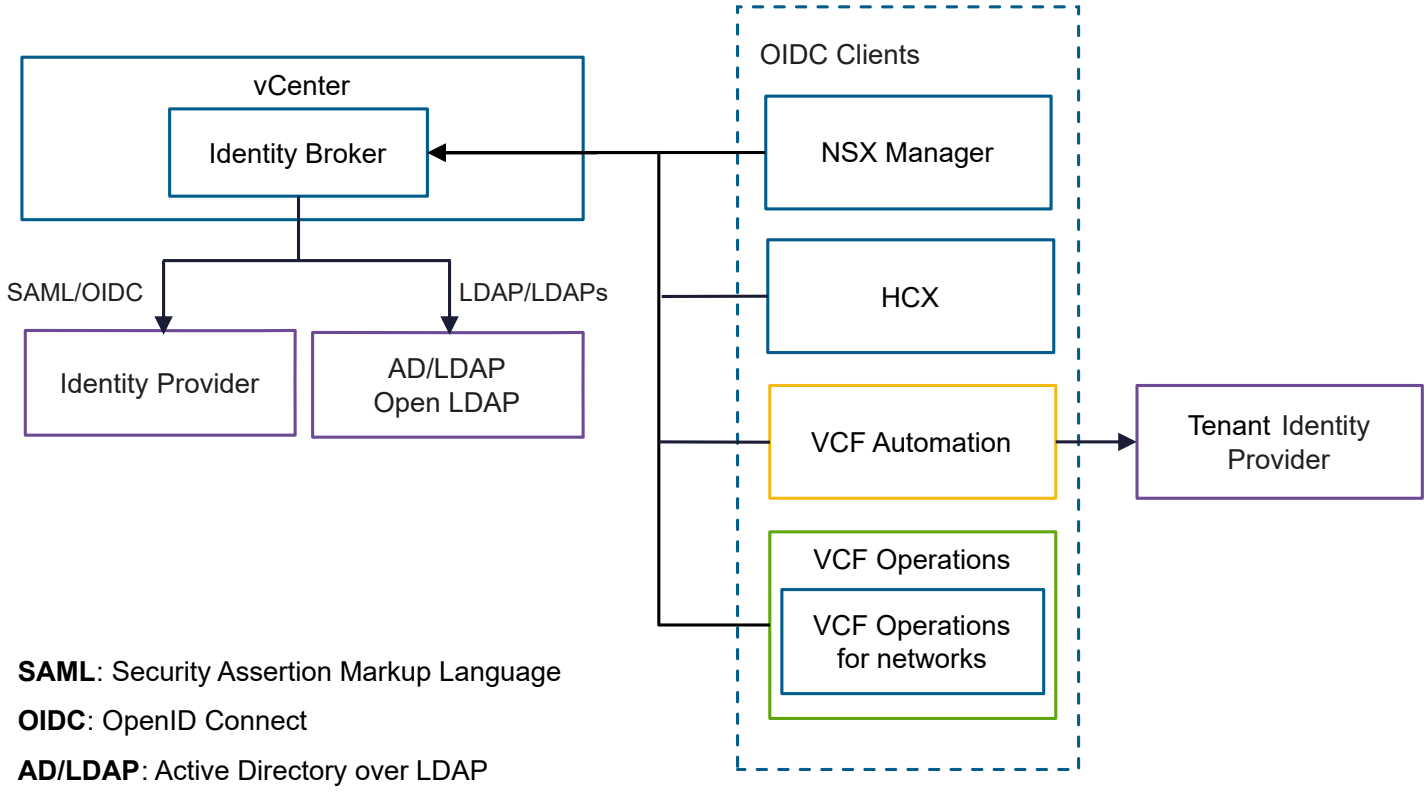
Use the embedded deployment mode if you want to use an identity broker that is embedded in the management domain vCenter. You use the embedded deployment mode typically within a single VCF Instance.

In the embedded deployment mode, the identity broker is configured in the management domain vCenter of the VCF Instance.

Figure 436: Embedded Deployment Mode

The following diagram shows the embedded deployment mode where the identity broker is configured in the management domain vCenter.

Figure 437:



For more information about design requirements, recommendations, and choices for each deployment mode, see Identity Broker Detailed Design.

Considerations and Prerequisites for VCF Single Sign-On

When you configure VCF single sign-on, it is important that you understand the following important points and ensure that you carry out the prerequisites.

Points to Consider While Setting Up VCF Single Sign-On

- **SDDC Manager UI Client authentication is not compatible with VCF single sign-on**
  After you configure VCF single sign-on in the management domain vCenter, administrators are no longer able to log in to the SDDC Manager UI client with VCF single sign-on users.
  - SDDC Manager UI supports logins only with local admin accounts such as `@vsphere.local users`.
  - The SDDC Manager APIs support logins with VCF single sign-on users.
- **VCF single sign-on is not compatible with Enhanced Linked Mode (ELM)**
  Before you configure VCF single sign-on, you must deactivate ELM on all participating vCenter Instances. For more information, see Deactivate Enhanced Link Mode for Standalone vCenter Instances Using the cmsso-util break-elm Utililty. If you upgraded from VCF 5.x to 9.0, see Deactivate Enhanced Link Mode for Management and Workload vCenter Instances Using the SDDC Manager API.
- **Manage VCF single sign-on only from VCF Operations and not from the individual components**
  After you configure VCF single sign-on from VCF Operations, you can no longer configure any identity sources in vCenter instances that are configured for VCF single sign-on.
- **PowerCLI impact**
  - PowerCLI automations continue to work if the identity provider is configured in the management domain vCenter in embedded mode.
  - If the identity provider is configured in an alternate management domain vCenter or in instance mode, the integration must be re-performed. Follow the steps in Using VCF Single Sign-On to Access vCenter using PowerCLI.

- **The users that configures VCF single sign-on in VCF Operations must have access to all objects**
  The 'All Objects' scope must be assigned in VCF Operations. Navigate to **Administration** > **Control Panel** > **Access Control** > **User Accounts/User Groups** and assign the 'All Objects' scope to the required user/group.
- **Behavior of identity and access management after setting up VCF single sign-on**
  - The management layer of VCF single sign-on is always in VCF Operations.
  - You can connect a standalone vCenter directly either to an identity provider that is configured within a vCenter or integrate them as part of VCF single sign-on using the identity broker in VCF Operations. However, after a vCenter is configured for single sign-on, you must manage its configuration in VCF Operations and not directly in vCenter.
- **VCF Components**
  - **vCenter:** After you configure VCF single sign-on, any existing identity provider configurations in the participating vCenter is overridden by the VCF single sign-on configuration. You might be required to perform role assignments to the required users and groups again to ensure continued access.
    Other VCF components: VCF single sign-on is an additional authentication source to the existing authentication sources such as VMware Identity Manager and the native authentication source of a component. You must provision users and groups again when you configure VCF single sign-on.
- **Single Component Log Out**
  - VCF single sign-on supports a single component logout mechanism. When you log out, the session ends exclusively for that particular VCF component within the designated browser.
- **Previously configured identity providers**
  - If you have previously configured an identity provider on a vCenter in the following combinations, the identity provider configurations are partially imported when you configure single sign-on. The combinations are:
    - Active Directory, OpenLDAP, or ADFS
      If Active Directory, OpenLDAP, or ADFS is the configured identity provider in vCenter, when configuring VCF single sign-on, the configuration is partially imported to identity broker. Only the configuration details that are not imported are required to be additionally configured.
    - Okta, Entra, or Ping
      The identity provider configuration is retained if the chosen deployment mode is embedded. If the deployment mode is instance, redo the configuration again.

    Note:

    The users and groups are not migrated. You must provision the users and groups again even if you have an existing identity provider configuration in place.

Prerequisites for Configuring VCF Single Sign-On

The identity broker is deployed in the management domain of the VCF Instance. Ensure that the following prerequisites are met before you configure VCF single sign-on.

Prerequisites for Embedded Identity Broker

For embedded deployment mode, ensure that the following prerequisites are met:

- The management domain vCenter must be at version 9.1 or later and must have a VCF license assigned.
- The management domain vCenter must NOT be part of Enhanced Linked Mode (ELM).

Do not activate ELM in the participating vCenter instances after configuring VCF Single Sign-On in embedded mode.

Note:

You can choose instance as the mode of deployment if the preceding criteria are not met. However, the integration of vCenter with VCF single sign-on is not possible at a later stage without first meeting the preceding criteria.

Prerequisites for Deploying Identity Broker as Instance

For instance mode, the VCF platform must meet the following minimum requirements:

- The initial identity broker instance is installed as part of VCF management services on the VCF services runtime nodes. Any additional identity broker instances that you deploy in additional VCF Instances require more compute resources to the VCF service runtime.
- The management domain vCenter must be at version 9.1 or later and must have a VCF license assigned.
- The management domain vCenter must not be part of Enhanced Linked Mode (ELM).

Deactivate Enhanced Link Mode for Standalone vCenter Instances Using the cmsso-util break-elm Utililty

VCF single sign-on is not compatible with vCenter Enhanced Linked Mode (ELM). Before you configure VCF single sign-on, you must deactivate ELM on all participating vCenter instances. As an administrator, if you want to deactivate ELM in vCenter Instances, you can do so without causing any loss to existing customer configurations such as global permissions.

If your vCenter instances are part not part of a VCF Instance, use the utility to deactivate ELM and related features, for example, vCenter linking and replication. Any shared configurations, such as tags, roles, and permissions that are in ELM, remain local to individual vCenter instances after you deactivate ELM.

> **Note:**
>
> The new **cmsso-util break-elm** utility works only if all vCenter instances in the ELM topology are up and running and vCenter instances are at version 9.0 or later. Back up all vCenter instances before you deactivate ELM. The only supported backup for vCenter instances in ELM is to take offline snapshots of them which results in environment downtime.

If your vCenter instances are part of a VCF Instance, see Deactivate Enhanced Link Mode for Management and Workload vCenter Instances Using the SDDC Manager API.

1. Take an offline backup of the participating vCenter instances in the ELM topology.

   This preparatory step requires downtime.
2. Connect over SSH to any vCenter instance that participates in ELM.
3. Run configuration pre-checks and fix any issues.
   a) Run `cmsso-util break-elm --mode pre-check --password <default administrator user_password>`.
   b) Fix any issues reported by the pre-check.

      Output of the break-elm precheck when successful:

      ```
      Command:cmsso-util break-elm --mode pre-check --password 'ZSIY+eDe-14dukrW'
      Output: Mode: pre-check This operation might take a few minutes to complete.
      [Pre-check] Invoking Break ELM topology workflow...
      [Pre-check] Fetch the peer nodes details...
      [Pre-check] Executing break ELM on local node...
      [Pre-check] Executing break ELM on all peer nodes...
      Break ELM pre-checks completed successfully.
      ```
4. Deactivate Enhanced Link Mode.
   a) Run `cmsso-util break-elm --mode execute --password <administrator_password>`.
   b) Log out from the vSphere Client and log in again for the changes to take effect.
   c) In case of a failure, fix the cause of the failure and rerun `cmsso-util break-elm` on all nodes where the failure was reported.
5. After successful validation that the break ELM process is successful, delete the offline backups.

You must resolve the ELM drift. For more information, see Resolve Enhanced Linked Mode (ELM) Drift for Upgraded vCenter Instances.

Deactivate Enhanced Link Mode for Management and Workload vCenter Instances Using the SDDC Manager API

If you upgraded your VCF Instance from 5.2 to 9.1, you must use the SDDC Manager API to deactivate Enhanced Link Mode (ELM) for upgraded vCenter instances that are part of your VCF fleet, before you can use certain 9.x features.

Minimum required version of the VCF components.

- SDDC Manager 9.0 and later
- vCenter upgraded to 9.0 and later

When you created a workload domain in VCF 5.x, you choose whether to join its vCenter to the management domain's vCenter single sign-on domain or to create a new vCenter single sign-on domain. If you joined it to the management domain's vCenter single sign-on domain, the workload domain vCenter was joined to the management domain vCenter in Enhanced Linked Mode (ELM).

ELM is deprecated in VCF 9.0. Before you can use new features in 9.0, such as VCF single sign-on and vCenter linking, you must deactivate ELM for all vCenter instances in your VCF Instance.

You can use the SDDC Manager API to remove all the vCenter instances in a VCF Instance from Enhanced Linked Mode (ELM), creating isolated workload domains.

> **Note:**
>
> This procedure deactivates ELM for all vCenter instances in your VCF Instance. You cannot select a subset of instances.

If you used the `cmsso-util break-elm` utility to remove some or all of your VCF-managed vCenter instances from ELM, see Resolve Enhanced Linked Mode (ELM) Drift for Upgraded vCenter Instances.

Back up all vCenter instances before you deactivate ELM. The only supported backup for vCenter instances in ELM is to take offline snapshots of them which results in environment downtime.

1. Take an offline backup of the participating vCenter instances in the ELM topology.

   This preparatory step requires downtime.

2. Use the SDDC Manager API to retrieve the management domain ID.

API	Description
GET /v1/domains	Find the management domain ID in the Response.

3. Acknowledge the offline backup of the participating vCenter instances in the ELM topology.

API	Description

PATCH /v1/domains/{domainId}	Replace *`{domainId}`* with the management domain ID.
	In the body parameter, enter:
	`{`
	`  "acknowledgmentSpec": {`
	`  "ackThatOfflineBackupsTaken": true`
	`  }`
	`}`

4. Validate the break ELM spec.

API	Description

POST /v1/domains/*{id}*/validations	Replace *{id}* with the management domain ID.
	In the body parameter, enter:
	`{`
	`"breakElmSpec": {`
	`"isReconcileWorkflow": false`
	`}`
	`}`
	Find the validation ID in the Response.

5. Get the validation response.

   Make sure that validation is successful.

API	Description
GET /v1/domains/*{id}*/validations/*{validation_id}*	Replace *{id}* with the management domain ID. Replace *{validation_id}* with the validation ID.

6. Break ELM across all joined domains.

API	Description

PATCH /v1/domains/*{id}*	Replace *{id}* with the management domain ID.
	In the body parameter enter:
	`{`
	`"breakElmSpec": {`
	`"isReconcileWorkflow": false`
	`}`
	`}`
	Find the task ID in the Response.

7. Monitor the task.

API	Description
GET /v1/tasks/*{id}*	Replace *{id}* with the task ID.

8. After successful validation that the break ELM process is successful, delete the offline backups.

Resolve Enhanced Linked Mode (ELM) Drift for Upgraded vCenter Instances

If you remove vCenter instances from ELM outside of SDDC Manager, some workflows are blocked until you reconcile the ELM drift using the SDDC Manager API.

Remove all the vCenter instances in your VCF Instance from ELM using the `cmsso-util break-elm` utility. See Deactivate Enhanced Link Mode for Standalone vCenter Instances Using the cmsso-util break-elm Utililty.

Until you resolve ELM drift, you might see error messages when running certain workflows. For example:
`Functionality CREATE_DOMAIN is not allowed because there is an ELM Drift. Ensure the ELM is completely broken, then run the reconcile workflow..`

1. Use the SDDC Manager API to retrieve the management domain ID.

API	Description
GET /v1/domains	Find the management domain ID in the Response.

2. Validate the break ELM reconcile workflow.

API	Description

POST /v1/domains/*{id}*/validations	Replace *{id}* with the management domain ID.
	In the body parameter, enter:
	`{`
	`"breakElmSpec": {`
	`"isReconcileWorkflow": true`
	`}`
	`}`
	Find the validation ID in the Response.

3. Get the validation response.

   Make sure that validation is successful.

API	Description
GET /v1/domains/*{id}*/validations/*{validation_id}*	Replace *{id}* with the management domain ID. Replace *{validation_id}* with the validation ID.

4. Resolve ELM drift across all joined domains.

API	Description

PATCH /v1/domains/*{id}*	Replace *{id}* with the management domain ID.
	In the body parameter, enter:
	`{`
	`"breakElmSpec": {`
	`"isReconcileWorkflow": true`
	`}`
	`}`
	Find the task ID in the Response.

5. Monitor the task.

API	Description
GET /v1/tasks/*{id}*	Replace *{id}* with the task ID.

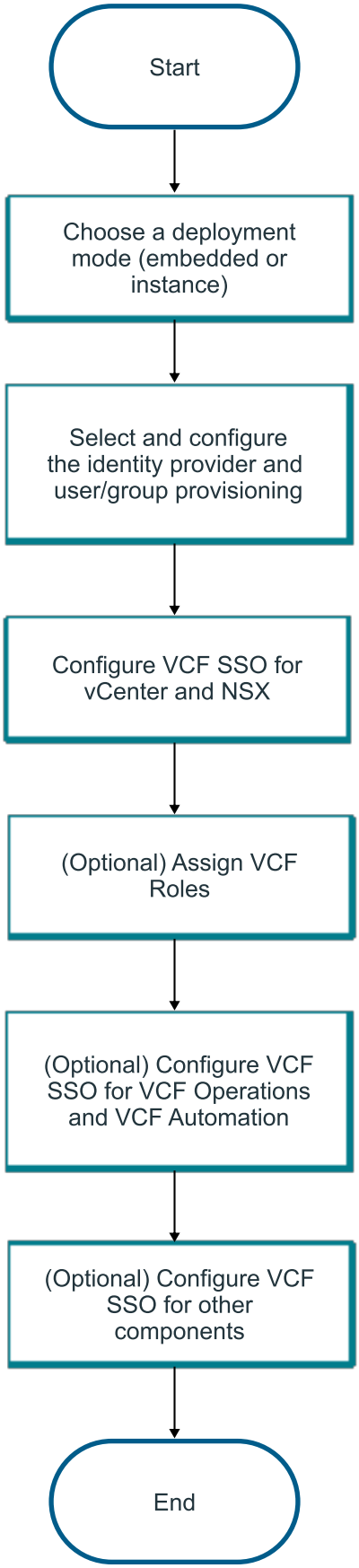
Configure a New VCF Single Sign-On for a VCF Instance

You can configure VCF single sign-on across components within a single VCF Instance, across a set of VCF Instances, and across the entire VCF fleet.

> **Important:**
>
> Ensure that you are familiar with the considerations and have completed the prerequisites before you set up VCF single sign-on. For more information, see Considerations and Prerequisites for VCF Single Sign-On.

Figure 438:

The following flowchart outlines the high-level steps to configure VCF single sign-on.


Start
Choose a deployment mode (embedded or instance)
Select and configure the identity provider and user/group provisioning
Configure VCF SSO for vCenter and NSX
(Optional) Assign VCF Roles
(Optional) Configure VCF SSO for VCF Operations and VCF Automation
(Optional) Configure VCF SSO for other components
End

Follow these high-level steps to configure VCF single sign-on:

1. **Select the identity broker deployment mode.**
   Select whether you want to deploy identity broker as embedded or instance. See, Step 1: Choose the Deployment Mode.
2. **Configure the identity provider.**
   Set up your preferred identity provider using modern or directory-based authentication. See Step 2: Configure an Identity Provider.
3. **Configure VCF single sign-on for NSX and vCenter.**
   See Step 3: Configure VCF Single Sign-On for NSX and vCenter.
4. **(Optional) Assign a VCF Role**
   See Step 4. (Optional) Assign a VCF Role.
5. **(Optional) Configure VCF single sign-on for VCF Operations and VCF Automation.**
   Connect the components to the identity broker for single sign-on-based access. See Step 5: (Optional) Configure VCF Single Sign-On for VCF Operations, VCF Automation Provider, and Salt RaaS.
6. **(Optional) Configure VCF single sign-on for other VCF Components.**
   Connect the components for single sign-on access. See, Step 6: (Optional) Configure VCF Single Sign-On for other VCF Components.

Step 1: Choose the Deployment Mode

Choose a deployment mode for the identity broker configuration. The options available are embedded or instance. For more information about the VCF single sign-on deployment modes, see Deployment Modes of the Identity Broker.

The identity broker instance is deployed by default in the first VCF Instance of a VCF fleet. For more information, see Start a New VCF Fleet or a New VCF Instance Deployment by Using the VCF Installer Deployment Wizard.

Procedure

1. Log in to the VCF Operations console at `https://<vcf_operations_fqdn>` with a user assigned the Administrator role.
2. In the VCF Operations navigation bar at the top, click **Manage** and in the left navigation pane, click **Fleet Management** > **Identity and Access**.
3. From the **Identity and Access** pane, click **VCF SSO Overview**, and then from the **Get Started** page on the right, click the **Get Started with SSO** tab.
4. In the **Get Started with SSO** tab, click **Configure SSO.**
5. In the **Prerequisites** tab, confirm each check box and click **Submit**. For more information about the prerequisites, see Considerations and Prerequisites for VCF Single Sign-On.
6. In the **Configure VCF SSO** page, click the **Start** button against the **Choose deployment mode** option.
7. Select either **Instance (Recommended)** or **Embedded**.
   a) If you choose **Instance**, from the **Choose Identity Broker** section, select an existing identity broker. To deploy a new identity broker instance, see Deploy Identity Broker as a Day-N Operation.
   b) If you choose **Embedded**, select a VCF Instance from the **Choose VCF Instance** section.

      Note:

      Identity broker embedded will be deprecated in later versions of VCF and you have to migrate to identity broker instance before identity broker embedded reaches end-of-support. For information about migration, see Migration of Identity Broker Embedded to Identity Broker Instance.
8. Click **Configure**.

- While configuring a new VCF single sign-on, you can change the deployment type only if you have not yet configured an identity provider.
- However, you can change the deployment mode as a Day N operation. For more information see, Change the Deployment Mode in VCF Single Sign-On.

Step 2: Configure an Identity Provider

After you select the deployment mode, you must select and configure the identity provider using SAML, OIDC, or LDAP as the authentication protocol.

For information about supported identity providers and protocols, see Identity Providers and Protocols Supported for VCF Single Sign-On.

- To configure Okta as an identity provider, see Configure Okta as an Identity Provider.
- To configure Ping Identity as an identity provider, see Configure Ping Identity as an Identity Provider.
- To configure Microsoft Entra ID as an identity provider, see Configure Microsoft Entra ID as an Identity Provider.
- To configure Microsoft ADFS as an identity provider, see Configure Microsoft ADFS as an Identity Provider.
- To configure Symantec Identity Security Platform as an identity provider, see Configure Symantec Identity Security Platform as an Identity Provider.
- To configure an identity provider using Generic SAML 2.0, see Configure an Identity Provider using Generic SAML 2.0.
- To configure an identity provider using Generic OIDC, see Configure an Identity Provider using Generic OIDC.
- To configure Active Directory as an identity provider using AD/LDAP, see Configure Active Directory as an Identity Provider Using AD/LDAP.
- To configure OpenLDAP as an identity provider, see Configure OpenLDAP as an Identity Provider.

> **Note:**
>
> For existing deployments, when an identity provider is configured on a vCenter instance, and is configured in the following combinations, you can use the preselected existing identity provider or choose a new identity provider in the UI workflows:
>
> - Active Directory, OpenLDAP, or ADFS
> - Okta, Microsoft Entra ID, or Ping Federate with OIDC

Configure Okta as an Identity Provider

You can configure Okta as an identity provider using SAML 2.0 or OIDC authentication protocols.

- For information about configuring Okta as an identity provider using SAML as the authentication protocol, see Configure Okta as an Identity Provider Using SAML.
- For information about configuring Okta as an identity provider using OIDC as the authentication protocol, see Configure Okta as an Identity Provider using OIDC.

Configure Okta as an Identity Provider Using SAML

After installing or upgrading to VCF 9.1, to configure VCF single sign-on, you can configure Okta as an identity provider using SAML as the authentication protocol.

To set up an identity provider, you must:

1. Choose an identity provider
2. Configure the identity provider
3. Configure user and group provisioning
4. Test login

> **Note:**
>
> You can test log in if the identity broker is at version 9.1.

Prerequisite

- Verify that you have chosen a deployment mode. For more information, see Step 1: Choose the Deployment Mode.

Procedure

Choose the identity provider

1. From the **Configure VCF SSO** page, click the **Start** button against the **Configure Identity Provider** option.
2. From the **Choose the identity provider** section, select **OKTA** from the list and click **Next**.

   Configure the identity provider
3. From the **Configure the Identity Provider** section, click **Configure** to configure the identity broker to integrate with the selected identity provider for user authentication.
4. In the **Authentication Method** screen, select **SAML** (Security Assertion Markup Language) and click **Next**.
5. In the **Service Provider Registration** screen, log in to your Okta admin console and follow the steps in Add a private single sign-on integration to create a SAML 2.0 application in Okta.

   While registering Okta, use the metadata provided in the **Service Provider Registration** screen for the **Entity ID/ Metadata URL** and **Assertion Consumer Service URL** options. If the identity provider requires the metadata to be copy-pasted, it is recommended to download this file to avoid formatting issues.

   Note:

   Ensure that you add the following default claims when you configure your external identity provider. Note that these claims are case-sensitive:

   - userName
   - email
   - domain
   - firstName
   - lastName
   - groups
6. Click **Next**.
7. In the **Identity Provider Configuration** screen, enter the relevant details or select the required options and click **Next**.

Option	Description
IDP Display Name	Enter a friendly name for your identity provider.
Metadata XML	Copy the Metadata XML of your identity provider and paste it in this field. The Metadata XML is a URL that points to an XML document containing metadata about the service provider (identity broker). This metadata includes important information required for establishing trust and facilitating secure communication between the service provider and the identity provider. **Note:** If you have configured VCF single sign-on, after you upgrade to 9.1.1, it is recommended that you re-import the identity broker service provider metadata into your identity provider. Re-importing ensures your identity provider reads the new encryption hint and switches to SHA-256 OAEP for all subsequent SAML assertion encryption.
Metadata URL	Copy the Metadata URL of your identity provider and paste it in this field. The Metadata URL is used to fetch identity provider metadata. This metadata includes important information required for establishing trust and facilitating secure communication between the service provider and the identity provider. **Note:** If your identity provider is not publicly accessible or if the certificate lacks a signature from a recognized Certificate Authority (CA), VCF Operations cannot validate the metadata URL. Provide the metadata in XML format instead.
Name ID format and Name ID value	Select the Name ID format and value. The Name ID format is the value in the SAML response to identify the authenticated user.

Option	Description
SAML Context	SAML context refers to the method of authentication that VCF requires the identity provider to support when verifying a user's identity. • **PasswordProtectedTransport**: Authentication with a password sent over a secure transport (For example, HTTPS). • **Password**: Authentication using a password without requiring transport security. • **Unspecified:** If you don't want to enforce any particular type of authentication mechanism, you can use this option.

8. In the **User/Group Provisioning Method** screen, select your preferred mode of provisioning the users and groups. You can select either:
   - System for Cross-domain Identity Management (SCIM), or
   - Just-In-Time Provisioning (JIT), or
   - Active Directory/Lightweight Directory Access Protocol (AD/LDAP)

   a) If you select **SCIM** and click **Next**, complete the configuration as follows.
      - In the **Domain(s)** screen, enter the domain that users from your enterprise will log in with, and click **Next**.

   b) If you select **JIT** and click **Next**, complete the configuration as follows:
      - In the **Group Provisioning** screen, select your preferred mode of provisioning the groups. The options are **Groups Pre-provisioning** or **Just-In-Time Group Provisioning** and click **Next**.

Option	Description
Groups Pre-Provisioning	This option creates the required groups in advance to perform role assignments. Users are assigned to these groups dynamically when they log in based on SAML assertion. Only the pre-provisioned groups are available for role assignments.
Just-In-Time Group Provisioning	This option creates groups dynamically based on the SAML assertion when the user logs in.

      - In the **Domain(s)** screen, add the domain details that users from your enterprise will log in with, and click **Next**.
      - In the **Pre-provisioning Domains and Groups** screen, add the groups from each domain that you want to provision in VCF for roles and assignments by clicking the **Add Group** button, and click **Next**.

      Note:

      The **Pre-provisioning Domains and Groups** screen is displayed only if you have chosen groups to be pre-provisioned which means that you chose **Groups pre-provisioning** in the previous screen.

      There should be at least one provisioned group added across the domains.
      - In the **Attributes** screen, the user attributes section displays a list of mandatory and non-mandatory user attributes that you view in the SAML response from the identity provider.
        You can add additional user attributes if required. In the group attributes section of the workflow you can add a group attribute to be called for in the SAML request. Attribute names are case-sensitive.

      Note:

      - If you choose JIT, it is recommended that you automatically disable inactive users.

   c) If you select **AD/LDAP** and click **Next**, complete the configuration as follows. To learn more about the benefits of using AD/LDAP as a provisioning method, see KB 386870.
      - In the **Directory Details** screen, enter the following details or select the required options.

Option	Description
Directory name	Enter a name for the directory.
DNS Server Location	Select this option if you want the identity broker to find and use optimal domain controllers by querying the DNS Service Record.

Option	Description
Global Catalog	Global catalog has a partial read-only replica of all objects across all domains. Some of the limitations with selecting the global catalog option include: • The Active Directory object attributes that are replicated to the global catalog are identified in the Active Directory schema as the Partial Attribute Set (PAS). Only these attributes are available for attribute mapping by the service. If necessary, edit the schema to add or remove attributes that are stored in the global catalog. • The global catalog stores the group membership (the member attributes) of only universal groups. Only universal groups are synced to the service. If necessary, change the scope of a group from a local domain or from global to universal. • The bind DN account that you define when you configure a directory in the service must have permissions to read the Token-Groups-Global-And-Universal (TGGAU) attribute. **Note:** When you use Global catalog with LDAP and multiple domains, the additional domains are discovered only after the user/group sync is complete.
Encrypted Connection (Displayed when you select the **DNS server location** option)	This option activates LDAPs for all connections to this directory.
Certificates for encrypted connection (Displayed when you select the **DNS server location** and **Encrypted Connections** options and if you enter a *ldaps://* string in the Primary domain controller field)	Upload the root certificate(s) required to access the domain controllers over SSL. All uploaded root certificates will be merged into a single file with the filename `{directory-name}-concatenated.pem` when you save the configuration. **Note:** Ensure that the certificate is in the (Privacy Enhanced Mail) PEM format and includes the "BEGIN CERTIFICATE" and "END CERTIFICATE" lines.
Directory search attribute	A specific LDAP attribute used in queries to search or identify entries in the directory. For example, `uid`, `sAMAccountName`.
Primary domain controller (Displayed when you select the **Global Catalog** option)	A server machine running Active Directory Domain Services (AD DS) responsible for managing and authenticating user access, enforcing policies, and storing directory data. For example, `hostname = DC01` for the domain called example.com.
Secondary domain controller (Displayed when you select the **Global Catalog** option)	A backup domain controller for the identity broker to point to if the primary domain controller is not reachable.
Base DN	The Distinguished Name (DN) of the top-level entry in the directory tree from which LDAP queries begin. For example, `dc=example`, `dc=com`.
Bind user name	The user name of a user account with sufficient permissions to authenticate and perform LDAP operations on the directory.
Bind user password	The password of a user account with sufficient permissions to authenticate and perform LDAP operations on the directory.

9. In the **Review** screen, review the details you have added for the configuration and click **Finish**.

Configure user and group provisioning

10. In the **Configure User and Group Provisioning** screen, depending on the option you choose, complete the following steps and click **Done**.
    a) If you have chosen SCIM, you see **Configure SCIM push in your identity provider** instead of **Configure User and Group Provisioning**. Use the values listed on the screen to configure the SCIM application in the identity provider.
       • If your identity provider supports authorizing the SCIM application using OAuth grant type, use the client ID, client secret, and token endpoint. Otherwise, use the SCIM bearer token to authorize this SCIM server in your identity provider.

- If required, you can regenerate these values after you have configured VCF Single Sign-On. To regenerate the values to configure the SCIM application in the identity provider, navigate to the respective VCF Instance and click on **Identity Source**. The SCIM related values are available under **Directory Information**.
- Ensure that the user attributes, username, and domain are pushed individually from the SCIM application by the identity provider. Similarly, ensure that the group attributes, group name, and domain are pushed individually as well.

b) If you have chosen JIT, there is no additional step to be performed under **User and Group Provisioning**.

c) If you have chosen AD/LDAP, review the directory information such as the domain controller and bind credentials that you had configured. Complete the following steps:

1. In the **Attributes Mappings** screen, review the attribute mappings and click **Next.**
   The attributes in the identity broker directory are mapped to the Active Directory attributes for accurate synchronization of user information. If the mappings are incorrect, select the correct attribute from the drop-down list.
2. In the **Group Provisioning** screen, enter the relevant details or select the required options and click **Next**.
   Use the **Group Provisioning** screen to search for the group names and select the corresponding Active Directory group Distinguished Names (DN) that you want to sync.

Option	Description
Specify the Base Group DN	Click the **Select Base Group DN** button if you have nested groups and want to specify the group names. You can search for the group name and select the group DNs that you want to sync from from the list in the datagrid below.
Sync Nested Groups	Select **Sync Nested Group** to sync all the users from the selected group and its nested groups without maintaining the group hierarchy.

3. From the **User Provisioning** screen, enter the relevant details or select the required options and click **Next**.
   Use the **User Provisioning** screen to search for the user names and select the corresponding Active Directory group Distinguished Names (DN) that you want to sync.

Option	Description
Specify the Base User DN	Click the **Select Base User DN** button to specify the user names that you want to sync. You can search for the user name and select the user DNs that you want to sync from from the list in the datagrid below. If you want to sync the users directly from the group, do not specify a base user DN.

4. From the **Review screen**, review the configuration details of the identity provider.
5. Click **Finish**. The sync starts in the background. After the initial sync, the sync is scheduled to run once a week at a scheduled time.
6. Click **Done**.

For more information about the various functionalities you can perform as part of the management operations in Active Directory, see Managing AD/LDAP.

(Optional) Test Login

11. Click **Test Login** to test the login using your single-sign on credentials.
12. Click **Next** in the identity provider screen.
13. After a successful validation, click **Done**.
14. Click **Finish Setup**. This completes the initial configuration.
15. Click **Continue** to view the **VCF SSO Overview** page.

Important:

Important Notes on Backup

It is highly advisable to create a backup of both VCF Operations and the identity broker instance. Furthermore, it is advisable to perform a similar backup following any substantial modifications to the initial identity source

configuration, such as when you 'Delete Single Sign-On'. This practice ensures that you can restore the VCF single sign-on configuration to a recognized and stable state if necessary. You can download the configuration as a JSON file.

Important Notes on Role Assignment

VCF roles can be created and mapped to individual component roles.

- As an admin you can assign built-in and custom VCF roles to groups and and users from the **Access Managment** tab (navigate to the VCF SSO Overview page, click on a VCF single sign-on configuration, and then click the Access Management tab).
- As an admin, you can also assign roles in the component console and enable VCF single sign-on from the **Component Configuration** tab (navigate to the VCF SSO Overview page, click on a VCF single sign-on configuration, and then click the Component Configuration).

Configure Okta as an Identity Provider using OIDC

After installing or upgrading to VCF 9.1, to configure VCF single sign-on, you can configure Okta as an identity provider using OIDC as the authentication protocol.

To set up an identity provider, you must:

1. Choose an identity provider
2. Configure the identity provider
3. Configure user and group provisioning
4. Test Login

Note:

You can test log in if the identity broker is at version 9.1.

Prerequisites

- Verify that you have chosen a deployment mode. For more information, see Step 1: Choose the Deployment Mode.
  If you configure VCF single sign-on using identity broker instance, and there are network restrictions that do not allow outbound public internet access, you must set up a proxy server. For more information, see Configure a Proxy Server for VCF Management Services Components and VCF Automation.

Procedure

Choose the identity provider

1. From the **Configure VCF SSO** page, click the **Start** button against the **Configure Identity Provider** option.

   If you have an existing identity provider configured in vCenter, see the information below. For a fresh configuration of the identity provider, continue from Step 6.

   For an existing source (existing identity provider configuration)

   - Choose either SAML or OIDC as the authentication protocol.

     Note:

     If you choose SAML, the VCF Single Sign-On configuration is considered as a new authentication source configuration. Continue from Step. 6.
   - If you choose OIDC:
     – If you choose the instance deployment mode, and select OIDC, the following fields are prepopulated in the **Identity Provider Configuration** screen.
       a. Review the details and provide the shared secret and click **Next**.
          - Identity provider name
          - Client identifier
          - OpenID address
          - Update Redirect URI in the Identity provider's OIDC application
          - Shared Secret: Paste the client secret of the OIDC application you created in the identity provider.
       b. Review the **User/Group Provisioning**, **Domain(s)**, and **Review** screens and click **Finish**.
       c. Review the SCIM push details and click **Done**. The identity provider configuration is complete.

- If you choose the embedded deployment mode, and select OIDC:
  a. The configuration details of the identity provider configuration are automatically imported from existing configuration into the identity broker with OIDC as the authentication protocol and SCIM as the user/group provisioning method.
  b. Review the configuration values in the **Identity Provider Configuration** screen and click **Finish**.
  c. Review the SCIM details, click **Done**, and then click **Finish Setup**. The identity provider configuration is complete.

2. For a fresh identity provider configuration, from the **Choose the identity provider** section, select **Okta** from the list and click **Next**.

   Configure the identity provider

   Note:

   Ensure that you add the following default claims when you configure your external identity provider. These claims are case-sensitive:

   - preferred_username
   - email
   - family_name
   - given_name
   - groups
   - domain

   If the default claims cannot be added, use alternative claim names and map them to the appropriate attributes in the VCF Operations **Identity & Access** page to support your JIT configurations.
3. From the **Configure the Identity Provider** section, click **Configure** to configure the identity broker to integrate with the selected identity provider for user authentication.
4. In the **Authentication Method** screen, select **OIDC** (OpenID Connect) and click **Next**.
5. In the **Identity Provider Configuration** screen, enter the relevant details or select the required options and click **Next**.

Option	Description
IDP Display Name	Enter a friendly name for your identity provider. Example, Okta.
Client Identifier	A unique ID assigned to your OIDC application by the identity provider for authentication.
Shared secret	A confidential key used by your OIDC application to authenticate itself with the identity provider. **Note:** Client secrets issued by external identity providers typically have expiration dates. When a client secret expires, it must be regenerated, and the corresponding OIDC configuration must be updated in VCF to ensure continued authentication functionality.
OpenID Address	The URL of the identity provider's OIDC configuration endpoint for initiating the authentication process.
Redirect URI	The URL where the identity provider sends the authentication response after a user logs in.

6. In the **User/Group Provisioning Method** screen, select your preferred mode of provisioning the users and groups. You can select either:
   - System for Cross-domain Identity Management (SCIM), or
   - Just-In-Time Provisioning (JIT), or
   - Active Directory/Lightweight Directory Access Protocol (AD/LDAP)

   a) If you select **SCIM** and click **Next**, complete the configuration as follows.
      - In the **Domain(s)** screen, enter the domains that users from your enterprise will log in with, and click **Next**.

b) If you select **JIT** and click **Next**, complete the configuration as follows:

- In the **Group Provisioning** screen, select your preferred mode of provisioning the groups. The options are **Groups Pre-provisioning** or **Just-In-Time Group Provisioning** and click **Next**.

Option	Description
Groups Pre-Provisioning	This option creates the required groups in advance to perform role assignments. Users are assigned to these groups dynamically when they log in based on OIDC token. Only the pre-provisioned groups are available for role assignments.
Just-In-Time Group Provisioning	This option creates groups dynamically based on the OIDC token when the user logs in. Alternatively, you can use group pre-provisioning if you know the group names from Active Directory that requires entitlements.

- In the **Domain(s)** screen, add the domain details that users from your enterprise will log in with, and click **Next**.
- In the **Pre-provisioning Domains and Groups** screen, add the groups from each domain that you want to provision in VCF for roles and assignments by clicking the **Add Group** button, and click **Next**.

Note:

The **Pre-provisioning Domains and Groups** screen is displayed only if you have chosen groups to be pre-provisioned which means that you chose **Groups pre-provisioning** in the previous screen.

There should be at least one provisioned group added across the domains.

- In the **Attributes** screen, the user attributes section displays a list of mandatory and non-mandatory user attributes that you view in the OIDC response from the identity provider.
  You can add additional user attributes if necessary. In the group attributes section of the workflow you can add a group attribute to be called for in the OIDC request. Attribute names are case-sensitive.

Note:

If you use public APIs to update identity provider attribute mappings, the following identity broker attributes are available and can be mapped to OpenID attributes:

- ExternalId
- userName
- firstName
- lastName
- email
- phone
- active
- employeeID
- distinguishedName
- userPrincipalName
- domain
- sourceAnchor
- displayName
- nickName
- profileUrl
- title
- organization
- costCenter
- department
- division
- managerId

Note:

If you choose JIT, it is recommended that you automatically disable inactive users.

c) If you select **AD/LDAP** and click **Next**, select **Unique Identifier in OIDC Identity Provider** and **Unique Identifier in VCF Identity Broker** and click **Next**. To learn more about the benefits of using AD/LDAP as a provisioning method, KB 386870.

Note:

The value of the identifier used to uniquely identify a user in your OIDC identity provider must exactly match with the value from Active Directory. The unique identifier in the identity broker is used to map the two systems.

Option	Description
Unique Identifier in OIDC Identity Provider	An identifier in your OIDC identity provider that is used to identify a user uniquely. You can select or enter a custom attribute (except 'sub'). For example, `name`.
Unique Identifier in VCF Identity Broker	Select the unique identifier that will be used to map to an attribute from Active Directory. The value of this identifier must exactly match the value of the previous field called **Unique Identifier in OIDC Identity Provider**.

- In the **Directory Details** screen, enter the relevant details or select the required options.

Option	Description
Directory name	Enter a name for the directory. The directory name should contain only letters, digits, spaces, hyphens, and underscores.
DNS Server Location	Select this option if you want the identity broker to find and use optimal domain controllers by querying the DNS Service Record.
Global Catalog	Global catalog has a partial read-only replica of all objects across all domains. Some of the limitations with selecting the global catalog option include: • The Active Directory object attributes that are replicated to the global catalog are identified in the Active Directory schema as the Partial Attribute Set (PAS). Only these attributes are available for attribute mapping by the service. If necessary, edit the schema to add or remove attributes that are stored in the global catalog. • The global catalog stores the group membership (the member attributes) of only universal groups. Only universal groups are synced to the service. If necessary, change the scope of a group from a local domain or from global to universal. • The bind DN account that you define when you configure a directory in the service must have permissions to read the Token-Groups-Global-And-Universal (TGGAU) attribute. **Note:** When you use Global catalog with LDAP and multiple domains, the additional domains are discovered only after the user/group sync is complete.
Encrypted Connection (Displayed when you select the **DNS server location** option)	Activates LDAPs for all the connections to this directory.

Certificates for encrypted connection (Displayed when you select the **DNS server location** and **Encrypted Connections** options and if you enter a *ldaps://* string in the Primary domain controller field)	Upload the root certificate(s) required to access the domain controllers over SSL. All uploaded root certificates will be merged into a single file with the filename `{directory-name}-concatenated.pem` when you save the configuration. **Note:** Ensure that the certificate is in the (Privacy Enhanced Mail) PEM format and includes the "BEGIN CERTIFICATE" and "END CERTIFICATE" lines.
Directory search attribute	A specific LDAP attribute used in queries to search or identify entries in the directory. For example, `uid`, `sAMAccountName`.
Primary domain controller	A server machine running Active Directory Domain Services (AD DS) responsible for managing and authenticating user access, enforcing policies, and storing directory data. For example, `hostname = DC01` for the domain called example.com.
Secondary domain controller	A backup domain controller for the identity broker to point to if the primary domain controller is not reachable.
Directory search attribute	A specific LDAP attribute used in queries to search or identify entries in the directory. For example, `uid`, `sAMAccountName`.
Base DN	The Distinguished Name (DN) of the top-level entry in the directory tree from which LDAP queries begin. For example, `dc=example`, `dc=com`.
Bind user name	The user name of a user account with sufficient permissions to authenticate and perform LDAP operations on the directory.
Bind user password	The password of a user account with sufficient permissions to authenticate and perform LDAP operations on the directory.

7. In the **Review** screen, review the details that you have added for the configuration and click **Finish**.

Configure user and group provisioning

8. In the **Configure User and Group Provisioning** screen, depending on the option you choose, complete the following steps and click **Done**.

   If you have chosen OIDC as the authentication protocol, and AD/LDAP as the user and group provisioning protocol, read KB 386870 to understand the finer details about configuring the setup.

   a) If you have chosen SCIM, you see **Configure SCIM push in your identity provider** instead of **Configure User and Group Provisioning**. Use the values listed on the screen to configure the SCIM application in the identity provider.

      - If your identity provider supports authorizing the SCIM application using OAuth grant type, use the client ID, client secret, and token endpoint. Otherwise, use the SCIM bearer token to authorize this SCIM server in your identity provider.
      - If required, you can regenerate these values after you have configured VCF single sign-on. To regenerate the values to configure the SCIM application in the identity provider, navigate to the respective VCF Instance and click on **Identity Source**. The SCIM related values are available under **Directory Information**.
      - Ensure that the user attributes, username, and domain are pushed individually from the SCIM application by the identity provider. Similarly, ensure that the group attributes, group name, and domain are pushed individually as well.

   b) If you have chosen JIT, there is no additional step to be performed under **User and Group Provisioning**.

   c) If you have chosen AD/LDAP, review the directory information such as the domain controller and bind credentials that you had configured. Complete the following steps:

      1. In the **Attributes Mappings** screen, review the attribute mappings and click **Next.**

The attributes in the identity broker directory are mapped to the Active Directory attributes for accurate synchronization of user information. If the mappings are incorrect, select the correct attribute from the drop-down list.

2. In the **Group Provisioning** screen, enter the relevant details or select the required options and click **Next**.
   Use the **Group Provisioning** screen to search for the group names and select the corresponding Active Directory group Distinguished Names (DN) that you want to sync.

Option	Description
Specify the Base Group DN	Click the **Select Base Group DN** button if you have nested groups and want to specify the group names. You can search for the group name and select the group DNs that you want to sync from from the list in the datagrid below.
Sync Nested Groups	Select **Sync Nested Group** to sync all the users from the selected group and its nested group without maintaining the group hierarchy.

3. From the **User Provisioning** screen, enter the relevant details or select the required options and click **Next**.
   Use the **User Provisioning** screen to search for the user names and select the corresponding Active Directory group Distinguished Names (DN) that you want to sync.

Option	Description
Specify the Base User DN	Click the **Select Base User DN** button to specify the user names that you want to sync. You can search for the user name and select the user DNs that you want to sync from from the list in the datagrid below. If you want to sync the users directly from the group, do not specify a base user DN.

4. From the **Review screen**, review the configuration details of the identity provider.
5. Click **Finish**. The sync starts in the background. After the initial sync, the sync is scheduled to run once a week at a scheduled time.
6. Click **Done**.

For more information about the various functionalities you can perform as part of the management operations in Active Directory, see Managing AD/LDAP.

(Optional) Test Login

9. Click **Test Login** to test the login using your single-sign on credentials.
10. Click **Next** in the identity provider screen.
11. After a successful validation, click **Done**.
12. Click **Finish Setup**. This completes the initial configuration.
13. Click **Continue** to view the **VCF SSO Overview** page.

Important:

Important Notes on Backup

It is highly advisable to create a backup of both VCF Operations and the identity broker instance. Furthermore, it is advisable to perform a similar backup following any substantial modifications to the initial identity source configuration, such as when you 'Delete Single Sign-On'. This practice ensures that you can restore the VCF single sign-on configuration to a recognized and stable state if necessary. You can download the configuration as a JSON file.

Important Notes on Role Assignment

VCF roles can be created and mapped to individual component roles.

- As an admin you can assign built-in and custom VCF roles to groups and and users from the **Access Managment** tab (navigate to the VCF SSO Overview page, click on a VCF single sign-on configuration, and then click the Access Management tab).

- As an admin, you can also assign roles in the component console and enable VCF single sign-on from the **Component Configuration** tab (navigate to the VCF SSO Overview page, click on a VCF single sign-on configuration, and then click the Component Configuration).

Configure Ping Identity as an Identity Provider
You can configure Ping Identity as an identity provider using SAML or OIDC authentication protocols.

- For information about configuring Ping Identity as an identity provider using SAML as the authentication protocol, see Configure Ping Identity as an Identity Provider Using SAML.
  For information about configuring Ping Identity as an identity provider using OIDC as the authentication protocol, see Configure Ping Identity as an Identity Provider Using OIDC.

Configure Ping Identity as an Identity Provider Using SAML

After installing or upgrading to VCF 9.1, to configure VCF single sign-on, you can configure Ping Identity as an identity provider using SAML as the authentication protocol.

To set up an identity provider, you must:

1. Choose an identity provider
2. Configure the identity provider
3. Configure user and group provisioning
4. Test Login

Note:

You can test log in if the identity broker is at version 9.1.

Prerequisite

- Verify that you have chosen a deployment mode. For more information, see Step 1: Choose the Deployment Mode.

Procedure

Choose the identity provider

1. From the **Configure VCF SSO** page, click the **Start** button against the **Configure Identity Provider** option.
2. From the **Choose the identity provider** section, select **Ping Identity** from the list and click **Next**.

Configure the identity provider

3. From the **Configure the Identity Provider** section, click **Configure** to configure the identity broker to integrate with the selected identity provider for user authentication.
4. In the **Authentication Method** screen, select **SAML** (Security Assertion Markup Language) and click **Next**.
5. In the **Service Provider Registration** screen, log in to your Ping Identity admin console and follow the steps in Ping Identity to create a SAML 2.0 application in Ping Identity.

   While registering Ping Identity, use the metadata provided in the **Service Provider Registration** screen for the **Entity ID/Metadata URL** and **Assertion Consumer Service URL** options. If the identity provider requires the metadata to be copy-pasted, it is recommended to download this file to avoid formatting issues.

   Note:

   Ensure that you add the following default claims when you configure your external identity provider. Note that these claims are case-sensitive:

   - userName
   - email
   - domain
   - firstName
   - lastName
   - groups

6. Click **Next**.
7. In the **Identity Provider Configuration** screen, enter the relevant details or select the required options and click **Next**.

Option	Description
IDP Display Name	Enter a friendly name for your identity provider.
Metadata	Select either URL or XML.
Metadata XML	Copy the Metadata XML of your identity provider and paste it in this field. The Metadata XML is a URL that points to an XML document containing metadata about the service provider (identity broker). This metadata includes important information required for establishing trust and facilitating secure communication between the service provider and the identity provider. **Note:** If you have configured VCF single sign-on, after you upgrade to 9.1.1, it is recommended that you re-import the identity broker service provider metadata into your identity provider. Re-importing ensures your identity provider reads the new encryption hint and switches to SHA-256 OAEP for all subsequent SAML assertion encryption.
Metadata URL	Copy the Metadata URL of your identity provider and paste it in this field. The Metadata URL is used to fetch identity provider metadata. This metadata includes important information required for establishing trust and facilitating secure communication between the service provider and the identity provider. **Note:** If your identity provider is not publicly accessible or if the certificate lacks a signature from a recognized Certificate Authority (CA), VCF Operations cannot validate the metadata URL. Provide the metadata in XML format instead.
Name ID format and Name ID value	Select the Name ID format and value. The Name ID format is the value in the SAML response to identify the authenticated user.
SAML Context	SAML context refers to the method of authentication that VCF requires the identity provider to support when verifying a user's identity. • **PasswordProtectedTransport**: Authentication with a password sent over a secure transport (For example, HTTPS). • **Password**: Authentication using a password without requiring transport security. • **Unspecified:** If you don't want to enforce any particular type of authentication mechanism, you can use this option.

8. In the **User/Group Provisioning Method** screen, select your preferred mode of provisioning the users and groups. You can select either:
   - System for Cross-domain Identity Management (SCIM), or
   - Just-In-Time Provisioning (JIT), or
   - Active Directory/Lightweight Directory Access Protocol (AD/LDAP)

   a) If you select **SCIM** and click **Next**, complete the configuration as follows.
      - In the **Domain(s)** screen, enter the domain that users from your enterprise will log in with, and click **Next**.

b) If you select **JIT** and click **Next**, complete the configuration as follows:

- In the **Group Provisioning** screen, select your preferred mode of provisioning the groups. The options are **Groups Pre-provisioning** or **Just-In-Time Group Provisioning** and click **Next**.

Option	Description
Groups Pre-Provisioning	This option creates the required groups in advance to perform role assignments. Users are assigned to these groups dynamically when they log in based on SAML assertion. Only the pre-provisioned groups are available for role assignments.
Just-In-Time Group Provisioning	This option creates groups dynamically based on the SAML assertion when the user logs in. Alternatively, you can use group pre-provisioning if you know the group names from Active Directory that requires entitlements.

- In the **Domain/s** screen, add the domain details that users from your enterprise will log in with, and click **Next**.
- In the **Pre-provisioning Domains and Groups** screen, add the groups from each domain that you want to provision in VCF for roles and assignments by clicking the **Add Group** button, and click **Next**.

Note:

The **Pre-provisioning Domains and Groups** screen is displayed only if you have chosen groups to be pre-provisioned which means that you chose **Groups pre-provisioning** in the previous screen.

There should be at least one provisioned group added across the domains.

- In the **Attributes** screen, the user attributes section displays a list of mandatory and non-mandatory user attributes that you view in the SAML response from the identity provider.
  You can add additional user attributes if required. In the group attributes section of the workflow you can add a group attribute to be called for in the SAML request. Attribute names are case-sensitive.

Note:

If you choose JIT, it is recommended that you automatically disable inactive users.

c) If you select **AD/LDAP** and click **Next**, complete the configuration as follows. To learn more about the benefits of using AD/LDAP as a provisioning method, KB 386870.

In the **Directory Details** screen, enter the following details.

Option	Description
Directory name	Enter a name for the directory.
DNS Server Location	Select this option if you want the identity broker to find and use optimal domain controllers by querying the DNS Service Record.

Global Catalog	Global catalog has a partial read-only replica of all objects across all domains. Some of the limitations with selecting the global catalog option include: • The Active Directory object attributes that are replicated to the global catalog are identified in the Active Directory schema as the Partial Attribute Set (PAS). Only these attributes are available for attribute mapping by the service. If necessary, edit the schema to add or remove attributes that are stored in the global catalog. • The global catalog stores the group membership (the member attributes) of only universal groups. Only universal groups are synced to the service. If necessary, change the scope of a group from a local domain or from global to universal. • The bind DN account that you define when you configure a directory in the service must have permissions to read the Token-Groups-Global-And-Universal (TGGAU) attribute. **Note:** When you use Global catalog with LDAP and multiple domains, the additional domains are discovered only after the user/group sync is complete.
Encrypted Connection (Displayed when you select the **DNS server location** option)	This option activates LDAPs for all connections to this directory.
Certificates for encrypted connection (Displayed when you select the **DNS server location** and **Encrypted Connections** options and if you enter a *ldaps://* string in the Primary domain controller field)	Upload the root certificate(s) required to access the domain controllers over SSL. All uploaded root certificates will be merged into a single file with the filename `{directory-name}-concatenated.pem` when you save the configuration. **Note:** Ensure that the certificate is in the (Privacy Enhanced Mail) PEM format and includes the "BEGIN CERTIFICATE" and "END CERTIFICATE" lines.
Directory search attribute	A specific LDAP attribute used in queries to search or identify entries in the directory. For example, `uid`, `sAMAccountName`.
Primary domain controller (Displayed when you select the **Global Catalog** option)	A server machine running Active Directory Domain Services (AD DS) responsible for managing and authenticating user access, enforcing policies, and storing directory data. For example, `hostname = DC01` for the domain called example.com.
Secondary domain controller (Displayed when you select the **Global Catalog** option)	A backup domain controller for the identity broker to point to if the primary domain controller is not reachable.
Directory search attribute	A specific LDAP attribute used in queries to search or identify entries in the directory. For example, `uid`, `sAMAccountName`.
Base DN	The Distinguished Name (DN) of the top-level entry in the directory tree from which LDAP queries begin. For example, `dc=example`, `dc=com`.
Bind user name	The user name of a user account with sufficient permissions to authenticate and perform LDAP operations on the directory.
Bind user password	The password of a user account with sufficient permissions to authenticate and perform LDAP operations on the directory.

9. In the **Review** screen, review the details you have added for the configuration and click **Finish**.

Configure user and group provisioning

10. In the **Configure User and Group Provisioning** screen, depending on the option you choose, complete the following steps and click **Done**.
    a) If you have chosen SCIM, you see **Configure SCIM push in your identity provider** instead of **Configure User and Group Provisioning**. Use the values listed on the screen to configure the SCIM application in the identity provider.
    - If your identity provider supports authorizing the SCIM application using OAuth grant type, use the client ID, client secret, and token endpoint. Otherwise, use the SCIM bearer token to authorize this SCIM server in your identity provider.
    - If required, you can regenerate these values after you have configured VCF single sign-on. To regenerate the values to configure the SCIM application in the identity provider, navigate to the respective VCF Instance and click on **Identity Source**. The SCIM related values are available under **Directory Information**.
    - Ensure that the user attributes, username, and domain are pushed individually from the SCIM application by the identity provider. Similarly, ensure that the group attributes, group name, and domain are pushed individually as well.

    b) If you have chosen JIT, there is no additional step to be performed under **User and Group Provisioning**.
    c) If you have chosen AD/LDAP, review the directory information such as the domain controller and bind credentials that you had configured. Complete the following steps:
    1. In the **Attributes Mappings** screen, review the attribute mappings and click **Next.**
       The attributes in the identity broker directory are mapped to the Active Directory attributes for accurate synchronization of user information. If the mappings are incorrect, select the correct attribute from the drop-down list.
    2. In the **Group Provisioning** screen, enter the relevant details or select the required options and click **Next**.
       Use the **Group Provisioning** screen to search for the group names and select the corresponding Active Directory group Distinguished Names (DN) that you want to sync.

Option	Description
Specify the Base Group DN	Click the **Select Base Group DN** button if you have nested groups and want to specify the group names. You can search for the group name and select the group DNs that you want to sync from from the list in the datagrid below.
Sync Nested Groups	Select **Sync Nested Group** to sync all the users from the selected group and its nested group without maintaining the group hierarchy.

    3. From the **User Provisioning** screen, enter the relevant details or select the required options and click **Next**.
       Use the **User Provisioning** screen to search for the user names and select the corresponding Active Directory group Distinguished Names (DN) that you want to sync.

Option	Description
Specify the Base User DN	Click the **Select Base User DN** button to specify the user names that you want to sync. You can search for the user name and select the user DNs that you want to sync from from the list in the datagrid below. If you want to sync the users directly from the group, do not specify a base user DN.

    4. From the **Review screen**, review the configuration details of the identity provider.
    5. Click **Finish**. The sync starts in the background. After the initial sync, the sync is scheduled to run once a week at a scheduled time.
    6. Click **Done**.

    For more information about the various functionalities you can perform as part of the management operations in Active Directory, see Managing AD/LDAP.

(Optional) Test Login

11. Click **Test Login** to test the login using your Single-Sign On credentials.
12. Click **Next** in the identity provider screen.
13. After a successful validation, click **Done**.
14. Click **Finish Setup**. This completes the initial configuration.
15. Click **Continue** to view the **VCF SSO Overview** page.

Important:

Important Notes on Backup

It is highly advisable to create a backup of both VCF Operations and the identity broker instance. Furthermore, it is advisable to perform a similar backup following any substantial modifications to the initial identity source configuration, such as when you 'Delete single sign-on'. This practice ensures that you can restore the VCF Single Sign-On configuration to a recognized and stable state if necessary. You can download the configuration as a JSON file.

Important Notes on Role Assignment

VCF roles can be created and mapped to individual component role.

- As an admin you can assign built-in and custom VCF roles to groups and and users from the **Access Managment** tab (navigate to the VCF SSO Overview page, click on a VCF single sign-on configuration, and then click the Access Management tab).
- As an admin, you can also assign roles in the component console and enable VCF single sign-on from the **Component Configuration** tab (navigate to the VCF SSO Overview page, click on a VCF single sign-on configuration, and then click the Component Configuration).

Configure Ping Identity as an Identity Provider Using OIDC

After installing or upgrading to VCF 9.1, to configure VCF single sign-on, you can configure Ping Identity as an identity provider using OIDC as the authentication protocol.

To set up an identity provider, you must:

1. Choose an identity provider
2. Configure the identity provider
3. Configure user and group provisioning
4. Test Login

Note:

You can test log in if the identity broker is at version 9.1.

Prerequisite

- Verify that you have chosen a deployment mode. For more information, see Step 1: Choose the Deployment Mode.
- If you configure VCF single sign-on using identity broker instance, and there are network restrictions that do not allow outbound public internet access, you must set up a proxy server. For more information, see Configure a Proxy Server for VCF Management Services Components and VCF Automation.

Procedure

Choose the identity provider

1. From the **Configure VCF SSO** page, click the **Start** button against the **Configure Identity Provider** option.

   If you have an existing identity provider configured in vCenter, see the information below. For a fresh configuration of the identity provider, continue from Step 6.

   For an existing source (existing identity provider configuration)

   - Choose either SAML or OIDC as the authentication protocol.

Note:

If you choose SAML, the VCF single sign-on configuration is considered as a new authentication source configuration. Continue from Step. 6.

- If you choose OIDC:
  - If you choose the instance deployment mode, and select OIDC, the following fields are prepopulated in the **Identity Provider Configuration** screen.
    a. Review the details and provide the shared secret and click **Next**.
       - Identity provider name
       - Client identifier
       - OpenID address
       - Update Redirect URI in the Identity provider's OIDC application
       - Shared Secret: Paste the client secret of the OIDC application you created in the identity provider.
    b. Review the **User/Group Provisioning**, **Domain(s)**, and **Review** screens and click **Finish**.
    c. Review the SCIM push details and click **Done**. The identity provider configuration is complete.
  - If you choose the embedded deployment mode, and select OIDC:
    a. The configuration details of the identity provider configuration are automatically imported from existing configuration into the identity broker with OIDC as the authentication protocol and SCIM as the user/group provisioning method.
    b. Review the configuration values in the **Identity Provider Configuration** screen and click **Finish**.
    c. Review the SCIM details, click **Done**, and then click **Finish Setup**. The identity provider configuration is complete.

2. For a fresh identity provider configuration, from the **Choose Identity Provider** section, select **Ping Identity** from the list and click **Next**.

Configure the identity provider

Note:

Ensure that you add the following default claims when you configure your external identity provider. These claims are case-sensitive:

- preferred_username
- email
- family_name
- given_name
- groups
- domain

If the default claims cannot be added, use alternative claim names and map them to the appropriate attributes in the VCF Operations **Identity & Access** page to support your JIT configurations.

3. From the **Configure the Identity Provider** section, click **Configure** to configure the identity broker to integrate with the selected identity provider for user authentication.
4. In the **Authentication Method** screen, select **OIDC** (OpenID Connect) and click **Next**.
5. In the **Identity Provider Configuration** screen, enter the relevant details or select the required options and click **Next**.

Option	Description
IDP Display Name	Enter the OpenID Connect provider name. Example, Ping Identity.
Client Identifier	A unique ID assigned to your OIDC application by the identity provider for authentication.

Shared secret	A confidential key used by your OIDC application to authenticate itself with the identity provider. **Note:** Client secrets issued by external identity providers typically have expiration dates. When a client secret expires, it must be regenerated, and the corresponding OIDC configuration must be updated in VCF to ensure continued authentication functionality.
OpenID Address	The URL of the identity provider's OIDC configuration endpoint for initiating the authentication process.
Redirect URI	The URL where the identity provider sends the authentication response after a user logs in.
SSL Certificate	A public key certificate to verify the OIDC identity.

6. In the **User/Group Provisioning Method** screen, select your preferred mode of provisioning the users and groups. You can select either:
   - System for Cross-domain Identity Management (SCIM), or
   - Just-In-Time Provisioning (JIT), or
   - Active Directory/Lightweight Directory Access Protocol (AD/LDAP)

   a) If you select **SCIM** and click **Next**, complete the configuration as follows.
      - In the **Domain(s)** screen, enter the domain that users from your enterprise will log in with, and click **Next**.

   b) If you select **JIT** and click **Next**, complete the configuration as follows:
      - In the **Group Provisioning** screen, select your preferred mode of provisioning the groups. The options are **Groups Pre-provisioning** or **Just-In-Time Group Provisioning** and click **Next**.

Option	Description
Groups Pre-Provisioning	This option creates the required groups in advance to perform role assignments. Users are assigned to these groups dynamically when they log in based on OIDC token. Only the pre-provisioned groups are available for role assignments.
Just-In-Time Group Provisioning	This option creates groups dynamically based on the OIDC assertion when the user logs in. Alternatively, you can use group pre-provisioning if you know the group names from Active Directory that requires entitlements.

      - In the **Domain(s)** screen, add the domain details that users from your enterprise will log in with, and click **Next**.
      - In the **Pre-provisioning Domains and Groups** screen, add the groups from each domain that you want to provision in VCF for roles and assignments by clicking the **Add Group** button, and click **Next**.

      Note:

      The **Pre-provisioning Domains and Groups** screen is displayed only if you have chosen groups to be pre-provisioned which means that you chose **Groups pre-provisioning** in the previous screen.

      There should be at least one provisioned group added across the domains.

   - In the **Attributes** screen, the user attributes section displays a list of mandatory and non-mandatory user attributes that you view in the OIDC response from the identity provider.

     You can add additional user attributes if necessary. In the group attributes section of the workflow you can add a group attribute to be called for in the OIDC request. Attribute names are case-sensitive.

     Note:

     If you use public APIs to update identity provider attribute mappings, the following identity broker attributes are available and can be mapped to OpenID attributes:

     - ExternalId

- userName
- firstName
- lastName
- email
- phone
- active
- employeeID
- distinguishedName
- userPrincipalName
- domain
- sourceAnchor
- displayName
- nickName
- profileUrl
- title
- organization
- costCenter
- department
- division
- managerId

Note:

If you choose JIT, it is recommended that you automatically disable inactive users.

c) If you select **AD/LDAP** and click **Next**, select **Unique Identifier in OIDC Identity Provider** and **Unique Identifier in VCF Identity Broker** and click **Next**. To learn more about the benefits of using AD/LDAP as a provisioning method, see KB 386870.

Note:

The value of the identifier used to uniquely identify a user in your OIDC identity provider must exactly match with the value from Active Directory. The unique identifier in the identity broker is used to map the two systems.

Option	Description
Unique Identifier in OIDC Identity Provider	An identifier in your OIDC identity provider that is used to identify a user uniquely. You can select or enter a custom attribute (except 'sub'). For example, `name`.
Unique Identifier in VCF Identity Broker	Select the unique identifier that will be used to map to an attribute from Active Directory. The value of this identifier must exactly match the value of the previous field called **Unique Identifier in OIDC Identity Provider**.

In the In the **Directory Details** screen, enter the relevant details or select the required options.

Option	Description
Directory name	Enter a name for the directory.
DNS Server Location	Select this option if you want the identity broker to find and use optimal domain controllers by querying the DNS Service Record.

Global Catalog	Global catalog has a partial read-only replica of all objects across all domains. Some of the limitations with selecting the global catalog option include: • The Active Directory object attributes that are replicated to the global catalog are identified in the Active Directory schema as the Partial Attribute Set (PAS). Only these attributes are available for attribute mapping by the service. If necessary, edit the schema to add or remove attributes that are stored in the global catalog. • The global catalog stores the group membership (the member attributes) of only universal groups. Only universal groups are synced to the service. If necessary, change the scope of a group from a local domain or from global to universal. • The bind DN account that you define when you configure a directory in the service must have permissions to read the Token-Groups-Global-And-Universal (TGGAU) attribute. **Note:** When you use Global catalog with LDAP and multiple domains, the additional domains are discovered only after the user/group sync is complete.
Encrypted Connection (Displayed when you select the **DNS server location** option)	Activates LDAPs for all the connections to this directory.
Certificates for encrypted connection (Displayed when you select the **DNS server location** and **Encrypted Connections** options and if you enter a *ldaps://* string in the Primary domain controller field)	Upload the root certificate(s) required to access the domain controllers over SSL. All uploaded root certificates will be merged into a single file with the filename `{directory-name}-concatenated.pem` when you save the configuration. **Note:** Ensure that the certificate is in the (Privacy Enhanced Mail) PEM format and includes the "BEGIN CERTIFICATE" and "END CERTIFICATE" lines.
Directory search attribute	A specific LDAP attribute used in queries to search or identify entries in the directory. For example, `uid`, `sAMAccountName`.
Primary domain controller	A server machine running Active Directory Domain Services (AD DS) responsible for managing and authenticating user access, enforcing policies, and storing directory data. For example, `hostname = DC01` for the domain called example.com.
Secondary domain controller	A backup domain controller for the identity broker to point to if the primary domain controller is not reachable.
Directory search attribute	A specific LDAP attribute used in queries to search or identify entries in the directory. For example, `uid`, `sAMAccountName`.
Base DN	The Distinguished Name (DN) of the top-level entry in the directory tree from which LDAP queries begin. For example, `dc=example`, `dc=com`.
Bind user name	The user name of a user account with sufficient permissions to authenticate and perform LDAP operations on the directory.
Bind user password	The password of a user account with sufficient permissions to authenticate and perform LDAP operations on the directory.

7. In the **Review** screen, review the details you have added for the configuration and click **Finish**.

Configure user and group provisioning

8. In the **Configure User and Group Provisioning** screen, depending on the option you choose, complete the following steps and click **Done**.

   If you have chosen OIDC as the authentication protocol, and AD/LDAP as the user and group provisioning protocol, read KB 386870 to understand the finer details about configuring the setup.

   a) If you have chosen SCIM, you see **Configure SCIM push in your identity provider** instead of **Configure User and Group Provisioning**. Use the values listed on the screen to configure the SCIM application in the identity provider.

      - If your identity provider supports authorizing the SCIM application using OAuth grant type, use the client ID, client secret, and token endpoint. Otherwise, use the SCIM bearer token to authorize this SCIM server in your identity provider.
      - If required, you can regenerate these values after you have configured VCF single sign-on. To regenerate the values to configure the SCIM application in the identity provider, navigate to the respective VCF Instance and click on **Identity Source**. The SCIM related values are available under **Directory Information**.
      - Ensure that the user attributes, username, and domain are pushed individually from the SCIM application by the identity provider. Similarly, ensure that the group attributes, group name, and domain are pushed individually as well.

   b) If you have chosen JIT, there is no additional step to be performed under **User and Group Provisioning**.

   c) If you have chosen AD/LDAP, review the directory information such as the domain controller and bind credentials that you had configured. Complete the following steps:

      1. In the **Attributes Mappings** screen, review the attribute mappings and click **Next.**
         The attributes in the identity broker directory are mapped to the Active Directory attributes for accurate synchronization of user information. If the mappings are incorrect, select the correct attribute from the drop-down list.
      2. In the **Group Provisioning** screen, enter the relevant details or select the required options and click **Next**.
         Use the **Group Provisioning** screen to search for the group names and select the corresponding Active Directory group Distinguished Names (DN) that you want to sync.

Option	Description
Specify the Base Group DN	Click the **Select Base Group DN** button if you have nested groups and want to specify the group names. You can search for the group name and select the group DNs that you want to sync from from the list in the datagrid below.
Sync Nested Groups	Select **Sync Nested Group** to sync all the users from the selected group and its nested group without maintaining the group hierarchy.

      3. From the **User Provisioning** screen, enter the relevant details or select the required options and click **Next**.
         Use the **User Provisioning** screen to search for the user names and select the corresponding Active Directory group Distinguished Names (DN) that you want to sync.

Option	Description
Specify the Base User DN	Click the **Select Base User DN** button to specify the user names that you want to sync. You can search for the user name and select the user DNs that you want to sync from from the list in the datagrid below. If you want to sync the users directly from the group, do not specify a base user DN.

      4. From the **Review screen**, review the configuration details of the identity provider.
      5. Click **Finish**. The sync starts in the background. After the initial sync, the sync is scheduled to run once a week at a scheduled time.
      6. Click **Done**.

   For more information about the various functionalities you can perform as part of the management operations in Active Directory, see Managing AD/LDAP.

(Optional) Test Login

9. Click **Test Login** to test the login using your single sign-on credentials.
10. Click **Next** in the identity provider screen.
11. After a successful validation, click **Done**.
12. Click **Finish Setup**. This completes the initial configuration.
13. Click **Continue** to view the **VCF SSO Overview** page.

Important:

Important Notes on Backup

It is highly advisable to create a backup of both VCF Operations and the identity broker instance. Furthermore, it is advisable to perform a similar backup following any substantial modifications to the initial identity source configuration, such as when you 'Delete Single Sign-On'. This practice ensures that you can restore the VCF single sign-on configuration to a recognized and stable state if necessary. You can download the configuration as a JSON file.

Important Notes on Role Assignment

VCF roles can be created and mapped to individual component roles.

- As an admin you can assign built-in and custom VCF roles to groups and and users from the **Access Managment** tab (navigate to the VCF SSO Overview page, click on a VCF single sign-on configuration, and then click the Access Management tab).
- As an admin, you can also assign roles in the component console and enable VCF Single Sign-On from the **Component Configuration** tab (navigate to the VCF SSO Overview page, click on a VCF single sign-on configuration, and then click the Component Configuration).

Configure Microsoft Entra ID as an Identity Provider

You can configure Microsoft Entra ID as an identity provider using SAML or OIDC authentication protocols.

- For information about configuring Microsoft Entra ID as an identity provider using SAML as the authentication protocol, see Configure Microsoft Entra ID as an Identity Provider Using SAML.
- For information about configuring Microsoft Entra ID as an identity provider using OIDC as the authentication protocol, see Configure Microsoft Entra ID as an Identity Provider Using OIDC.

Configure Microsoft Entra ID as an Identity Provider Using SAML

After installing or upgrading to VCF 9.1, to configure VCF single sign-on, you can configure Microsoft Entra ID as an identity provider using SAML as the authentication protocol.

To set up an identity provider, you must:

1. Choose an identity provider
2. Configure the identity provider
3. Configure user and group provisioning
4. Test Login

Note:

You can test log in if the identity broker is at version 9.1.

Prerequisite

- Verify that you have chosen a deployment mode. For more information, see Step 1: Choose the Deployment Mode.

Procedure

Choose the identity provider

1. From the **Configure VCF SSO** page, click the **Start** button against the **Configure Identity Provider** option.
2. From the **Choose the identity provider** section, select **Microsoft Entra ID** from the list and click **Next**.

Configure the identity provider

3. From the **Configure the Identity Provider** section, click **Configure** to set up the identity broker to integrate with the selected identity provider for user authentication.
4. In the **Authentication Method** pane, select **SAML** (Security Assertion Markup Language).
5. In the **Service Provider Registration** pane, log in to your Microsoft Entra ID admin console and follow the steps in Register a SAML App in your external client.

   While registering Microsoft Entra ID, use the metadata provided in the **Service Provider Registration** screen for **Entity ID/Metadata URL** and **Assertion Consumer Service URL**. If the identity provider requires the metadata to be copy-pasted, it is recommended to download this file to avoid formatting issues.

   Note:

   Ensure that you add the following default claims when you configure your external identity provider. These claims are case-sensitive:

   - userName
   - email
   - domain
   - firstName
   - lastName
   - groups
6. Click **Next**.
7. In the **Identity Provider Configuration** screen, enter the relevant details or select the required options and click **Next**.

Option	Description
IDP Display Name	Enter a friendly name for your identity provider. For example, Microsoft Entra ID.
Metadata	Select either URL or XML.
Metadata XML	Copy the Metadata XML of your identity provider and paste it in this field. The Metadata XML is a URL that points to an XML document containing metadata about the service provider (identity broker). This metadata includes important information required for establishing trust and facilitating secure communication between the service provider and the identity provider. **Note:** If you have configured VCF single sign-on, after you upgrade to 9.1.1, it is recommended that you re-import the identity broker service provider metadata into your identity provider. Re-importing ensures your identity provider reads the new encryption hint and switches to SHA-256 OAEP for all subsequent SAML assertion encryption.

Metadata URL	Copy the Metadata URL of your identity provider and paste it in this field. The Metadata URL is used to fetch identity provider metadata. This metadata includes important information required for establishing trust and facilitating secure communication between the service provider and the identity provider. **Note:** If your identity provider is not publicly accessible or if the certificate lacks a signature from a recognized Certificate Authority (CA), VCF Operations cannot validate the metadata URL. Provide the metadata in XML format instead.
Name ID format and Name ID value	Select the Name ID format and value. The Name ID format is the value in the SAML response to identify the authenticated user.
SAML Context	SAML context refers to the method of authentication that VCF requires the identity provider to support when verifying a user's identity. • **PasswordProtectedTransport**: Authentication with a password sent over a secure transport (For example, HTTPS). • **Password**: Authentication using a password without requiring transport security. • **Unspecified:** If you don't want to enforce any particular type of authentication mechanism, you can use this option.

8. In the **User/Group Provisioning Method** screen, select your preferred mode of provisioning the users and groups. You can select either:
   - System for Cross-domain Identity Management (SCIM), or
   - Just-In-Time Provisioning (JIT), or
   - Active Directory/Lightweight Directory Access Protocol (AD/LDAP)

   a) If you select **SCIM** and click **Next**, complete the configuration as follows.
      - In the **Domain(s)** screen, enter the domain that users from your enterprise will log in with, and click **Next**.

   b) If you select **JIT** and click **Next**, complete the configuration as follows:

      In the **Group Provisioning** screen, select your preferred mode of provisioning the groups. The options are **Groups Pre-provisioning** or **Just-In-Time Group Provisioning** and click **Next**.

      Note:

      Add group claims. For more information, see the section about adding group claims to tokens in the page called Configure group claims for applications by using Microsoft Entra ID.

Option	Description
Groups Pre-Provisioning	This option creates the required groups in advance to perform role assignments. Users are assigned to these groups dynamically when they log in based on SAML assertion. Only the pre-provisioned groups are available for role assignments.
Just-In-Time Group Provisioning	This option creates groups dynamically based on the SAML assertion when the user logs in. Alternatively, you can use group pre-provisioning if you know the group names from Active Directory that requires entitlements.

      - In the **Domain/s** screen, add the domain details that users from your enterprise will log in with, and click **Next**.
      - In the **Pre-provisioning Domains and Groups** screen, add the groups from each domain that you want to provision in VCF for roles and assignments by clicking the **Add Group** button, and click **Next**.

Note:

The **Pre-provisioning Domains and Groups** screen is displayed only if you have chosen groups to be pre-provisioned which means that you chose **Groups pre-provisioning** in the previous screen.

There should be at least one provisioned group added across the domains.

- In the **Attributes** screen, the user attributes section displays a list of mandatory and non-mandatory user attributes that you view in the OIDC response from the identity provider.

  You can add additional user attributes if necessary. In the group attributes section of the workflow you can add a group attribute to be called for in the SAML request. Attribute names are case-sensitive.

Note:

If you choose JIT, it is recommended that you automatically disable inactive users.

c) If you select **AD/LDAP** and click **Next**, complete the configuration as follows. To learn more about the benefits of using AD/LDAP as a provisioning method, KB 386870.

In the **Directory Details** screen, enter the following details.

Option	Description
Directory name	Enter a name for the directory.
DNS Server Location	Select this option if you want the identity broker to find and use optimal domain controllers by querying the DNS Service Record.
Global Catalog	Global catalog has a partial read-only replica of all objects across all domains. Some of the limitations with selecting the global catalog option include: • The Active Directory object attributes that are replicated to the global catalog are identified in the Active Directory schema as the Partial Attribute Set (PAS). Only these attributes are available for attribute mapping by the service. If necessary, edit the schema to add or remove attributes that are stored in the global catalog. • The global catalog stores the group membership (the member attributes) of only universal groups. Only universal groups are synced to the service. If necessary, change the scope of a group from a local domain or from global to universal. • The bind DN account that you define when you configure a directory in the service must have permissions to read the Token-Groups-Global-And-Universal (TGGAU) attribute. **Note:** When you use Global catalog with LDAP and multiple domains, the additional domains are discovered only after the user/group sync is complete.
Encrypted Connection (Displayed when you select the **DNS server location** option)	This option activates LDAPs for all connections to this directory.
Certificates for encrypted connection (Displayed when you select the **DNS server location** and **Encrypted Connections** options and if you enter a `ldaps://` string in the Primary domain controller field)	Upload the root certificate(s) required to access the domain controllers over SSL. All uploaded root certificates will be merged into a single file with the filename `{directory-name}-concatenated.pem` when you save the configuration. **Note:** Ensure that the certificate is in the (Privacy Enhanced Mail) PEM format and includes the "BEGIN CERTIFICATE" and "END CERTIFICATE" lines.
Directory search attribute	A specific LDAP attribute used in queries to search or identify entries in the directory. For example, `uid`, `sAMAccountName`.

Primary domain controller (Displayed when you select the **Global Catalog** option)	A server machine running Active Directory Domain Services (AD DS) responsible for managing and authenticating user access, enforcing policies, and storing directory data. For example, `hostname = DC01` for the domain called example.com.
Secondary domain controller (Displayed when you select the **Global Catalog** option)	A backup domain controller for the identity broker to point to if the primary domain controller is not reachable.
Directory search attribute	A specific LDAP attribute used in queries to search or identify entries in the directory. For example, `uid`, `sAMAccountName`.
Base DN	The Distinguished Name (DN) of the top-level entry in the directory tree from which LDAP queries begin. For example, `dc=example`, `dc=com`.
Bind user name	The user name of a user account with sufficient permissions to authenticate and perform LDAP operations on the directory.
Bind user password	The password of a user account with sufficient permissions to authenticate and perform LDAP operations on the directory.

9. In the **Review** screen, review the details you have added for the configuration and click **Finish**.

Configure user and group provisioning

10. In the **Configure User and Group Provisioning** screen, depending on the option you choose, complete the following steps and click **Done**.
    a) If you have chosen SCIM, you see **Configure SCIM push in your identity provider** instead of **Configure User and Group Provisioning**. Use the values listed on the screen to configure the SCIM application in the identity provider.
       - If your identity provider supports authorizing the SCIM application using OAuth grant type, use the client ID, client secret, and token endpoint. Otherwise, use the SCIM bearer token to authorize this SCIM server in your identity provider.
       - If required, you can regenerate these values after you have configured VCF single sign-on. To regenerate the values to configure the SCIM application in the identity provider, navigate to the respective VCF Instance and click on **Identity Source**. The SCIM related values are available under **Directory Information**.
       - Ensure that the user attributes, username, and domain are pushed individually from the SCIM application by the identity provider. Similarly, ensure that the group attributes, group name, and domain are pushed individually as well.

    b) If you have chosen JIT, there is no additional step to be performed under **User and Group Provisioning**.
    c) If you have chosen AD/LDAP, review the directory information such as the domain controller and bind credentials that you had configured. Complete the following steps:
       1. In the **Attributes Mappings** screen, review the attribute mappings and click **Next.**
          The attributes in the identity broker directory are mapped to the Active Directory attributes for accurate synchronization of user information. If the mappings are incorrect, select the correct attribute from the drop-down list.
       2. In the **Group Provisioning** screen, enter the relevant details or select the required options and click **Next**.
          Use the **Group Provisioning** screen to search for the group names and select the corresponding Active Directory group Distinguished Names (DN) that you want to sync.

Option	Description
Specify the Base Group DN	Click the **Select Base Group DN** button if you have nested groups and want to specify the group names. You can search for the group name and select the group DNs that you want to sync from from the list in the datagrid below.
Sync Nested Groups	Select **Sync Nested Group** to sync all the users from the selected group and its nested group without maintaining the group hierarchy.

       3. From the **User Provisioning** screen, enter the relevant details or select the required options and click **Next**.

Use the **User Provisioning** screen to search for the user names and select the corresponding Active Directory group Distinguished Names (DN) that you want to sync.

Option	Description
Specify the Base User DN	Click the **Select Base User DN** button to specify the user names that you want to sync. You can search for the user name and select the user DNs that you want to sync from from the list in the datagrid below. If you want to sync the users directly from the group, do not specify a base user DN.

4. From the **Review screen**, review the configuration details of the identity provider.
5. Click **Finish**. The sync starts in the background. After the initial sync, the sync is scheduled to run once a week at a scheduled time.
6. Click **Done**.

For more information about the various functionalities you can perform as part of the management operations in Active Directory, see Managing AD/LDAP.

(Optional) Test Login

11. Click **Test Login** to test the login using your single-sign on credentials.
12. Click **Next** in the identity provider screen.
13. After a successful validation, click **Done**.
14. Click **Finish Setup**. This completes the initial configuration.
15. Click **Continue** to view the **VCF SSO Overview** page.

Important:

Important Notes on Backup

It is highly advisable to create a backup of both VCF Operations and the identity broker instance. Furthermore, it is advisable to perform a similar backup following any substantial modifications to the initial identity source configuration, such as when you 'Delete Single Sign-On'. This practice ensures that you can restore the VCF single sign-on configuration to a recognized and stable state if necessary. You can download the configuration as a JSON file.

Important Notes on Role Assignment

VCF roles can be created and mapped to individual component role.

- As an admin you can assign built-in and custom VCF roles to groups and and users from the **Access Managment** tab (navigate to the VCF SSO Overview page, click on a VCF single sign-on configuration, and then click the Access Management tab).
- As an admin, you can also assign roles in the component console and enable VCF Single Sign-On from the **Component Configuration** tab (navigate to the VCF SSO Overview page, click on a VCF single sign-on configuration, and then click the Component Configuration tab).

Configure Microsoft Entra ID as an Identity Provider Using OIDC

After installing or upgrading to VCF 9.1, to configure VCF single sign-on, you can configure Microsoft Entra ID as an identity provider using OIDC as the authentication protocol.

To set up an identity provider, you must:

1. Choose an identity provider
2. Configure the identity provider
3. Configure user and group provisioning
4. Test Login

Note:

You can test log in if the identity broker is at version 9.1.

Prerequisite

- Verify that you have chosen a deployment mode. For more information, see Step 1: Choose the Deployment Mode.
  If you configure VCF single sign-on using identity broker instance, and there are network restrictions that do not allow outbound public internet access, you must set up a proxy server. For more information, see Configure a Proxy Server for VCF Management Services Components and VCF Automation.

Procedure

Choose the identity provider

1. From the **Configure VCF SSO** page, click the **Start** button against the **Configure Identity Provider** option.

   If you have an existing identity provider configured in vCenter, see the information below. For a fresh configuration of the identity provider, continue from Step 6.

   For an existing source (existing identity provider configuration)

   - Choose either SAML or OIDC as the authentication protocol.

     Note:

     If you choose SAML, the VCF single sign-on configuration is considered as a new authentication source configuration. Continue from Step. 6.
   - If you choose OIDC:
     – If you choose the instance deployment mode, and select OIDC, the following fields are prepopulated in the **Identity Provider Configuration** screen.
       a. Review the details and provide the shared secret and click **Next**.
          - Identity provider name
          - Client identifier
          - OpenID address
          - Update Redirect URI in the Identity provider's OIDC application
          - Shared Secret: Paste the client secret of the OIDC application you created in the identity provider.
       b. Review the **User/Group Provisioning**, **Domain(s)**, and **Review** screens and click **Finish**.
       c. Review the SCIM push details and click **Done**. The identity provider configuration is complete.

     - If you choose the embedded deployment mode, and select OIDC:
       a. The configuration details of the identity provider configuration are automatically imported from existing configuration into the identity broker with OIDC as the authentication protocol and SCIM as the user/group provisioning method.
       b. Review the configuration values in the **Identity Provider Configuration** screen and click **Finish**.
       c. Review the SCIM details, click **Done**, and then click **Finish Setup**. The identity provider configuration is complete.

2. For a fresh identity provider configuration, from the **Choose the identity provider** section, select **Microsoft Entra ID** from the list and click **Next**.

   Configure the identity provider

   Note:

   Ensure that you add the following default claims when you configure your external identity provider. These claims are case-sensitive:

   - preferred_username
   - email
   - family_name
   - given_name
   - groups
   - domain

   If the default claims cannot be added, use alternative claim names and map them to the appropriate attributes in the VCF Operations **Identity & Access** page to support your JIT configurations.

3. From the **Configure the Identity Provider** section, click **Configure** to configure the identity broker to integrate with the selected identity provider for user authentication.
4. In the **Authentication Method** screen, select **OIDC** (OpenID Connect) and click **Next**.
5. In the **Identity Provider Configuration** screen, enter the relevant details or select the required options and click **Next**.

Option	Description
IDP Display Name	Enter the OpenID Connect provider name. Example, Microsoft Entra ID.
Client Identifier	A unique ID assigned to your OIDC application by the identity provider, for authentication.
Shared secret	A confidential key used by your OIDC application to authenticate itself with the identity provider. **Note:** Client secrets issued by external identity providers typically have expiration dates. When a client secret expires, it must be regenerated, and the corresponding OIDC configuration must be updated in VCF to ensure continued authentication functionality.
OpenID Address	The URL of the identity provider's OIDC configuration endpoint for initiating the authentication process.
Redirect URI	The URL where the identity provider will send the authentication response after a user logs in.

6. In the **User/Group Provisioning Method** screen, select your preferred mode of provisioning the users and groups. You can select either:
   - System for Cross-domain Identity Management SCIM, or
   - Just-In-Time Provisioning (JIT), or
   - Active Directory/Lightweight Directory Access Protocol (AD/LDAP)

   a) If you select **SCIM** and click **Next**, complete the configuration as follows.
      - In the **Domain(s)** screen, enter the domain that users from your enterprise will log in with, and click **Next**.

   b) If you select **JIT** and click **Next**, complete the configuration as follows:
      - In the **Group Provisioning** screen, select your preferred mode of provisioning the groups. The options are **Groups Pre-provisioning** or **Just-In-Time Group Provisioning** and click **Next**.

Option	Description
Groups Pre-Provisioning	This option creates the required groups in advance to perform role assignments. Users are assigned to these groups dynamically when they log in based on OIDC token. Only the pre-provisioned groups are available for role assignments.
Just-In-Time Group Provisioning	This option creates groups dynamically based on the OIDC assertion when the user logs in. Alternatively, you can use group pre-provisioning if you know the group names from Active Directory that requires entitlements.

      - In the **Domain(s)** screen, add the domain details that users from your enterprise will log in with, and click **Next**.
      - In the **Pre-provisioning Domains and Groups** screen, add the groups from each domain that you want to provision in VCF for roles and assignments by clicking the **Add Group** button, and click **Next**.

      Note:

      The **Pre-provisioning Domains and Groups** screen is displayed only if you have chosen groups to be pre-provisioned which means that you chose **Groups pre-provisioning** in the previous screen.

There should be at least one provisioned group added across the domains.

- In the **Attributes** screen,
  the user attributes section displays a list of mandatory and non-mandatory user attributes that you view in the OIDC response from the identity provider.
  You can add additional user attributes if necessary. In the group attributes section of the workflow you can add a group attribute to be called for in the OIDC request. Attribute names are case-sensitive.

Note:

If you use public APIs to update identity provider attribute mappings, the following identity broker attributes are available and can be mapped to OpenID attributes:

- ExternalId
- userName
- firstName
- lastName
- email
- phone
- active
- employeeID
- distinguishedName
- userPrincipalName
- domain
- sourceAnchor
- displayName
- nickName
- profileUrl
- title
- organization
- costCenter
- department
- division
- managerId

Note:

If you choose JIT, it is recommended that you automatically disable inactive users.

c) If you select **AD/LDAP**, select the **Unique Identifier in OIDC Identity Provider** and **Unique Identifier in VCF Identity Broker**, and click **Next**. To learn more about the benefits of using AD/LDAP as a provisioning method, KB 386870.

Note:

The value of the identifier used to uniquely identify a user in your OIDC identity provider must exactly match with the value from Active Directory. The unique identifier in the identity broker is used to map the two systems.

Option	Description
Unique Identifier in OIDC Identity Provider	An identifier in your OIDC identity provider that is used to identify a user uniquely. You can select or enter a custom attribute (except 'sub'). For example, `name`.
Unique Identifier in VCF Identity Broker	Select the unique identifier that will be used to map to an attribute from Active Directory. The value of this identifier must exactly match the value of the previous field called **Unique Identifier in OIDC Identity Provider**.

In the In the **Directory Details** screen, enter the relevant details or select the required options.

Option	Description
Directory name	Enter a name for the directory.
DNS Server Location	Select this option if you want the identity broker to find and use optimal domain controllers by querying the DNS Service Record.
Global Catalog	Global catalog has a partial read-only replica of all objects across all domains. Some of the limitations with selecting the global catalog option include: • The Active Directory object attributes that are replicated to the global catalog are identified in the Active Directory schema as the Partial Attribute Set (PAS). Only these attributes are available for attribute mapping by the service. If necessary, edit the schema to add or remove attributes that are stored in the global catalog. • The global catalog stores the group membership (the member attributes) of only universal groups. Only universal groups are synced to the service. If necessary, change the scope of a group from a local domain or from global to universal. • The bind DN account that you define when you configure a directory in the service must have permissions to read the Token-Groups-Global-And-Universal (TGGAU) attribute. **Note:** When you use Global catalog with LDAP and multiple domains, the additional domains are discovered only after the user/group sync is complete.
Encrypted Connection (Displayed when you select the option **DNS server location**)	Activates LDAPs for all connections to this directory.
Certificates for encrypted connection (Displayed when you select the **DNS server location** and **Encrypted Connections** options and if you enter a *ldaps://* string in the Primary domain controller field)	Upload the root certificate(s) required to access the domain controllers over SSL. All uploaded root certificates will be merged into a single file with the filename `{directory-name}-concatenated.pem` when you save the configuration. **Note:** Ensure that the certificate is in the (Privacy Enhanced Mail) PEM format and includes the "BEGIN CERTIFICATE" and "END CERTIFICATE" lines.
Directory search attribute	A specific LDAP attribute used in queries to search or identify entries in the directory. For example, `uid`, `sAMAccountName`.
Primary domain controller	A server machine running Active Directory Domain Services (AD DS) responsible for managing and authenticating user access, enforcing policies, and storing directory data. For example, `hostname = DC01` for the domain called example.com.
Secondary domain controller	A backup domain controller for the identity broker to point to if the primary domain controller is not reachable.
Directory search attribute	A specific LDAP attribute used in queries to search or identify entries in the directory. For example, `uid`, `sAMAccountName`.
Base DN	The Distinguished Name (DN) of the top-level entry in the directory tree from which LDAP queries begin. For example, `dc=example`, `dc=com`.
Bind user name	The user name of a user account with sufficient permissions to authenticate and perform LDAP operations on the directory.

Bind user password	The password of a user account with sufficient permissions to authenticate and perform LDAP operations on the directory.

7. In the **Review** screen, review the details you have added for the configuration and click **Finish**.

Configure user and group provisioning

8. In the **Configure User and Group Provisioning** screen, depending on the option you choose, complete the following steps and click **Done**.

   If you have chosen OIDC as the authentication protocol, and AD/LDAP as the user and group provisioning protocol, read KB 386870 to understand the finer details about configuring the setup.

   a) If you have chosen SCIM, you see **Configure SCIM push in your identity provider** instead of **Configure User and Group Provisioning**. Use the values listed on the screen to configure the SCIM application in the identity provider.

      - If your identity provider supports authorizing the SCIM application using OAuth grant type, use the client ID, client secret, and token endpoint. Otherwise, use the SCIM bearer token to authorize this SCIM server in your identity provider.
      - If required, you can regenerate these values after you have configured VCF single sign-on. To regenerate the values to configure the SCIM application in the identity provider, navigate to the respective VCF instance and click on **Identity Source**. The SCIM related values are available under **Directory Information**.
      - Ensure that the user attributes, username, and domain are pushed individually from the SCIM application by the identity provider. Similarly, ensure that the group attributes, group name, and domain are pushed individually as well.

   b) If you have chosen JIT, there is no additional step to be performed under **User and Group Provisioning**.

   c) If you have chosen AD/LDAP, review the directory information such as the domain controller and bind credentials that you had configured. Complete the following steps:

      1. In the **Attributes Mappings** screen, review the attribute mappings and click **Next.**
         The attributes in the identity broker directory are mapped to the Active Directory attributes for accurate synchronization of user information. If the mappings are incorrect, select the correct attribute from the drop-down list.
      2. In the **Group Provisioning** screen, enter the relevant details or select the required options and click **Next**.
         Use the **Group Provisioning** screen to search for the group names and select the corresponding Active Directory group Distinguished Names (DN) that you want to sync.

Option	Description
Specify the Base Group DN	Click the **Select Base Group DN** button if you have nested groups and want to specify the group names. You can search for the group name and select the group DNs that you want to sync from from the list in the datagrid below.
Sync Nested Groups	Select **Sync Nested Group** to sync all the users from the selected group and its nested group without maintaining the group hierarchy.

      3. From the **User Provisioning** screen, enter the relevant details or select the required options and click **Next**.
         Use the **User Provisioning** screen to search for the user names and select the corresponding Active Directory group Distinguished Names (DN) that you want to sync.

Option	Description
Specify the Base User DN	Click the **Select Base User DN** button to specify the user names that you want to sync. You can search for the user name and select the user DNs that you want to sync from from the list in the datagrid below. If you want to sync the users directly from the group, do not specify a base user DN.

      4. From the **Review screen**, review the configuration details of the identity provider.

5. Click **Finish**. The sync starts in the background. After the initial sync, the sync is scheduled to run once a week at a scheduled time.
6. Click **Done**.

For more information about the various functionalities you can perform as part of the management operations in Active Directory, see Managing AD/LDAP.

(Optional) Test Login

9. Click **Test Login** to test the login using your single-sign on credentials.
10. Click **Next** in the identity provider screen.
11. After a successful validation, click **Done**.
12. Click **Finish Setup**. This completes the initial configuration.
13. Click **Continue** to view the **VCF SSO Overview** page.

Important:

Important Notes on Backup

It is highly advisable to create a backup of both VCF Operations and the identity broker instance. Furthermore, it is advisable to perform a similar backup following any substantial modifications to the initial identity source configuration, such as when you 'Delete Single Sign-On'. This practice ensures that you can restore the VCF single sign-on configuration to a recognized and stable state if necessary. You can download the configuration as a JSON file.

Important Notes on Role Assignment

VCF roles can be created and mapped to individual component roles.

- As an admin you can assign built-in and custom VCF roles to groups and and users from the **Access Managment** tab (navigate to the VCF SSO Overview page, click on a VCF single sign-on configuration, and then click the Access Management tab).
- As an admin, you can also assign roles in the component console and enable VCF Single Sign-On from the **Component Configuration** tab (navigate to the VCF SSO Overview page, click on a VCF single sign-on configuration, and then click the Component Configuration tab).

Configure Microsoft ADFS as an Identity Provider

You can configure Microsoft ADFS as an identity provider using SAML or OIDC authentication protocols.

- For information about configuring Microsoft ADFS as an identity provider using the SAML authentication protocol, see Configure Microsoft ADFS as an Identity Provider Using SAML.
  For information about configuring Microsoft ADFS as an identity provider using the OIDC authentication protocol, see Configure Microsoft ADFS as an Identity Provider Using OIDC.

Configure Microsoft ADFS as an Identity Provider Using SAML

After installing or upgrading to VCF 9.1, to configure VCF single sign-on, you can configure Microsoft ADFS as an identity provider using SAML as the authentication protocol.

To set up an identity provider, you must:

1. Choose an identity provider
2. Configure the identity provider
3. Configure user and group provisioning
4. Test Login

Note:

You can test log in if the identity broker is at version 9.1.

Prerequisite

- Verify that you have chosen a deployment mode. For more information, see Step 1: Choose the Deployment Mode.

Procedure

Choose the identity provider

1. From the **Configure VCF SSO** page, click the **Start** button against the **Configure Identity Provider** option.
2. From the **Choose the identity provider** section, select **Microsoft Entra ADFS** from the list and click **Next**.

Configure the identity provider

3. From the **Configure the Identity Provider** section, click **Configure** to configure the identity broker to integrate with the selected identity provider for user authentication.
4. In the **Authentication Method** pane, select **SAML** (Security Assertion Markup Language).
5. In the **Service Provider Registration** pane, log in to your Microsoft ADFS admin console and follow the steps in Set up a SAML 2.0 provider with ADFS to create a SAML 2.0 in Microsoft ADFS.

   While registering Microsoft Entra ADFS, use the metadata provided in the **Service Provider Registration** screen for **Entity ID/Metadata URL** and **Assertion Consumer Service URL**. If the identity provider requires the metadata to be copy-pasted, it is recommended to download this file to avoid formatting issues.

   Note:

   Ensure that you add the following default claims when you configure your external identity provider. These claims are case-sensitive:

   - userName
   - email
   - domain
   - firstName
   - lastName
   - groups
6. Click **Next**.
7. In the **Identity Provider Configuration** screen, enter the relevant details or select the required options and click **Next**.

Option	Description
IDP Display Name	Enter a friendly name for your identity provider.
Metadata	Select either URL or XML.
Metadata XML	Copy the Metadata XML of your identity provider and paste it in this field. The Metadata XML is a URL that points to an XML document containing metadata about the service provider (identity broker). This metadata includes important information required for establishing trust and facilitating secure communication between the service provider and the identity provider. **Note:** If you have configured VCF single sign-on, after you upgrade to 9.1.1, it is recommended that you re-import the identity broker service provider metadata into your identity provider. Re-importing ensures your identity provider reads the new encryption hint and switches to SHA-256 OAEP for all subsequent SAML assertion encryption.

Metadata URL	Copy the Metadata URL of your identity provider and paste it in this field. The Metadata URL is used to fetch identity provider metadata. This metadata includes important information required for establishing trust and facilitating secure communication between the service provider and the identity provider. **Note:** If your identity provider is not publicly accessible or if the certificate lacks a signature from a recognized Certificate Authority (CA), VCF Operations cannot validate the metadata URL. Provide the metadata in XML format instead.
Name ID format and Name ID value	Select the Name ID format and value. The Name ID format is the value in the SAML response to identify the authenticated user.
SAML Context	SAML context refers to the method of authentication that VCF requires the identity provider to support when verifying a user's identity. • **PasswordProtectedTransport**: Authentication with a password sent over a secure transport (For example, HTTPS). • **Password**: Authentication using a password without requiring transport security. • **Unspecified:** If you don't want to enforce any particular type of authentication mechanism, you can use this option.

8. In the **User/Group Provisioning Method** screen, select your preferred mode of provisioning the users and groups. You can select either:
   - Just-In-Time Provisioning (JIT), or
   - Active Directory/Lightweight Directory Access Protocol (AD/LDAP)

   a) If you select **JIT** and click **Next**, complete the configuration as follows:
      - In the **Group Provisioning** screen, select your preferred mode of provisioning the groups. The options are **Groups Pre-provisioning** or **Just-In-Time Group Provisioning** and click **Next**.

Option	Description
Groups Pre-Provisioning	This option creates the required groups in advance to perform role assignments. Users are assigned to these groups dynamically when they log in based on SAML assertion. Only the pre-provisioned groups are available for role assignments.
Just-In-Time Group Provisioning	This option creates groups dynamically based on the SAML assertion when the user logs in. To assign entitlements to users before their first login, add them as users to a local group and assign permissions. Alternatively, you can use group pre-provisioning if you know the group names from Active Directory that requires entitlements.

      - In the **Domain(s)** screen, add the domain details that users from your enterprise will log in with, and click **Next**.
      - In the **Pre-provisioning Domains and Groups** screen, add the groups from each domain that you want to provision in VCF for roles and assignments by clicking the **Add Group** button, and click **Next**.

      Note:

      The **Pre-provisioning Domains and Groups** screen is displayed only if you have chosen groups to be pre-provisioned which means that you chose **Groups pre-provisioning** in the previous screen.

There should be at least one provisioned group added across the domains.

- In the **Attributes** screen, the user attributes section displays a list of mandatory and non-mandatory user attributes that you view in the SAML response from the identity provider.

  You can add additional user attributes if required. In the group attributes section of the workflow you can add a group attribute to be called for in the SAML request. Attribute names are case-sensitive.

Note:

If you choose JIT, it is recommended that you automatically disable inactive users.

b) If you select **AD/LDAP** and click **Next**, complete the configuration as follows. To learn more about the benefits of using AD/LDAP as a provisioning method, KB 386870.

In the **Directory Details** screen, enter the following details.

Option	Description
Directory name	Enter a name for the directory.
DNS Server Location	Select this option if you want the identity broker to find and use optimal domain controllers by querying the DNS Service Record.
Global Catalog	Global catalog has a partial read-only replica of all objects across all domains. Some of the limitations with selecting the global catalog option include: • The Active Directory object attributes that are replicated to the global catalog are identified in the Active Directory schema as the Partial Attribute Set (PAS). Only these attributes are available for attribute mapping by the service. If necessary, edit the schema to add or remove attributes that are stored in the global catalog. • The global catalog stores the group membership (the member attributes) of only universal groups. Only universal groups are synced to the service. If necessary, change the scope of a group from a local domain or from global to universal. • The bind DN account that you define when you configure a directory in the service must have permissions to read the Token-Groups-Global-And-Universal (TGGAU) attribute. **Note:** When you use Global catalog with LDAP and multiple domains, the additional domains are discovered only after the user/group sync is complete.
Encrypted Connection (Displayed when you select the **DNS server location** option)	Activates LDAPs for all connections to this directory.
Certificates for encrypted connection (Displayed when you select the **DNS server location** and **Encrypted Connections** options and if you enter a *ldaps://* string in the Primary domain controller field)	Upload the root certificate(s) required to access the domain controllers over SSL. All uploaded root certificates will be merged into a single file with the filename `{directory-name}-concatenated.pem` when you save the configuration. **Note:** Ensure that the certificate is in the (Privacy Enhanced Mail) PEM format and includes the "BEGIN CERTIFICATE" and "END CERTIFICATE" lines.
Directory search attribute	A specific LDAP attribute used in queries to search or identify entries in the directory. For example, `uid`, `sAMAccountName`.

Primary domain controller (Displayed when you select the **Global Catalog** option)	A server machine running Active Directory Domain Services (AD DS) responsible for managing and authenticating user access, enforcing policies, and storing directory data. For example, `hostname = DC01` for the domain called example.com.
Secondary domain controller (Displayed when you select the **Global Catalog** option)	A backup domain controller for the identity broker to point to if the primary domain controller is not reachable.
Directory search attribute	A specific LDAP attribute used in queries to search or identify entries in the directory. For example, `uid`, `sAMAccountName`.
Base DN	The Distinguished Name (DN) of the top-level entry in the directory tree from which LDAP queries begin. For example, `dc=example`, `dc=com`.
Bind user name	The user name of a user account with sufficient permissions to authenticate and perform LDAP operations on the directory.
Bind user password	The password of a user account with sufficient permissions to authenticate and perform LDAP operations on the directory.

9. In the **Review** screen, review the details you have added for the configuration and click **Finish**.

Configure user and group provisioning

10. In the **Configure User and Group Provisioning** screen, click **Done**.
    a) If you have chosen JIT, there is no additional step to be performed under **User and Group Provisioning**.
    b) If you have chosen AD/LDAP, review the directory information such as the domain controller and bind credentials that you had configured. Complete the following steps:
        1. In the **Attributes Mappings** screen, review the attribute mappings and click **Next.**
           The attributes in the identity broker directory are mapped to the Active Directory attributes for accurate synchronization of user information. If the mappings are incorrect, select the correct attribute from the drop-down list.
        2. In the **Group Provisioning** screen, enter the relevant details or select the required options and click **Next**.
           Use the **Group Provisioning** screen to search for the group names and select the corresponding Active Directory group Distinguished Names (DN) that you want to sync.

Option	Description
Specify the Base Group DN	Click the **Select Base Group DN** button if you have nested groups and want to specify the group names. You can search for the group name and select the group DNs that you want to sync from from the list in the datagrid below.
Sync Nested Groups	Select **Sync Nested Group** to sync all the users from the selected group and its nested group without maintaining the group hierarchy.

        3. From the **User Provisioning** screen, enter the relevant details or select the required options and click **Next**.
           Use the **User Provisioning** screen to search for the user names and select the corresponding Active Directory group Distinguished Names (DN) that you want to sync.

Option	Description
Specify the Base User DN	Click the **Select Base User DN** button to specify the user names that you want to sync. You can search for the user name and select the user DNs that you want to sync from from the list in the datagrid below. If you want to sync the users directly from the group, do not specify a base user DN.

        4. From the **Review screen**, review the configuration details of the identity provider.
        5. Click **Finish**. The sync starts in the background. After the initial sync, the sync is scheduled to run once a week at a scheduled time.
        6. Click **Done**.

For more information about the various functionalities you can perform as part of the management operations in Active Directory, see Managing AD/LDAP.

(Optional) Test Login

11. Click **Test Login** to test the login using your single-sign on credentials.
12. Click **Next** in the identity provider screen.
13. After a successful validation, click **Done**.
14. Click **Finish Setup**. This completes the initial configuration.
15. Click **Continue** to view the **VCF SSO Overview** page.

Important:

Important Notes on Backup

It is highly advisable to create a backup of both VCF Operations and the identity broker appliance. Furthermore, it is advisable to perform a similar backup following any substantial modifications to the initial identity source configuration, such as when you 'Delete Single Sign-On'. This practice ensures that you can restore the VCF single sign-on configuration to a recognized and stable state if necessary. You can download the configuration as a JSON file.

Important Notes on Role Assignment

VCF roles can be created and mapped to individual component roles.

- As an admin you can assign built-in and custom VCF roles to groups and and users from the **Access Managment** tab (navigate to the VCF SSO Overview page, click on a VCF single sign-on configuration, and then click the Access Management tab).
- As an admin, you can also assign roles in the component console and enable VCF single sign-on from the **Component Configuration** tab (navigate to the VCF SSO Overview page, click on a VCF single sign-on configuration, and then click the Component Configuration tab).

Configure Microsoft ADFS as an Identity Provider Using OIDC

After installing or upgrading to VCF 9.1, to configure VCF single sign-on, you can configure Microsoft ADFS as an identity provider using OIDC as the authentication protocol.

To set up an identity provider, you must:

1. Choose an identity provider
2. Configure the identity provider
3. Configure user and group provisioning
4. Test Login

Note:

You can test log in if the identity broker is at version 9.1.

Prerequisite

- Verify that you have chosen a deployment mode. For more information, see Step 1: Choose the Deployment Mode.
- If you configure VCF single sign-on using identity broker instance, and there are network restrictions that do not allow outbound public internet access, you must set up a proxy server. For more information, see Configure a Proxy Server for VCF Management Services Components and VCF Automation.

Procedure

Choose the identity provider

1. From the **Configure VCF SSO** page, click the **Start** button against the **Configure Identity Provider** option.

   If you have an existing identity provider configured in vCenter, see the information below. For a fresh configuration of the identity provider, continue from Step 6.

   For an existing source (existing identity provider configuration)

- Choose either SAML or OIDC as the authentication protocol.

  Note:

  If you choose SAML, the VCF single sign-on configuration is considered as a new authentication source configuration. Continue from Step. 6.
- If you choose OIDC:
  – If you choose the instance deployment mode, and select OIDC, the following fields are prepopulated in the **Identity Provider Configuration** screen.
    a. Review the details and provide the shared secret and click **Next**.
       - Identity provider name
       - Client identifier
       - OpenID address
       - Update Redirect URI in the Identity provider's OIDC application
       - Shared Secret: Paste the client secret of the OIDC application you created in the identity provider.
    b. Review the **User/Group Provisioning**, **Domain(s)**, and **Review** screens and click **Finish**.
    c. Review the SCIM push details and click **Done**. The identity provider configuration is complete.

  - If you choose the embedded deployment mode, and select OIDC:
    a. The configuration details of the identity provider configuration are automatically imported from existing configuration into the identity broker with OIDC as the authentication protocol and SCIM as the user/group provisioning method.
    b. Review the configuration values in the **Identity Provider Configuration** screen and click **Finish**.
    c. Review the SCIM details, click **Done**, and then click **Finish Setup**. The identity provider configuration is complete.

2. For a fresh identity provider configuration, from the **Choose the identity provider** section, select **Microsoft Entra ADFS** from the list and click **Next**.

   Configure the identity provider

   Note:

   Ensure that you add the following default claims when you configure your external identity provider. These claims are case-sensitive:

   - preferred_username
   - email
   - family_name
   - given_name
   - groups
   - domain

   If the default claims cannot be added, use alternative claim names and map them to the appropriate attributes in the VCF Operations **Identity & Access** page to support your JIT configurations.
3. From the **Configure the Identity Provider** section, click **Configure** to configure the identity broker to integrate with the selected identity provider for user authentication.
4. In the **Authentication Method** screen, select **OIDC** (OpenID Connect) and click **Next**.
5. In the **Identity Provider Configuration** screen, enter the relevant details or select the required options and click **Next**.

Option	Description
IDP Display Name	Enter the OpenID Connect provider name. Example, Microsoft Entra ADFS.
Client Identifier	A unique ID assigned to your OIDC application by the identity provider, for authentication.

Shared secret	A confidential key used by your OIDC application to authenticate itself with the identity provider. **Note:** Client secrets issued by external identity providers typically have expiration dates. When a client secret expires, it must be regenerated, and the corresponding OIDC configuration must be updated in VCF to ensure continued authentication functionality.
OpenID Address	The URL of the identity provider's OIDC configuration end point for initiating the authentication process.
Redirect URI	The URL where the identity provider will send the authentication response after a user logs in.
SSL Certificate	Public key certificate to verify OIDC identity. This field is required only if a self-signed certificate is used to sign the identity provider.

6. In the **User/Group Provisioning Method** screen, select your preferred mode of provisioning the users and groups. You can select either:
   - Just-In-Time Provisioning (JIT), or
   - Active Directory/Lightweight Directory Access Protocol (AD/LDAP)

   a) If you select **JIT** and click **Next**, complete the configuration as follows:
      - In the **Group Provisioning** screen, select your preferred mode of provisioning the groups. The options are **Groups Pre-provisioning** or **Just-In-Time Group Provisioning** and click **Next**.

Option	Description
Groups Pre-Provisioning	This option creates the required groups in advance to perform role assignments. Users are assigned to these groups dynamically when they log in based on OIDC token. Only the pre-provisioned groups are available for role assignments.
Just-In-Time Group Provisioning	This option creates groups dynamically based on the OIDC assertion when the user logs in. Alternatively, you can use group pre-provisioning if you know the group names from Active Directory that requires entitlements.

      - In the **Domain(s)** screen, add the domain details that users from your enterprise will log in with, and click **Next**.
      - In the **Pre-provisioning Domains and Groups** screen, add the groups from each domain that you want to provision in VCF for roles and assignments by clicking the **Add Group** button, and click **Next**.

      Note:

      The **Pre-provisioning Domains and Groups** screen is displayed only if you have chosen groups to be pre-provisioned which means that you chose **Groups pre-provisioning** in the previous screen.

      There should be at least one provisioned group added across the domains.
      - In the **Attributes** screen, the user attributes section displays a list of mandatory and non-mandatory user attributes that you view in the OIDC response from the identity provider.
        You can add additional user attributes if necessary. In the group attributes section of the workflow you can add a group attribute to be called for in the OIDC request. Attribute names are case-sensitive.

      Note:

      If you choose JIT, it is recommended that you automatically disable inactive users.

b) If you select **AD/LDAP**, select the **Unique Identifier in OIDC Identity Provider** and **Unique Identifier in VCF Identity Broker**, and click **Next**. To learn more about the benefits of using AD/LDAP as a provisioning method, see KB 386870.

Note:

The value of the identifier used to uniquely identify a user in your OIDC identity provider must exactly match with the value from Active Directory. The unique identifier in the identity broker is used to map the two systems.

Option	Description
Unique Identifier in OIDC Identity Provider	An identifier in your OIDC identity provider that is used to identify a user uniquely. You can select or enter a custom attribute (except 'sub'). For example, `name`.
Unique Identifier in VCF Identity Broker	Select the unique identifier that will be used to map to an attribute from Active Directory. The value of this identifier must exactly match the value of the previous field called **Unique Identifier in OIDC Identity Provider**.

In the In the **Directory Details** screen, enter the relevant details or select the required options.

Option	Description
Directory name	Enter a name for the directory.
DNS Server Location	Select this option if you want the identity broker to find and use optimal domain controllers by querying the DNS Service Record.
Global Catalog	Global catalog has a partial read-only replica of all objects across all domains. Some of the limitations with selecting the global catalog option include: • The Active Directory object attributes that are replicated to the global catalog are identified in the Active Directory schema as the Partial Attribute Set (PAS). Only these attributes are available for attribute mapping by the service. If necessary, edit the schema to add or remove attributes that are stored in the global catalog. • The global catalog stores the group membership (the member attributes) of only universal groups. Only universal groups are synced to the service. If necessary, change the scope of a group from a local domain or from global to universal. • The bind DN account that you define when you configure a directory in the service must have permissions to read the Token-Groups-Global-And-Universal (TGGAU) attribute. **Note:** When you use Global catalog with LDAP and multiple domains, the additional domains are discovered only after the user/group sync is complete.
Encrypted Connection (Displayed when you select the **DNS server location** option)	Activates LDAPs for all the connections to this directory.
Certificates for encrypted connection (Displayed when you select the **DNS server location** and **Encrypted Connections** options and if you enter a *`ldaps://`* string in the Primary domain controller field)	Upload the root certificate(s) required to access the domain controllers over SSL. All uploaded root certificates will be merged into a single file with the filename `{directory-name}-concatenated.pem` when you save the configuration. **Note:** Ensure that the certificate is in the (Privacy Enhanced Mail) PEM format and includes the "BEGIN CERTIFICATE" and "END CERTIFICATE" lines.

Directory search attribute	A specific LDAP attribute used in queries to search or identify entries in the directory. For example, `uid`, `sAMAccountName`.
Primary domain controller	A server machine running Active Directory Domain Services (AD DS) responsible for managing and authenticating user access, enforcing policies, and storing directory data. For example, `hostname = DC01` for the domain called example.com.
Secondary domain controller	A backup domain controller for the identity broker to point to if the primary domain controller is not reachable.
Directory search attribute	A specific LDAP attribute used in queries to search or identify entries in the directory. For example, `uid`, `sAMAccountName`.
Base DN	The Distinguished Name (DN) of the top-level entry in the directory tree from which LDAP queries begin. For example, `dc=example`, `dc=com`.
Bind user name	The user name of a user account with sufficient permissions to authenticate and perform LDAP operations on the directory.
Bind user password	The password of a user account with sufficient permissions to authenticate and perform LDAP operations on the directory.

7. In the **Review** screen, review the details you have added for the configuration and click **Finish**.
8. **Configure user and group provisioning**
9. In the **Configure User and Group Provisioning** screen, click **Done**.
   a) If you have chosen JIT, there is no additional step to be performed under **User and Group Provisioning**.
   b) If you have chosen AD/LDAP, review the directory information such as the domain controller and bind credentials that you had configured. Complete the following steps:
      1. In the **Attributes Mappings** screen, review the attribute mappings and click **Next.**
         The attributes in the identity broker directory are mapped to the Active Directory attributes for accurate synchronization of user information. If the mappings are incorrect, select the correct attribute from the drop-down list.
      2. In the **Group Provisioning** screen, enter the relevant details or select the required options and click **Next**.
         Use the **Group Provisioning** screen to search for the group names and select the corresponding Active Directory group Distinguished Names (DN) that you want to sync.

Option	Description
Specify the Base Group DN	Click the **Select Base Group DN** button if you have nested groups and want to specify the group names. You can search for the group name and select the group DNs that you want to sync from from the list in the datagrid below.
Sync Nested Groups	Select **Sync Nested Group** to sync all the users from the selected group and its nested group without maintaining the group hierarchy.

      3. From the **User Provisioning** screen, enter the relevant details or select the required options and click **Next**.
         Use the **User Provisioning** screen to search for the user names and select the corresponding Active Directory group Distinguished Names (DN) that you want to sync.

Option	Description
Specify the Base User DN	Click the **Select Base User DN** button to specify the user names that you want to sync. You can search for the user name and select the user DNs that you want to sync from from the list in the datagrid below. If you want to sync the users directly from the group, do not specify a base user DN.

      4. From the **Review screen**, review the configuration details of the identity provider.

5. Click **Finish**. The sync starts in the background. After the initial sync, the sync is scheduled to run once a week at a scheduled time.
6. Click **Done**.

For more information about the various functionalities you can perform as part of the management operations in Active Directory, see Managing AD/LDAP.

(Optional) Test Login

10. Click **Test Login** to test the login using your single sign-on credentials.
11. Click **Next** in the identity provider screen.
12. After a successful validation, click **Done**.
13. Click **Finish Setup**. This completes the initial configuration.
14. Click **Continue** to view the **VCF SSO Overview** page.

Important:

Important Notes on Backup

It is highly advisable to create a backup of both VCF Operations and the identity broker instance. Furthermore, it is advisable to perform a similar backup following any substantial modifications to the initial identity source configuration, such as when you 'Delete Single Sign-On'. This practice ensures that you can restore the VCF single sign-on configuration to a recognized and stable state if necessary. You can download the configuration as a JSON file.

Important Notes on Role Assignment

VCF roles can be created and mapped to individual component roles.

- As an admin you can assign built-in and custom VCF roles to groups and and users from the **Access Managment** tab (navigate to the VCF SSO Overview page, click on a VCF single sign-on configuration, and then click the Access Management tab).
- As an admin, you can also assign roles in the component console and enable VCF Single Sign-On from the **Component Configuration** tab (navigate to the VCF SSO Overview page, click on a VCF single sign-on configuration, and then click the Component Configuration tab).

Configure Symantec Identity Security Platform as an Identity Provider

You can configure Symantec Identity Security Platform as an identity provider using SAML or OIDC authentication protocols.

- For information about configuring Symantec Identity Security Platform as an identity provider using the SAML authentication protocol, see Configure Symantec Identity Security Platform as an Identity Provider Using SAML.
- For information about configuring Symantec Identity Security Platform as an identity provider using the OIDC authentication protocol, see Configure Symantec Identity Security Platform as an Identity Provider Using OIDC.

Configure Symantec Identity Security Platform as an Identity Provider Using SAML

After installing or upgrading to VCF 9.1, to configure VCF Single Sign-On, you can configure Symantec Identity Security Platform as an identity provider using SAML as the authentication protocol.

To set up an identity provider, you must:

1. Choose an identity provider
2. Configure the identity provider
3. Configure user and group provisioning
4. Test Login

Note:

You can test log in if the identity broker is at version 9.1.

Prerequisite

- Verify that you have chosen a deployment mode. For more information, see Step 1: Choose the Deployment Mode.

Procedure

Choose the identity provider

1. From the **Configure VCF SSO** page, click the **Start** button against the **Configure Identity Provider** option.
2. From the **Choose the identity provider** section, select **Symantec Identity Security Platform** from the list and click **Next**.

Configure the identity provider

3. From the **Configure the Identity Provider** section, click **Configure** to configure the identity broker to integrate with the selected identity provider for user authentication.
4. In the **Authentication Method** screen, select **SAML** (Security Assertion Markup Language) and click **Next**.
5. In the **Service Provider Registration** screen, log in to your Symantec Identity Security Platform admin console and follow the steps in SAML to create a SAML 2.0 application in Symantec Identity Security Platform.

   While registering Symantec Identity Security Platform, use the metadata provided in the **Service Provider Registration** screen for the **Entity ID/Metadata URL** and **Assertion Consumer Service URL** options. If the identity provider requires the metadata to be copy-pasted, it is recommended to download this file to avoid formatting issues.

   Note:

   Ensure that you add the following default claims when you configure your external identity provider. Note that these claims are case-sensitive:

   - userName
   - email
   - domain
   - firstName
   - lastName
   - groups
6. Click **Next**.
7. In the **Identity Provider Configuration** screen, enter the relevant details or select the required options and click **Next**.

Option	Description
IDP Display Name	Enter a friendly name for your identity provider.
Metadata XML	Copy the Metadata XML of your identity provider and paste it in this field. The Metadata XML is a URL that points to an XML document containing metadata about the service provider (identity broker). This metadata includes important information required for establishing trust and facilitating secure communication between the service provider and the identity provider.
Entity URL/Metadata URL	Copy the Metadata URL of your identity provider and paste it in this field. The Metadata URL is used to fetch identity provider metadata. This metadata includes important information required for establishing trust and facilitating secure communication between the service provider and the identity provider. **Note:** If your identity provider is not publicly accessible or if the certificate lacks a signature from a recognized Certificate Authority (CA), VCF Operations cannot validate the metadata URL. Provide the metadata in XML format instead.
Name ID format and Name ID value	Select the Name ID format and value. The Name ID format is the value in the SAML response to identify the authenticated user.

Option	Description
SAML Context	SAML context refers to the method of authentication that VCF requires the identity provider to support when verifying a user's identity. • **PasswordProtectedTransport**: Authentication with a password sent over a secure transport (For example, HTTPS). • **Password**: Authentication using a password without requiring transport security. • **Unspecified:** If you don't want to enforce any particular type of authentication mechanism, you can use this option.

8. In the **User/Group Provisioning Method** screen, select your preferred mode of provisioning the users and groups. You can select either:
   - Just-In-Time Provisioning (JIT), or
   - Active Directory/Lightweight Directory Access Protocol (AD/LDAP)

   a) If you select **JIT** and click **Next**, complete the configuration as follows:
      - In the **Group Provisioning** screen, select your preferred mode of provisioning the groups. The options are **Groups Pre-provisioning** or **Just-In-Time Group Provisioning** and click **Next**.

Option	Description
Groups Pre-Provisioning	This option creates the required groups in advance to perform role assignments. Users are assigned to these groups dynamically when they log in based on SAML assertion. Only the pre-provisioned groups are available for role assignments.
Just-In-Time Group Provisioning	This option creates groups dynamically based on the SAML assertion when the user logs in.

      - In the **Domain(s)** screen, add the domain details that users from your enterprise will log in with, and click **Next**.
      - In the **Pre-provisioning Domains and Groups** screen, add the groups from each domain that you want to provision in VCF for roles and assignments by clicking the **Add Group** button, and click **Next**.

        Note:

        The **Pre-provisioning Domains and Groups** screen is displayed only if you have chosen groups to be pre-provisioned which means that you chose **Groups pre-provisioning** in the previous screen.

        There should be at least one provisioned group added across the domains.
      - In the **Attributes** screen, the user attributes section displays a list of mandatory and non-mandatory user attributes that you view in the SAML response from the identity provider.

        You can add additional user attributes if required. In the group attributes section of the workflow you can add a group attribute to be called for in the SAML request. Attribute names are case-sensitive.

        Note:

        If you choose JIT, it is recommended that you automatically disable inactive users.

   b) If you select **AD/LDAP** and click **Next**, complete the configuration as follows. To learn more about the benefits of using AD/LDAP as a provisioning method, see KB 386870.
      - In the **Directory Details** screen, enter the following details or select the required options.

Option	Description
Directory name	Enter a name for the directory.
DNS Server Location	Select this option if you want the identity broker to find and use optimal domain controllers by querying the DNS Service Record.

Option	Description
Global Catalog	Global catalog has a partial read-only replica of all objects across all domains. Some of the limitations with selecting the global catalog option include: • The Active Directory object attributes that are replicated to the global catalog are identified in the Active Directory schema as the Partial Attribute Set (PAS). Only these attributes are available for attribute mapping by the service. If necessary, edit the schema to add or remove attributes that are stored in the global catalog. • The global catalog stores the group membership (the member attributes) of only universal groups. Only universal groups are synced to the service. If necessary, change the scope of a group from a local domain or from global to universal. • The bind DN account that you define when you configure a directory in the service must have permissions to read the Token-Groups-Global-And-Universal (TGGAU) attribute. **Note:** When you use Global catalog with LDAP and multiple domains, the additional domains are discovered only after the user/group sync is complete.
Encrypted Connection (Displayed when you select the **DNS server location** option)	This option activates LDAPs for all connections to this directory.
Certificates for encrypted connection (Displayed when you select the **DNS server location** and **Encrypted Connections** options and if you enter a *ldaps://* string in the Primary domain controller field).	Upload the root certificate(s) required to access the domain controllers over SSL. All uploaded root certificates will be merged into a single file with the filename `{directory-name}-concatenated.pem` when you save the configuration. **Note:** Ensure that the certificate is in the (Privacy Enhanced Mail) PEM format and includes the "BEGIN CERTIFICATE" and "END CERTIFICATE" lines.
Directory search attribute	A specific LDAP attribute used in queries to search or identify entries in the directory. For example, `uid`, `sAMAccountName`.
Primary domain controller (Displayed when you select the **Global Catalog** option)	A server machine running Active Directory Domain Services (AD DS) responsible for managing and authenticating user access, enforcing policies, and storing directory data. For example, `hostname = DC01` for the domain called example.com.
Secondary domain controller (Displayed when you select the **Global Catalog** option)	A backup domain controller for the identity broker to point to if the primary domain controller is not reachable.
Base DN	The Distinguished Name (DN) of the top-level entry in the directory tree from which LDAP queries begin. For example, `dc=example`, `dc=com`.
Bind user name	The user name of a user account with sufficient permissions to authenticate and perform LDAP operations on the directory.
Bind user password	The password of a user account with sufficient permissions to authenticate and perform LDAP operations on the directory.

9. In the **Review** screen, review the details that you have added for the configuration and click **Finish**.

Configure user and group provisioning

10. In the **Configure User and Group Provisioning** screen, depending on the option you choose, complete the following steps and click **Done**.
    a) If you have chosen JIT, there is no additional step to be performed under **User and Group Provisioning**.
    b) If you have chosen AD/LDAP, review the directory information such as the domain controller and bind credentials that you had configured. Complete the following steps:
        1. In the **Attributes Mappings** screen, review the attribute mappings and click **Next.**
           The attributes in the identity broker directory are mapped to the Active Directory attributes for accurate synchronization of user information. If the mappings are incorrect, select the correct attribute from the drop-down list.
        2. In the **Group Provisioning** screen, enter the relevant details or select the required options and click **Next**.

Use the **Group Provisioning** screen to search for the group names and select the corresponding Active Directory group Distinguished Names (DN) that you want to sync.

Option	Description
Specify the Base Group DN	Click the **Select Base Group DN** button if you have nested groups and want to specify the group names. You can search for the group name and select the group DNs that you want to sync from from the list in the datagrid below.
Sync Nested Groups	Select **Sync Nested Group** to sync all the users from the selected group and its nested group without maintaining the group hierarchy.

3. From the **User Provisioning** screen, enter the relevant details or select the required options and click **Next**.
   Use the **User Provisioning** screen to search for the user names and select the corresponding Active Directory group Distinguished Names (DN) that you want to sync.

Option	Description
Specify the Base User DN	Click the **Select Base User DN** button to specify the user names that you want to sync. You can search for the user name and select the user DNs that you want to sync from from the list in the datagrid below. If you want to sync the users directly from the group, do not specify a base user DN.

4. From the **Review screen**, review the configuration details of the identity provider.
5. Click **Finish**. The sync starts in the background. After the initial sync, the sync is scheduled to run once a week at a scheduled time.
6. Click **Done**.

For more information about the various functionalities you can perform as part of the management operations in Active Directory, see Managing AD/LDAP.

(Optional) Test Login

11. Click **Test Login** to test the login using your single-sign on credentials.
12. Click **Next** in the identity provider screen.
13. After a successful validation, click **Done**.
14. Click **Finish Setup**. This completes the initial configuration.
15. Click **Continue** to view the **VCF SSO Overview** page.

Important:

Important Notes on Backup

It is highly advisable to create a backup of both VCF Operations and the identity broker instance. Furthermore, it is advisable to perform a similar backup following any substantial modifications to the initial identity source configuration, such as when you 'Delete Single Sign-On'. This practice ensures that you can restore the VCF Single Sign-On configuration to a recognized and stable state if necessary. You can download the configuration as a JSON file.

Important Notes on Role Assignment

VCF roles can be created and mapped to individual component roles.

- As an admin you can assign built-in and custom VCF roles to groups and and users from the **Access Managment** tab (navigate to the VCF SSO Overview page, click on a VCF single-sign on configuration, and then click the Access Management tab).
- As an admin, you can also assign roles in the component console and enable VCF single-sign on from the **Component Configuration** tab (navigate to the VCF SSO Overview page, click on a VCF single-sign on configuration, and then click the Component Configuration tab).

Configure Symantec Identity Security Platform as an Identity Provider Using OIDC

After installing or upgrading to VCF 9.1, to configure VCF single sign-on, you can configure Symantec Identity Security Platform as an identity provider using OIDC as the authentication protocol.

To set up an identity provider, you must:

1. Choose an identity provider
2. Configure the identity provider
3. Configure user and group provisioning
4. Test Login

Note:

You can test login if the identity broker is at version 9.1.

Prerequisites

- Verify that you have chosen a deployment mode. For more information, see Step 1: Choose the Deployment Mode.
- If you configure VCF single sign-on using identity broker instance, and there are network restrictions that do not allow outbound public internet access, you must set up a proxy server. For more information, see Configure a Proxy Server for VCF Management Services Components and VCF Automation.

Choose the identity provider

1. From the **Configure VCF SSO** page, click the **Start** button against the **Configure Identity Provider** option.

   If you have an existing identity provider configured in vCenter, see the information below. For a fresh configuration of the identity provider, continue from Step 6.

   For an existing source (existing identity provider configuration)

   - Choose either SAML or OIDC as the authentication protocol.

     Note:

     If you choose SAML, the VCF single sign-on configuration is considered as a new authentication source configuration. Continue from Step. 6.
   - If you choose OIDC:
     – If you choose the instance deployment mode, and select OIDC, the following fields are prepopulated in the **Identity Provider Configuration** screen.
       a. Review the details and provide the shared secret and click **Next**.
          - Identity provider name
          - Client identifier
          - OpenID address
          - Update Redirect URI in the Identity provider's OIDC application
          - Shared Secret: Paste the client secret of the OIDC application you created in the identity provider.
       b. Review the **User/Group Provisioning**, **Domain(s)**, and **Review** screens and click **Finish**.
       c. Review the SCIM push details and click **Done**. The identity provider configuration is complete.
     – If you choose the embedded deployment mode, and select OIDC:
       a. The configuration details of the identity provider configuration are automatically imported from existing configuration into the identity broker with OIDC as the authentication protocol and SCIM as the user/group provisioning method.
       b. Review the configuration values in the **Identity Provider Configuration** screen and click **Finish**.
       c. Review the SCIM details, click **Done** and then click **Finish Setup**. The identity provider configuration is complete.

2. For a fresh identity provider configuration, from the **Choose the identity provider** section, select **Symantec Identity Security Platform** from the list and click **Next**.

Configure the identity provider

Note:

Ensure that you add the following default claims when you configure your external identity provider. These claims are case-sensitive:

- preferred_username
- email
- family_name
- given_name
- groups
- domain

If the default claims cannot be added, use alternative claim names and map them to the appropriate attributes in the VCF Operations **Identity & Access** page to support your JIT configurations.

3. From the **Configure the Identity Provider** section, click **Configure** to configure the identity broker to integrate with the selected identity provider for user authentication.
4. In the **Authentication Method** screen, select **OIDC** (OpenID Connect) and click **Next**.
5. In the **Identity Provider Configuration** screen, enter the relevant details or select the required options and click **Next**.

Option	Description
IDP Display Name	Enter the OpenID Connect provider name. Example, Ping Identity.
Client Identifier	A unique ID assigned to your OIDC application by the identity provider for authentication.
Shared secret	A confidential key used by your OIDC application to authenticate itself with the identity provider. **Note:** Client secrets that are issued by external identity providers typically have expiration dates. When a client secret expires, it must be regenerated, and the corresponding OIDC configuration must be updated in VCF to ensure continued authentication functionality.
OpenID Address	The URL of the identity provider's OIDC configuration endpoint for initiating the authentication process. For example, `https://<hostname>/<tenant>/.well-known/openid-configuration`.
Redirect URI	The URL where the identity provider sends the authentication response after a user logs in.
List of domains	Domains for a user to log in to your company.

6. In the **User/Group Provisioning Method** screen, select your preferred mode of provisioning the users and groups. You can select either:
   - Just-In-Time Provisioning (JIT), or
   - Active Directory/Lightweight Directory Access Protocol (AD/LDAP)

   a) If you select **JIT** and click **Next**, complete the configuration as follows:
      - In the **Group Provisioning** screen, select your preferred mode of provisioning the groups. The options are **Groups Pre-provisioning** or **Just-In-Time Group Provisioning** and click **Next**.

Option	Description
Groups Pre-Provisioning	This option creates the required groups in advance to perform role assignments. Users are assigned to these groups dynamically when they log in based on OIDC token. Only the pre-provisioned groups are available for role assignments.
Just-In-Time Group Provisioning	This option creates groups dynamically based on the OIDC token when the user logs in. Alternatively, you can use group pre-provisioning if you know the group names from Active Directory that requires entitlements.

- In the **Domain(s)** screen, add the domain details that users from your enterprise will log in with, and click **Next**.
- In the **Pre-provisioning Domains and Groups** screen, add the groups from each domain that you want to provision in VCF for roles and assignments by clicking the **Add Group** button, and click **Next**.

Note:

The **Pre-provisioning Domains and Groups** screen is displayed only if you have chosen groups to be pre-provisioned which means that you chose **Groups pre-provisioning** in the previous screen.

There should be at least one provisioned group added across the domains.

- In the **Attributes** screen, the user attributes section displays a list of mandatory and non-mandatory user attributes that you view in the OIDC response from the identity provider.
  You can add additional user attributes if necessary. In the group attributes section of the workflow you can add a group attribute to be called for in the OIDC request. Attribute names are case-sensitive.

Note:

If you use public APIs to update identity provider attribute mappings, the following identity broker attributes are available and can be mapped to OpenID attributes:

- ExternalId
- userName
- firstName
- lastName
- email
- phone
- active
- employeeID
- distinguishedName
- userPrincipalName
- domain
- sourceAnchor
- displayName
- nickName
- profileUrl
- title
- organization
- costCenter
- department
- division
- managerId

Note:

If you choose JIT, it is recommended that you automatically disable inactive users.

b) If you select **AD/LDAP** and click **Next**, select **Unique Identifier in OIDC Identity Provider** and **Unique Identifier in VCF Identity Broker** and click **Next**. To learn more about the benefits of using AD/LDAP as a provisioning method, see KB 386870.

Note:

The value of the identifier used to uniquely identify a user in your OIDC identity provider must exactly match with the value from Active Directory. The unique identifier in the identity broker is used to map the two systems.

Option	Description
Unique Identifier in OIDC Identity Provider	An identifier in your OIDC identity provider that is used to identify a user uniquely. You can select or enter a custom attribute (except 'sub'). For example, `name`.

Unique Identifier in VCF Identity Broker	Select the unique identifier that will be used to map to an attribute from Active Directory. The value of this identifier must exactly match the value of the previous field called **Unique Identifier in OIDC Identity Provider**.

In the **Directory Details** screen, enter the relevant details or select the required options.

Option	Description
Directory name	Enter a name for the directory.
DNS Server Location	Select this option if you want the identity broker to find and use optimal domain controllers by querying the DNS Service Record.
Global Catalog	Global catalog has a partial read-only replica of all objects across all domains. Some of the limitations with selecting the global catalog option include: • The Active Directory object attributes that are replicated to the global catalog are identified in the Active Directory schema as the Partial Attribute Set (PAS). Only these attributes are available for attribute mapping by the service. If necessary, edit the schema to add or remove attributes that are stored in the global catalog. • The global catalog stores the group membership (the member attributes) of only universal groups. Only universal groups are synced to the service. If necessary, change the scope of a group from a local domain or from global to universal. • The bind DN account that you define when you configure a directory in the service must have permissions to read the Token-Groups-Global-And-Universal (TGGAU) attribute. **Note:** When you use Global catalog with LDAP and multiple domains, the additional domains are discovered only after the user/group sync is complete.
Encrypted Connection (Displayed when you select the **DNS server location** option)	Activates LDAPs for all the connections to this directory.
Certificates for encrypted connection (Displayed when you select the **DNS server location** and **Encrypted Connections** options and if you enter a *ldaps://* string in the Primary domain controller field)	Upload the root certificate(s) required to access the domain controllers over SSL. All uploaded root certificates will be merged into a single file with the filename `{directory-name}-concatenated.pem` when you save the configuration. **Note:** Ensure that the certificate is in the (Privacy Enhanced Mail) PEM format and includes the "BEGIN CERTIFICATE" and "END CERTIFICATE" lines.
Directory search attribute	A specific LDAP attribute used in queries to search or identify entries in the directory. For example, `uid`, `sAMAccountName`.
Primary domain controller	A server machine running Active Directory Domain Services (AD DS) responsible for managing and authenticating user access, enforcing policies, and storing directory data. For example, `hostname = DC01` for the domain called example.com.
Secondary domain controller	A backup domain controller for the identity broker to point to if the primary domain controller is not reachable.
Directory search attribute	A specific LDAP attribute used in queries to search or identify entries in the directory. For example, `uid`, `sAMAccountName`.

Base DN	The Distinguished Name (DN) of the top-level entry in the directory tree from which LDAP queries begin. For example, `dc=example, dc=com`.
Bind user name	The user name of a user account with sufficient permissions to authenticate and perform LDAP operations on the directory.
Bind user password	The password of a user account with sufficient permissions to authenticate and perform LDAP operations on the directory.

7. In the **Review** screen, review the details that you have added for the configuration and click **Finish**.

Configure user and group provisioning

8. In the **Configure User and Group Provisioning** screen, depending on the option you choose, complete the following steps and click **Done**.

   If you have chosen OIDC as the authentication protocol, and AD/LDAP as the user and group provisioning protocol, read KB 386870 to understand the finer details about configuring the setup.

   a) If you have chosen JIT, there is no additional step to be performed under **User and Group Provisioning**.
   b) If you have chosen AD/LDAP, review the directory information such as the domain controller and bind credentials that you had configured. Complete the following steps:

      1. In the **Attributes Mappings** screen, review the attribute mappings and click **Next.**
         The attributes in the identity broker directory are mapped to the Active Directory attributes for accurate synchronization of user information. If the mappings are incorrect, select the correct attribute from the drop-down list.
      2. In the **Group Provisioning** screen, enter the relevant details or select the required options and click **Next**.
         Use the **Group Provisioning** screen to search for the group names and select the corresponding Active Directory group Distinguished Names (DN) that you want to sync.

Option	Description
Specify the Base Group DN	Click the **Select Base Group DN** button if you have nested groups and want to specify the group names. You can search for the group name and select the group DNs that you want to sync from from the list in the datagrid below.
Sync Nested Groups	Select **Sync Nested Group** to sync all the users from the selected group and its nested group without maintaining the group hierarchy.

      3. From the **User Provisioning** screen, enter the relevant details or select the required options and click **Next**.
         Use the **User Provisioning** screen to search for the user names and select the corresponding Active Directory group Distinguished Names (DN) that you want to sync.

Option	Description
Specify the Base User DN	Click the **Select Base User DN** button to specify the user names that you want to sync. You can search for the user name and select the user DNs that you want to sync from from the list in the datagrid below. If you want to sync the users directly from the group, do not specify a base user DN.

      4. From the **Review screen**, review the configuration details of the identity provider.
      5. Click **Finish**. The sync starts in the background. After the initial sync, the sync is scheduled to run once a week at a scheduled time.
      6. Click **Done**.

   For more information about the various functionalities you can perform as part of the management operations in Active Directory, see Managing AD/LDAP.

(Optional) Test Login

9. Click **Test Login** to test the login using your single-sign on credentials.
10. Click **Next** in the identity provider screen.
11. After a successful validation, click **Done**.
12. Click **Finish Setup**. This completes the initial configuration.
13. Click **Continue** to view the **VCF SSO Overview** page.

Important:

Important Notes on Backup

It is highly advisable to create a backup of both VCF Operations and the identity broker instance. Furthermore, it is advisable to perform a similar backup following any substantial modifications to the initial identity source configuration, such as when you 'Delete Single Sign-On'. This practice ensures that you can restore the VCF single sign-on configuration to a recognized and stable state if necessary. You can download the configuration as a JSON file.

Important Notes on Role Assignment

VCF roles can be created and mapped to individual component roles.

- As an admin you can assign built-in and custom VCF roles to groups and and users from the **Access Managment** tab (navigate to the VCF SSO Overview page, click on a VCF Single Sign-On configuration, and then click the **Access Management** tab).
- As an admin, you can also assign roles in the component console and enable VCF single sign-on from the **Component Configuration** tab (navigate to the VCF SSO Overview page, click on a VCF single sign-on configuration, and then click the Component Configuration tab).

Configure an Identity Provider using Generic SAML 2.0
After installing or upgrading to VCF 9.1, to configure VCF single-sign on, you can configure an identity provider using Generic SAML 2.0 as the authentication protocol.

To set up an identity provider, you must:

1. Choose an identity provider
2. Configure the identity provider
3. Configure user and group provisioning
4. Test Login

Note:

You can test login if the identity broker is at version 9.1.

Prerequisite

- Verify that you have chosen a deployment mode. For more information, see Step 1: Choose the Deployment Mode.

Procedure

Choose the identity provider

1. From the **Configure VCF SSO** page, click the **Start** button against the **Configure Identity Provider** option.
2. From the **Choose the identity provider** section, select **Generic SAML 2.0** from the list and click **Next**.

Configure the identity provider

3. From the **Configure the Identity Provider** section, click **Configure** to configure the identity broker to integrate with the selected identity provider for user authentication.
4. In the **Authentication Method** screen, select **SAML** (Security Assertion Markup Language).
5. In the **Service Provider Registration** screen, log in to your Identity Provider admin console and create a SAML 2.0 application.

   While configuring the application, use the metadata provided in the **Service Provider Registration** screen for the **Entity ID/Metadata URL** and **Assertion Consumer Service URL** options. If the identity provider requires the metadata to be copy-pasted, it is recommended to download this file to avoid formatting issues.

   Note:

   Ensure that you add the following default claims when you configure your external identity provider. These claims are case-sensitive:

   - userName
   - email
   - domain
   - firstName
   - lastName
   - groups

   Click **Next**.
6. In the **Identity Provider Configuration** screen, enter the relevant details or select the required options and click **Next**.

Option	Description
IDP Display Name	Enter a friendly name for your identity provider.
Metadata XML	Copy the Metadata XML of your identity provider and paste it in this field. The Metadata XML is a URL that points to an XML document containing metadata about the service provider (identity broker). This metadata includes important information required for establishing trust and facilitating secure communication between the service provider and the identity provider. **Note:** If you have configured VCF single sign-on, after you upgrade to 9.1.1, it is recommended that you re-import the identity broker service provider metadata into your identity provider. Re-importing ensures your identity provider reads the new encryption hint and switches to SHA-256 OAEP for all subsequent SAML assertion encryption.
Metadata URL	Copy the Metadata URL of your identity provider and paste it in this field. The Metadata URL is used to fetch identity provider metadata. This metadata includes important information required for establishing trust and facilitating secure communication between the service provider and the identity provider. **Note:** If your identity provider is not publicly accessible or uses a self-signed certificate, VCF Operations cannot validate the metadata URL. Provide the metadata in XML format instead.
Name ID format and Name ID value	Select the Name ID format and value. The Name ID format is the value in the SAML response to identify the authenticated user.

SAML Context	SAML context refers to the method of authentication that VCF requires the identity provider to support when verifying a user's identity. • **PasswordProtectedTransport**: Authentication with a password sent over a secure transport (For example, HTTPS). • **Password**: Authentication using a password without requiring transport security. • **Unspecified:** If you don't want to enforce any particular type of authentication mechanism, you can use this option.

Table 1133: Keycloak Options

Enter the following details if you are configuring Keycloak as an identity provider.

Option	Description
Entity ID/Metadata URL	Copy the Entity ID/Metadata XML of your identity provider and paste it in this field. The Entity ID/Metadata XML is a URL that points to an XML document containing metadata about the service provider (identity broker). This metadata includes important information required for establishing trust and facilitating secure communication between the service provider and the identity provider.
Assertion Consumer Service URL	An endpoint on the service provider (identity broker) that is responsible for receiving and processing the SAML assertions sent by the identity provider.
Name ID format and Name ID value	Select the Name ID format and value. The Name ID format is the value in the SAML response to identify the authenticated user.
SAML Context	SAML context refers to the method of authentication that VCF requires the identity provider to support when verifying a user's identity. • **PasswordProtectedTransport**: Authentication with a password sent over a secure transport (e.g., HTTPS). • **Password**: Authentication using a password without requiring transport security. • **Unspecified:** Select this if you don't want to enforce any particular type of authentication mechanism

7. In the User/Group Provisioning Method screen, select your preferred mode of provisioning the users and groups. You can select either:
   - System for Cross-domain Identity Management (SCIM), or
   - Just-In-Time Provisioning (JIT), or
   - Active Directory/Lightweight Directory Access Protocol (AD/LDAP)

   a) If you select SCIM and click Next, complete the configuration as follows.
      - In the Domain(s) screen, enter the domain that users from your enterprise will log in with, and click Next.

   b) If you select JIT and click Next, complete the configuration as follows:
      - In the Group Provisioning screen, select your preferred mode of provisioning the groups. The options are Groups Pre-provisioning or Just-In-Time Group Provisioning and click Next.

Option	Description
Groups Pre-Provisioning	This option creates the required groups in advance to perform role assignments. Users are assigned to these groups dynamically when they log in based on SAML assertion. Only the pre-provisioned groups are available for role assignments.

Just-In-Time Group Provisioning	This option creates groups dynamically based on the SAML assertion when the user logs in. Alternatively, you can use group pre-provisioning if you know the group names from Active Directory that requires entitlements.

- In the **Domain(s)** screen, add the domain details that users from your enterprise will log in with, and click Next.
- In the **Pre-provisioning Domains and Groups** screen, add the groups from each domain that you want to provision in VCF for roles and assignments by clicking the **Add Group** button, and click **Next**.

Note:

The **Pre-provisioning Domains and Groups** screen is displayed only if you have chosen groups to be pre-provisioned which means that you chose Groups pre-provisioning in the previous screen.

There should be at least one provisioned group added across the domains.

- In the **Attributes** screen, the user attributes section displays a list of mandatory and non-mandatory user attributes that you view in the SAML response from the identity provider.
  You can add additional user attributes if required. In the group attributes section of the workflow you can add a group attribute to be called for in the SAML request. Attribute names are case-sensitive.

Note:

If you choose JIT, it is recommended that you automatically disable inactive users/groups.

c) If you select AD/LDAP and click **Next**, complete the configuration as follows. To learn more about the benefits of using AD/LDAP as a provisioning method, see KB 386870.

- In the **Directory Details** screen, enter the following details or select the required options.

Option	Description
Directory name	Enter a name for the directory.
DNS Server Location	Select this option if you want the identity broker to find and use optimal domain controllers by querying the DNS Service Record.
Global Catalog	Global catalog has a partial read-only replica of all objects across all domains. Some of the limitations with selecting the global catalog option include: • The Active Directory object attributes that are replicated to the global catalog are identified in the Active Directory schema as the Partial Attribute Set (PAS). Only these attributes are available for attribute mapping by the service. If necessary, edit the schema to add or remove attributes that are stored in the global catalog. • The global catalog stores the group membership (the member attributes) of only universal groups. Only universal groups are synced to the service. If necessary, change the scope of a group from a local domain or from global to universal. • The bind DN account that you define when you configure a directory in the service must have permissions to read the Token-Groups-Global-And-Universal (TGGAU) attribute. **Note:** When you use Global catalog with LDAP and multiple domains, the additional domains are discovered only after the user/group sync is complete.
Encrypted Connection (Displayed when you select the DNS server location option)	This option activates LDAPs for all connections to this directory.

Certificates for encrypted connection (Displayed when you select the **DNS server location** and **Encrypted Connections** options and if you enter a *ldaps://* string in the Primary domain controller field)	Upload the root certificate(s) required to access the domain controllers over SSL. All uploaded root certificates will be merged into a single file with the filename {directory-name}-concatenated.pem when you save the configuration. **Note:** Ensure that the certificate is in the (Privacy Enhanced Mail) PEM format and includes the "BEGIN CERTIFICATE" and "END CERTIFICATE" lines.
Directory search attribute	A specific LDAP attribute used in queries to search or identify entries in the directory. For example, uid, sAMAccountName.
Primary domain controller	A server machine running Active Directory Domain Services (AD DS) responsible for managing and authenticating user access, enforcing policies, and storing directory data. For example, hostname = DC01 for the domain called example.com.
Secondary domain controller	A backup domain controller for the identity broker to point to if the primary domain controller is not reachable.
Directory search attribute	A specific LDAP attribute used in queries to search or identify entries in the directory. For example, uid, sAMAccountName.
Base DN	The Distinguished Name (DN) of the top-level entry in the directory tree from which LDAP queries begin. For example, dc=example, dc=com.
Bind user name	The user name of a user account with sufficient permissions to authenticate and perform LDAP operations on the directory.
Bind user password	The password of a user account with sufficient permissions to authenticate and perform LDAP operations on the directory.

8. In the **Review** screen, review the details you have added for the configuration and click Finish.

Configure user and group provisioning

9. In the **Configure User and Group Provisioning** screen, depending on the option you choose, complete the following steps and click **Done**.
   a) If you have chosen SCIM, you see Configure SCIM push in your identity provider instead of Configure User and Group Provisioning. Use the values listed on the screen to configure the SCIM application in the identity provider.
      - If your identity provider supports authorizing the SCIM application using OAuth grant type, use the client ID, client secret, and token endpoint. Otherwise, use the SCIM bearer token to authorize this SCIM server in your identity provider.
      - If required, you can regenerate these values after you have configured VCF single sign-on. To regenerate the values to configure the SCIM application in the identity provider, navigate to the respective VCF Instance and click on Identity Source. The SCIM related values are available under Directory Information.
      - Ensure that the user attributes, username, and domain are pushed individually from the SCIM application by the identity provider. Similarly, ensure that the group attributes, group name, and domain are pushed individually as well.
   b) If you have chosen JIT, there is no additional step to be performed under User and Group Provisioning.
   c) If you have chosen AD/LDAP, review the directory information such as the domain controller and bind credentials that you had configured. Complete the following steps:
      1. In the **Attributes Mappings** screen, review the attribute mappings and click **Next**.
         The attributes in the identity broker directory are mapped to the Active Directory attributes for accurate synchronization of user information. If the mappings are incorrect, select the correct attribute from the drop-down list.
      2. In the **Group Provisioning** screen, enter the relevant details or select the required options and click **Next**.

Use the **Group Provisioning** screen to search for the group names and select the corresponding Active Directory group Distinguished Names (DN) that you want to sync.

Option	Description
Specify the Base Group DN	Click the **Select Base Group DN** button if you have nested groups and want to specify the group names. You can search for the group name and select the group DNs that you want to sync from from the list in the datagrid below.
Sync Nested Groups	Select **Sync Nested Group** to sync all the users from the selected group and its nested group without maintaining the group hierarchy. **Note:** For large AD groups, do not select this option. Instead, you must manually select the required groups.

3. From the **User Provisioning** screen, enter the relevant details or select the required options and click Next.
   Use the **User Provisioning** screen to search for the user names and select the corresponding Active Directory group Distinguished Names (DN) that you want to sync.

Option	Description
Specify the Base User DN	Click the **Select Base User DN** button to specify the user names that you want to sync. You can search for the user name and select the user DNs that you want to sync from from the list in the datagrid below. If you want to sync the users directly from the group, do not specify a base user DN.

4. From the **Review** screen, review the configuration details of the identity provider.
5. Click **Finish**. The sync starts in the background. After the initial sync, the sync is scheduled to run once a week at a scheduled time.
6. Click **Done**.

For more information about the various functionalities you can perform as part of the management operations in Active Directory, see Managing AD/LDAP.

(Optional) Test Login

10. Click **Test Login** to test the login using your single-sign on credentials.
11. Click **Next** in the identity provider screen.
12. After a successful validation, click **Done**.
13. Click **Finish Setup**. This completes the initial configuration.
14. Click **Continue** to view the **VCF SSO Overview** page.

Important:

Important Notes on Backup

It is highly advisable to create a backup of both VCF Operations and the identity broker instance. Furthermore, it is advisable to perform a similar backup following any substantial modifications to the initial identity source configuration, such as when you 'Delete Single Sign-On'. This practice ensures that you can restore the VCF single-sign on configuration to a recognized and stable state if necessary. You can download the configuration as a JSON file.

Important Notes on Role Assignment

VCF roles can be created and mapped to individual component roles.

- As an admin you can assign built-in and custom VCF roles to groups and and users from the Access Managment tab (navigate to the VCF SSO Overview page, click on a VCF single-sign on configuration, and then click the Access Management tab).
- As an admin, you can also assign roles in the component console and enable VCF single-sign on from the Component Configuration tab (navigate to the VCF SSO Overview page, click on a VCF single-sign on configuration, and then click the Component Configuration tab).

Configure SiteMinder as an Identity Provider Using SAML 2.0

After installing or upgrading to VCF 9.1, to configure VCF single-sign on, you can configure SiteMinder using SAML 2.0 as the authentication protocol.

To set up an identity provider, you must:

1. Choose an identity provider
2. Configure the identity provider
3. Configure user and group provisioning
4. Test Login

Note:

You can test login if the identity broker is at version 9.1.

Prerequisite

- Verify that you have chosen a deployment mode. For more information, see Step 1: Choose the Deployment Mode.

Procedure

Choose the identity provider

1. From the **Configure VCF SSO** page, click the **Start** button against the **Configure Identity Provider** option.
2. From the **Choose the identity provider** section, select **SAML 2.0** from the list and click **Next**.

Configure the identity provider

3. From the **Configure the Identity Provider** section, click **Configure** to configure the identity broker to integrate with the selected identity provider for user authentication.
4. In the **Authentication Method** screen, select **SAML** (Security Assertion Markup Language).
5. In the **Service Provider Registration** screen, log in to your Identity Provider admin console and create a SAML 2.0 application.

   While configuring the application, use the metadata provided in the **Service Provider Registration** screen for the **Entity ID/Metadata URL** and **Assertion Consumer Service URL** options. If the identity provider requires the metadata to be copy-pasted, download this file to avoid formatting issues.

   Note:

   Ensure that you add the following default claims when you configure your external identity provider. These claims are case-sensitive:

   - userName
   - email
   - domain
   - firstName
   - lastName
   - groups

6. Click **Next**.
7. In the **Identity Provider Configuration** screen, enter the relevant details or select the required options and click **Next**.

Table 1134: SiteMinder Options

Option	Description
Entity ID/Metadata URL	Copy the Entity ID/Metadata XML of your identity provider and paste it in this field. The Entity ID/Metadata XML is a URL that points to an XML document containing metadata about the service provider (identity broker). This metadata includes important information required for establishing trust and facilitating secure communication between the service provider and the identity provider.
Assertion Consumer Service URL	An endpoint on the service provider (identity broker) that is responsible for receiving and processing the SAML assertions sent by the identity provider.
Name ID format and Name ID value	Select the Name ID format and value. The Name ID format is the value in the SAML response to identify the authenticated user.
SAML Context	SAML context refers to the method of authentication that VCF requires the identity provider to support when verifying a user's identity. • **PasswordProtectedTransport**: Authentication with a password sent over a secure transport (For example, HTTPS). • **Password**: Authentication using a password without requiring transport security. • **Unspecified:** If you don't want to enforce any particular type of authentication mechanism, you can use this option.

8. In the **User/Group Provisioning Method** screen, select your preferred mode of provisioning the users and groups. You can select either:
   - System for Cross-domain Identity Management (SCIM), or
   - Just-In-Time Provisioning (JIT), or
   - Active Directory/Lightweight Directory Access Protocol (AD/LDAP)

   a) If you select **SCIM** and click **Next**, complete the configuration as follows.
      - In the **Domain(s)** screen, enter the domain that users from your enterprise will log in with, and click **Next**.

   b) If you select **JIT** and click **Next**, complete the configuration as follows:
      - In the **Group Provisioning** screen, select your preferred mode of provisioning the groups. The options are **Groups Pre-provisioning** or **Just-In-Time Group Provisioning** and click **Next**.

Option	Description
Groups Pre-Provisioning	This option creates the required groups in advance to perform role assignments. Users are assigned to these groups dynamically when they log in based on SAML assertion. Only the pre-provisioned groups are available for role assignments.
Just-In-Time Group Provisioning	This option creates groups dynamically based on the SAML assertion when the user logs in. Alternatively, you can use group pre-provisioning if you know the group names from Active Directory that requires entitlements.

      - In the **Domain(s)** screen, add the domain details that users from your enterprise will log in with, and click **Next**.
      - In the **Pre-provisioning Domains and Groups** screen, add the groups from each domain that you want to provision in VCF for roles and assignments by clicking the **Add Group** button, and click **Next**.

Note:

The **Pre-provisioning Domains and Groups** screen is displayed only if you have chosen groups to be pre-provisioned which means that you chose **Groups pre-provisioning** in the previous screen.

There should be at least one provisioned group added across the domains.

- In the **Attributes** screen, the user attributes section displays a list of mandatory and non-mandatory user attributes that you view in the SAML response from the identity provider.

  You can add additional user attributes if required. In the group attributes section of the workflow you can add a group attribute to be called for in the SAML request. Attribute names are case-sensitive.

Note:

If you use public APIs to update identity provider attribute mappings, the following identity broker attributes are available and can be mapped to OpenID attributes:

- ExternalId
- userName
- firstName
- lastName
- email
- phone
- active
- employeeID
- distinguishedName
- userPrincipalName
- domain
- sourceAnchor
- displayName
- nickName
- profileUrl
- title
- organization
- costCenter
- department
- division
- managerId

Note:

If you choose JIT, it is recommended that you automatically disable inactive users.

c) If you select **AD/LDAP** and click **Next**, complete the configuration as follows. To learn more about the benefits of using AD/LDAP as a provisioning method, see KB 386870.

- In the **Directory Details** screen, enter the following details or select the required options.

Option	Description
Directory name	Enter a name for the directory.
DNS Server Location	Select this option if you want the identity broker to find and use optimal domain controllers by querying the DNS Service Record.

Global Catalog	Global catalog has a partial read-only replica of all objects across all domains. Some of the limitations with selecting the global catalog option include: • The Active Directory object attributes that are replicated to the global catalog are identified in the Active Directory schema as the Partial Attribute Set (PAS). Only these attributes are available for attribute mapping by the service. If necessary, edit the schema to add or remove attributes that are stored in the global catalog. • The global catalog stores the group membership (the member attributes) of only universal groups. Only universal groups are synced to the service. If necessary, change the scope of a group from a local domain or from global to universal. • The bind DN account that you define when you configure a directory in the service must have permissions to read the Token-Groups-Global-And-Universal (TGGAU) attribute. **Note:** When you use Global catalog with LDAP and multiple domains, the additional domains are discovered only after the user/group sync is complete.
Encrypted Connection (Displayed when you select the **DNS server location** option)	This option activates LDAPs for all connections to this directory.
Certificates for encrypted connection (Displayed when you select the **DNS server location** and **Encrypted Connections** options and if you enter a *ldaps://* string in the Primary domain controller field)	Upload the root certificate(s) required to access the domain controllers over SSL. All uploaded root certificates will be merged into a single file with the filename `{directory-name}-concatenated.pem` when you save the configuration. **Note:** Ensure that the certificate is in the (Privacy Enhanced Mail) PEM format and includes the "BEGIN CERTIFICATE" and "END CERTIFICATE" lines.
Directory search attribute	A specific LDAP attribute used in queries to search or identify entries in the directory. For example, `uid`, `sAMAccountName`.
Primary domain controller	A server machine running Active Directory Domain Services (AD DS) responsible for managing and authenticating user access, enforcing policies, and storing directory data. For example, `hostname = DC01` for the domain called example.com.
Secondary domain controller	A backup domain controller for the identity broker to point to if the primary domain controller is not reachable.
Directory search attribute	A specific LDAP attribute used in queries to search or identify entries in the directory. For example, `uid`, `sAMAccountName`.
Base DN	The Distinguished Name (DN) of the top-level entry in the directory tree from which LDAP queries begin. For example, `dc=example`, `dc=com`.
Bind user name	The user name of a user account with sufficient permissions to authenticate and perform LDAP operations on the directory.
Bind user password	The password of a user account with sufficient permissions to authenticate and perform LDAP operations on the directory.

9. In the **Review** screen, review the details you have added for the configuration and click **Finish**.

Configure user and group provisioning

10. In the **Configure User and Group Provisioning** screen, depending on the option you choose, complete the following steps and click **Done**.
    a) If you have chosen SCIM, you see **Configure SCIM push in your identity provider** instead of **Configure User and Group Provisioning**. Use the values listed on the screen to configure the SCIM application in the identity provider.
    - If your identity provider supports authorizing the SCIM application using OAuth grant type, use the client ID, client secret, and token endpoint. Otherwise, use the SCIM bearer token to authorize this SCIM server in your identity provider.
    - If required, you can regenerate these values after you have configured VCF single-sign on. To regenerate the values to configure the SCIM application in the identity provider, navigate to the respective VCF Instance and click on **Identity Source**. The SCIM related values are available under **Directory Information**.
    - Ensure that the user attributes, username, and domain are pushed individually from the SCIM application by the identity provider. Similarly, ensure that the group attributes, group name, and domain are pushed individually as well.

    b) If you have chosen JIT, there is no additional step to be performed under **User and Group Provisioning**.
    c) If you have chosen AD/LDAP, review the directory information such as the domain controller and bind credentials that you had configured. Complete the following steps:
    1. In the **Attributes Mappings** screen, review the attribute mappings and click **Next.**
       The attributes in the identity broker directory are mapped to the Active Directory attributes for accurate synchronization of user information. If the mappings are incorrect, select the correct attribute from the drop-down list.
    2. In the **Group Provisioning** screen, enter the relevant details or select the required options and click **Next**.
       Use the **Group Provisioning** screen to search for the group names and select the corresponding Active Directory group Distinguished Names (DN) that you want to sync.

Option	Description
Specify the Base Group DN	Click the **Select Base Group DN** button if you have nested groups and want to specify the group names. You can search for the group name and select the group DNs that you want to sync from from the list in the datagrid below.
Sync Nested Groups	Select **Sync Nested Group** to sync all the users from the selected group and its nested group without maintaining the group hierarchy. **Note:** For large AD groups, do not select this option. Instead, you must manually select the required groups.

    3. From the **User Provisioning** screen, enter the relevant details or select the required options and click **Next**.
       Use the **User Provisioning** screen to search for the user names and select the corresponding Active Directory group Distinguished Names (DN) that you want to sync.

Option	Description
Specify the Base User DN	Click the **Select Base User DN** button to specify the user names that you want to sync. You can search for the user name and select the user DNs that you want to sync from from the list in the datagrid below. If you want to sync the users directly from the group, do not specify a base user DN.

    4. From the **Review screen**, review the configuration details of the identity provider.
    5. Click **Finish**. The sync starts in the background. After the initial sync, the sync is scheduled to run once a week at a scheduled time.
    6. Click **Done**.

For more information about the various functionalities you can perform as part of the management operations in Active Directory, see Managing AD/LDAP.

(Optional) Test Login

11. Click **Test Login** to test the login using your single-sign on credentials.
12. Click **Next** in the identity provider screen.
13. After a successful validation, click **Done**.
14. Click **Finish Setup**. This completes the initial configuration.
15. Click **Continue** to view the **VCF SSO Overview** page.

Important:

Important Notes on Backup

It is highly advisable to create a backup of both VCF Operations and the identity broker instance. Furthermore, it is advisable to perform a similar backup following any substantial modifications to the initial identity source configuration, such as when you 'Delete Single Sign-On'. This practice ensures that you can restore the VCF single-sign on configuration to a recognized and stable state if necessary. You can download the configuration as a JSON file.

Important Notes on Role Assignment

VCF roles can be created and mapped to individual component roles.

- As an admin you can assign built-in and custom VCF roles to groups and and users from the **Access Managment** tab (navigate to the VCF SSO Overview page, click on a VCF single-sign on configuration, and then click the Access Management tab).
- As an admin, you can also assign roles in the component console and enable VCF single-sign on from the **Component Configuration** tab (navigate to the VCF SSO Overview page, click on a VCF single-sign on configuration, and then click the Component Configuration tab).

Configure an Identity Provider using Generic OIDC

After installing or upgrading to VCF 9.1, to configure VCF single sign-on, you can configure an identity provider using Generic OIDC as the authentication protocol.

To set up an identity provider, you must:

1. Choose an identity provider
2. Configure the identity provider
3. Configure user and group provisioning
4. Test Login

Note:

You can test login if the identity broker is at version 9.1.

Prerequisite

- Verify that you have chosen a deployment mode. For more information, see Step 1: Choose the Deployment Mode.
- If you configure VCF single sign-on using identity broker instance, and there are network restrictions that do not allow outbound public internet access, you must set up a proxy server. For more information, see Configure a Proxy Server for VCF Management Services Components and VCF Automation.

Procedure

Choose the identity provider

1. From the **Configure VCF SSO** page, click the **Start** button against the **Configure Identity Provider** option.

   If you have an existing identity provider configured in vCenter, see the information below. For a fresh configuration of the identity provider, continue from Step 6.

   For an existing source (existing identity provider configuration)

- Choose either SAML or OIDC as the authentication protocol.

  Note:

  If you choose SAML, the VCF Single Sign-On configuration is considered as a new authentication source configuration. Continue from Step. 6.
- If you choose OIDC:
  - If you choose the instance deployment mode, and select OIDC, the following fields are prepopulated in the **Identity Provider Configuration** screen.
    a. Review the details and provide the shared secret and click **Next**.
       - Identity provider name
       - Client identifier
       - OpenID address
       - Update Redirect URI in the Identity provider's OIDC application
       - Shared Secret: Paste the client secret of the OIDC application you created in the identity provider.

    b. Review the **User/Group Provisioning**, **Domain(s)**, and **Review** screens and click **Finish**.
    c. Review the SCIM push details and click **Done**. The identity provider configuration is complete.
  - If you choose the embedded deployment mode, and select OIDC:
    a. The configuration details of the identity provider configuration are automatically imported from existing configuration into the identity broker with OIDC as the authentication protocol and SCIM as the user/group provisioning method.
    b. Review the configuration values in the **Identity Provider Configuration** screen and click **Finish**.
    c. Review the SCIM details, click **Done** and then click **Finish Setup**. The identity provider configuration is complete.

2. For a fresh identity provider configuration, from the **Choose the identity provider** section, select **OIDC** from the list and click **Next**.

   Configure the identity provider

   Note:

   Ensure that you add the following default claims when you configure your external identity provider. These claims are case-sensitive:

   - preferred_username
   - email
   - family_name
   - given_name
   - groups
   - domain

   If the default claims cannot be added, use alternative claim names and map them to the appropriate attributes in the VCF Operations **Identity & Access** page to support your JIT configurations.
3. From the **Configure the Identity Provider** section, click **Configure** to configure the identity broker to integrate with the selected identity provider for user authentication.
4. In the **Identity Provider Configuration** screen, enter the relevant details or select the required options and click **Next**.

Option	Description
IDP Display Name	Enter the OpenID Connect provider name. Example, Microsoft Entra ADFS.
Client Identifier	A unique ID assigned to your OIDC application by the identity provider, for authentication.

Shared secret	A confidential key used by your OIDC application to authenticate itself with the identity provider. **Note:** Client secrets issued by external identity providers typically have expiration dates. When a client secret expires, it must be regenerated, and the corresponding OIDC configuration must be updated in VCF to ensure continued authentication functionality.
OpenID Address	The URL of the identity provider's OIDC configuration endpoint for initiating the authentication process.
Redirect URI	The URL where the identity provider will send the authentication response after a user logs in.
List of domains	Domains for user to login to your company.

Table 1135: Keycloak Options

Enter the following details if you are configuring Keycloak as an identity provider.

Option	Description
Entity ID/Metadata URL	Copy the Entity ID/Metadata XML of your identity provider and paste it in this field.The Entity ID/Metadata XML is a URL that points to an XML document containing metadata about the service provider (identity broker). This metadata includes important information required for establishing trust and facilitating secure communication between the service provider and the identity provider.
Assertion Consumer Service URL	An endpoint on the Service Provider (identity broker) that is responsible for receiving and processing the SAML assertions sent by the identity provider.
Name ID format and Name ID value	The Name ID format is the value in the SAML response to identify the authenticated user.
SAML Context	SAML context refers to the type of user authentication that VCF needs to be mandated in the identity provider. • **PasswordProtectedTransport**: Authentication with a password sent over a secure transport (For example, HTTPS). • **Password**: Authentication using a password without requiring transport security. • **Unspecified:** Select this option if you do not want to enforce any particular type of authentication mechanism.

5. In the **User/Group Provisioning Method** screen, select your preferred mode of provisioning the users and groups. You can select either:
   - System for Cross-domain Identity Management (SCIM), or
   - Just-In-Time Provisioning (JIT), or
   - Active Directory/Lightweight Directory Access Protocol (AD/LDAP)

   a) If you select SCIM and click **Next**, complete the configuration as follows.
      - In the **Domain(s)** screen, enter the domain that users from your enterprise will log in with, and click **Next**.

b) If you select **JIT** and click **Next**, complete the configuration as follows:

- In the **Group Provisioning** screen, select your preferred mode of provisioning the groups. The options are **Groups Pre-provisioning** or **Just-In-Time Group Provisioning** and click **Next**.

Option	Description
Groups Pre-Provisioning	This option creates the required groups in advance to perform role assignments. Users are assigned to these groups dynamically when they log in based on OIDC token. Only the pre-provisioned groups are available for role assignments.
Just-In-Time Group Provisioning	This option creates groups dynamically based on the OIDC token when the user logs in. Alternatively, you can use group pre-provisioning if you know the group names from Active Directory that requires entitlements.

- In the **Domain(s)** screen, add the domain details that users from your enterprise will log in with, and click **Next**.
- In the **Pre-provisioning Domains and Groups** screen, add the groups from each domain that you want to provision in VCF for roles assignments by clicking the **Add Group** button, and click **Next**.

Note:

The **Pre-provisioning Domains and Groups** screen is displayed only if you have chosen groups to be pre-provisioned which means that you chose **Groups pre-provisioning** in the previous screen.

There should be at least one provisioned group added across the domains.

- In the **Attributes** screen, the user attributes section displays a list of mandatory and non-mandatory user attributes that you view in the OIDC response from the identity provider.
  You can add additional user attributes if necessary. In the group attributes section of the workflow you can add a group attribute to be called for in the OIDC request. Attribute names are case-sensitive.

Note:

If you use public APIs to update identity provider attribute mappings, the following identity broker attributes are available and can be mapped to OpenID attributes:

- ExternalId
- userName
- firstName
- lastName
- email
- phone
- active
- employeeID
- distinguishedName
- userPrincipalName
- domain
- sourceAnchor
- displayName
- nickName
- profileUrl
- title
- organization
- costCenter
- department
- division
- managerId

Note:

If you choose JIT, it is recommended that you automatically disable inactive users.

c) If you select **AD/LDAP**, select the **Unique Identifier in OIDC Identity Provider** and **Unique Identifier in VCF Identity Broker**, and click **Next**. To learn more about the benefits of using AD/LDAP as a provisioning method, see KB 386870.

Note:

The value of the identifier used to uniquely identify a user in your OIDC identity provider must exactly match with the value from Active Directory. The unique identifier in the identity broker is used to map the two systems.

Option	Description
Unique Identifier in OIDC Identity Provider	An identifier in your OIDC identity provider that is used to identify a user uniquely. You can select or enter a custom attribute (except 'sub'). For example, `name`.
Unique Identifier in VCF Identity Broker	Select the unique identifier that will be used to map to an attribute from Active Directory. The value of this identifier must exactly match the value of the previous field called **Unique Identifier in OIDC Identity Provider**.

In the **Directory Details** screen, enter the relevant details or select the required options.

Option	Description
Directory name	Enter a name for the directory.
DNS Server Location	Select this option if you want the identity broker to find and use optimal domain controllers by querying the DNS Service Record.
Global Catalog	Global catalog has a partial read-only replica of all objects across all domains. Some of the limitations with selecting the global catalog option include: • The Active Directory object attributes that are replicated to the global catalog are identified in the Active Directory schema as the Partial Attribute Set (PAS). Only these attributes are available for attribute mapping by the service. If necessary, edit the schema to add or remove attributes that are stored in the global catalog. • The global catalog stores the group membership (the member attributes) of only universal groups. Only universal groups are synced to the service. If necessary, change the scope of a group from a local domain or from global to universal. • The bind DN account that you define when you configure a directory in the service must have permissions to read the Token-Groups-Global-And-Universal (TGGAU) attribute. **Note:** When you use Global catalog with LDAP and multiple domains, the additional domains are discovered only after the user/group sync is complete.
Encrypted Connection (Displayed when you select the **DNS server location** option)	Activates LDAPs for all the connections to this directory.

Certificates for encrypted connection (Displayed when you select the **DNS server location** and **Encrypted Connections** options and if you enter a *ldaps://* string in the Primary domain controller field)	Upload the root certificate(s) required to access the domain controllers over SSL. All uploaded root certificates will be merged into a single file with the filename `{directory-name}-concatenated.pem` when you save the configuration. **Note:** Ensure that the certificate is in the (Privacy Enhanced Mail) PEM format and includes the "BEGIN CERTIFICATE" and "END CERTIFICATE" lines.
Directory search attribute	A specific LDAP attribute used in queries to search or identify entries in the directory. For example, `uid`, `sAMAccountName`.
Primary domain controller	A server machine running Active Directory Domain Services (AD DS) responsible for managing and authenticating user access, enforcing policies, and storing directory data. For example, `hostname = DC01` for the domain called example.com.
Secondary domain controller	A backup domain controller for the identity broker to point to if the primary domain controller is not reachable.
Directory search attribute	A specific LDAP attribute used in queries to search or identify entries in the directory. For example, `uid`, `sAMAccountName`.
Base DN	The Distinguished Name (DN) of the top-level entry in the directory tree from which LDAP queries begin. For example, `dc=example`, `dc=com`.
Bind user name	The user name of a user account with sufficient permissions to authenticate and perform LDAP operations on the directory.
Bind user password	The password of a user account with sufficient permissions to authenticate and perform LDAP operations on the directory.

6. In the **Review** screen, review the details that you have added for the configuration and click **Finish**.
7. **Configure user and group provisioning**
8. In the **Configure User and Group Provisioning** screen, depending on the option you choose, complete the following steps and click **Done**.

   If you have chosen OIDC as the authentication protocol, and AD/LDAP as the user and group provisioning protocol, read KB 386870 to understand the finer details about configuring the setup.

   a) If you have chosen JIT, there is no additional step to be performed under **User and Group Provisioning**.
   b) If you have chosen AD/LDAP, review the directory information such as the domain controller and bind credentials that you had configured. Complete the following steps:

      1. In the **Attributes Mappings** screen, review the attribute mappings and click **Next.**
         The attributes in the identity broker directory are mapped to the Active Directory attributes for accurate synchronization of user information. If the mappings are incorrect, select the correct attribute from the drop-down list.
      2. In the **Group Provisioning** screen, enter the relevant details or select the required options and click **Next**.
         Use the **Group Provisioning** screen to search for the group names and select the corresponding Active Directory group Distinguished Names (DN) that you want to sync.

Option	Description
Specify the Base Group DN	Click the **Select Base Group DN** button if you have nested groups and want to specify the group names. You can search for the group name and select the group DNs that you want to sync from from the list in the datagrid below.
Sync Nested Groups	Select **Sync Nested Group** to sync all the users from the selected group and its nested group without maintaining the group hierarchy.

3. From the **User Provisioning** screen, enter the relevant details or select the required options and click **Next**. Use the **User Provisioning** screen to search for the user names and select the corresponding Active Directory group Distinguished Names (DN) that you want to sync.

Option	Description
Specify the Base User DN	Click the **Select Base User DN** button to specify the user names that you want to sync. You can search for the user name and select the user DNs that you want to sync from from the list in the datagrid below. If you want to sync the users directly from the group, do not specify a base user DN.

4. From the **Review screen**, review the configuration details of the identity provider.
5. Click **Finish**. The sync starts in the background. After the initial sync, the sync is scheduled to run once a week at a scheduled time.
6. Click **Done**.

For more information about the various functionalities you can perform as part of the management operations in Active Directory, see Managing AD/LDAP.

(Optional) Test Login

9. Click **Test Login** to test the login using your single-sign on credentials.
10. Click **Next** in the identity provider screen.
11. After a successful validation, click **Done**.
12. Click **Finish Setup**. This completes the initial configuration.
13. Click **Continue** to view the **VCF SSO Overview** page.

Important:

Important Notes on Backup

It is highly advisable to create a backup of both VCF Operations and the identity broker instance. Furthermore, it is advisable to perform a similar backup following any substantial modifications to the initial identity source configuration, such as when you 'Delete Single Sign-On'. This practice ensures that you can restore the VCF single-sign on configuration to a recognized and stable state if necessary. You can download the configuration as a JSON file.

Important Notes on Role Assignment

VCF roles can be created and mapped to individual component roles.

- As an admin you can assign built-in and custom VCF roles to groups and and users from the **Access Managment** tab (navigate to the VCF SSO Overview page, click on a VCF single-sign on configuration, and then click the Access Management tab).
- As an admin, you can also assign roles in the component console and enable VCF single-sign on from the **Component Configuration** tab (navigate to the VCF SSO Overview page, click on a VCF single-sign on configuration, and then click the Component Configuration tab).

Configure Active Directory as an Identity Provider Using AD/LDAP

After installing or upgrading to VCF 9.1, to configure VCF single-sign on, you can configure Active Directory as an identity provider using AD/LDAP.

To set up an identity provider, you must:

1. Choose an identity provider
2. Configure the identity provider
3. Configure user and group provisioning
4. Test Login

Note:

You can test login if the identity broker is at version 9.1.

Prerequisite

- Verify that you have chosen a deployment mode. For more information, see Step 1: Choose the Deployment Mode.

Procedure

Choose the identity provider

1. From the **Configure VCF SSO** page, click the **Start** button against the **Configure Identity Provider** option.

   If you have an existing identity provider configured in vCenter, see the information below. For a fresh configuration of the identity provider, continue from Step 6.

   For an existing source (existing identity provider configuration)

   Depending on the protocol you have chosen, the following fields are prepopulated from the existing configuration in the vCenter.

   1. AD/LDAP
      a. Directory name
      b. Primary domain controller: Host and port
      c. Base DN
      d. Bind user name

      Review the prepopulated values. If there are any values that are empty, enter the details. Proceed to Step 7 to complete configuring the identity provider.
2. For a fresh identity provider configuration, from the **Choose the identity provider** section, select **AD/LDAP** from the list and click **Next**.

Configure the identity provider

3. From the **Configure the identity provider** section, click **Configure** to configure the identity broker to integrate with the selected identity provider for user authentication.
4. In the **Directory information** screen, enter the relevant details or select the required options and click **Next**.

Option	Description
Directory name	Enter a name for the directory.
DNS Server Location	Select this option if you want the identity broker to find and use optimal domain controllers by querying the DNS Service Record.

Global Catalog (In version 9.1.1 and later, this option is inactive when the **On-demand look-up** option is displayed)	Global catalog has a partial read-only replica of all objects across all domains. Some of the limitations with selecting the global catalog option include: • The Active Directory object attributes that are replicated to the global catalog are identified in the Active Directory schema as the Partial Attribute Set (PAS). Only these attributes are available for attribute mapping by the service. If necessary, edit the schema to add or remove attributes that are stored in the global catalog. • The global catalog stores the group membership (the member attributes) of only universal groups. Only universal groups are synced to the service. If necessary, change the scope of a group from a local domain or from global to universal. • The bind DN account that you define when you configure a directory in the service must have permissions to read the Token-Groups-Global-And-Universal (TGGAU) attribute. **Note:** When you use Global catalog with LDAP and multiple domains, the additional domains are discovered only after the user/group sync is complete.
Encrypted Connection (Displayed when you select the **DNS server location** option)	Activates LDAPs for all the connections to this directory.
Certificate(s) for encrypted connection (Displayed when you select the **DNS server location** and **Encrypted Connections** options and if you enter a `ldaps://` string in the Primary domain controller field)	Upload the root certificate(s) required to access the domain controllers over SSL. All uploaded root certificates will be merged into a single file with the filename `{directory-name}-concatenated.pem` when you save the configuration. **Note:** Ensure that the certificate is in the (Privacy Enhanced Mail) PEM format and includes the "BEGIN CERTIFICATE" and "END CERTIFICATE" lines.
Primary domain controller (Removed from display when you select DNS Server Location)	A server machine running Active Directory Domain Services (AD DS) responsible for managing and authenticating user access, enforcing policies, and storing directory data. For example, `hostname = DC01` for the domain called example.com.
Certificate for primary domain controller	The root certificate required to access the primary domain controller over SSL. **Note:** Ensure that the certificate is in PEM format and includes the "BEGIN CERTIFICATE" and "END CERTIFICATE" lines. This field is displayed if you type 'LDAP' in the **Primary domain controller** field.
Secondary domain controller (Removed from display when you select DNS Server Location)	A backup domain controller for the identity broker to point to if the primary domain controller is not reachable.
Certificate for secondary domain controller	The root certificate required to access secondary domain controller over SSL. **Note:** Ensure the certificate is in PEM format and include the "BEGIN CERTIFICATE" and "END CERTIFICATE" lines. This field is displayed if you type 'LDAP' in the **Secondary domain controller** field.

Directory search attribute	A specific LDAP attribute used in queries to search or identify entries in the directory. For example, `uid`, `sAMAccountName`.
Base DN	The Distinguished Name (DN) of the top-level entry in the directory tree from which LDAP queries begin. For example, `dc=example`, `dc=com`.
Bind user name	The user name of a user account with sufficient permissions to authenticate and perform LDAP operations on the directory. **Note:** Use a bind user account with sufficient Active Directory privileges so that VCF SSO can properly enumerate and retrieve all required AD users during configuration.
Bind user password	The password of a user account with sufficient permissions to authenticate and read the directory.
On-demand lookup	In identity broker version 9.1.1 and later, you can choose on-demand lookup for user and group provisioning. Users and groups are provisioned either on demand or upon login.

Note:

An additional Active Directory server can be created after the first Active Directory server is configured.

5. In the **Review** screen, review the details that you have added for the configuration and click **Finish**.

Configure user and group provisioning

Note:

In version 9.1.1 and later, if you have enabled on-demand lookup, configure attribute mappings in this step. Users are dynamically authenticated against Active Directory based on their group membership and authorization rights at login.

6. In the **Configure user and group provisioning** screen, click **Configure** to provision the users and groups from the Active Directory to the identity broker.
7. In the **Review Directory Information** screen, review the directory information and click **Next**.
8. In the **Attributes Mappings** screen, review the attribute mappings and click **Next**.

   The attributes in the identity broker directory are mapped to the Active Directory attributes for accurate synchronization of user information. If the mappings are incorrect, select the correct attribute from the drop-down list.
9. In the **Group Provisioning** screen, enter the relevant details or select the required options and click **Next**.
   - Use the **Group Provisioning** screen to search for the group names and select the corresponding Active Directory group Distinguished Names (DN) that you want to sync.

- In version 9.1.1 and later, when on-demand lookup is enabled, group and user provisioning steps are not required and are unavailable.

Option	Description
Sync Nested Groups	Select **Sync Nested Group** to sync all the users from the selected group and its nested group without maintaining the group hierarchy.
Specify the base group DN	Enter the required base group DN and click the **Select Base Group DN** button. The list of groups that belong to the base group DN is listed in the grid below. You can search for the group name and select the group DNs that you want to provision. **Note:** VCF Single Sign-On allows for the specification of a single base group DN for conducting group searches. If you need to provision groups from multiple locations within Active Directory, it is recommended that you set the base group DN at a higher level in the directory hierarchy. This should encompass all the required groups, such as a parent Organizational Unit (OU) or domain root, and then selecting the necessary groups.

Note:

Any change to the provisioned user/group DN) in active directory causes a discrepancy in VCF single sign-on which leads to the deletion of the user/group. Existing role assignments and the API token that is generated is lost in this process. Re-provision the user/group with the updated DN and role assignment to complete the DN change.

10. From the **User Provisioning** screen, enter the relevant details or select the required options and click **Next**.

    Use the **User Provisioning** screen to search for the user names and select the corresponding Active Directory group Distinguished Names (DN) that you want to sync.

Option	Description
Specify the base user DN	Click the **Select Base User DN** button to specify the user names that you want to sync. You can search for the user name and select the user DNs that you want to sync from from the list in the datagrid below. **Note:** If you want to sync the users directly from the group, do not specify a base user DN.

11. From the **Review** screen, review the configuration details of the identity provider.

    In versions 9.1.1 and later, user and group details are not displayed if you had selected **On-demand lookup** while configuring directory information.

12. Click **Finish**. The sync starts in the background. After the initial sync, the sync is scheduled to run once a week at a scheduled time. To modify the sync schedule, see Managing AD/LDAP.

13. Click **Done**.

(Optional) Test Login

14. Click **Test Login** to test the login using your single-sign on credentials.
15. Click **Next** in the identity provider screen.
16. After a successful validation, click **Done**.

Managing AD/LDAP

You can integrate Active Directory with VCF single-sign on. After Active Directory is connected, you can perform several operations to manage the Active Directory connection such as view a list of active directories, add additional directories, modify sync settings, and view the sync log.

To start managing the AD/LDAP connection, navigate to the corresponding VCF Instance in which VCF single-sign on was configured. The **Identity Source** tab contains the relevant information pertaining to the identity provider configuration. If more than one VCF Instance is connected to the Active Directory, use the link in the **Identity Source** tab to navigate to the relevant VCF Instance. Reset VCF single-sign on to manage the Active Directory connection.

Prerequisite

- Configure an identity provider for AD/LDAP before you start managing it. For more information, see Configure Active Directory as an Identity Provider Using AD/LDAP.

Procedure

1. In the VCF Operations navigation bar at the top, click **Manage** and in the left navigation pane, click **Fleet Management** > **Identity & Access**.
2. In the **Identity & Access** pane, click **Configure VCF SSO**.
3. In the **VCF Overview SSO** page, in the section that is called **VCF SSO**, click the identity broker that you have configured for VCF single-sign on.
4. In the **Identity Source** tab you see different sections such as Deployment Mode, Authentication Information, and Directory Information.

The **Deployment Mode** and **Authentication Information** sections indicate the type of deployment and connection respectively.

Directory Information

The datagrid lists the directories that have been configured. You can add, edit, or delete the directories.

Column Name	Description
Directory Name	User friendly name of the directory that was configured.
Domain	The domain name of the domain controller that was configured as part of the directory configuration.
Synced groups	The total number of groups that have been synced from the directory to the identity broker. For example, as an enterprise admin, if you have selected 10 groups to provision, but only 8 groups were synced, then only 8 groups are displayed.
Synced users	The total number of users that have been synced from the directory to the identity broker. For example, as an enterprise admin, if you have selected 10 groups to provision, but, only 8 groups were synced, then only 8 groups are displayed.
Alerts	The number of alert(s) that might have occurred during the last performed successful sync.
Sync on request	Click the link to trigger an on-demand sync from the corresponding directory in Active Directory.
Last sync status	The day and time of the last performed successful sync. If the last sync fails, the status reflects the same. Click on Learn More for more information. See the table about #unique_2216/unique_2216_Connect_42_id-657241ba-294f-494e-cd00-820752f147dd for more information about sync failure errors and the reasons.

Sync Failure Messages

Error Message	Reason/Remediation
LDAP server is not reachable	This could be due to network issues, firewall blocks, an incorrect host name, or invalid BIND credentials. Verify the configuration and retry the sync.
An internal error occurred during a sync with the LDAP server.	Retry sync.
The users don't have the required object UUID property	Ensure that the attribute mapping is correct and all the objects have the attribute.
The groups don't have the required object UUID property	Ensure that the attribute mapping is correct and all the objects have the attribute.
Group Query Failed	The group is not found in Active Directory.

Add Additional Directories

You can add a new directory by clicking on the **Add** button. You can add a new LDAP-based directory with different modes of on-demand or pre-provisioning user/group settings. To configure a directory in AD/LDAP, see Configure Active Directory as an Identity Provider Using AD/LDAP .

Edit Existing Directories

You can edit the directory by clicking on the **Edit** button. From the Directory Details page, click the **Edit** button at the top of each of the following pages to modify the existing configuration details:

- Details
- Attribute mappings
- Group provisioning
- User provisioning
- Sync Settings - For more information, see the section called 'Modify Sync Settings' below.
- Sync Log - For more information, see the section called 'View Sync Log' below.

> **Note:**
>
> After you modify configurations in the Attribute Mappings, Group provisioning, User provisioning, and Sync Settings screens, you are prompted with an option to trigger an on-demand sync. If you do not trigger the on-demand sync, changes to users and groups take place in the next scheduled sync.

Modify Sync Settings

Editing the directories includes modifying the sync settings. The sync schedule is set to once per week, however you can modify the sync schedule by clicking the **Edit** button in the **Sync Settings** screen. Sync schedule is configured in UTC.

View Sync Log

The sync log under each directory contains the log of each sync between identity broker. and the configured domain controller in Active Directory. As part of this sync, users and groups are added or removed. You can view the time at which the sync was completed, the number of groups and users that were modified because of the sync, the status of the sync, and alerts. Here is a list of error messages.

Error Message	Possible Reason(s)
Invalid group attribute name	The group object has an invalid or a missing required attribute from Active Directory.
Invalid user attribute name	The user object has an invalid or missing required attribute from Active Directory.
Multiple attribute values	The attribute that was mapped to the user/group contains more than one value in Active Directory.
User or Group Query Failed	The user/group was not found in Active Directory.

Configure OpenLDAP as an Identity Provider

After installing or upgrading to VCF 9.1, to configure VCF single-sign on, you can configure OpenLDAP as an identity provider.

To set up an identity provider, you must:

1. Choose an identity provider
2. Configure the identity provider
3. Configure user and group provisioning
4. Test Login

Note:

You can test login if the identity broker is at version 9.1.

Prerequisite

- Verify that you have chosen a deployment mode. For more information, see Step 1: Choose the Deployment Mode.
- Ensure that the `memberOf overlay` is activated in the OpenLDAP server. For information about activating `memberOf overlay`, see OpenLDAP Overlays.

Procedure

Choose the identity provider

1. From the **Configure VCF SSO** page, click the **Start** button against the **Configure Identity Provider** option.
2. From the **Choose the identity provider** section, select **OpenLDAP** from the list and click **Next**.

Configure the identity provider

3. From the **Configure the Identity Provider** section, click **Configure** to configure the identity broker to integrate with the selected identity provider for user authentication.
4. In the **Directory Details** screen, enter the relevant details or select the required options and click **Next**.

Option	Description
Directory name	Enter a name for the directory.
Primary domain controller	A server machine responsible for managing and authenticating user access, enforcing policies, and storing directory data. For example, `hostname = DC01` for the domain called example.com.
Certificate for primary domain controller	The root certificate required to access primary domain controller over SSL. **Note:** Ensure that the certificate is in PEM format and include the "BEGIN CERTIFICATE" and "END CERTIFICATE" lines.
Secondary domain controller	A backup domain controller for the identity broker to point to if the primary domain controller is not reachable.
Certificate for secondary domain controller	The root certificate required to access secondary domain controller over SSL. **Note:** Ensure that the certificate is in PEM format and include the "BEGIN CERTIFICATE" and "END CERTIFICATE" lines.
Directory search attribute	A specific LDAP attribute used in queries to search or identify entries in the directory. **Note:** By default 'Custom Attribute' is chosen.

Custom directory search attribute for Users	The LDAP attribute used in queries to search or identify users in the directory.
Custom directory search attribute for Groups	The LDAP attribute used in queries to search or identify groups in the directory.
Base DN	The Distinguished Name (DN) of the top-level entry in the directory tree from which LDAP queries begin. For example, `dc=example, dc=com`.
Bind user name	The user name of a user account with sufficient permissions to authenticate and perform LDAP operations on the directory.
Bind user password	The password of a user account with sufficient permissions to authenticate and perform LDAP operations on the directory.

Note:

An additional OpenLDAP server can be created after the first OpenLDAP server is configured.

5. In the **LDAP Configuration** screen, enter the relevant details or select the required options and click **Next**.

Option	Description
Groups	The search filter for obtaining group objects. For example: (objectClass=group)
Bind user	The search filter for obtaining the bind user object, that is, the user that can bind to the directory. For example: (objectClass=person)
Users	The search filter for obtaining users to sync. For example: (&(objectClass=user)(objectCategory=person))
Object UUID	The attribute that is used in your LDAP directory to define the UUID of a user or group. For example: entryUUID.

6. In the **Review** screen, review the details you have added for the configuration and click **Finish**.

Configure user and group provisioning

7. In the **Configure User and Group Provisioning** screen, click **Configure** to provision the users and groups from the identity provider to the identity broker .
8. In the **Directory Review Information** screen, review the information directory information and click **Next**.
9. In the **Attributes Mappings** screen, review the attribute mappings and click **Next**.

   The attributes in the identity broker directory are mapped to the LDAP Directory attributes for accurate synchronization of user information. If the mappings are incorrect, select the correct attribute from the drop-down list.
10. In the **Group Provisioning** screen, enter the relevant details or select the required option and click **Next**.

Option	Description
Specify the base group DN	Enter the details of the base group DN.
View Groups	Click this option if you want to specify the group names. You can search for the group name and select the group DNs that you want to sync from from the list in datagrid below. **Note:** You can specify only one Base DN for group searches. To provision groups from multiple locations within Active Directory, it is recommended that you set the Base DN at a higher level in the directory hierarchy that includes all the required groups such as a parent Organizational Unit (OU) or domain root and then select the necessary groups.

Only groups with a valid DN, a unique identifier, and the required attributes (such as group name and object GUID) that match the configured directory search attribute are provisioned.

11. From the **User Provisioning** screen, enter the relevant details or select the required options and click **Next**.

    Use the **User Provisioning** screen to search for the user names and select the corresponding Active Directory group Distinguished Names (DN) that you want to sync.

Option	Description
Specify the Base User DN	Click this option to specify the user names that you want to sync. You can search for the user name and select the user DNs that you want to sync from from the datagrid. **Note:** If you want to sync the users directly from the group, do not specify a base user DN.
View Users	Click this option if you want to specify the user names. You can search for the user name and select the user DNs that you want to sync from the list in the datagrid below.

12. From the **Review** screen, review the configuration details of the identity provider.
13. Click **Finish**. The sync starts in the background. After the initial sync, the sync is scheduled to run once a week at a scheduled time.
14. Click **Done**.

(Optional) Test Login

15. Click **Test Login** to test the login using your single-sign on credentials.
16. Click **Next** in the identity provider screen.
17. After a successful validation, click **Done**.

Managing OpenLDAP Configurations

You can integrate OpenLDAP with VCF single-sign on. After OpenLDAP is integrated, you can perform several operations to manage the OpenLDAP connection such as view a list of directories, add additional directories, modify sync settings, and view the sync log.

To start managing the OpenLDAP connection, navigate to the corresponding VCF Instance in which VCF single-sign on was configured. The **Identity Source** tab contains the relevant information pertaining to the identity provider configuration. If more than one VCF Instance is connected to the OpenLDAP, use the link in the **Identity Source** tab to navigate to the relevant VCF Instance.

Procedure

1. In the VCF Operations navigation bar at the top, click **Manage** and in the left navigation pane, click **Fleet Management** > **Identity & Access**.
2. From the **Identity & Access** pane, click **Configure VCF SSO**.
3. In the **VCF Overview SSO** page, in the section that is called **VCF SSO**, click the identity broker that you have configured for VCF Single Sign-On.
4. From the **Identity Source** tab on the right side, you see different sections such as Deployment Mode, Authentication Information, and Directory Information.

The **Deployment Mode** and **Authentication Information** sections indicate the type of deployment and connection respectively.

Directory Information

The datagrid lists the directories that have been configured. You can add, edit, or delete the directories.

Column Name	Description
Directory Name	User friendly name of the directory that was configured.
Domain	The domain name of the domain controller that was configured as part of the directory configuration.
Synced groups	The total number of groups that have been synced from the directory to the identity broker. For example, as an enterprise admin, if you have selected 10 groups to provision, but only 8 groups were synced, then only 8 groups are displayed.
Synced users	The total number of users that have been synced from the directory to the identity broker. For example, as an enterprise admin, if you have selected 10 groups to provision, but only 8 groups were synced, then only 8 groups are displayed.
Alerts	The number of alert(s) that might have occurred during the last performed successful sync.
Sync on request	Click the link to trigger an on-demand sync from the corresponding directory in OpenLDAP.
Last sync status	The day and time of the last performed successful sync. If the last sync fails, the status reflects the same. Click on Learn More for more information. See the table about #unique_2226/unique_2226_Connect_42_id-c5c5876a-fdc0-4c16-a353-94fee016b16a for more information about sync failure errors and the reasons.

Sync Failure Messages

Error Message	Reason/Remediation
LDAP server is not reachable	This could be due to network issues, firewall blocks, an incorrect host name, or invalid BIND credentials. Verify the configuration and retry the sync.
An internal error occurred during a sync with the LDAP server.	Retry sync.
The users don't have the required object UUID property	Ensure that the attribute mapping is correct and all the objects have the attribute.
The groups don't have the required object UUID property	Ensure that the attribute mapping is correct and all the objects have the attribute.
Group Query Failed	The group is not found in Active Directory.

Add Additional Directories

You can add a new directory by clicking on the **Add** button. To configure a directory in OpenLDAP, see Configure OpenLDAP as an Identity Provider.

Edit Existing Directories

You can edit the directory by clicking on the **Edit** button. From the **Directory Details** page, click the **Edit** button at the top of each of the following pages to modify the existing configuration details:

- Details
- Attribute mappings
- Group provisioning
- User provisioning
- Sync Settings - For more information see, #unique_2226/unique_2226_Connect_42_id-996abb49-c8b4-4f35-b844-6d96047c2822.

- Sync Log - For more information, see #unique_2226/unique_2226_Connect_42_id-50b722c2-2b70-41c1-aa2f-ff7e5459f083.

  Note:

  After you modify configurations in the Attribute Mappings, Group provisioning, User provisioning, and Sync Settings screens, you are prompted with an option to trigger an on-demand sync. If you do not trigger the on-demand sync, changes to users and groups take place in the next scheduled sync.

Modify Sync Settings

Editing the directories includes modifying the sync settings. The sync schedule is set to once per week, however you can modify the sync schedule by clicking the **Edit** button in the **Sync Settings** screen. Sync schedule is configured in UTC.

View Sync Log

The sync log under each directory contains the log of each sync between identity broker. and the configured domain controller in OpenLDAP. As part of this sync, users and groups are added or removed. You can view the time at which the syn was completed, the number of groups and users that were modified because of the sync, the status of the sync, and alerts. Here is a list of error messages.

Error Message	Possible Reason(s)
Invalid group attribute name	The group object has an invalid or a missing required attribute from OpenLDAP.
Invalid user attribute name	The user object has an invalid or missing required attribute from OpenLDAP.
Multiple attribute values	The attribute that was mapped to the user/group contains more than one value in OpenLDAP.
User or Group Query Failed	The user/group was not found in OpenLDAP.

Step 3: Configure VCF Single Sign-On for NSX and vCenter

After you have configured an identity provider for VCF single-sign on, connect vCenter and NSX components to the identity broker for single-sign on-based access. You can select only those vCenter and NSX components that belong to the VCF Instance that hosts the identity broker. In the case of an embedded identity broker, the management domain vCenter is automatically configured and you cannot configure/unjoin the management domain vCenter.

Prerequisites

- Verify that you have configured the identity provider for VCF single-sign on. For more information, see Step 2: Configure an Identity Provider.

Procedure

1. In the **Configure VCF SSO** page, click the **Start** button against the **Enable SSO for vCenter and NSX** option.
2. In the **Enable SSO for vCenter and NSX** page, from the grid, select the vCenter or NSX components for which you want to enable VCF Single Sign-On.

   Note:

   - Only vCenter or NSX components that are of version 9.0 or later and are not part of the ELM ring are supported for configuring VCF single-sign on access.
3. Click **Configure**.
4. In the **Confirmation required** box, read the information, select the checkbox, and click **Finish**.

Role Assignments

You can assign VCF roles to the provisioned users and groups from VCF Operations or you can assign roles and permissions from the individual components.

- **Assign VCF roles from VCF Operations.** After the components are configured as part of the initial setup, you can create VCF roles and can assign the VCF roles to users/groups. For information about assigning VCF roles to users/groups, see Step 4. (Optional) Assign a VCF Role. For information about VCF roles, see Managing VCF Roles.
- **Assign Roles and Permissions from Individual Components.** Alternatively, after the components are configured as part of the initial setup, log in with the local admin account to each component and assign the necessary roles to the provisioned users and groups. You can launch the components from the **Component Configuration** tab. For information about assigning roles, see Assigning Roles and Permissions from Individual Components.

Step 4. (Optional) Assign a VCF Role

As an administrator, when you configure VCF single-sign on in VCF Operations, you can assign VCF roles to provisioned users and groups from VCF Operations for components such as vCenter, VCF Operations, VCF Automation, VCF Operations orchestrator, VCF Operations HCX, and NSX.

> **Note:**
>
> Alternatively, you can also assign the required roles and permissions for users or groups from each individual component. To assign roles and permissions for users and groups from each individual component, see Assigning Roles and Permissions from Individual Components.

Procedure

1. From the **Configure VCF SSO** page, click the **Start** button against the **(Optional) Assign VCF Roles** option.
2. In the **Assign VCF Roles** page, you see a list of users and groups that are provisioned under the **Users** and **Groups** tabs.
3. Select the group and/or user that you would like to assign the role to, and click **Assign**.
4. In the **Assign VCF Roles** page, under the section that is called **Roles**, enter the following details and then click **Apply**. If you would like to create roles, see Managing VCF Roles.

Option	Description
Scope	• All components. All the components configured with the relevant identity broker. • Individual VCF Instances. You can scope the role assignment to one or more VCF Instances. Scoping is applicable to vCenter and NSX components that are part of the VCF Instance. **Note:** As a user, you automatically get access to any new vCenter or NSX instances that are added.
Role	Select a role from the drop-down list. If you selected VCF Instance as the scope, only roles that contain vCenter or NSX component roles are listed.
Valid Until	• Never Expires. Select this option if you do not want to set an expiry date for the role. • Custom Date (UTC). Enter a date until when you would like the role to be valid.

> **Note:**
>
> You can add more than one role.

5. Click **Assign** and **Done**.
6. In the **Configure VCF SSO** page, click **Finish Setup**.

   For information about managing VCF role assignments, see Manage VCF Role Assignments After Configuring VCF Single Sign-On.

(Optional) Assign VCF Roles to Groups and Users using On-Demand Lookup

In versions 9.1.1 and later, you can search groups and users dynamically from active directories that are configured with on-demand lookup for role assignment. If an Active Directory is set up with on-demand lookup, you can search its groups and users dynamically for provisioning.

> **Note:**
>
> On-demand lookup is supported only if you are configuring Active Directory as an identity provider using AD/LDAP. For more information, see Configure Active Directory as an Identity Provider Using AD/LDAP.

1. From the **Configure VCF SSO** page, click the **Start** button against the **(Optional) Assign VCF Roles** option.
2. In the **Assign VCF Roles** page, from the **Groups/Users** section, select **Look Up**.

   > **Note:**
   >
   > The **On-Demand Search** button is visible if you have selected on-demand lookup while configuring Active Directory as an identity provider using AD/LDAP

3. In the **On-Demand Search** page, you can search for groups/users dynamically from active directories that are configured for on-demand lookup.

Option	Description
Choose Directory	Select the source directory for the group/user search from the drop-down list. **Note:** Only on-demand directories are displayed. The first directory in the list is pre-selected.
Add Groups/Add Users	Select the groups/users from the drop-down list. **Note:** Groups/Users with role assignments are not available for selection. You can edit the assigned roles from the **Access Management** tab.
(Optional) Assign Roles	**Scope** • All components. All the components configured with the relevant identity broker. • Individual VCF Instances. You can scope the role assignment to one or more VCF Instances.
	Role Select a role from the drop-down list. If you selected VCF Instance as the scope, only roles that contain vCenter or NSX component roles are listed.
	Valid Until • Never Expires. Select this option if you do not want to set an expiry date for the role. • Custom Date (UTC). Enter a date until when you would like the role to be valid.

4. Click **Finish**, and then **Done**.

Step 5: (Optional) Configure VCF Single Sign-On for VCF Operations, VCF Automation Provider, and Salt RaaS

After you have configured an identity provider for VCF single sign-on, you can configure VCF single sign-on for VCF Operations and VCF Automation provider. Select the identity broker with which you wish to establish a connection for VCF single sign-on.

- To add VCF Operations and VCF Automation Provider components to the identity broker for single sign-on access, see Configure VCF Single Sign-On for VCF Operations and VCF Automation Provider.
- For versions 9.1.1 and later, to add the Salt RaaS component to the identity broker for single sign-on access, see Configure VCF Single Sign-On for Salt RaaS.

Configure VCF Single Sign-On for VCF Operations and VCF Automation Provider

You can connect VCF Operations and/or VCF Automation Provider to an identity broker for single sign-on access.

Configure VCF Single Sign-On for VCF Operations

Procedure

1. In the VCF Operations navigation bar at the top, click **Manage** and in the left navigation pane, click **Fleet Management** > **Identity & Access**.
2. In the **Identity & Access** pane, click **VCF SSO Overview**.
3. On the **VCF SSO Overview** page, from the grid under the **VCF Management** tab, select **operations appliance**.
4. Click **Join VCF SSO**.
5. On the **Join VCF SSO** page, select an identity broker and click **Join**.
6. Read the information about role assignments and click **Finish**.

Note:

- Create and assign VCF roles to grant users and groups permissions to individual roles. Role assignments are required to enable single sign-on for the selected component. For information about creating and managing VCF roles, see Create a Custom VCF Role and Assign Custom and Built-In VCF Roles.
- Alternatively, you can assign roles from each individual component. For more information, see Assigning Roles and Permissions from Individual Components.

Configure VCF Single Sign-On for VCF Automation Provider

Procedure

- In the VCF Operations navigation bar at the top, click **Manage** and in the left navigation pane, click **Fleet Management** > **Identity & Access**.
- In the **Identity & Access** pane, click **VCF SSO Overview**.
- On the **VCF SSO Overview** page, from the data grid under the **VCF Management** tab, select **Automation appliance**.
- Click **Join VCF SSO**.
- On the **Join VCF SSO** page, select an identity broker and click **Join**.
- Read the information about role assignments and click **Finish**.

Note:

- Create and assign VCF roles that to grant users and groups permissions to individual roles. Role assignments are required to activate Single Sign-On for the selected component. For information about creating and managing VCF roles, see Create a Custom VCF Role and Assign Custom and Built-In VCF Roles.
- Alternatively, you can assign roles from each individual component. For more information, see Assigning Roles and Permissions from Individual Components.

Configure VCF Single Sign-On for Salt RaaS

In versions 9.1.1 and later, you can connect Salt RaaS to an identity broker for single sign-on access.

Procedure

1. In the VCF Operations navigation bar at the top, click **Manage** and in the left navigation pane, click **Fleet Management** > **Identity & Access**.
2. In the **Identity & Access** pane, click **VCF SSO Overview**.
3. On the **VCF SSO Overview** page, from the grid under the **VCF Management** tab, select **Salt RaaS**.
4. Click **Join VCF SSO**.
5. On the **Join VCF SSO** page, select an identity broker and click **Join**. Read the information about role assignments and click **Finish**.

- Create and assign VCF roles to grant users and groups permissions to individual roles. Role assignments are required to enable single sign-on for the selected component. For information about creating and managing VCF roles, see Create a Custom VCF Role and Assign Custom and Built-In VCF Roles.
- Alternatively, you can assign roles from each individual component. For more information, see Assigning Roles and Permissions from Individual Components.

Step 6: (Optional) Configure VCF Single Sign-On for other VCF Components

Configure VCF single sign-on for other VCF Components such as VCF Operations HCX, VCF Operations for networks, VCF Operations orchestrator, and VCF Automation tenant.

- To configure VCF single sign-on for VCF Operations HCX, Configure VCF SSO.
- To configure VCF single sign-on for VCF Operations for networks, see Configuring VCF SSO for VCF Operations for networks.
- To configure VCF single sign-on for CLI tools like PowerCLI, see Using VCF Single Sign-On to Access vCenter using PowerCLI.
- To configure VCF single sign-on for VCF Automation tenant, see Configure Your VCF Automation Organization to Use an OpenID Identity Provider.

CAUTION:

- VCF Automation tenant support is restricted to a single segment.
- Multiple VCF Automation instances are unsupported.
- The configured user and group scale must not exceed the published identity broker configuration maximum. For more information, see identity broker maximum limits.
- Configuring VCF single sign-on using other components is supported in enterprise environments only. Cloud service provider environments are not supported.
- Configuring VCF single sign-on using other components is limited to the identity broker instance deployment mode only, and is not supported on identity broker embedded deployment mode.
- Configuring VCF single sign-on using other components, is supported on VCF products only and is not available for non-VCF products.

Using VCF Single Sign-On to Access vCenter using PowerCLI

You can configure CLI and automation tools like PowerCLI to use VCF single sign-on. You can configure CLI and automation tools like PowerCLI to access vCenter only, using VCF single sign-on. If you are an existing customer upgrading to VCF 9.1, existing automation scripts using CLI tools will not be impacted if VCF single sign-on is configured in the same management domain vCenter in embedded mode. However, if single sign-on is configured in instance mode, or in an alternate management domain vCenter, a new OAuth client must be generated for PowerCLI.

Prerequisite

Verify that VCF single sign-on is configured for at least one VCF Instance. For more information, see Configure a New VCF Single Sign-On for a VCF Instance.

1. From the VCF Operations navigation pane, select **Fleet Management** > **Identity & Access** > **VCF Other Components**.
2. Click **Add** and enter the following information:
   - Name - Custom name for the OIDC client.

- Identity broker - Choose the identity broker with which you want to configure Single Sign-On.
- Redirect URI - The endpoint to which the identity broker needs to redirect after authentication. The URI must follow the HTTP schema, and be on localhost with a free port on the client machine. For example: "http://localhost:8844/auth."
- Post logout redirect URI - Not applicable for Single Sign-On with PowerCLI.

3. Click **Generate OIDC Client**.
4. Copy and store the Client ID, client secret, and Identity Broker issuer URL securely in a safe place.
5. Click **Save**.
6. Create an OAuth security context object for PowerCLI. For example, see the following script.

```
$oauthSecContext = New-OAuthSecurityContext -TokenEndpointUrl "https://<vcenter_fqdn>/acs/t/CUSTOMER/
token" -
AuthorizationEndpointUrl "https://<vcenter_fqdn>/acs/t/CUSTOMER/authorize" -RedirectUrl "http://
localhost:8844/auth" -
ClientId "powercli-native" -ClientSecret "<secret_value>"
TokenEndpointUrl
AuthorizationEndpointUrl
RedirectUrl
ClientId
ClientSecret
```

   Note:

   The Redirect URL should be the same as the value entered in Step 2 - Redirect URI.

7. Exchange the OAuth 2.0 security context for a SAML security context.

```
$samlSecContext = New-VISamlSecurityContext -VCenterServer '<vcenter_fqdn>' -OAuthSecurityContext
 $oauthSecContext
```

8. Connect to your vCenter using the SAML security context.

```
Connect-VIServer -Server '<vcenter_fqdn>' -SamlSecurityContext $samlSecContext
```

   Note:

   Single sign-on through CLI tools is possible only through vCenter in VCF 9.1.

Manage VCF Instances after Configuring VCF Single Sign-On

After you have configured VCF single sign-on, you can connect to an existing single sign-on configuration and add a VCF Instance to the pre-configured identity broker. After you have configured single sign-on from VCF Operations, you can also choose to change or remove VCF Instances from/to an existing VCF single sign-on Configuration.

- To connect or add a VCF Instances to an existing VCF single sign-on configuration, see Connect VCF Instances to an Existing VCF Single Sign-On.
- To change or modify VCF Instances in an existing VCF single sign-on configuration, see Modify VCF Instances in an Existing VCF Single Sign-On Configuration.
- To remove VCF Instances from an existing VCF single sign-on configuration, see Remove VCF Instances from an Existing VCF Single Sign-On Configuration.

Connect VCF Instances to an Existing VCF Single Sign-On

After you have configured VCF single sign-on from VCF Operations, you can choose to join additional VCF Instances to an existing VCF single sign-on configuration.

Prerequisites

- Ensure that you have added a VCF Instance. For more information, see Start a New VCF Fleet or a New VCF Instance Deployment by Using the VCF Installer Deployment Wizard.

- Ensure that you have configured an identity provider for VCF Single Sign-On. For more information, see Step 2: Configure an Identity Provider.

Procedure

1. In the VCF Operations navigation bar at the top, click **Manage** and in the left navigation pane, click **Fleet Management** > **Identity & Access**.
2. In the **Identity & Access** pane, click **VCF SSO Overview**.
3. In the **VCF SSO Overview** page, in the grid under **VCF Instances**, select the VCF Instances that you would like to add to VCF single sign-on and click **Join VCF SSO**.
4. In the **Join VCF SSO** page, follow these steps and click **Next**:
   a) In the grid under **Choose SSO**, select the identity broker you would like to connect to.
   b) In the grid under the **Configure Components** section, click the drop-down arrow against the relevant VCF Instances and verify if you want to add all the components that are selected.

      Note:

      - By default, all the components in the selected VCF Instance are part of the single sign-on configuration. You can choose to customize the list and de-select the components that you wish to exclude from the configuration.
      - vCenter and NSX 9.0 and later versions are supported and displayed.
5. Read and accept the information about role assignment that is required to activate VCF single sign-on for the selected components and then click **Finish**.

   You can log in as a local admin to each component and can assign service roles to provisioned users and groups. For more information about assigning service roles, see Assigning Roles and Permissions from Individual Components.
6. In the **Configure VCF SSO Overview** page, navigate to the **VCF Instances** tab to see the VCF Instances you have added to VCF single sign-on configuration.

Modify VCF Instances in an Existing VCF Single Sign-On Configuration

After you have configured VCF single sign-on from VCF Operations, you can choose to change or modify VCF Instances in an existing VCF single sign-on.

Prerequisites

- Ensure that you have added a VCF Instance. For more information, see Start a New VCF Fleet or a New VCF Instance Deployment by Using the VCF Installer Deployment Wizard.
- Ensure that you have configured an identity provider for VCF single sign-on. For more information, see Step 2: Configure an Identity Provider.

Procedure

1. In the VCF Operations navigation bar at the top, click **Manage** and in the left navigation pane, click **Fleet Management** > **Identity & Access**.
2. In the **Identity & Access** pane, click **VCF SSO Overview**.
3. In the **VCF SSO Overview** page, in the grid under **VCF Instances**, select the VCF Instances that you would like to change in the VCF single sign-on configuration, and then click the **horizontal ellipsis** > **Change VCF SSO** option.
4. In the **Change VCF SSO** page, follow these steps:
   a) In the grid under **Choose SSO**, select a identity broker.
   b) In the grid under the **Configure Components** section, click the drop-down arrow against the relevant VCF Instances to view the components that were previously configured.
      - All previously configured components in the selected VCF Instance are migrated to the new VCF single sign-on Configuration.
      - vCenter and NSX 9.0 and later versions are supported and displayed.

c) Click **Next** and read and confirm the information about role assignments.

Modifying VCF single sign-on removes all role assignments that are scoped to the VCF Instance. To reassign roles, see Assign VCF Roles After Configuring VCF Single Sign-On.

Remove VCF Instances from an Existing VCF Single Sign-On Configuration

After you have configured VCF single sign-on from VCF Operations, you can choose to remove VCF Instances in an existing VCF single sign-on.

Prerequisites

- Ensure that you have added a VCF Instance. For more information, see Start a New VCF Fleet or a New VCF Instance Deployment by Using the VCF Installer Deployment Wizard.
- Ensure that you have configured an identity provider for VCF single sign-on. For more information, see Step 2: Configure an Identity Provider.

Procedure

1. In the VCF Operations navigation bar at the top, click **Manage** and in the left navigation pane, click **Fleet Management** > **Identity & Access**.
2. In the **Identity & Access** pane, click **VCF SSO Overview**.
3. In the **VCF SSO Overview** page, in the grid under **VCF Instances**, select the VCF Instances that you would like to remove in the VCF single sign-on configuration, and then click the **horizontal ellipsis** > **Unjoin VCF SSO** option.
4. Read the message about role assignments and click **Unjoin**.

   If you remove a VCF Instance from an existing VCF single sign-on configuration, all role assignments that are scoped to the VCF Instance are removed.

Managing VCF Single Sign-On

After you configure and start using VCF single sign-on, you can view the existing identity brokers, reset VCF single sign-on, change an identity provider, change the deployment mode, edit the configuration of an identity provider, add or deregister existing component configurations, manage API and access tokens, and so on.

- To view configured identity brokers, see Overview of VCF Single Sign-On Configurations.
- To delete VCF single sign-on, see Delete a VCF Single Sign-On Configuration.
- To change the identity provider in VCF single sign-on, see Change the Identity Provider in VCF Single Sign-On.
- To change the deployment mode, see Change the Deployment Mode in VCF Single Sign-On.
- To change the provisioning method, see Change the Provisioning Method in VCF Single Sign-On.
- To migrate identity broker embedded to identity broker instance, see Migration of Identity Broker Embedded to Identity Broker Instance.
- To edit the configuration of an identity provider, see Edit the Configuration of an Identity Provider.
- To edit the display name of the identity broker, see Edit the Display Name of the Identity Broker.
- To manage VCF role assignments after configuring VCF Single Sign-On, see Manage VCF Role Assignments After Configuring VCF Single Sign-On.
- To test login after configuring VCF single sign-on, see Test Login after Configuring VCF Single Sign On.
- To change or remove VCF Operations and VCF Automation components, see Change or Remove VCF Operations and VCF Automation Components.
- To add, change, or remove other VCF components, see Add, Change, or Remove Other VCF Components.
- To assign roles and permissions from individual components, see Assigning Roles and Permissions from Individual Components.
- For versions 9.1.1 and later, to search for on-demand users and groups and assign roles, see Search for Users and Groups Dynamically and Assign Roles using On-Demand Lookup.
- For versions 9.1.1 and later, to verify if on-demand lookup is enabled, see Review Directory Details to Verify if On-demand Lookup is Enabled.
- For versions 9.1.1 and later, to add a directory for identity providers that are configured using AD/LDAP, see Add a Directory for Identity Providers that are Configured using AD/LDAP.

Overview of VCF Single Sign-On Configurations

Use the **VCF SSO Overview** page for a holistic overview of the VCF fleet that has been configured with VCF single sign-on.

Multiple VCF instances can participate in a VCF single sign-on configuration managed by a single identity broker. To view details of a specific VCF single sign-on configuration and its participating VCF Instances, in the **VCF Overview SSO** page, in the section that is called **VCF SSO**, click the identity broker that you have configured for VCF single sign-on. From the **Overview** tab, click the **VCF Instances** tab.

In VCF Operations, you can monitor the status of VCF identity brokers based on parameters such as health and readiness. Health indicates if the identity broker is up and running or not. Readiness indicates if the identity broker is ready to accept incoming requests or not. From VCF Operations navigation bar at the top, click **Operate**, and then navigate to **Inventory** > **Integrations icon** > **Infrastructure Management** > **Identity Broker health**.

Delete a VCF Single Sign-On Configuration

You can delete a VCF single sign-on configuration that is associated with one or more VCF Instance.

When you delete a VCF single sign-on configuration, the entire VCF single sign-on configuration including the identity provider configuration, provisioned users/groups, and component configuration are deleted.

Note:

- The delete action cannot be reverted.
- If the user performing the deletion of VCF single sign-on is part of the particular VCF single sign-on configuration being deleted, then the session terminates immediately post deletion. The user has to log in again as a local admin account user. You might want to consider logging in as a local admin user to perform the reset activity.

Procedure

1. In the VCF Operations navigation bar at the top, click **Manage** and in the left navigation pane, click **Fleet Management** > **Identity & Access**.
2. From the **Identity & Access** pane, click **VCF SSO Overview**.
3. In the **VCF Overview SSO** page, in the section that is called **VCF SSO**, click the identity broker that you have configured for VCF single sign-on.
4. Click **Actions** > **Delete SSO** from the top-right corner.
5. Select the following option and click **Delete**.

   I confirm to delete of all SSO-related configurations, including the identity provider configuration, the provisioned users and groups, and the component configuration.

   Note:

   If you delete VCF single sign-on for the VCF Instance, you must configure VCF single sign-on again.
6. You are directed to the **Configure VCF SSO** page from where you can complete the steps to configure VCF single sign-on. For more information about configuring VCF single sign-on, see, Configure a New VCF Single Sign-On for a VCF Instance.

   Note:

   If problems arise after deleting the VCF single sign-on configuration, refer to KB 393161 to restore a clean state.

Change the Identity Provider in VCF Single Sign-On

If you have configured VCF single sign-on using a certain identity provider and protocol, you can change the identity provider in the future.

For example, you have configured VCF single sign-on using Active Directory, and wish to change to a modern identity provider like Okta, Ping Identity, or Microsoft Entra ID. To change the identity provider, you must reset VCF single sign-on and reconfigure the identity provider.

> **Note:**
>
> Since this is a partial reset, the deployment mode and client component configuration are retained. Therefore, you do not have to reconfigure the client for vCenter, VCF Operations HCX,, VCF Operations, VCF Automation, VCF Operations for networks, or log management to the specific identity broker.

Procedure

1. In the VCF Operations navigation bar at the top, click **Manage** and in the left navigation pane, click **Fleet Management** > **Identity & Access**.
2. In the **Identity & Access** pane, click **VCF SSO Overview**.
3. In the **VCF Overview SSO** page, in the section that is called **VCF SSO**, click the identity broker that you have configured for VCF Single Sign-On.
4. Click **Actions** > **Change the Identity Provider** from the top-right corner.
5. In the **Change Identity Provider** dialog box that is displayed, select the following option and click **Change**.

   I confirm to delete of all SSO-related configurations, including the identity provider configuration, the provisioned users and groups, and the component configuration.

   > **Note:**
   >
   > The components that are configured against this identity broker are retained. After you configure a new identity provider, you might need to assign roles to the newly provisioned users and groups. For information about assigning roles, see Step 4. (Optional) Assign a VCF Role.

6. You are directed to the **Configure VCF SSO** page. In the **Configure VCF SSO** screen, click the second step, **Configure Identity Provider**, to reconfigure the identity provider. For more information, see Step 2: Configure an Identity Provider.

Change the Deployment Mode in VCF Single Sign-On

If you have configured VCF single sign-on using a certain deployment mode, you can choose another deployment mode in the future.

For example, you have chosen the identity broker embedded deployment mode to configure VCF single sign-on for a VCF Instance. If necessary, you can change the deployment mode to identity broker instance for the specific VCF Instance. Conversely, you can change the deployment mode from identity broker instance to identity broker embedded.

Procedure

1. In the VCF Operations navigation bar at the top, click **Manage** and in the left navigation pane, click **Fleet Management** > **Identity & Access**.
2. In the **Identity & Access** pane, click **VCF SSO Overview**.
3. In the **VCF SSO Overview** page, in the section that is called **VCF SSO**, click the identity broker that you have configured for VCF Single Sign-On.
4. Click **Actions** > **Delete SSO** from the top-right corner. For more information about deleting VCF Single Sign-On, see Delete a VCF Single Sign-On Configuration.
5. Click **Delete**. The **Configure VCF SSO** screen is displayed.
6. In the **Configure VCF SSO** screen, click the **Start** button against the **Choose deployment mode** option. For more information, see Step 1: Choose the Deployment Mode and Configure a New VCF Single Sign-On for a VCF Instance.

Ensure that the users and groups are assigned the required VCF roles after you complete configuring the identity provider. For information about assigning VCF roles, see Step 4. (Optional) Assign a VCF Role.

If you configured VCF Single Sign-On in VCF 9.0 using identity broker embedded and you want to migrate to identity broker instance, see Migration of Identity Broker Embedded to Identity Broker Instance.

Change the Provisioning Method in VCF Single Sign-On

If you have configured VCF single sign-on using SCIM or JIT as the provisioning method, you can choose to change the provisoning method to AD/LDAP or JIT.

The following migrations are supported:

- SCIM to LDAP. For more information, see Change the Provisioning Method from SCIM to AD/LDAP.
- SCIM to JIT. For more information, see Change the Provisioning Method from SCIM to JIT.
- JIT to LDAP. For more information, see Change the Provisioning Method from JIT to LDAP.

Change the Provisioning Method from SCIM to AD/LDAP

If you have configured VCF single sign-on with SCIM as the provisioning method, you can change to AD/LDAP provisioning.

Prerequisites

- Verify that VCF single sign-on was configured using SCIM as the provisioning method.

1. In the VCF Operations navigation bar at the top, click **Manage** and in the left navigation pane, click **Fleet Management** > **Identity & Access**.
2. In the **Identity & Access** pane, click **VCF SSO Overview**.
3. In the **VCF SSO Overview** page, in the section that is called **VCF SSO**, click the identity broker that you have configured for VCF single sign-on.
4. In the **Identity Source** tab, against **Provisioning Method**, click **Change**.
5. In the **Pre-read** screen, read and understand the information, and then click **Next**.
6. In the **User/Group Provisioning Method** screen, click **AD/LDAP (Active Directory/Lightweight Directory Access Protocol)**, and then click **Next**.
7. In the **LDAP Lookup** screen, select the type of domain controller configuration you use in Active Directory. Select one of the following options, complete the relevant steps, and then click **Next**.
   - Global Catalog, or
   - Domain Specific Controller

a) If you select Global Catalog, complete the following steps.

In the **Directory Configuration** screen, enter the following details or select the required options and click **Validate and Next.**

Option	Description
Directory name	Enter a name for the directory.
Global Catalog	Global catalog has a partial read-only replica of all objects across all domains. Some of the limitations with selecting the global catalog option include: • The Active Directory object attributes that are replicated to the global catalog are identified in the Active Directory schema as the Partial Attribute Set (PAS). Only these attributes are available for attribute mapping by the service. If necessary, edit the schema to add or remove attributes that are stored in the global catalog. • The global catalog stores the group membership (the member attributes) of only universal groups. Only universal groups are synced to the service. If necessary, change the scope of a group from a local domain or from global to universal. • The bind DN account that you define when you configure a directory in the service must have permissions to read the Token-Groups-Global-And-Universal (TGGAU) attribute. **Note:** When you use Global catalog with LDAP and multiple domains, the additional domains are discovered only after the user/group sync is complete.
Primary domain controller (Displayed when you select the **Global Catalog** option)	A server machine running Active Directory Domain Services (AD DS) responsible for managing and authenticating user access, enforcing policies, and storing directory data. For example, `hostname = DC01` for the domain called example.com.
Secondary domain controller (Displayed when you select the **Global Catalog** option)	A backup domain controller for the identity broker to point to if the primary domain controller is not reachable.
Directory search attribute	A specific LDAP attribute used in queries to search or identify entries in the directory. For example, `uid`, `sAMAccountName`.
Base DN	The Distinguished Name (DN) of the top-level entry in the directory tree from which LDAP queries begin. For example, `dc=example`, `dc=com`.
Bind user name	The user name of a user account with sufficient permissions to authenticate and perform LDAP operations on the directory.
Bind user password	The password of a user account with sufficient permissions to authenticate and perform LDAP operations on the directory.

b) If you select **Domain Specific Controllers**, complete the following steps:

In the **Choose Domains** screen, the domains that were identified from the existing configuration are displayed.

- Select the domain/s that you want to retain. Users and groups from the selected domain are retained to support migration.

Note:

If a domain is not configured, the list of users/groups are deleted permanently and not referenced for migration.

- Click the check box to confirm your understanding, and then click **Next**.

8. In the **Choose Directory** screen, configure the directory details for each listed domain and click **Test Connection**. If successful, click **Next**.
9. In the **Grace Period** screen, enter a date. The mismatched users and groups from the existing configuration are retained till the date specified. The cleanup of the mismatched users and groups begins on or after the date specified.
10. Click **Finish** and then click **Continue**. You see that the provisioning method has changed to AD/LDAP in the **Identity Source** tab.

    If there are mismatched user and groups, you can view them by clicking the **View More** link in the following banner at the top of the page: `The previously configured users and groups are available for x days.`

Provisioning users and groups

11. In the **Identity Source** tab, navigate to the **Directory Information** section. In the data grid under **Provisioning Method**, click **Setup Provisioning** in the required row.
12. In the **Review Directory Information** page, review the information and click **Next**.
13. In the **Attribute Mappings** screen, ensure that the correct attributes have been mapped to the Active Directory attributes, and then click **Next**.
14. In the **Group Provisioning** screen, complete the following steps, and click **Next**.
    1. **Sync Nested Group**. Click this option if you want all the users from the selected group and its nested group to sync without maintaining the group hierarchy. Do not select this option for large AD setups and manually select the required groups.
    2. **Specify the Base Group DN**. Click **Add Base DN** to select the base groups.
15. In the **Users Provisioning** screen, specify the base user group DN by clicking the **Select Base User DN** option to select the base users, and click **Next**.
16. In the **Review** page, review the configuration details of the identity provider and then click **Finish**.

After the sync is complete, you can view the users and groups that have been provisioned by clicking the **View More** link from the banner at the top of the **Identity Source** tab.

Change the Provisioning Method from SCIM to JIT

If you have configured VCF single sign-on with SCIM as the provisioning method, you can change to JIT provisioning.

Prerequisites

- Verify that VCF single sign-on was configured using SCIM as the provisioning method.

1. In the VCF Operations navigation bar at the top, click **Manage** and in the left navigation pane, click **Fleet Management** > **Identity & Access**.
2. In the **VCF SSO Overview** page, in the section that is called **VCF SSO**, click the identity broker that you have configured for VCF Single Sign-On.
3. In the **Identity Source** tab, against **Provisioning Method**, click **Change**.
4. In the **Pre-read** screen, read and understand the information, and then click **Next**.
5. In the **Provisioning Method** screen, click **JIT (Just-in-Time Provisioning)**, and then click **Next**.
6. In the **Group Provisioning** screen, select one of the following options, complete the relevant steps, and then click **Next**.
   - **Groups Pre-provisioning.** Select this option if you want a restricted list of groups for role assignments.
   - **Just-in-Time Group Provisioning.** Select this option to retain all the existing users and groups from the previous provisioning group.

   a) If you select **Groups Pre-provisioning**, complete the following steps.
      - In the **Pre-provisioning Domains and Groups** screen, add the groups from each domain that you want to provision in VCF for roles and assignments by clicking the **Add Group** button, and then click **Next**.
      - In the **Attributes** screen, the user and group attributes are listed. Enter the user/group attributes for JIT.

b) If you select **Just-in-Time Provisioning**, complete the following steps.

- In the **Attributes** screen, the user and group attributes are listed. Enter the user/group attributes for JIT.
- Click **Finish**.

After the sync is complete, you can view the users and groups that have been provisioned by clicking the **View More** link from the banner at the top of the **Identity Source** tab.

Change the Provisioning Method from JIT to LDAP

If you have configured VCF single sign-on with JIT as the provisioning method, you can change to LDAP provisioning.

Prerequisites

- Verify that VCF single sign-on was configured using JIT as the provisioning method.

1. In the VCF Operations navigation bar at the top, click **Manage** and in the left navigation pane, click **Fleet Management** > **Identity & Access**.
2. In the **VCF SSO Overview** page, in the section that is called **VCF SSO**, click the identity broker that you have configured for VCF Single Sign-On.
3. In the **Identity Source** tab, against **Provisioning Method**, click **Change**.
4. In the **Pre-read** screen, read and understand the information, and then click **Next**.
5. In the **User/Group Provisioning Method** screen, click **AD/LDAP (Active Directory/Lightweight Directory Access Protocol)**, verify the details, and then click **Next**.
6. In the **LDAP Lookup** screen, select the type of domain controller configuration you use in Active Directory. Select one of the following options, complete the relevant steps, and then click **Next**.
   - Global Catalog, or
   - Domain Specific Controller

   a) If you select Global Catalog, complete the following steps.

   In the **Directory Configuration** screen, enter the following details or select the required options and click **Validate and Next**.

Option	Description
Directory name	Enter a name for the directory.
Global Catalog	Global catalog has a partial read-only replica of all objects across all domains. Some of the limitations with selecting the global catalog option include: • The Active Directory object attributes that are replicated to the global catalog are identified in the Active Directory schema as the Partial Attribute Set (PAS). Only these attributes are available for attribute mapping by the service. If necessary, edit the schema to add or remove attributes that are stored in the global catalog. • The global catalog stores the group membership (the member attributes) of only universal groups. Only universal groups are synced to the service. If necessary, change the scope of a group from a local domain or from global to universal. • The bind DN account that you define when you configure a directory in the service must have permissions to read the Token-Groups-Global-And-Universal (TGGAU) attribute. **Note:** When you use Global catalog with LDAP and multiple domains, the additional domains are discovered only after the user/group sync is complete.
Primary domain controller (Displayed when you select the **Global Catalog** option)	A server machine running Active Directory Domain Services (AD DS) responsible for managing and authenticating user access, enforcing policies, and storing directory data. For example, `hostname = DC01` for the domain called example.com.

Option	Description
Secondary domain controller (Displayed when you select the **Global Catalog** option)	A backup domain controller for the identity broker to point to if the primary domain controller is not reachable.
Directory search attribute	A specific LDAP attribute used in queries to search or identify entries in the directory. For example, `uid`, `sAMAccountName`.
Base DN	The Distinguished Name (DN) of the top-level entry in the directory tree from which LDAP queries begin. For example, `dc=example`, `dc=com`.
Bind user name	The user name of a user account with sufficient permissions to authenticate and perform LDAP operations on the directory.
Bind user password	The password of a user account with sufficient permissions to authenticate and perform LDAP operations on the directory.

b) If you select **Domain Specific Controllers**, complete the following steps:

1. In the **Choose Domains** screen, the domains that were identified from the existing configuration are displayed.
   – Select the domain/s that you want to retain. Users and groups from the selected domain are retained to support migration.

Note:

If a domain is not configured, the list of users/groups are deleted permanently and not referenced for migration.

   – Click the check box to confirm your understanding, and then click **Next**.
2. In the **Directory Configuration** screen, configure the directory details for each listed domain and click **Test Connection**. If successful, click **Next**.

7. In the **Grace Period** screen, enter a date. The mismatched users and groups from the existing configuration are retained till the date specified. The cleanup of the mismatched users and groups begins on or after the date specified.
8. Click **Finish** and then click **Continue**. You see that the provisioning method has changed to AD/LDAP in the **Identity Source** tab.

   If there are mismatched user and groups, you can view them by clicking the **View More** link in the following banner at the top of the page: `The previously configured users and groups are available for x days`.

Provisioning users and groups

9. In the **Identity Source** tab, navigate to the **Directory Information** section. In the grid under **Provisioning Method**, click **Setup Provisioning** in the required row.
10. In the **Review Directory Information** page, review the information and click **Next**.
11. In the **Attribute Mappings** screen, ensure that the correct attributes have been mapped to the Active Directory attributes, and then click **Next**.
12. In the **Group Provisioning** screen, complete the following steps, and click **Next**.
    1. **Sync Nested Group**. Click this option if you want all the users from the selected group and its nested group to sync without maintaining the group hierarchy. Do not select this option for large AD setups and manually select the required groups.
    2. **Specify the Base Group DN**. Click **Add Base DN** to select the base groups.
13. In the **Users Provisioning** screen, specify the base user group DN by clicking the **Select Base User DN** option to select the base users, and click **Next**.
14. In the **Review** page, review the configuration details of the identity provider and then click **Finish**.

After the sync is complete, you can view the users and groups that have been provisioned by clicking the **View More** link from the banner at the top of the **Identity Source** tab.

Migration of Identity Broker Embedded to Identity Broker Instance

If you configured VCF single sign-on in VCF 9.0 using identity broker embedded and you want to migrate to identity broker instance, you can do so seamlessly. This transition ensures no data loss and maintains full access to your components following the migration.

Prerequisites

- Verify that you configured VCF single sign-on in VCF 9.0 using identity broker embedded.
- Verify that the identity broker embedded is at version 9.1. Identity Broker embedded is updated with the vCenter instance. For more information, see Upgrading vCenter and NSX Manager.

Procedure

1. In the VCF Operations navigation bar at the top, click **Manage** and in the left navigation pane, click **Fleet Management** > **Identity & Access**.
2. From the **Identity & Access** pane, click **VCF SSO Overview**.
3. In the **VCF SSO Overview** page, in the section that is called **VCF SSO**, click the identity broker that you have configured for VCF Single Sign-On.
4. Click **Actions** > **Migrate from Embedded to Instance** from the top-right corner.
5. Read and accept the information about the impact of migrating identity broker embedded to identity broker instance.
6. From the **Identity Broker Instance Target** section, in the data grid, select the target identity broker instance that you wish to migrate to.

   Note:

   - Only those VCF Instances with identity broker instance deployed are listed in the data grid.
   - You can deploy a new identity broker instance if necessary and can use that as the target. For information about deploying an identity broker instance, see Deploy Identity Broker as a Day-N Operation.
7. Enter the following SFTP details:

Option	Description
SFTP Server Host Name	Enter the name of the SFTP server host name.
SFTP Port	Enter the SFTP port number.
Username	Enter the name used to log in using SFTP.
Password	Enter the password used to log in using SFTP.
Path	Enter the file path where the data should be stored on the SFTP server. The default value is set to `/path`.

   Note:

   For the duration of the data migration, create a dedicated, single-purpose SFTP user account. For security purposes, this account's permissions must be restricted to access only the specific folder defined for the specific account and must be deleted after the migration process is successfully completed.
8. Click **Pre-check**. If the pre-check is successful, click **Start Migration**.

   If there is an error, pre-check displays the error that must be fixed. For example, if the SFTP server is not reachable, an appropriate error message is displayed.
9. Data transfer follows a series of steps such as data export, import of data from SFTP, and restarting of services.
   - During migration, identity provider configuration and user/group provisioning are suspended.
   - Data transfer might take some time to complete.
10. Click **Confirm** to proceed to the next step of migration.

    If you click **Cancel**, the migration process is aborted and configuration is deleted in the identity broker instance.

11. In the **Update Identity Provider** section, copy the configuration details of the identity broker instance, navigate to the admin console of the identity provider, and update the service provider details.

    You can get service provider details from your third-party identity system.

12. Select the check box to confirm that you have updated the configuration details in the admin console of the external identity provider.

    In versions 9.1.1 and later, the **Directory Information** section displays a list of directories that have been configured. The column that is called **Provisioning Method** is visible if Active Directory is configured as an identity provider using AD/LDAP with on-demand search. The **Provisioning Method** column indicates whether users and groups are pre-provisioned or configured through on-demand search.

13. Click **Next**.
14. In the **Test Login** section, you can validate log in to your identity provider connection using the **Test Login** link.
15. Click **Next** to validate log in to the corporate account using the identity provider details. After a successful login, you will be directed back to the **Change Deployment Mode** page.
16. Click **Next**.
17. In the **Update Components** section, click **Update Components** to update the components so that they connect to the identity broker instance and perform a sanity test. All component configurations are modified.
18. Click **Next**.
19. In the **Summary** section, click the check box to confirm that you will carry out the next steps which are:
    - Manually update the configuration details for other components such as VCF Operations HCX, log management, and VCF Operations for networks.
    - Manually update the automation scripts that use the API clients.
20. Click **Done**.
21. In the **Overview** tab, you see the updated identity broker instance details.

Edit the Configuration of an Identity Provider

You can change the identity provider configuration after you have configured VCF single sign-on.

The **Identity Source** tab that you access for each VCF single sign-on configuration displays the deployment mode, authentication information, and directory information.

Procedure

1. In the VCF Operations navigation bar at the top, click **Manage** and in the left navigation pane, click **Fleet Management** > **Identity & Access**.
2. In the **Identity & Access** pane, click **VCF SSO Overview**.
3. In the **VCF Overview SSO** page, in the section that is called **VCF SSO**, click the identity broker that you have configured for VCF Single Sign-On.
4. In the **Identity Source** tab, click **Edit**.

   To configure identity providers, see Step 2: Configure an Identity Provider. Navigate to the page depending on which identity provider you have configured.

   If you configure Active Directory as an identity provider using AD/LDAP and need more information about sync settings, see **Edit Existing Directories** and **Modify Sync Settings** in the topic called Managing AD/LDAP.

Edit the Display Name of the Identity Broker

You can change the display name of the identity broker after you have configured VCF single sign-on.

Procedure

1. In the VCF Operations navigation bar at the top, click **Manage** and in the left navigation pane, click **Fleet Management** > **Identity & Access**.
2. In the **Identity & Access** pane, click **VCF SSO Overview**.
3. In the **Overview** tab, click **EDIT** against **Identity Broker Details**.
4. Enter a name against the **Display Name** field.
5. Click **Save**.

Manage VCF Role Assignments After Configuring VCF Single Sign-On

After you configure VCF single sign-on, you can assign VCF roles, and edit and delete existing VCF roles assignments.

- To assign VCF roles after configuring VCF single sign-on, see Assign VCF Roles After Configuring VCF Single Sign-On.
- To edit and delete existing VCF role assignments after configuring VCF single sign-on, see Edit and Delete Existing Role Assignments After Configuring VCF Single Sign-On.
- To reset and assign role assignments, see Reset and Assign Role Assignments.

Assign VCF Roles After Configuring VCF Single Sign-On

You can assign custom and built-in VCF roles after you have configured VCF single sign-on. VCF level role functionality is available starting with VCF 9.1.

Prerequisites

- Verify that you have configured VCF single sign-on.

  Note:

  - You can also assign component level roles from the associated component. For more information, see Assigning Roles and Permissions from Individual Components
  - Users and groups that have been assigned roles both at the component level and through VCF roles have privileges that are combined.

Procedure:

1. In the VCF Operations navigation bar at the top, click **Manage** and in the left navigation pane, click **Fleet Management** > **Identity and Access**.
2. In the **Identity & Access** pane, click **VCF SSO Overview**.
3. In the **VCF Overview SSO** page, in the section that is called **VCF SSO**, click the identity broker that you have configured for VCF Single Sign-On.
4. In the **Access Management** tab, you see a list of users and groups that are provisioned. Select the group and/or user that you would like to assign the role to, and click **Assign**.
   - The list displays only those users and groups that have been provisioned and might not be complete.
   - With JIT provisioning, users/groups can be assigned only after the first login.
5. In the **Assign VCF Roles** page, in the section called **Roles**, you can scope the role by selecting one of the following options as the **Scope**, select a role from the drop drop-down list, and enter the validity of the role, and then click **Apply**.

Option	Description
Components with SSO	Covers all the components that are configured with the relevant identity broker.
Individual VCF Instances	You can scope the role to one or more VCF Instances.

Note:

- You can add more than one role.

- If you select **Individual VCF Instance** as the scope, the **VCF Viewer**, **VCF Administrator**, and **VCF Operations Administrator** roles are grayed out in the roles drop-down list. These roles apply only to vCenter and NSX instances that are part of the selected VCF Instance.

6. Click **Assign**.

Edit and Delete Existing Role Assignments After Configuring VCF Single Sign-On
You can make changes and delete existing VCF roles assignments to users or groups after configuring VCF single sign-on.

Edit VCF Role Assignments

Procedure

1. In the VCF Operations navigation bar at the top, click **Manage** and in the left navigation pane, click **Fleet Management** > **Identity and Access**.
2. In the **Identity & Access** pane, click **VCF SSO Overview**.
3. In the **VCF SSO Overview** page, from the **Users** tab or **Group** tab, click the double-arrow against the VCF role you want to edit, and then click **Edit Roles** on the right.

   Note:

   If the Roles are by Group Assignment, you must navigate to the Groups section.
4. After you make changes, click **Save**.

Delete VCF Role Assignments That Have Been Assigned to a User/Group

Procedure

1. In the VCF Operations navigation bar at the top, click **Manage** and in the left navigation pane, click **Fleet Management** > **Identity and Access**.
2. In the **Identity & Access** pane, click **VCF SSO Overview**.
   Delete the role assignments
3. In the **VCF SSO Overview** page, from the **Custom Role** section, click the custom role that you want to delete. The role components and role assignments are displayed for the selected custom VCF role.
4. In the **Role Assignments** section, click the VCF Identity Broker. The groups and users who are assigned to that role are displayed.
5. Select the groups or users associated with the VCF role that you want to delete. Click the **horizontal ellipsis** next to the **Assign** button and click **Delete**.
   Delete the Custom VCF role
6. In the **VCF Roles** page, from the **Custom Roles** section, against the role you want to delete, click the **vertical ellipsis** > **Delete**.

Bulk Deletion of VCF Role Assignments

Procedure

1. In the VCF Operations navigation bar at the top, click **Manage** and in the left navigation pane, click **Fleet Management** > **Identity and Access**. In the **Identity & Access** pane, click **VCF SSO Overview**.
2. From the **Users** tab or **Group** tab, select the VCF roles you want to edit, and then next to the **Assign** button, click the **horizontal ellipsis** > **Delete Role Assignments**.

Click **Delete** in the confirmation message.

Note:

- All role assignments are deleted. Bulk deletion cannot be reversed.
- Updates to role assignments take effect in the next authentication cycle.

Reset and Assign Role Assignments

You can reset and reassign role assignments to establish a uniform approach to role assignments. All the previous role assignments for the selected users/groups are reset and overwritten.

Note:

All the previous role assignments for the selected principals are reset and overwritten. To add role assignments to the selected principals, use the Edit feature instead.

Procedure

1. In the VCF Operations navigation bar at the top, click **Manage** and in the left navigation pane, click **Fleet Management** > **Identity and Access**.
2. In the **Identity & Access** pane, click **VCF SSO Overview**.
3. In the **VCF SSO Overview** page, under the section that is called **VCF SSO**, click the relevant identity broker.
4. Click the **Access Management** tab, you see a list of users and groups that are provisioned.

For Groups

5. From the **Groups** section, select the required groups, click the horizontal ellipsis next to **Assign**, and then click **Reset and Reassign** from the drop-down options.
6. From the **Reset and Reassign** page, you see the list of selected groups. Add necessary roles from the **Roles** section.
7. Click **Assign**. Updates to the role assignment take place in the next authentication.

For Users

8. From the **Users** section, select the required users, click the horizontal ellipsis next to **Assign**, and then click **Reset and Reassign** from the drop-down options.
9. From the **Reset and Reassign** page, you see the list of selected users. Add necessary roles from the **Roles** section.

Note:

If any of the users who are selected are part of a group, you cannot reset and assign the role assignments. Navigate to the Groups section to reset role assignments. See steps from 5 through 7 in this page.

10. Click **Reset and Assign**. Updates to the role assignment take place in the next authentication.

Test Login after Configuring VCF Single Sign On

You can test login using your single sign-on credentials. Any account that is used while testing login remains signed in after the process.

Procedure:

1. In the VCF Operations navigation bar at the top, click **Manage** and in the left navigation pane, click **Fleet Management** > **Identity & Access**.
2. In the **Identity & Access** pane, click **VCF SSO Overview**.
3. In the **VCF SSO Overview** page, in the section that is called **VCF SSO**, click the identity broker that you have configured for VCF single sign-on.
4. In the **Identity Source** tab, scroll to the end of the page and click **Test Login**.

Change or Remove VCF Operations and VCF Automation Components

You can connect additional VCF Operations and VCF Automation components to the identity broker for single sign-on access. You can also change or remove existing VCF Operations and VCF Automation components for single sign-on-based access.

- To add additional VCF Operations and VCF Automation components to the identity broker for single sign-on access, see Configure VCF Single Sign-On for VCF Operations and VCF Automation Provider.

- To modify and remove VCF Operations and VCF Automation components from the identity broker for single sign-on access, see Modify or Remove VCF Operations and/or VCF Automation Components in an Existing VCF Single Sign-On Configuration.

Modify or Remove VCF Operations and/or VCF Automation Components in an Existing VCF Single Sign-On Configuration

You can modify or remove existing VCF Operations and VCF Automation components from the identity broker in an existing VCF single sign-on configuration.

> **Note:**
>
> If you have configured VCF single sign-on for VCF Operations and/or VCF Automation, you can choose to change the identity broker for these components. The change deactivates the previous authentication source and the existing provisioned users and groups cannot be used in VCF Automation. You have to reassign the VCF Automation and/or VCF Operations roles to the newly provisioned users and groups from the new authentication source. This action cannot be reverted.

Modify or change VCF Single Sign-On for VCF Operations

Procedure

1. In the VCF Operations navigation bar at the top, click **Manage** and in the left navigation pane, click **Fleet Management** > **Identity & Access**.
2. In the **Identity & Access** pane, click **VCF SSO Overview**.
3. In the **VCF SSO Overview** page, from the data grid under the **VCF Management** tab, select **operations appliance** and then click the **horizontal ellipsis** > **Change SSO** option.
4. In the **Change VCF SSO** page, select a different identity broker and click **Change**.
5. From the pop-up box that is displayed, read about role assignments and click **Confirm**.

> **Note:**
>
> Changing VCF single sign-on might impact user access to the operations appliance. You must review role assignments and carry out relevant updates.

Modify or change VCF Single Sign-On for VCF Automation

Procedure

1. In the VCF Operations navigation bar at the top, click **Manage** and in the left navigation pane, click **Fleet Management** > **Identity & Access**.
2. In the **Identity & Access** pane, click **VCF SSO Overview**.
3. In the **VCF SSO Overview** page, from the data grid under the **VCF Management** tab, select **Automation appliance** and then click the **horizontal ellipsis** > **Change SSO** option.
4. In the **Change VCF SSO** page, select a different identity broker and click **Change**.
5. From the pop-up box that is displayed, read about role assignments and click **Confirm**.

> **Note:**
>
> Changing VCF single sign-on might impact user access to the Automation appliance. You must review role assignments and carry out relevant updates.

Remove VCF Operations and/or VCF Automation Components from an Existing VCF Single Sign-On Configuration

If you have configured VCF single sign-on for VCF Operations and/or VCF Automation, you can choose to remove these components from VCF single sign-on.

To remove VCF Operations and/or VCF Automation Components from an existing VCF single sign-on Configuration, perform the following steps:

Procedure

1. In the VCF Operations navigation bar at the top, click **Manage** and in the left navigation pane, click **Fleet Management** > **Identity & Access**.

2. In the **Identity & Access** pane, click **VCF SSO Overview**.
3. In the **VCF SSO Overview** page, from the data grid under the **VCF Management** tab, select **Automation appliance/ operations appliance**.
4. Click the **horizontal ellipsis** > **Unjoin SSO**.
5. Read and understand the information in the pop-up that is displayed and click **Unjoin**.

   Note:

   Removing the VCF Operations and/or VCF Automation components might impact user access to the VCF Operations and/or VCF Automationc omponents.

Add, Change, or Remove Other VCF Components

You can add, change, or delete other components such as VCF Operations for networks, VCF Operations HCX, and VCF Operations orchestrator from an existing VCF single sign-on configuration from VCF Operations.

Add Other VCF Components for Single Sign-On Access

You can add other components such as log management, VCF Operations for networks, VCF Operations HCX, and VCF Operations orchestrator, to an existing VCF single sign-on Configuration.

Procedure

1. In the VCF Operations navigation bar at the top, click **Manage** and in the left navigation pane, click **Fleet Management** > **Identity & Access**.
2. In the **Identity & Access** pane, click **VCF SSO Overview**.
3. In the **VCF SSO Overview** page, from the grid under the **Other Components** tab, select the component from the grid and click **Add Component**.
4. Use the values generated in the **Add Component** page in the respective component console to enable VCF single sign-on.

   Note:

   The configuration details are exclusive for that component.
5. Click **Save**.

   Note:

   Store the configuration details as this information cannot be retrieved at a later date.

   CAUTION:

   - VCF Automation tenant support is restricted to a single segment.
   - The configured user and group scale must not exceed the published identity broker configuration maximum. For more information, see identity broker maximum limits.
   - Configuring VCF single sign-on using other components is supported in enterprise environments only. Cloud service provider environments are not supported.
   - Configuring VCF single sign-on using other components is limited to the identity broker instance deployment mode only, and is not supported on identity broker embedded deployment mode.
   - Configuring VCF single sign-on using other components, is supported on VCF products only and is not available for non-VCF products.

Modify Other VCF Components in an Existing VCF Single Sign-On for

If you have configured VCF single sign-on for other components such as log management, VCF Operations for networks, VCF Operations HCX, and VCF Operations orchestrator, you can choose to modify the other VCF component in an existing VCF single sign-on Configuration.

Procedure

1. In the VCF Operations navigation bar at the top, click **Manage** and in the left navigation pane, click **Fleet Management** > **Identity & Access**.

2. In the **Identity & Access** pane, click **VCF SSO Overview**.
3. In the **VCF SSO Overview** page, from the grid under the **Other Components** tab, select the component that you want to change.
   The list of generic components for which you have configured VCF Single Sign-On is listed in the grid.
4. Click the **horizontal ellipsis** > **Change Component** option.
5. Modify the required values in the relevant component page.

   Note:

   The configuration details are exclusive for that component.
6. Click **Save**.

Remove VCF Single Sign-On for Other VCF Components

If you have configured VCF single sign-on for other components such as log management, VCF Operations for networks, VCF Operations HCX, and VCF Operations orchestrator, you can choose to remove other VCF components in an existing VCF Single Sign-On Configuration.

Procedure

1. In the VCF Operations navigation bar at the top, click **Manage** and in the left navigation pane, click **Fleet Management** > **Identity & Access**.
2. In the **Identity & Access** pane, click **VCF SSO Overview**.
3. In the **VCF SSO Overview** page, from the grid under the **Other Components** tab, select the component that you want to delete.
   The list of generic components for which you have configured VCF single sign-on is listed in the grid.
4. Click the **horizontal ellipsis** > **Delete Component** option.
5.

   Note:

   Depending on the component for which this OIDC client was created, you can choose to delete the VCF single sign-on configuration from the respective component console for complete removal. For example, if this OIDC client was used to configure VCF Single Sign-On for VCF Operations HCX, after the client is deleted under **Other Components**, log in to VCF Operations HCX as a local admin user and delete the VCF single sign-on configuration.

Assigning Roles and Permissions from Individual Components

As an administrator, after you complete configuring VCF single sign-on in VCF Operations, you can optionally log in to the individual components, such as, vCenter, NSX, VCF Automation, and the VCF Operations and assign the required roles and permissions for users or groups. Permissions assigned in each individual component apply only to the specific individual component.

- **Assign roles in vCenter:** To know more about assigning roles in vCenter, see Assigning Permissions.
- **Assign roles in NSX**: To know more about assigning roles in NSX, see Manage Local User Accounts .
- **Assign roles in VCF Operations:** To know more about assigning roles in VCF Operations, see Assigning Roles in VCF Operations.
- **Assign roles in VCF Automation**: To know more about assigning roles in VCF Automation, see Assigning Roles in VCF Automation.

  Note:

  - From VCF 9.1, when you configure VCF single sign-on in VCF Operations, you can assign VCF roles to provisioned users and groups from VCF Operations for components such as vCenter, VCF Operations, VCF Automation, log management, VCF Operations for networks, VCF Operations orchestrator, VCF Operations HCX, and NSX. For more information, see Step 4. (Optional) Assign a VCF Role.

Assigning Roles in VCF Operations

After you complete configuring VCF single sign-on in the VCF Operations console, log in to VCF Operations as a local admin to assign the required roles and permissions for users or groups.

Assign roles in VCF Operations if configured for VCF single sign-on. Roles have to be assigned to the provisioned users/groups to be able to log in with the configured identity provider to achieve single sign-on between the VCF components. Without these roles, users will not be able to access the respective component consoles with their enterprise credentials.

> **Note:**
>
> If you had chosen JIT as the user/group provisioning method without specifying any groups, then attempt login from the the Active Directory domain that was configured while configuring the VCF single sign-on. This is essential before you can assign the roles.

Procedure

1. Log in to VCF Operations as a local admin.
2. In the navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Control Panel**, and then click the **Access Control** tile.
3. From the **Access Control** page, navigate to the **User Accounts and/or User Groups** tabs and click the **horizontal ellipsis** next to the **Add** button, and select **Import from Source**.
4. From the **Import from User Groups and/or Import from Users** page, select **VCF SSO** from the drop-down options against the **Import From** field.
5. Use the **Search Prefix** field to search for users and/or groups and click **Finish**.
6. After the user and/or group has been added, from the **Access Control** page, select the **User and/or Group**, and click the **Vertical ellipsis** and click **Edit**.
7. From the section called **Assign Roles and Scope**, assign the required roles and scope to the use and/or groups and click **Save**.

Assigning Roles in VCF Automation

After you complete configuring VCF single sign-on in VCF Operations, log in to the VCF Automation Provider Management Portal to assign the required roles and permissions for users or groups.

Assign roles in the VCF Automation Provider Management Portal, if configured for VCF single sign-on. Roles have to be assigned to the provisioned users/groups to be able to log in with the configured identity provider to achieve single sign-on between VCF components. Without these roles, users will not be able to access the respective component consoles with their enterprise credentials.

> **Note:**
>
> If you had chosen JIT as the user/group provisioning method without specifying any groups, then attempt login from the the Active Directory domain that was configured while configuring the VCF single sign-on for the user and groups to be dynamically provisioned when you log in. This is essential before you can assign the service roles.

Procedure

1. Log in to the VCF Automation Provider Management Portal with the **system** organization and **admin** account.
2. From the left menu, click **Access Control**.
3. From the **Users** tab, click the **Import Users** option.
4. From the **Source** drop-down options, select **VCF SSO**.
5. Search for and select the required users. Assign the required roles from the **Assign Role** option.

   Repeat the steps to assign roles for groups.

Search for Users and Groups Dynamically and Assign Roles using On-Demand Lookup

In versions 9.1.1 and later, if Active Directory is configured as an identity provider using AD/LDAP with on-demand lookup, you can search its groups and users dynamically for role provisioning.

Procedure

1. In the VCF Operations navigation bar at the top, click **Manage** and in the left navigation pane, click **Fleet Management** > **Identity and Access**.
2. In the **Identity & Access** pane, click **VCF SSO Overview**.
3. In the **VCF SSO Overview** page, under the section that is called **VCF SSO**, click the relevant identity broker.
4. Click the **Access Management** tab, and in the **Groups** /**Users** section, click **On-Demand Search**.

   Note:

   The **On-Demand Search** button is visible if you have selected on-demand lookup while configuring Active Directory as an identity provider using AD/LDAP. For more information, see Configure Active Directory as an Identity Provider Using AD/LDAP.
5. In the **On-Demand Search** page, you can search for groups/users dynamically from active directories that are configured for on-demand lookup for provisioning. Select and complete the following options.

Option	Description
Choose Directory	Select the source directory for the group/user search from the drop-down list. **Note:** Only on-demand directories are displayed. The first directory in the list is pre-selected.
Add Groups/Add Users	Select the groups/users from the drop-down list. **Note:** Groups/Users with role assignments are not available for selection. You can edit the assigned roles from the **Access Management** tab.
(Optional) Assign Roles	**Scope** • All components. Components with the chosen identity broker. • Individual VCF Instances. You can scope the role assignment to one or more VCF Instances. Scoping is applicable to vCenter and NSX components that are part of the VCF Instance.
	Role Select a role from the drop-down list. If you selected VCF Instance as the scope, only roles that contain vCenter or NSX component roles are listed.
	Valid Until • Never Expires. Select this option if you do not want to set an expiry date for the role. • Custom Date (UTC). Enter a date until when you would like the role to be valid.

6. Click **Finish**. The on-demand search completes with the following possible statuses:
   - **Partially successful**. On-demand search has failed for some of the groups/users.
   - **Failed**. On-demand search has failed for all the groups/users.

For partially successful and failed statuses, in the **Access management** tab, in the banner at the top, click **View Details**. You see the groups/users that have failed provisioning and the associated error message from the **Search Groups and Assign Roles** or **Search Users and Assign Roles** dialog box.

- **Successful**. On-demand search is successful for all the groups/users. In the **Groups**/**Users** section, click the double-arrow against a group/user name to see the roles that are assigned to the group/user.

Note:

You can remove a group/user by selecting the group/user you want to remove from the **Access Management** tab, and then click the horizontal ellipsis and select **Remove**. The group/user is permanently deleted from on-demand lookup and all associated roles are revoked.

Review Directory Details to Verify if On-demand Lookup is Enabled

In versions 9.1.1 and later, if Active Directory is configured as an identity provider using AD/LDAP, you can review the directory details to find out if on-demand lookup is enabled.

Procedure

1. In the VCF Operations navigation bar at the top, click **Manage** and in the left navigation pane, click **Fleet Management** > **Identity and Access**.
2. In the **Identity & Access** pane, click **VCF SSO Overview**.
3. In the **VCF SSO Overview** page, under the section that is called **VCF SSO**, click the relevant identity broker.
4. In the **Identity Source** tab, under the section that is called **Directory Information**, click the relevant Directory link.
5. In the **Directory** page, under the **Details** section, you can view if the **On-demand lookup** option is enabled.

Add a Directory for Identity Providers that are Configured using AD/LDAP

In versions 9.1.1 and later, you can add an additional directory for identity providers that are configured using AD/LDAP.

Procedure:

1. In the VCF Operations navigation bar at the top, click **Manage** and in the left navigation pane, click **Fleet Management** > **Identity and Access**.
2. In the **Identity & Access** pane, click **VCF SSO Overview**.
3. In the **VCF SSO Overview** page, under the section that is called **VCF SSO**, click the relevant identity broker.
4. In the **Identity Source** tab, under the section that is called **Directory Information**, click **Add**. Enter the following information and click **Next**.

Option	Description
Directory name	Enter a name for the directory.
DNS Server Location	Select this option if you want the identity broker to find and use optimal domain controllers by querying the DNS Service Record.

Global Catalog (Inactive when the **On-demand look-up** option is displayed)	Global catalog has a partial read-only replica of all objects across all domains. Some of the limitations with selecting the global catalog option include: • The Active Directory object attributes that are replicated to the global catalog are identified in the Active Directory schema as the Partial Attribute Set (PAS). Only these attributes are available for attribute mapping by the service. If necessary, edit the schema to add or remove attributes that are stored in the global catalog. • The global catalog stores the group membership (the member attributes) of only universal groups. Only universal groups are synced to the service. If necessary, change the scope of a group from a local domain or from global to universal. • The bind DN account that you define when you configure a directory in the service must have permissions to read the Token-Groups-Global-And-Universal (TGGAU) attribute. **Note:** When you use Global catalog with LDAP and multiple domains, the additional domains are discovered only after the user/group sync is complete.
Encrypted Connection (Displayed when you select the **DNS server location** option)	Activates LDAPs for all the connections to this directory.
Certificate(s) for encrypted connection (Displayed when you select the **DNS server location** and **Encrypted Connections** options and if you enter a *ldaps://* string in the Primary domain controller field)	Upload the root certificate(s) required to access the domain controllers over SSL. All uploaded root certificates are merged into a single file with the filename `{directory-name}-concatenated.pem` when you save the configuration. **Note:** Ensure that the certificate is in the (Privacy Enhanced Mail) PEM format and includes the "BEGIN CERTIFICATE" and "END CERTIFICATE" lines.
Primary domain controller (Removed from display when you select DNS Server Location)	A server machine running Active Directory Domain Services (AD DS) responsible for managing and authenticating user access, enforcing policies, and storing directory data. For example, `hostname = DC01` for the domain called example.com.
Certificate for primary domain controller	The root certificate required to access the primary domain controller over SSL. **Note:** Ensure that the certificate is in PEM format and includes the "BEGIN CERTIFICATE" and "END CERTIFICATE" lines. This field is displayed if you type 'LDAP' in the **Primary domain controller** field.
Secondary domain controller (Removed from display when you select DNS Server Location)	A backup domain controller for the identity broker to point to if the primary domain controller is not reachable.
Certificate for secondary domain controller	The root certificate required to access secondary domain controller over SSL. **Note:** Ensure the certificate is in PEM format and include the "BEGIN CERTIFICATE" and "END CERTIFICATE" lines. This field is displayed if you type 'LDAP' in the **Secondary domain controller** field.

Directory search attribute	A specific LDAP attribute used in queries to search or identify entries in the directory. For example, `uid`, `sAMAccountName`.
Base DN	The Distinguished Name (DN) of the top-level entry in the directory tree from which LDAP queries begin. For example, `dc=example`, `dc=com`.
Bind user name	The user name of a user account with sufficient permissions to authenticate and perform LDAP operations on the directory. **Note:** Use a bind user account with sufficient Active Directory privileges so that VCF SSO can properly enumerate and retrieve all required AD users during configuration.
Bind user password	The password of a user account with sufficient permissions to authenticate and perform LDAP operations on the directory.
On-demand lookup	In versions 9.1.1 and later, use on-demand lookup for user and group provisioning. After successful authentication, user accounts and group memberships are queried dynamically from Active Directory, ensuring access decisions reflect current directory state.

5. In the **Directory Information** page, review the information and click **Finish**.
6. In the **User/Group Provisioning** dialog box, you can choose to provision users and groups for the directory that has been added.
   a) Click **Later** if you do not want to configure user and group provisioning immediately.
   b) Click **Configure Now** to configure users/groups provisioning. See **Step 6** onwards in Configure Active Directory as an Identity Provider Using AD/LDAP for more instructions.

Managing VCF Roles

VCF roles have been introduced to provide simplified access management for VCF users across VCF components such as vCenter, NSX, VCF Operations, VCF Automation, VCF Operations HCX, and VCF Operations orchestrator. As an admin you can define user access using VCF roles to manage access across the VCF fleet from VCF Operations, instead of assigning the component roles in each individual component. VCF roles are mapped to the individual VCF component roles. The two kinds of VCF roles are built-in VCF roles and custom VCF roles. You can continue to assign the roles/permissions for individual components. VCF roles do not aim to replace the individual components roles or their Authorization model.

Note:

If a user has VCF roles and component roles assigned, then the overall access is a combination of both, and not the replacement of one over the other.

Note:

To use VCF roles, the vCenter instance creates local groups and new global permissions to support the intended VCF role.

CAUTION:

Do not modify the new global permissions that have been added and which are for principals whose names start with `vcf_reserved_`. Modifying global permissions could break your system as it causes VCF authorization issues.

Benefits of VCF Roles:

As an administrator, you can:

- Visualize VCF roles and the corresponding VCF component roles.
- Assign VCF roles to users and groups that are provisioned from the identity provider.
- Assign VCF roles to API clients.

- Specify expiry date/time-to-live during VCF roles assignments.

Built-In VCF Roles

You can view built-in VCF roles from VCF Operations. For information about role assignment, see Manage VCF Role Assignments After Configuring VCF Single Sign-On.

Table 1136: Built-In VCF Roles and Their Mapping

Built-in or predefined VCF roles are shipped with the product and with component role mapping. Built-in VCF roles cannot be modified.

Role Name	vCenter	NSX	VCF Operations	VCF Automation	VCF Operations HCX	VCF Operations orchestrator
VCF Administrator	Admin*	enterprise_admin	Administrator	System Administrator	Migration Admin	Orchestrator Administrator
VCF Viewer	ReadOnly	auditor	ReadOnly	-	-	
SDDC Administrator	Admin	enterprise_admin	-	-	Migration Appliance Admin	Orchestrator Viewer
SDDC Viewer	ReadOnly	auditor	-	-	-	

Note:

- * This role honors the existing vCenter Administrator role. To grant vSphere Supervisor Management permissions, the assigned principal must be manually added to local vCenter groups (such as Administrators, SupervisorOperationsAdministrator).
- * From vCenter 9.1.1 onwards, the VCF Administrator role, and any custom VCF role containing the vCenter Administrator role, automatically maps to the local vCenter Administrators group. For vCenter instances earlier than 9.1.1, the behavior stays the same as explained in the preceding point, even if VCF Operations is upgraded to 9.1.1.

Custom VCF Roles

You can create additional VCF roles called custom VCF roles. You can create custom VCF roles from VCF Operations using components such as vCenter, VCF Operations, VCF Automation, VCF Operations orchestrator, VCF Operations HCX, and NSX.

For information about creating custom roles and role assignments, see Create a Custom VCF Role and Assign Custom and Built-In VCF Roles.

Create a Custom VCF Role and Assign Custom and Built-In VCF Roles

As an administrator, you can create custom VCF roles from VCF Operations using components such as vCenter, VCF Operations, VCF Automation, VCF Operations orchestrator, VCF Operations HCX, and NSX.

VCF components that are at 9.1 and later versions support VCF role assignments. For more information about built-in VCF roles, see Built-In VCF Roles.

For VCF role assignments recommendations, see VCF Role Assignment.

Note:

- You can also assign component level roles from the associated component. For more information, see Assigning Roles and Permissions from Individual Components.
- If a user has VCF roles and component roles assigned, then the overall access is a combination of both, and not the replacement of one over the other.

- When a user or group has object-level permissions that are limited to a specific inventory subset, assigning them a VCF role overrides that restriction, granting visibility across all objects. Do not assign a VCF role to a user/group when inventory-level permissions are required.

Create a Custom VCF Role and Assign Custom and Built-In VCF Roles in VCF Operations After Configuring VCF Single Sign-On

Procedure

Create a Custom VCF Role

1. In the VCF Operations navigation bar at the top, click **Manage** and in the left navigation pane, click **Fleet Management** > **Identity and Access**.
2. In the **Identity & Access** pane, click **VCF Roles**.
3. In the **VCF Roles** page, click **Create New Custom Role**.
4. In the **Name Role** section, add the following details, and click **Next**.

Option Name	Description
Role Name	Add a name for the custom role.
Role ID	Automatically populated based on the role name and cannot be edited.
Description	Optionally, add a description for the role.

5. In the **Choose Role** section, based on your requirements, select the roles from the categories listed such as, vCenter, NSX, and so on. For example, you can choose the Administrator role for full access rights to vCenter and VCF Operations.
6. Click **Next** and review the details of the roles you have selected in the **Summary** page.
7. Click **Create**. The role that you created is listed under **Custom Roles**.

(Optional) Assign a Custom VCF Role

8. In the **Identity & Access** pane, click **VCF SSO Overview**.
9. In the **VCF SSO Overview** page, under the section that is called **VCF SSO**, click the relevant identity broker.
10. Click the **Access Management** tab, you see a list of users and groups that are provisioned.

    Note:

    In versions 9.1.1 and later, the column that is called **Provisioning Method** in the users and groups data grid is visible if Active Directory is configured as an identity provider using AD/LDAP with on-demand lookup. The **Provisioning Method** column indicates whether users and groups are pre-provisioned or configured through on-demand lookup.

11. Select the group and/or user that you would like to assign the role to, and click **Assign**.

    Note:

    You are not allowed to assign a role if the user or group has role assignments.

12. In the **Assign VCF Roles** page, under the section called **Roles**, you can scope the role by selecting one of the following options as the **Scope**, then select a Role from the drop-down down list, and enter the validity of the role, and then click **Apply**.

Option	Description
Components configured within the identity broker's display name (FQDN)	Covers all the components that are configured with the relevant identity broker .
Individual VCF Instances	You can scope the role to one or more VCF Instances.

Note:

You can add more than one role.

13. Click **Assign**.

Note:

Updates to the role assignment take place in the next authentication cycle.

Edit and Delete Custom VCF Roles

You can make changes, delete, and assign custom VCF roles and role assignments while configuring VCF single sign-on.

Edit a Custom VCF Role

Procedure

1. In the VCF Operations navigation bar at the top, click **Manage** and in the left navigation pane, click **Fleet Management** > **Identity and Access**.
2. In the **Identity & Access** pane, click **VCF Roles**.
3. In the **VCF Roles** page, from the **Custom Roles** section, click the **vertcial ellipsis** > **Edit** option against the custom role you want to edit.
4. Edit the name and/or roles and click **Next**.
5. Review the details of the role that you edited in the **Summary** page and click **Save**.

Delete a Custom VCF Role

You can delete a custom VCF role in the following scenarios:

- **Scenario #1:** The custom VCF role has not been assigned to a user/group.
- **Scenario #2:** The custom VCF role has been assigned to at least one user/group. Depending on the situation, you can:
  - Scenario #2a: Delete role assignments in VCF Single Sign-On. For more information, see Scenario #2a.
  - Scenario #2b: Bulk delete all role assignments. For more information, see Scenario #2b.

Scenario #1

Delete a custom VCF role that has not been assigned to a user/group

Procedure

1. In the VCF Operations navigation bar at the top, click **Manage** and in the left navigation pane, click **Fleet Management** > **Identity and Access**.
2. In the **Identity & Access** pane, click **VCF Roles**.
   Delete the custom VCF role
3. In the **VCF Roles** page, from the **Custom Roles** section, click the **vertical ellipsis** > **Delete** option against the custom VCF role you want to delete.

Note:

Deleting a custom VCF role cannot be reversed.

Scenario #2

Delete a custom VCF role that has been assigned to at least one user or group

Scenario #2a

You can delete a custom VCF role that has been assigned to at least one user or group.

Procedure

1. In the VCF Operations navigation bar at the top, click **Manage** and in the left navigation pane, click **Fleet Management** > **Identity and Access**.

2. In the **Identity & Access** pane, click **VCF Roles**.

Delete the custom VCF role assignment

3. In the **VCF Roles** page, from the **Custom Roles** section, click the double arrow against custom role you want to delete. The role components and role assignments are displayed for the selected VCF role.
4. In the **Role Assignments** section, click the role assignment that you would like to delete. The groups and users who are assigned to that role are displayed in the **Access Management** tab.
5. Select the groups and users who are associated with the custom VCF role that you want to delete. Click the **horizontal ellipsis** next to the **Assign** button and click **Delete**.

Scenario #2b

You can delete a custom VCF role that has been assigned to at least one user or group by deleting all the role assignments in bulk.

Procedure

1. In the VCF Operations navigation bar at the top, click **Manage** and in the left navigation pane, click **Fleet Management** > **Identity and Access**.
2. In the **Identity & Access** pane, click **VCF Roles**.
3. In the **VCF Roles** page, from the **Custom Role** section, click the double arrow against custom role you want to delete. The role components and role assignments are displayed for the selected VCF role.
4. In the **Role Assignments** section, click **Delete All Role Assignments**.
5. Click **Delete** in the confirmation message.

Provisioning vCenter Custom Roles

You can choose to provision a vCenter custom role to all the vCenter instances in a VCF Instance. Later, when new vCenter instances are added to the VCF Instance, the roles are assigned to the new vCenter instance thus streamlining user access and management. You do not have to log in to the relevant vCenter and assign the required role. This automatic provisioning also applies to single sign-on (SSO) configurations.

Provision vCenter Custom Roles to vCenter Instances and Remediate Drifts

Procedure

Provision vCenter Custom Roles

1. In the VCF Operations navigation bar at the top, click **Manage** and in the left navigation pane, click **Fleet Management** > **Identity and Access**.
2. In the **Identity & Access** pane, click **vCenter Custom Roles**.
3. In the section that is called **Provisions for vCenter Roles,** click **Import & Provision Roles**.
4. In the **Select Source and Role** section, select a source VCF Instance, vCenter , and the role you want to provision.

   Note:

   Only vCenter instances that are created in VCF 9.1 and later are supported for custom roles and displayed.
5. Click **Next**.
6. In the **Select Target** section, select the VCF Instance/s to which you want to assign the custom role.

   Note:

   - All the vCenter instances that have been created in VCF 9.1 and later in the selected VCF Instance receive the provisioned vCenter custom role.
   - All the VCF Instances are listed as targets. You cannot provision a vCenter custom role to a VCF Instance if a vCenter custom role was previously provisioned to the VCF Instance. The VCF Instance is unavailable for selection.
7. Click **Next**.

   A banner is displayed stating that if there are conflicts, the source role overrides the existing vCenter custom roles in the vCenter instances.

8. Click **Confirm Role Provision**.

   If provisioning fails, you see a banner stating that role provisioning has failed or failed partially.

   Remediate Drifts for vCenter Custom Roles
9. Click the vCenter custom role that you provisioned. You see the vCenter custom role details, such as, the status which can be Failed, Provisioned, Drifted, and so on, the provisioned vCenter, and the target VCF Instance.

   For vCenter custom roles that have failed provisioning click **Retry** in the grid.
10. Click **Remediate Drifts** from the banner above the grid.
11. In the **Remediate Drifts** page, the current master role definition under VCF Operations is displayed along with the vCenter custom role/s with a conflict. Select the role that you want to make as master.
    - You can continue to override the drift with the master role definition or select one of the role/s that has a conflict and make it the master.
    - After you select the role you want to make as master, the role is assigned to all vCenter instances in the provisioned VCF Instance/s. The existing role definitions are overwritten.

After you complete provisioning a vCenter custom role to other vCenter instances, you can:

- Manually Remediate Drift. For more information, see Remediate Drifts for vCenter Custom Roles by Manually Running a Check.
- If there are vCenter custom roles assigned to the VCF Instance you have provisioned, you can de-provision the role. For more information, see De-provision a vCenter Custom Role from VCF Instances.
- Add more VCF Instances to the provision. For more information, see Provision a vCenter Custom Role to Additional VCF Instances.
- Delete a VCF custom role if it is part of a role definition. For more information, see Delete a vCenter Custom Role.
- For information about role provisioning scope, see Managing vCenter Role Consistency.

Remediate Drifts for vCenter Custom Roles by Manually Running a Check

At a later date, you can manually remediate drifts for vCenter custom roles that have failed provisioning.

For more information about vCenter custom role drift management, see Managing vCenter Role Consistency.

Prerequisites

- Verify that you have provisioned vCenter custom roles in VCF 9.1. For more information, see Provisioning vCenter Custom Roles.

Procedure

1. In the VCF Operations navigation bar at the top, click **Manage** and in the left navigation pane, click **Fleet Management** > **Identity and Access**.
2. In the **Identity & Access** pane, click **vCenter Custom Roles**.
3. Click a vCenter custom role that you provisioned. You see the vCenter custom role details, such as, the status which can be Failed, Provisioned, Drifted, and so on, the provisioned vCenter, and the target VCF Instance.
4. If drifts are detected, click **Remediate Drifts** from the banner above the grid.
5. In the **Remediate Drifts** page, the current master role definition under VCF Operations is displayed along with the role/s that have a conflict. Select the role that you want to make as master.
   - You can continue to override the drift with the master role definition or select one of the role/s that has a conflict and make it the master.
   - After you select the role you want to make as master, the role is assigned to all vCenter instances in the provisioned VCF Instance/s. The existing role definitions are overwritten.

De-provision a vCenter Custom Role from VCF Instances

You can de-provision a vCenter custom role from multiple VCF Instances. When you de-provision a vCenter custom role, drifts are not monitored and role definitions are not updated.

Prerequisites

- Verify that you have provisioned vCenter custom roles in VCF 9.1. For more information, see Provisioning vCenter Custom Roles.

Procedure

1. In the VCF Operations navigation bar at the top, click **Manage** and in the left navigation pane, click **Fleet Management** > **Identity and Access**.
2. In the **Identity & Access** pane, click **vCenter Custom Roles**.
3. In the **vCenter Custom Roles** page, click the vCenter custom role you that you want to de-provision from VCF Instances.
4. Click **Actions** > **De-provision from VCF Instances** from the top-right corner of the page.
5. Select the VCF Instances that you want to de-provision and click **Confirm**.

Provision a vCenter Custom Role to Additional VCF Instances

You can provision a vCenter custom role to additional VCF Instances.

Prerequisites

- Verify that you have provisioned a vCenter custom role in VCF 9.1. For more information, see Provisioning vCenter Custom Roles.

Procedure

1. In the VCF Operations navigation bar at the top, click **Manage** and in the left navigation pane, click **Fleet Management** > **Identity and Access**.
2. In the **Identity & Access** pane, click **vCenter Custom Roles**.
3. In the **vCenter Custom Roles** page, click the vCenter custom role you that you want to provision to additional VCF Instances.
4. Click **Actions** > **Provision to More VCF Instances** from the top-right corner of the page.

   You cannot provision the vCenter custom role if the role is either already provisioned to all available VCF Instances or if eligible VCF Instances do not exist. VCF Instances are ineligible because of an invalid license, VCF version is earlier than 9.0, or the VCF Instance is running in ELM mode.
5. In the **Select Target** section, select the VCF Instances to which you want to assign the vCenter custom role.
   - All the vCenter instances that have been created in VCF 9.1 and later in the selected VCF Instance receive the provisioned vCenter custom role.
6. Click **Next**. You see details of the vCenter custom role and the VCF Instances that the role has been provisioned to.

Delete a vCenter Custom Role

If you created a vCenter custom role, provisioned it to other vCenter instances, and then deleted the provisioned vCenter custom role from VCF Operations, there is no impact to the provisioned roles. You cannot use the role in the future.

Prerequisites

- Verify that you have provisioned vCenter custom roles in VCF 9.1. For more information, see Provisioning vCenter Custom Roles.

Procedure

1. In the VCF Operations navigation bar at the top, click **Manage** and in the left navigation pane, click **Fleet Management** > **Identity and Access**.
2. In the **Identity & Access** pane, click **vCenter Custom Roles**.
3. In the **vCenter Custom Roles** page, click the vCenter custom role you that you want to delete.
4. Click **Actions** > **Delete Role**.
5. Click **Delete** in the message dialog that is displayed.

   Note:

   The action cannot be reversed.

Managing API Clients and Tokens

VCF single sign-on API tokens can be generated either by a single sign-on user or under an API client, with lifespans ranging from 30 to 180 days.

Within a VCF fleet, these two token types serve distinct purposes:

- **API client tokens:** Highly recommended and utilized for system-to-system integrations (such as automation pipelines, backups, or monitoring setups). VCF single sign-on provides a token regeneration API to automate the lifecycle of API client tokens, ensuring uninterrupted system integration and preventing production downtime.
- **Single sign-on user tokens:** Linked directly to individual single sign-on user accounts for manual, on-demand operations. Since these tokens are not used for automated system integrations, their expiration does not affect background infrastructure operations. When a token expires, single sign-on users can generate a new token through the VCF component user interface.

API Client Tokens

As a VCF Operations admin user, you can create an API client and can generate an API token to make API calls on behalf of an authenticated user without requiring repeated interactive login. You can also view and manage the token lifecycle and security.

- To create an API client and generate an API token, see Create an API Client and Generate an API Token.
- To edit, delete, and regenerate an API token, see Edit, Delete, and Regenerate an API Token.
- To create an emergency API client, see Create an Emergency Access Client.
- To view and manage the token lifecycle and security, see Identity and Access Management Settings.

Single sign-on User Tokens

- To generate a single sign-on user token, see Generate a Single Sign-On User Token.

Create an API Client and Generate an API Token

Starting with VCF 9.1, you can log in to VCF Operations and create an API client and generate an API token.

Create an API Client and Assign User Roles

Procedure

1. In the VCF Operations navigation bar at the top, click **Manage** and in the left navigation pane, click **Fleet Management** > **Identity & Access**.
2. In the **Identity & Access** pane, click **VCF SSO Overview**.
3. In the **VCF SSO Overview** page, in the section that is called **VCF SSO**, click the identity broker that you have configured for VCF Single Sign-On.
4. Click the **API Access** tab and then click the **API Clients** tab.
5. In the **API Clients** tab, click **Create**.
6. Enter a name for the client. The client ID is autopopulated.
7. Optionally enter a description for the API client.
8. Click **Create API Client**.

9. In the **Roles** section, scope the role by selecting one of the options as the **Scope**, select a role from the drop-down options, and enter the validity of the role.
10. Click **Save**.

Note:

Use the drop-down options from the vertical ellipsis to edit and delete API clients.

Generate an API Token

Procedure

1. In the VCF Operations navigation bar at the top, click **Manage** and in the left navigation pane, click **Fleet Management** > **Identity & Access**.
2. In the **Identity & Access** pane, click **VCF SSO Overview** and then from the page on the right, click the **API Access** tab.
3. In the **API Clients** tab, against the API client for which you want to generate an API token, click the **vertical ellipsis** > **Generate API Token.**
4. In the **Generate API Token** page, add the following information:

Option	Description
API Token Name	Enter a name for the token.
API Token TTL	Select a valid time-to-live period for the API token. API Token TTL refers to the duration for which an API token remains valid and usable before it expires. The default is 30 days. **Note:** The maximum value is limited by the value that you set as the **API Token Expiry** under **Fleet Settings > IAM Setting**. For more information, see Identity and Access Management Settings.
Access Token TTL	Select a valid time-to-live for the access token. Access Token TTL is the maximum period for which the token is valid, after which it expires and can no longer be used to authenticate or authorize requests. The default acess token TTL is 30 mins. **Note:** The maximum value is limited by the value that you set as the **Access Token Expiry** under **Fleet Settings > IAM Setting**. For more information, see Identity and Access Management Settings.
Description	Enter a meaningful description.

5. Click **Generate API Token**. The new API token is generated.
6. Click **Copy** against the token and then click **Continue**.

Note:

If you are not going to use the API token immediately, store the API token. After you click **Continue**, you cannot retrieve the API token that was generated.

Note:

Use the API client to automate API token generation instead of generating it manually.

If an API token is generated by a human user, there is a possibility that the API token might remain valid after the JIT user is deleted from the identity provider, remaining valid until the API tokens time-to-live (TTL) expires. As a VCF Operations admin, you can manage the API tokens of JIT users by navigating to the **API Access** tab, and then selecting **API Tokens**. You can then filter the list by setting **Token Type** as **User** and **Principal status** as **INACTIVE**.

For more information about setting the retention period for an expired API token and to view and manage API and access token lifecycle and security, see Identity and Access Management Settings.

Edit, Delete, and Regenerate an API Token

You can choose to edit, delete, or regenerate an API token.

Edit an API Token

Procedure

1. In the VCF Operations navigation bar at the top, click **Manage** and in the left navigation pane, click **Fleet Management** > **Identity & Access**.
2. In the **Identity & Access** pane, click **VCF SSO Overview**.
3. In the **VCF SSO Overview** page, in the section that is called **VCF SSO**, click the identity broker that you have configured for VCF Single Sign-On.
4. Click the **API Access** tab and then click the **API Clients** tab.
5. In the **API Clients** tab, click on the name of an API client, find the token in the resulting dialog box that is displayed, and then select the **vertical ellipsis** > **Edit**.

Delete an API Token

Procedure

1. In the VCF Operations navigation bar at the top, click **Manage** and in the left navigation pane, click **Fleet Management** > **Identity & Access**.
2. In the **Identity & Access** pane, click **VCF SSO Overview**.
3. In the **VCF SSO Overview** page, in the section that is called **VCF SSO**, click the identity broker that you have configured for VCF Single Sign-On.
4. Click the **API Access** tab.
5. To delete a single API token, click the **API Token** tab, and against the relevant API token, select the **vertical ellipsis** > **Delete** option.

   Note:

   To delete multiple API tokens, in the **API Token** tab, select the relevant API tokens, and click **Delete**.

Regenerate an API Token

Procedure

1. In the VCF Operations navigation bar at the top, click **Manage** and in the left navigation pane, click **Fleet Management** > **Identity & Access**.
2. In the **Identity & Access** pane, click **VCF SSO Overview**.
3. In the **VCF SSO Overview** page, in the section that is called **VCF SSO**, click the identity broker that you have configured for VCF Single Sign-On.
4. Click the **API Access** tab and then click the **API Clients** tab.
5. In the **API Clients** tab, click on the name of an API client, find the token in the resulting dialog box that is displayed, and then select the **vertical ellipsis** > **Regenerate API Token**.
6. Click **Confirm Regenerate**. The API token is regenerated.
   Click **Copy** against the token and then click **Continue**.

   Note:

   If you are not going to use the API token immediately, store the API token. After you click **Continue**, you cannot retrieve the API token that was generated.

Create an Emergency Access Client

You can create emergency API clients for break-glass scenarios that provide high-privilege and long-lived access tokens to critical systems when standard methods fail.

Procedure

1. In the VCF Operations navigation bar at the top, click **Manage** and in the left navigation pane, click **Fleet Management** > **Identity & Access**.
2. In the **Identity & Access** pane, click **VCF SSO Overview**.
3. In the **VCF SSO Overview** page, in the section that is called **VCF SSO**, click the identity broker that you have configured for VCF Single Sign-On.
4. Click the **API Access** tab.
5. Click the **Emergency Access** tab, click **Create**.
6. Enter a name for the client. The client ID is autopopulated.
7. Optionally enter a description for the API client.
8. Click **Create Emergency Client**. The Emergency Access Token is created.

   Note:

   Emergency access clients are assigned the highest role with the largest scope. The role cannot be edited.
9. Click **Copy** against the token and then click **Continue**.

   Note:

   - The default TTL for an emergency access token is 180 days.
   - You can create only one emergency access token at a time.
   - Use the drop-down options from the vertical ellipsis to regenerate and delete the emergency access token.

Generate a Single Sign-On User Token

You can generate a single sign-on user token from component UIs. The generated API token will inherit the permission of the single sign-on user.

Generate a Single Sign-On User Token from VCF Operations

Procedure

1. Log in to VCF Operations as a single sign-on user.
2. In the **Home** page, navigate to the top-right corner of the screen and click the Profile icon, and click **Generate API token** from the drop-down list.
3. In the **Generate API Token** page, from the **Identity Broker** UI, add the following information to configure token information.

Option	Description
API Token Name	Enter a name for the token.
API Token TTL	Select a valid time-to-live period for the API token. API Token TTL refers to the duration for which an API token remains valid and usable before it expires. The default is 30 days.
Access Token TTL	Select a valid time-to-live for the access token. Access Token TTL is the maximum period for which the token is valid, after which it expires and can no longer be used to authenticate or authorize requests. The default access token TTL is 30 mins.

4. Click **Generate API Token**.

Similarly, you can create a single sign-on user token from component UIs such as vCenter, VCF Automation, VCF Operations for networks, VCF Operations orchestrator, VCF Operations HCX, and NSX. For comprehensive information about token management, see VCF API Token Management.

Identity and Access Management Settings

You can view and set the maximum time-to-live for API and access tokens. You can also set the retention period to retain an expired API token before it is deleted permanently.

Prerequisites

- Verify that VCF Single Sign-On is configured.

Procedure

1. In the VCF Operations navigation bar at the top, click **Manage** and in the left navigation pane, click **Fleet Management** > **Fleet Settings**.
2. From the **Fleet Settings** page, click **IAM Settings**.
3. To make changes, click **Edit** at the top of the page and configure the following settings using the drop-down options.

Option	Description
API Token Expiry	Set the maximum time-to-live for API tokens. The maximum that is allowed is 180 days. Minimum is 30 days.
Access Token Expiry	Set the maximum time-to-live for access tokens. The maximum that is allowed is 480 minutes. Minimum is 30 minutes.
Expired API Token Retention Period	Set the retention period to retain an expired API token before it is deleted permanently. The maximum that is allowed is 90 days.
Groups and Users > ALL_USERS Group	Activate if you want to include the ALL_USERS group for role assignments. **Note:** • The ALL_USERS group includes only the users available within the identity broker. • The activate/deactivate action applies to all configured identity brokers. • When the setting is activated, it activates the ALL_USERS group on a configured identity broker automatically. • The ALL_USERS group is isolated per identity broker and contains only users who are known to that specific identity broker. • The ALL_USERS group does not aggregate users across multiple identity brokers and cannot function as a global cross identity broker group.
Identity Provider Policies > JIT User Inactivity Period	Set the inactivity period for JIT users. If JIT users do not sign in during this period, they become inactive and previously issued API tokens cannot be used to get access tokens until the user logs in again.

4. Click **Save**.

Migrating VMware Identity Manager to Identity Broker

In VCF 9.1, you can migrate from VMware Identity Manager to identity broker. Read and understand the scope, prerequisites, limitations, and steps before you attempt migration.

Scope of Migration

- **Users and Groups**: Users and groups are migrated from VMware Identity Manager to identity broker.
- **Users and Groups Sync Settings:** The sync settings for existing identity providers in VMware Identity Manager and identity broker are compared and displayed, but not migrated. You can view the comparison and can make changes to the sync settings in the identity broker if necessary.
- **Component Updates**: If VCF Operations, VCF Automation Provider, or NSX currently use VMware Identity Manager for single sign-on, their identity source type is updated to point to the new identity broker. The identity broker must be fully set up before this update. The migration script will not remove the VMware Identity Manager identity provider from the components.
- **Data Formats**: Users and groups can be exported in either CSV or JSON format.
- **Validation**: Comprehensive data integrity and compatibility checks are carried out.

Prerequisites

Read and understand the prerequisites before you attempt migration.

System Requirements: Source System (VMware Identity Manager)	• VMware Identity Manager (3.3.7 GA version/3.3.7 GA with the latest patch version). • Ensure administrative credentials with read access to users, groups, and sync settings. • Export scripts work with FQDN only. • Ensure network connectivity to VMware Identity Manager APIs. • Take a backup of the VMware Identity Manager. • **CAUTION:** To ensure data consistency, directory sync must not be running during export. For more information, see Directory Sync Scheduling.
System Requirements: Target System (identity broker)	• Ensure that identity broker is at version 9.1 or later. • Deployment type supported is Instance only. • Ensure administrative credentials for VCF Operations APIs. • Third-party identity provider and AD/LDAP should be configured for the domains that you want to migrate. • Ensure network connectivity to identity broker APIs. • Take a backup of the identity broker configuration. • **CAUTION:** To prevent data conflicts, directory sync must not be running during import. For more information, see #unique_2216/unique_2216_Connect_42_id-e7aaf190-aff4-42fe-9641-1a3f249d2a05.
System Requirements: Component Systems	Verify that VCF single sign-on enablement for VCF Operations, VCF Automation, and NSX is complete. • **VCF Operations**: Administrative credentials are required. **Note:** The script supports local (system) admin credentials only. • **NSX**: Administrative credentials with role binding management permissions are required. • **VCF Automation**: Administrative credentials for provider organization are required.
System Requirements: Client System	• Supported operating systems: – Windows x86_64 – macOS ARM64 (Apple Silicon) – Linux x86_64 • Network connectivity to all systems. • Sufficient disk space for exported data.

Network Requirements	• HTTPS connectivity to all systems. • DNS resolution for all hostnames. • Firewall rules allowing API access.
Credentials Requirements	Here are the required credentials for the following components: • **VMware Identity Manager:** Administrative username and password. • **Identity Broker:** Target host details. • **VCF Operations:** Username and password. • **NSX:** Username and password. • **VCF Automation:** Username and password for VCF Automation Provider Organization.

Migration Limitations

Before you proceed with migration, review these critical limitations to ensure that migration aligns with your environment and requirements.

1. If you have a single VMware Identity Manager instance connected to multiple components, such as, VCF Operations, VCF Automation, or NSX , all the components will point to the new identity broker after migration if you carry out component migration.
2. Identity provider configurations in VMware Identity Manager are not migrated automatically by the script and must be configured manually in identity broker.
3. OAuth clients are not migrated automatically. You must manually regenerate the client and secret using identity broker and configure accordingly.
4. VMware Identity Manager local accounts and local accounts with multifactor authentication are not supported in identity broker.
5. Multifactor authentication with Active Directory is not supported in identity broker.
6. This script only exports single tenant data from VMware Identity Manager. If you have multiple tenants, each tenant should be exported separately.
7. If you have multiple VMware Identity Manager instances under the same domain, for example, one VMware Identity Manager instance with VCF Operations and another VMware Identity Manager instance with NSX, you must run the export script twice and subsequently import it into the same identity broker instance.
8. Supported Components:
   – **VCF Operations**. Migrates identity source types from VMware Identity Manager to identity broker for all configured domains. The migration is asynchronous and processes all users/groups that match between VMware Identity Manager and identity broker.
   – **NSX**: Updates NSX user/groups type from VMware Identity Manager to identity broker identity source (OIDC). The migration filters users/groups based on the domains configured by the identity broker (if VCF Operations credentials are provided), or migrates all VMware Identity Manager users/groups (with user confirmation if VCF_OPS credentials are not provided).

     Note:

     Migration is not supported for Global Manager NSX.
   – **VCF Automation**: Imports all identity broker users and groups to VCF Automation. Each imported user/group is assigned the `Defer to Identity Provider` role (mandatory in VCF Automation). You can optionally choose to also assign the `System Administrator` role to all imported users/groups during the migration process. Previous roles will not be migrated.

     Note:

     Migration is supported exclusively for the VCF Automation Provider Organization. VCF Automation tenant organizations are not supported.

Procedure

Complete the following steps to complete migration.

1. Download the `VIDM-VIDB-Migration-9.1` zip file from the following location: Broadcom Support.
   a) Unzip the file and select the file compatible with your operating system.
   b) Execute the following files after navigating to the respective OS folder.
2. To extract and export users, groups, and sync settings from VMware Identity Manager, run the `vidm-export` file and follow the instructions on-screen. To run the commands using a terminal, run `./vidm-export --help`.

   Review the `Migration_Readiness_Report` before importing users/groups.
3. To import exported users and groups to identity broker with comprehensive validation and conflict resolution, run the `vidb-import` file and follow the instructions on-screen. To run the commands using a terminal, run `./vidb-import --help`.
4. To update component identity source configurations from VMware Identity Manager to identity broker, run the `component-update` file and choose the relevant component. To run the commands using a terminal, run `./component-update --help`.

   Note:

   Migration is supported only for the following three components: VCF Operations, VCF Automation, and NSX.

Troubleshooting While Migrating from VMware Identity Manager to Identity Broker

Here are a few scenarios that you might encounter while migrating from VMware Identity Manager to identity broker. Review the cause and solution to solve the problem.

Table 1137: Export Phase Issues

Issue	Cause	Solution
Invalid hostname 'hostname'. Must be a fully qualified domain name.	VMware Identity Manager hostname is not in FQDN format.	Verify that the VMware Identity Manager URL uses FQDN. For example, `vidm.company.com` not `vidm`.
Connection failed to the VMware Identity Manager server	Network connectivity or credential issues.	• Verify network connectivity to VMware Identity Manager. • Verify username and password. • Use `--non_ssl` if SSL certificate issues exist.
The export path is not accessible	Insufficient permissions or invalid path.	• Verify that the export directory exists and is writable. • Verify the file system permissions.

Table 1138: Import Phase Issues

Issue	Cause	Solution
Identity broker system validation failed: deployment_type is not APPLIANCE	The target identity broker is not an instance deployment.	Migration only supports identity broker instance deployments.
Identity broker system validation failed: version 8.x is not supported	Identity broker version is below 9.1.	Upgrade identity broker to version 9.1 or later.
Import shows 100% conflicts (all entities skipped)	All exported entities exist in the target identity broker.	• Verify that you are importing to the correct identity broker instance. • Verify if a previous import was already completed. • Review the existing data in identity broker to confirm that it is correct. • Verify if this is an incremental import scenario.

Table 1139: Component Update Issues

Issue	Cause	Solution
NSX authentication failed	Invalid NSX credentials or connectivity issues.	• Verify the NSX hostname and credentials. • Verify that there is network connectivity. • Verify that the user has role binding management permissions.
VMware Identity Manage identity source not found in NSX	NSX is not configured with a VMware Identity Manage identity source.	• Verify that the VMware Identity Manage identity source configuration in NSX. • Verify the identity source name and configuration.
Identity broker identity source not found in NSX	NSX is not configured with a identity broker identity source.	• Configure identity broker identity source in NSX before running the component update. • Verify the OIDC configuration for identity broker.
No VMware Identity Manage role bindings found.	No role bindings configured with VMware Identity Manager identity source.	• Verify that role bindings exist in NSX. • Verify if the role bindings are using the correct identity source.

Managing Certificates in VMware Cloud Foundation

Manage certificates for VCF components according to industry standards and the requirements of your organization. You use VCF Operations to view and manage the certificates for all components.

For additional information on certificate management design, see Certificate Management.

Access to all management component interfaces must be carried over a Secure Socket Layer (SSL) connection. During deployment, each component is assigned a certificate from a default signing Certificate Authority (VMware Certificate Authority CA).

Certificate Categories in VCF

In a VCF environment, certificates fall into three categories based on how they are managed. The categories determine whether VCF Operations provides full lifecycle management, monitoring and alerting, or no management or monitoring capabilities.

Certificate Category	Description
User-managed	User-managed certificates are certificates that users can fully manage through VCF Operations. VCF Operations provides full lifecycle management for these certificates. Users can view, monitor, generate certificate signing requests (CSRs), import, replace, and enable automatic renewal for these certificates.
Externally managed	Externally managed certificates are certificates that VCF Operations can view and monitor but cannot manage. The individual components own the management of these certificates. Users must replace or renew these certificates through the component's own UI, API, or CLI. VCF Operations provides visibility into the certificate state and sends expiry alerts so that users can take action before the certificate expires. Certificate status for these certificates is available through the VCF Operations API.

Certificate Category	Description
System-managed	System-managed certificates are internal component certificates that are automatically managed by the individual components. The components auto-renew these certificates during routine operations such as upgrades. VCF Operations does not provide a view or monitoring interface for these certificates, and certificate status is not available through the VCF Operations API. When unrecoverable issues occur, the component triggers alerts directly in VCF Operations with a link to a Knowledge Base (KB) article for remediation guidance.

VCF Operations provides different levels of management and monitoring depending on the certificate category.

Certificate Category	View	Monitor	Manage
User-managed	Yes	Yes	Yes, full lifecycle
Externally managed	Yes	Yes, expiry alerts	No
System-managed	No	No, alerts for unrecoverable issues only	No

Certificate Management Capabilities in VCF Operations

Use VCF Operations to:

- View certificates and certificate alerts.
- Enable automatic renewal of certificates.
- Configure a certificate authority (Microsoft CA and OpenSSL CA).
- Generate certificate signing requests (CSRs).
- Import certificates.
- Replace certificates with a configured CA-signed certificate, an external CA-signed certificate, or a VMCA-signed certificate.
- Monitor certificate expiry for system-managed certificates.

Starting with VCF Operations 9.1, you can generate certificate signing requests, renew, import, and replace multiple certificates simultaneously.

User-managed certificates are certificates that a user can replace with third-party CA-signed certificates or customize with VMware Certificate Authority (VMCA). Users are expected to manage the lifecycle of these certificates, including renewal, import, and replacement. VCF Operations cannot auto-renew user-managed certificates that are signed by a third-party CA.

System-managed certificates are certificates that are managed by the component or system. Usually, no API, CLI, or UI is available for customers to manage these certificates directly. The auto-renewal of system-managed certificates is handled by the individual components. VCF Operations monitors these certificates for expiry and raises alerts when auto-renewal fails.

Certificate Operations by Certificate Authority Type

The certificate operations available in VCF Operations depend on whether the certificate is signed by a configured CA or an external CA.

A configured CA is a certificate authority that you configure in VCF Operations. VCF Operations supports the following configured CA types:

- VMware Certificate Authority (VMCA)
- Microsoft Certificate Authority (MSCA)
- OpenSSL Certificate Authority

Before you can replace a certificate with a CA-signed certificate or activate auto-renewal for Microsoft Certificate Authority or OpenSSL certificates, you must configure a Certificate Authority (CA). VCF supports automatic renewal for all VCF

Operations managed TLS certificates that are signed by VMware Certificate Authority CA, Microsoft CA, or OpenSSL CA and support non-disruptive certificate updates.

Self-signed certificates and certificates signed by an external CA cannot be renewed from VCF Operations. You perform a 3-step manual replace (generate CSR, import the signed certificate, and replace).

Starting with VCF Operations 9.1.1, VCF Operations detects and displays component self-signed CAs and self-signed CA certificates as self-signed in the VCF Operations UI. In VCF Operations 9.1, these certificates are categorized as external CA certificates. Self-signed certificates do not support manual renew or automatic renew from VCF Operations. Component certificates managed by VCF Operations cannot be self-signed. If existing component certificates are self-signed or issued by a private internal CA, replace them through VCF Operations using VMCA, Microsoft CA, OpenSSL CA, or an external CA.

Operation	Configured CA	External CA	Self-signed CA
Generate CSR	Yes	Yes	Yes
Import certificate	Yes	Yes	Yes
Auto-renew	Yes	No	No
Manual renew	Yes	No	No
Manual replace with configured CA-signed certificate	Yes	Yes	Yes
Manual replace with external CA-signed certificate	Yes	Yes	Yes
Replace with VMCA-signed certificate	Yes (new in 9.1.1)	Yes (new in 9.1.1)	Yes (new in 9.1.1)
Non-disruptive certificate updates	Yes (for components that support NDC)	Yes (for components that support NDC)	Yes (for components that support NDC)
View certificate	Yes	Yes	Yes
Monitor for expiry	Yes	Yes	Yes

VCF Components with Certificate Management Support

You can manage certificates for the VCF management components and for the components in a VCF instance with version 5.2 and later. You can also manage certificates for standalone vCenter instances and ESX hosts with version 8.0 Update 3 and later.

VCF Management Components	VCF Instance / Domain Components	Standalone Components
• VCF Operations • VCF Automation • VCF Operations for networks • Log management • License server • Identity broker • VCF management services • Cloud proxy	• ESX • vCenter • NSX Manager • SDDC Manager • VCF Operations HCX * • VMware AVI Load Balancer ** • vSphere Supervisor	• ESX *** • vCenter • vSphere Supervisor

*To manage the VCF Operations HCX certificates, you must install and configure the VCF Operations Management Pack for HCX. See Management Pack for VCF Operations HCX.

** To manage the VMware AVI Load Balancer certificates, you must install and configure the VCF Operations Management Pack for VMware Avi Load Balancer. See Management Pack for VMware Avi Load Balancer.

*** Using VCF Operations to changes ESX certificates changes the vCenter `vpxd.certmgmt.mode` setting to *`Custom`*.

Starting with VCF 9.1.1 and later, certificate management coverage expands to include the following components:

- Cloud proxy

- License server
- vSphere Supervisor

Certificate Signing Requirements

VCF Operations supports the following certificate signing algorithms and key sizes:

- RSA with 2048, 3072, and 4096-bit key sizes.

Component certificates that are managed by VCF Operations must be signed by a trusted Certificate Authority (CA) and cannot be self-signed. If existing component certificates are self-signed or issued by a private internal CA, replace them through VCF Operations using VMCA (default), Microsoft CA, OpenSSL CA, or a third-party CA.

Subject Alternative Name Requirements

VCF 9.1 enforces strict certificate validation and requires an FQDN for the VCF component in the Subject Alternative Name (SAN) field. If a certificate does not include the FQDN in the SAN field, the automated inventory migration and registration into VCF management services fails during the VCF 9.1 upgrade process. Optionallty, the SAN field can also contain the IP of the component.

When you generate a certificate signing request (CSR) or import a certificate through VCF Operations, the SAN field must contain the FQDN of the component. Optionally, the SAN field can also contain the IP of the component.

VCF Operations performs TLS hostname verification, as defined in RFC 6125, when it connects to a component over HTTPS. The hostname verification matches the component FQDN against the DNS SAN entries in the certificate. If the certificate does not include the FQDN as a DNS SAN entry, the hostname match fails and VCF Operations cannot establish a connection to the component. A certificate that contains only an IP address in the SAN field does not satisfy the hostname verification requirement.

VCF Operations supports the following SAN entry types:

- FQDNs — fully qualified, dotted domain names (`fqdn.domain.org`)
- IP addresses

Dot-less shortnames in the SAN field are not supported. A dot-less shortname is a bare hostname label without a domain suffix. The SAN validation requires at least one dot in each SAN entry. If a dot-less shortname is included alongside an FQDN or an IP address, the entire CSR generation fails.

Some VCF components, such as vSphere Supervisor, can be deployed using only an IP address and might not have an FQDN assigned. You can use one of the following in the Subject Alternative Name field:

- A domain-structured FQDN string, for example, `supervisor.local` or `component.internal`
- A Punycode-encoded label
- The string `localhost`

Use a value that your issuing Certificate Authority (CA) accepts.

Configure a Certificate Authority for VMware Cloud Foundation

Before you can replace a certificate with a CA-configured certificate or activate auto-renewal for Microsoft Certificate Authority or OpenSSL certificates, you must configure a Certificate Authority (CA).

For configuring a Microsoft CA:

- Prepare the Microsoft Certificate Authority. See Prepare Your Microsoft Certificate Authority to Allow VMware Cloud Foundation to Manage Certificates.
- Verify connectivity between VCF Operations and the Microsoft Certificate Authority Server. See VMware Ports and Protocols.

VCF Operations supports the following configured CA types:

- VMware Certificate Authority (VMCA)
- Microsoft Certificate Authority (MSCA)
- OpenSSL Certificate Authority

VCF supports automatic renewal of Transport Layer Security (TLS) certificates for components with certificates that are signed by VMware Certificate Authority CA, Microsoft CA, or Open SSL CA and support non-disruptive certificate updates.

If existing component certificates are self-signed or issued by a private internal CA, replace them through VCF Operations using VMCA, Microsoft CA, OpenSSL CA, or a third-party CA. Certificates signed by an external CA and self-signed certificates cannot be renewed from VCF Operations. For these certificates, you perform a 3-step manual replace: generate a certificate signing request (CSR), import the signed certificate, and replace.

VMCA is the default signing CA for VCF. During deployment, each component receives a VMCA-signed certificate. You can replace the default VMCA-signed certificates with certificates signed by Microsoft CA or OpenSSL CA.

Microsoft Certificate Authority

Microsoft Certificate Authority (MSCA) is an enterprise CA that runs on Windows Server with Active Directory Certificate Services. VCF Operations connects to MSCA through a web enrollment interface to request and sign certificates automatically. MSCA is suitable for enterprise environments with existing Microsoft PKI infrastructure.

Before configuring MSCA in VCF Operations, you must prepare the Microsoft Certificate Authority server by installing the required roles, configuring basic authentication, creating a certificate template, and assigning certificate management privileges to a service account. For more information, see Prepare Your Microsoft Certificate Authority to Allow VMware Cloud Foundation to Manage Certificates.

With VCF Operations 9.1 and later, VCF Operations can request and sign certificates automatically if the Certificate Authority role and the Certificate Authority Web Enrollment role are installed on different machines. In earlier versions, both roles must be installed on the same server.

OpenSSL Certificate Authority

OpenSSL Certificate Authority is a certificate service hosted within VCF Operations. VCF Operations generates and manages the OpenSSL CA root certificate internally, so unlike Microsoft CA, OpenSSL CA does not require external infrastructure or preparation steps. OpenSSL CA is suitable for testing or development environments where an external PKI infrastructure is not available.

OpenSSL CA functions independently and cannot be chained into an existing external CA hierarchy. To integrate with an enterprise PKI infrastructure, use Microsoft CA or a third-party CA instead.

When you configure OpenSSL CA, VCF Operations generates a self-signed root CA certificate using the organization details you provide.

1. Log in to VCF Operations.
2. From the navigation bar at the top, click **Manage**.
3. In the left pane, navigate to **Fleet Management** > **Certificates**.
4. Click **VCF Management** and then click the **Configure CA for Fleet** button.
5. Configure the settings for the fleet certificate authority and click **Save**.

Table 1140: Microsoft Certificate Authority

Setting	Description
CA Server URL	Specify the URL for the issuing certificate authority. This address must begin with `https://` and end with `certsrv`. For example, `https://ca.example.com/certsrv`.
User Name	Enter a least privileged service account. For example, `svc-vcf-ca`.
Password	Enter the password for the least privileged service
Template Name	Enter the issuing certificate template name. You must create this template in Microsoft Certificate Authority. For example, VMware.

Table 1141: OpenSSL

Setting	Description
Common Name	Specify the FQDN of the SDDC Manager appliance.
Country	Select the country where your company is registered.
Locality Name	Specify the city or the locality where your company is legally registered.
Organization Name	Specify the name under which your company is known. The listed organization must be the legal registrant of the domain name in the certificate request.
Organization Unit Name	Use this field to differentiate between the divisions within your organization with which this certificate is associated.
State	Enter the full name (do not abbreviate) of the state, province, region, or territory where your company is legally registered.

Prepare Your Microsoft Certificate Authority to Allow VMware Cloud Foundation to Manage Certificates

To generate signed certificates with Microsoft CA for your VCF components, the Microsoft Certificate Authority must be configured to integrate with VCF.

Install Microsoft Certificate Authority Roles

To facilitate certificate generation from VCF Operations, install the Certificate Authority and Certificate Authority Web Enrollment roles on the Microsoft Certificate Authority server.

For this procedure, the **Web Enrollment** role is installed on the same machine where the **Certificate Authority** role is installed. With VCF Operations 9.1 and later, VCF Operations can request and sign certificates automatically if the two roles (**Certificate Authority** and **Web Enrollment** roles) are installed on different machines.

1. Log in to the Microsoft Certificate Authority server by using a Remote Desktop Protocol (RDP) client.

FQDN	*`Active Directory Host`*
User	Active Directory administrator
Password	*`ad_admin_password`*

2. Add roles to Microsoft Certificate Authority server.
   a) Click **Start** > **Run**, enter `ServerManager`, and click **OK**.
   b) From the **Dashboard**, click **Add roles and features** to start the **Add Roles and Features** wizard.
   c) On the **Before you begin** page, click **Next**.
   d) On the **Select installation type** page, click **Next**.
   e) On the **Select destination server** page, click **Next**.
   f) On the **Select server roles** page, under **Active Directory Certificate Services**, select **Certification Authority** and **Certification Authority Web Enrollment** and click **Next**.
   g) On the **Select features** page, click **Next**.
   h) On the **Confirm installation selections** page, click **Install**.

Configure the Microsoft Certificate Authority for Basic Authentication

Configure the Microsoft Certificate Authority with basic authentication to allow VCF Operation the ability to manage signed certificates.

For VCF Operations 9.1.0.x The Microsoft Certificate Authority and IIS must be installed on the same server.

With VCF Operations 9.1.1 and later, VCF Operations can request and sign certificates automatically if the two roles (**Certificate Authority** and **Web Enrollment** roles) are installed on different machines.

1. Log in to the Active Directory server by using a Remote Desktop Protocol (RDP) client.

FQDN	*Active Directory Host*
User	Active Directory administrator
Password	*ad_admin_password*

2. Add Basic Authentication to the Web Server (IIS).
   a) Click **Start** > **Run**, enter `ServerManager`, and click **OK**.
   b) From the **Dashboard**, click **Add roles and features** to start the **Add Roles and Features** wizard.
   c) On the **Before you begin** page, click **Next**.
   d) On the **Select installation type** page, click **Next**.
   e) On the **Select destination server** page, click **Next**.
   f) On the **Select server roles** page, under **Web Server (IIS)** > **Web Server** > **Security**, select **Basic Authentication** and click **Next**.
   g) On the **Select features** page, click **Next**.
   h) On the **Confirm installation selections** page, click **Install**.
3. Configure the certificate service template and CertSrv web site, for basic authentication.
   a) Click **Start** > **Run**, enter `Inetmgr.exe` and click **OK** to open the **Internet Information Services Application Server Manager**.
   b) Navigate to ***your_server*** > **Sites** > **Default Web Site** > **CertSrv**.
   c) Under **IIS**, double-click **Authentication**.
   d) On the **Authentication** page, right-click **Basic Authentication** and click **Enable**.
   e) In the navigation pane, select **Default Web Site**.
   f) In the **Actions** pane, under **Manage Website**, click **Restart** for the changes to take effect.

Create and Add a Microsoft Certificate Authority Template

You must set up a certificate template in the Microsoft Certificate Authority. The template contains the certificate authority attributes for signing certificates for the VMware Cloud Foundation components. After you create the template, you add it to the certificate templates of the Microsoft Certificate Authority.

1. Log in to the Active Directory server by using a Remote Desktop Protocol (RDP) client.

FQDN	*Active Directory Host*
User	Active Directory administrator
Password	*ad_admin_password*

2. Click **Start** > **Run**, enter `certtmpl.msc`, and click **OK**.
3. In the **Certificate Template Console** window, under **Template Display Name**, right-click **Web Server** and select **Duplicate Template**.
4. In the **Properties of New Template** dialog box, click the **Compatibility** tab and configure the following values.

Setting	Value
Certification Authority	Windows Server 2008 R2
Certificate recipient	Windows 7 / Server 2008 R2

5. In the **Properties of New Template** dialog box, click the **General** tab and enter a name for example, `VMware` in the **Template display name** text box.
6. In the **Properties of New Template** dialog box, click the **Extensions** tab and configure the following.
   a) Click **Application Policies** and click **Edit**.
   b) Click **Server Authentication**, click **Remove**, and click **OK**.
   c) Click **Basic Constraints** and click **Edit**.
   d) Click the **Enable this extension** check box and click **OK**.
   e) Click **Key Usage** and click **Edit**.
   f) Click the **Signature is proof of origin (nonrepudiation)** check box, leave the defaults for all other options and click **OK**.
7. In the **Properties of New Template** dialog box, click the **Subject Name** tab, ensure that the **Supply in the request** option is selected, and click **OK** to save the template.
8. Add the new template to the certificate templates of the Microsoft CA.
   a) Click **Start** > **Run**, enter `certsrv.msc`, and click **OK**
   b) In the **Certification Authority** window, expand the left pane, right-click **Certificate Templates**, and select **New** > **Certificate Template to Issue**.
   c) In the **Enable Certificate Templates** dialog box, select **VMware**, and click **OK**.

Assign Certificate Management Privileges to the VMware Cloud Foundation Service Account

Before you can use the Microsoft Certificate Authority and the pre-configured template, it is recommended to configure least privilege access to the Microsoft Active Directory Certificate Services using an Active Directory user account as a restricted service account.

- Create a user account in Active Directory with Domain Users membership. For example, `svc-vcf-ca`.

1. Log in to the Microsoft Certificate Authority server by using a Remote Desktop Protocol (RDP) client.

FQDN	*`Active Directory Host`*
User	Active Directory administrator
Password	*`ad_admin_password`*

2. Configure least privilege access for a user account on the Microsoft Certificate Authority.
   a) Click **Start** > **Run**, enter `certsrv.msc`, and click **OK**.
   b) Right-click the certificate authority server and click **Properties**.
   c) Click the **Security** tab, and click **Add**.
   d) Enter the name of the user account and click **OK**.
   e) In the **Permissions for ....** section configure the permissions and click **OK**.

Setting	Value (Allow)
Read	Deselected
Issue and Manage Certificates	Selected
Manage CA	Deselected
Request Certificates	Selected

3. Configure least privilege access for the user account on the Microsoft Certificate Authority Template.
   a) Click **Start** > **Run**, enter `certtmpl.msc`, and click **OK**.
   b) Right-click the VMware template and click **Properties**.
   c) Click the **Security** tab, and click **Add**.
   d) Enter the `svc-vcf-ca` service account and click **OK**.
   e) In the **Permissions for ....** section configure the permissions and click **OK**.

Setting	Value (Allow)
Full Control	Deselected
Read	Selected
Write	Deselected
Enroll	Selected
Autoenroll	Deselected

Set Up Automatic Renewal of Certificates in VMware Cloud Foundation

You can activate automatic renewal of certificates for VCF management components or a VCF instance. Automatic renewal uses the configured Certificate Authority to renew the certificate for each component.

- If you are using Microsoft CA or OpenSSL root certificates for auto-renewal, configure a Certificate Authority before activating auto-renewal. See Configure a Certificate Authority for VMware Cloud Foundation .
- Verify that the certificates you want to auto-renew are signed by VMCA, Microsoft CA, or OpenSSL CA. VCF Operations cannot auto-renew certificates signed by an external CA.
- Verify that the certificates are not root or intermediate certificates. You cannot auto-renew root or intermediate certificates.

VCF supports automatic renewal of Transport Layer Security (TLS) certificates for components with certificates that are signed by VMware Certificate Authority, Microsoft CA, or Open SSL CA and support non-disruptive certificate updates.

You configure automatic renewal separately for VCF management components and VCF Instance components.

Automatic renewal happens 60 days before a certificate expires.

You must manually renew a certificate before turning on auto-renewal if that certificate is due to expire in 60 days or less.

1. Log in to VCF Operations.
2. From the navigation bar at the top, click **Manage**.
3. In the left pane, navigate to **Fleet management** > **Certificates**.
4. On the **Certificates** page, select **VCF Management** or expand **VCF Instances**, and select a VCF Instance.

   You can enable or disable automatic renewal of certificates for VCF management components, VCF Instances, VCF domains, and standalone vCenter instances.
5. Click the **Activate Auto-renewal** toggle to turn on automatic renewal of certificates.
6. In the **Activate Auto-renewal** dialog box, review the information and click **Confirm**.

Manually Renew Certificates in VMware Cloud Foundation

You can manually renew Transport Layer Security (TLS) certificates for components that have certificates signed by VMware Certificate Authority CA, Microsoft CA, or Open SSL CA.

- If you are using Microsoft CA or OpenSSL certificates to manually renew certificates, configure a Certificate Authority. See Configure a Certificate Authority for VMware Cloud Foundation .
- You can renew multiple certificates simultaneously.
- You cannot renew root or intermediate certificates.
- You cannot renew self-signed certificates.

- You cannot renew expired certificates. To renew an expired certificate, you can use the VCF Certificate Management Tool. See Broadcom knowledge base article 445742.

1. Log in to VCF Operations.
2. From the navigation bar at the top, click **Manage**.
3. In the left pane, navigate to **Fleet management** > **Certificates**.
4. Click **VCF Management** or click **VCF Instances** and click a VCF Instance or a VCF domain name.

   You can manually renew certificates for VCF management components, VCF Instances, VCF domains, and standalone vCenter instances.
5. Select a component with a supported certificate type and click **Renew certificate**.
6. In the **Renew certificate** dialog box, click the check-box and click **Confirm**.

Generate Certificate Signing Requests in VMware Cloud Foundation

Use VCF Operations UI to generate a certificate signing request (CSR) for a VCF component to replace the TLS certificate with a CA-signed certificate.

You can generate certificate signing requests for a single or multiple VCF components in VCF Operations simultaneously.

When generating a CSR, the public key and Subject Alternative Names (SANs) are embedded in the CSR, while the corresponding private key is retained in the VCF Operations keystore. When the CA-signed certificate is imported in VCF Operations, the certificate is validated with the stored private key.

1. Log in to VCF Operations.
2. From the navigation bar at the top, click **Manage**.
3. In the left pane, navigate to **Fleet management** > **Certificates**.
4. On the **VCF Management** page, filter for the required VCF component.

   You can generate CSRs for VCF management components, VCF Instances, VCF domains, and standalone vCenter instances.
5. Select the VCF component and, from the ellipsis drop-down menu, select **Generate CSR**.
6. In the **Generate CSR**, configure the settings and click **Generate**.

Option	Description
Organization	Type the name under which your company is known. The listed organization must be the legal registrant of the domain name in the certificate request.
Organizational Unit	Use this field to differentiate between divisions within your organization with which this certificate is associated.
Country	Select the country name where your company is legally registered.
State/Province	Type the full name (do not abbreviate) of the state, province, region, or territory where your company is legally registered.
Locality	Type the city or locality where your company is legally registered.
Email Address	Optionally, enter a contact email address.
Key Size	Select the key size (2048 bit, 3072 bit, or 4096 bit) from the drop-down menu.
Common Name	Enter the common name.

Option	Description
Subject Alternative Name	Enter the subject alternative name(s). **Note:** VCF 9.1 enforces strict certificate validation and requires an FQDN for the VCF component in the SAN field. VCF Operations supports the following SAN entries: • Fully qualified, dotted domain names, for example `fqdn.domain.org` • A domain-structured FQDN string, for example, `supervisor.local` or `component.internal` • A Punycode-encoded label • The string `localhost`
IP Name	Enter the IP addresses of the component (comma-separated list). To avoid issues with certificate validation, do not leave this field blank.

Note:

If using an optional external load balancer for a VCF management component, the `Subject Alternative Name` field of the certificate signing request must contain the FQDNs of the load balancer and all component nodes. In addition, the `IP Name` field must contain the IP addresses of the load balancer and all component nodes.

Note:

Some VCF components, such as VMware vSphere Supervisor, can be deployed using only an IP address and might not have an FQDN assigned. You can use one of the following in the `Subject Alternative Name` field:

- A domain-structured FQDN string, for example, `supervisor.local` or `component.internal`
- A Punycode-encoded label
- The string `localhost`

Use a value that your issuing Certificate Authority (CA) accepts.

From the ellipsis drop-down menu, you can view and download the generated CSRs.

Import Certificates in VMware Cloud Foundation

After you generate a certificate signing request for one or multiple VCF components, you can import the certificates, signed by an external CA.

You can import multiple certificates simultaneously. The maximum number of files you can import simultaneously is 20. Files with `.crt`, `.cer`, `.pem`, `.p7b`, and `.p7c` extensions are supported.

1. Log in to VCF Operations.
2. From the navigation bar at the top, click **Manage**.
3. In the left pane, navigate to **Fleet management** > **Certificates**.
4. On the **VCF Management** page, from the ellipsis drop-down menu, select **Import Certificates**.
5. In the **Import Certificates** dialog, configure the settings and click **Validate**.

Setting	Description
Certificate name	Enter the certificate name, for example, the hostname for the VCF component.
Source	Select whether you want to paste the content or add the files for the root CA and component certificate.

Paste Text	Copy and paste the:
	• Server Certificate
	• Certificate Authority
	Paste the server certificate and the certificate authority in PEM format (base64-encoded) . For example:
	`-----BEGIN CERTIFICATE-----`
	`<certificate content>`
	`-----END CERTIFICATE------`
	
	If the Certificate Authority includes intermediate certificates, it should be in the following format:
	`-----BEGIN CERTIFICATE-----`
	`<Intermediate certificate content>`
	`-----END CERTIFICATE------`
	`-----BEGIN CERTIFICATE-----`
	`<Root certificate content>`
	`-----END CERTIFICATE-----`

Certificate Chain	Click **Browse** to upload the certificate chain. Files with `.crt`, `.cer`, `.pem`, `.p7b`, and `.p7c` extensions are supported. Each file must be validated separately. Only `base64` encoding format is supported. The certificate chain must be ordered - first, import the leaf certificate, then all intermediate certificates, and finally the root CA.

6. To import additional certificates, click **Add more**.
7. Click **Import**.

Replace a Certificate with a Configured CA-Signed Certificate

After you configure a certificate authority (CA) in VCF Operations, you can generate a certificate signing request (CSR) and then replace an existing certificate for a component with a CA-signed certificate.

- Configure a Certificate Authority. See Configure a Certificate Authority for VMware Cloud Foundation .
- You must first generate a certificate signing requests for each certificate you are replacing.
- You can replace multiple certificates simultaneously.
- You cannot replace root or intermediate certificates.
- You cannot replace expired certificates. To replace an expired certificate, you can use the VCF Certificate Management Tool. See Broadcom knowledge base article 445742.

1. Log in to VCF Operations.
2. From the navigation bar at the top, click **Manage**.
3. In the left pane, navigate to **Fleet** > **Certificates**.
4. Click **VCF Management** or click **VCF Instances** and click a VCF Instance or a VCF domain name.

   You can replace a certificate with a configured CA-signed certificate for VCF management components, VCF Instances, VCF domains, and standalone vCenter instances.

5. Select the VCF Component, and, from the ellipsis drop-down menu, select **Replace with configured CA certificate**.
6. In the **Replace with configured CA certificate** dialog box, select your CA and click **Confirm**.

Replace a Certificate with an External CA-Signed Certificate

If you do not want to configure a Certificate Authority in VCF Operations, you can generate a CSR, request a signed certificate from your external Certificate Authority, import the certificate, and then use it to replace an existing certificate for a VMware Cloud Foundation component.

- You cannot replace root or intermediate certificates.
- You cannot replace expired certificates.
- You must first import the certificates that are signed by an external CA.
- You can replace multiple certificates simultaneously.

You can replace multiple certificates simultaneously.

1. Log in to VCF Operations.
2. From the navigation bar at the top, click **Manage**.
3. In the left pane, navigate to **Fleet** > **Certificates**.
4. Click **VCF Management** or click **VCF Instances** and click a VCF Instance or a VCF domain name.

   You can replace a certificate with an external CA-signed certificate for VCF management components, VCF Instances, VCF domains, and standalone vCenter instances.
5. Select the VCF components and click **Replace with imported certificate**.
6. On the **Replace with imported certificate** dialog box, select the imported certificates and click **Confirm**.

Managing Certificates in VMware vSphere Foundation

If you are using VMware vSphere Foundation, you can use VCF Operations to view and manage certificates for VCF Operations.

With VMware vSphere Foundation 9.1, the log management component runs on a dedicated Kubernetes cluster, you can manage certificates for the log management component by using VCF Operations API.

Managing VCF Operations Certificates for vSphere Foundation

VCF Operations includes a central page where you can review authentication certificate contents.

How the Certificates Page Works

The Certificates page lets you examine certificate details without the need to open the certificate outside of VCF Operations.

Where You Find Certificates

In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Control Panel**, and then click the **Trusted Certificates** tile.

Certificate Tabs

The certificate tab describes columns of exceptions tabs.

Note:

The Certificate Revocation List (CRL) tab is activated only when you select the **Activate Standard Certificate Validation** under **Global Settings** > **System Settings**. For more information, see List of Global Settings.

Tabs	Description
Exceptions	Lists the certificate that is accepted by the VCF Operations administrator but is not certified by the Certificate Authority (CA).
CRL	A Certificate Revocation List (CRL) is a list of digital certificates that have been revoked by the issuing Certificate Authority (CA) before their scheduled expiration date and should no longer be trusted. Click the **Add** icon to upload the certificates.

Certificate Options

The options include a data grid for examining certificate contents.

Table 1142: Certificate Options

Option	Description
Certificate Thumbprint	A unique alphanumeric string associated with the certificate
Issued By	Content associated with the issuer of the certificate, such as organization name and location
Issued To	Typically, content associated with the issuer, plus the certificate object Identifier (OID)
Expires	The date after which the certificate cannot be used for successful authentication

Importing CA Certificates in VCF Operations

Certificate Authority (CA) or root certificates are used for establishing the outgoing connections from VCF Operations. CA Certificates imported by the users are used in the following VCF Operations domains: Authentication Sources (Active Directory (AD), Open LDAP, VMware Identity Manager), Outbound Plugins, and Adapter Endpoint.

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Control Panel**, and then click the **Trusted Certificates** tile.
2. Click **Import**.

   The Import CA Certificate(s) dialog box appears. You can only import certificates that are encoded in the PEM format.
3. Click **Browse**.
4. Locate the certificate `.pem` file and click **Open** to load the file in the Import CA Certificate(s) dialog box.

   The certificate information box appears with the certificate thumbprint, issued by, issued to and expiry date. For example, if you select a certificate that expires in 10 days, you receive a notification that the certificate is expiring soon.

   Note:
   If a certificate is close to its expiry date, a corresponding notification is displayed on the **Home** page.
5. Click **Preview** and then click **Import**.

Removing an Adapter Certificate in VCF Operations

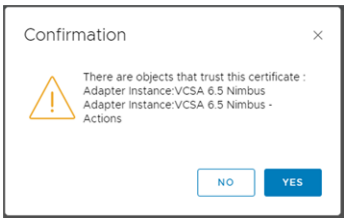
You can delete an old or expired certificate associated with an adapter.

1. Log in to VCF Operations.
2. From the navigation bar at the top, click **Operate**
3. In the left pane, navigate to **Administration** > **Control Panel**.
4. Click the **Trusted Certificates** tile.
5. On the **Trusted Certificates** page, click the ellipsis menu next to the certificate you want to delte and click **Delete**.

   Note:

   You must own the credential used to configure the adapter to be able to delete the certificate. For more information, see Configuring Credentials in Integrations.

   If the certificate is being used by the adapter, then the following message comes up:

   A certificate can be configured for one or more adapters if it is the same destination system.
6. If you delete a certificate which is already being used by another adapter, the adapter fails to connect or start. As a workaround, perform the following steps:
   a) In the left pane, click **Administration** > **Integrations**.

      The Accounts tab appears.
   b) Select the particular integration and click the vertical ellipsis and then click **Edit**.

      The Account Information page opens.
   c) Click **Validate Connection**.

      Note:
      If the adapter instance is saved using a credential that is owned by another user, you must apply your own credentials to validate the connection. For more information, see Adding Credentials.
   d) A prompt comes up asking the user to import the associated certificate. Click **OK**.
   e) Restart the adapter from the **Accounts** tab

Reestablish Endpoint Connectivity in VCF Operations

When a VCF Operations adapter instance stops collecting data, resulting in blank reports and empty dashboard widgets, this indicates a loss of connectivity between the configured endpoint and VCF Operations. To reestablish connectivity, check the validity of the trusted certificate for the adapter and renew the certificate.

An endpoint certificate can become invalid due to:

- Trusted certificate expiration. See Managing VCF Operations Certificates for vSphere Foundation.
- Misconfigured account credentials. See Configuring Credentials in Integrations.

To correct the problem, perform the following steps to renew the endpoint certificate and delete the invalid certificate.

1. Check for trusted certificates that have expired.
   a) Log in to the VCF Operations interface at `https://<vcf_operations_fqdn>` as a user assigned the Administrator role.
   b) Under **Operate** > **Administration** > **Control Panel**, click **Trusted Certificates**.
   c) In the table that appears, click to sort on the **Expires** column and identify the certificates that have expired.

      For any certificate that has expired, hover over entries in the row and record the following information for that certificate:

      - Name of the certificate thumbprint.

- Endpoint Issued to.
- Expiration date.
- Certificate type: ADAPTER_INSTANCE

2. To renew and delete an expired certificate, click the vertical ellipsis (⋮) menu for the certificate and select **Delete**.

   If a confirmation window appears with multiple objects that trust the certificate listed, that indicates that multiple adapter instances are using the same certificate. Deleting the certificate can cause the other instances to lose their connections, so click **No**. Then validate credentials on the adapter instances to import a new certificate before deleting the expired certificate.

   - For an ADAPTER_INSTANCE certificate type:

     1. Navigate to **Operate** > **Administration** > **Integrations**.
     2. For each Account instance that is correlated to the information collected for the expired certificate, click the vertical ellipsis menu and select **Edit**.
     3. On the Account Information page that appears, click **Validate Connection** to validate credentials.
     4. If prompted to import a new certificate, click **OK** and **Save**.
     5. Return to **Operate** > **Administration** > **Control Panel**, click **Trusted Certificates**, and then delete the certificate.

Custom VCF Operations Certificates

For secure VCF Operations operation, you might need to perform maintenance on authentication certificates.

Authentication certificates are for a secure machine-to-machine communication within VCF Operations itself or between VCF Operations and other systems.

By default, VCF Operations includes its own authentication certificates. The default certificates cause the browser to display a warning when you connect to the VCF Operations user interface.

Your site security policies might require that you use another certificate, or you might want to avoid the warnings caused by the default certificates. In either case, VCF Operations supports the use of your own custom certificate. You can upload your custom certificate during the initial primary node configuration or later.

Custom VCF Operations Web Certificate Requirements

A certificate used with VCF Operations must conform to certain requirements. Using a custom certificate is optional and does not affect VCF Operations features.

Requirements for Custom Certificates

Custom VCF Operations certificates must meet the following requirements.

- The certificate file must include the terminal (leaf) server certificate, a private key, and all issuing certificates if the certificate is signed by a chain of other certificates.
- In the file, the leaf certificate must be first in the order of certificates. After the leaf certificate, the order does not matter.
- In the file, all certificates and the private key must be in PEM format. VCF Operations does not support certificates in PFX, PKCS12, PKCS7, or other formats.
- In the file, all certificates and the private key must be PEM-encoded. VCF Operations does not support DER-encoded certificates or private keys.

  PEM-encoding is base-64 ASCII and contains legible BEGIN and END markers, while DER is a binary format. Also, file extension might not match encoding. For example, a generic `.cer` extension might be used with PEM or DER. To verify encoding format, examine a certificate file using a text editor.
- The file extension must be `.pem`.
- The private key must be generated by the RSA or DSA algorithm.
- The private key can be encrypted by a pass phrase. The generated certificate can be uploaded using the primary node configuration wizard or the administration interface.
- The REST API in this VCF Operations release supports private keys that are encrypted by a pass phrase.

- The VCF Operations certificate must have IPs and Hostnames (wildcard certificates are supported) in the Subject Alternative Name (SAN) extension.
  For example, Subject Alternative Name comprises of the DNS Name: localhost and the IP Address: 127.0.0.1.
- The VCF Operations Web server on all nodes have the same certificate file, so it must be valid for all nodes. One way to make the certificate valid for multiple addresses is with multiple Subject Alternative Name (SAN) entries.
- SHA1 certificates create browser compatibility issues. Therefore, ensure that all certificates that are created and being uploaded to VCF Operations are signed using SHA2 or newer.
- The VCF Operations supports custom security certificates with key length up to 8192 bits. An error is displayed when you try to upload a security certificate generated with a stronger key length beyond 8192 bits.

  Note:
  Fill the certificate extension fields using the UTF-8 encoding.

If VCF Operations fails to accept and apply Custom CA Certificate. For more information, see the following KB article 2046591.

Configure a Custom Web Certificate

You can use OpenSSL to configure an authentication certificate for use with VCF Operations. You must first generate a Certificate PEM for VCF Operations, then install the Certificate PEM in VCF Operations. The certificates applied through the VCF Operations Admin UI will be used only for securely connecting and serving the user interfaces to (external) clients. We do not update the SSL certificates used for establishing a secure connection from VCF Operations to other services like VMware Identity Manager, vCenter, and Log management.

Take your cluster offline before uploading the custom web certificate.

Note:

If you are using VCF Operations 9.0 or later, update your certificates from the **Fleet Management** > **Certificates** page.

1. Generate a Certificate PEM file for use with VCF Operations.
   1. Generate a key pair by running this command:
      ```
      openssl genrsa -out key_filename.key 2048
      ```
   2. Use the key to generate a certificate signing request by running this command:
      ```
      openssl req -new -key key_filename.key -out certificate_request.csr
      ```
   3. Submit the CSR file to your Certificate Authority (CA) to obtain a signed certificate.
   4. From your Certificate Authority, download the certificate and the complete issuing chain (one or more certificates). Download them in Base64 format.
   5. Enter the command to create a single PEM file containing all certificates and the private key. In this step, the example certificate is *server_cert.cer* and the issuing chain is *cacerts.cer*.

      Note:
      The order of CA's certs in the .PEM file: Cert, Private Key, Intermediate Cert and then Root Cert.

      cat *server_cert.cer key_filename.key cacerts.cer* > *multi_part.pem*

      In Windows replace cat with type.

   The finished PEM file should look similar to the following example, where the number of CERTIFICATE sections depends on the length of the issuing chain:

   ```
   -----BEGIN CERTIFICATE-----
   (Your Primary SSL certificate: your_domain_name.crt)
   -----END CERTIFICATE-----
   -----BEGIN RSA PRIVATE KEY-----
   (Your Private Key: your_domain_name.key)
   -----END RSA PRIVATE KEY-----
   -----BEGIN CERTIFICATE-----
   (Your Intermediate certificate: DigiCertCA.crt)
   -----END CERTIFICATE-----
   ```

```
-----BEGIN CERTIFICATE-----
(Your Root certificate: TrustedRoot.crt)
-----END CERTIFICATE-----
```

2. Install a PEM in VCF Operations.
    1. In a Web browser, navigate to the VCF Operations administration interface.
       `https://vrops-node-FQDN-or-ip-address/admin`
    2. Log in with the admin user name and password.
    3. At the upper right, click the yellow **SSL Certificate** icon.
    4. In the **SSL Certificate** window, click **Install New Certificate**.
    5. Click **Browse** for certificate.
    6. Locate the certificate .pem file, and click Open to load the file in the **Certificate Information** text box. The certificate file must contain a valid private key and a valid certificate chain.
    7. Click **Install**.

Verifying a Custom VCF Operations Web Certificate
When you upload a custom certificate file, the VCF Operations interface displays summary information for all certificates in the file.

For a valid custom certificate file, you should be able to match issuer to subject, issuer to subject, back to a self-signed certificate where the issuer and subject are the same.

In the following example, `OU=MBU,O=VMware\, Inc.,CN=vc-ops-slice-32` is issued by `OU=MBU,O=VMware\, Inc.,CN=vc-ops-intermediate-32`, which is issued by `OU=MBU,O=VMware\, Inc.,CN=vc-ops-cluster-ca_33717ac0-ad81-4a15-ac4e-e1806f0d3f84`, which is issued by itself.

```
Thumbprint: 80:C4:84:B9:11:5B:9F:70:9F:54:99:9E:71:46:69:D3:67:31:2B:9C
Issuer Distinguished Name: OU=MBU,O=VMware\, Inc.,CN=vc-ops-intermediate-32
Subject Distinguished Name: OU=MBU,O=VMware\, Inc.,CN=vc-ops-slice-32
Subject Alternate Name:
PublicKey Algorithm: RSA
Valid From: 2015-05-07T16:25:24.000Z
Valid To: 2020-05-06T16:25:24.000Z

Thumbprint: 72:FE:95:F2:90:7C:86:24:D9:4E:12:EC:FB:10:38:7A:DA:EC:00:3A
Issuer Distinguished Name: OU=MBU,O=VMware\, Inc.,CN=vc-ops-cluster-ca_33717ac0-ad81-4a15-ac4e-e1806f0d3f84
Subject Distinguished Name: OU=MBU,O=VMware\, Inc.,CN=vc-ops-intermediate-32
Subject Alternate Name: localhost,127.0.0.1
PublicKey Algorithm: RSA
Valid From: 2015-05-07T16:25:19.000Z
Valid To: 2020-05-06T16:25:19.000Z

Thumbprint: FA:AD:FD:91:AD:E4:F1:00:EC:4A:D4:73:81:DB:B2:D1:20:35:DB:F2
Issuer Distinguished Name: OU=MBU,O=VMware\, Inc.,CN=vc-ops-cluster-ca_33717ac0-ad81-4a15-ac4e-e1806f0d3f84
Subject Distinguished Name: OU=MBU,O=VMware\, Inc.,CN=vc-ops-cluster-ca_33717ac0-ad81-4a15-ac4e-e1806f0d3f84
Subject Alternate Name: localhost,127.0.0.1
PublicKey Algorithm: RSA
Valid From: 2015-05-07T16:24:45.000Z
Valid To: 2020-05-06T16:24:45.000Z
```

Sample Contents of Custom VCF Operations Web Certificates
For troubleshooting purposes, you can open a custom certificate file in a text editor and inspect its contents.

PEM Format Certificate Files

A typical PEM format certificate file resembles the following sample.

```
-----BEGIN CERTIFICATE-----
```

```
MIIF1DCCBLygAwIBAgIKFYXYUwAAAAAGTANBgkqhkiG9w0BAQ0FADBhMRMwEQYK
CZImiZPyLGQBGRYDY29tMRUwEwYKCZImiZPyLGQBGRYFdm13Y3MxGDAWBgoJkiaJ
<snip>
vKStQJNr7z2+pTy92M6FgJz3y+daL+9ddbaMNp9fVXjHBoDLGGaLOvyD+KJ8+xba
aGJfGf9ELXM=
-----END CERTIFICATE-----
-----BEGIN RSA PRIVATE KEY-----
MIIEowIBAAKCAQEA4l5ffX694riI1RmdRLJwL6sOWa+Wf70HRoLtx21kZzbXbUQN
mQhTRiidJ3Ro2gRbj/btSsI+OMUzotz5VRT/yeyoTC5l2uJEapld45RroUDHQwWJ
<snip>
DAN9hQus3832xMkAuVP/jt76dHDYyviyIYbmxzMalX7LZy1MCQVg4hCH0vLsHtLh
M1rOAsz62Eht/iB61AsVCCiN3gLrX7MKsYdxZcRVruGXSIh33ynA
-----END RSA PRIVATE KEY-----
-----BEGIN CERTIFICATE-----
MIIDnTCCAoWgAwIBAgIQY+j29InmdYNCs2cK1H4kPzANBgkqhkiG9w0BAQ0FADBh
MRMwEQYKCZImiZPyLGQBGRYDY29tMRUwEwYKCZImiZPyLGQBGRYFdm13Y3MxGDAW
<snip>
ukzUuqX7wEhc+QgJWgl41mWZBZ09gfsA9XuXBL0k17IpVHpEgwwrjQz8X68m4I99
dD5Pflf/nLRJvR9jwXl62yk=
-----END CERTIFICATE-----
```

Private Keys

Private keys can appear in different formats but are enclosed with clear BEGIN and END markers.

Valid PEM sections begin with one of the following markers.

```
-----BEGIN RSA PRIVATE KEY-----
-----BEGIN PRIVATE KEY-----
```

Encrypted private keys begin with the following marker.

```
-----BEGIN ENCRYPTED PRIVATE KEY-----
```

Bag Attributes

Microsoft certificate tools sometimes add Bag Attributes sections to certificate files. VCF Operations safely ignores content outside of BEGIN and END markers, including Bag Attributes sections.

```
Bag Attributes
Microsoft Local Key set: <No Values>
localKeyID: 01 00 00 00
Microsoft CSP Name: Microsoft RSA SChannel Cryptographic Provider
friendlyName: le-WebServer-8dea65d4-c331-40f4-aa0b-205c3c323f62
Key Attributes
X509v3 Key Usage: 10
-----BEGIN PRIVATE KEY-----
MIICdwIBADANBgkqhkiG9w0BAQEFAASCAmEwggJdAgEAAoGBAKHqyfc+qcQK4yxJ
om3PuB8dYZm34Qlt81GAAnBPYe3B4Q/0ba6PV8GtWG2svIpcl/eflwGHgTU3zJxR
gkKh7I3K5tGESn81ipyKTkPbYebh+aBMqPKrNNUEKlr0M9sa3WSc0o3350tCc1ew
5ZkNYZ4BRUVYWm0HogeGhOthRn2fAgMBAAECgYABhPmGN3FSZKPDG6HJlARvTlBH
KAGVnBGHd0MOmMAbghFBnBKXa8LwD1dgGBng1oOakEXTftkIjdB+uwkU5P4aRrO7
vGujUtRyRCU/4fjLBDuxQL/KpQfruAQaof9uWUwh5W9fEeW3g26fzVL8AFZnbXS0
7Z0AL1H3LNcLd5rpOQJBANnI7vFu06bFxVF+kq6ZOJFMx7x3K4VGxgg+PfFEBEPS
UJ2LuDH5/Rc63BaxFzM/q3B3Jhehvgw61mMyxU7QSSUCQQC+VDuW3XEWJjSiU6KD
gEGpCyJ5SBePbLSukljpGidKkDNlkLgbWVytCVkTAmuoAz33kMWfqIiNcqQbUgVV
UnpzAkB7d0CPO0deSsy8kMdTmKXLKf4qSF0x55epYK/5MZhBYuA1ENrR6mmjW8ke
TDNc6IGm9sVvrFBz2n9kKYpWThrJAkEAk5R69DtW0cbkLy5MqEzOHQauP36gDi1L
WMXPvUfzSYTQ5aM2rrY2/1FtSSkqUwfYh9sw8eDbqVpIV4rc6dDfcwJBALiiDPT0
tz86wySJNeOiUkQm36iXVF8AckPKT9TrbC3Ho7nC8OzL7gEllETa4Zc86Z3wpcGF
```

```
BHhEDMHaihyuVgI=
-----END PRIVATE KEY-----
Bag Attributes
localKeyID: 01 00 00 00
1.3.6.1.4.1.311.17.3.92: 00 04 00 00
1.3.6.1.4.1.311.17.3.20: 7F 95 38 07 CB 0C 99 DD 41 23 26 15 8B E8
 D8 4B 0A C8 7D 93
friendlyName: cos-oc-vcops
1.3.6.1.4.1.311.17.3.71: 43 00 4F 00 53 00 2D 00 4F 00 43 00 2D 00
 56 00 43 00 4D 00 35 00 37 00 31 00 2E 00 76 00 6D 00 77 00 61 00
  72 00 65 00 2E 00 63 00 6F 00 6D 00 00 00
1.3.6.1.4.1.311.17.3.87: 00 00 00 00 00 00 00 00 02 00 00 00 20 00
00 00 02 00 00 00 6C 00 64 00 61 00 70 00 3A 00 00 00 7B 00 41 00
45 00 35 00 44 00 44 00 33 00 44 00 30 00 2D 00 36 00 45 00 37 00
30 00 2D 00 34 00 42 00 44 00 42 00 2D 00 39 00 43 00 34 00 31 00
2D 00 31 00 43 00 34 00 41 00 38 00 44 00 43 00 42 00 30 00 38 00
42 00 46 00 7D 00 00 00 70 00 61 00 2D 00 61 00 64 00 63 00 33 00
2E 00 76 00 6D 00 77 00 61 00 72 00 65 00 2E 00 63 00 6F 00 6D 00
5C 00 56 00 4D 00 77 00 61 00 72 00 65 00 20 00 43 00 41 00 00 00
31 00 32 00 33 00 33 00 30 00 00 00
subject=/CN=cos-oc-vcops.eng.vmware.com
issuer=/DC=com/DC=vmware/CN=VMware CA
-----BEGIN CERTIFICATE-----
MIIFWTCCBEGgAwIBAgIKSJGT5gACAAAwKjANBgkqhkiG9w0BAQUFADBBMRMwEQYK
CZImiZPyLGQBGRYDY29tMRYwFAYKCZImiZPyLGQBGRYGdm13YXJlMRIwEAYDVQQD
EwlWTXdhcmUgQ0EwHhcNMTQwMjA1MTg1OTM2WhcNMTYwMjA1MTg1OTM2WjAmMSQw
```

Add a Custom Web Certificate to VCF Operations

If you did not add your own SSL/TLS certificate when configuring the VCF Operations primary node, you can still add a certificate after VCF Operations is installed.

- Create and configure the primary node.
- Verify that your certificate file meets the requirements for VCF Operations .

1. In a Web browser, navigate to the VCF Operations administration interface at https://*node-FQDN-or-ip-address*/admin.
2. Log in with the admin user name and password.
3. At the upper right, click the SSL certificate icon.
4. In the certificate window, click **Install New Certificate**.
5. Click **Browse for certificate**.
6. Locate the certificate `.pem` file, and click **Open** to load the file in the Certificate Information text box.
7. Click **Install**.

Managing Passwords for VMware Cloud Foundation Components

Manage passwords in your VCF fleet according to industry standards and the requirements of your organization. You can use VCF Operations to manage the passwords for VCF components, create, apply, and detach password policies to a VCF fleet or VCF instances, and view password alerts for managed accounts.

Managing Passwords

VCF Operations does not store a copy of an account password in a local vault. This offers a robust approach to managing passwords - changing an account password where the account resides does not affect the ability of VCF Operations to update that password (on both the client and the server). Using VCF Operations to update passwords is the recommended way to manage passwords in your environment. Use the VCF Operations UI and API to:

- Update passwords.
- Create, assign, and detach password policies.
- View password alerts and information about the components whose passwords are managed by VMware Cloud Foundation.
- View the last password update date for managed accounts.

With VCF Operations 9.1, you can manage passwords for the VCF components for VCF version 5.2 and later. You can also manage passwords for standalone vCenter instances and managed ESX hosts with version 8.0 Update 3 and later.

For additional information on account and password management, see Account Management Design.

You can also manage passwords by using SDDC Manager. VCF Operations is the preferred method for managing password in VCF. Use SDDC Manager to manage local or application account passwords in case your environment requires it. See Using SDDC Manager to Manage Passwords.

Password Policies

Starting with VCF 9.1, you can manage account passwords settings across VCF components by using password policies. A password policy defines password length, password complexity, password expiration, account lockout, password change interval, and password history.

You create password policies in VCF Operations. You can then assign the password policy to the entire VCF fleet, the VCF management components, or separate VCF Instances. Every VCF fleet object can have only one active password policy that applies. An instance-level password or management component level policy takes precedence over a fleet-level policy when both are assigned.

With VCF 9.1.1 and later, if you remove an instance-level policy, the instance falls back to the fleet policy if one is configured.

VCF Operations periodically checks if VCF components are compliant with their assigned password policy. If components are not compliant (i.e. the password settings differ from the assigned password policy), you can use VCF Operations to remediate the password settings of the respective component to the assigned password policy.

Password expiration and account lockout attributes are applied immediately when the password policy is applied to a VCF component. The password expiration is calculated based on the last password update. If your password policy specifies an N-day password expiration interval, if the password update occurred N+1 or more days ago, the password expires when the policy is applied. To avoid password expiries and account lockouts, when applying a password policy, ensure that all passwords have their last update within the new password policy expiration interval.

By default, password policy attributes (such as password expiration and account lockout) are not enforced for the vCenter `administrator@vsphere.local` account. You can use the vCenter API `com.vmware.identity.admin.administrator.PasswordPolicy` to enable password policy for the `administrator@vsphere.local` account.

> **Note:**
>
> Password policy tasks use the Salt master and Salt RaaS components. Password policy operations might fail if they are started right after VCF management services deployment completes. After deployment, the Salt RaaS component requires approximately 30 minutes to establish connections to all managed nodes.

CyberArk Integration

CyberArk enterprise vault integration in VCF Operations 9.1 introduces a centralized, policy-driven approach to managing OS and application account credentials across the VCF environment.

Through a dedicated CyberArk marketplace plugin, VCF Operations acts as a secure gateway between VCF components and the CyberArk vault, enabling administrators to store and protect application and OS accounts, automate password rotation, and enforce enterprise password policies. The integration supports modern API-based authentication, automated credential lifecycle management, and audit-ready reporting, transforming manual password handling into a streamlined, compliant, and scalable process while strengthening security across the entire VCF infrastructure.

Accounts in VMware Cloud Foundation

VMware Cloud Foundation comprises a number of local (OS), application, and service accounts.

Account Types in VCF

Account Type	Intended Use
Local (OS) accounts	Local or OS accounts are provisioned directly in the operating system of the virtual appliance and are primarily intended to facilitate secure shell (SSH) access to that appliance.
Application accounts	Application accounts are provisioned for interactive logins and have access to the application's UI and API. Some VCF components use application accounts (for example, the vCenter Single Sign-On account `administrator@vsphere.local`) for inter-component communication by default.
Service accounts	Service accounts are system-managed and provisioned for inter-component communication and use automated methods for password management.

Local (OS) and Application Accounts in VCF

Local and application accounts are managed according to the specific methods outlined in the table. Not all local and application accounts are managed by SDDC Manager or VCF Operations. User-created local accounts are not managed by SDDC Manager or VCF Operations.

VCF Component	Account	Description	Managed by
ESX	`root`	• Local appliance account • OS level access • Default expiry: Never • Password automatically generated during deployment	• VCF Operations UI or API – Update • SDDC Manager UI or API – Update – Rotate – Remediate – Password look up
vCenter	`root`	• Local appliance account • OS level access • VAMI access • Default expiry: 90 days • Password automatically generated during deployment	• VCF Operations UI or API – Update • SDDC Manager UI or API – Update – Rotate – Remediate – Password look up – Automatic scheduled password rotation
	`administrator@vsphere.local`	• vCenter Single Sign-On account • Application and API access • Additional VMware Cloud Foundation **Admin** account required to perform manual password rotation. • Default expiry: never • Password automatically generated during deployment • By default, password policy is not enforced for this account	

VCF Component	Account	Description	Managed by
NSX Global Manager	`root`	• Local appliance account • OS level access • Default expiry: 90 days • Password automatically generated during deployment	Manually by using the NSX Global Manager UI or API.
	`admin`	• Local appliance account • OS level, API, and application access • Default expiry: 90 days • Password automatically generated during deployment	
	`audit`	• Local appliance account • OS level, API, and application access • Default expiry: 90 days • Password automatically generated during deployment	
NSX Local Manager	`root`	• Local appliance account • OS level access • Default expiry: 90 days • Password automatically generated during deployment	• VCF Operations UI or API – Update • SDDC Manager UI or API – Update – Rotate – Remediate – Password look up – Automatic scheduled password rotation
	`admin`	• Local application account • OS level, API, and application access • Default expiry: 90 days • Password automatically generated during deployment	
	`audit`	• Local application account • Read-only application level access • Default expiry: 90 days • Password automatically generated during deployment	
NSX Edge	`root`	• Local appliance account • OS level access • Default expiry: 90 days	• VCF Operations UI or API – Update • SDDC Manager UI or API – Update – Rotate – Remediate – Password look up – Automatic scheduled password rotation
	`admin`	• Local appliance account • OS level, API, and application access • Default expiry: 90 days	
	`audit`	• Local appliance account • Read-only application level access • Default expiry: 90 days	

VCF Component	Account	Description	Managed by
NSX Virtual Network Appliance	`root`	• Local appliance account • OS level access • Default expiry: 90 days	• VCF Operations UI or API – Update • SDDC Manager UI or API – Update – Rotate – Remediate – Password look up – Automatic scheduled password rotation
	`admin`	• Local appliance account • OS level, API, and application access • Default expiry: 90 days	
	`audit`	• Local appliance account • OS level access • Read-only application level access • Default expiry: 90 days	
SDDC Manager	`vcf`	• Local appliance account • OS level access • Default expiry: 365 days • Password automatically generated during deployment	• In VCF 9.1, manually by using the OS. • In VCF 9.1.1, VCF Operations UI or API
	`root`	• Local appliance account • OS level access • Default expiry: 365 days • Password automatically generated during deployment	• In VCF 9.1, manually by using the OS. • In VCF 9.1.1, VCF Operations UI or API
	`backup`	• Local appliance account • OS level access • Default expiry: 365 days • Password automatically generated during deployment	• VCF Operations UI or API – Update • SDDC Manager UI or API – Update – Rotate – Remediate – Password look up – Automatic scheduled password rotation
	`admin@local`	• Application account • API access • Default expiry: Never • Password automatically generated during deployment	• In VCF 9.1, manually by SDDC Manager API. • In VCF 9.1.1, VCF Operations UI or API
VCF Operations	`root`	• Local appliance account • OS level access • Default expiry: 90 days • Password automatically generated during deployment	VCF Operations UI or API
	`admin`	• Application account • API and application access • Default expiry: Never • Password automatically generated during deployment	VCF Operations UI or API
Cloud proxy	`root`	• Local appliance account • OS level access • Default expiry: 90 days • Password automatically generated during deployment	VCF Operations UI or API

VCF Component	Account	Description	Managed by
VCF Operations orchestrator (standalone)	`root`	• Local appliance account • OS level access • Default expiry: Never	Manually by using the OS. **Note:** The `root` and `admin` accounts listed for VCF Operations orchestrator apply only to standalone instances manually deployed using the VCF Operations orchestrator appliance OVA. In the default VCF deployment, VCF Operations orchestrator runs as an embedded component of VCF Automation. These standalone appliance accounts are not applicable to the embedded deployment.
	`admin`	• Local appliance account • API, and application access • Default expiry: Never	
VCF Operations for networks	`root`	• Local appliance account • OS level access • Default expiry: Never	Manually by using the OS.
	`admin@local`	• API and application access • Default expiry: Never	VCF Operations UI or API
	`support`	• Local appliance account • OS level access • Default expiry: Never	
	`consoleuser`	• Local appliance account • CLI level access • Default expiry: Never	
VCF Operations for networks collector	`support`	• Local appliance account • OS level access • Default expiry: Never	• In VCF 9.1, Manually by using the OS. • In VCF 9.1.1, VCF Operations UI or API
	`consoleuser`	• Local appliance account • CLI level access • Default expiry: Never	
VCF Operations HCX	`root`	• Local appliance account • OS level access • Default expiry: Never	VCF Operations UI or API To manage the VCF Operations HCX local accounts passwords, you must install and configure the VCF Operations Management Pack for HCX, see Management Pack for VCF Operations HCX.
	`admin`	• Local appliance account • API and application access • Default expiry: Never	
• VCF management services – VCF services runtime – Fleet lifecycle – SDDC lifecycle – Salt RAAS – Salt Master – Telemetry • Log management • Identity broker • License server	`admin@vsp.local`	• Local appliance account • OS and application level access • Default expiry: 365 days • Password is the same as for the `vmware-system-user` account	Manually by using VCF management services
	`vmware-system-user`	• Local appliance account • OS and application level access • Default expiry: 365 days • Password generated during deployment	

VCF Component	Account	Description	Managed by
VCF Automation	admin	• Administrative account for the VCF Automation system organization • Default expiry: Never • Password automatically generated during deployment	VCF Operations UI or API
	admin@vsp.local	• Local appliance account • OS and application level access • Default expiry: 365 days • Password automatically generated during deployment	
	vmware-system-user	• Local appliance account • OS level access • Default expiry: 365 days • Password automatically generated during deployment	
vSphere Supervisor	vmware-system-user	• Local appliance account • OS level access • Encoded in the control plane namespace secret • Default expiry: Never	Manually by using the Kubernetes Secrets.
Protection and Recovery	root	• Local appliance account • OS level access • Default expiry: Never	Manually by using the OS.
	admin	• Local appliance account • API and application access • Default expiry: Never	Manually by using the VAMI.
VMware Avi Load Balancer	admin	• Local appliance account • API and application access • Default expiry: Never	VCF Operations UI or API
	vcfops-admin	• Local application account • API and application access • Default expiry: Never • Used for VCF Operations and AVI Load Balancer integration.	

Service Accounts in VCF

Service accounts are used for inter-component communication and are managed according to the specific methods outlined in the table.

Service accounts generated and utilized internally by VCF components must not be rotated, changed, or managed by external Privileged Access Management (PAM) tools. Modifying or forcing password rotation on these internal service accounts outside of VCF Operations or SDDC Manager workflows might break critical internal product dependencies, resulting in loss of inter-component communications, loss of lifecycle management capabilities, and potential environment

outages. Password rotation for supported components must always be orchestrated natively through VCF Operations or SDDC Manager.

VCF Component	Account	Description	Managed by
ESX	`svc-vcf-`*`ESX-hostname`*	• Service account between SDDC Manager (client) and the ESX host (server).	SDDC Manager UI or API
SDDC Manager	`svc-`*`sddc-manager-hostname-hcx-hostname-uuid`*	• Service account between VCF Operations HCX (client) and the SDDC Manager (server). • Default expiry: Never	System managed
	`svc-vcf-sddc-lcm-SDDC_MANAGER-`*`uuid`*	• Service account between SDDC lifecycle (client) and the SDDC Manager (server). • **Administrator** role • Automatically rotated every 30 days by default • Default expiry: 90 days • Initially created as `svc-sddclcm-sddcm-`*`uuid`* and is renamed after the first automated rotation.	System managed
vCenter	`svc-`*`sddc-manager-hostname-vcenter-hostname`*`@vsphere.local`	• Service account between SDDC Manager (client) and vCenter (server). • System managed • Automatically rotated every 30 days by default • **Administrator** role • Default expiry: 90 days	• SDDC Manager UI or API • Supports automatic, scheduled rotation (SDDC Manager UI or API)
	`svc-`*`nsx-manager-hostname-vcenter-hostname`*`@vsphere.local`	• Service account between NSX Manager (client) and vCenter (server). • **Administrator** role • System managed • Automatically rotated every 30 days by default • Default expiry: 90 days	System managed
	`svc-`*`fleet-management-hostname-vcenter-hostname`*`@vsphere.local`	• Service account between VCF Operations fleet management (client) and vCenter (server) • **Administrator** role • System managed • Automatically rotated every 30 days by default	System managed
	`svc-ops-vc-`*`ops-cluster-uid`*`@vsphere.local`	• Service account between VCF Operations fleet management (client) and vCenter (server) • **Administrator** role • System managed. • Automatically rotated every 30 days by default.	System managed
	`svc-`*`vcfa_uid`*`@vsphere.local`	• Service account between VCF Operations (client) and vCenter (server) • **Administrator** role • System managed • Automatically rotated every 30 days by default	System managed

VCF Component	Account	Description	Managed by
	svc-vcfsp-*vc-uid*@vsphere.local	• Service account between VCF Services Platform (client) and vCenter (server) • **VCF Services Runtime** role • System managed • Automatically rotated every 30 days by default.	System managed
	svc-vcfsp-*vc-admin-uid*@vsphere.loca	• Service account between VCF Services Platform (client) and vCenter (server) • **VCF Services Platform Admin** role • Automatically rotated every 30 days by default	System managed
	svc-*hcx-hostname-vc-hostname-hcx-uuid*	• Service account between VCF Operations HCX (client) and vCenter (server) • Administrator role.	System managed
NSX Local Manager	svc-*sddc-manager-hostname-nsx-vip-hostname-uid*	• Service account between SDDC Manager (client) and NSX (server) • **Enterprise Admin** role • System managed • Automatically rotated every 30 days by default • Default expiry: 90 days	System managed
	svc-ops-nsx-*uid*	• Service account between VCF Operations (client) and NSX (server). • **Enterprise Admin** role	System managed
	svc-ni-nsx-*uid*	• Service account between VCF Operations for networks (client) and NSX (server)	System managed
	svc-vcfa_*uid*	• Service account between VCF Automation (client) and NSX (server)	System managed
VCF Operations	svc-salt-ops-*uid*	• Service Account between Salt RaaS (client) and VCF Operations (server).	System managed
	svc-ops-li-ops-*uid*	• Service Account between Log Management (client) and VCF Operations (server).	System managed
	svc-ops-ni-ops-*uid*	• Service Account between VCF Operations for Networks (client) and VCF Operations (server).	System managed
Log Management	svc-ops-ops-li-*uid*	• Service Account between VCF Operations (client) and Log Management (server).	System managed
VCF Operations for Networks	svc-ops-ops-ni-*uid*	• Service Account between VCF Operations (client) and VCF Operations for Networks (server).	System managed
Salt RaaS	svc-ops-salt-*uid*	• Service Account between VCF Operations (client) and Salt RaaS (server).	System managed

Default Password Requirements for VCF Components

VCF components have default password requirements for user-managed local (OS), application, and service accounts. Use the information in the table below to change component password requirements. With VCF 9.1, you can manage these setting for all components by using password policies in VCF Operations.

Special characters may vary across VCF components. Use the following set of special characters which are valid for all VCF components:

`!, @, #, $, ^`

VCF Component	Account	Settings	Configured in

SDDC Manager	`vcf`
	`root`
	`backup`
		Minimum characters: 12
		Uppercase characters: 1
		Lowercase characters: 1
		Numerical characters: 1
		Special characters: 1
		Alphabetic characters: 2
		Reject dictionary word: Yes
			SDDC Manager uses the Linux Pluggable Authentication Modules (PAM) to enforce password settings. To change the password settings, edit the corresponding OS files directly.

	`admin@local`	Minimum characters: 15 Maximum characters: 127 Uppercase characters: 1 Lowercase characters: 1 Numerical characters: 1 Special characters: 1 A single character cannot be repeated consecutively: 3	Cannot be modified
ESX	`root`	Minimum characters: 7 Maximum characters: 40 A mix of at least three out of four character classes: lowercase letters, uppercase letters, numbers, and special characters: Yes Reject dictionary word: Yes Failed log-in attempts: 5 Lockout period (seconds): 900	Use the vSphere Client, the ESX host Client, or the ESX CLI to configure the password settings.
	`svc-vcf-`*`ESX-hostname`*	Minimum characters: 8 Maximum characters: 20 Uppercase characters: 1 Lowercase characters: 1 Numerical characters: 1 Special characters: 1 Non-ASCII characters are allowed.	vCenter Single Sign-On password policy. Administrators can change the default password policy.
vCenter	`root`	Minimum characters: 15 Uppercase characters: 1 Lowercase characters: 1 Numerical characters: 1 Special characters: 1 Failed log-in attempts: 3 Reject dictionary word: Yes Number of different characters: 8 Password history: 5 Lockout period (seconds): 300	vCenter uses the Linux Pluggable Authentication Modules (PAM) to enforce password settings. To change the password settings, you can use the vCenter VAMI or you can edit the corresponding OS files directly.
	`administrator@vsp`	Minimum characters: 8	vCenter Single Sign-On password

Password Policies in VMware Cloud Foundation

A password policy defines the rules that govern password complexity, expiration, history, and account lockout for application and local (OS) accounts for VCF components.

Password Policy Attributes

Policy Attribute	Default Value	Value Range	Additional Information
Minimum password length	At least 15 characters	8-64	DISA STIG requires 14 characters. FedRAMP requires 15 characters.
Upper case	At least 1	0-10	Must have at least 1 upper case character.
Lower case	At least 1	0-10	Must have at least 1 lower case character.
Special characters	At least 1	0-10	Must have at least 1 special character.
Numeric character	At least 1	0-10	Must have at least 1 numeric character.
Password history	Reject re-using any of previous 5 passwords	1-10	Number of unique passwords before passwords can be repeated.
Password expiry	365 days	30-730	Password expiration and lockout policies are not enforced on non-expiring accounts by default.
Account lockout	3 unsuccessful (failed) attempts	0-5	If unsuccessful password attempts are made consecutively, the account is locked out. With VCF 9.1.1, when the account is locked, you can update the password from VCF Operations. Password update does not impact the lockout state.
Account lockout evaluation duration	900 seconds	300-1800	Account lockout criteria must occur within this time window. For example, 2 authentication failures in 900 seconds trigger the lockout.
Account lockout duration	900 seconds	600-1800	The account remains locked for this period following consecutive unsuccessful attempts.

Non-expiring Accounts

Password expiration and account lockout policies are not enforced on non-expiring accounts by default. This ensures that upgrade or patching to VCF 9.1.1 does not distrupt the oerations of your SDDC. Of you require strict compliance, you can enforce password expiration and account lockout policies on specific non-expiring account by using the VCF Operations API in 9.1.1.

Account Lockout Behavior

With VCF 9.1.1 and later, when an account is locked due to a number of failed login attempts, the following behavior applies:

- An alert is generated in VCF Operations.
- You can update the password for a locked account from VCF Operations.
- Password update does not unlock the account. The account follows the lockout duration before unlocking, irrespective of the success of the password update operation.
- An expired account supports the update password operation from VCF Operations.

Create a Password Policy for VMware Cloud Foundation Components

Use VCF Operations to create a password policy that defines password complexity, history, expiration, and lockout requirements for managed accounts.

- VCF Operations is version 9.1 or later.
- VCF management domain components are at version 9.1 or later.
- VCF services runtime is deployed in the management domain.

9.1 Procedure

1. Log in to VCF Operations.
2. On the top navigation bar, click **Manage**.
3. In the left pane, navigate to **Fleet management** > **Fleet settings**.
4. On the **Fleet settings** page, click **Password policy**.
5. On the **Password policy** page, click **Create pocliy**.
6. On the **Create new policy** page, configure the password policy settings and click **Save**.

9.1.1 Procedure

1. Log in to VCF Operations.
2. On the top navigation, click **Manage**.
3. In the left pane, click **Passwords**.
4. On the **Overview** page, click the**Password Policy** tab.
5. Click **Create Policy**.
6. On the **Create New Policy** page, configure the password settings and click **Create**.

The password policy is created and appears in the password policy list. You can now apply the policy to the VCF fleet, VCF management components, or individual VCF Instances.

Apply a Password Policy for VMware Cloud Foundation Components

You can assign a password policy to the entire VCF fleet, the VCF management components, or separate VCF Instances. Every VCF fleet object can have one active password policy applied. You can assign a password policy on an instance level even if a fleet policy is already assigned. The instance-level policy takes precedence over the fleet-levet policy.

- VCF Operations is version 9.1 or later.
- VCF management domain components are at version 9.1 or later.
- VCF services runtime is deployed in the management domain.

Password expiration and account lockout attributes are applied immediately when the password policy is applied to a VCF component. The password expiration is calculated based on the last password update. If your password policy specifies an N-day password expiration interval and the last password update occurred N+1 or more days ago, the password expires when the policy is applied. To avoid password expiries and account lockouts, when applying a password policy, ensure that all passwords have their last update within the new password policy expiration interval.

In VCF 9.1, VCF Automation does not support password policies. With VCF 9.1.1 and later, VCF Automation supports password policies.

9.1 Procedure

In VCF 9.1, assigning a policy to a VCF component does not change passwords or password settings configured in that component. After you assign the password policy, you run a compliance check and if the VCF component is not compliant (the password settings on the component differ from the assigned policy), you can apply the password policy to that VCF component by running the remediation workflow.

The compliance check runs automatically every six hours. You can run the compliance check manually. It might take up to 30 minutes for the check to complete, avoid running multiple compliance checks simultaneously.

1. Log in to VCF Operations.

2. On the top navigation bar, click **Manage**.
3. In the left pane, navigate to **Fleet management** > **Fleet settings**.
4. On the **Fleet settings** page, click **Password policy**.
5. Assign a password policy to your VCF fleet.
   a. In the **Password Policies** section, select the password policy to assign.
   b. From the ellipsis drop-down menu, select **Assign Policy to Fleet**.
   c. On the **Assign Policy to Fleet** dialog box, click **Proceed**.
6. Assign a password policy to a VCF Instance or your management components.
   a. In the **Password Policies** section, select the password policy to assign.
   b. From the ellipsis drop-down menu, select **Assign Policy to Components**.
   c. In the **Assign Policy to Components** dialog box, select VCF Management or a VCF Instance and click **Assign**.
      You can assign a policy to VCF management components and single or multiple VCF Instances.
7. Apply a password policy.
   a. In the **Compliance status** section, from the ellipsis drop-down menu, select **Run compliance check**.
      You can select VCF Management and one or multiple VCF Instances. If you do not select any VCF Management or VCF Instances, the compliance check runs for the entire environment.
   b. From the ellipsis drop-down menu, select **Assign Policy to Components**.
   c. In the **Assign Policy to Components** dialog box, select VCF Management or a VCF Instance and click **Assign**.
      Ensure that all passwords have their last update within the new password policy expiration interval.
      After remediation, the password policy is applied to VCF Management and the VCF Instances.

9.1.1 Procedure

1. Log in to VCF Operations.
2. On the top navigation, click **Manage**.
3. In the left pane, click **Passwords**.
4. On the **Overview** page, click the**Password Policy** tab.
5. Apply a password policy to a VCF fleet.
   a. Click the ellipsis drop-down menu next to the password policy you want to apply and select **Apply Policy to Fleet**.
   b. Wait for the process to complete.
6. Apply a password policy to a VCF Instance or your management components.
   a. Click the ellipsis drop-down menu next to the password policy you want to apply and select **Apply Policy to Component Groups**.
   b. In the **Apply Policy to Component Groups** dialog box, select the VCF instances and VCF Management to apply the policy to and click **Apply**.

Detach a Password Policy from VMware Cloud Foundation Components

With VCF 9.1.1 and later, you can detach a password policy from a VCF fleet, VCF Instance, or the VCF management components. Detaching a password policy stops VCF Operations from enforcing the policy, detecting drift, and remediating the VCF components.

When you detach an instance-level policy from a component group, if a fleet-level policy is applied, the component group falls back to the fleet-level policy. If no fleet-level policy is applied, the detached component group falls back to its local component password policies.

1. Log in to VCF Operations.
2. On the top navigation bar, click **Manage**.
3. In the left pane, click **Passwords**.
4. On the **Overview** page, click the**Password Policy** tab.
5. Detach a fleet-level policy.
   a) Click the ellipsis drop-down menu next to the password policy you want to detach and select **Detach Fleet Policy**.
   b) Wait for the process to complete.
6. Detach a policy from a component group.
   a) Click the ellipsis drop-down menu next to the password policy you want to detach and select **Detach Policy from Component Groups**.
   b) In the **Detach Policy from Component Groups** dialog box, select the VCF instances and VCF Management to detach the policy from and click **Detach**.

Update Passwords for VMware Cloud Foundation Components

For security reasons, you can change passwords for the accounts that are used by VMware Cloud Foundation components. Changing these passwords periodically or when certain events occur, such as an administrator leaving your organization, reduces the likelihood of security vulnerabilities.

- Verify that there are no failed workflows. To check for failed workflows, in VCF Operations., click **Fleet Management** > **Tasks**.
- Verify that no active workflows are running or are scheduled to run during the password update.
- Verify that the VCF Operations user has the following permissions:
  - `Fleet Management: Passwords:Manage`
  - `Fleet Management: Passwords:View`

Using the VCF Operations UI to update a VCF component password is the preferred method for changing a password. If the VCF component has a password policy applied, VCF Operations automatically validates the new password against the password policy settings. In addition, this method does not create a configuration drift between VCF Operations and the environment state.

1. Log in to VCF Operations.
2. From the navigation bar at the top, click **Manage**.
3. In the left pane, navigate to **Fleet Management** > **Passwords**.
4. Click **VCF Management** or click **VCF Instances** and click a VCF Instance or a VCF domain name.
5. Select a component and click **Update password**.
6. In the **Update Password** dialog box, enter the current password, enter and confirm the new password.
7. Click **Update password**.

Reset Local Account Passwords for VMware Cloud Foundation Components

You can reset a forgotten or an expired local account password for some VCF components.

Use the following table to reset a forgotten or an expired local or application account password.

VCF Component	Account	Expired Password	Forgotten Password
SDDC Manager	`vcf`	Manually by using the OS with **root** permissions.	Manually by using the OS with **root** permissions.
	`root`	Password never expires.	Password cannot be reset.
	`backup`	Manually by using the OS with **root** permissions.	Manually by using the OS with **root** permissions.

VCF Component	Account	Expired Password	Forgotten Password
	admin@local	Password never expires.	See Broadcom Knowledge Base Article 403099.
ESX	root	Password never expires.	• See Broadcom Knowledge Base Article 323617. • See Broadcom Knowledge Base Article 376979.
vCenter	root	See Broadcom Knowledge Base Article 322247.	See Broadcom Knowledge Base Article 322247.
	administrator@vsphere.local	Password never expires.	See Broadcom Knowledge Base Article 326189.
NSX Manager NSX Edge NSX Virtual Network Appliance	root	See Resetting Expired Passwords.	See Resetting User Passwords.
	admin		
	audit		
VCF Operations	root	See Broadcom Knowledge Base Article 325005.	See Broadcom Knowledge Base Article 325005.
	admin (OS)	Password never expires.	See Broadcom Knowledge Base Article 326391.
	admin (application)	Password never expires.	• For UI method, see Broadcom Knowledge Base Article 326391 or Reset the VCF Operations Administrator Password from the Administration UI. • For CLI method, see Reset the VCF Operations Administrator Password from CLI.
VCF Operations cloud proxy	root	See Broadcom Knowledge Base Article 325005.	See Broadcom Knowledge Base Article 325005.
VCF Operations orchestrator	root	Password never expires.	Password cannot be reset.
VCF management services	admin@vsp.local	See Broadcom Knowledge Base Article 444456.	See Broadcom Knowledge Base Article 444456.
	vmware-system-user	See Broadcom Knowledge Base Article 444453.	See Broadcom Knowledge Base Article 444453.
VCF Automation	admin	See Broadcom knowledge base article 445470.	See Broadcom knowledge base article 445470.
	admin@vsp.local	See Broadcom Knowledge Base Article 444456.	See Broadcom Knowledge Base Article 444456.
	vmware-system-user	See Broadcom knowledge base article 424323.	See Broadcom knowledge base article 424323.

Using SDDC Manager to Manage Passwords

VCF Operations is the preferred method for managing password in VCF. SDDC Manager will be deprecated in a future VCF release. Use SDDC Manager to manage local or application account passwords in case your environment requires it.

In the context of password management actions, server refers to the component where the account resides, and client refers to the component where the account is used. SDDC Manager offers the following password management actions:

Action	Description
Remediate	You use the **Remediate** action to provide the new password on the client after the password is changed on the server. **Note:** Only available in SDDC Manager. Use VCF Operations to manage passwords for your environment.
Rotate	You use the **Rotate** action to change the password with an auto-generated password on the server and on the client. **Note:** Only available in SDDC Manager. Use VCF Operations to manage passwords for your environment.

SDDC Manager stores a copy of an account password in a local, internal vault. Changing an account password on the VCF component itself causes a mismatch between SDDC Manager's vault and the new password, and requires a remediation step, before the inter-component communication can be restored. Using SDDC Manager to manage passwords is not advised. Use SDDC Manager UI to:

- Rotate multiple passwords simultaneously.
- Remediate passwords for VCF core components.
- Schedule automatic password rotation.

You can also use the SDDC Manager API to look up and manage credentials.

Rotate Passwords for VMware Cloud Foundation Components

You can rotate passwords for the core components in your VCF Instance by using the SDDC Manager UI. The process of password rotation generates randomized passwords for the selected accounts. You can rotate passwords manually or can set up automatic rotation for accounts that are managed by SDDC Manager.

- Verify that there are no failed workflows in SDDC Manager. To check for failed workflows, click **Dashboard** in the navigation pane and expand the **Tasks** pane at the bottom of the page.
- Verify that no active workflows are running or are scheduled to run during the time period that the password rotation process runs. It is recommended that you schedule password rotation for a time when you expect to have no running workflows.

You can rotate passwords for the following VCF core components.

- ESX

  Note:
  Auto-rotate is not supported for ESX.
- vCenter Server

  Note:
  Auto-rotate is automatically activated for vCenter service accounts. It might take up to 24 hours to configure the service account auto-rotate policy for a newly deployed vCenter.
- NSX

When rotating a password, SDDC Manager generates the new password that complies with the passwords settings or password policy of the component.

1. Log in to SDDC Manager with a user assigned the **Administrator** role.
2. In the navigation pane, click **Security** > **Password Management**.
3. Select one or more accounts and click one of the following operations.

Action	Description
Rotate	Generates randomized passwords for the selected accounts.
Schedule rotation	You can set the password rotation interval to 30, 60, or 90 days. You can also deactivate the automatic rotation. You can use the SDDC Manager API to create a custom rotation interval. The automated rotation runs at midnight on the scheduled date. If auto-rotate could not start due to any technical issue, an auto-retry occurs every hour until the start of the next day. In case schedule rotation is missed due to technical issues, the UI shows a global notification with failed task status. The status of the schedule rotation can also be checked on the Tasks panel.
Rotate all	Generates randomized passwords for the all accounts.

   A message appears at the top of the page showing the progress of the operation. The tasks panel also shows detailed status for the password rotation operation. To view sub-tasks, click the task name. As each of these tasks is run, the status is updated. If the task fails, you can click **Retry**.

Password rotation completes when all sub-tasks complete successfully.

Remediate Passwords for VMware Cloud Foundation Components

To resolve any errors that might occur during password changes, you use password remediation. Password remediation synchronizes the password of the account that is stored in the SDDC Manager with the updated password in the component. After you reset the password in a component, you must remediate the password in VCF Operations.

Note:
You can remediate the password for only one account at a time.

For information on updating passwords manually, see Update Passwords for VMware Cloud Foundation Components .

1. Log in to SDDC Manager.
2. In the left pane, navigate to **Security** > **Password Management**
3. From the navigation bar at the top, click **Manage**.
4. Select a component and, from the ellipsis drop-down menu, select **Remediate Password**.
5. Enter and confirm the password that was set manually on the component.
6. Click **Remediate Password**.

If VCF components lose connectivity between them due to synchronization issues (for example, the password for the service account changes on the server side, while the client side is not updated), you can remediate the service account on the client side by following the procedures below.

Created in	Used by	Procedure

vCenter	SDDC-M	To manually remediate the SDDC Manager service account in vCenter, use the vCenter Password API to reset the SDDC Manager service account password. You then use the remediate password functionality in SDDC Manager.
		1. Create a session to vCenter.
		`session=$(curl -ks https://<vcenter_fqdn>/api/session -u 'administrator@vsphere.local:password' -X POST | jq -r .)`
		2. Reset the service account password.
		`curl -ks https://<vcenter_fqdn>/api/vcenter/svc-account-mgmt/password?action=reset -H "vmware-api-session-id: $session" -H 'Content-Type: application/json' -d '{ "account_name": "sddc_m_to_vcenter_service_account" }' | jq`
		The command returns the new password.
		3. Log in to SDDC Manager at `https://<sddc_manager_fqdn>` with a user assigned the **Administrator** role.
		4. In the left pane, navigate to **Security** > **Password management**.
		5. On the **Password management** page, select the vCenter service account, click the ellipsis icon, and select **Remediate password**.
		6. In the **Remediate password** dialog box, enter the new password and click **Remediate password**.
		Contact Broadcom Global Support for additional assistance.

	NSX	To manually remediate the NSX service account in vCenter, you must enter the service account credentials in the NSX UI and NSX automatically repairs the integration. 1. Log in to NSX Manager at `https://<nsx_manager_fqdn>/login.jsp` with a user assigned the **Administrator** role. 2. On the main navigation bar, click **System**. 3. In the left pane, navigate to **Fabric** > **Compute managers**. 4. Select the compute manager that has a `Not registered` status. 5. Click the message and click **Resolve**. 6. In the dialog box, enter the vCenter service account credentials and click **Resolve**.

	VCF Operations VCF Operations for networks VCF Operations for logs VCF Operations fleet management	To manually remediate the VCF Operations service account in vCenter, you use the VCF Operations UI. 1. Log in to the VCF Operations interface at `https://<vcf_operations_fqdn>` with a user assigned the **Administrator** role. 2. In the left pane, navigate to **Administration** > **Integrations**. 3. On the **Accounts** tab, expand the **VMware Cloud Foundation** section. 4. Expand your VCF Instance and click the domain. 5. On the domain page, click the **vCenter** tab. 6. Next to **System managed credentials**, click **Rotate**. The existing service account gets deleted and a new service account is created. The integration between VCF components is not affected during the creation of the new service account. VCF Operations synchronizes the inventory and generates a new service account in vCenter through SDDC Manager.

	VCF Operations orchestrator	To remediate the VCF Operations orchestrator service account in vCenter, log in to the VCF Operations orchestrator pod as **root** and reset the authentication provider by using the `vracli vro authentication` CLI command.
		`vracli vro authentication set -p vsphere \`
		`--hostname https://`*`FQDN_SPHERE`*`\`
		`--username`*`administrator@vsphere.local`*`\`
		`--tenant`*`vsphere.local`*`\`
		`--admin-group`*`Administrators`*`\`
		`--admin-group-domain`*`vsphere.local`*
		Restart the VCF Operations orchestrator pod.

	VCF Automation	To manually remediate the VCF Automation service account in vCenter, use the VCF Automation UI. 1. Log in to the VCF Automation interface at `https://<vcf_automation_fqdn>` with a user assigned the **Administrator** role. 2. In the left pane, navigate to **VCF Instances**. 3. On the **VCF Instance** page, click the vertical ellipsis next to the vCenter instance, and, from the drop-down menu, select **Rotate service account**. VCF Automation deletes and recreates the service account.

NSX	SDDC Manager	To manually remediate the SDDC Manager service account in NSX, use the NSX API to update the SDDC Manager service account password. You then use the remediate password functionality in SDDC Manager.
		1. Find the user ID for the SDDC Manager service account from the list of service accounts.
		`curl -sk 'https://vip_nsx_manager/api/v1/node/users?filter=service' -u 'admin:admin_password' | jq`
		2. Reset the service account password.
		`curl -ik 'https://vip_nsx_manager/api/v1/node/users/user_id?action=reset_password' -u 'admin:admin_password' -H 'Content-Type: application/json' -d '{ "password": "new_service_account_password" }'`
		3. Log in to SDDC Manager at `https://<sddc_manager_fqdn>` with a user assigned the **Administrator** role.
		4. In the left pane, navigate to **Security** > **Password management**.
		5. On the **Password management** page, select the NSX service account, click the elipsis icon, and select **Remediate password**.
		6. In the **Remediate password** dialog box, enter the password you configured and click **Remediate password**.

	VCF Operations VCF Operations for networks	To manually remediate the VCF Operations service account in NSX, you use the VCF Operations UI. 1. Log in to the VCF Operations interface at `https://<vcf_operations_fqdn>` with a user assigned the **Administrator** role. 2. In the left pane, navigate to **Administration** > **Integrations**. 3. On the **Accounts** tab, expand the **VMware Cloud Foundation** section. 4. Expand your VCF Instance and click the domain. 5. On the domain page, click the **NSX** tab. 6. Next to **System managed credentials**, click **Rotate**. The existing service account gets deleted and a new service account is created. The integration between VCF components is not affected during the creation of the new service account. VCF Operations synchronizes the inventory and generates a new service account in vCenter through SDDC Manager.
	VCF Automation	To manually remediate the VCF Automation service account in NSX, use the VCF Automation UI. VCF Automation deletes and recreates the service account. 1. Log in to the VCF Automation interface at `https://<vcf_automation_fqdn>` with a user assigned the **Administrator** role. 2. In the left pane, navigate to **VCF Instances**. 3. On the **VCF Instance** page, click the vertical ellipsis next to the NSX instance, and, from the drop-down menu, select **Rotate service account**.
SDDC Manager API keys	VCF Operations VCF Operations fleet management	To manually remediate the SDDC Manager API key in VCF Operations, you use the VCF Operations UI. 1. Log in to the VCF Operations interface at `https://<vcf_operations_fqdn>` with a user assigned the **Administrator** role. 2. In the left pane, navigate to **Administration** > **Integrations**. 3. On the **Accounts** tab, expand the **VMware Cloud Foundation** section. 4. Click the VCF Instance. 5. On the **Account** tab, in the **VCF Credentials** section, click **Rotate**. The existing API key is deleted and an API key is created. The integration between VCF components is not affected during the creation of the new service account. VCF Operations synchronizes the inventory and generates a new service account in vCenter through SDDC Manager.
	VCF Automation	To manually remediate the SDDC Manager API key in VCF Automation, you use the VCF Automation API to delete the SDDC Manager endpoint. VCF Automation creates an API key and adds the SDDC Manager instance during the next synchronization. This action usually takes 24 hours, during which period VCF Automation is not able to register new workload domains.

Update SDDC Manager Root and Super User Passwords

For security reasons, you can change passwords for the SDDC Manager `root` and the super user `vcf` accounts. Changing these passwords periodically or when certain events occur, such as an administrator leaving your organization, reduces the security risks.

The SDDC Manager `root` password expires after 90 days. To update an expired `root` password, see Update an Expired SDDC Manager Root Password.

1. SSH in to the SDDC Manager appliance using the `vcf` user account.
2. Enter `su` to switch to the root user.
3. Enter one of the following commands:

Option	Description
`passwd vcf`	To change the super user password.
`passwd root`	To change the root password.

4. Enter and reenter the new password. For example:

```
root@sddc-manager [ /home/vcf ]# passwd vcf
```

```
New password:
Retype new password:
passwd: password updated successfully
```

The password is updated.

Updating Passwords for SDDC Manager Accounts

The process for updating SDDC Manager passwords varies, depending on which account you are updating.

The following table lists the SDDC Manager user accounts and links to the procedures for how to update them.

SDDC Manager Account	Update Procedure
backup	Update Passwords for VMware Cloud Foundation Components
vcf root	Update SDDC Manager Root and Super User Passwords
admin@local	Use the `PATCH /v1/users/local/admin` API to update the local user password. See Update password for local account for more information.

Update an Expired SDDC Manager Root Password

If the SDDC Manager root (`root`) user password expires, you must update the password using the vSphere Client.

1. In a web browser, log in to the management domain vCenter using the vSphere Client (`https://<vcenter_fqdn>/ui`).
2. In the inventory, browse to and select the SDDC Manager virtual machine.
3. On the **Summary** tab, click **Launch Remote Console**.
4. Click within the console window and press **Enter** on the Login menu item.
5. Type **root** as the user name and enter the current password for the root user.
6. Type `passwd root`.
7. When prompted for a new password, enter a different password than the previous one and click **Enter**.

Use the SDDC Manager Appliance to Look Up Account Credentials

To look up the account credentials for the built-in accounts that are managed by SDDC Manager, you can log in to the SDDC Manager appliance.

This task returns information about the components for a specific VCF Instance and does not include VCF Management components.
If you prefer, you can use the `GET /v1/credentials` API to get the same information.

1. SSH in to the SDDC Manager appliance using the `vcf` user account.
2. Enter the following command:

   ```
   lookup_passwords
   ```

3. Enter an entity type from the displayed list.

   For example: `VCENTER`.

4. When prompted, enter a user name and password for an SDDC Manager account assigned the **ADMIN** role.
5. Optional: (Optional) Save the command output to a secure location with encryption so that you can access it later and use it to log in to the accounts as needed.

Managing Passwords for VMware vSphere Foundation Components

If you are using VMware vSphere Foundation, you can use the VCF Operations console to update and remediate the passwords for the VCF Operations component of VMware vSphere Foundation.

With VMware vSphere Foundation 9.1, the log management component runs on a dedicated Kubernetes cluster, you can manage passwords for the log management component by using VCF Operations API.

Reset the VCF Operations Administrator Password from the Administration UI

Passwords are used by users to access the product interfaces or to console sessions on cluster nodes. You might need to reset the VCF Operations administrator password as part of securing or maintaining your deployment and if you forget the admin account password.

1. Log in to the VCF Operations administration interface at https://*<primary-node-name> or <primary-node-ip-address>*/admin.
2. Log in with the admin user name and password for the primary node.
3. In the left pane, click **Administrator Settings**, and then expand the **Password Recovery Settings** section.
4. Enter the current password, and enter the new password twice to ensure its accuracy.

   Note:
   You cannot change the administrator user name.
5. Click **Save**.
6. Optionally, to recover a forgotten password, configure the **Password Recovery Settings**.

Table 1143: Password Recovery Settings

Password Recovery Settings Options	Description
Your E-mail	Email id to which you want to receive the recovery email.
SMTP Server	DNS name or IP address of the SMTP server that is used to send the password recovery email.
Port	Port used for the communication. By default, 25 is used for a non-secure port and 465 for a secure port.
SSL (SMTPS)	Activate to protect the communication using the secure socket layer.
STARTTLS Encryption	Activate to switch the insecure communication starting with the TLS handshake.
Sender E-mail	The email id from which the password recovery email is sent.
User name	User name for the STMP server account, as some servers require authentication.
Password	Password for the SMTP server account.
Test	To verify the mandatory fields and make an attempt to communicate with the given SMTP server.

7. Click **Save**. Optionally, click **Reset** to enter the details again.

Reset the VCF Operations Administrator Password from CLI

You must reset the password if the admin account password is lost.

This procedure requires root account credentials.

- In VCF Operations vApp deployments, when you log in to the console of the virtual application for the first time, you are forced to set a root password.
- The VCF Operations console root password can be different than the admin account password that you set when configuring the VCF Operations primary node.

When the VCF Operations password for the built-in admin account is lost, follow these steps to reset it on vApp clusters.

1. Log in to the master node command-line console as `root`.
2. Enter the following command, and follow the prompts.

   ```
   $VMWARE_PYTHON_BIN $VCOPS_BASE/../vmware-vcopssuite/utilities/sliceConfiguration/bin/vcopsSetAdminPassword.py --reset
   ```

Generate a VCF Operations Passphrase

When users need to add a node to the VCF Operations cluster, you can generate a temporary passphrase instead of giving them the primary administrator login credentials, which might be a security issue.

Create and configure the primary node.

A temporary passphrase is good for one use only.

1. In a Web browser, navigate to the VCF Operations administration interface at https://*master-node-name-or-ip-address*/admin.
2. Log in with the admin user name and password for the master node.
3. In the list of cluster nodes, select the master node.
4. From the toolbar above the list, click the option to generate a passphrase.
5. Enter the number of hours before the passphrase expires.
6. Click **Generate**.

   A random alphanumeric string appears, which you can send to a user who needs to add a node.

Have the user supply the passphrase when adding a node.

Tags and Categories Overview

You can create, manage, and assign tag categories and tags in your VCF Operations instance and use them to organize and manage your inventory objects at scale.

What Are Tags and Categories

A tag consists of a name and description that you specify. To group related tags together and manage inventory objects at scale, you create categories and tags inside the categories. When you create a category, you can specify the object types for its tags, and whether more than one tag in the category can be applied to an object. After you create the category and tags, you can apply the tags to objects in your vSphere Client inventory or in your VCF Operations inventory to organize and manage the objects more easily, and to tag specific inventory objects when you configure different vSphere policies.

For example, if you want to tag inventory objects with memory over-commitment, you can create a category named Compute Memory, specify that the category applies to just ESX hosts as an object type, and then add a tag named Memory Over-Commitment. After you create the category and tag, you can assign the tag to the specific hosts from your inventory.

VCF Operations Tags and Categories

By using VCF Operations, you can manage tags from a central place for all VCF Instances that are managed by the VCF Operations instance. You can create tags and categories, assign tags to objects in the VCF Operations inventory, import existing tags and categories from vCenter instances. You can assign tags from VCF Operations to inventory objects that are managed by vCenter instances. For more information, see Assign a Tag to an Object in VCF Operations.

After you create or import categories and tags in VCF Operations, you can push them to selected vCenter instances. You can then use the tags and categories in existing automation scripts or to tag objects in the vSphere Client. This ensures scalability and prevents loss of tags and their associations to objects during vMotion. For more information, see How Do You Assign or Remove a vSphere Tag.

For more information on how to select a tag management approach and define the provisioning scope, see VCF Tag Management for vSphere.

Create a Tag Category in VCF Operations

When you create a tag category, you must also create at least one tag in that category. After you create the category, you can create and add as many tags in it as you need.

- Verify that you have **Tags** > **Manage** permissions in VCF Operations.

You cannot create an empty category, you must create at least one tag in the category. For each category that you create, you must specify whether a single or multiple tags are allowed per object, and which object types can be associated with this category. After you create the category, you can edit the tags added to the category. You can add more tags, delete some of the existing tags, or edit them.

> **Note:**
>
> You can also import existing tags and categories from vCenter instances. For more information, see Import Tags and Categories to VCF Operations from a vCenter Instance.

1. Log in to VCF Operations.
2. From the navigation bar at the top, click **Manage** and in the left navigation pane, select **Fleet Management** > **Tags** .
3. Click **Create**.

   The **Create Category** dialog box appears.
4. Enter a name for the category. The name can consist of up to 128 characters.

   Once you create the category, you cannot change its name.
5. Optional: Enter a description for the category. The description can consist of up to 1024 characters.
6. Optional: To edit the tags per object and the object types that can be associated with the category, click **Advanced Settings**.
   a) Select the number of tags that can be assigned per associated object.

   > **CAUTION:**
   >
   > If you select **Multiple**, after you create the category, you cannot change the value to **Single**.

   a) Select to which objects types the tags from the category can be assigned to.

   > **CAUTION:**
   >
   > After you create the category, you can only add more object types to be associated with the category, you cannot remove the object types that you selected when creating the category. If you select **All object types**, after you create the category, you cannot change the value.
7. Create a tag to add to the category:
   a) Enter a name for the tag.

   After you create the tag, you cannot change its name.

   a) Enter a description for the tag.
8. Optional: To create more tags and add them to the category, click **Add Tag**, and enter a unique name and description for each tag.
9. Click **Create**.

The category appears in the table with available categories for this VCF Operations instance.

You can edit, delete, or push the newly created category to vCenter instances or custom groups. When you delete a category, you also delete all tags added to it. For backup purposes, you can also export all tag categories and tags available in the VCF Operations instance in JSON file format.

Create a Tag in VCF Operations

In VCF Operations, you can create tags only inside an existing tag category or at the time when you create a category.

- Verify that you have **Tags** > **Manage** permissions in VCF Operations.

1. Log in to VCF Operations.
2. From the navigation bar at the top, click **Manage** and in the left navigation pane, select **Fleet Management** > **Tags**.
3. Click the arrow icon next to the category in which you want to create a tag.

   A table with all tags added to this category appears.
4. Click **Create**.

   The **Create Tag** dialog box appears.
5. Enter a unique name for the tag.

   The name can consist of up to 128 characters.
6. Optional: Enter a description for the tag.

   The description can consist of up to 1024 characters.
7. Optional: To create more tags and add them to the category, click **Add Tag**, and enter a name and description for each tag.
8. Click **Create**.

The tag appears in the table with the available tags for the category.

You can edit the tag description or can delete the tag. You can also export all definitions for tag categories and tags available in the VCF Operations instance in JSON file format.

> **Note:**
>
> When an export is in progress, initiated by you or another user, you cannot initiate the export again before the first export completes.

You can assign tags to specific inventory objects in VCF Operations. For more information, see Assign a Tag to an Object in VCF Operations.

Import Tags and Categories to VCF Operations from a vCenter Instance

You can import existing tags and tag categories from your VCF domain vCenter instances or standalone vCenter instances to VCF Operations. Importing tags into VCF Operations enables you to manage tags and categories for more than one vCenter instance from a single place.

- Verify that the vCenter instance from which you want to import tags and tag categories is of version 9.0 or later.
- Verify that the vCenter instance from which you want to import tags has a valid VCF 9.0 or later license.
- Verify that the vCenter instance from which you want to import tags and tag categories is integrated with this VCF Operations instance.
- Verify that you have **Tags** > **Manage** permissions in VCF Operations.

After you import tags and categories to your VCF Operations instance, you can push them to other vCenter instances to have consistent tagging across your VCF fleet.

> **Important:**
>
> If you cannot see the conflict details, verify that you have the REST API read privileges assigned.

When a task is in progress, you cannot initiate the same task again. For example, if another user or you initiate an import, you cannot initiate import again before the first task completes.

1. Log in to VCF Operations.
2. From the navigation bar at the top, click **Manage** and in the left navigation pane, select **Fleet Management** > **Tags**.
3. In the **Tags** page, at the top-right corner, click **Import**.

   The **Import Categories** dialog box appears, and you can see a list of all VCF domains that are integrated with your VCF Operations instance.
4. Select a VCF domain from which you want to import tags and tag categories, and click **Import**.

   You cannot initiate an import task if another import task is in progress. You must wait for the task to complete before you can initiate it again.

   If another user initiated an import task and it is in progress, a warning appears.

The tags and categories are imported from the vCenter instance.

If some of the categories cannot be imported, click **View Conflict Details** or **View Import Details** in the banner to view a list with the import conflict or error reasons and possible resolutions. After you dismiss the banner notification, you cannot view it again.

You cannot see banner notifications for tasks that are performed by another user.

Assign a Tag to an Object in VCF Operations

You can assign tags to objects connected to vCenter instances that are added to your VCF Operations instance.

- Verify that you have **Tags** > **Attach/Detach** permissions in VCF Operations.
- Verify that the VCF Operations instance is of version 9.1.

If you want to assign a tag to more than one object, the objects that you select must all meet the criteria for the tag assignment. For example, if you want to assign the tag GOLD from a tag category with associable object type cluster, the objects you select before you assign the tag must all be clusters. If you select both hosts and clusters, the tag GOLD does not appear in the list.

1. Log in to VCF Operations.
2. Assign a tag to an object.

Assign a tag from the inventory view.	1. From the navigation bar at the top, click **Operate** and in the left navigation pane, select **Inventory**. 2. Click an inventory object that is connected to a vCenter instance, and click **Actions** > **Edit Tag Assignment**. 3. Click **Add Tag**. 4. Enter the tag or category name, and from the drop-down menu, select the tag that you want to assign. 5. Click **Save**.
Assign a tag from the tags view.	1. From the navigation bar at the top, click **Manage** and in the left navigation pane, select **Fleet Management** > **Tags**. 2. Click the **Tag Assignments** tab. 3. Before you can assign tags, use the filters to find the objects to which you want to assign tags. **Note:** If you cannot see the filter Custom Groups, verify that you have the REST API read privileges assigned. 4. Select the check box next to the object to which you want to assign a tag, and click **Assign Tags**. 5. The **Assign Tags** dialog box appears. 6. Optional: If you want to view the selected object, click the arrow next to Selected objects. 7. Click **Add Tags**. A text field appears. 8. Enter the name of the tag, or the name of the tag category in which the tag is added. 9. From the drop-down menu, select the tag, and click **Assign**.

The tag is assigned to the object. If the tag cannot be assigned, click **View Details** in the banner to learn more about the reasons and possible resolutions. After you dismiss the banner notification, you cannot view it again. After the tag assignment task completes, there might be a delay before the updated assignment appears when you search and filter by tag in the VCF Operations instance.

Unassign a Tag from an Object in VCF Operations

You can only unassign tags that are managed by the VCF Operations instance.

- Verify that you have **Tags** > **Attach/Detach** permissions in VCF Operations.

You cannot unassign tags that are created and assigned by using the vSphere Client and not imported in VCF Operations. Those tags are displayed in the table with a lock icon.

1. Log in to VCF Operations.
2. Unassign a tag from an object.

Unassign a tag from the inventory view.	1. From the navigation bar at the top, click **Operate** and in the left navigation pane, select **Inventory**. 2. Click an object in the inventory, and click **Actions** > **Edit Tag Assignment**. 3. Under Tags Assigned, click the times icon **Figure 439:**  next to the tag you want to unassign from the object. The tag appears under Tags to Unassign. 4. Click **Save**.
Unassign a tag from the tags view.	1. From the navigation bar at the top, click **Manage** and in the left navigation pane, select **Fleet Management** > **Tags**. 2. Click the **Tag Assignments** tab. 3. Before you can unassign tags, you must use the filters to find the objects from which you want to unassign the tag. To unassign a tag, you must select only objects that have the tag that is assigned to them. You must select the checkbox**Exclude Objects** > **With No Tags** . **Note:** If you cannot see the filter Custom Groups, verify that you have the REST API read privileges assigned. 4. Select the check box next to the object to which you want to assign a tag, and click **Unassign Tags**. You can only remove tags that are assigned to all objects that you select. For example, if you want to remove the tag GOLD from objects, and you select 15 objects but 1 of them does not have the GOLD tag assigned, you cannot unassign the tag from the remaining 14 objects. The Unassign Tags dialog box appears. 5. Optional: If you want to view the selected object, click the arrow next to Selected objects. 6. To remove a tag, under Common Tags, click the times icon **Figure 440:**  next to the tag name. 7. Click **Unassign**.

The tag is removed from the objects. After the tag removal task completes, there might be a delay before the tag assignment disappears when you search and filter by tag in the VCF Operations instance.

Edit a Tag Category in VCF Operations

You can edit the description of any category in VCF Operations. To edit advanced settings of the category, special conditions apply. You cannot change the category name.

- Verify that you have **Tags** > **Manage** permissions in VCF Operations.

1. Log in to VCF Operations.
2. From the navigation bar at the top, click **Manage** and in the left navigation pane, select **Fleet Management** > **Tags**.
3. In the table with available categories, select the check box next to the category you want to edit.

   Note:

   You can edit only one category at a time.
4. Click **Edit**.
5. Enter a new description for the category.
6. Optional: To change the number of tags allowed per object and the object types to be associated with tags from the category, click **Advanced Settings**, and update the values.

   Important:

   - If during the creation of a category, the tags allowed per object are set to Multiple, you cannot change that value to Single.
   - If during the creation of a category, the associable object types are set to All object types, you cannot change that value at a later time. Any object type that you select when you create the category cannot be deselected later.
7. Click **Save**.

Edit a Tag in VCF Operations

You can edit the description of any tag in VCF Operations. You cannot change the tag names.

- Verify that you have **Tags** > **Manage**permissions in VCF Operations.

1. Log in to VCF Operations.
2. From the navigation bar at the top, click **Manage** and in the left navigation pane, select **Fleet Management** > **Tags**.
3. In the table with available categories, click the double arrow button next to the category that contains the tag you want to edit.

   A table with all tags added to this category appears on the right.
4. In the table with available tags, select the check box next to the tag you want to edit.

   Note:

   You can edit only one tag at a time.
5. Click **Edit**.
6. Enter a new description for the tag, and click **Save**.

Remove a vCenter Instance from a Tag Category

After you remove a vCenter instance from a tag category, you cannot use VCF Operations to unassign any tags from that category that are assigned to inventory objects added to that vCenter instance. If you delete the tag or tag category from VCF Operations, it remains available on the vCenter instance.

- Verify that you have **Tags** > **Manage**permissions in VCF Operations.

After you perform a tag management task from VCF Operations, VCF Operations assumes the tag management role for the tags and categories related to that task. For example, when you push categories from VCF Operations to vCenter instances, or when you import tags and categories from vCenter instances to the VCF Operations instance. When VCF Operations assumes the management roles for a tag category, you can use VCF Operations for all tag management tasks, including the following tasks:

- Unassign any tags from that category from the objects added to a vCenter instance. This applies even if the tag is assigned through the vSphere Client.

- Delete a tag or a tag category. When you delete the tag or category in VCF Operations, the tag or category is also deleted from the vCenter instance.

When you remove the vCenter instance from a tag category, the tag category is no longer managed by VCF Operations for that vCenter instance. When VCF Operations no longer manages a tag category:

- You cannot use VCF Operations to unassign the tags from that category from any of the inventory objects added to that vCenter instance. A lock icon is displayed next to the tag.
- If you delete the tag category or any tag added to it by using VCF Operations, the tag or category is not deleted from the removed vCenter instance.

If you perform a tag management task for that tag category from VCF Operations, VCF Operations regains the tag management role for the tag category. Examples of such tag management tasks are:

- Assigning a tag from that category to any object in the removed vCenter instance by using VCF Operations,
- Pushing that category from VCF Operations to the vCenter instance,
- Importing the category from the vCenter instance to VCF Operations.

1. Log in to VCF Operations.
2. From the navigation bar at the top, click **Manage** and in the left navigation pane, select **Fleet Management** > **Tags**.
3. Click the **Tag Definitions** tab.
4. In the table with available categories, click the double arrow button next to the category from which you want to remove the vCenter instance.
5. Click the **Available In** tab.
6. In the table with the vCenter instances, select the check box next to the vCenter instance you want to remove, and click **Remove**.
7. Select the check box to confirm that you understand the impact from removing the vCenter instance from the tag category.
8. Click **Remove**.

The vCenter instance is removed from that tag category.

Delete a Tag Category in VCF Operations

You can delete a category if no tags are added to it.

- Verify that you have **Tags** > **Manage** permissions in VCF Operations.

1. Log in to VCF Operations.
2. From the navigation bar at the top, click **Manage** and in the left navigation pane, select **Fleet Management** > **Tags**.
3. In the table with available categories, click the double arrow button next to the category that you want to delete.

   A table with all tags added to this category appears on the right.
4. Select all tags and click **Delete**.

   Note:

   You cannot delete tags if they are assigned to objects. For information on how to unassign a tag from an object, see Unassign a Tag from an Object in VCF Operations.

5. Select the check box to confirm that you understand the impact from deleting the tag.
6. Click **Delete**.

   If no tags are added to the category but the Delete button is deactivated, click the refresh button

   at the top-right corner.
7. In the table with available categories, select the check box next to the category that you want to delete, and click **Delete**.
8. Select the check box to confirm that you understand the impact from deleting the tag category.
9. Click **Delete**.

Push Tag Categories to a vCenter Instance or a Custom vCenter Group

After you create or import tag categories and tags into a VCF Operations instance, you can push these tag categories to other vCenter instances. To push categories to vCenter instances part of custom groups, the custom groups must be of type *Tag Group*.

- Verify that you have vSphere Tagging privileges for the target vCenter instance.
- Verify that the target vCenter instance is of version 9.0 or later.
- Verify that you have **Tags** > **Manage** permissions in VCF Operations.
- If you want to push tag categories to vCenter instances that are part of custom groups, verify that there are custom groups of type *Tag Group*created in the VCF Operations instance, and vCenter instances are added to those custom groups. Even if other object types are also added to the custom groups, tags and categories are pushed only to vCenter instances. For more information about creating custom object groups in VCF Operations, see Managing Custom Object Groups in VCF Operations.

By using VCF Operations, you create and manage tags and categories from a single place, at scale. You can then push the created categories and the tags added to them to as many vCenter instance as you need, instead of creating the categories manually in each vCenter instance.

When you try to push a tag category to vCenter instances, if the tag category you try to push, has some of the same properties as a tag category in the target vCenter instance, but differences in other properties, the operations fails and the categories are displayed as conflicts. Similarly, if a tag added to the category have some of the same properties as a tag in the same category in the target vCenter instance, but differences in other properties, the operation fails and the categories are displayed as conflicts.

> **Important:**
>
> If you cannot see the conflict details, verify that you have the REST API read privileges assigned for the VCF Operations instance.

To proceed with pushing the categories, you must first manually resolve the conflicts by following the guidance provided in the details for each conflict. Alternatively, you can select to overwrite the category properties in the target vCenter instance wherever a conflict is detected. The following types of conflicts can only be resolved manually, even if you select the check box:

Conflict	Resolution
A category or tag with the same name and ID exists both in VCF Operations, and in the vCenter instance selected for the push operation. The value for **Tags per object** is set to **One Tag** in VCF Operations, but it is set to **Multiple Tags** in the vCenter instance.	In VCF Operations, change the value for the **Tags per object** attribute to **Multiple Tags**.
A category or tag with the same name and ID exists both in VCF Operations, and in the vCenter instance selected for the push operation. The **Associable object types** attribute in the vCenter instance contains more object types than the **Associable object types** attribute in VCF Operations.	In VCF Operations, edit the values for the **Associable object types** attribute and verify they are the same as in the vCenter instance.

Conflict	Resolution
A category or tag with the same name but a different ID exists both in VCF Operations, and in the vCenter instance selected for the push operation.	Rename the category or tag in the vCenter instance, or delete the conflicting category or tag from VCF Operations or from the vCenter instance.

When a task is in progress, you cannot initiate the same task again. For example, if another user or you initiate a push of categories, you cannot initiate push again before the first task completes.

1. Log in to VCF Operations.
2. From the navigation bar at the top, click **Manage** and in the left navigation pane, select **Fleet Management** > **Tags**.
3. From the table with available tag categories, select the check box next to the tag categories you want to push.

   Note:

   You can push up to 20 tag categories at a time.
4. Click **Push Categories**.

   The **Push Categories** dialog box appears.
5. Select the vCenter instances or custom groups to which you want to push categories:
   - To push categories to specific VCF domain vCenter instances, click the **VCF Domain vCenter Instances** tab, and select VCF Instances or a VCF domain vCenter instances from the list.
   - To push categories to custom groups, click the Custom Groups tab, and select a custom group from the list.
6. Optional: If you want to overwrite the category properties in the target vCenter instance when a conflict is detected, select the check box at the bottom of the dialog box.
7. Click **Push**.

The tag categories are pushed to the selected vCenter instances or vCenter instances in a custom group.

You cannot see banner notifications for tasks performed by another user.

You can assign the tags to objects in the VCF Operations inventory. For more information, see Assign a Tag to an Object in VCF Operations.

You can also use the tags and categories with existing automation scripts or assign tags to inventory objects by using the vSphere Client. For more information about assigning tags to specific inventory object, see How Do You Assign or Remove a vSphere Tag.

Managing the Configuration of your Environment

You can use VCF Operations configuration management feature to monitor and view the configuration settings of vCenter instances and vSphere clusters that have drifted from the desired configurations, without the need to manually track each change.

Configuration management shows the changes in product configurations at a point in time and allows you to compare the changes to the assigned desired values. It provides a proactive method to monitor configuration changes.

Why use Configuration Management

Configuration management offers several benefits, including:

1. Consistency: Ensures that vCenter instances and vSphere clusters are configured consistently. This helps reduce potential issues from drifts.
2. Compliance: Helps ensure that vCenter instances and vSphere clusters are configured to meet organizational policies and regulations.
3. Scalability: Allows for infrastructure configuration at scale.

For more information, see Recommendations for VCF Configuration Management.

Managing Configuration Templates in VCF Operations

You can define a configuration state based on your current environment using configuration templates. You can create, edit, or delete configuration templates for vCenter instances and for vSphere clusters in VCF Operations .

A configuration template is a collection of product configuration settings representing desired values that you are monitoring. A configuration template enables an administrator to define and monitor the configurations. For more information, see Template Management and Versioning.

The **Configuration Templates** tab provides an overview of all configuration templates, including the object type (vCenter or cluster) assigned to each template. It also includes the description, number of policies assigned to each template, template author, and template creation date.

Optionally, you can manage configuration templates using a supported source control solution. For more information, refer to Managing Configuration Templates with Source Control.

Creating Configuration Templates for vCenter Instances

You can create configuration templates for vCenter instances and assign policies. You can then use these templates to detect drift for vCenter instances.

- If the target vCenter is not deployed as a part of the same VCF instance as the VCF Operations console, ensure that the user is added to the vCenter Administrators group and that the account used to add vCenter to VCF Operations has administrator privileges.

  Note:

  This applies only if you select user credentials when integrating a vCenter cloud account in VCF Operations. For more information, see Configuring a vCenter Account in VCF Operations.
- Verify that the user has the required **Configuration Management** roles to access templates. For more information, see the Access Control: Roles Tab.
  - **Manage Privilege**: For creating templates, assigning them to policy, and running drift check.
  - **View Privilege**: For viewing template content and viewing existing drifts.
- Verify that your cloud account is registered to a cloud proxy or a cloud proxy group.

1. In the VCF Operations navigation bar at the top, click **Manage** and in the left navigation pane, click **Fleet Management**, and then click **Configuration Management**.

   The vCenter Instances tile is selected by default.
2. Click **Create Configuration Template**.
3. On the **Template Setup** tab, enter a name and description for the template and ensure that the **Object Type** is set to **vCenter**.
4. Select a **Configuration Source** vCenter from which you want to extract the configuration settings and click **Next**.

   Note:

   The **Configuration Source** vCenter must meet the following criteria:

   - vCenter 9.0 or later.
   - Possess a VCF Configuration Management license.
   - vCenter must be registered to a cloud proxy or a cloud proxy group.

   VCF automatically applies the required license and cloud proxy configuration to newly deployed vCenter instances (version 9.0 and later). For vCenter instances (version 9.0 and later) that are imported into VCF , you must manually apply the license and configure the cloud proxy. For more information, see Assign a Primary License to a vCenter Instance and Add a Cloud Proxy.

5. On the **Configuration Settings Selection** tab, select the configuration settings that you want to include in your template and click **Next**.
6. Optional: On the **Assign vCenters using Operations Policy (Optional)** tab select one or more VCF Operations policies to assign to your template and then click **Next**.

   Note:

   Templates can be created without selecting a policy. You can assign to policies after template creation. For more information, see Assign Configuration Templates to Policies.
7. Optional: On the **Review** tab, a preview of your configuration template is displayed in the right pane. Review the template and then click **Create**.

   The configuration template for the vCenter is created and listed on the **Configuration Templates** tab.

After creating the configuration template for vCenter, you can perform the following steps:

- Click the configuration template to view template details and assigned objects.
- Click the vertical ellipsis to edit, assign objects using policy, or delete the configuration template.

Creating Configuration Templates for Clusters

You can create configuration templates for vSphere clusters and assign objects using policies. You can then use these templates to detect drift for clusters in VCF Operations.

- Verify that you have vCenter version 9.0 or later.
- Verify that you have registered your vCenter with a cloud proxy.
- Clusters are managed with vSphere Configuration Profiles (VCP) in individual vCenter instances. Enable vSphere Configuration Profiles for clusters in vSphere. For more information, see vSphere Configuration Profiles.
- Verify that the user has the required **Configuration Management** roles to access configuration templates. For more information, see the Access Control: Roles Tab.
  - **Manage Privilege**: For creating templates, assigning them to policy, and running drift check.
  - **View Privilege**: For viewing template content and viewing existing drifts.
- Verify that your cloud account has been assigned a collector group or a collector.
- Ensure that all the required settings are configured in the source cluster's VCP configuration. Only settings defined in the source appear in this template.

1. In the VCF Operations navigation bar at the top, click **Manage** and in the left navigation pane, click **Fleet Management**, and then click **Configuration Management**.
2. Click the **Clusters** tile.
3. Click **Create Configuration Template**.
4. In the **Template Setup** tab, enter a name and description for the template and ensure that the **Object Type** is set to **Cluster**.

   Note:

   Only settings defined in the source cluster's VCP configuration appear in this template.
5. For **Configuration Source**, use the drop downs to view and select the configuration sources from which you want to extract the configuration settings, and then click **Next**.
   a) **Select a vCenter**: Select a vCenter.
   b) **Select a Cluster**: Select a cluster. Only vSphere Configuration Profiles enabled clusters are listed.

   After you select the vCenter, the **Source Version** of the vCenter is displayed.

6. On the **Configuration Settings Selection** tab, select the configuration settings that you want to include in your template and click **Next**.
7. Optional: On the **Assign Objects using Operations Policy (Optional)** tab select one VCF Operations policies to assign to your template and then click **Next**.

   Note:

   Only a single cluster configuration template can be assigned to a policy. To activate a different template, you must first set the current one to Inherited or Deactivated. You cannot assign a new template to a policy if another template is still active. You can also assign to policies after template creation. For more information, see Assign Configuration Templates to Policies.
8. Optional: On the **Review** tab, a preview of your configuration template is displayed in the right pane. Review the template and then click **Create**.

   The configuration template for the cluster is created and listed on the **Configuration Templates** tab.

After creating the configuration template for a cluster you can perform the following steps:

- Click the cluster configuration template to view template details and assigned objects.
- Click the vertical ellipsis to edit, assign objects using policy, or delete the cluster configuration template.

Assign Configuration Templates to Policies

You can assign your vCenter or cluster configuration templates to policies in VCF Operations.

If you have not assigned your vCenter or cluster configuration templates to policies at the time of template creation, you can do so after the template creation process.

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations**, and then click the **Policy Definition** tile.
2. Select the policy you want to assign.
3. Select **Configuration Templates** and then click the **Edit** icon.
4. For the **Select Object Type** drop-down, the vCenter object type is selected by default and the vCenter configuration templates are listed. To view the cluster configuration templates, use the drop-down and select the **Cluster Compute Resource** object type.
5. Select the vCenter or cluster configuration template you want to assign to this policy.
6. Click the drop-down and click **Activated** to assign the template for the policy.

   Note:

   Multiple vCenter configuration templates can be assigned to a single policy. But, only one cluster configuration template can be assigned to a single policy. If you want to activate a different cluster configuration template, you need to set the current active template to either inherited or deactivated. For example, if you have two cluster configuration templates: Template 1 and Template 2, and Template 1 is assigned to the selected policy in VCF Operations, you cannot assign Template 2 for the same policy. To activate Template 2, you must deactivate Template 1 or change its assignment to 'Inherited'. Once Template 1 is set to 'Inherited', you can then activate Template 2 for the same policy. For more information, see Default Policy and Select the Inherited Policy Details.
7. Click **Save**.

Managing Configuration Templates with Source Control

Integrating the VCF Operations configuration templates with a source control repository allows you to control versioning, auditing, and the management of your configuration templates.

Integrating with Source Control

Integrating with a source control repository enables you to manage configuration templates under source control. This functionality also facilitates accurate configuration updates, supports change control with reviews, and integrates continuous integration and continuous development for automated testing and validation.

GitLab and GitHub are supported source control repositories.

The source control repository system must support the following Console requirements:

- Reachable by FQDN
- Access Token can be generated by the source control and used by the Console
- Accepts a Branch input field
- Accepts a Repository input field

This integration has been tested with the following Git-based source control solutions:

- **GitLab**: Version 15.x, 16.x, 17-17.6, and 18.8.1.
- **GitHub**: Version 3.13.13.

Generating Access Tokens in GitLab (Example)

The following is a generic workflow for generating an access token in GitLab. Your experience may vary based on your environment.

1. Log in to your GitLab account.
2. Click the **Profile** icon and click **Edit profile**.
3. Click **Access Tokens**.
4. Enter the **Token name** and select the **Expiration date**.
5. Select the following scopes. Scopes set the permission levels granted to the token.
   a. api
   b. read_api
   c. read_repository
   d. write_repository
6. Click **Create personal access token**.

Generating Access Tokens in GitHub (Example)

The following is a generic workflow for generating an access token in GitHub. Your experience may vary based on your environment.

1. Log in to your GitHub account.
2. Click the **Profile** icon and click **Settings**.
3. Click **Developer settings**.
4. Expand **Personal Access Tokens** and then select **Tokens (classic)**.
5. Click **Generate new token** and then click **Generate new token (classic)**.
6. Enter the **Name** and select the **Expiration** date.
7. Select the following scopes. Scopes define the access for personal tokens.
   a. repo
   b. Admin:repo_hook
   c. project
8. Click **Generate token**.

Configuring Source Control in VCF Operations

You can configure source control settings in VCF Operations to synchronize configuration templates to and from a repository.

- Verify that your VCF Operations fleet management can establish a connection to the appropriate the source control solution.
- Ensure bi-directional HTTPS connectivity between the VCF Operations *management appliance and the Git repository.*
- Verify that you are using a supported source control solution.
- Generate access tokens to authenticate and authorize a secure connection to the repositories.
  - **GitLab**: Generate an access token for your repository following the documentation from GitLab.

    Note:

    The access token must have the Maintainer or Owner role. Ensure the access token has an expiration date and the following scopes:

    - Read access to the repository. (`read_repository` in GitLab)
    - Write access to the repository. (`write_repository` in GitLab)
    - Read access to the project. (`read_api` in GitLab)
    - Read and write access to the project. (`api` in Gitlab)
  - **GitHub**: Generate an access token for your repository following the documentation from GitHub.

After you configure a source control solution, it becomes the primary record holder for the configuration templates.

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Control Panel**, and then click **Source Control**.
2. For **Source Control Type**, select a source control solution.
3. Enter the **Source Control Server** address where the configuration templates will be stored.
4. Enter the access token from your source control type in the **Access Key**.
5. Specify the **Branch** for storing or reading content.
6. Provide the **Repository** path for storing or retrieving contents.
7. By default, **Code Review** is enabled by default. The code review process creates a merge request for any operation on a configuration template. The content syncs to the repository after the merge request is approved.

   Note:

   To disable the code review process, uncheck the **Code Review** checkbox.
8. Click **Test Connection** to verify if the integration is successful. If the **Connection Result** is successful, click **Save** to complete the integration.

After the setup is complete, you can manage the configuration templates from your source control solution.

Note:

If your VCF Operations instance uses a self-signed certificate, disable the SSL verification setting of the webhook for the Git repository.

Template Organization and Repository Structure: Configuration templates are automatically created and organized in a defined folder structure: `Operations-ConfigTemplate/config-templates/vCenter`. Each template must be stored in a unique folder in the source control repository.

Template Name and Storage: Each configuration template must have a unique name to avoid duplication.

Template Content and File Structure: For each configuration template, three files are automatically created in the repository:

1. `<templatename>-desiredState.json`: Contains the desired state of the template.
2. `<templatename>-metaData.json`: Contains metadata information like template name, version, and selected configuration.
3. `<templatename>-manifest.json`: Used to manage and maintain the template's structure and metadata.

Syncing Configuration Templates with Source Control

After you configure the source control solution in VCF Operations, sync the configuration templates with the source control repository. This topic outlines the different workflows that you can follow to manage the configuration templates and integrate them with the repository.

Review, Sync, and Merge Templates from VCF Operations

Review Process Enabled

When you create, edit, or delete a template in VCF Operations, a **merge request** is created in source control and is visible in the configuration template list view. You must review and approve the request in your source control solution. After the request is approved, the template status changes to **OUT_OF_SYNC**. Sync the template using the **Sync** option in VCF Operations. The template status changes to **IN_SYNC** after the sync is complete.

> **Note:**
>
> The edited or deleted content is reflected only after the merge request is approved and the content is synced.

Review Process Disabled

When you edit a template, the template is directly edited in the source control repository without a request. If you delete a template from VCF Operations, the template is marked **OUT_OF_SYNC** and is directly deleted from the source control during the next sync.

Sync Existing Templates in VCF Operations

Code Review Process Enabled

When source control integration is enabled, merge **requests** are created for syncing all existing templates from VCF Operations to the source control repository. After the request is approved, the template status changes to **OUT_OF_SYNC**. Use the **Pull from GitHub** option to sync content from the repository. The status of the template changes to **IN_SYNC** after the sync is successful.

Code Review Process Disabled

The status of all existing templates reflects as **OUT_OF_SYNC**: No request is raised. You must use the **Sync** option to sync content from source control. The status changes to **IN_SYNC** after the sync is complete.

Sync Remote Templates to VCF Operations

If you edit or delete existing templates in the source control repository, the status of the corresponding templates reflects as **OUT_OF_SYNC** in VCF Operations.

When you edit a template in the source control repository, you must **Pull from GitHub**, and after the sync is successful, the updated template content gets reflected in VCF Operations and the status changes to **IN_SYNC**.

When you delete a template in the source control repository, the template is removed from VCF Operations.

When you configure a source control in VCF Operations, existing templates without any content in the source control repository are created in VCF Operations as placeholders and reflect the **OUT_OF_SYNC** status. You must sync the content to populate the templates. The status changes to **IN_SYNC** after the sync is complete.

Additional Information

Merge Request Discarded

A merge request can be discarded if the changes proposed in the merge request are no longer valid or required. If a merge request is discarded or closed by the user, a warning message is displayed informing the user that the changes made to the corresponding template in VCF Operations are not merged and the current template might not reflect the desired state in source control

You can edit the template to make the necessary updates and address the issues that caused the merge request to be discarded. After you make the update, a new request is raised. This ensures that the latest template changes are submitted for review.

Drift Detection

Drift detection is skipped if the template does not exist in the source control repository. This might occur in scenarios where the template has not been synced with the source control.

Updating Source Control Configurations

Editing or Updating Branch or Repositories

You can update the branch or repository or switch to a new branch or repository. Once done, the templates from the previous source control repository displays the **OUT_OF_SYNC** status. The existing templates in VCF Operations will sync to the new branch or repository. For more information, see Sync Existing Templates in VCF Operations section.

If you delete a source control configuration, the templates stored locally in VCF Operations remain unchanged, however the status changes to **OUT_OF_SYNC**.

> **Note:**
>
> If you want to update your source control type, you must first delete the existing configuration.

Using the Configuration Management Overview Page

The configuration management overview dashboard allows you to view and monitor vCenter and vSphere cluster drift statuses.

You can view and compare drifts and identify settings that have changed.

A drift notifies you of any deviation from the assigned template values. Use the **Configuration Management Enabled Objects Summary** bar to view the total number of objects and their enabled statues. Enabled objects are composed of all the vCenter and vSphere clusters that have a configuration template attached to them.

The **vCenter Instances Drift Status** and **Cluster Drift Status** charts represent the state of the enabled vCenter instances and vSphere clusters at a given point in time. All the potential drift statuses and their definitions are listed after this:

- **In Progress**: This state shows that active drift detection is currently executing for vCenter instances or vSphere clusters.
- **Not Drifted**: This state shows vCenter instances and vSphere clusters with no drifts from the configuration state.
- **Drift Detected**: This state shows when vCenter instances or vSphere clusters have deviated from their desired template configuration.
- **Drift Check Failed**: This state occurs under the following scenario's
  - vCenter instances or vSphere clusters are unavailable.
  - The drift has not been computed against that specific resource.
  - An internal error has occurred.
- **Unavailable**: This state shows a list of vCenter instances and vSphere clusters that are either
  - Below version 9.0
  - Not registered with the cloud proxy.

Additionally, you can click the **Objects** tab to view the objects assigned to your vCenter instances and clusters and view all **Objects**.

Scheduling Drift Detection

You can automate the drift detection process and schedule drift detection in VCF Operations.

Scheduling drift detection enables you to run automatic drift checks on your vCenter instances or clusters at an interval of your choice.

1. In the VCF Operations navigation bar at the top, click **Manage** and in the left navigation pane, click **Fleet Management** > **Configuration Management**, and then click **Schedule Drift Detection**.
2. For step 1 of scheduling drift detection, enter the **Configuration Drift Check Information**.
   a) Provide a name for the job. This information is displayed in the calendar in the **Automation Central** page.
   b) Provide a description for the job.
   c) Select the **Adapter Type**.

      The vCenter adapter type is the only available adapter type for both vCenter and cluster templates.
   d) Select the **Object Type**.

      You can schedule drift detection for the vCenter or the Cluster Compute Resource object type.
   e) Click **Next**.
3. For step 2, **Select an object**. **Select Scope** of the object which contains the vCenter on which VCF Operations must run the automation jobs.

   For example, if you click and drag a VCF instance from the right hand Scope box to the left hand Scope selection box, it will include all vCenter instances or clusters, depending on your template, into the **Scope Selection**. You can filter out objects in the next step if needed.

   a) Drag and drop the object type to the left pane.
   b) To delete members, click the X icon.
   c) Click **Next**.
4. Optional: For step 3, you can **Filter Criteria**, and add multiple criteria for the vCenter object type.
5. Optional: Click **Preview Scope** at the bottom-right corner to see on which vCenter instances the jobs are executed.
6. Optional: Close the **Preview Scope** and click **Next**.
7. Set the **Schedule** and click **Create**.

Property	Description
Start Date	Set a start date which is on, or after the current date, for the job to start. Select a date from the calendar displayed on the page. **Note:** There is a delay of up to five-minutes for a job to run. The actual start time for an action also depends on the action itself and the number of target objects involved.
Time zone	For the start time and date to be calculated for the job, select a time zone. Choices are: • Browser - based on the current location reported by the browser. • Host - based on the current location of the host machine. • GMT based time zone - Based on a location calculated as per UTC+0.
Start Time	Select a start time for the job. The drop-down menu provides options in five-minute intervals, starting closest to your current browser reported time.
Recurrence	Set a recurrence for the job. The choices are: • **One-Time**. No further configuration is possible. • **Daily**. Set the recurrence in days. Set how frequently it must run by providing a value in **Run Every** field. • **Weekly**. Select the days of the week when you want the job to run, by clicking the day of the week abbreviation. • **Monthly**. Select the months when you want the job to run, by clicking the month abbreviation. By default, all the months are selected. For the months that you select, you can configure the job to run:

Property	Description
	– On specific days of the month, by clicking the number. Or, the last day of the month, without specifying the exact date, by clicking **Last**. – On the specific number of the week (first, second, third, fourth or the last) in the month, combined with the specific day of the week. You can make multiple selections for each of the options in the drop down menu. For the **Daily**, **Weekly**, **Monthly** options, you can set when the job must end based on a date, or the number of occurrences.

A new job is created to run scheduled drift detection in the automation central page. For more information, see Automation Central.

Detecting Drifts and Downloading Drift Reports for vCenter Instances

You can detect drifts for each configuration template you have created for a specific vCenter.

To view vCenter configuration setting values and detect drifts, you must:

- Ensure that you have created at least one configuration template for your vCenter.
- Ensure that the user is configured with the **Administrator** role in VCF Operations. For more information, see Privileges Required for Configuring a vCenter Account in VCF Operations.
- Verify that you have **Configuration Management** privileges to access configuration templates. For more information, see Managing Users and Access Control.
  - **Manage Privilege**: For creating templates, assigning them to an operations policy, and running drift check.
  - **View Privilege**: For viewing template content and viewing existing drift.
- Verify that your vCenter instance is configured, registered with the cloud proxy, and has one or more operations policy assigned to the configuration template.

1. In the VCF Operations navigation bar at the top, click **Manage** and in the left navigation pane, click **Fleet Management** > **Configuration Management**, and then click the **Drifts** tab.

   On the **Drifts Details** page, you can view the number of enabled vCenter instances on the **vCenter Instances** tile. You can also view the number of vSphere Configuration Profiles enabled clusters on the **Clusters** tile. For more information, see Detecting Drifts and Downloading Drift Reports for Clusters.

   By default, the vCenter instances tile is selected.
2. Select one or more vCenter instances.
3. Click **Detect Drift** to detect configuration drift.

   This detects drifts across all the configuration templates and displays the result in the Drifts table.
4. To view the drift for each vCenter, click the vCenter instance.
5. Select a **Configuration Template** to view the drift details.

   If a drift is detected, the right pane displays the drifts between the **Current vCenter Value** and the **Configuration Template Value**. If no drift is detected for the selected template, the message **No drift found for this configuration template since last configuration drift check** status is displayed.

   - **Detect Drift**: To further detect drift for a particular template in a vCenter instance.
   - **View Template**: To view the template.
6. Navigate to the **Drifts Details** page, select one or more vCenter instances, and click **Download Drift Status Report** to download the historical drift detection reports PDF or XLS format.

   The drift report displays the date, resource, template, and drift status for each vCenter instance. By default the report shows data for the last seven days. You can update the duration to up to 30 days from the Global Settings page. For more information, see List of Global Settings.

Detecting Drifts and Downloading Drift Reports for Clusters

You can detect drifts for each configuration template you have created for specific clusters in VCF Operations. The drift status is visible only for vSphere clusters that have vSphere Configuration Profiles enabled.

- Verify that the target clusters are being managed by vCenter version 9.0 or later.
- Verify that vCenter managing your target clusters is registered with a cloud proxy.
- Target clusters are managed with vSphere Configuration Profiles. Ensure that vSphere Configuration Profiles are enabled for all target clusters in vSphere. For more information, see vSphere Configuration Profiles.

1. In the VCF Operations navigation bar at the top, click **Manage** and in the left navigation pane, click **Fleet Management** > **Configuration Management**, and then click the **Drifts** tab.
2. Click the **Clusters** tile.

   The drift status of all vSphere Configuration Profiles enabled clusters that are associated with the **VCF Instance** is displayed. You can also view the number of clusters that are managed by configuration templates.
3. Select one or more clusters and then click **Detect Drift** to detect configuration drift.

   This detects drifts across all the cluster configuration templates and displays the result in the Drifts table.
4. Click **Download Drift Status Report** to download the historical drift data.

   The drift report displays the date, resource, template, and drift status for each cluster. By default the report shows data for the last seven days. You can update the duration to up to 30 days from the Global Settings page. For more information, see List of Global Settings.
5. To view the Effective Cluster Configuration for individual clusters, click the number that is displayed in the **Templates (Including Inhertited)** column.
6. To view the drift for each cluster, click a cluster.

   The **Drifts Details** page opens and lists the total number of Hosts and the number of drifted Host in the cluster.
7. Switch between Hosts to view the associated cluster configuration templates, the drift status, and the drifted values.
8. Click **Download Drift Status Report** to download cluster drift report for individual hosts.

   You can download the drift status report as a pdf or .xls file format.

Viewing Drifts and Manual Remediation Steps for Clusters

You can view the drift status and manually remediate a cluster that has drifted from the configuration template and its expected configuration values.

- Ensure that you have created a configuration template for the cluster and assigned it to a policy. For more information, see Creating Configuration Templates for Clusters.
- Ensure that you have run drift detection for the cluster. For more information, see Detecting Drifts and Downloading Drift Reports for Clusters.

1. In the VCF Operations navigation bar at the top, click **Manage** and in the left navigation pane, click **Fleet Management** > **Configuration Management**, and then click the **Drifts** tab.
2. Click the **Clusters** tile.

   The drift status of all vSphere Configuration Profiles enabled clusters that are associated with the **VCF Instance** is displayed. You can also view the number of clusters that are managed by configuration templates.
3. View the **Drift Status** in Drifts table.
   - Not Drifted: The cluster has not drifted from the configuration template.
   - Drift Detected: The cluster has drifted from the configuration template and requires remediation.

4. Click a cluster that displays the **Drift Detected** status to open the **Drifts Details** page.
5. Click **Download Remediation JSON** to download the remediation draft for the cluster.

   This JSON file contains the configuration template JSON file that can be applied to each cluster manually using the vCenter vSphere Configuration Profiles API.
6. The following outlines the high-level steps to use the downloaded JSON file to manually apply to a cluster through the vSphere Client UI. Navigate to the destination vSphere Client and create a draft using the downloaded json file. For more information, see Create a Draft Configuration in vSphere Client.
7. Run the remediation for the drifted cluster. For more information, see Remediate a Cluster Against the Desired or Draft Configuration.
8. Once done, log in to VCF Operations to the **Clusters** tile in Configuration Management.
9. Select the cluster, and click **Detect Drift** to detect configuration drift.

   The drift status changes to **Not Detected** for the cluster.

Configuration Management FAQ

This topic covers some frequently asked questions about configuration management in VCF Operations.

- What are the prerequisites for setting up a configuration template?
  - To view the prerequisites for setting up configuration template, see Creating Configuration Templates for vCenter Instances.
- When does a configuration template creation fail?
  - Configuration template creation can fail, if the template size exceeds 9 MB. You can copy the configuration template contents to a file to check the file size and ensure it is under 9 MB.
  - vCenter configuration template creation can fail, if a connection issues occurs between vCenter and the configuration adapter. You can check the connection status of vCenter and the configuration adapter to ensure that a proper connection exists between the two. To troubleshoot, see Cloud Proxy Troubleshooting and vSphere Management Pack Issues.
- What should I do if there is an error while displaying drift details?
  - If you encounter an error while attempting to view drift details, it may be due to connectivity issues between vCenter and the configuration adapter. To troubleshoot, see Cloud Proxy Troubleshooting and vSphere Management Pack Issues.
- Why is the drift report PDF generated with "No drift details available for the selected resource/resources" message?
  - This message may appear if the **Detect Drift** operation has never been triggered for the selected resources. To resolve this issue, run the **Detect Drift** process on the relevant resources first, and then try downloading the report again.
- How can I remediate a vCenter if I receive a notification indicating that it has drifted from a configuration template?
  - Remediation cannot be done at the VCF Operations console at this time. You must log in to the vCenter and manually revert the drifted configurations to their original values as defined in the template to make the vCenter compliant again.
  - If you want the drifted values to be the configuration template values, edit the template under configuration management to match the current configuration of the vCenter.
- How do I troubleshoot if a vCenter Drift status is in an **Error** state?
  - Check the connection between VCF Operations and vCenter to ensure that vCenter is in a **Collecting** state within VCF Operations. If it is not collecting, start or resume the data collection for vCenter. This will restore proper drift detection and clear the error. To troubleshoot, see vSphere Management Pack Issues.
- How can I debug when the configuration template sync fails to and from a remote repository?
  - In the VCF Operations navigate to **Administration** > **Control Panel** > **Source Control** and click **Test Connection** to verify if connection with the Git Repository is successful or not. Ensure that the connection is successful before performing any action related to configuration drifts using Git. If the test fails, check the repository URL, credentials, and network connectivity.
- Starting with version 9.1, why does the cluster show an old or incorrect drift status (like "Drifted" or "Not Drifted") immediately after I assign a new template?

– The drift status of the cluster is not reset to "Unavailable" when previous templates are deleted. The system retains those historical drift records and accordingly shows that old drift status for the cluster.
  To clear this stale data and view the accurate status, manually trigger a new drift analysis:
  a. Navigate to the affected cluster object.
  b. Select **Run Drift Detection**.
  c. Allow the analysis to complete to see the current, accurate drift state.

Managing Infrastructure Services for VMware Cloud Foundation

Name resolution and time synchronization are critical services on which your environment relies for connectivity, authentication, certificate management, logging, and so on. You use VCF Operations to update the DNS and NTP configuration for multiple VCF or VVF components.

Update the DNS Server Configuration for VCF Components

VCF Operations uses DNS servers to provide name resolution for the VCF components. You can update the DNS server information that you provided during VMware Cloud Foundation deployment. You can update the DNS server information for single or multiple VCF Instances in a single operation.

- Verify that both forward and reverse DNS resolution is functional for each component using the updated DNS server information.
- Verify that the new DNS server is reachable from each of the VMware Cloud Foundation components.
- Verify that SDDC Manager can reach all VMware Cloud Foundation components over SSH.
- Verify that all VMware Cloud Foundation components are in an active state.

When you update the DNS server configuration, VCF Operations performs DNS configuration updates for the following components:

- SDDC Manager
- vCenter
- ESX
- NSX
- VCF management services
- VCF Automation

If the update fails, VCF Operations rolls back the DNS settings for the failed component. Fix the underlying issue and retry the operation.

Updating the DNS server configuration can take some time to complete, depending on the size of your environment. Schedule DNS updates at a time that minimizes the impact to the system users.

1. Log in to VCF Operations.
2. From the navigation bar at the top, click **Manage**.
3. In the left pane, navigate to **Fleet management** > **Fleet settings**.
4. On the **Fleet settings** page, click the **DNS settings** card.
5. To create a DNS configuration that can be applied to VCF Instances, on the **DNS settings** page, click **Create setting**.
6. On the **Create DNS setting** page, configure the name of the setting, the description, provide a DNS server, and click **Create**.

   For multiple DNS servers, use a comma-separated list.

7. On the **DNS settings** page, in the **Instances** section, select the VCF Instance to update and click **Apply**.
8. On the **Apply setting to instance** page, select the DNS configuration to be assigned to the VCF Instance and click **Assign**.

Update the NTP Server Configuration for VCF Components

VCF Operations uses NTP servers to synchronize time between the VCF core components. You can update the NTP server information that you provided during VMware Cloud Foundation deployment.

- Verify the new NTP server is reachable from all components.
- Time skew between new NTP servers is less than 5 minutes.
- Verify that SDDC Manager can reach all VMware Cloud Foundation components over SSH.

You must have at least one NTP server. When you update the NTP server configuration, SDDC Manager performs NTP configuration updates for the following components:

- SDDC Manager
- vCenter
- ESX
- NSX
- VCF Operations
- VCF Automation
- VCF management services
- VCF Operations for networks

If the update fails, VCF Operations rolls back the NTP settings for the failed component. Fix the underlying issue and retry the operation.

Updating the NTP server configuration can take some time to complete, depending on the size of your environment. Schedule NTP updates at a time that minimizes the impact to the system users.

1. Log in to VCF Operations.
2. From the navigation bar at the top, click **Manage**.
3. In the left pane, navigate to **Fleet management** > **Fleet settings**.
4. On the **Fleet settings** page, click the **NTP settings** card.
5. To create an NTP configuration that can be applied to VCF Instances, on the **NTP settings** page, click **Create setting**.
6. On the **Create NTP setting** page, configure the name of the setting, the description, provide an NTP server, and click **Create**.

   When using multiple NTP servers, use a comma-separated list.
7. On the **NTP settings** page, in the **Instances** section, select the VCF Instance to update and click **Apply**.
8. On the **Apply setting to instance** page, select the NTP configuration to be assigned to the VCF Instance and click **Assign**.

Configuring the NTP Settings for VMware vSphere Foundation Components

To configure the NTP configuration for your VMware vSphere Foundation component, you can use the VCF Operations administration UI.

The Network Time Protocol (NTP) setting is used to synchronize the time of your VCF Operations cluster nodes. Configure the NTP setting in VCF Operations to ensure that all the cluster nodes follow the same time protocol. Following the same time protocol allows your cluster's services to run smoothly. The primary node serves as the NTP server by default and the time of all nodes are synchronized to the primary node. You can continue using the primary node as the NTP server and add another NTP server to serve as a failover if the primary node fails. In case you do not want to use the primary node, you can add the NTP server you wish to use.

1. Log in to the VCF Operations administration interface at https://*primary-node-name-or-ip-address*/admin.
2. Click the **NTP Settings** icon.

   The **Global Network Time Protocol** settings wizard opens. The primary node is listed as the existing NTP server.

   Note:
   You must configure at least two NTP servers to have a back-up if one server fails.
3. Add a secondary NTP server. Enter the IP of FDQN of the NTP server you wish to use in **NTP Server Address** field.
4. Click **Add**.

   The new NTP server is displayed along with the primary node.
5. Ensure that the status of the NTP server is green which means its reachable and working.

Configure Dual-Stack Networking for SDDC Manager

Starting with VCF 9.1, SDDC Manager supports dual-stack (IPv4 and IPv6) networking. You configure dual-stack networking on SDDC Manager to deploy and manage VCF components with dual-stack networking (wherever supported).

- The VCF components in your management domain for this VCF Instance must be on version 9.1 or later.
- IPv6 must be enabled for all ESX hosts in the management domain for this VCF instance.
- SDDC Manager supports static, DHCP, or SLAAC IPv6 address assignment for storage and vMotion vmkernel adapters.
- The FQDN for the SDDC Manager instance must resolve to an IPv6 address.

Native support for IPv6 networking resolves the IPv4 address exhaustion issue. With IPv6 you can avoid stop-gap measures such as Network Address Translation (NAT) and extensive use of private RFC1918 address space.

After you convert to dual-stack networking, SDDC Manager retains backwards compatibility to manage IPv4-only workload domains. However, IPv4-only SDDC Manager cannot manage dual-stack workload domains.

1. Log in to VCF Operations.
2. In the left pane, navigate to **Administration** > **SDDC Manager**.
3. Click **VCF Management** or click **VCF Instances** and click a VCF Instance.
4. On the **Network settings** tab, click **Dual stack networking** and click **Enable dual stack**.

   The **Enable dual stack** wizard opens.
5. On the **Introduction** page, click **Next**.
6. On the **IPv6 configuration** page, enter the IPv6 address of the management network gateway.
7. On the **Validate** page, remediate any issues and click **Next**.
8. On the **Enable dual stack** page, click **Finish**.

   During the configuration process, SDDC Manager becomes unavailable.

Adding or Removing VCF Components Post Deployment

After the automated deployment of components for your VMware Cloud Foundation platform, you can manually add VCF Automation, Identity broker, or a secondary software depot to a VCF instance.

Removing VCF Components Post Deployment

If the manual deployment of a component fails and the component appears listed in the UI, there is no option to delete the listing. To remove the component from the UI, see Broadcom knowledge base article 441333.

Deploying VCF Automation as a Day-N Operation

If VCF Automation was not deployed as part of the initial VMware Cloud Foundation deployment or import of an Aria Automation 8.x instance, you can add it as a Day-N operation using Fleet lifecycle.

Deployment of only one VCF Automation instance per VCF fleet is supported.

Note:

To deploy VCF Automation on custom networking, see Deploying Deferred Components on Custom Networking.

VCF services runtime uses an internal CIDR block of 198.18.0.0/15. If this subnet is currently in use on your network, you must change the internal CIDR block for VCF management services to 240.0.0.0/15 or 250.0.0.0/15. You can specify the internal network to be used during the deployment by using a JSON specification file for VCF and vSphere Foundation. See Use a JSON Specification File to Deploy VMware Cloud Foundation or vSphere Foundation. Alternatively, you can specify the internal network by using the SDDC ManagerAPI for VCF management services, or by using the fleet management API for VCF Automation.

1. Log in to the VCF Operations interface at `https://<vcf_operations_fqdn>` as a user assigned the Administrator role.
2. Navigate to **Build** > **Lifecycle** > **VCF Management** > **Add Components**.
3. On the Components tab, select VCF Automation to start the deployment wizard.
   a) Provide the Deployment type.
      - New: Deploys a new component. If selected, you will be able to choose a different profile for the deployment.
      - Import VMware Aria Automation 8.x or VCF Automation 9.0.x: If upgrading from VCF 5.x to VCF 9.1 or converging from vSphere to VCF 9.1, choose this option to import the VMware Aria Automation 8.x appliance that you plan to upgrade.

      Note:

      The VMware Aria Automation 8.x primary node must reside in the management domain.
   b) Select the version for the VCF Automation instance.
   c) Select a small, medium, or large deployment size. The deployment size determines the compute and storage resources allocated to each node.

      Note:

      Regardless of the deployment size, each VCF services runtime node is allocated five IP addresses.
   d) Click **Next** to configure the remaining parameters for the deployment.

Network Option	Description
VCF services runtime nodes CIDR or IPs	CIDR notation or individual IPs. **Note:** Because a /29 CIDR block provides eight IP addresses and VCF Automation requires five IP addresses, three addresses are free.
VCF Automation FQDN	Enter the Fully Qualified Domain Name to assign to the VCF Automation instance. **Note:** Do not use capital letters in the FQDN.
VCF services runtime FQDN	Enter the Fully Qualified Domain Name for VCF services runtime

Network Option	Description
VCF Automation and VCF services runtime password	Type a custom password or click **Regenerate** to create one. Click **Copy to Clipboard** to save the credentials. The password applies to: • VCF Automation: admin account • VCF services runtime : `vmware-system-user` and `admin@vsp.local` accounts **Note:** Ensure that the password is safely stored. If lost, it cannot be retrieved.

e) Click **Next** to proceed to the Summary step for final review before deployment.

4. To start the deployment after reviewing the summary, click **Confirm**.

Deploy Identity Broker as a Day-N Operation

When you run VCF Installer on Day-0, the identity broker component is automatically installed on your first VCF Instance along with other management components such as Fleet lifecycle and SDDC Manager. However, if you want to install identity broker on a subsequent VCF Instance or after an SDDC Manager upgrade, you must deploy it manually because Fleet lifecycle will not deploy an additional identity broker if it detects that identity broker is already present in the environment.

To manually deploy identity broker , perform the following steps.

1. Log in to the VCF Operations interface at `https://<vcf_operations_fqdn>` as a user assigned the Administrator role.
2. Navigate to **Build** > **Lifecycle** > **VCF Management** > **Add Components**.
3. On the Components tab, select **VCF Identity Broker**.
   a) Provide the information required for deployment.
      - Version: Select the version of the identity broker component.
      - Size: Small, Medium, Large.
      - VCF Instance: Select the VCF Instance where you want to have identity broker deployed.
      - FQDN: Enter the Fully Qualified Domain Name of the identity broker component. The new FQDN must point to a unique IP that is not assigned to VCF services runtime.

      Note:

      Do not use capital letters in the FQDN.
      - Display Name: Provide a user-friendly name for the identity broker deployment on the VCF instance.

   b) Click **Next** to proceed to the Summary step for final review before deployment.
   c) Click to **Install** or **Cancel** the identity broker component deployment.

Deploy a Secondary Software Depot Instance

During the deployment and upgrade procedures of your management domain, software depot (part of VCF management services) is deployed on the first VCF Instance as a fleet-level component. You can also deploy other software depot instances as instance-level components. For example, when your environment requires a software depot instance in the same region as your other VCF Instances or the primary software depot instance experiences bandwidth issues.

- Verify that the VCF services runtime component is deployed and running.

By default, a VCF fleet is operated by using a centralized software depot that is located in the first VCF Instance. By default, your VCF Instances within the same VCF fleet access installation, upgrade, and patch binaries from this source. While a single software depot centralizes management, network latency between geographically distant instances can impact the performance of downloads and lifecycle management operations.

To ensure optimal performance and reliability during lifecycle management operations for multi-region deployments, if the network latency between the software depot of the first VCF Instance and another VCF Instance exceeds 150 ms, you

can deploy a secondary local software depot within the other VCF Instance. After the deployment completes, this software depot is automatically assigned to the VCF Instance on which it is deployed.

For more information, see Binary Management for VMware Cloud Foundation.

1. Log in to VCF Operations at `https://<vcf_operations_fqdn>` with a user assigned the **Administrator** role.
2. Select **Build** > **Lifecycle** > **VCF Management**.
3. On the **Components** tab, select **Add Component** > **Software depot**.
4. Deploy the component to a VCF Instance.
   a) Select a version.
   b) Select a VCF Instance to deploy the component to.
   c) Click **Deploy**.

The secondary software depot instance is deployed and assigned to the VCF Instance on which it is deployed.

To assign the new software depot instance to other VCF Instances, for example, in the same region, see Assign a Software Depot to VCF Instances.

Deploy a License Server Version 9.1.0

A license server is deployed as part of a new deployment or upgrade of VCF or vSphere Foundation. You need at least one license server and a VCF Operations instance to license your environment. If you need another license server, you can download and install a license server manually.

- Verify that you have administrator privileges in the VCF Operations instance that you use for license management.
- Verify that you have administrator privileges in the vCenter instance that you use to deploy the license server.
- Verify that you have a Broadcom Support Portal role with sufficient privileges to download the license server binaries.
- Define a unique Fully Qualified Domain Name (FQDN) for the license server and add an A record and PTR record to your DNS server.
- Download the license server binaries from the Broadcom Support Portal.

> **Important:**
>
> For version 9.1.1, see Deploy a License Server Version 9.1.1.

You must deploy a license server by using an OVA file in the following cases:

- You do not use the VCF Operations UI/SDDC Manager API to upgrade VCF, or VCF Installer UI/ VCF Installer API to upgrade vSphere Foundation.
- You need an additional license server for your environment.
- You need to replace a license server.

You must add at least one license server to each VCF Operations instance that you use for license management. One license server can be used to license many VCF instances and vCenter instances in the same VCF fleet.

> **Note:**
>
> You cannot deploy a license server on a standalone ESX host. You must deploy the license server in a vCenter instance as an OVF template.
>
> The license server does not support IPv6.

1. Obtain a unique registration key for the license server from the VCF Operations instance.
   a) Log in to VCF Operations.
   b) From the navigation bar at the top, click **Manage** and in the left navigation pane, click **Licensing** > **Licenses & Registration**.
   c) In the **Registration and License Server Status** card, click **Manage** > **License Servers**.

      The VCF Operations Registration page appears, and the License Servers tab is selected.

d) Click **Add License Server**.

The **Add License Server** page appears.

e) Copy and save the unique registration key.

2. Import and install the OVA file into a vCenter instance that is connected to this VCF Operations instance.
   a) Log in to the vSphere Client, and navigate to the vCenter instance.
   b) Right-click the vCenter instance, and click **Deploy OVF template**.

   The **Deploy OVF Template** wizard appears.
   c) On the **Select and OVF template** page, click **Local file** > **Upload Files** and upload the license server OVA file that you downloaded from the Broadcom Support Portal.
   d) Click **Next**.
   e) On the **Select a name and folder** page, enter a name for the license server virtual machine, and select a location from the list.
   f) On the **Select a compute resource** page, select the destination compute resource for the virtual machine.
   g) At the bottom of the page, select the check box **Automatically power on deployed VM**, and click **Next**.
   h) Review the details for the OVF template, and click **Next**.
   i) On the **Select storage** page, define where and how to store the files for the deployed template.
   j) On the **Select networks** page, from the Destination Network drop-down menu, select a destination network for the template, and click **Next**.
   k) On the **Customize template** page, customize the required deployment properties.

Property	Definition
Hostname	The hostname for this license server. As a best practice, set the hostname as an FQDN that uses the same domain suffix populated in the **Domain Name** field. **Note:** Add the license server FQDN to your DNS server. All vCenter systems that use licenses from this license server must be able to resolve the license server FQDN through DNS.
Unique Registration Key	An encoded, time-bound key that instructs the license server on how to connect to VCF Operations. This key is in not related to the registration of the license server with the VCF Business Services console. Enter the unique registration key that you copied from VCF Operations.
Domain Name	The primary DNS domain suffix for this license server. For example, corp.local. Leave this field blank if you use Dynamic Host Configuration Protocol (DHCP).
Domain Name Search	The domain search paths for this license server. This includes your local domain and parent domains. Leave this field blank unless instructed otherwise.
Default Gateway	The default gateway IPv4 address for this license server. Leave this field blank if you use DHCP.
Domain Name Servers	The domain name server IPv4 addresses for this license server. Leave this field blank if you use DHCP.
Network 1 IP Address	The IPv4 address for this license server. Leave this field blank if you use DHCP.
Network 1 IP Netmask	The netmask for this license server. Leave this field blank if you use DHCP. You must use a dotted-decimal format. For example, 255.255.255.0.
Egress proxy IP	The egress proxy IP address of the HTTP/HTTPs egress proxy between the license server and VCF Operations. Leave this field blank if a proxy is not required to connect to VCF Operations. **Note:** The license server does not connect to the Internet. An egress proxy is used only if a proxy is required to connect to VCF Operations.
Egress proxy port	The port number of the HTTP/HTTPs egress proxy between the license server and VCF Operations. Leave this field blank if a proxy is not required to connect to VCF Operations.

Property	Definition
Egress proxy username	The username for the HTTP/HTTPS egress proxy between the license server and VCF Operations. Leave this field blank if a proxy is not required to connect to VCF Operations or if a username is not required to connect to the proxy.
Egress proxy password	The password for the HTTP/HTTPS egress proxy between the license server and VCF Operations. Leave blank if a proxy is not required to connect to VCF Operations or a password is not required to connect to the proxy.
API Key	In limited cases, API interaction might be required to configure the license server before you register it with the VCF Business Services console. Do not set the API Key unless instructed otherwise.

l) Review the details, and click **Finish**.

The license server deployment takes up to 20 minutes. When the deployment completes, and the license server virtual machines powers on, the license server automatically connects to the VCF Operations instance. On the **License Server Details** page, **Connectivity to VCF Operations** shows **Connected**.

3. In the VCF Operations instance, click **Review License Servers**.

   The table with license servers added to this VCF Operations instance appears. A banner is displayed with information about unregistered license servers.

After you deploy the license server and add it to VCF Operations, you must register the license server with the VCF Business Services console.

If you add the license server to a VCF Operations instance that is registered with the VCF Business Services console, see Register an Additional License Server with the VCF Business Services Console.

If you add the license server to a VCF Operations instance that is not registered with the VCF Business Services console, see Registering VCF Operations and a License Server with the VCF Business Services Console.

Note:

The lifecycle management of the license server and the cloud proxy components is bundled with the lifecycle management of VCF Operations. Upgrading or patching VCF Operations automatically upgrades or patches license server and cloud proxy.

Deploy a License Server Version 9.1.1

A license server is deployed as part of a new deployment or upgrade of VCF or vSphere Foundation. You need at least one license server and a VCF Operations instance to license your environment. If you need another license server, you can download and install a license server manually.

- Verify that you have administrator privileges in the VCF Operations instance that you use for license management.
- Verify that you have administrator privileges in the vCenter instance that you use to deploy the license server.
- Verify that you have a Broadcom Support Portal role with sufficient privileges to download the license server binaries.
- Define a unique Fully Qualified Domain Name (FQDN) for the license server and add an A record and PTR record to your DNS server.
- Download the license server binaries from the Broadcom Support Portal.

Important:

For version 9.1.0, see Deploy a License Server Version 9.1.0.

You must deploy a license server by using an OVA file in the following cases:

- You do not use the VCF Operations UI/SDDC Manager API to upgrade VCF, or VCF Installer UI/ VCF Installer API to upgrade vSphere Foundation.
- You need an additional license server for your environment.
- You need to replace a license server.

You must add at least one license server to each VCF Operations instance that you use for license management. One license server can be used to license many VCF instances and vCenter instances in the same VCF fleet.

Note:

You cannot deploy a license server on a standalone ESX host. You must deploy the license server in a vCenter instance as an OVF template.

1. Obtain a unique registration key for the license server from the VCF Operations instance.
   a) Log in to VCF Operations.
   b) From the navigation bar at the top, click **Manage** and in the left navigation pane, click **Licensing** > **Licenses & Registration**.
   c) In the **Registration and License Server Status** card, click **Manage** > **License Servers**.

      The VCF Operations Registration page appears, and the License Servers tab is selected.
   d) Click **Add License Server**.

      The **Add License Server** page appears.
   e) Copy and save the unique registration key.
2. Import and install the OVA file into a vCenter instance that is connected to this VCF Operations instance.
   a) Log in to the vSphere Client, and navigate to the vCenter instance.
   b) Right-click the vCenter instance, and click **Deploy OVF template**.

      The **Deploy OVF Template** wizard appears.
   c) On the **Select and OVF template** page, click **Local file** > **Upload Files** and upload the license server OVA file that you downloaded from the Broadcom Support Portal.
   d) Click **Next**.
   e) On the **Select a name and folder** page, enter a name for the license server virtual machine, and select a location from the list.
   f) On the **Select a compute resource** page, select the destination compute resource for the virtual machine.
   g) At the bottom of the page, select the check box **Automatically power on deployed VM**, and click **Next**.
   h) Review the details for the OVF template, and click **Next**.
   i) On the **Select storage** page, define where and how to store the files for the deployed template.
   j) On the **Select networks** page, from the Destination Network drop-down menu, select a destination network for the template, and click **Next**.
   k) On the **Customize template** page, customize the required deployment properties.

      Note:

      Starting with version 9.1.1, the license server supports IPv6 networking using static addressing, or DHCPv6/SLAAC. You can configure the license server for IPv4-only, IPv6-only, or dual-stack networking.

      - IPv4-only. You do not need to configure the properties in the section IPv6 Configuration.

- IPv6-only. You do not need to configure the properties in the section IPv4 Configuration, and you must activate the property `Force IPv6 Only`.
- Dual-stack. You must configure the properties in the sections IPv6 Configuration and IPv4 Configuration. Do not activate the property `Force IPv6 Only`.

Category	Property	Definition
License Server Configuration	`Unique Registration Key`	An encoded, time-bound key that instructs the license server on how to connect to VCF Operations. This key is not related to the registration of the license server with the VCF Business Services console. Enter the unique registration key that you copied from VCF Operations.
Networking Properties	`Hostname`	The hostname for this license server. As a best practice, set the hostname as an FQDN that uses the same domain suffix populated in the `Domain Name` field. **Note:** Add the license server FQDN to your DNS server. All vCenter systems that use licenses from this license server must be able to resolve the license server FQDN through DNS.
	`Domain Name`	The primary DNS domain suffix for this license server. For example, corp.local. Leave this field blank if you use Dynamic Host Configuration Protocol (DHCP).
	`Domain Search Path`	The domain search paths for this license server. This includes your local domain and parent domains. Leave this field blank unless instructed otherwise.
	`Domain Name Servers`	The domain name server IPv4 addresses for this license server. Leave this field blank if you use DHCP.
IPv4 Configuration	`IP Address`	The IPv4 address for this license server. Leave this field blank if you use DHCP.
	`Default Gateway`	The default gateway IPv4 address for this license server. Leave this field blank if you use DHCP.
	`IP Netmask`	The netmask for this license server. Leave this field blank if you use DHCP. You must use a dotted-decimal format. For example, 255.255.255.0.
IPv6 Configuration	`Force IPv6 Only`	If you want to disable IPv4 and restrict the license server to IPv6-only, select the check box.
	`IPv6 Type`	The IPv6 configuration type. You can choose from: • Disabled. The network mode of the license server is IPv4-only. • Static. The network mode of the license server is dual-stack. If both IPv4 and IPv6 are available, IPv6 is preferred. • DHCP/SLAAC. The network mode of the license server is dual-stack. If both IPv4 and IPv6 are available, IPv6 is preferred. To enable IPv6-only mode, you must select Static or DHCP/SLAAC, and you must select the check box next to the property Force IPv6 Only. If you do not select the check box, after you configure IPv6, the license server operates in dual-stack mode.
	`IP Address`	The IPv6 address for this license server. Leave this field blank if you selected DHCP/SLAAC.
	`Default Gateway`	The default gateway IPv6 address for this license server. Leave this field blank if you selected DHCP/SLAAC.
	`Prefix Length`	The IPv6 prefix length for this license server (e.g., 64). You must use an integer between 1-128. Leave this field blank if you selected DHCP/SLAAC.

Category	Property	Definition
Network Proxy Between License Server and VCF Operations	`Network Proxy IP Address`	IP address of the network proxy between the license server and VCF Operations. Leave this field blank if a proxy is not required to connect to VCF Operations. **Note:** The license server does not connect to the Internet. A network proxy is used only if a proxy is required to connect to VCF Operations.
	`Network Proxy Port`	The port number of the network proxy between the license server and VCF Operations. Leave this field blank if a proxy is not required to connect to VCF Operations.
	`Network Proxy Username`	The username for the network proxy between the license server and VCF Operations. Leave this field blank if a proxy is not required to connect to VCF Operations or if a username is not required to connect to the proxy.
	`Network Proxy Password`	The password for the network proxy between the license server and VCF Operations. Leave this field blank if a proxy is not required to connect to VCF Operations or a password is not required to connect to the proxy.
Advanced Configuration	`API Key`	In limited cases, API interaction might be required to configure the license server before you register it with the VCF Business Services console. Do not set the API Key unless instructed otherwise.

l) Review the details, and click **Finish**.

The license server deployment takes up to 20 minutes. When the deployment completes, and the license server virtual machines powers on, the license server automatically connects to the VCF Operations instance. On the **License Server Details** page, **Connectivity to VCF Operations** shows **Connected**.

3. In the VCF Operations instance, click **Review License Servers**.

The table with license servers added to this VCF Operations instance appears. A banner is displayed with information about unregistered license servers.

After you deploy the license server and add it to VCF Operations, you must register the license server with the VCF Business Services console.

If you add the license server to a VCF Operations instance that is registered with the VCF Business Services console, see Register an Additional License Server with the VCF Business Services Console.

If you add the license server to a VCF Operations instance that is not registered with the VCF Business Services console, see Registering VCF Operations and a License Server with the VCF Business Services Console.

Note:

The lifecycle management of the license server and the cloud proxy components is bundled with the lifecycle management of VCF Operations. Upgrading or patching VCF Operations automatically upgrades or patches license server and cloud proxy.

Linking vCenter instances in VCF Operations

vCenter linking in VCF Operations provides you with a single pane of glass view of all the linked vCenter instances from a single vSphere Client.

- Verify that you have vCenter version 9.0 or later.
- Ensure that there is functioning forward and reverse DNS resolution between all vCenter instances in a vCenter linking group.
- Ensure that the user used to register the vCenter account in VCF Operations has Administrator privileges.

Note:

The Administrator privileges apply only when you manually configure the vCenter cloud account with user credentials. When you use system managed credentials, these requirements are handled automatically. For more information, see Configuring a vCenter Account in VCF Operations.

- Verify that the user has the required **vCenter Linking** role assigned to access the vCenter instances. For more information, see the Access Control: Roles Tab.
  - **Manage**: To add, edit, or delete the vCenter linking group.
  - **View Privilege**: To view the vCenter linking group and its members.
- Ensure that common Identity Providers (IDP) is configured on all vCenter systems. For more information, see Identity Providers and Protocols Supported for VCF Single Sign-On.
- Ensure that the IDP user has the required permission for each vCenter system.

  Note:

  IDP user permissions are not shared between vCenter systems and must configured on each vCenter individually.
- Verify that the vCenter systems are not part of an Enhanced Link Mode group.

  Note:

  From VMware Cloud Foundation 9.0 onward, its recommended to use vCenter linking for vSphere Client single pane of glass functionality instead of Enhanced Linked Mode (ELM). If a vCenter system is part of ELM, it cannot be added to a vCenter linking group. You must deactivate ELM before proceeding. To deactivate ELM, see Deactivate Enhanced Link Mode for Standalone vCenter Instances Using the cmsso-util break-elm Utililty.

  Functionalities previously provided by Enhanced Linked Mode in vCenter like synchronization of roles, tags, global permissions across the linked vCenter instances are not available in VCF Operations based vCenter linking.

  - To configure SSO for VCF components, see Configuring VCF Single Sign-On.
  - To manage tags for a fleet of vCenter instances, see Tags and Categories Overview.

You can link a minimum of two and a maximum of 15 vCenter instances in a group. All the vCenter instances in a linked group must be configured with common IDP users. Once linked, you can log into any one of the linked vCenter instances using the vSphere Client, and monitor and manage the inventories of all the linked vCenter instances.

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations**.
2. Under **Logical Groupings**, click the **vCenter Linking** tile.
3. Click **Create Group**.
4. Enter the **Group Name** and **Description**, and then click **Next**.
5. Select the compatible vCenter instances that you want to add to the group and then click **Next**.

   Note:

   You can view the list of compatible and incompatible vCenter systems.

   - Compatible: vCenter instances that meet the version prerequisite and are not part of any other vCenter linking group.
   - Incompatible: vCenter instances that do not meet the version prerequisite or belong to another vCenter linking group.
6. Review your selection and click **Finish**.

   The vCenter linking group is created.
7. If the status of any of the vCenter instances shows an error, fix the cause, and then click **Reset Linking** to troubleshoot.

   Note:

   Some of the common errors you might face and fix before you click **Reset Linking** as as follows:

- vCenter is not reachable from VCF Operations: You must fix the connection between VCF Operations and vCenter.
- vCenter is part of Enhanced Linked Mode (ELM): You must deactivate ELM.

8. After the group is created, you can add or remove members from a group and you can also edit the group details or delete the group.

You can log in to any vSphere Client for any one of the linked vCenter instances using the IDP user that has the required permissions for all vCenter instances and view the inventory of all the vCenter instances in the group.

Collecting Data with Cloud Proxy in VCF Operations

Cloud proxies are used to collect and monitor data from your on-premises data centers across various geographical locations within a VCF environment. They act as data collection points, ensuring continuous monitoring and operational visibility.

Cloud proxies provide high availability within your VCF Operations environment, you can group two or more cloud proxies to form a collector group. The cloud proxy collector group ensures that there is no single point of failure in your cloud environment. If one of the cloud proxies experiences a network interruption or becomes unavailable, the other cloud proxy from the collector group takes charge and ensures that there is no downtime. All other user-initiated manual operations on the collector, such as to stop or restart the collector manually, do not result in automated re-balancing. For more information, see Configuring Cloud Proxies in VCF Operations.

To configure cloud proxy using VCF Operations Fleet Management, see Add Nodes to VCF Operations.

Note:

When a node within the collector group goes down, high availability failover occurs, so that data collection is not impacted. However some data loss is expected before the failover is complete.

You can also use cloud proxies to re-balance the resources in your collector group. The re-balance option is available as part of the Edit menu in the Collector Groups page.

Note:

You can use the re-balance option before vCenter initiates data collection. After the data collection starts, the re-balance option is disabled.

FIPS mode is supported in a cloud proxy. To leverage this functionality, make sure your cluster is in FIPS mode.

You can upgrade your cloud proxy. For more information, see Upgrading.

You can deploy a cloud proxy based on your sizing requirements. For more information, see KB article 324340.

Table 1144: Cloud Proxy Types

Cloud Proxy Type	VM Scale Requirements	CPU Requirements	Memory Requirements
Classic Small Cloud Proxy	Up to 16,000 VMs	2 vCPUs	8GB
Classic Standard Cloud Proxy	Between 16,000 to 80,000 VMs	4 vCPUs	32GB
Unified Small Cloud Proxy	Up to 16,000 VMs	4 vCPUs	16GB
Unified Standard Cloud Proxy	Between 16,000 to 80,000 VMs	8 vCPUs	48 GB

Use a classic cloud proxy in VCF Operations, to collect and monitor data from your remote data centers. In addition to data collection, you can use your cloud proxy to collect logs into the Log Cluster and to collect and upload support bundles to the Broadcom support portal site. Use unified cloud proxy for log collection or log assist. For more information on log assist, see Log Assist. For more information on log collection, see Configuring and Analyzing Logs.

Monitoring the Health of Cloud Proxies

You can view the status and health of your cloud proxy after you add it in VCF Operations. You can then monitor the health and view alerts and metrics of your cloud proxy using the VCF Operations cloud proxy object.

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Cloud Proxies**.

   The list of cloud proxies is displayed.

Option	Description
Name	The name of the cloud proxy.
IP	The IP address of the cloud proxy.
Status	Status of the cloud proxy. For example, the Going Online status is displayed for a few minutes when you add a new cloud proxy. After the cloud proxy is connected to VCF Operations, the status changes to Online. If the VCF Operations is not connected, the Offline status is displayed.
Version	The version used to install the cloud proxy.
Accounts	The number of accounts that are created and associated with the cloud proxy.
Type	Displays the cloud proxy type.
Cloud Proxy Group	Displays if the cloud proxy is part of the cloud proxy group. If the cloud proxy is part of the cloud proxy group, the name is displayed. Click the name to view the cloud proxy group details. **Note:** No data is displayed if the cloud proxy is a standalone cloud proxy and not a part of the cloud proxy group.
Network Proxy Address	The network proxy address of the cloud proxy.
Network Proxy Port	The network proxy port number of the cloud proxy.
Target	Displays the target location on which the cloud proxy is deployed.
Data Persistence	Displays the status of data persistence for the cloud proxy. Data persistence activates the cloud proxy to store data if the connection fails between the cloud proxy and VCF Operations. • **Activated**. Cloud proxy will store data. • **Deactivated**. Cloud proxy will not store data. To activate/deactivate data persistence for multiple cloud proxies, select the cloud proxies, click the horizontal ellipsis, and then select **Activate Data Persistence** or **Deactivate Data Persistence**. To activate/deactivate data persistence for a single cloud proxy, select the cloud proxy, click the vertical ellipsis, and then select **Activate Data Persistence** or **Deactivate Data Persistence**. **Note:** When the connection is restored, the cloud proxy sends the stored data to VCF Operations. Data appears in chronological order, with stored data displayed before real-time data. Fore more information, see Activating Data Persistence in Cloud Proxy.
Log Assist	Displays if log assist is activated or deactivated for the cloud proxy.
Time Estimation	Displays the estimated time duration for which the cloud proxy persists data. Cloud proxy can store data for a maximum duration of one hour. If there is a lack of space or if the connection fail lasts for more than an hour, the cloud proxy rotates the stored data by deleting the oldest stored data and replacing it with the most recently collected data. **Note:** The time estimate value is displayed only if data persistence is activated.

2. Optional: Click the vertical ellipsis.
   - **Activate/Deactivate Data Persistence**: Use to activate or deactivate data persistence for the cloud proxy. For more information
   - **Activate/Deactivate Log Assist**: Use to activate or deactivate log assist for the cloud proxy.

- **Delete**: Use to delete the cloud proxy.

3. Click a **Cloud Proxy** name.
   The **Cloud Proxy Details** open.

Table 1145: Cloud Proxy Details

Option	Description
Rename	Click the **Rename Cloud Proxy** icon to update the cloud proxy name.
Proxy ID	ID of the cloud proxy.
IP Address	IP address of the cloud proxy.
OVA Version	The OVA file version used to install the cloud proxy.
Status	Status of the cloud proxy. For example, the Getting Online status is displayed for a few minutes when you add a cloud proxy. After the cloud proxy is connected to VCF Operations, the status changes to Online. If the VCF Operations is not connected, the Offline status is displayed.
Creation Date	Date of creation of the cloud proxy.
Last Heartbeat	Last time stamp when VCF Operations ran a Health Check for this cloud proxy. When you click a cloud proxy to view its details, VCF Operations sends a heartbeat to check if the cloud proxy is still reachable.
CPU	CPU usage.
Memory	Memory usage.
Number of Cloud Accounts	Displays the number of cloud accounts using the cloud proxy. Each cloud proxy might have one or more cloud accounts. You can also view the health and status of these cloud accounts from this details pane.

4. If your cloud proxy is not collecting data, you can view the health of the cloud proxy. From the left menu, click **Inventory**, select the **VCF Operations Cloud Proxy Object** from the list, and then click **Show Detail**.

   For more details, see Inventory : List of Objects.

5. After you locate the VCF Operations cloud proxy object, you can view the object details using the Summary tab. For more information, see Summary Tab.

6. Use the Alerts Tab to monitor the health of the cloud proxy. If there are any issues, troubleshoot them using the Metrics tab.

   If your cloud proxy is not working properly, an alert is displayed.

   ```
   One or more VCF Operations services on a cloud proxy are down.
   ```

   To clear this alert, perform the following steps:

   - Check the network connectivity and configuration for the cloud proxy.
   - Take the cloud proxy offline and then bring it back online.

   For more information, see Cloud Proxy FAQ and Cloud Proxy Troubleshooting.

   Note:
   It is recommended that you create a notification rule for this alert so that, quick remediation steps can be taken, if necessary.

7. Optional: You can use the cloud proxy command-line interface for other cloud proxy related actions. For more details, see Using the Cloud Proxy Command-Line Interface.

Click the **Cloud Proxy Groups** tab to view the cloud proxies that are part of the cloud proxy groups. For more information on cloud proxy groups, see Configuring Collector Groups.

Activating Data Persistence in Cloud Proxy

Use data persistence, to avoid data gaps if there are temporary connectivity issues. You can activate data persistence to allow the cloud proxy to store data that is sent to VCF Operations, if the connection fails between the cloud proxy and VCF Operations. The cloud proxy stores all the persisted data that includes metrics and properties, along with time stamps.

Cloud proxy can store data for a maximum duration of one hour. If storage is full or a connection failure exceeds one hour, the cloud proxy rotates data by replacing the oldest records with the most recent collections. If there is lack of space, you can add additional storage, for more information, see Broadcom knowledge base article 318768.

After the connection is restored, the cloud proxy sends the stored data to VCF Operations before the real-time data. Data appears in chronological order, with stored data displayed before real-time data. If the connection loss was more than one hour, there can also be a gap in the displayed data.

You can activate data persistence after you deploy a new cloud proxy. For more information, see Add a Cloud Proxy. You can also activate data persistence in an existing cloud proxy. For more information, see Monitoring the Health of Cloud Proxies.

Monitoring the Health of Cloud Proxies from the Administration UI

After you configure your cloud proxy, you can view the status, health, and upgrade history of your cloud proxy in the VCF Operations administration interface.

1. Log in to the VCF Operations administration interface at https://*primary-node-fqdn*/admin.
2. Click **Cloud Proxies**.

Table 1146: Cloud Proxies Page Options

Option	Description
IP Address	IP address of the cloud proxy.
Name	Name of the cloud proxy.
Network Proxy Configuration	Determines whether the network proxy setting is configured or not.
Health Status	Determines the health of the cloud proxy.
Upgrade Status	Determines whether the upgrade is complete, in progress, or failed.
Last Upgrade Time	Determines when was the last upgrade done.
Version	Version number of the cloud proxy.

Click the **Expand** icon to view the upgrade history.

Table 1147: Upgrade History Options

Option	Description
ID	The conventional name used to identify the PAK file. It is usually the name of the PAK file and its version numbers joined together without extensions. For example, VMware Aria-Operations-Cloud-Proxy-<number>.
Type	The type of upgrade used for the cloud proxy. The cloud proxy can be upgraded either automatically or manually using the command-line interface. For more information, see Using the Cloud Proxy Command-Line Interface.
Start Time	Time stamp when the upgrade started.
End Time	Time stamp when the upgrade ended.

Option	Description
Upgrade Status	Determines whether the upgrade is complete, in progress, or failed. **Note:** For a detailed view of the upgrade status, click the row to open the **Info** pop-up. If the cloud proxy upgrade fails due to low disk space, the `FAIL_NO_SPACE_FOR_PAK_DOWNLOAD` message appears.
Version	Version number of the cloud proxy PAK file.

Deleting Cloud Proxies

You can delete a cloud proxy if there are issues with your cloud proxy. For example, you can delete the cloud proxy if it was deployed incorrectly, needs to be redeployed to another cluster, or is no longer required. After you delete a cloud proxy, data collection stops.

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Cloud Proxies**.
2. Click the vertical ellipsis in front of the cloud proxy you want to delete and then click **Delete**.

   The confirmation wizard appears.
3. Select the **I understand the risk** check box and then click **OK**.
4. The cloud proxy is deleted.

Configuring Outbound Network Proxy Settings on an Upgraded Cloud Proxy

After upgrading the unified cloud proxy to version 9.1.1, you can manually configure the outbound network proxy OVF properties. Since these are not exposed in vCenter vApp Options, create them only if your deployment requires a dedicated proxy to route Log Assist traffic to the Broadcom Portal.

- Verify that the unified cloud proxy is upgraded to version 9.1.1.
- Verify that you have access to the vCenter instance that manages the cloud proxy.
- Verify that you have the outbound network proxy values (IP address, port, user name, password, SSL, and SSL verification) available from your network administrator.

Starting with version 9.1.1, you can optionally set up Outbound Network Proxy Settings to route Log Assist traffic to the Broadcom Portal through a dedicated proxy.

When you upgrade a cloud proxy from any lower version to version 9.1.1, any existing network proxy values are automatically applied to the outbound network proxy properties, but you cannot view or modify them in the user interface. If the outbound network proxy requires different values from the existing network proxy, you must manually create the outbound network proxy properties and set their values so that the outbound network proxy feature functions correctly.

1. Log in to the vCenter instance that manages the cloud proxy.
2. In the **Inventory**, locate and select the cloud proxy virtual machine.
3. Power off the cloud proxy virtual machine.

   You must power off the virtual machine before you can add new OVF properties.
4. Click the **Configure** tab, and under **Settings**, select **vApp Options**.
5. Click **ADD** to add each property listed in the Outbound network proxy properties table. You must enter the **Key**, **Label**, **Category**, and **Type** values exactly as shown, and click **OK**.

Key	Label	Category	Type
outbound_network_proxy_ip	Outbound Network Proxy IP Address	Outbound Network Proxy Settings	string
outbound_network_proxy_port	Outbound Network Proxy Port	Outbound Network Proxy Settings	string

outbound_network_proxy_username	Outbound Network Proxy User Name	Outbound Network Proxy Settings	tring
outbound_network_proxy_password	Outbound Network Proxy Password	Outbound Network Proxy Settings	password
outbound_network_proxy_ssl	Outbound Network Proxy SSL	Outbound Network Proxy Settings	boolean
outbound_network_proxy_ssl_verification	Outbound Network Proxy SSL Verification	Outbound Network Proxy Settings	boolean

6. Select each property you created, click **SET VALUE**, enter the appropriate value for your environment, and click **OK**.
7. Power on the cloud proxy virtual machine.
8. To verify that the outbound network proxy properties are applied, use SSH to connect to the cloud proxy and run the following command: `ovfenv`

   The command output lists all OVF environment properties. Confirm that the newly added properties display the values you configuredfor the following:

   - outbound_network_proxy_ip
   - outbound_network_proxy_port
   - outbound_network_proxy_username
   - outbound_network_proxy_password
   - outbound_network_proxy_ssl
   - outbound_network_proxy_ssl_verification

The outbound network proxy properties are created and configured on the cloud proxy. When you power on the cloud proxy, it reads the property values from the OVF environment and applies the outbound network proxy settings.

Verify that the outbound network proxy settings are applied correctly by confirming that the cloud proxy communicates through the configured outbound network proxy.

Using the Cloud Proxy Command-Line Interface

Use the cloud proxy command-line interface to upgrade your cloud proxy, generate support bundles, and view the cloud proxy health status.

You can use SSH to access the cloud proxy instance and use its command-line Interface to run the following actions:

- Manually upgrade your cloud proxy in case the automatic download of the latest binary fails. When the automatic download fails, you see a notification on the VCF Operations user interface. To manually upgrade your cloud proxy instance to the latest version, see the following Broadcom knowledge base article 344168.
- Generate the support bundle.
- Gather the status of the cloud proxy's health and connectivity details.

Command Line	Description
`cprc-cli -h, --help`	Displays the help message and use of the command-line interface.
`cprc-cli -s, --status`	Prints the cloud proxy life-cycle status, configuration details, upgrade-related information, and more. It is useful to catch necessary information related to support and troubleshooting, or check the connection to VCF Operations, or check the product version number, and so on.
`cprc-cli -u PRODUCT_PAK, --upgrade PRODUCT_PAK`	The cloud proxy instance is enabled for an automated upgrade by default. But if the automated upgrade fails due to any exceptional issue, use this command line to upgrade your cloud proxy instance to the desired version.

Command Line	Description

`cprc-cli -hsb`
`cprc-cli --generate-heavy-support-bundle`
`cprc-cli -sb [{false,true}]`
`cprc-cli --generate-support-bundle [{false,true}]`
	Generates the cloud proxy support bundle which is a package of logs, configurations, and status files. The support bundles are necessary for product support and troubleshooting. Generated support bundles can be found at the `/storage/core/vmware-vrops-cprc/support` location.
	You can generate light or heavy support bundles. Heavy support bundles include journal.ctl logs. To generate heavy support bundles, the following commands must be set to true.
	`cprc-cli -sb [{true}]`
	`cprc-cli --generate-support-bundle [{true}]`
	With the true option, the heavy support bundle is generated with journal.ctl logs. With the false option, the light support bundle is generated without journal.ctl logs. If no option is specified here, the light support bundle will be generated by default.
	To exclude docker logs from the support bundles during generation, include the following options in the support bundle generation commands:
	`-ed`
	`--exclude-docker-support-bundle`

`cprc-cli -ssb` `cprc-cli --status-support-bundle`	Displays the support bundle statuses.
`cprc-cli -rsb SUPPORT_BUNDLE, --remove-support-bundle SUPPORT_BUNDLE`	Removes any specified support bundle. Although generated support bundle packages can be removed using the system-embedded commands, it is recommended to use this command for that action.
`cprc-cli -fm, --enable-fips-mode`	Enables FIPS mode for cloud proxy.
`cprc-cli -v -d all | install | network | services | connection`	Runs the troubleshoot process to check for all or any one of the following issues: • Installation: Checks if VM deployment and cloud proxy installation is successful. • Network: Checks IP configurations, DNS configurations, and Host (hostname) configurations. • Service: Checks whether the `vmware-casa.service`, `collector.service`, `haproxy.service`, `httpd-north.service`, and `httpd-south.service` are active and loaded based on the `systemctl show` command. • Connection: Checks the connection to VCF Operations cluster nodes using `curl`. For the proxy server cases, checks that connection via the proxy server using its specified configuration. Use the `-v` command to generate more verbose outputs. This command generates a log file that gets stored in `/var/log/support/cprc.analysis.log` location.
`-cs, --certificates-status`	Displays the status of the imported certificates and if the certificates have been configured correctly or not.
`cprc-cli -rc, --replace-certificate`	Allows you to replace the self signed certificates for cloud proxy. Add the full path to the new certificate to the command. The function expects a certificate full chain containing the certificate and private key. For more information, see Broadcom knowledge base article 342838.
`cprc-cli -cc, --connectivity-check`	Allows you to run the connectivity check. You can check the HTTPS connectivity to a specified url. In case the url is not specified, it checks the HTTPS connectivity to the default prerequisites For on-prem configurations, the default prerequisites are the cluster nodes.

Command Line	Description
`cprc-cli -cc -r, --response`	Displays the response from the server during the connectivity check.
`cprc-cli -cc -rt, --routing`	Displays the routing information during the connectivity check.
`cprc-cli -env, --ovfenv`	Displays the content of the ovfEnv.xml file.
`cprc-cli -rec, --recovery`	Runs the CPRC recovery script.
`cprc-cli -rh, --run-on-hosts`	Runs a specific script on hosts according to the configurations provided.
`cprc-cli -gt, --generate-template`	Generates the template.json files for -rh options configurations.

Using APIs with VCF Operations Cloud Proxy

You can use a browser or an HTTP client program to send requests and receive responses.

REST Client Programs

Any client application that can send HTTPS requests is an appropriate tool for developing REST applications with the VCF Operations API. REST client plug-ins are available for most browsers and many IDEs. The following open-source programs are commonly used:

- cURL. http://curl.haxx.se
- Postman application. http://www.getpostman.com

In addition, VMware provides language-specific client bindings for the VCF Operations API.

About the Schema Reference

The VCF Operations REST API documentation includes reference material for all elements, types, queries, and operations in the VCF Operations API. It also includes the schema definition files.

Swagger-based API documentation is available with the product, with the capability of making REST API calls right from the landing page.

To access the API documentation, use the URL of your VCF Operations cloud proxy. For example, if the URL of your cloud proxy is https://cloudproxy.vmware.com, the API reference is available from: https://cloudproxy.vmware.com/suite-api/doc/swagger-ui.html

> **Note:**
> You must use your cloud proxy IP or FDQN in the url to access the API documentation.

Language-specific client bindings are available from:

```
https://<your-cloud-proxy-fqdn>/suite-api/
```

Authorize VCF Operations API

To perform API calls, you must authorize your VCF Operations cloud proxy. You can perform basic authorization and enter the **Username**, **Password**, and the **Auth Source**, or perform token based authorization and enter the **Ops Token (apikey)** value, and then click **Authorize**.

About the VCF Operations API Examples

You can view the VCF Operations API from **Developer Center** > **APIs & SDKs** > **Operations API** and then click **API Documentation**. Browse the VCF Operations API and try the different API calls to automate available features. For example to get all the Collectors registered with the VCF Operations system, use the following API:

`GET/api/collectors/{id}/adapters`

All examples include HTTP requests and responses. These examples show the workflow and content associated with operations such as creating and querying for information about objects in your monitored environment.

Example request bodies are in JSON. Request headers required by the VCF Operations API are included in example requests that are not fragments of a larger example.

Most example responses show only those elements and attributes that are relevant to the operation being discussed. Ellipses (...) indicate omitted content within response bodies.

Cloud Proxy FAQ

This topic covers some frequently asked questions about VCF Operations cloud proxy.

Configuration

1. What are the prerequisites for setting up a cloud proxy account?
   Prerequisites are given in the topic, Add a Cloud Proxy.
2. What does one-way connection mean?
   Outbound connections are initiated from cloud proxy to VCF Operations, over `https/443`. Cloud proxy can also facilitate vCenter actions.
3. Which ports should be opened?
   The most up-to-date technical information about ports can be found on Ports and Protocols.
4. Which ports should be allowed for incoming traffic to cloud proxy?
   Allow port 443 https protocol for push model adapters like application monitoring or Suite-API on cloud proxy. Allow ports 4505 and 4506 via TCP protocol for application monitoring. Allow the VRRP protocol for intercommunication between cloud proxies in a application monitoring high availability activated cloud proxy group.
5. How do I edit environment settings for cloud proxy?
   You can edit vApp options. For more information, see Edit OVF Details for a Virtual Machine.
6. How are certificates managed?
   Certificates are managed by cloud proxies. But for any additional proxy servers with SSL communication, you need to provide certificate(s).
7. What credential is used to login to cloud proxy?
   You can login as the “root” user. You are expected to set a new password on the first login to cloud proxy VM.
   SSH access is disabled by default, so the first login must be done via the vCenter console. You can run the following command to start SSH service:
   ```
   systemctl start sshd
   systemctl enable sshd
   ```
   To reset password, see the Broadcom knowledge base article 325005.
8. Will I be notified if the connection between cloud proxy and VCF Operations breaks down?
   You can configure alerts/notifications on the *VCF Operations cloud proxy* object. For more information, see Monitoring the Health of Cloud Proxies.
   VCF Operations automatically generates notifications for the following scenarios:
   - Cloud proxy is not reachable.
   - Cloud proxy is nearing sizing limits.
9. How do I change account for cloud proxy?
   You can edit vApp options. For more information, see Edit OVF Details for a Virtual Machine.
10. How can I check the status of cloud proxy?
    For more information, see Monitoring the Health of Cloud Proxies.
11. Should I use Remote Collector or cloud proxy for monitoring?
    VMware recommends that you use cloud proxy to take advantage of the latest enhancements. Also, application monitoring, HA of cloud proxy groups, and data persistence are only supported through cloud proxy.

Sizing

1. How should I size the cloud proxy?
   For information on sizing, see the Broadcom knowledge base article 318402.

Upgrade

1. How do I upgrade cloud proxy?
   Cloud proxy is upgraded automatically. In case the upgrade fails, see the Broadcom knowledge base article 344168.

High Availability

1. Is high availability supported?
   Cloud proxy supports high availability. You can add multiple cloud proxies to a cloud proxy group. If the collecting cloud proxy fails or gets disconnected, collection can be picked up by another proxy in the group.

   Note:
   Since the failover is initiated after a period of 10 minutes, few collection cycles are lost.

To troubleshoot cloud proxy issues, see Cloud Proxy Troubleshooting.

Cloud Proxy Troubleshooting

Cloud proxy troubleshooting steps are provided to help you easily resolve issues that you might come across in VCF Operations.

Before you proceed with troubleshooting, see the Cloud Proxy FAQ.

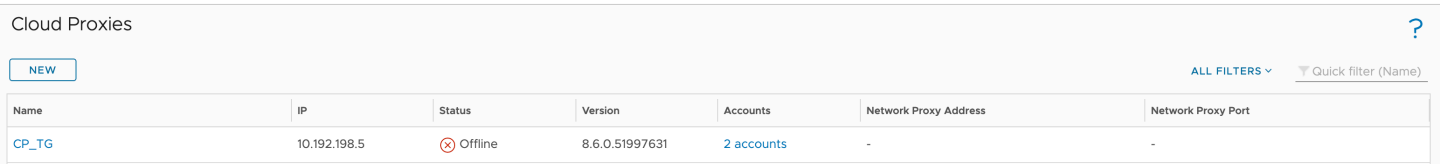
Installation and/or First Boot Failure

`To verify the issue, check if /var/log/firstbootcontains a file named "Succeeded".`

If not, the following problems could result in VCF Operations installation and/or first boot failure:

1. Unique Registration Key used while deploying Cloud Proxy is invalid. To verify, check the cloud proxy console.
   Solution: Redeploy cloud proxy.

Cloud Proxy VM is running, but the status is Offline in VCF Operations.



To verify the connection, use the following commands: (For the complete list of commands, see Using the Cloud Proxy Command-Line Interface.

```
# Overall status of cloud proxy:cprc-cli -s

# Ping itself:
ip addr
ping <address>

# Ping gateway:
ip route
ping <gateway>
```

```
# Verify the connection outside the cloud proxy.
ping 8.8.8.8

Note: If you are using a network proxy,
use the /opt/vmware/share/vami/vami_config_net option#5 command
to ensure you have the correct configuration for the testings.
```

The following problems could result in VCF Operations displaying the status of cloud proxy as offline.

1. Incorrect network proxy information in cloud proxy configuration.
   To verify the connection via a network proxy, use the following:

   ```
   curl -vvv --proxy http(s)://proxy_user:proxy_pass@proxy_ip:proxy_port -H 'Accept: application/json' -H
    'Content-Type: application/json' -X GET https://<gateway url>/casa/security/ping (gateway url example -
    10238.gw.dev.vrops-ops.com)

   To ignore SSL validation for a proxy server,
   use curl --proxy-insecure. With SSL validation the customer can provide Proxy Server certificate during
    cloud proxy deployment or re-configuration
   so that provided certificate from customer can be used to check the connection with curl with SSL
    certificate validation.
   ```

   Solution:

   1. SSH to the cloud proxy VM and set the `connectretry` to `0` in `/storage/db/vmware-vrops-cprc/configuration/cprc.configuration` to ensure that the cloud proxy retries to connect.
   2. Shutdown the cloud proxy VM.
   3. Update the network proxy configurations from the vCenter VM options using the vApp options.
   4. Boot the cloud proxy VM.
2. Required ports are not open.
   To verify:

   ```
   openssl s_client -showcerts -connect {address}:443

   curl -v telnet://{address}:443

   # Or, change the address to the machine you want to check:
   python -c "import socket; print(socket.socket(socket.AF_INET, socket.SOCK_STREAM).connect_ex(('127.0.0.1',
    443)))"

   # If you get a !=0 response, the server is not listening to the port.
   ```

   Solution:

   1. SSH to the cloud proxy VM and set `connectretry` to `0` in `/storage/db/vmware-vrops-cprc/configuration/cprc.configuration` to ensure that the cloud proxy retries to connect.
   2. Provide port access as mentioned in the prerequisite section of Deploy Cloud Proxy as Part of a VCF 9.1 Upgrade
   3. Boot the cloud proxy VM.
3. Invalid certificate.
   To verify:

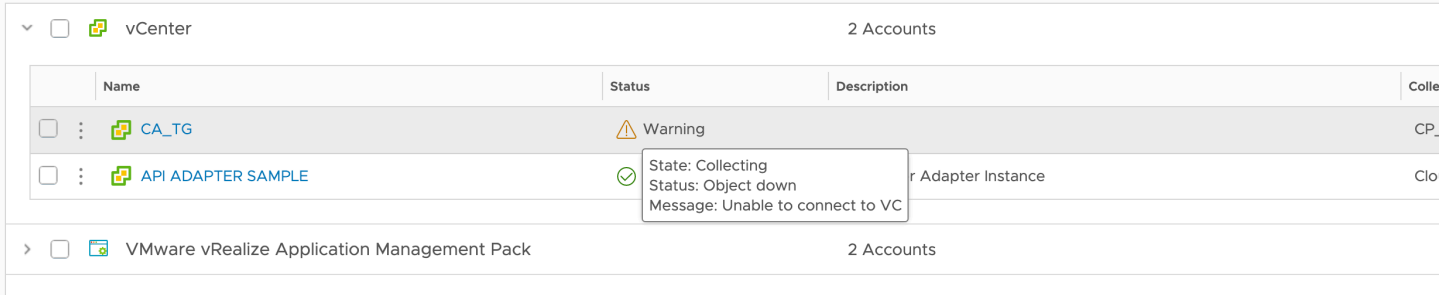
   ```
   openssl s_client -showcerts -connect {address}:443
   ```

   Solution:

   1. SSH to the cloud proxy VM and set `connectretry` to `0` in `/storage/db/vmware-vrops-cprc/configuration/cprc.configuration` to ensure that the cloud proxy retries to connect.
   2. Follow the steps mentioned in Broadcom knowledge base article 320344.
4. The logs folder `/storage/log` is running out of partition space.
   Solution: Remove log files to ensure that enough space is available. Note that this is an exceptional case. In normal conditions, log files are auto archived.

5. One or more of the following services are down: `httpd-north.service`, `haproxy.service` and `collector.service`.
   Solution:
   - Check service status by running the following command: `systemctl status <service name>`.
   - To start service, use the following command: `systemctl start <service name>`.
6. Unique Registration Key expired.
   Solution: Redeploy cloud proxy with new Unique Registration Key.

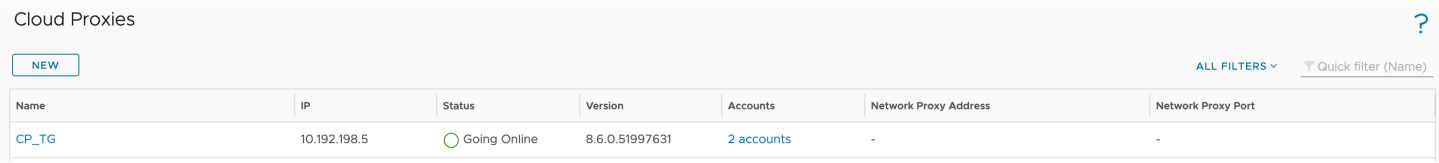
Cloud proxy is online, and state of Cloud Account is Collecting, but the status is Object Down.



The following problem could result in VCF Operations displaying the state of Cloud Account as `Collecting`, while the status is `Object Down`.

1. Incorrect account credentials.
   Solution: Check and update the credentials used while setting up the cloud account.

Cloud proxy status is stuck in Going Online.



It can take up to 20 minutes on first reboot, for the cloud proxy to be registered and come online. Wait for the specified time to see if cloud proxy comes online. If it still does not come online, one or more of the following services are down: `httpd-north.service`, `haproxy.service`, and `collector.service`.

Solution:

1. Check service status by running the following command: `systemctl status <service name>`
2. To start service, use the following command: `systemctl start <service name>`.

Cloud proxy does not upgrade automatically, after the upgrade of VCF Operations

There could be a few possible reasons why cloud proxy does not upgrade automatically after an upgrade of VCF Operations.

1. High network latency leading to PAK download failure. Latency of >500 ms not supported.
   Solution: See the Broadcom knowledge base article 344168 on how to manually upgrade cloud proxy via CLI.
2. Upgrade status is stuck at `Running` since the previous upgrade had failed.
   Solution: Follow the steps given below to change the upgrade status.
   1. Stop the casa service: `systemctl stop vmware-casa.service`.
   2. Change the upgrade status from `RUNNING` to `NONE` in the following files:
      `./storage/db/vmware-vrops-cprc/status/cprc.upgrade.status`
      `./storage/db/vmware-vrops-cprc/status/cprc.pak.status`
   3. See the Broadcom knowledge base article 344168 and run the manual upgrade.

Cloud proxy gets disconnected at regular intervals

There could be a few possible reasons why cloud proxy gets disconnected at regular intervals.

1. Check the network connectivity and latency.
2. Check if the cloud proxy VM can reach the DNS and use NSlookup to validate the DNS connectivity.

Configuring Collector Groups

VCF Operations uses collectors like cloud proxies to manage cloud account processes such as gathering metrics from objects. You can select a single cloud proxy or a cloud proxy group when configuring a cloud account.

If there are cloud proxies in your environment, you can create a collector group, and add the cloud proxies to the group. When you assign an account to a collector group, that account can use any collector in the group. Use collector groups to achieve resiliency in cases where the collector experiences network interruption or becomes unavailable. If this occurs, and the collector is part of a group, the total workload is redistributed among all the collectors in the group, reducing the workload on each collector. For more information, see Configuring Cloud Proxies in VCF Operations.

You can add, edit, or remove collector groups in VCF Operations, and re-balance your cloud accounts.

Re-balancing a Cloud Account

Re-balancing of your cloud accounts is not intended to provide equally distributed cloud accounts across each collector in the collector group. The re-balancing action considers the number of resources that each cloud account collects to determine the re-balancing placement. The re-balancing happens at the cloud account, which can result in several small cloud accounts on a single collector, and a single huge cloud account on another collector, in your VCF Operations instance.

Re-balancing your collector groups can add a significant load on the entire cluster. Moving cloud accounts from one collector to another collector requires that VCF Operations stops the cloud account and all its resources on the source collector, then starts them on the target collector.

If a collector fails to respond or loses connectivity to the cluster, VCF Operations starts automated re-balancing in the collector group. All other user-initiated manual operations on the collector, such as stopping or restarting the collector manually, do not result in automated re-balancing.

If one of the collectors fails to respond, or if it loses network connectivity, VCF Operations performs automated re-balancing. In cases of automated re-balancing, to properly re-balance the collector group, you must have spare capacity on the collectors in the collector group.

Where You Manage Collector Groups

In the VCF Operations navigation bar at the top, click **Home** and in the left navigation pane, click **Administration** > **Cloud Proxies**, and then click the **Collector Groups** tab.

Table 1148: Collector Group Summary Grid

Options	Description
Collector Group toolbar	To manage collector groups, use the toolbar icons. • **Add**: Add a collector group • Click the **Vertical Ellipses** to perform any one of the following actions: – **Edit**: Modify the collector group by adding or removing cloud proxies. You can also edit a collector group to activate or deactivate application monitoring high availability. – **Delete**: Remove the selected collector group. – **Re-balance collector group**: Re-balance one collector group at a time. If you have permission to manage clusters, you can re-balance the workload across the collectors and the cloud proxies in the collector group. The re-balance action moves objects from one collector group to another to re-balance the number of objects on each collector in the collector group. If a disk re-balance is already in progress, the collector re-balance does not run – **Retry collector group configuration**: In case the application monitoring high availability activated collector group configuration fails, you can retry the configuration. – The actions are disabled when the application monitoring high availability activated collector group configuration is in progress. After the configuration is complete you can perform the actions.
Name	The name given to the collector group when the collector group is created or updated.
Collector Group ID	The collector group ID created when the collector group is created.
Description	Description given to the collector group when the collector group is created or updated.
Application Monitoring HA	Displays the application monitoring high availability status of the collector group. • Activated: The collector group can be used for application monitoring. • Deactivated: The collector group cannot be used for application monitoring. **Note:** For default collector groups, high availability for application monitoring is deactivated by default.
HA Status	Displays the application monitoring high availability of the collector group. • OK: The collector group high availability configuration is successful. • In Progress: The collector group high availability configuration is in progress. This status is displayed when adding, deleting, updating, or deactivating the collector group. • Empty: If high availability is not configured, no status is displayed. **Note:** If the high availability status of the cloud proxy fails, the HA status of the failed cloud proxy is displayed. In case there are multiple cloud proxies in a collector group with failed HA status, the first failed message is displayed in the Collectors Group page.
Virtual IP	Virtual IP address of the collector group. **Note:** Pick the Virtual IP address that any of the cloud proxies in a collector group can own and receive traffic from. The Virtual IP address must be in the same subnet as the physical address of the cloud proxies. After you configure the Virtual IP address, ping it from a different network to ensure that it is routable.

Options	Description
Filters	Search the list of collector groups according to the following criteria: • Name • Description • Collector Name • Collector IP Address • Application Monitoring HA • HA Status • Group Virtual IP • Collector Group Id

Click a collector group to view the collector group details.

Table 1149: Collector Group Details Grid

Detail Grid Options	Description
Collector Group Name	Displays the name of the collector group. Click **Edit** to modify the collector group details. Click **View Description** to view the collector group description.
Application Monitoring High Availability Status	Displays the status of the application monitoring high availability.
Virtual IP	Displays the virtual IP added during the collector group creation.
Name	Name given to the cloud proxy when the cloud proxy was created. Click the name to view the cloud proxy details. For more information on cloud proxies, see Monitoring the Health of Cloud Proxies.
IP Address	IP address of the cloud proxy
Status	Status of the cloud proxy: online or offline.
HA Status	Displays the cloud proxy high availability status. • Ok: The cloud proxy application monitoring high availability configuration is successful. • Activation In Progress: Displayed when the cloud proxy is added to a application monitoring HA activated collector group or when the creation of a new application monitoring HA activated collector group is in progress. • Deactivation In Progress: Displayed when the deactivation of a application monitoring HA activated collector group is in progress. • Removal In Progress: Displayed when the removal of a cloud proxy from a application monitoring HA activated collector group is in progress. • Group Delete In Progress: Displayed when the deletion of the entire application monitoring HA activated collector group is in progress. • Empty: If the cloud proxy is not part of a collector group with high availability, no status is displayed. • If the cloud proxy high availability configuration fails, the failure reason is displayed. For example, "Keepalived is failed" or "Apache south is failed".
Type	Displays the cloud proxy type.
Number of Objects	Displays the number of objects collected by the cloud proxy.
Number of cloud accounts	Displays the number of cloud accounts using the cloud proxy.

Adding a Collector Group

Create a new cloud proxy group from the available cloud proxies in your VCF Operations environment. A cloud proxy can only be added to only one group at a time.

Where You Add New Collector Groups?

In the VCF Operations navigation bar at the top, click **Home** and in the left navigation pane, click **Administration** > **Cloud Proxies**, and then click the **Collector Groups** tab. In the Collectors Group tab, click **Add**.

Table 1150: Add New Collector Group

Option	Description
Name	Name of the collector group.
Description	Description of the collector group.
Application Monitoring High Availability	Activate this option to use high availability in application monitoring using collector groups.
Virtual IP	If you activate high availability for application monitoring, enter the virtual IP address. **Note:** Pick the Virtual IP address that any of the cloud proxies in a collector group can own and receive traffic from. The Virtual IP address must be in the same subnet as the physical address of the cloud proxies. After you configure the Virtual IP address, ping it from a different network to ensure that it is routable.
Activate for load balancing	Select this option to activate load balancing for HA activated collector group for open source Telegraf agents.
Collectors added to this collector group	Displays the collectors that are assigned to this collector group. To assign collectors, double click a collector, or drag and drop it from the **Select Collectors** section to add it to the collector group. **Note:** You cannot assign a collector that is part of another collector group. To reassign a collector to another collector group, you must remove it from the existing collector group and then reassign it. To remove a collector from the collector group, click the **Remove Collector** icon.
Select Collectors	Displays a list of cloud proxies in your VCF Operations environment together with their collector name, IP address, type, collector group, and status. Collectors that are assigned to this collector group appear with a check mark before the collector name. **Note:** Cloud proxies with log forwarding cannot be added to collector groups.
Filters	You can search the list of collectors according to the following criteria: • Collector Name • IP address • Collector Group Name • Status

Editing Cloud Proxy Groups

Edit a cloud proxy group in VCF Operations by adding or removing the cloud proxies that you no longer require from an existing group.

Where You Edit a Cloud Proxy Group?

In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Cloud Proxies**, and then click the **Cloud Proxy Groups** tab. Select a cloud proxy group, click the vertical ellipsis, and then select **Edit**. On the Edit Cloud Proxy Group page, update the cloud proxy group details and then click **Save**.

Table 1151: Edit Cloud Proxy Group Options

Option	Description
Name	Name given to the cloud proxy group when the group is created.
Description	Description given to the cloud proxy group when the group is created.
Application Monitoring High Availability	Activate this option to use high availability in application monitoring using cloud proxy groups. If high availability in application monitoring is activated, you can deactivate it.
Virtual IP	If you activate high availability for application monitoring, enter the virtual IP. **Note:** You cannot edit the virtual IP address for a cloud proxy group with high availability for application monitoring activated.
Activate for load balancing	Select this option to activate load balancing for HA activated cloud proxy groups for open source Telegraf agents and product managed Telegraf agents.
Cloud Proxies added to this cloud proxy group	Displays the cloud proxies that are assigned to this group. To assign cloud proxies, double-click a cloud proxy, or drag and drop it from the **Select Cloud Proxy** section to add it to the group. **Note:** You cannot assign a cloud proxy that is part of another group. To reassign a cloud proxy to another group, you must remove it from the existing cloud proxy group and then reassign it. To remove a cloud proxy from the cloud proxy group, click the **Remove cloud proxy** icon.
Select Cloud Proxies	Displays a list of the available cloud proxies in your VCF Operations environment together with their name, IP address, type, cloud proxy group, and status. Cloud proxies that are assigned to this cloud proxy group appear with a check mark before their name. **Note:** Cloud proxies with log forwarding cannot be added to cloud proxy groups.

Managing Cloud Proxy Support Bundles from the Administration UI

VCF Operations support bundles contain log and configuration files that help troubleshoot a VCF Operations cloud proxy issue.

Use the Support Bundles (Cloud Proxy) in VCF Operations Administration UI to create the support bundle on cloud proxy.

Log in to the VCF Operations administration interface at https://*master-node-name-or-ip-address*/admin and then click **Support**.

Support Bundle (Cloud Proxy) Options

The options include toolbar and data grid options.

Use the toolbar options to add, download, or remove items.

Table 1152: Support Bundle (Cloud Proxy) Toolbar Options

Option	Description
Create Support Bundle	Open a dialog box that guides you through the process of creating a support bundle on cloud proxy. Select the cloud proxy and then click **OK** to create a support bundle on the selected cloud proxy. The support bundle is created under the following directory / storage/core/vmware-vrops-cprc/support.
Delete Support Bundle	Remove the selected support bundle.
Download Support Bundle	Download the support bundle in ZIP format.
Reload Support Bundles	Refresh the list of support bundles.

Use the data grid options to view item details.

Table 1153: Support Bundle (Cloud Proxy) Data Grid Options

Option	Description
Bundle	System-generated identifier for the support bundle.
Cloud Proxy Name	The name of the cloud proxy on which the support bundle is created.
Date and Time Created	Time when support bundle creation began.
Status	Progress of support bundle creation.
File Size	The size of the support bundles.

Note:
Generation and download of support bundles through the Support Bundles (Cloud Proxy) page works only if the cloud proxy is connected to the cluster.

If there is a disconnect between the cloud proxy and VCF Operations, you can generate a support bundle on cloud proxy manually.

Open an SSH connection with the cloud proxy appliance and run the followng command:

- For 8.5 and later versions: `$> cprc-cli -sb IS_HEAVY`, where `IS_HEAVY` should be specified as `true` or `false`

With `cprc-cli -sb true`, the support bundle is generated with journal.ctl logs. With `cprc-cli -sb false` , the support bundle is generated without journal.ctl logs. The support bundle is created in the following path: `directory / storage/db/vmware-vrops-cprc/support`.

For adding some extra files into the support bundle, you need to modify configuration file available here: `/storage/db/vmware-vrops-cprc/configuration/cprc.support.bundle.configuration` . Add the path of needed file into `cprc.support.bundle.configuration` in the `files:` section. Then generate the support bundle with the `cprc-cli -sb` or `cprc-cli -sb false` command, based on the version you are on.

For example if you want to add the `/var/log/firstboot/vcopssuitevm.log` file into the support bundle, add this path into the `files:` section of `/storage/db/vmware-vrops-cprc/configuration/cprc.support.bundle.configuration`. Then generate the support bundle with either the `cprc-cli -sb` or the `cprc-cli -sb false` command, based on the version you are on.

Once done, you can download or delete these support bundles from the Support Bundles (Cloud Proxy) page.

Shutdown and Startup of VMware Cloud Foundation

Shutting down VMware Cloud Foundation, for example, during hardware or power maintenance, and then starting it up must be done in a way that prevents data loss or appliance malfunction, and supports collection of troubleshooting data.

You follow a strict order and steps for shutdown and startup of the VMware Cloud Foundation management components.

Shutting Down VMware Cloud Foundation

To avoid data loss and maintain the SDDC components operational, you follow a specific order when shutting down the management virtual machines in VCF.

- Verify that you have complete backups of all management components. See Component Backup and Restore of VMware Cloud Foundation.
- Verify that the management virtual machines are not running on snapshots.
- If a vSphere Storage APIs for Data Protection (VADP) based backup solution is running on the management clusters, verify that the solution is properly shut down by following the vendor guidance.
- To reduce the startup time before you shut down the management virtual machines, migrate the vCenter instance for the management domain to the first VMware ESX host in the default management cluster in the management domain.

You shut down the customer workloads and the management components for the workload domains before you shut down the components for the management domain. If you have multiple VCF Instances, you start with the instances that do not run VCF Operations and VCF Automation. The VCF Instance that is running VCF Operations must be the last instance to shut down.

If the NSX Manager cluster and NSX Edge cluster are shared with other workload domains, shut down the NSX Manager and NSX Edge clusters as part of the shutdown of the first workload domain.

In this chapter:

- Shut Down a Workload Domain
- Shut Down the Management Domain

Shut Down a Workload Domain

You shut down the components of a workload domain that runs virtualized workloads in VCF in a specific order to keep components operational by maintaining the necessary infrastructure, networking, and management services as long as possible before shutdown.

You shut down the management components for the workload domains before you shut down the components for the management domain.

If the NSX Manager cluster and NSX Edge or Virtual Network Appliance (VNA) cluster are shared with other workload domains, follow this general order:

1. Shut down the consumer workloads in all workload domains that share the NSX Manager instance. Otherwise, all NSX networking services in the consumer workloads will be interrupted when you shut down NSX Manager.
2. Shut down the workload domain that runs the shared NSX Edge nodes.
3. Shut down the other workload domains.

Shutdown Order for a Workload Domain

Table 1154: Shutdown Order for a Workload Domain

Shutdown Order	SDDC Component
1	Virtualized customer workloads
2	Protection and recovery for the workload domain *
4	NSX Edge or VNA nodes in the workload domain
5	NSX Manager nodes for the workload domain
7	ESX hosts in the workload domain
8	vCenter for the workload domain

* The information on the shutdown steps is available in another documentation.

Shut Down the NSX Edge or VNA Nodes in the Workload Domain

You begin shutting down the NSX infrastructure in a workload domain in VCF by shutting down the NSX Edge nodes that provide north-south traffic connectivity between the physical data center networks and the NSX SDN networks

1. Identify the vCenter instance that runs NSX Edge or VNA nodes.
2. Log in to vCenter for the workload domain at `https://<vcenter_fqdn>/ui` as a user with the **Administrator** role.
3. In the **VMs and Templates** inventory, locate the NSX Edge or VNA appliance.
4. Right-click the NSX Edge or VNA appliance and select **Power** > **Shut down Guest OS**.
5. Repeat the steps for the remaining NSX Edge or VNA appliances in the VCF domain.

Shut Down the NSX Manager Nodes in a Workload Domain

You continue shutting down the NSX infrastructure for the workload domain by shutting down the NSX Manager VM or cluster by using the vSphere Client.

1. Log in to the management domain vCenter at `https://<vcenter_fqdn>/ui` as **administrator@vsphere.local**.
2. In the **VMs and Templates** inventory, locate an NSX Manager appliance.
3. Right-click the NSX Manager appliance and select **Power** > **Shut down Guest OS**.
4. In the confirmation dialog box, click **Yes**.

   This operation takes several minutes to complete.
5. Repeat the steps for the remaining NSX Manager appliances.

Shut Down the ESX Hosts in a Workload Domain

You follow specific steps according to the principal storage of the clusters in the workload domain.

Save the Credentials for the ESX Hosts and vCenter for the Workload Domain

Before you shut down the workload domain, get the credentials for the workload domain hosts and vCenter and save them. You might need these credentials to shut down the ESX hosts and then to start them and vCenter back up.

Shut Down vSAN and the ESX Hosts in a Workload Domain

You use the vSAN shutdown cluster wizard in the vSphere Client to shut down gracefully the vSAN clusters in a workload domain. The wizard shuts down the vSAN storage and the ESX hosts added to the cluster.

You perform this operation on all vSAN clusters in all workload domains.

1. Identify the vCenter instance that runs the vCenter VM for this workload domain.

   CAUTION:

   If the vCenter VM is running on the infrastructure that it manages, note down the cluster in which it is running. This cluster must be stopped last.

2. Log in to vCenter for the workload domain at `https://<vcenter_fqdn>/ui` as a user with the **Administrator** role.
3. For a vSAN cluster, verify the vSAN health and resynchronization status.
   a) Select the cluster and click the **Monitor** tab.
   b) In the left pane, navigate to **vSAN** > **Skyline health** and verify the status of each vSAN health check category.
   c) In the left pane, under **vSAN** > **Resyncing objects**, verify that all synchronization tasks are complete.
4. If any member host is in lockdown mode, add the host's **root** account to the Exception Users list.
   a) Select the host in the inventory and click the **Configure** tab.
   b) In the left pane, select **System** > **Security Profile**.
   c) In the **Lockdown Mode** pane, click the **Edit** button.
   d) On the **Exception Users** page, enter `root` and click **Add User**.
   e) Click **OK**.
5. Shut down the vSAN cluster.
   a) In the inventory, right-click the vSAN cluster and select **vSAN** > **Shutdown cluster**.
   b) In the **Shutdown Cluster** wizard, verify that all pre-checks are green and click **Next**.
   c) Review the **vCenter Server notice** and click **Next**.
   d) If vCenter is running on the selected cluster, note the orchestration host details.

      Connection to vCenter is lost because the vSAN shutdown cluster wizard shuts it down. The shutdown operation is complete after all ESXi hosts are stopped.
   e) Enter a reason for performing the shutdown, and click **Shutdown**.
6. Repeat Step 3 to Step 5 for other vSAN clusters in the domain.

Shut Down ESX Hosts with NFS or Fibre Channel Storage in a Workload Domain
You use the vSphere Client to shut down gracefully the clusters with non-vSAN in a workload domain.

1. Identify the vCenter instance that runs the vCenter appliance for this workload domain.

   CAUTION:

   This cluster must be shut down last.

2. Log in to vCenter for the workload domain at `https://<vcenter_fqdn>/ui` as a user with the **Administrator** role.
3. In the **Hosts and Clusters** inventory, for a cluster that is not running the vCenter appliance for the workload domain, put all hosts in the cluster in maintenance mode.

   These operations take several minutes to complete.
4. If a cluster is running the vCenter appliance, note down the host on which it is running and put all hosts in the cluster but this one in maintenance mode.

   You need the details about this host during the startup process too.
5. Shut down the hosts in maintenance mode
   a) Select the cluster and click the **Hosts** tab.
   b) Select all hosts that are in maintenance mode, right-click **Power** > **Shut Down**.

      This operation takes several minutes to complete

6. Repeat Step 3 to Step 4 for all clusters with non-vSAN storage in the domain.
7. If the vCenter appliance is running on one of the ESX hosts in the cluster, shut down the vCenter appliance and the host from the VMware Host Client.
   a) Log in to the ESX host at `https://<esx_fqdn>` as a user with the **Administrator** role.
   b) In the left plane, click **Virtual Machines**.
   c) Locate the vCenter appliance for the workload domain.
   d) Right-click the appliance and select **Power** > **Shut down Guest OS**.
   e) In the confirmation dialog box, click **Yes**.

      This operation takes several minutes to complete.
   f) In the left plane, right-click on **Host** > **Shut down**.
   g) Read the warning and click **Shut down**.

      You lose connectivity to the ESX host. The shut down operation takes several minutes to complete.
8. Shut down the storage for the workload domain if there are no other workloads that use it.

   Follow vendor 's documentation to check if there are active clients and how to shut it down in a safe manner.

Shut Down the vCenter Appliance for a Workload Domain

If the vCenter appliance is running on a different infrastructure than the one it manages, to shut down the vCenter appliance for a workload domain in VCF, you use the vSphere Client.

Verify that all ESX hosts in all clusters are stopped and are disconnected.

If the vCenter appliance is running on the infrastructure that it manages, you stop it as part of the shutdown operation for ESX hosts.

1. Identify the vCenter instance that runs the vCenter appliance for this workload domain.
2. Log in to vCenter for the management or workload domain at `https://<vcenter_fqdn>/ui` as a user with the **Administrator** role.
3. Shut down vCenter Server.
   a) Locate the vCenter virtual machine for the workload domain.
   b) Right-click the virtual machine and select **Power** > **Shut down Guest OS**.
   c) In the confirmation dialog box, click **Yes**.

   This operation takes several minutes to complete.

Shut Down the Management Domain

You shut down the components of the management domain in VCF in a specific order to keep components operational by maintaining the necessary infrastructure, networking, and management services as long as possible before shutdown.

After you shut down the components in all workload domains, you begin shutting down the management domain.

Shutdown Order for the Management Domain

> **Note:**
>
> If your VCF Instance is deployed without workload domains, shut down any customer workloads or additional virtual machines in the management domain before you proceed with the shutdown order of the management components.

You shut down VMware Live Site Recovery after you shut down the management components that can be failed over between the VCF Instances. You also shut VMware Live Site Recovery down as late as possible to have the management virtual machines that are protected as long as possible if a disaster event occurs. The virtual machines in the paired VCF Instance become unprotected after you shut downVMware Live Site Recovery in the current VCF Instance.

CAUTION:

If you plan to shut down all VCF Instances in a VCF fleet, you must shut down the first VCF Instance, which is hosting VCF Operations, last.

Table 1155: Shutdown Order for the Management Domain

Shutdown Order	SDDC Component
1	VCF Automation
2	VCF Operations for networks
3	Cloud proxy
4	License server
5	VCF management services
6	VCF Operations
7	Protection and recovery for the management domain *
8	NSX Edge nodes
9	NSX Manager
10	SDDC Manager
11	• Shut Down vSAN and the ESX Hosts in the Management Domain • Shut Down ESX Hosts with NFS or Fibre Channel Storage in the Management Domain

* The information on the shutdown steps is available in another documentation.

Save the Credentials for the ESX Hosts and vCenter for the Management Domain

Before you shut down the management domain, get the credentials for the management domain hosts and vCenter and save them. You need these credentials to shut down the ESX hosts and then to start them and vCenter back up.

Shutting Down a Management Domain with Infrastructure Services VMs

If the management domain contains virtual machines that are running infrastructure services like Active Directory, NTP, DNS and DHCP servers, stop these virtual machines last, that is, after the vCenter appliance.

- For a vSAN cluster, select **I confirm all VMs below are infrastructure VMs** in the vSAN shutdown wizard.
- For a non-vSAN cluster, migrate all such virtual machines to the first host with vCenter and shut them down after the vCenter appliance is shut down.

Shut Down VCF Automation

To gracefully power off the VCF Automation, use the VCF Operations user interface.

1. Log in to VCF Operations at the `https://<vcf_operations_fqdn>` URL as a user with the **Administrator** role.
2. On the main navigation bar, click **Build** and, in the left navigation pane, click **Lifecycle** > **VCF Management** > **Components**.
3. On the **Components** tab, click the component that you are starting or stopping, and from the **Actions** > **State management** menu, select the power operation.
4. Confirm the action and wait for the task to complete.

Shut Down VCF Operations for Networks

To gracefully power off the VCF Operations for Networks, use the VCF Operations user interface.

1. Log in to VCF Operations at the `https://<vcf_operations_fqdn>` URL as a user with the **Administrator** role.
2. On the main navigation bar, click **Build** and, in the left navigation pane, click **Lifecycle** > **VCF Management** > **Components**.
3. On the **Components** tab, click the component that you are starting or stopping, and from the **Actions** > **State management** menu, select the power operation.
4. Confirm the action and wait for the task to complete.

Shut Down the Cloud Proxy Appliances

Shut down the cloud proxy nodes in the management domain of the VCF Instance by using the vSphere Client.

1. Log in to vCenter for the management domain at `https://<vcenter_fqdn>/ui` as a user with the **Administrator** role.
2. In the **VMs and Templates** inventory, locate a cloud proxy appliance.
3. Right-click the appliance and select **Power** > **Shut down Guest OS**.
4. In the confirmation dialog box, click **Yes**.
5. Repeat the steps for the remaining cloud proxy appliances.

Shut Down the License Server

Shut down the license server appliance in the management domain by using the vSphere Client.

1. Log in to vCenter for the management domain at `https://<vcenter_fqdn>/ui` as a user with the **Administrator** role.
2. In the **VMs and Templates** inventory, locate a license server appliance.
3. Right-click the license server appliance and select **Power** > **Shut down Guest OS**.
4. In the confirmation dialog box, click **Yes**.

Shut Down VCF Management Services

To gracefully power off VCF management services, you shut down the VCF services runtime by using a script.

> **Warning:**
>
> To perform the steps for shutting down VCF management services, contact Broadcom Technical Support.

- Verify that you have the IP address of one of the control plane nodes of the VCF services runtime.
  To locate the control plane nodes, follow these steps:
  a. Log in to VCF Operations at the `https://<vcf_operations_fqdn>` URL as a user with the **Administrator** role.
  b. Navigate to **Build** > **Lifecycle** > **VCF Management** > **Components**.
  c. Click the VCF services runtime component for the VCF instance that you are stopping.
  d. Locate the IP addresses of the control plane nodes in the **Nodes** table.
  e. Note the IP addresses and VM names of all control plane nodes.
- Verify that you have the password of the **vmware-system-user** local account for VCF services runtime.

For information about the components that are available as VCF management services, see Components in VCF and vSphere Foundation.

1. Download the `vcf_services_runtime_shutdown.sh` script from Broadcom knowledge base article 440874 and copy it to the SDDC Manager appliance.
2. Log in to the SDDC Manager appliance over SSH at `<sddc_manager_fqdn>:22` as the**vcf** user.
3. On the SDDC Manager appliance, in the folder where you copied the `vcf_services_runtime_shutdown.sh` script, make the script executable by running the following command.

   ```
   chmod +x vcf_services_runtime_shutdown.sh
   ```

4. Perform a dry run of the shutdown operation on one of the control plane nodes.

   ```
   ./vcf_services_runtime_shutdown.sh --node-ip <control_node_ip_address> --dry-run
   ```

5. Shut down the VCF services runtime on the control plane node.

   ```
   ./vcf_services_runtime_shutdown.sh --node-ip <control_node_ip_address>
   ```

   After the script completes, it shows a list of VMs that must be shut down in the management vCenter instance.
6. Log in to vCenter for the management domain at `https://<vcenter_fqdn>/ui` as a user with the **Administrator** role.
7. Locate the VM in the vCenter inventory.
   a) Right-click the VM and select **Power** > **Shut down Guest OS**.
   b) Repeat for all VMs provided by the script.
8. Once all the VMs hosting the VCF services runtime are in the desired state, proceed to the next component.

Shut Down VCF Operations

You shut down VCF Operations by first taking the cluster offline and then shutting down the appliances of the VCF Operations cluster.

1. Log in to the VCF Operations administration UI at the `https://<vcf_operations_fqdn>/admin` URL as the **admin** local user.
2. Take the VCF Operations cluster offline.
   a) On the **System status** page, click **Take cluster offline**.
   b) In the **Take cluster offline** dialog box, provide the reason for the shutdown and click **OK**.

      This operation might take about an hour to complete.
3. Log in to vCenter for the management domain at `https://<vcenter_fqdn>/ui` as a user with the **Administrator** role.
4. Shut down VCF Operations.

   Important:

   Shut down the guest OS of the VCF Operations appliances by following the order in Broadcom knowledge base article 341964.

   a) In the **VMs and Templates** inventory, locate a VCF Operations appliance.
   b) Right-click the appliance and select **Power** > **Shut down Guest OS**.
   c) In the confirmation dialog box, click **Yes**.

      This operation takes several minutes to complete.
   d) Repeat the steps for the remaining VCF Operations appliances by following the order in Broadcom knowledge base article 341964.

Shut Down the NSX Edge or VNA Nodes in the Management Domain

You begin shutting down the NSX infrastructure in the management domain VMware Cloud Foundation by shutting down the NSX Edge or VNA nodes that provide north-south traffic connectivity between the physical data center networks and the NSX SDN networks

1. Log in to vCenter for the management domain at `https://<vcenter_fqdn>/ui` as a user with the **Administrator** role.
2. In the **VMs and Templates** inventory, locate the NSX Edge or VNA appliance.
3. Right-click the NSX Edge or VNA appliance and select **Power** > **Shut down Guest OS**.
4. In the confirmation dialog box, click **Yes**.

   This operation takes several minutes to complete.
5. Repeat the steps for the remaining NSX Edge or VNA nodes for the domain.

Shut Down the NSX Manager Nodes

You continue shutting down the NSX infrastructure in the management domain and a workload domain by shutting down the three-node NSX Manager cluster by using the vSphere Client.

1. Log in to vCenter for the management domain at `https://<vcenter_fqdn>/ui` as a user with the **Administrator** role.
2. Identify the vCenter instance that runs NSX Manager.
3. In the **VMs and Templates** inventory, locate an NSX Manager appliance.
4. Right-click the NSX Manager appliance and select **Power** > **Shut down Guest OS**.
5. In the confirmation dialog box, click **Yes**.

   This operation takes several minutes to complete.
6. Repeat the steps for the remaining NSX Manager appliances.

Shut Down the SDDC Manager Appliance

Shut down the SDDC Manager appliance in the management domain by using the vSphere Client.

1. Log in to vCenter for the management domain at `https://<vcenter_fqdn>/ui` as a user with the **Administrator** role.
2. In the **VMs and templates** inventory, expand the management domain vCenter Server tree and expand the management domain data center.
3. Expand the **Management VMs** folder.
4. Right-click the SDDC Manager appliance and click **Power** > **Shut down Guest OS**.
5. In the confirmation dialog box, click **Yes**.

   This operation takes several minutes to complete.

Shut Down the ESX Hosts and vCenter in the Management Domain

In the shutdown order, you must down the ESX hosts and management vCenter last.

> **Important:**
>
> You perform this operation on all clusters in the management domain. The primary cluster which hosts the management vCenter must be stopped last.

Shut Down vSAN and the ESX Hosts in the Management Domain

You use the vSAN shutdown cluster wizard in the vSphere Client to shut down gracefully the vSAN clusters in a management domain. The wizard shuts down the vSAN storage and the ESX hosts added to the cluster.

1. Identify the cluster that hosts the management vCenter for this management domain.

   > **Important:**
   >
   > This cluster must be shut down last.

2. Log in to vCenter for the management domain at `https://<vcenter_fqdn>/ui` as a user with the **Administrator** role.
3. For a vSAN cluster, verify the vSAN health and resynchronization status.
   a) In the **Hosts and Clusters** inventory, select the cluster and click the **Monitor** tab.
   b) In the left pane, navigate to **vSAN** > **Skyline health** and verify the status of each vSAN health check category.
   c) In the left pane, under **vSAN** > **Resyncing objects**, verify that all synchronization tasks are complete.
4. If any member host is in lockdown mode, add the host's **root** account to the Exception Users list.
   a) Select the host in the inventory and click the **Configure** tab.
   b) In the left pane, select **System** > **Security Profile**.
   c) In the **Lockdown Mode** pane, click the **Edit** button.
   d) On the **Exception Users** page, enter `root` and click **Add User**.
   e) Click **OK**.
5. Shut down the vSAN cluster.
   a) In the inventory, right-click the vSAN cluster and select **vSAN** > **Shutdown cluster**.
   b) In the **Shutdown Cluster** wizard, verify that all pre-checks are green and click **Next**.
   c) Review the **vCenter Server notice** and click **Next**.
   d) If vCenter is running on the selected cluster, note the orchestration host details.

      Connection to vCenter is lost because the vSAN shutdown cluster wizard shuts it down. The shutdown operation is complete after all ESXi hosts are stopped.
   e) Enter a reason for performing the shutdown, and click **Shutdown**.
6. Repeat Step 3 to Step 5 for the other vSAN clusters in the domain.

Shut Down ESX Hosts with NFS or Fibre Channel Storage in the Management Domain
You use the vSphere Client to shut down gracefully the clusters in a management domain with a principal storage that is different from vSAN.

1. Identify the cluster that hosts the management vCenter for this management domain.

   CAUTION:

   This cluster must be shut down last.
2. Log in to vCenter for the management domain at `https://<vcenter_fqdn>/ui` as a user with the **Administrator** role.
3. In the **Hosts and Clusters** inventory, for a cluster that is not running the management vCenter, put all hosts in the cluster in maintenance mode.

   These operations take several minutes to complete.
4. If a cluster is running the management vCenter, note down the host on which it is running and put all hosts in the cluster but this one in maintenance mode.

   You need the details about this host during the startup process too.
5. Shut down the hosts in maintenance mode.
   a) Select the cluster and click the **Hosts** tab.
   b) Select all hosts that are in maintenance mode, right-click **Power** > **Shut Down**

      This operation takes several minutes to complete.

6. Repeat Step 3 to Step 5 for all clusters with NFS and Fibre Channel storage in the domain.
7. If vCenter is running on one of the ESX hosts in the cluster, shut down the vCenter appliance and the host from the VMware Host Client.
   a) Log in to the ESX host at `https://<esx_fqdn>` as a user with the **Administrator** role.
   b) In the left pane, click **Virtual Machines**.
   c) Locate the vCenter appliance for the domain.
   d) Right-click the virtual machine and select **Power** > **Shut down Guest OS**.
   e) In the confirmation dialog box, click **Yes**.

      This operation takes several minutes to complete.
   f) In the left plane, right-click on **Host** > **Shut down**.
   g) Read the warning and click **Shut down**.

      You lose connectivity to the ESX host. The shutdown operation takes several minutes to complete.
8. Shut down the storage for the management domain if there are no other workloads that use it.

   Follow vendor 's documentation to check if there are active clients and how to shut it down in a safe manner.

Starting Up VMware Cloud Foundation

To maintain the components integration and avoid operation faults, you follow a specified order to start up the management virtual machines in VMware Cloud Foundation.

- Verify that external services such as Storage, Active Directory, DNS, NTP, SMTP, and FTP or SFTP are available.
- If a vSphere Storage APIs for Data Protection (VADP) based backup solution is deployed on the default management cluster, verify that the solution is properly started and operational according to the vendor guidance.

You start the management components for the management domain first. Then, you start the management components for the workload domains and the customer workloads.

If the NSX Manager cluster and NSX Edge cluster are shared with other workload domains, start the other workload domains first. Start up NSX Manager and NSX Edge nodes as part of the startup of the last workload domain.

In this chapter:

- Start the Management Domain
- Start a Workload Domain

Start the Management Domain

You start the management components for the management domain in a specific order to provide the necessary infrastructure, networking, and management services before powering on the components for cloud management.

You start the management components for the management domain first. Then, you start the management components for the workload domains and the customer workloads.

> **CAUTION:**
>
> If you have shut down all VCF Instances in a VCF fleet, you must first start the VCF Instance, which is running VCF Operations.

Startup Order for the Management Domain

You start the virtual infrastructure of the management domain that hosts VCF Operations and VCF Automation first.

Table 1156: Startup Order for the Management Domain

Startup Order	SDDC Component
1	• Start vSAN and the ESX Hosts in the Management Domain • Start ESX Hosts with NFS or Fibre Channel Storage in the Management Domain
2	SDDC Manager
3	NSX Manager
4	NSX Edge
5	Protection and recovery
6	VCF Operations
7	VCF management services
8	License server
9	Cloud proxy
10	VCF Operations for networks
11	VCF Automation

Start the ESX Hosts and vCenter in the Management Domain

You start the ESX hosts using an out-of-band management interface, such as, ILO or iDRAC to connect to the hosts and power them on.

> **Important:**
>
> You perform this operation on all clusters in the management domain. Start the cluster where the management vCenter must be running first.

Start vSAN and the ESX Hosts in the Management Domain

You start the ESX hosts using an out-of-band management interface, such as, ILO or iDRAC to connect to the hosts and power them on. Then, restarting the vSAN cluster starts automatically vCenter and vSAN.

You perform this operation on all vSAN clusters in the management domain. You must start with the cluster that hosts the management vCenter.

1. Power on the first ESX host in the management domain.
   a) Log in to the first ESX host in the domain by using the out-of-band management interface.
   b) Power on the ESX host according to the hardware vendor guide.
2. Repeat the previous step to start all the remaining ESX hosts in the management domain.

   This operation takes several minutes to complete.

   vCenter is started automatically. Wait until vCenter is running and the vSphere Client is available again.
3. Log in to vCenter for the management domain at `https://<vcenter_fqdn>/ui` as a user with the **Administrator** role.
4. Restart the vSAN cluster.
   a) In the **Hosts and Clusters** inventory, right-click the vSAN cluster and select **vSAN** > **Restart cluster**.
   b) In the **Restart Cluster** dialog box, click **Restart**.

   The **vSAN Services** page on the **Configure** tab changes to display information about the restart process.

5. After the cluster has been restarted, check the vSAN health service and resynchronization status, and resolve any outstanding issues.
   a) Select the cluster and click the **Monitor** tab.
   b) In the left pane, under **vSAN** > **Resyncing objects**, verify that all synchronization tasks are complete.
   c) In the left pane, navigate to **vSAN** > **Skyline health** and verify the status of each vSAN health check category.
6. If you have added the **root** user of the ESXi hosts to the Exception Users list for lockdown mode during shutdown, remove the user from the list on each host.
   a) Select the host in the inventory and click the **Configure** tab.
   b) In the left pane, select **System** > **Security Profile**.
   c) In the **Lockdown Mode** pane, click the **Edit** button.
   d) On the **Exception Users** page, from the vertical ellipsis menu in front of the **root** user, select **Remove User** and click **OK**.

Start ESX Hosts with NFS or Fibre Channel Storage in the Management Domain
You start the ESX hosts using an out-of-band management interface, such as, ILO or iDRAC to connect to the hosts and power them on. Then, restart the cluster from vCenter.

External storage is running and accessible.

You perform this operation on all non-vSAN clusters in the management domain. You must start with the cluster that hosts the management vCenter.

1. Power on the first ESX host in the management domain.
   a) Log in to the first ESX host in the domain by using the out-of-band management interface.
   b) Power on the ESX host according to the hardware vendor guide.
2. Repeat the previous step to start all the remaining ESX hosts in the management domain.

   This operation takes several minutes to complete.
3. Log in into the ESX host that hosts vCenter and any infrastructure components.

   Important:

   You must first start infrastructure components that runs in the management cluster before you start the vCenter.

   a) Log in into ESX host at `https://<esx_fqdn>` as a user with the **Administrator** role.
   b) Locate the management vCenter appliance.
   c) Right-click the appliance and select **Power** > **Power on**.
4. Log in to vCenter for the management domain at `https://<vcenter_fqdn>/ui` as a user with the **Administrator** role.
5. Verify that all hosts in the cluster are connected to the management vCenter.
6. Exit maintenance mode for all hosts in the cluster.

   This operations takes several minutes to complete.
7. Repeat Step 5 and Step 6 for all clusters with non-vSAN storage in the domain.

Start the SDDC Manager Appliance

Start the SDDC Manager appliance by using the vSphere Client.

1. Log in to vCenter for the management domain at `https://<vcenter_fqdn>/ui` as a user with the **Administrator** role.
2. In the **VMs and templates** inventory, expand the management domain vCenter Server tree and expand the management domain data center.
3. Expand the **Management VMs** folder.
4. Right-click the SDDC Manager appliance and click **Power** > **Power On**.
5. In the confirmation dialog box, click **Yes**.

   This operation takes several minutes to complete.

Start the NSX Manager Nodes

You begin powering on the NSX infrastructure in the management domain by starting the NSX Manager appliance or cluster by using the vSphere Client.

1. Log in to vCenter for the management domain at `https://<vcenter_fqdn>/ui` as a user with the **Administrator** role.
2. In the **VMs and templates** inventory, expand the management domain vCenter Server tree and expand the management domain data center.
3. Power on the NSX Manager nodes for the management domain.
   a) Right-click NSX Manager node and select **Power** > **Power on**.
   b) Repeat the steps to power on the remaining NSX Manager nodes.

   This operation takes several minutes to complete until the NSX Manager cluster becomes fully operational again and its user interface - accessible.
4. Log in to NSX Manager for the management domain at `https://<nsxt_manager_cluster_fqdn>` as **admin**.
5. Verify the system status of NSX Manager cluster.
   a) On the main navigation bar, click **System**.
   b) In the left pane, navigate to **Configuration** > **Appliances**.
   c) On the **Appliances** page, verify that the NSX Manager cluster has a *`Stable`* status and all NSX Manager nodes are available.

Start the NSX Edge or VNA Nodes

You continue powering on the NSX infrastructure in the management domain by starting the NSX Edge or VNA nodes by using the vSphere Client.

1. Log in to vCenter for the management domain at `https://<vcenter_fqdn>/ui` as a user with the **Administrator** role.
2. In the **VMs and templates** inventory, expand the management domain vCenter Server tree and expand the management domain data center.
3. Right-click an NSX Edge node from the edge cluster or the VNA node from the VNA cluster and select **Power** > **Power on**.

   This operation takes several minutes to complete.
4. Repeat these steps to power on the remaining NSX Edge or VNA nodes.

Start VCF Operations

You start VCF Operations by first starting the appliances of the VCF Operations cluster, then bring the cluster online.

1. Log in to vCenter for the management domain at `https://<vcenter_fqdn>/ui` as a user with the **Administrator** role.
2. Start VCF Operations.

   Important:

   Power on the guest OS of the VCF Operations appliances by following the order in Broadcom knowledge base article 341964.

   a) Locate a VCF Operations appliance.
   b) Right-click the appliance and select **Power** > **Power On**.
   c) In the confirmation dialog box, click **Yes**.

      This operation takes several minutes to complete.
   d) Repeat the steps for the remaining VCF Operations appliances by following the order described in Broadcom knowledge base article 341964.
3. Log in to the VCF Operations administration UI at the `https://<vcf_operations_fqdn>/admin` URL as the **admin** local user.
4. Bring the VCF Operations cluster online.
   a) Log in to the VCF Operations administration interface at the `https://<vcf_operations_fqdn>/admin` URL as the **admin** local user.
   b) On the **System status** page, click **Bring Cluster Online**.

      This operation might take about an hour to complete.

Start VCF Management Services

To start VCF management services, power on the nodes of the VCF services runtime by using the vSphere Client.

1. Log in to vCenter for the management domain at `https://<vcenter_fqdn>/ui` as a user with the **Administrator** role.
2. In the **VMs and templates** inventory, expand the management domain vCenter Server tree and expand the management domain data center.
3. Locate the `vcf-management-services` folder which hosts the VCF services runtime nodes.
4. Identify the control nodes, noted during the shutdown, and start them.

   The control nodes are the VMs with less CPU and memory in the folder.

   a) Right-click a control node and select **Power** > **Power on**.
   b) Repeat for all control nodes.

      This operation takes several minutes to complete.
5. Once all VCF services runtime control nodes are up and running, start the VCF services runtime worker nodes.

   The worker nodes are the remaining VMs in the `vcf-management-services` folder.

   a) Right-click a worker node and select **Power** > **Power on**.
   b) Repeat for all worker nodes.

      Worker nodes startup takes about 20 minutes to complete.
6. Verify that all the worker nodes are up and running and no new worker nodes are in the process of deployment.

   The startup operation might redeploy some of the worker nodes.

7. Log in to VCF Operations at the `https://<vcf_operations_fqdn>` URL as a user with the **Administrator** role.
8. On the main navigation bar, click **Build** and, in the left navigation pane, click **Lifecycle** > **VCF Management** > **Components**.
9. Verify that the UI is loaded.
10. Click the VCF services runtime component and verify the Status column of all services.
11. If the UI is not loading or some services are not shown as running, follow the manual recovery procedure in Broadcom knowledge base article 440862.

Start the License Server

Start the license server node in the management domain by using the vSphere Client.

1. Log in to vCenter for the management domain at `https://<vcenter_fqdn>/ui` as a user with the **Administrator** role.
2. In the **VMs and Templates** inventory, locate a license server appliance.
3. Right-click the license server appliance and select **Power** > **Power On**.
4. In the confirmation dialog box, click **Yes**.

Start the Cloud Proxy Appliances

Start the cloud proxy nodes in the management domain by using the vSphere Client.

1. Log in to vCenter for the management domain at `https://<vcenter_fqdn>/ui` as a user with the **Administrator** role.
2. In the **VMs and Templates** inventory, locate a cloud proxy appliance.
3. Right-click thecloud proxy appliance and select **Power** > **Power On**.
4. In the configuration dialog box, click **Yes**.
5. Repeat the steps on the remaining cloud proxy appliances.

Start VCF Operations for Networks

To start VCF Operations for Networks, use the VCF Operations user interface.

1. Log in to VCF Operations at the `https://<vcf_operations_fqdn>` URL as a user with the **Administrator** role.
2. On the main navigation bar, click **Build** and, in the left navigation pane, click **Lifecycle** > **VCF Management** > **Components**.
3. On the **Components** tab, click the component that you are starting or stopping, and from the **Actions** > **State management** menu, select the power operation.
4. Confirm the action and wait for the task to complete.

Start VCF Automation

To start VCF Automation, use the VCF Operations user interface.

1. Log in to VCF Operations at the `https://<vcf_operations_fqdn>` URL as a user with the **Administrator** role.
2. On the main navigation bar, click **Build** and, in the left navigation pane, click **Lifecycle** > **VCF Management** > **Components**.
3. On the **Components** tab, click the component that you are starting or stopping, and from the **Actions** > **State management** menu, select the power operation.
4. Confirm the action and wait for the task to complete.

Start a Workload Domain

You start the management components for a workload domain in a specific order to provide the necessary infrastructure, networking, and management services.

You start the management components for the management domain first. Then, you start the management components for the workload domains and the customer workloads.

If the NSX Manager cluster and NSX Edge or Virtual Network Appliance (VNA) cluster are shared across workload domains, follow this general order:

1. Start the other workload domains.
2. Start the workload domain that runs the shared NSX Edge nodes.
3. Start the customer workloads that rely on NSX services.

Startup Order for a Workload Domain

Table 1157: Startup Order for a Workload Domain

Startup Order	SDDC Component
1	vCenter for the workload domain
2	ESX hosts for the workload domain
3	Restart Clusters for the workload domain
5	NSX Manager nodes for the workload domain
6	NSX Edge nodes for the workload domain
8	Protection and recovery for the workload domain *
9	Virtualized customer workloads

* The information on the startup steps is available in another documentation.

Start the vCenter Appliance for a Workload Domain

Use the vSphere Client to power on the vCenter appliance for the workload domain. If the workload domain contains a vSAN cluster, check its health status too.

vCenter startup is done:

- If vCenter is running on a vSAN cluster, automatically from the orchestration host.
- If vCenter is running on a non-vSAN storage, manually following the steps below.

If the vCenter appliance is running on the infrastructure, you must start the ESX hosts first.

1. Log in into the resource that hosts the vCenter appliance.
   a) For an ESX host, log in to the ESX host at `https://<esx_fqdn>` as a user with the **Administrator** role.
   b) For the management vCenter, log in to vCenter at the `https://<vcenter_fqdn>/ui` URL as a user with the **Administrator** role.
2. Start vCenter Server.
   a) Locate the vCenter virtual machine for the workload domain.
   b) Right-click the virtual machine and select **Power** > **Power on**.

   The startup of the virtual machine and the vSphere services takes some time to complete.

Start the ESX Hosts in a Workload Domain

You start the ESX hosts using an out-of-band management interface, such as ILO or iDRAC to connect to the hosts and power them on.

If you are using non-vSAN storage, the storage must be started and accessible.

1. Power on the first ESX host in the workload domain.
   a) Log in to the first ESX host in the workload domain by using the out-of-band management interface.
   b) Power on the ESX host according to the documentation from the hardware vendor.
2. Repeat the step to start all the remaining ESX hosts in the workload domain.

   This operation takes several minutes to complete.

Restart Clusters in a Workload Domain

After you start the ESX hosts and the vCenter appliance for a workload domain, restart all clusters to re-initiate the integration with the principal storage.

Restart the vSAN Clusters in a Workload Domain

You start vSAN in a workload domain by restarting the vSAN cluster.

All hosts are started.

Verify that all ESX hosts in the cluster are connected to the workload domain vCenter.

You perform this operation on all vSAN clusters in all workload domains.

1. Log in to vCenter Server for the management or workload domain at `https://<vcenter_fqdn>/ui` as **administrator@vsphere.local**.
2. In the **Hosts and clusters** inventory, expand the tree of the workload domain vCenter and expand the data center for the workload domain.
3. Restart the vSAN cluster.
   a) Right-click the vSAN cluster and select **vSAN** > **Restart cluster**.
   b) In the **Restart Cluster** dialog box, click **Restart**.

      The **vSAN Services** page on the **Configure** tab changes to display information about the restart process.

4. After the cluster has restarted, check the vSAN health service and resynchronization status, and resolve any outstanding issues.
   a) Select the cluster and click the **Monitor** tab.
   b) In the left pane, navigate to **vSAN** > **Skyline health** and verify the status of each vSAN health check category.
   c) In the left pane, under **vSAN** > **Resyncing objects**, verify that all synchronization tasks are complete.
5. If you have added the **root** user of the ESXi hosts to the Exception Users list for lockdown mode during shutdown, remove the user from the list on each host.
   a) Select the host in the inventory and click the **Configure** tab.
   b) In the left pane, select **System** > **Security Profile**.
   c) In the **Lockdown Mode** pane, click the **Edit** button.
   d) On the **Exception Users** page, from the vertical ellipsis menu in front of the **root** user, select **Remove User** and click **OK**.

Restart the Clusters in a Workload Domains with NFS or Fibre Channel Storage Storage
You use the vSphere Client to restart the clusters with non-vSAN storage in a workload domain.

All hosts are started

Verify that all ESXi hosts in the cluster are connected to the workload domain vCenter Server.

1. Log in to vCenter for the workload domain at `https://<vcenter_fqdn>/ui` as a user with the **Administrator** role.
2. Check that all hosts in the cluster are connected to vCenter.
3. Exit maintenance mode for all hosts in the cluster.

   This operations takes several minutes to complete.
4. Repeat Step 2 and Step 3 for all clusters with non-vSAN storage in the domain.

Start the NSX Manager Nodes

You begin powering on the NSX infrastructure in the workload domain by starting the NSX Manager appliance or cluster by using the vSphere Client.

1. Identify the vCenter instance that runs NSX Manager.
2. Log in to vCenter for the management or workload domain at `https://<vcenter_fqdn>/ui` as a user with the **Administrator** role.
3. In the **VMs and templates** inventory, expand the management domain vCenter Server tree and expand the management domain data center.
4. Power on the NSX Manager nodes for the workload domain.
   a) Right-click NSX Manager node and select **Power** > **Power on**.
   b) Repeat the steps to power on the remaining NSX Manager nodes.

   This operation takes several minutes to complete until the NSX Manager cluster becomes fully operational again and its user interface - accessible.
5. Log in to NSX Manager for the workload domain at `https://<nsx_manager_cluster_fqdn>` as **admin**.
6. Verify the system status of the NSX Manager cluster.
   a) On the main navigation bar, click **System**.
   b) In the left pane, navigate to **Configuration** > **Appliances**.
   c) On the **Appliances** page, verify that the NSX Manager cluster has a *`Stable`* status and all the listed NSX Manager nodes are available.

Start the NSX Edge or VNA Nodes

You continue powering on the NSX infrastructure in the workload domain by starting the NSX Edge or VNA nodes by using the vSphere Client.

1. Identify the vCenter instance that runs NSX Edge or VNA nodes.
2. Log in to vCenter for the workload domain at `https://<vcenter_fqdn>/ui` as a user with the **Administrator** role.
3. In the **VMs and templates** inventory, expand the tree of workload domain vCenter and expand data center for the workload domain.
4. Right-click an NSX Edge from the edge cluster or a VNA node from the VNA cluster and select **Power** > **Power on**.

   This operation takes several minutes to complete.
5. Repeat these steps to power on the remaining NSX Edge or VNA nodes.

Component Backup and Restore of VMware Cloud Foundation

Regularly backing up the management components of a VMware Cloud Foundation platform is essential to ensure the resilience and continuity of your VCF private cloud. In the event of data loss or system failure, having up-to-date backups allows for efficient restoration of critical services, minimizing downtime and maintaining operational integrity. Implementing a consistent backup strategy safeguards your infrastructure against unforeseen disruptions and supports effective disaster.

You implement scheduled backups to prepare for:

- A critical failure of a management component.
- An upgrade of a management component.
- A certificate update of a management component.

In addition, as a best practice, you can take on-demand manual backups in the following use cases:

- After a successful recovery operation.
- After resolving asynchronously reported errors in SDDC components.
- After resolving an incomplete workflow in SDDC Manager.
- After noting the failure of a scheduled backup of an SDDC component.
- Immediately before performing a system upgrade.

It is recommended that you configure an external SFTP server as a backup location for the following reasons:

- An external SFTP server is a prerequisite for restoring SDDC Manager file-based backups.
- Using an external SFTP server provides better protection against failures because it decouples NSX backups from SDDC Manager.

Prerequisites

Before performing backup and restore operations in your VMware Cloud Foundation platform, ensure the following requirements are met:

General Requirements:

- Administrator permissions to VMware Cloud Foundation platform components.
- A stable network connection between the management domain and backup storage.
- Sufficient storage capacity to accommodate backup data.
- A defined backup and retention policy aligned with best practices or organizational requirements.
- Access to an image-level backup solution for required components.
- Sufficient time allowed between backup operations so that a new backup request is not submitted when another backup of the same environment is in progress.

Backup-Specific Requirements:

- Configure and validate an external backup target based on individual management component requirements.
- Ensure all management components, including SDDC Manager, vCenter, NSX, VCF Operations, and VCF Automation, are configured with a regular backup schedule and retention policy.

Restore-Specific Requirements:

- Access to the most recent backup files.

- In place infrastructure for restoring.
- Proper sequencing and verification of component restoration to maintain dependencies and system integrity.

Backup Methods for VMware Cloud Foundation

Category	Component	Supported Backup Method(s)
Management Appliance	VCF Operations	Image-Based
	VCF Operations for networks	Image-Based
Management Services	VCF services runtime	File-Based
	SDDC lifecycle	File-Based
	Fleet Lifecycle	File-Based
	Log Management	File-Based **Important:** Log data is not included in the backup and must be archived
	Software Depot	File-based
	Identity Broker	File-based
	Salt RaaS	File-based
	Salt Master	File-based
	Telemetry	File-based **Important:** Only captured as part of a scheduled backup
	Real-Time Metrics	File-based **Important:** Only captured as part of a scheduled backup
	Real-Time Metrics Store	File-based **Important:** Only captured as part of a scheduled backup
	SDDC Manager	File-based or Image-Based*
Consumption	VCF Automation	File-based
	Migration service engine	File-based **Important:** Only captured as part of a scheduled backup
	VCF Automation services runtime	File-based **Important:** Only captured as part of a scheduled backup
Core	vCenter	File-based or Image-Based*
	NSX Manager	File-based
	NSX Edge	Edge configuration is included in NSX Manager backup

* This document uses file-based backup for this component.

Configure Backup Target and Schedules in VMware Cloud Foundation

Following a succesful deployment of a VMware Cloud Foundation platform to ensure that you are able to recover from failues you must configure a backup target and schedule across components.

- Configure an External SFTP Server for VCF Management Services and VCF Automation Backups
- Configure an External SFTP Server for SDDC Manager and NSX Manager Backups
- Configure Image-Based Backup of VCF Operations and VCF Operations for Networks
- Configure File-Based Backup Schedules in VMware Cloud Foundation
  - Configure File-Based Backup Schedule for VCF Management Services and VCF Automation
  - Configure File-Based Backup Schedule for SDDC Manager
  - Configure File-Based Backup Schedule for NSX Manager
  - Configure File-Based Backup Schedule and External SFTP Server for vCenter

Configure an External SFTP Server for VCF Management Services and VCF Automation Backups

To allow for backup and restore of VCF management services and VCF Automation, you configure an SFTP target in VCF Operations. This designated backup location is also used for automatic backups that are generated during patching and upgrade procedures. This configuration is a prerequisite for patching or upgrading VCF management services components, and for scheduled backups.

Verify that no post-deployment actions are in progress for VCF Automation or any of the following VCF management services:

- Log management
- Identity broker
- Software depot
- Fleet lifecycle
- SDDC lifecycle
- Real-time metrics
- Salt RaaS

Running tasks in parallel can cause the SFTP configuration to fail.

1. Log in to the VCF Operations interface at `https://<vcf_operations_fqdn>` as a user assigned the Administrator role.
2. Navigate to **Build** > **Lifecycle**.
3. In the **Lifecycle** navigation, select **VCF Management**.
4. On the **VCF Management** page, select the **Backup & Restore** tab.
5. Click **Add Backup Location**.
6. Enter the following information, click **Add** to fetch the fingerprint, then click **Confirm Fingerprint**, and click **Add** to save the configuration.

Table 1158: SFTP Configuration

Configuration	Value
VCF Instance	Select the target VCF Instance.
SFTP Host FQDN or IP Address	The FQDN or IP Address of the SFTP server.
Port	22 **Note:** With VCF 9.1.1 and later, you can configure a custom SFTP port.

Configuration	Value
Username	A service account with privileges to the SFTP server. For example: `svc-vcf-bck`.
Password	The password for the username provided.
Backup directory	The directory on the SFTP server where backups are saved. For example: `/backups/`. **Important:** The SFTP user must have read access to all sub-directories of the backup directory path.
Encryption passphrase	The encryption passphrase used to encrypt the backup data. **Important:** The encryption passphrase should be stored safely as it is required during the restore process.

Note:

Before saving, ensure that the SFTP host, port, username, and password are correct and validated. Any incorrect information that you enter propagates to all VCF management services components and causes the SFTP configuration to fail.

The SFTP server configuration is automatically pushed to the VCF management services and VCF Automation instances.

Note:

If there is a failure, an option to **Retry** appears. Click **Refresh** to fetch the status of the SFTP configuration and display it in the UI.

Configure an External SFTP Server for SDDC Manager and NSX Manager Backups

By default, SDDC Manager and NSX Manager backups are stored on the SDDC Manager appliance. Change the destination of the backups to an external SFTP server.

- The external SFTP server must support a 256-bit length ECDSA SSH public key.
- The external SFTP server must support a 2048-bit length RSA SSH public key
- You will need the SHA256 fingerprint of RSA key of the SFTP server.
- Host Key algorithms: At least one of rsa-sha2-512 or rsa-sha2-256 and one of ecdsa-sha2-nistp256, ecdsa-sha2-nistp384, or ecdsa-sha2-nistp521.

Additional pre-requisites when FIPS Security Mode is enabled on SDDC Manager:

Algorithms and Ciphers	Required when FIPS Security Mode is Enabled
Kex Algorithms	At least one of: • diffie-hellman-group-exchange-sha256 • ecdh-sha2-nistp256 • ecdh-sha2-nistp384 • ecdh-sha2-nistp521
Message Authentication Key (MAC) Algorithms	• hmac-sha2-256-etm@openssh.com • hmac-sha2-256

Algorithms and Ciphers	Required when FIPS Security Mode is Enabled
Ciphers	At least one of: • TLS_DHE_DSS_WITH_AES_128_CBC_SHA256 • TLS_DHE_DSS_WITH_AES_256_CBC_SHA256 • TLS_ECDHE_ECDSA_WITH_AES_128_CBC_SHA256 • TLS_ECDHE_ECDSA_WITH_AES_256_CBC_SHA384 • TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA256 • TLS_RSA_WITH_AES_128_CBC_SHA256 • TLS_RSA_WITH_AES_256_CBC_SHA256 • TLS_ECDHE_ECDSA_WITH_AES_256_GCM_SHA384 • TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384 • TLS_ECDHE_ECDSA_WITH_AES_128_GCM_SHA256 • TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256 • TLS_AES_128_GCM_SHA256 • TLS_AES_256_GCM_SHA384

Note:
SHA1 algorithms are not supported.

1. Log in to the VCF Operations interface at `https://<vcf_operations_fqdn>` with a user assigned the Administrator role.
2. Navigate to **Operate** > **Administration** > **SDDC Manager**.
3. Select the VCF Instance to configure backup.
4. In the right hand pane, click the **Backup Settings** tab and then click **Site Settings**.
5. On the **Backup Settings** page, click **Edit**.
6. Enter the settings and click **Save**.

Setting	Value
Host FQDN or IP	The FQDN or IP Address of the SFTP server.
Port	22 **Note:** With VCF 9.1.1 and later, you can configure a custom SFTP port.
Transfer Protocol	SFTP
Username	A service account with read and write privileges to the SFTP server. For example: `svc-vcf-bck`.
Password	The password for the username provided.
Backup Directory	The directory on the SFTP server where backups are saved. For example: `/backups/`.
SSH Fingerprint	The SSH Fingerprint is automatically retrieved from the SFTP server, verify the SSH Fingerprint.
Confirm Fingerprint	Selected
Encryption Passphrase	The encryption passphrase used to encrypt the backup data. **Note:** The encryption passphrase should be stored safely as it is required during the restore process.

7. In the **Confirm your changes to the backup settings** dialog box, click **Confirm**.

Configure Image-Based Backup of VCF Operations and VCF Operations for Networks

To back up VCF Operations and VCF Operations for networks, you create full virtual machine image-based backup jobs by using your VMware vSphere Storage APIs - Data Protection compatible backup solution.

To ensure that all management components are backed up correctly, you must create backup jobs that capture the state of a set of related components at a common point in time. With some components, simultaneous backups of the component nodes ensure that you can restore the component to a state where the nodes are logically consistent with each other and eliminate the necessity for further logical integrity remediation of the component.

Component	Deployment Model	Nodes	Quiescing	Notes
VCF Operations	Simple	• Primary Node • Cloud Proxies	Disabled	• You must back up all cluster nodes at the same time. • It is recommend to take the cluster offline during backup. If the cluster cannot be taken offline, do not quiesce the file system. • Do not perform a backup when dynamic threshold (DT) calculations are running because the backup might lead to performance problems or loss of nodes. By default, DT calculations run at 2 a.m.
	High Availability	• Primary Node • Replica Node(s) • Data Node(s) • Cloud Proxies	Disabled	
VCF Operations for networks	Simple	• Platform Node • Collector(s)	Enabled	
	High Availability	• Platform Nodes • Collector(s)	Enabled	

1. Log in to your backup solution by using a set of credentials with permissions to create backup jobs.
2. Create backup jobs by using the configurations from the table above.
3. Schedule the backups to run on a daily basis.
4. Retain each backup for a minimum of three days.
5. Perform an initial on-demand backup to ensure the backups complete successfully.

Configure File-Based Backup Schedules in VMware Cloud Foundation

You can use the native file-based backup capabilities of VCF management services, VCF Automation, SDDC Manager, vCenter and NSX Manager to schedule regular backups. SDDC Manager configure NSX Manager to backup at 1 hour intervals during the bring-up process. You must manually configure scheduled file-based backup for VCF management services, VCF Automation, SDDC Manager and vCenter.

Verify that you have an SFTP server on the network to serve as a target of the file-based backups.

To ensure that all management components are backed up correctly, you must create a series of backup jobs that capture the state of a set of related components at a common point in time. With some components, simultaneous backups of the component nodes ensure that you can restore the component a state where the nodes are logically consistent with each other and eliminate the necessity for further logical integrity remediation of the component.

Table 1159: File-Based Backup Jobs

Component	Recommended Frequency	Recommended Retention	Notes
VCF management services	Daily	7 days	The following components are captured as part of a scheduled backup, but not by an on-demand backup: • Telemetry • Real-Time Metrics • Real-Time Metrics Store
VCF Automation	Daily	7 days	The following components are captured as part of a scheduled backup, but not by an on-demand backup: • Migration Service Engine
SDDC Manager	Daily	7 days	You must configure the backup jobs for the SDDC Manager instance and all vCenter instances to start within the same 5-minute window.
vCenter	Daily	7 days	
vSphere Distributed Switch	On-demand	Retain last 3 configurations.	-
NSX Manager	Hourly	7 days	Configured by SDDC Managerduring the bring-up process.

Note:

- You must monitor the space utilization on the SFTP server to ensure that you have sufficient storage space to accommodate all backups taken within the retention period.
- Do not make any changes to the `/opt/vmware/vcf` directory on the SDDC Manager appliance. If this directory contains any large files, backups might fail.

Configure File-Based Backup Schedule for VCF Management Services and VCF Automation

You configure an automated backup schedule per component to ensure the regular backup of VCF components at predefined intervals. To view the status of backups or backup failures, see the VCF Operations Health dashboard.

The SFTP server must support rsa-sha2-512 or rsa-sha2-256 SSH key algorithms.

Key components include:

- VCF services runtime
- Fleet lifecycle
- SDDC lifecycle
- Identity broker
- Salt RaaS
- Software depot
- Salt master
- Real-time metrics
- Telemetry
- Log management
- VCF Automation

Note:

Backup schedule considerations for VCF Automation , log management, or identity broker:

- To properly configure an existing backup schedule that was set up on a VCF Instance before VCF Automation was deployed or upgraded to 9.1, you must reconfigure the backup schedule by re-entering the password.
- If log management or identity broker are deployed or upgraded to 9.1 after setting up a backup schedule, the schedule will automatically be applied.

To configure a backup schedule for your VCF component, perform the following steps:

1. Log in to the VCF Operations interface at `https://<vcf_operations_fqdn>` as a user assigned the Administrator role.
2. Navigate to **Build** > **Lifecycle**.
3. In the **Lifecycle** navigation, select **VCF Management**.
4. On the **VCF Management** page, select the **Backup & Restore** tab.
5. Select **Create Schedule**.
6. Set a backup schedule and retention for the component in the VCF Instance.
   a) Select a VCF Instance that will be used to schedule backup for the management components.
   b) Choose the frequency.
      - For **Daily Schedule**, choose the hour of the day when backups will be taken.
      - For **Weekly**, choose the day of the week and the hour of the day when backups will be taken.
   c) Optional: Select and configure incremental backup if hourly backups are needed for fleet lifecycle, SDDC lifecycle, identity broker, or Salt RaaS.

      Software depot, Salt master, log management, VCF Automation, and VCF services runtime, only follow a full backup schedule.

      Note:

      Full backup is a complete backup of the component configuration and data. Incremental backup captures the configuration and data changes that have changed or been added since the most recent full or incremental backup.
   d) Optional: To choose how many backups should be kept, choose the retention policy.
   e) Click **Next**.
7. If an SFTP backup location is not already configured, you are prompted to configure one.

   Enter the following information.

Configuration	Value
Host FQDN or IP	The FQDN or IP Address of the SFTP server.
Port	22 **Note:** With VCF 9.1.1 and later, you can configure a custom SFTP port.
Transfer protocol	SFTP
Username	A service account with privileges to the SFTP server. For example: `svc-vcf-bck`.
Password	The password for the username provided.
Backup Directory	The directory on the SFTP server where backups are saved. For example: `/backups/`. **Note:** The SFTP user must have read access to all sub-directories of the backup directory path.

Configuration	Value
Encryption Passphrase	The encryption passphrase used to encrypt the backup data. **Important:** The encryption passphrase should be stored safely as it is required during the restore process.

Note:

Before saving, ensure that the SFTP host, username, and password are correct and validated. Any incorrect information that is entered will propagate to VCF services runtime and cause the SFTP configuration to fail.

Click **Save**.

The SFTP server configuration is automatically pushed to the VCF Automation and VCF services runtime instances. In the SFTP Status table below the SFTP Settings settings, the configuration status for each VCF Instance appears.

8. Click **Next** to review the configuration.
9. Click **Finish** and monitor the task from the Task panel
10. Repeat the procedure for other VCF Instances.

Configure File-Based Backup Schedule for SDDC Manager

You configure file-based daily backups of the SDDC Manager instances using VCF Operations.

Only a user with the **Admin** role can perform this task.

1. Log in to the VCF Operations interface at `https://<vcf_operations_fqdn>` with a user assigned the Administrator role.
2. Navigate to **Operate** > **Administration** > **SDDC Manager**.
3. Select the VCF Instance to configure a backup schedule.
4. In the right-hand pane, click the **Backup Settings** tab and then click **SDDC Manager Configurations**.
5. Under **Backup Schedule**, click **Edit**.
6. On the **Backup Schedule** page, enter the settings and click **Save**.

Setting	Value
Automatic Backup	Enabled
Backup Frequency	Weekly
Days of the Week	All selected
Schedule Time	23:55 PM
Take Backup on State Change	Enabled
Retain Last Backups	7
Retain Hourly Backups for Days	1
Retain Daily Backups for Days	7

7. To verify the backup, click **Backup Now**.

The status and the start time of the backup is displayed on the UI. You have set the SDDC Manager backup schedule to run daily at 23:55 PM and after each change of state.

If the backup is unsuccessful, verify if the SFTP server is available and able to provide its SSH fingerprint:

- SSH to the SDDC Manager appliance run the following command as the root user:

  ```
  sftp username@IP of sftp server
  ```

Enter the SFTP user password when prompted. The following message indicates a successful connection:

```
Connected to.
```

- To check that the SFTP server SSH fingerprint is available, run:

```
ssh-keygen -lf <(ssh-keyscan -t ssh-rsa -p port_number IP of sftp server 2>/dev/null)
```

Configure File-Based Backup Schedule for NSX Manager

You configure file-based daily backups of the NSX Manager instances using the NSX Manager user interface.

1. Log in to the NSX Manager user interface at `https://<nsx_manager-fqdn>` with the **admin** user.
2. On the main navigation bar, click **System**.
3. In the left navigation pane, under **Lifecycle Management**, click **Backup & Restore**.
4. In the **Backup & Restore** pane, under **Configuration & History**, click **Edit**.
5. In the **Schedule Recurring Backup** dialog, enter the settings and click **Save**.

Setting	Value
Recurring Backup	Selected
Frequency	Weekly
Select day of the Week	All selected
Select time	23:55 PM
Take Backup on State Change	Enabled
Retain Last Backups	7
Retain Hourly Backups for Days	1
Retain Daily Backups for Days	7

6. To verify the backup, click **Start Backup**.
7. Repeat the procedure for all other NSX Manager instances.

Configure File-Based Backup Schedule and External SFTP Server for vCenter

You configure file-based daily backups of the vCenter instances by using the vCenter management interface of each vCenter instance.

1. Log in to the vCenter management interface at `https://<vcenter_fqdn>:5480` using the root user.
2. In the left navigation pane, click **Backup**.
3. In the **Backup schedule** pane, click **Configure**.
4. In the **Create Backup Schedule** dialog box, enter the settings and click **Create**.

Setting		Value
Backup location		Enter the backup location from SFTP server. For example: `sftp://192.168.1.2/backups/`
Backup server credentials	User name	A service account with privileges to the SFTP server. For example: `svc-vcf-bck`.
	Password	Enter the password for the username provided.
Schedule		Daily 11:55 PM
Encrypt backup	Encryption password	*encryption_password*
	Confirm password	*encryption_password*
Number of backups to retain		Retain last 7 backups
Data	Supervisors Control Plane	Select if running vSphere Supervisor on this vCenter.

Setting		Value
	Stats, events, and tasks	Selected
	Inventory and configuration	Selected

Note:

Leave all three data options selected for a backup intended for disaster recovery.

- **Inventory and configuration** is mandatory and cannot be deselected. It carries the critical configuration and inventory data the vCenter needs to become functional again after a restore.
- **Stats, events, and tasks** is the optional historical data. It controls both the content and the method of the database backup: selected, the backup contains a full copy of the vCenter database; cleared, it contains a partial export that omits historical performance statistics, events, and tasks. Alarm definitions and alarm state are retained either way.
- **Supervisors Control Plane** must be selected if any vSphere Supervisor runs on this vCenter. A backup taken without it cannot be used to recover a Supervisor cluster.

Important: Record the encryption password in a secure location outside this VCF instance, and keep it for as long as you keep the backups it protects. The password is not stored in the backup and cannot be recovered from it. A backup whose encryption password has been lost cannot be restored.

The backup schedule information appears in the **Backup schedule** pane.

5. To verify the backup, click **Back up Now**.
   a) In the **Back up Now** dialog, click **Use backup location and user name from backup schedule**.
   b) Enter the password for the backup server credentials.
   c) Enter the encryption password if required.
   d) Click **Start**.
6. Repeat the procedure for all other vCenter instances.

Any complete and in-progress backup appears in the **Activity** pane.

What to do next

Record the version and build number of each vCenter instance from the **Summary** tab of its Management Interface, and download the matching vCenter appliance image from the Broadcom Support Portal. Store the image and the recorded version alongside your backups.

Restoring a vCenter requires deploying a replacement appliance at the same version as the backup. If the instance is later patched, repeat this step so the stored image continues to match.

On-Demand Backups in VMware Cloud Foundation

Whilst scheduled backups are critical in the recovery of a VMware Cloud Foundation platform, its also possible to run on-demand backups for other planned activities like pre-upgrades. There are also some elements of the system that should be backed up for recovery purposes that can not be scheduled and must be run on-demand.

- On-Demand Backup of VCF Management Services and VCF Automation
- On-Demand Backup of SDDC Manager
- On-Demand Backup of SDDC Manager Credential Store
- On-Demand Backup of NSX Manager
- On-Demand Backup of vCenter
- On-Demand Backup of vSphere Distributed Switch Configuration

On-Demand Backup of VCF Management Services and VCF Automation

For critical operations or before performing maintenance activities, such as upgrades, you should initiate manual, on-demand backups of specific VCF component. This provides an immediate recovery point, offering an extra layer of protection.

1. Log in to the VCF Operations interface at `https://<vcf_operations_fqdn>` as a user assigned the Administrator role.
2. Navigate to **Build** > **Lifecycle** > **VCF Management**.
3. On the VCF Management page, select the **Backup and Restore** tab.
4. Click **Backup** > **Backup Now.**
5. To backup your VCF component, select the VCF Instance.
   a) Select the component that you want to backup.
   b) Click **Backup Now**.

On-Demand Backup of SDDC Manager

For critical operations or before performing maintenance activities, such as upgrades, you should initiate manual, on-demand backups of yourSDDC Manager instances. This provides an immediate recovery point, offering an extra layer of protection. You use VCF Operations to perform this operation.

1. Log in to the VCF Operations interface at `https://<vcf_operations_fqdn>` with a user assigned the Administrator role.
2. Navigate to **Operate** > **Administration** > **SDDC Manager**.
3. Select the VCF Instance to configure a backup schedule.
4. In the right-hand pane, click the **Backup Settings** tab and then click **SDDC Manager Configurations**.
5. Under **Last Backup Status**, click **Backup Now**.

On-Demand Backup of SDDC Manager Credential Store

If you do not have an external password vault, you must backup the SDDC Manager Credential Store using the SDDC Manager API.

You should configure a scheduled job using external automation to backup the credential store at least once a day. The steps below use the SDDC Manager developer center but you can also use the REST API.

1. Log into the SDDC Manager UI as a user with the ADMIN role.
2. Navigate to the developer center.
3. Expand **Credentials**.
4. Expand **GET**.
5. Click **Execute** and download the resulting JSON.
6. Store the JSON in a secure location.
7. Repeat the above steps or leverage the SDDC Manager REST API to backup the credentials store daily.

On-Demand Backup of NSX Manager

For critical operations or before performing maintenance activities, such as upgrades, you should initiate manual, on-demand backups of yourNSX Manager instances. This provides an immediate recovery point, offering an extra layer of protection. You use NSX Manager user interface to perform this operation.

1. Log in to the NSX Manager user interface at `https://<nsx_manager-fqdn>` with the **admin** user.
2. On the main navigation bar, click **System**.
3. In the left navigation pane, under **Lifecycle Management**, click **Backup & Restore**.
4. In the **Backup & Restore** pane, under **Configuration & History**, click **Start Backup**.

On-Demand Backup of vCenter

For critical operations or before performing maintenance activities, such as upgrades, you should initiate manual, on-demand backups of yourvCenter instances. This provides an immediate recovery point, offering an extra layer of protection. You use vCenter management interface to perform this operation

- In the vSphere Client, for each vSphere cluster that is managed by the vCenter, note the current vSphere DRS Automation Level setting and then change the setting to **Manual**. After the vCenter backup is complete, you can change the vSphere DRS Automation Level setting back to its original value.
- Ensure that there are no active vSphere vMotion tasks.

1. Log in to the vCenter management interface at `https://<vcenter_fqdn>:5480` using the root user.
2. In the left navigation pane, click **Backup**.
3. In the Activity pane, click **Backup Now**.
4. If you already have a backup schedule set up, select **Use backup location and user name from backup schedule** and click **Start**.
5. If you do not already have a backup schedule, enter the following information and click **Start**.

Setting		Value
Backup location		Enter the backup location from SFTP server. For example: `sftp://172.16.11.60/backups/`
Backup server credentials	User name	A service account with privileges to the SFTP server. For example: `svc-vcf-bck`.
	Password	Enter the password for the username provided.
Encrypt backup	Encryption password	*`encryption_password`*
	Confirm password	*`encryption_password`*
Number of backups to retain	Choose to retain all backups or a certain number of backups.	
Data	Supervisors Control Plane	Selected (If using Supervisor Control Plane in this vCenter)
	Stats, events, and tasks	Selected
	Inventory and configuration	Selected

Note:

Leave all three data options selected for a backup intended for disaster recovery.

- **Inventory and configuration** is mandatory and cannot be deselected. It carries the critical configuration and inventory data the vCenter needs to become functional again after a restore.
- **Stats, events, and tasks** is the optional historical data. It controls both the content and the method of the database backup: selected, the backup contains a full copy of the vCenter database; cleared, it contains a partial export that omits historical performance statistics, events, and tasks. Alarm definitions and alarm state are retained either way.
- **Supervisors Control Plane** must be selected if any vSphere Supervisor runs on this vCenter. A backup taken without it cannot be used to recover a Supervisor cluster.

Important: Record the encryption password in a secure location outside this VCF instance, and keep it for as long as you keep the backups it protects. The password is not stored in the backup and cannot be recovered from it. A backup whose encryption password has been lost cannot be restored.

In order to restore vCenter, you will need the vCenter Appliance ISO file that matches the version you backed up.

- Identify the required vCenter version. In the vCenter Management Interface, click **Summary** in the left navigation pane to see the vCenter version and build number.
- Download the vCenter Appliance ISO file for that version from the Broadcom Support Portal.

On-Demand Backup of vSphere Distributed Switch Configuration

The vCenter backup includes the configuration of the entire vCenter instance. To have a backup only of the vSphere Distributed Switch and distributed port group configurations, you export a configuration file that includes the validated network configurations. If you want to recover only the vSphere Distributed Switch, you can import this configuration file to the vCenter instance.

You can use the exported file to create multiple copies of the vSphere Distributed Switch configuration on an existing deployment, or overwrite the settings of existing vSphere Distributed Switch instances and port groups.

You must backup the configuration of a vSphere Distributed Switch immediately after each change in configuration of that switch.

1. Log in to the vCenter interface at `https://<vcenter_fqdn>/ui` with a user assigned the Administrator role.
2. Select the **Networking** inventory.
3. In the inventory expand the vCenter and Datacenter.
4. Right click the vSphere Distributed Switch you want to backup, and select **Settings** > **Export Configuration**.
5. In the **Export configuration** dialog box, select **Distributed switch and all port groups**.
6. In the **Description** text box enter the date and time of export, and click **OK**.
7. Copy the backup zip file to a secure location from where you can retrieve the file and use it if a failure of the appliance occurs.
8. Repeat the procedure for the other vSphere Distributed Switches.

Restoring VMware Cloud Foundation

Recovering your VMware Cloud Foundation platform from an unexpected outage or failure is a critical operation. Depending on your failure scenario and what components of your environemnt is affected could involve any number of steps. This section provides guidance across all components of the VMware Cloud Foundation platform

- Image-Based Restore of VCF Operations
- File-Based Restore for SDDC Manager, vCenter, and NSX
- File-Based Restore of VCF Management Services
- File-Based Restore of VCF Automation

Image-Based Restore of VCF Operations

Restore a simple or high availability deployment of VCF Operations by using an image-based backup tool.

- **Backup Data Available**: Ensure that you have a recent backup of your VCF Operations cluster.
- **VCF Operations Cluster Configuration**: Ensure you know the details of your VCF Operations cluster configuration, such as cluster nodes, user credentials.
- **Backup Software**: Ensure that the backup solution you are using is compatible with your VCF Operations version.

Table 1160: VCF Operations Failure Scenarios

Component	Deployment Model	Failure Scenario	Recovery Process
VCF Operations	Simple	Primary node lost	Restore primary node from image-based backup.
		Cloud proxy lost	Restore cloud proxy from image-based backup.
	High Availability	Primary node lost	Restore primary node from image-based backup.

Component	Deployment Model	Failure Scenario	Recovery Process
		Replica node lost	Restore replica node from image-based backup.
		Data node lost	Restore data node from image-based backup.
		Cloud proxy lost	Restore cloud proxy from image-based backup.

1. If there are existing VCF Operations virtual machines, power them off and rename or delete.
2. Access your backup software interface.
3. Locate the backup taken of the VCF Operations nodes.
4. Initiate the restore process.
5. After the virtual machine is restored, power it on.

Verify the Operational State of VCF Operations

After a successful restore of VCF Operations, you verify the state of the analytics cluster and each integration adapter.

The cluster should be online and all integration adapters should be in a collecting state.

1. Verify the VCF Operations cluster state.
   a) Log in to the VCF Operations admin console at `https://<vcf-operations-primary-node-fqdn>/admin` with the **admin** user.
   b) On the **System Status** page, verify that all nodes are in a **Running** state and their status is **Online**.
   c) Navigate to the **Cloud Proxies** page and verify that the **Health Status** for each cloud proxy is **Healthy**.
2. Verify the VCF Operations integration adapters state.
   a) Log in to the VCF Operations interface at `https://<vcf-operations-primary-node-fqdn>/` as a user assigned the **Administrator** role.
   b) Navigate to **Administration** > **Integrations**.
   c) Expand each integration and ensure that its status is **OK**.

Image Based Restore of VCF Operations for Networks

- **Backup Data Available**: Ensure that you have a recent backup of your VCF Operations for networks cluster.
- **VCF Operations Cluster Configuration**: Ensure you know the details of your VCF Operations for networks cluster configuration, such as cluster nodes, user credentials.
- **Backup Software**: Ensure that the backup solution you are using is compatible with your VCF Operations for networks version.

Table 1161: VCF Operations for networks Failure Scenarios

Component	Deployment Model	Failure Scenario	Recovery Process
VCF Operations for networks	Simple	Platform node lost	Restore primary node from image-based backup.
		Collector lost	Restore Collector from image-based backup.
	High Availability	Primary Platform node lost	Restore primary node from image-based backup.
		Additional Platform node lost	Restore replica node from image-based backup.
		Collector lost	Restore Collector from image-based backup.

1. If there are existing VCF Operations for networks virtual machines, power them off and rename or delete.
2. Access your backup software interface.
3. Locate the backup taken of the VCF Operations for networks nodes.
4. Initiate the restore process.
5. After each virtual machine is restored, power it on.

File-Based Restore for SDDC Manager, vCenter, and NSX

When SDDC Manager, vCenter or NSX Manager in your VMware Cloud Foundation platform fails, you can restore the component to a fully operational state by using their file-based backups. When an NSX Edge node fails, you redeploy the node from the NSX Manager instance.

Use this guidance as appropriate based on the exact nature of the failure encountered within your environment. Sometimes, you can recover localized logical failures by restoring individual components. In more severe cases, such as a complete and irretrievable hardware failure, to restore the operational status of your VMware Cloud Foundation platform, you must perform a complex set of manual deployments and restore sequences. In failure scenarios where there is a risk of data loss, there has already been data loss or where it involves a catastrophic failure, contact Broadcom Support to review your recovery plan before taking any steps to remediate the situation.

Restore SDDC Manager

If SDDC Manager fails, you can restore it from its file-based backup.

- Power off and rename the failed SDDC Manager instance.
- Verify that you have a valid file-based backup of the failed SDDC Manager instance.
  To be valid, the backup must be of the same version as the version of the SDDC Manager appliance on which you plan to restore the instance.
- Verify that you have the SFTP server details:
  - SFTP Server IP
  - SFTP Server Username
  - SFTP Server Password
  - Encryption Password
- The following components must be available before proceeding with the restore of SDDC Manager:
  - vCenter

After a successful recovery, securely delete the decrypted backup files.

Decrypting the SDDC Manager Backup
Should you need to decrypt the SDDC Manager, you must download and decrypt the encrypted backup file from the SFTP server.

- OpenSSL 1.1.1 or later must be available on your Linux machine that you use to access the environment.
- Download the SDDC Manager that corresponds to your VCF version from the Broadcom support portal.

The backup file contains sensitive data about your SDDC Manager instance. As a best practice, you must control access to the decrypted files and securely delete them after you complete the restore operation.

> **Note:**
>
> The procedures have been written based on the host machine being a Linux-based operating system.

1. Identify the backup file for the restore and download it from the SFTP server to your host machine.
2. Replace the values in the sample code with your values, these will be used in later steps.

```
export backupFile='vcf-backup-sfo-vcf01-sfo-rainpole-io-xxxx-xx-xx-xx-xx-xx.tar.gz'
export encryptionPassword='***************'
```

3. On your host machine, open a terminal and extract the content of the backup file.
   a) For VCF 9.1.1 decrypt and extract the content of the backup file by running the following command.

```
tail -c +5 "$backupFile" | OPENSSL_FIPS=1 openssl enc -d -aes-256-cbc -pbkdf2 -iter 600000 -md sha256 -pass env:encryptionPassword | tar -xz
```

   a) For VCF 9.1.0 decrypt and extract the content of the backup file by running the following command.

```
OPENSSL_FIPS=1 openssl enc -d -aes-256-cbc -md sha256 -in "$backupFile" -pass env:encryptionPassword 2>/dev/null | tar -xz
```

Restore SDDC Manager from a File-Based Backup

First, you deploy a new SDDC Manager appliance by using the OVA file that you downloaded during the preparation for the restore. After that, you restore the file-based backup on the newly deployed SDDC Manager appliance.

1. Log in to the vCenter interface at `https://<vcenter_fqdn>/ui` with a user assigned the Administrator role.
2. Select **Menu** > **VMs and templates**.
3. In the inventory expand **vCenter** > **Datacenter**.
4. Right-click the management folder and select **Deploy OVF template**.
5. On the **Select an OVF template** page, select **Local file**, click **Upload files**, browse to the location of the SDDC Manager OVA file, click **Open**, and click **Next**.
6. On the **Select a name and folder** page, in the **Virtual machine name** text box, enter a virtual machine name, and click **Next**.
7. On the **Select a compute resource** page, click **Next**.
8. On the **Review details** page, review the settings and click **Next**.
9. On the **License agreements** page, accept the license agreement and click **Next**.
10. On the **Select storage** page, select the vSAN datastore and click **Next.**

    The datastore must match the `vsan_datastore` value in the `metadata.json` file that you downloaded during the preparation for the restore.
11. On the **Select networks** page, from the **Destination network** drop-down menu, select the management network distributed port group and click **Next**.

    The distributed port group must match the `port_group` value in the `metadata.json` file that you downloaded during the preparation for the restore.
12. On the **Customize template** page, enter the following values and click **Next**.

Setting	Description
Enter root user password	You can use the original **root** user password or a new password.
Enter Local user password	You can use the original **Local user** password or a new password.
Host Name	The FQDN must match the `hostname` value in the `metadata.json` file that you downloaded during the preparation for the restore.
NTP Servers	The NTP server details for the appliance.
Enable FIPs	Selected
Network 1 IP address	The IP address for the appliance.
Network 1 Subnet Mask	The subnet mask for the appliance.
Network Default Gateway	The default gateway for the appliance.
DNS Domain Name	The domain name for the appliance.
Domain search path	The domain search path(s) for the appliance.
Domain name servers	The DNS servers for the appliance.

13. On the **Ready to complete** page, click **Finish** and wait for the process to complete.
14. Once the SDDC Manager appliance deployment completes, expand the management folder.
15. Right-click the SDDC Manager appliance and select **Snapshots** > **Take Snapshot**.
16. Right-click the SDDC Manager appliance, select **Power** > **Power On**.
17. Replace the values in the sample code with your values, these will be used in later steps.

```
export sddcManagerFqdn='sfo-vcf01.sfo.rainpole.io'
export sddcManagerUsername='admin@local'
export sddcManagerPassword='***************'

export tmpFolderPath='/tmp/'
export backupFile='vcf-backup-sfo-vcf01-sfo-rainpole-io-xxxx-xx-xx-xx-xx-xx.tar.gz'
export encryptionPassword='***************'
```

18. On the host machine, copy the encrypted backup file to the `/tmp` folder on the newly deployed SDDC Manager appliance by running the following command. When prompted, enter the *`vcf_user_password`*.

```
scp $backupFile vcf@$sddcManagerFqdn:$tmpFolderPath
```

19. Retrieve an access token from the SDDC Manager instance by running the following command.

```
sddcManagerToken=$(curl -k -s -X POST "https://${sddcManagerFqdn}/v1/tokens" \
    --header "Content-Type: application/json" \
    -d "{\"username\":\"${sddcManagerUsername}\", \"password\":\"${sddcManagerPassword}\"}" \
| jq -r '.accessToken' | tee /dev/tty )
```

20. Create the JSON payload for the restore by running the following command.

```
sddcRestorePayload=$(jq -n \
  --arg file "${tmpFolderPath}${backupFile}" \
  --arg pass "${encryptionPassword}" \
  '{
    "elements": [ { "resourceType": "SDDC_MANAGER" } ],
    "backupFile": $file,
    "encryption": { "passphrase": $pass }
  }' | tee /dev/tty )
```

21. Perform a restore of the SDDC Manager instance by running the following command.

```
restoreId=$(curl -k -s -X POST "https://${sddcManagerFqdn}/v1/restores/tasks" \
    --header "Authorization: Bearer ${sddcManagerToken}" \
    --header "Content-Type: application/json" \
    --header "Accept: application/json" \
    --data "$sddcRestorePayload" \
| jq -r '.id' | tee /dev/tty )
```

22. Monitor the restore task by running the following command.

```
curl -k -s -X GET "https://${sddcManagerFqdn}/v1/restores/tasks/${restoreId}" \
    --header "Authorization: Bearer ${sddcManagerToken}" \
    --header "Content-Type: application/json" \
    --header "Accept: application/json" \
| jq
```

Refresh the SSH keys that are stored in the SDDC Manager inventory. See Broadcom knowledge base article 313210.

Validate the Status of SDDC Manager

After a successful restore of SDDC Manager you must validate its status. You run the health checks by using the `sos` tool.

1. Open an SSH connection to the SDDC Manager appliance, connect using the `vcf` user.
2. Switch to the root user and change to the `/opt/vmware/sddc-support` folder by running the following command.

```
su -
```

```
cd /opt/vmware/sddc-support
```

3. Run the `sos` health check utility by running the following command.

```
./sos --health-check --domain-name ALL --skip-cert-check
```

4. Remediate any issues reported by the health check.

   All tests show green when SDDC Manager is in healthy state.

5. Manually delete the snapshot created in Restore SDDC Manager from a File-Based Backup.

Refresh the SSH keys that are stored in the SDDC Manager inventory. See Broadcom knowledge base article 313210.

Restoring a vCenter Instance

If a vCenter instance fails, you can restore it from its file-based backup.

- Power off and rename the failed vCenter instance.
- Verify that you have a valid file-based backup of the failed vCenter instance.
  To be valid, the backup must be of the version of the vCenter appliance on which you plan to restore the instance.
- Verify that you have the SFTP server details:
  - SFTP Server IP
  - SFTP Server Username
  - SFTP Server Password
  - Encryption Password

Prepare for Restoring a vCenter Instance

Before restoring a vCenter instance, you must retrieve the vCenter build number and deployment details, as well as vCenter and ESX credentials from the SDDC Manager inventory.

- The following components must be available before proceeding with the restore of vCenter:
  - SDDC Manager

Retrieve the vCenter Intance Deployment Details

Before restoring a vCenter instance, you must retrieve the vCenter build number and deployment details from the SDDC Manager inventory. The vCenter instances in your system might be running different build numbers if the backups are taken during an upgrade process. You must restore each vCenter instance to its correct version.

Because the management domain vCenter might be unavailable to authenticate the login, you use the SDDC Manager API via the shell to retrieve this information.

1. Open an SSH connection to the SDDC Manager appliance, connect using the `vcf` user.
2. Get the list of vCenter instances by running the following command.

```
curl http://localhost/inventory/vcenters -k | json_pp
```

3. For each vCenter instance, record the values of these settings.

Setting	Value
domainId	Id of the domain
vmName	VM name of the vCenter instance
hostName	FQDN of the vCenter instance
version	*version_number-build_number*

4. Verify that the vCenter version retrieved from SDDC Manager is the same as the version associated with the backup file that you plan to restore.

Retrieve the Credentials for Restoring a vCenter Instance

Before restoring a vCenter instance, you must retrieve the vCenter root and vCenter Single Sign-On administrator credentials from the SDDC Manager inventory. Before restoring the management domain vCenter instance, you must also retrieve the credentials of a healthy management domain ESX host.

Note:

If SDDC Manager is not operational, you can retrieve the required vCenter root, vCenter Single Sign-On administrator, and ESX root credentials from the file-based backup of SDDC Manager. See Decrypting the SDDC Manager Backup.

Before you can query the SDDC Manager API, you must obtain an API access token by using **admin@local** account.

1. Log in to your host machine with access to the SDDC and open a terminal.
2. Replace the values in the sample code with your values, these will be used in later steps

```
export sddcManagerFqdn='sfo-vcf01.sfo.rainpole.io'
export sddcManagerUsername='admin@local'
export sddcManagerPassword='***************'
```

3. Retrieve an access token from the SDDC Manager instance by running the following command.

```
sddcManagerToken=$(curl -k -s -X POST "https://${sddcManagerFqdn}/v1/tokens" \
    --header "Content-Type: application/json" \
    -d "{\"username\":\"${sddcManagerUsername}\", \"password\":\"${sddcManagerPassword}\"}" \
| jq -r '.accessToken' | tee /dev/tty )
```

4. Retrieve the vCenter **root** credentials.
   a) Retrieve the vCenter **root** credentials by running the following command.

```
curl -k -s -X GET "https://${sddcManagerFqdn}/v1/credentials?resourceType=VCENTER&accountType=USER" \
    --header "Authorization: Bearer ${sddcManagerToken}" \
    --header "Content-Type: application/json" \
    --header "Accept: application/json" \
| jq
```

   The command returns the vCenter **root** credentials.

Setting	Value
domainName	Name of the domain
resourceName	FQDN of the vCenter instance
username	root
password	*vcenter_root_password*

   b) Record the vCenter **root** credentials.

5. Retrieve the vCenter Single Sign-On administrator credentials.
   a) Retrieve the vCenter Single Sign-On administrator credentials by running the following command.

```
curl -k -s -X GET "https://${sddcManagerFqdn}/v1/credentials?resourceType=PSC" \
    --header "Authorization: Bearer ${sddcManagerToken}" \
    --header "Content-Type: application/json" \
    --header "Accept: application/json" \
| jq
```

   The command returns the **administrator@vsphere.local** credentials.

Setting	Value
domainName	Name of the domain
resourceName	FQDN of the vCenter instance
username	administrator@vsphere.local
password	*vcenter_sso_admin_password*

b) Record the **administrator@vsphere.local** credentials.

6. If you plan to restore the management domain vCenter instance, retrieve the credentials for a healthy management domain ESX host.
   a) Retrieve the credentials for a management domain ESX host by running the following command.

```
curl -k -s -X GET "https://${sddcManagerFqdn}/v1/credentials?resourceType=ESXI" \
    --header "Authorization: Bearer ${sddcManagerToken}" \
    --header "Content-Type: application/json" \
    --header "Accept: application/json" \
| jq
```

   The command returns the ESX **root** credentials.

Setting	Value for first ESX host
domainName	management domain name
resourceName	FQDN of the first ESX host
username	root
password	*esx_root_password*

   b) Record the ESX **root** credentials.

Restore a vCenter Instance from a File-Based Backup

If a vCenter instance fails, you can restore it from its file-based backup. If the management domain vCenter instance and the workload domain vCenter instance are both in a failed state, you must restore the management domain vCenter instance before restoring the workload domain vCenter instance.

- Download the vCenter ISO file for the version of the failed instance. See Retrieve the vCenter Intance Deployment Details.
- If you are recovering a workload domain vCenter instance, verify that the management domain vCenter instance is available.

You deploy a new vCenter appliance and perform a file-based restore. If you are restoring the management domain vCenter instance, you deploy the new appliance on a healthy ESX host in the management domain cluster. If you are restoring the workload domain vCenter instance, you deploy the new appliance in the management domain vCenter instance.

1. Mount the vCenter ISO image to your host machine with access to the SDDC and run the UI installer for your operating system.

   For example, for a Windows host machine, open the *dvd-drive*:\vcsa-ui-installer\win32\installer application file.

2. Click **Restore**.
3. Complete the **Restore - Stage 1: Deploy vCenter** wizard.
   a) On the **Introduction** page, click **Next**.
   b) On the **End user license agreement** page, select the **I accept the terms of the license agreement** check box and click **Next**.
   c) On the **Enter backup details** page, enter these values and click **Next**.

Setting	Value
Location or IP/hostname	sftp://sftp_server_ip/backups/vCenter/sn_vc_fqdn/*backup_folder*/
User name	SFTP Server Username
Password	*sftp-user-password*

   d) On the **Review backup information** page, review the backup details, record the **vCenter configuration** information, and click **Next**.

   You use the vCenter configuration information at a later step to determine the deployment size for the new vCenter appliance.

e) On the **vCenter deployment target** page, enter the values by using the information that you retrieved during the preparation for the restore, and click **Next**.

Setting	Value for Management Domain vCenter Instance	Value for Workload Domain vCenter Instance
ESX host or vCenter name	FQDN of the first ESX host	FQDN of the management vCenter instance
HTTPS port	443	443
User name	root	administrator@vsphere.local
Password	*esx_root_password*	*vcenter_sso_admin_password*

f) In the **Certificate warning** dialog box, click **Yes** to accept the host certificate.

g) On the **Set up target vCenter VM** page, enter the values by using the information that you retrieved during the preparation for the restore, and click **Next**.

Setting	Value
VM name	vCenter instance VM name
Set root password	*vcenter_root_password*
Confirm root password	*vcenter_root_password*

Important:

Enter the vCenter root password retrieved in *Retrieve the Credentials for Restoring a vCenter Instance* — the password the failed vCenter had when the backup was taken. The restore preserves the password you enter here; it does not restore the original from the backup. SDDC Manager and VCF Operations store the original password and fail to connect to the restored vCenter if it differs.

The Single Sign-On administrator password behaves differently and **is** restored from the backup: after the restore, **administrator@<sso_domain>** has the password it had at backup time.

h) On the **Select deployment size** page, select the deployment size that corresponds with the vCenter configuration information from Step 3.d and click **Next**.

The size should match the size found in backup-metadata.json (from the field **DeploymentSize)**

Example

large=lstorage

i) On the **Select datastore** page, select these values, and click **Next**.

Setting	Value
Datastore	Datastore name
Enable thin disk mode	Selected

j) On the **Configure network settings** page, enter the values by using the information that you retrieved during the preparation for the restore, and click **Next**.

Setting	Value
Network	Name of the vSphere distributed switch portgroup
IP version	IPV4
IP assignment	static
FQDN	FQDN of the vCenter
IP address	IP address of the vCenter
Subnet mask or prefix length	24
Default gateway	Default gateway IP address
DNS servers	Comma separated list of DNS server IP addresses

k) On the **Ready to complete stage 1** page, review the restore settings and click **Finish**.
l) When stage 1 of the restore process completes, click **Continue**.

4. Complete the **Restore - Stage 2: vCenter** wizard.
   a) On the **Introduction** page, click **Next**.
   b) On the **Backup details** page, in the **Encryption password** text box, enter the encryption password of the SFTP server and click **Next**.
   c) On the **Ready to complete** page, review the restore details and click **Finish**.
   d) In the **Warning** dialog box, click **OK** to confirm the restore.
   e) When stage 2 of the restore process completes, click **Close**.

Refresh the SSH keys that are stored in the SDDC Manager inventory. See VMware Cloud Foundation SDDC Manager Recovery Scripts (79004).

The restored appliance has new SSH host keys. In addition to SDDC Manager, refresh the stored host key for this vCenter in any other system that connects to it over SSH — cloud proxy, monitoring systems, automation hosts, and administrator workstations.

Key-based SSH access to the appliance is also lost: the contents of **/root/.ssh** are not included in a vCenter file-based backup. Re-provision any authorized keys the failed instance had.

Validate the vCenter Instance State
After restoring a vCenter instance, you must validate the state of the vCenter and vCenter Single Sign-On.

1. Log in to the vCenter interface at `https://<vcenter_fqdn>/ui` with a user assigned the Administrator role.
2. In the inventory, click the management domain vCenter inventory, click the **Summary** tab, and verify that there are no unexpected vCenter alerts.
3. **Note:**

   This step only applies to upgraded environments that still have Enhanced Linked Mode (ELM) configured. VCF 9.x deployed vCenter instances do not leverage ELM.

   For vCenter instances participating in a shared vCenter Single Sign-On domain, do the following:
   a) Click the **Linked vCenter systems** tab and verify that the list contains all other vCenter instances in the vCenter Single Sign-On domain.
   b) Log in to the recovered vCenter instance by using a Secure Shell (SSH) client.
   c) Navigate to the bin directory by running the following command.

      ```
      cd /usr/lib/vmware-vmdir/bin
      ```

   d) List the current replication partners of the vCenter instance with the current replication status between the nodes by running the following command.

      ```
      ./vdcrepadmin -f showpartnerstatus -h localhost -u administrator -w vsphere_admin_password
      ```

e) Verify that for each partner, the `vdcrepadmin` command output contains `Host available: Yes`, `Status available: Yes`, and `Partner is 0 changes behind`.
f) If you observe significant differences, because the resyncing might take some time, wait five minutes and repeat this step.

4. Repeat the procedure for the other vCenter instance.
5. Review the restore logs. Check **/var/log/vmware/applmgmt/restore.log** and **/var/log/vmware/applmgmt/reconciliation.log** for **ERROR** entries. Per-adapter network failures and skipped configuration are recorded here only; they do not cause the restore to report failure.
6. Verify and re-apply the network configuration. Compare the running configuration against the values recorded during preparation:
   a) **ip addr** — every address on every adapter, both IPv4 and IPv6. In a dual-stack deployment only one address family is restored from the backup; the other keeps the value entered when the replacement appliance was deployed.
   b) **ip route** — static routes are not restored**.** Re-apply them with **appliancesh** > **networking.routes.add**.
   c) **cat /etc/resolv.conf** (or **resolvectl status**) — DNS servers and the **search** line. **DNS search domains are not restored.** Re-apply them with **appliancesh** > **networking.dns.domains.set**, or from the Direct Console User Interface.
   d) Confirm every adapter the failed instance had is present and addressed.
7. Verify time synchronization. Confirm the appliance clock is correct and the configured NTP servers are reachable. Single Sign-On token validation fails if the clock has drifted.
8. Verify identity sources and Active Directory authentication.
   a) Under **Administration > Single Sign On > Configuration > Identity Provider**, confirm each identity source connects. If an LDAPS source fails, its certificate may have been renewed after the backup was taken; re-import the current certificate.
   b) If the vCenter is joined to an Active Directory domain, confirm domain authentication works. If it fails, the computer account password may have rotated since the backup was taken; reset the machine account or rejoin the domain.
   c) Log in as **administrator@<sso_domain>** using the password that was in effect **when the backup was taken**.
   d) Roles, permissions, privileges, tags, and categories are restored with the database. Do not re-create them.
9. Verify SSH access and trust.
   a) The appliance has new SSH host keys. Refresh the stored host key wherever this vCenter is reached over SSH.
   b) **/root/.ssh** is not restored. Re-provision any authorized keys.
   c) Confirm SDDC Manager and VCF Operations can authenticate to the vCenter. If they cannot, verify the root password matches the one they hold.
10. Verify certificates.
    a) If the VMCA root certificate was replaced or renewed after the backup was taken, the restored vCenter carries the earlier root and does not trust certificates issued from the newer one. Check for host connection failures reporting certificate errors and re-issue certificates if required.
11. Reconcile the inventory against the environment.
    a) The restored vCenter reflects the environment as it was when the backup was taken. Hosts added after the backup are absent and must be re-added; virtual machines created after the backup appear as orphaned or unregistered; cluster membership, resource pool hierarchy, DRS configuration and virtual machine resource settings revert; storage policy providers registered after the backup must be re-registered. Compare the vSphere Distributed Switch configuration against the export produced by *Export the Configuration of the vSphere Distributed Switches*

and, if it is newer, restore it using *Restore the Configuration of a vSphere Distributed Switch*. For the full list, see Considerations and Limitations for File-Based Backup and Restore in the vSphere documentation.

12. Verify content libraries.
    a) The content library service remounts remote storage backings and reconciles its database at startup. Confirm each library is accessible. Items deleted after the backup are no longer accessible; items created after the backup remain on the storage backing without a library record.
13. Verify alarms.
    a) Alarm definitions and their state are restored. Scripts stored on the appliance and invoked by alarm actions are **not** restored unless they reside in a directory included in the backup. Confirm every alarm action that runs a script still points at a script that exists.
14. Verify vSphere Supervisor, if in use.
    a) Confirm the Supervisor cluster reaches a running state. Supervisor control-plane state is restored only if **Supervisors Control Plane** was selected when the backup was taken.
15. Redeploy vCenter High Availability, if it was in use.
    a) The restore removes the vCenter HA configuration.
16. Re-point and re-run the backup.
    a) The restored appliance carries the backup schedule that was in effect when the backup was taken, including its backup location. Update the backup configuration, then take an on-demand backup immediately so a current recovery point exists.
17. If ESX root passwords were automatically rotated by SDDC Manager *after* the backup was taken, or if new hosts were commissioned, they may appear as "Disconnected" or "Not Responding" in the restored vCenter.
    a) In the vCenter UI, check the hosts state. Reconnect any disconnected hosts. You may need to provide the current root credentials if the passwords have been changed.

Validate the SDDC Manager State After a vCenter Instance Restore
After a successful vCenter restore, verify that the SDDC Manager inventory is consistent with the recovered virtual machine and that the vCenter instances are healthy. You use the Supportability and Serviceability tool (`SoS`) and the SDDC Manager patch/upgrade precheck function.

1. Open an SSH connection to the SDDC Manager appliance, connect using the `vcf` user.
2. Switch to the root user and change to the `/opt/vmware/sddc-support` folder by running the following command.

   ```
   su -
   cd /opt/vmware/sddc-support
   ```

3. Run the `sos` health check utility by running the following command.

   ```
   ./sos --health-check
   ```

   All tests show green when SDDC Manager is in a healthy state.
4. Log in to the SDDC Manager user interface at `https://<sddc-manager-fqdn>/admin` with a user assigned the Administrator role.
5. In the navigation pane, click **Inventory** > **Workload Domains**.
6. For each workload domain, validate the vCenter status.
   a) Click the workload domain name and click the **Updates/Patches** tab.
   b) Click **Precheck**.
   c) Click **View status** to review the precheck result for the vCenter instance and verify that the status is `Succeeded`.

Restore the Configuration of a vSphere Distributed Switch

To recover the configuration of a vSphere Distributed Switch, you can restore its settings from the configuration file that you previously exported.

This procedure restores only the vSphere Distributed Switch configuration of a vCenter instance.

The restore operation changes the settings on the vSphere Distributed Switch back to the settings saved in the configuration file. The operation overwrites the current settings of the vSphere Distributed Switch and its port groups. The operation does not delete existing port groups that are not a part of the configuration file.

The vSphere Distributed Switch configuration is part of the vCenter backup. If you want to restore the entire vCenter vCenter, see Restoring a vCenter Instance.

1. Log in to the vCenter interface at `https://<vcenter_fqdn>/ui` with a user assigned the Administrator role.
2. Select **Menu** > **Networking**.
3. In the inventory expand **vCenter** > **Datacenter**.
4. Right-click the distributed switch and select **Settings** > **Restore configuration**.
5. On the **Restore switch configuration** page, click **Browse**, navigate to the location of the configuration file for the distributed switch, and click **Open**.
6. Select the **Restore distributed switch and all port groups** radio-button and click **Next**.
7. On the **Ready to complete** page, review the changes and click **Finish**.
8. Repeat these steps for the other vSphere Distributed Switch.
9. Review the switch configuration to verify that it is as you expect after the restore.

Restore an NSX Manager Cluster Node

If an NSX Manager instance fails, you can restore it from its file-based backup.

- Verify that you have a valid file-based backup of the failed NSX Manager instance.
- Verify that you have the SFTP server details:
  - SFTP Server IP
  - SFTP Server Username
  - SFTP Server Password
  - Encryption Password
- The following components must be available before proceeding with the restore of NSX Manager:
  - vCenter
  - SDDC Manager

Prepare for Restoring an NSX Manager Cluster Node

Before restoring an NSX Manager node, you must retrieve the NSX Manager build number and deployment details, as well as the credentials from the SDDC Manager inventory.

Retrieve the NSX Manager Version

Before restoring a failed NSX Manager instance, you must retrieve its version.

1. Navigate to the secure SFTP location that stores your NSX Manager backups, and determine the backup that you intend to restore. Normally you will be using the most recent backup.
2. The name of the backup folder is made up of four parts
   a) The NSX Manager version, for example, 9.1.0.0.25318227.
   b) The IP Address format, for example, IPv4.
   c) The UUID of the NSX Manager node, for example, 0e3f2442-1d82-5727-cbaf-5b8894186c59.
   d) The IP Address of the NSX Manager node, for example, 10.11.10.72.
3. Ensure you that the source NSX Manager Unified Appliance OVA file matches the version found in Step 2a.

   This is what you deploy before restoring the selected backup.

Retrieve the Credentials for Restoring NSX Manager from SDDC Manager

Before restoring a failed NSX Manager instance, you must retrieve the NSX Manager **root** and **admin** credentials from the SDDC Manager inventory.

Before you can query the SDDC Manager API, you must obtain an API access token by using an API service account.

1. Log in to your host machine with access to the SDDC and open a terminal.
2. Replace the values in the sample code with your values, these will be used in later steps

```
export sddcManagerFqdn='sfo-vcf01.sfo.rainpole.io'
export sddcManagerUsername='admin@local'
export sddcManagerPassword='***************'
```

3. Retrieve an access token from the SDDC Manager instance by running the following command.

```
sddcManagerToken=$(curl -k -s -X POST "https://${sddcManagerFqdn}/v1/tokens" \
    --header "Content-Type: application/json" \
    -d "{\"username\":\"${sddcManagerUsername}\", \"password\":\"${sddcManagerPassword}\"}" \
| jq -r '.accessToken' | tee /dev/tty )
```

4. Retrieve the **root** and **admin** credentials for the NSX Manager instance.
   a) Run the command to retrieve the credentials.

```
curl -k -s -X GET "https://${sddcManagerFqdn}/v1/credentials?resourceType=NSXT_MANAGER" \
    --header "Authorization: Bearer ${sddcManagerToken}" \
    --header "Content-Type: application/json" \
    --header "Accept: application/json" \
| jq
```

   b) Save the **root** and **admin** credentials for the instance you are restoring.

Restore the First Node of a Failed NSX Manager Cluster
If all three NSX Manager nodes in an NSX Manager cluster are in a failed state, you begin the restore process by restoring the first cluster node.

Important:

This procedure is not applicable in use cases when there are operational NSX Manager cluster nodes.

- If two of the three NSX Manager nodes in the NSX Manager cluster are in a failed state, you begin the restore process by deactivating the cluster. See Deactivate the NSX Manager Cluster.
- If only one of the three NSX Manager nodes in the NSX Manager cluster is in a failed state, you directly restore the failed node to the cluster. See Restore an NSX Manager Node to an Existing NSX Manager Cluster.

Redeploy the First Node of a Failed NSX Manager Cluster
You deploy a new NSX Manager instance by using the configuration of the first NSX Manager cluster node.

- Contact Broadcom support to download the NSX Manager OVA file for the version of the failed NSX Manager cluster. See Retrieve the NSX Manager Version.
- Verify that the backup file that you plan to restore is associated with the version of the failed NSX Manager cluster.
- **Note:**

  The first node that you deploy in the cluster must be the node that was running the NSX Cluster VIP when the last backup was taken. Retrieve the IP address from the backup folder name on the SFTP server, and restore this NSX manager node first.

1. In a web browser, log in to the management domain vCenter by using the vSphere Client (https://<vcenter_server_fqdn>/ui).
2. Select **Menu** > **VMs and Templates**.
3. In the inventory, expand **vCenter** > **Datacenter**.
4. Right-click the NSX folder and select **Deploy OVF Template**.
5. On the **Select an OVF template** page, select **Local file**, click **Upload files**, navigate to the location of the NSX Manager OVA file, click **Open**, and click **Next**.
6. On the **Select a name and folder** page, enter the VM name and click **Next**.
7. On the **Select a compute resource** page, click **Next**.
8. On the **Review details** page, click **Next**.
9. On the **Configuration** page, select the appropriate size and click **Next**.

   For the management domain, select **Medium** and for workload domains, select **Large** unless you changed these defaults during deployment.
10. On the **Select storage** page, select the vSAN datastore, and click **Next**.
11. On the **Select networks** page, from the **Destination network** drop-down menu, select the management distributed port group, and click **Next**.
12. On the **Customize template** page, enter these values and click **Next**.

Setting	Value for first NSX Manager cluster node
System root user password	*nsx_root_password*
CLI admin user password	*nsx_admin_password*
CLI audit user password	*nsx_audit_password*
Hostname	Enter the hostname for the appliance using FQDN format.
Rolename	NSX Manager
Management network IPv4 address	Enter the IP Address for the appliance.
Management network netmask	Enter the subnet mask for the appliance.
Default IPv4 gateway	Enter the default gateway for the appliance.
DNS server list	Enter the DNS servers for the appliance.
NTP server list	Enter the NTP server for the appliance.
Enable SSH	Deselected
Allow root SSH logins	Selected

13. On the **Ready to complete** page, review the deployment details and click **Finish**.

Restore the First Node in a Failed NSX Manager Cluster from a File-Based Backup

You restore the file-based backup of the first NSX Manager cluster node to the newly deployed NSX Manager instance.

1. In a web browser, log in to the NSX Manager node for the domain by using the user interface (https://<nsx_manager_node_fqdn>/login.jsp?local=true)
2. Accept the EULA and click **Continue**.
3. On the **Customer Experience Improvement Program** screen, select**Join the VMware Customer Experience Improvement Program** and click **Save**.
4. On the **Welcome to NSX** quick tour page, click **Skip**.
5. On the main navigation bar, click **System**.
6. In the left navigation pane, under **Lifecycle management**, click **Backup and Restore**.
7. In the **NSX configuration** pane, under **SFTP server**, click **Edit**.
8. In the **Backup configuration** dialog box, enter these values, and click **Save**.

Setting	Value
FQDN or IP address	IP address of SFTP server
Protocol	SFTP
Port	22
Directory path	• For 9.1.1 and beyond – <path_to_backup_dir>/<nsx_cluster_vip_fqdn> • For 9.1.0 and beyond – <path_to_backup_dir>/
Username	Service account user name For example, svc-vcf-bck@rainpole.io
Password	*service_account_password*
SSH fingerprint	*SFTP_ssh_fingerprint*

9. Under **Backup History**, select the target backup, and click **Restore**.
10. On the **Restoring NSX Managers** dialog box, click **Continue**.

    Note:

    The NSX Manager services will restart during the restore. You will need to refresh the NSX Manager UI.
11. During the restore, when prompted, reject adding NSX Manager nodes by clicking **I understand** and **Resume**.

A progress bar displays the status of the restore operation with the current step of the process.

Validate the Status of the First NSX Manager Cluster Node

After you restored the first NSX Manager cluster node, you validate the services state from the VM Web console of the restored node.

1. Log in to the vCenter interface at https://<vcenter_fqdn>/ui with a user assigned the Administrator role.
2. Select **Menu** > **VMs and Templates**.
3. In the inventory expand **vCenter** > **Datacenter** > **NSX Folder**.
4. Click the VM name of the newly deployed first NSX Manager cluster node, click **Launch Web Console**, and log in by using administrator credentials.

Setting	Value
User name	admin
Password	*nsx_admin_password*

5. Run the command to view the cluster status.

```
get cluster status
```

The services on the single-node NSX Manager cluster appear as `UP`.

6. Log in to the NSX Manager UI as a user with the NSX Administrator role.
7. Navigate to System > Backup & Restore and verify that the restore completed successfully.

Deactivate the NSX Manager Cluster

If two of the three NSX Manager cluster nodes are in a failed state or if you restored the first node of a failed NSX Manager cluster, you must deactivate the cluster.

Important:

- This procedure is not applicable in use cases when there are two healthy NSX Manager cluster nodes.
  If only one of the three nodes in the NSX Manager cluster is in a failed state, after you prepare for the restore, you directly restore the failed node to the cluster. See Restore an NSX Manager Node to an Existing NSX Manager Cluster.
- You must power off and delete the two failed NSX Manager nodes from the inventory before performing with this procedure.

1. Log in to the vCenter interface at https://<vcenter_fqdn>/ui with a user assigned the Administrator role.
2. Select **Menu** > **VMs and Templates**.
3. In the inventory expand **vCenter** > **Datacenter** > **NSX Folder**.
4. Click the VM of the operational NSX Manager node in the cluster, click **Launch Web Console**, and log in by using administrator credentials.

Setting	Value
User name	admin
Password	*nsx_admin_password*

5. Run the command to deactivate the cluster

```
deactivate cluster
```

6. On the **Are you sure you want to remove all other nodes from this cluster? (yes/no)**prompt, enter `yes`.

   You deactivated the cluster.

Restore an NSX Manager Node to an Existing NSX Manager Cluster

If only one of the three NSX Manager cluster nodes is in a failed state, you restore the failed node to the existing cluster. If two of the three NSX Manager cluster nodes are in a failed state, you repeat this process for each of the failed nodes.

Detach the Failed NSX Manager Node from the NSX Manager Cluster

Before you recover a failed NSX Manager node, you must detach the failed node from the NSX Manager cluster.

1. Log in to the vCenter interface at https://<vcenter_fqdn>/ui with a user assigned the Administrator role.
2. Select **Menu** > **VMs and Templates**.
3. In the inventory expand **vCenter** > **Datacenter** > **NSX Folder**.
4. Click the VM of an operational NSX Manager node in the cluster, click **Launch Web Console**, and log in by using administrator credentials.

Setting	Value
User name	admin
Password	*nsx_admin_password*

5. Retrieve the UUID of the failed NSX Manager node.
   a) Run the command to view the details of the cluster members.

```
get cluster status
```

The status of the failed node is `Down`.

b) Record the UUID of the failed NSX Manager node.

6. Run the command to detach the failed node from the cluster

```
detach node faild_node_uuid
```

The detach process might take some time.

7. When the detaching process finishes, run the command to view the cluster status.

```
get cluster status
```

The status of all cluster nodes is `Up`.

Redeploy the Failed NSX Manager Node

You deploy a new NSX Manager instance by using the configuration of the failed node.

Download the NSX Manager OVA file for the version of the failed NSX Manager instance. See Retrieve the NSX Manager Version.

1. Log in to the vCenter interface at https://<vcenter_fqdn>/ui with a user assigned the Administrator role.
2. Select **Menu** > **VMs and Templates**.
3. In the inventory expand **vCenter** > **Datacenter**.
4. Right-click the NSX folder and select **Deploy OVF Template**.
5. On the **Select an OVF template** page, select **Local file**, click **Upload files**, navigate to the location of the NSX Manager OVA file, click **Open**, and click **Next**.
6. On the **Select a name and folder** page, in the **Virtual machine name** text box, enter VM name of the failed node, and click **Next**.
7. On the **Select a compute resource** page, click **Next**.
8. On the **Review details** page, click **Next**.
9. On the **Configuration** page, select **Medium**, and click **Next**.
10. On the **Select storage** page, select the vSAN datastore, and click **Next**.
11. On the **Select networks** page, from the **Destination network** drop-down menu, select the management network distributed port group, and click **Next**.
12. On the **Customize template** page, enter these values and click **Next**.

Setting	Value
System root user password	*nsx_root_password*
CLI admin user password	*nsx_admin_password*
CLI audit password	*nsx_audit_password*
Hostname	*failed_node_FQDN*
Default IPv4 gateway	Enter the default gateway for the appliance.
Management network IPv4 address	*failed_node_IP_address*
Management network netmask	Enter the subnet mask for the appliance.
DNS server list	Enter the DNS servers for the appliance.
NTP servers list	Enter the NTP services for the appliance.
Enable SSH	Deselected
Allow root SSH logins	Selected

13. On the **Ready to complete** page, review the deployment details and click **Finish**.

The NSX Manager virtual machine begins to deploy.

Join the New NSX Manager Node to the NSX Manager Cluster

You join the newly deployed NSX Manager node to the cluster by using the virtual machine web console from the vSphere Client.

1. In a web browser, log in to the management domain vCenter by using the vSphere Client (https://<vcenter_server_fqdn>/ui).
2. Select **Menu** > **VMs and Templates**.
3. In the inventory expand **vCenter** > **Datacenter** > **NSX Folder**.
4. Click the VM of an operational NSX Manager node in the cluster, click **Launch web console**, and log in by using administrator credentials.

Setting	Value
User name	admin
Password	*nsx_admin_password*

5. Retrieve the ID of the NSX Manager cluster.
   a) Run the command to view the cluster ID.

      ```
      get cluster config | find Id:
      ```

   b) Record the cluster ID.
6. Retrieve the API thumbprint of the NSX Manager API certificate.
   a) Run the command to view the certificate API thumbprint.

      ```
      get certificate api thumbprint
      ```

   b) Record the certificate API thumbprint.
7. Exit the VM Web console.
8. In the vSphere Client, click the VM of the newly deployed NSX Manager node, click **Launch Web console**, and log in by using administrator credentials.

Setting	Value
User name	admin
Password	*nsx_admin_password*

9. Run the command to join the new NSX Manager node to the cluster.

   ```
   join existing_node_ip cluster-id cluster_id thumbprint api_thumbprint username admin
   ```

10. At the prompt, **Data on this node will be lost. Are you sure?**, enter **yes** and press enter.
11. Enter the password for the API user and press enter.

    The node will join the cluster and restart all services.

Validate the Status of the NSX Manager Cluster

After you added the new NSX Manager node to the cluster, you must validate the operational state of the NSX Manager cluster.

To view the state of the NSX Manager cluster, you log in to the NSX Manager for the particular domain.

1. In a web browser, log in to the NSX Manager cluster for the domain by using the user interface (https://<nsx_manager_cluster_fqdn>/login.jsp?local=true)
2. On the main navigation bar, click **System**.
3. In the left pane, under **Configuration**, click **Appliances**.
4. Verify that each cluster node is `Available`.

   Important:

   It can take some time for the cluster to reach a stable state after adding a new node.

Validate the Status of the NSX Manager Cluster

After restoring an NSX Manager node, you must validate the system status of the NSX Manager cluster.

To view the system status of the NSX Manager cluster, you log in to the NSX Manager for the particular domain.

1. In a web browser, log in to the NSX Manager cluster for the domain by using the user interface (https://<nsx_manager_cluster_fqdn>/login.jsp?local=true)
2. On the **Home** page, click **System Health**.
3. Verify that all components are healthy.
4. If the host transport nodes are in a `Pending` state, run **Configure NSX** on these nodes to refresh the UI.

Refresh the SSH keys that are stored in the SDDC Manager inventory. See VMware Cloud Foundation SDDC Manager Recovery Scripts (79004).

Update or Recreate the VM Anti-Affinity Rule for the NSX Manager Cluster Nodes

During the NSX Manager bring-up process, SDDC Manager creates a VM anti-affinity rule to prevent the VMs of the NSX Manager cluster from running on the same ESX host. If you redeployed all NSX Manager cluster nodes, you must recreate this rule. If you redeployed one or two nodes of the cluster, you must add the new VMs to the existing rule.

1. Log in to the vCenter interface at https://<vcenter_fqdn>/ui with a user assigned the Administrator role.
2. Select **Menu** > **Hosts and Clusters**.
3. In the inventory expand **vCenter** > **Datacenter**.
4. Click the cluster object.
5. Click the **Configure** tab and click **VM/Host Rules**.
6. Update or recreate the VM anti-affinity rule.
   - If you redeployed one or two nodes of the cluster, add the new VMs to the existing rule.
     a. Click the VM anti-affinity rule name and click **Edit**.
     b. Click **Add VM/Host rule member**, select the new NSX Manager cluster nodes, and click **Add**.
   - If you redeployed all NSX Manager cluster nodes, click **Add VM/Host rule**, enter these values to create the rule, and click **OK**.

Setting	Value
Name	Enter the name of the anti-affinity rule
Type	Separate virtual machines
Members	Click **Add VM/Host rule member**, select the NSX Manager cluster nodes, and click **Add**.

Validate the SDDC Manager Inventory State

After a successful restore of an NSX Manager cluster, you must verify that the SDDC Manager inventory is consistent with the recovered virtual machines. You run this verification by using the `sos` tool.

1. Log in to SDDC Manager by using a Secure Shell (SSH).
2. Verify the SDDC Manager health.
   a) Run the command to view the details about the VMware Cloud Foundation system.

      ```
      sudo /opt/vmware/sddc-support/sos --get-vcf-summary
      ```

   b) When prompted, enter the *vcf_password*.

   All tests show green state.
3. Run the command to collect the log files from the restore of the NSX Manager cluster.

   ```
   sudo /opt/vmware/sddc-support/sos --domain-name domain_name --nsx-logs
   ```

Refresh the SSH keys that are stored in the SDDC Manager inventory. See VMware Cloud Foundation SDDC Manager Recovery Scripts (79004).

Restoring NSX Edge Cluster Nodes

If one or both NSX Edge cluster nodes fail due to a hardware or software issue, you must redeploy the failed NSX Edge instances. You do not restore the NSX Edge nodes from a backup.

Redeploy an NSX Edge Cluster for VCF Component Backup and Restore
To restore an NSX Edge cluster, use the NSX API to redeploy the edge nodes.

Redeploy an NSX Edge Node Using The NSX Manager UI

1. From a browser, log in with admin privileges toNSX Manager.
2. Select**System > Fabric > Nodes > Edge Transport Nodes**.
3. In the list, select the edge node that you want to redeploy.
4. Select **Actions > Redeploy Edge**

   Note:

   The Redeploy Edge Node workflow is identical to the Add Edge Node workflow that was originally used to auto-deploy the node. The editable fields in the Redeploy Edge Node workflow are pre-filled with the existing configuration settings of the node. You can modify these settings as needed.
5. Proceed through the workflow steps and either accept the existing settings or modify the settings as needed. See "Create an NSX Edge Transport Node" in the **Installation Guide**

   Important:

   You must reenter the CLI credentials for the node VM, as the Credentials fields are left blank for security. Also, do not modify the Compute Manager setting unless you intend to change the compute manager for the node. Once you modify the setting, the configuration mappings associated with the original compute manager are cleared and cannot be retrieved, even if you re-select the original compute manager.
6. a. After finishing the Redeploy Edge Node workflow, allow the node redeployment to proceed and reach completion.
   b. Verify the node configuration status and connection status on the**Edge Transport Nodes**page. Alternatively, you can verify the transport node status by running theGet edge-cluster-status | get managers | get controllers | get host-switchCLI command.
   c. Perform the postrequisites described later on this page.

Redeploy an NSX Edge Node Using The NSX Manager API

1. Send a **GET** request to **https://<nsx-manager>/api/v1/transport-nodes/** and note every result whose **node_deployment_info.resource_type** is **EdgeNode**.
2. Retrieve the edge node configuration by running the following API call and copy the output payload of this API.

   ```
   GET /<NSX-Manager-IPaddress>/api/v1/transport-nodes/<edgenode_id>
   "resource_type": "EdgeNode",
            "id": "9f34c0ea-4aac-4b7f-a02c-62f306f96649",
            "display_name": "Edge_TN2",
            "description": "EN",
   ```

```
            "external_id": "9f34c0ea-4aac-4b7f-a02c-62f306f96649",
            "ip_addresses": [
                "10.170.94.240"
            ],
            "_create_user": "admin",
            "_create_time": 1600106319056,
            "_last_modified_user": "admin",
            "_last_modified_time": 1600106907312,
            "_system_owned": false,
            "_protection": "NOT_PROTECTED",
            "_revision": 2
        },
        "is_overridden": false,
        "failure_domain_id": "4fc1e3b0-1cd4-4339-86c8-f76baddbaafb",
        "resource_type": "TransportNode",
        "id": "9f34c0ea-4aac-4b7f-a02c-62f306f96649",
        "display_name": "Edge_TN2",
        "_create_user": "admin",
        "_create_time": 1600106319399,
        "_last_modified_user": "admin",
        "_last_modified_time": 1600106907401,
        "_system_owned": false,
        "_protection": "NOT_PROTECTED",
        "_revision": 1
    }
```

3. Redeploy the edge node using the following API call, passing the retrieved JSON data .

    Note:

    The node_id is retrieved in step 2

You do not need to pass any passwords in the JSON file.

```
POST /<NSX-Manager-IPaddress>/api/v1/transport-nodes/<edgenode_id>?action=redeploy
```

Note:

An example of the payload for POST https://<manager-ip>/api/v1/transport-nodes/<transport-node-id>?action=redeploy:

```
{
    "node_id": "9f34c0ea-4aac-4b7f-a02c-62f306f96649",
    "host_switch_spec": {
        "host_switches": [
            {
                "host_switch_name": "nsxvswitch_overlay",
                "host_switch_id": "c0a4a83e-c8b8-4324-a4d7-dbbc07b30b53",
                "host_switch_type": "NVDS",
                "host_switch_mode": "STANDARD",
                "host_switch_profile_ids": [
                    {
                        "key": "UplinkHostSwitchProfile",
                        "value": "f9a2a2fa-b49d-498f-abaf-2fdc81917716"
                    },
                    {
                        "key": "LldpHostSwitchProfile",
                        "value": "9e0b4d2d-d155-4b4b-8947-fbfe5b79f7cb"
                    }
                ],
                "pnics": [
```

```
                    {
                        "device_name": "fp-eth0",
                        "uplink_name": "uplink1"
                    }
                ],
                "is_migrate_pnics": false,
                "ip_assignment_spec": {
                    "ip_pool_id": "647d9b0d-0143-4903-91f5-930d9ab011e8",
                    "resource_type": "StaticIpPoolSpec"
                },
                "cpu_config": [],
                "transport_zone_endpoints": [
                    {
                        "transport_zone_id": "0b33b078-6438-4d9b-a1ec-33211fd36822",
                        "transport_zone_profile_ids": [
                            {
                                "resource_type": "BfdHealthMonitoringProfile",
                                "profile_id": "52035bb3-ab02-4a08-9884-18631312e50a"
                            }
                        ]
                    },
                    {
                        "transport_zone_id": "a0133574-48de-4e3a-9407-7db1a68bae41",
                        "transport_zone_profile_ids": [
                            {
                                "resource_type": "BfdHealthMonitoringProfile",
                                "profile_id": "52035bb3-ab02-4a08-9884-18631312e50a"
                            }
                        ]
                    }
                ],
                "vmk_install_migration": [],
                "pnics_uninstall_migration": [],
                "vmk_uninstall_migration": [],
                "not_ready": false
            }
        ],
        "resource_type": "StandardHostSwitchSpec"
    },
    "transport_zone_endpoints": [],
    "maintenance_mode": "DISABLED",
    "node_deployment_info": {
        "deployment_type": "VIRTUAL_MACHINE",
        "deployment_config": {
            "vm_deployment_config": {
                "vc_id": "cc82da39-b119-4869-a7fe-a54621cb4d3d",
                "compute_id": "domain-c9",
                "storage_id": "datastore-14",
                "host_id": "host-12",
                "compute_folder_id": "group-v5",
                "management_network_id": "network-16",
                "hostname": "EdgeSmallFactor",
                "data_network_ids": [
                    "5638c577-e142-4a50-aed3-a7079dc3b08c",
                    "5638c577-e142-4a50-aed3-a7079dc3b08c",
                    "5638c577-e142-4a50-aed3-a7079dc3b08c"
                ],
```

```
                "search_domains": [
                    "eng.vmware.com",
                    "vmware.com"
                ],
                "enable_ssh": true,
                "allow_ssh_root_login": true,
                "reservation_info": {
                    "memory_reservation": {
                        "reservation_percentage": 100
                    },
                    "cpu_reservation": {
                        "reservation_in_shares": "HIGH_PRIORITY",
                        "reservation_in_mhz": 0
                    }
                },
                "resource_allocation": {
                    "cpu_count": 4,
                    "memory_allocation_in_mb": 8192
                },
                "placement_type": "VsphereDeploymentConfig"
            },
            "form_factor": "MEDIUM",
            "node_user_settings": {
                 "cli_username": "admin",
                    "root_password":"Admin!23Admin",
                    "cli_password":"Admin!23Admin"
            }
        },
        "node_settings": {
            "hostname": "EdgeSmallFactor",
            "search_domains": [
                "eng.vmware.com",
                "vmware.com"
            ],
            "enable_ssh": true,
            "allow_ssh_root_login": true
        },
        "resource_type": "EdgeNode",
        "id": "9f34c0ea-4aac-4b7f-a02c-62f306f96649",
        "display_name": "Edge_TN2",
        "description": "EN",
        "external_id": "9f34c0ea-4aac-4b7f-a02c-62f306f96649",
        "ip_addresses": [
            "10.170.94.240"
        ],
        "_create_user": "admin",
        "_create_time": 1600106319056,
        "_last_modified_user": "admin",
        "_last_modified_time": 1600106907312,
        "_system_owned": false,
        "_protection": "NOT_PROTECTED",
        "_revision": 2
    },
    "is_overridden": false,
    "failure_domain_id": "4fc1e3b0-1cd4-4339-86c8-f76baddbaafb",
    "resource_type": "TransportNode",
    "id": "9f34c0ea-4aac-4b7f-a02c-62f306f96649",
```

```
        "display_name": "Edge_TN2",
        "_create_user": "admin",
        "_create_time": 1600106319399,
        "_last_modified_user": "admin",
        "_last_modified_time": 1600106907401,
        "_system_owned": false,
        "_protection": "NOT_PROTECTED",
        "_revision": 1
    }
```


1. Repeat the steps for the remaining edge cluster nodes.
2. In NSX Manager, monitor the configuration status of the new NSX Edge nodes, until they show **Success**.
3. If you encounter an error **`Edge redeploy is blocked as an active alarm is found for Edge VM present in NSX Inventory but missing in`** `vCenter`, perform the following steps.
   a. In the NSX Manager UI, navigate to the **Alarms** section.
   b. Select all alarms whose **Event Type** is **Edge VM Present In NSX Inventory Not Present In vCenter**.
   c. From the vertical ellipsis menu, select **Resolve**.
   d. Retry the redeployment operation for the NSX Edge cluster.

Verify the State of the NSX Edge Cluster Nodes for VCF Instance Backup and Restore

After completing all NSX Edge node redeployments, you must verify the state of the NSX Edge cluster nodes.

1. Log in to the NSX Manager user interface at `https://<nsx_manager-fqdn>` with the **admin** user.
2. On the main navigation bar, click **System**.
3. In the left pane, under **Configuration**, click **Fabric** > **Edges**.
4. Click the **Edge clusters** tab.
5. Verify that all edge transport nodes show these values.

Setting	Value
Configuration state	Success
Node status	Up
Tunnels	Up

File-Based Restore of VCF Management Services

Choose a recovery process depending upon your failure scenario. In-place restoration applies when VMs exist. Redeployment applies when VMs do not exist and must be redeployed as part of the restoration.

Table 1162: VCF management services Failure Scenarios

Failure Scenario	Recovery Process
VCF management services component corruption	In Place Restore of VCF Management Services Components
Single node lost	• No action required • The failed node is automatically redeployed
Two nodes lost	Redeploy VCF Management Services Cluster and Restore Initial Components
Three nodes lost	Redeploy VCF Management Services Cluster and Restore Initial Components

In Place Restore of VCF Management Services Components

The following procedure can be used to perform an in place restore of a VCF management services instance if there is a failure at the application level of one or more components.

- VCF management services is in a failed state but the virtual machines are still running.
- You must have at least one recent backup of the component that you are restoring.
- Before proceeding with restoration of VCF management services, the following components must be available:
  - vCenter
  - VCF Operations
  - SDDC Manager

    Note:

    Considerations for restoring Fleet lifecycle or SDDC lifecycle:

    - Restoring Fleet lifecycle and SDDC lifecycle interrupts any active tasks and causes them to disappear from the Tasks panel. In addition, the status of the restoration is not reflected on the Tasks panel after the components are restored.
    - Restore Fleet lifecycle and SDDC lifecycle individually. Do not restore Fleet lifecycle and SDDC lifecycle with other components.

1. Log in to the VCF Operations interface at `https://<vcf_operations_fqdn>` as a user assigned the Administrator role.
2. Navigate to **Build** > **Lifecycle** > **VCF Management**.
3. On the VCF Management page, click **Backup & Restore** > **Restore**.
4. To restore your VCF component, select the VCF Instance.
5. On the Restore pane, select the VCF component from a specific backup or the latest backup.
   - Restore from the latest backup on the cluster.
   - Specify the date range to filter the backups available, select the backup and optionally select the restore point by selecting the incremental backup.
6. Optional: Enter the Encryption passphrase.
7. Click **Restore**.
8. Monitor the restore task from the **Tasks** pane.

Retrieve Component Details from VCF Operations for Component Backup and Restore

Before you can redeploy a services runtime cluster, you must find the current version from the VCF Operations inventory.

You use the VCF Operations API to retrieve the component IDs and versions.

1. Retrieve a VCF Operations access token.

```
# VCF_OPS_FQDN should be the VCF Operations master node FQDN
# VCF_OPS_ADMIN_PASSWORD should be the VCF Operations local admin password

export VCF_OPS_FQDN='flt-ops01a.rainpole.io'
export VCF_OPS_ADMIN_PASSWORD='****************'

OPS_TOKEN=$(curl -k -s -X POST "https://$VCF_OPS_FQDN/suite-api/api/auth/token/acquire" \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d "{
    \"username\": \"admin\",
    \"password\": \"$VCF_OPS_ADMIN_PASSWORD\",
    \"authSource\": \"local\"
```

```
  }" | jq -r .token)
```

2. Retrieve a list of component IDs and versions from VCF Operations and save them for later use.

```
curl -k -s -X GET \
  "https://${VCF_OPS_FQDN}/suite-api/internal/components?_no_links=true" \
  -H 'accept: application/json' \
  -H 'X-Ops-API-use-unsupported: true' \
  -H "Authorization: OpsToken ${OPS_TOKEN}" \
| jq '
  .components as $all
  | ( $all
    | map(
        select(
          (.componentType | ascii_upcase) as $t
          | $t == "FLEET_LCM" or $t == "SALT_RAAS" or $t == "VIDB" or $t == "LI"
        )
      )
    ) as $filtered
  | ( $all
    | map(select((.componentType | ascii_upcase) == "VSP"))
    | map({
        id: .componentUuid,
        componentVersion: .componentVersion,
        fqdn: .properties.fqdn,
        fleetFqdn: .properties.fleetFqdn,
        ip: (.properties.ip // .vcfInstance.ip),
        instanceName: .vcfInstance.instanceName,
        vcfInstanceFqdn: .vcfInstance.fqdn
      })
    | unique_by(.id)
    ) as $vsp
  | ( $all
    | map(select((.componentType | ascii_upcase) == "VCFA"))[0]
    | if . then {
        fqdn: .properties.fqdn,
        componentVersion: .componentVersion,
        platformFqdn: .properties.platformFqdn,
        vspComponentUuid: .properties.vspComponentUuid
      } else null end
    ) as $vcfa
  | {
      components: (
        $filtered
        | map({
            componentType,
            id: .componentUuid,
            componentVersion: .componentVersion,
            fqdn: .properties.fqdn
          })
      ),
      vsp: $vsp,
      vcfa: $vcfa
    }
'
```

Sample output

```
{
```

```
  "components": [
    {
      "componentType": "FLEET_LCM",
      "id": "7bee057f-a164-4bce-8eb6-dea66f916008",
      "componentVersion": "9.1.1.0.25626340",
      "fqdn": "flt-fc01.rainpole.io"
    },
    {
      "componentType": "SALT_RAAS",
      "id": "1786c678-8565-43a0-a6e7-42405a24c3d8",
      "componentVersion": "9.1.1.0.25628391",
      "fqdn": "flt-fc01.rainpole.io"
    },
    {
      "componentType": "LI",
      "id": "7555d644-5d8c-4695-8b54-6b741ac9bafb",
      "componentVersion": "9.1.0.0400.25544947",
      "fqdn": "flt-logs01.rainpole.io"
    },
    {
      "componentType": "VIDB",
      "id": "b90d023a-b33a-469a-a4b6-096776bb732e",
      "componentVersion": "9.1.1.0.25629297",
      "fqdn": "flt-idb01.rainpole.io"
    }
  ],
  "vsp": [
    {
      "id": "6dfa787b-f697-44b4-8b13-7cc4b35bd8b7",
      "componentVersion": "9.1.1.0.25628383",
      "fqdn": "lax-sr01.lax.rainpole.io",
      "fleetFqdn": null,
      "ip": "10.21.10.13",
      "instanceName": "Los Angeles",
      "vcfInstanceFqdn": "lax-vcf01.lax.rainpole.io"
    },
    {
      "id": "c722868a-3db6-4991-9c22-d6610c2bed5d",
      "componentVersion": "9.1.1.0.25628383",
      "fqdn": "sfo-sr01.sfo.rainpole.io",
      "fleetFqdn": "flt-fc01.rainpole.io",
      "ip": "10.11.10.13",
      "instanceName": "San Francisco",
      "vcfInstanceFqdn": "sfo-vcf01.sfo.rainpole.io"
    }
  ],
  "vcfa": {
    "fqdn": "flt-auto01.rainpole.io",
    "componentVersion": "9.1.1.0.25629010",
    "platformFqdn": "flt-vcfa-sr01.rainpole.io",
    "vspComponentUuid": "c7862cc3-2857-4df8-a985-50467545d9cd"
  }
}
```


Redeploy and Restore VCF Management Services Components

You redeploy when VCF management services VMs do not exist and must be redeployed as part of the restoration

Extract Services Runtime Configuration from the Services Runtime Backup for VCF Component Backup and Restore

Using the VCF services runtime backup, you extract the package deployment configuration.

- OpenSSL 1.1.1 or later must be available on your Linux machine that you use to access the environment.
- Download and extract the scripts from Broadcom knowledge base article 443506 to the machine you use to access your VMware Cloud Foundation platform.
- A linux workstation with access to the SFTP server.

The VCF services runtime package deployment configuration is captured as part of the VCF management services backup. You use this extracted data in a later step.

1. Browse the SFTP server and locate the VCF management services backup that you plan to use for the restore operation.
2. Replace the values in the sample code with your values, these will be used in later steps.

```
export sftpHost='192.168.1.2'
export sftpPort='22'
export sftpDirectory='/media/backups/'
export sftpUsername='svc-vcf-bck'
export sftpPassword='***************'
export sftpEncryptionPassphrase='***************'
export remoteArchive='/media/backups/vcf/backups/5d29166a-780d-4481-81df-59cc3cefb697/9.1.1.0.25714471/
vsp/5d29166a-780d-4481-81df-59cc3cefb697/9.1.1.0.25714471/2026-09-01T14-53-53Z/2026-09-01T14-53-53Z.base.tgz'
 # remote-archive should be the path to the vsp backup for the VCF Management Services instance
```

3. Run the following command to execute the script to extract the configuration from the backup.

```
./vmsp-backup-sftp-to-yaml.sh \
    --sftp-host ${sftpHost} \
    --sftp-port ${sftpPort} \
    --sftp-user ${sftpUsername} \
    --sftp-password ${sftpPassword} \
    --sftp-dir ${sftpDirectory} \
    --remote-archive ${remoteArchive} \
    --passphrase ${sftpEncryptionPassphrase} \
    --output-dir "./tmp/backup-yaml"
```

Sample Output

```
vmsp-backup-sftp-to-yaml.sh: using local file: /media/backups/vcf/
backups/5d29166a-780d-4481-81df-59cc3cefb697/9.1.1.0.25714471/
vsp/5d29166a-780d-4481-81df-59cc3cefb697/9.1.1.0.25714471/2026-09-01T14-53-53Z/2026-09-01T14-53-53Z.base.tgz
tmp/backup-yaml/vmsp-platform.yaml
tmp/backup-yaml/ingress-fleet-tls-ndc.yaml
tmp/backup-yaml/ingress-instance-tls-ndc.yaml
tmp/backup-yaml/ingress-platform-tls-ndc.yaml
tmp/backup-yaml/ingress-fleet-tls.yaml
tmp/backup-yaml/ingress-instance-tls.yaml
tmp/backup-yaml/ingress-platform-tls.yaml

vmsp-backup-sftp-to-yaml.sh: YAML written under: ./tmp/backup-yaml
total 148
drwxr-xr-x 2 root root  4096 Sep  2 14:23 .
drwxr-xr-x 3 root root  4096 Sep  2 14:23 ..
-rw-r--r-- 1 root root 19019 Sep  2 14:23 ingress-fleet-tls-ndc.yaml
-rw-r--r-- 1 root root 19232 Sep  2 14:23 ingress-fleet-tls.yaml
-rw-r--r-- 1 root root 19046 Sep  2 14:23 ingress-instance-tls-ndc.yaml
-rw-r--r-- 1 root root 19262 Sep  2 14:23 ingress-instance-tls.yaml
-rw-r--r-- 1 root root 19054 Sep  2 14:23 ingress-platform-tls-ndc.yaml
```

```
-rw-r--r-- 1 root root 19356 Sep  2 14:23 ingress-platform-tls.yaml
-rw-r--r-- 1 root root 17409 Sep  2 14:23 vmsp-platform.yaml

# The file you require is backup-yaml/vmsp-platform.yaml
```

Redeploy VCF Management Services Cluster and Restore Initial Components

The following procedure can be used to redeploy and restore a VCF management services instance in case of a failure.

- VCF management services is in a failed state and cannot be restored using the in place procedure.
- Existing VCF management services virtual machines must be deleted from vCenter.
- You must have at least one recent scheduled backup of the components that you are restoring.
- Before proceeding with restoration of VCF management services, the following components must be available:
  - vCenter
  - SDDC Manager
- VCF Installer with the correct binary versions for the release you are running for VCF services runtime, Fleet lifecycle, Software depot, SDDC lifecycle, Salt master, Salt RaaS, and Identity broker.
- Retrieve the correct component version using Retrieve Component Details from VCF Operations for Component Backup and Restore

To restore VCF management services, you restore the components from the table below depending on your scenario.

VCF Management Services in the First VCF Instance	VCF Management Services in an Additional VCF Instance
• VCF services runtime • Fleet lifecycle • SDDC lifecycle • Software depot • Salt RaaS • Salt master • Identity broker	• VCF services runtime • SDDC lifecycle • Software depot * • Salt master • Identity broker *

* Optional components in an additional VCF Instance

1. Open an SSH connection to the SDDC Manager appliance, connect using the **vcf** user.
2. Switch to the **root** user.

   ```
   su -
   ```

3. Connect to the Postgres database by running the following command.

   ```
   psql -U postgres -h localhost -d platform;
   ```

4. Query the *vsp_cluster* table and record the **vsp_cluster_id** of the entry with type *MANAGEMENT* by running the following command.

   ```
   select * from vsp_cluster;
   ```

5. Delete the *MANAGEMENT* entry by running the following command.

   ```
   delete from vsp_cluster where type='MANAGEMENT';
   ```

6. Delete the corresponding credential by running the following command.

   ```
   delete from credential where username='vsp/<MANAGEMENT-vsp_cluster_id>/svc-sddc-manager-admin';
   ```

   Example

   ```
   delete from credential where username='vsp/1f5c79fe-e3aa-41b1-a5cf-774a6497fa3d/svc-sddc-manager-admin';
   ```

7. Retrieve token from SDDC Manager.

   ```
   # sddcmFQDN should be the FQDN for SDDC Manager
   # adminUsername should be a username with the ADMIN role in SDDC Manager
   # adminPassword should be the password for the user with the ADMIN role in SDDC Manager
   ```

```
export sddcmFQDN='sfo-vcf01.sfo.rainpole.io'
export adminUsername='administrator@vsphere.local'
export adminPassword='VMw@re1!VMw@re1!'

# generate token
token=$(curl -k https://"$sddcmFQDN"/v1/tokens \
  -X POST \
  -H "Content-type:application/json" \
  -d "{\"username\":\"$adminUsername\", \"password\":\"$adminPassword\"}" | jq -r '.accessToken')
```

8. If SDDC Manager was previously restored, or VCF Management Services was upgraded since initial deployment, then SDDC Manager will need to be pointed to an offline depot to source binares for the redeployment.
   a) To repoint to an offline depot, refer to Repoint SDDC Manager Depot to an Offline Depot for VCF Instance Backup and Restore
   b) To download the required binaries, refer to Download Binaries for VCF Management Services Runtime for VCF Instance Backup and Restore
9. Deploy a new VCF management services runtime instance.

```
# Domain id should be the management domain id from SDDC Manager. Retrive using the v1/domains API in the
 SDDC Manager developer center
# FQDNs must match the original
# clusterId must match the original retrieved in step 4
# networkMoId must be from the portgroup you are deploying to
curl -k -X POST -H 'accept: application/json' -H 'Content-Type: application/json' -H "Authorization:
 Bearer $token" https://"$sddcmFQDN"/v1/vsp-clusters \
-d '{
  "domainId": "0810c87d-3758-4c28-95fc-458b1196f4eb",
  "platformFqdn": "sfo-sr01.sfo.rainpole.io",
  "instanceFqdn": "sfo-ic01.sfo.rainpole.io",
  "fleetFqdn": "flt-fc01.rainpole.io",
  "systemUserPassword": "VMw@re1!VMw@re1!",
  "type": "MANAGEMENT",
  "ipv4Pool": {
    "addresses": [
      "10.11.99.29",
      "10.11.99.30",
      "10.11.99.31",
      "10.11.99.32",
      "10.11.99.33",
      "10.11.99.34",
      "10.11.99.35",
      "10.11.99.36",
      "10.11.99.37",
      "10.11.99.38",
      "10.11.99.39",
      "10.11.99.40"
    ]
  },
  "size": "medium",
  "version": "9.1.1.0.25714471",
  "networkMoId": "dvportgroup-28",
  "gatewayCidrIpv4": "10.11.99.1/24",
  "clusterId": "1f5c79fe-e3aa-41b1-a5cf-774a6497fa3d",
  "internalClusterCidrIpv4": "198.18.0.0/15"
}' | jq
```

10. Monitor the deployment until it completes.

```
curl -k -s -X GET "https://${sddcManagerFqdn}/v1/tasks/${TASK_ID}" \
```

```
     --header "Authorization: Bearer ${sddcManagerToken}" \
     --header "Content-Type: application/json" \
     --header "Accept: application/json" \
  | jq
```

11. Trust the VCF Installer (or SDDC Manager if the VCF installer was converted to SDDC manager during bringup) Certificate on the new VCF management services instance
    a) Retrieve the VCF Installer Certificate

```
CERT_DATA=$($echo | openssl s_client -connect <VCF-Installer-FQDN>:443 2>/dev/null | \
openssl x509 -outform pem)
```

    b) Retrieve a VCF management services services runtime token

```
export VCFMS_SR='sfo-sr01.sfo.rainpole.io'
export VCFMS_SR_PASSWORD='VMw@re1!VMw@re1!'

VCFMS_SR_TOKEN=$(curl -ks --request POST \
  --url "https://$VCFMS_SR/api/v1/identity/token" \
  --header 'Content-Type: application/x-www-form-urlencoded' \
  --data-urlencode grant_type=password \
  --data-urlencode "username=admin@vsp.local" \
  --data-urlencode "password=$VCFMS_SR_PASSWORD" \
  | jq -r '.access_token')
```

    c) Trust the VCF installer certificate

```
jq -n --arg cert "$CERT_DATA" '{cert: $cert}' | \
curl -k -X POST "https://sfo-sr01.sfo.rainpole.io/api/v1/system/trusted-certificates?action=add" \
     -H "Content-Type: application/json" \
     -H "Authorization: Bearer $VCFMS_SR_TOKEN" \
     -d @-
```

12. Update SFTP Settings on the new VCF management services Instance.
    a) Retrieve the SFTP server fingerprint

```
ssh-keyscan -p 22 <sftp_server_ip> | ssh-keygen -lf -
# SSH fingerprints of type RSA and ED25519 are supported
# Enter the full fingerprint in the payload of step 9c. Example:
 SHA256:yA3FHIQMbiMnansI4+lONYBASLP2IQhGZkGJIMZ2Pi8
```

    b) Retrieve a VCF management services services runtime token

```
export VCFMS_SR='sfo-sr01.sfo.rainpole.io'
export VCFMS_SR_PASSWORD='VMw@re1!VMw@re1!'

VCFMS_SR_TOKEN=$(curl -ks --request POST \
  --url "https://$VCFMS_SR/api/v1/identity/token" \
  --header 'Content-Type: application/x-www-form-urlencoded' \
  --data-urlencode grant_type=password \
  --data-urlencode "username=admin@vsp.local" \
  --data-urlencode "password=$VCFMS_SR_PASSWORD" \
  | jq -r '.access_token')
```

    c) Apply the SFTP server settings

```
# componentId in the POST URI should match the clusterId retrieved in step 4
curl -ks -X POST "https://$VCFMS_SR/api/v1/components/<componentId>?action=apply" \
  -H "Authorization: Bearer $VCFMS_SR_TOKEN" \
  -d  '{
    "spec": {
      "configuration": {
         "backups": {
            "destination": "sftp",
            "encryptionPassphrase": "VMw@re1!VMw@re1!",
```

```
                "storage": {
                  "sftp": {
                    "directory": "/media/backups/",
                    "host": "10.167.173.126",
                    "port": "22",
                    "username": "svc-vcf-bck",
                    "password": "VMw@re1!",
                    "fingerprint": "<sftpFingerprint>"
                  }
                }
              }
            }
          },
          "options": {
          }
        }' | jq
```

d) Monitor the task to apply the SFTP server settings

```
curl -X GET "https://$VCFMS_SR/api/v1/tasks/puwvsedyfzb7zaaoz546pyh6le"   -k -H "Authorization: Bearer
 $VCFMS_SR_TOKEN" | jq

# Wait until the response shows "Succeeded" at the end of the task output
],
  "startTime": "2026-04-02T02:52:16Z",
  "status": "Succeeded",
  "type": "com.vmware.vcfms.task.ConfigureComponent"
}
```

13. Retrieve list of backups for VCF management services runtime, fleet lifecycle, software depot, and SDDC lifecycle.

```
curl -k "https://$VCFMS_SR/api/v1/system/backups" \
  -H "Authorization: Bearer $VCFMS_SR_TOKEN" \
  -H 'accept: application/json' | \
  jq -r '
    ["vsp", "vcf-fleet-lcm", "vcf-fleet-depot", "vcf-sddc-lcm", "salt", "salt-raas", "vidb", "vcfms-
metrics-store", "vcf-obs-data-platform", "telemetry"] as $order
    | .backups as $all
    | $order[] as $current_type
    | $all[]
    | select(.component.type == $current_type)
    | (
        (.name | sub("T(?<h>\\d{2})-(?<m>\\d{2})-(?<s>\\d{2})Z"; "T\(.h):\(.m):\(.s)Z") | fromdateiso8601)
      ) as $timestamp
    | ( (now - $timestamp) / 86400 | floor ) as $days_old
    | [ $current_type, .component.version, .name, ($days_old | tostring + " days ago"), .path ]
    | @tsv' | \
    column -t -s $'\t'
```

14. Perform the Restore.

```
curl -k -X POST "https://$VCFMS_SR/api/v1/system/backups?action=restore" \
  -H "Authorization: Bearer $VCFMS_SR_TOKEN" \
  -H 'accept: application/json' \
  -H "Content-type: application/json" \
  -d '{
  "components": [
    {
      "point": "2026-09-14T15-33-55Z",
```

```
      "path": "sftp://svc-vcf-bck@10.167.173.134:22/media/backups/vcf/backups/
f365cb61-c1a4-42f1-b0c3-e77a21edabd6/9.1.1.0.25714471/vsp/f365cb61-c1a4-42f1-b0c3-
e77a21edabd6/9.1.1.0.25714471/2026-09-14T15-33-55Z"
    },
    {
      "point": "2026-09-14T14-56-46Z",
      "path": "sftp://svc-vcf-bck@10.167.173.134:22/media/backups/vcf/backups/f365cb61-
c1a4-42f1-b0c3-e77a21edabd6/9.1.1.0.25714471/vcf-fleet-lcm/aaf5baa7-0dde-43f4-8dc3-
d3ac5243c63f/9.1.1.0.25713934/2026-09-14T14-56-46Z"
    },
    {
      "point": "2026-09-14T15-16-43Z",
      "path": "sftp://svc-vcf-bck@10.167.173.134:22/media/backups/vcf/backups/f365cb61-
c1a4-42f1-b0c3-e77a21edabd6/9.1.1.0.25714471/vcf-fleet-depot/6bc3a1a1-c045-431d-
bc12-193ae61a08cc/9.1.1.0.25713941/2026-09-14T15-16-43Z"
    },
    {
      "point": "2026-09-14T15-37-28Z",
      "path": "sftp://svc-vcf-bck@10.167.173.134:22/media/backups/vcf/backups/f365cb61-
c1a4-42f1-b0c3-e77a21edabd6/9.1.1.0.25714471/vcf-sddc-lcm/68161b94-63b5-4843-be76-
a99dcc7b07ff/9.1.1.0.25713940/2026-09-14T15-37-28Z"
    },
    {
      "point": "2026-09-14T15-12-09Z",
      "path": "sftp://svc-vcf-bck@10.167.173.134:22/media/backups/
vcf/backups/f365cb61-c1a4-42f1-b0c3-e77a21edabd6/9.1.1.0.25714471/
salt/12571e84-16fe-4daa-94c1-2385b6ceebf6/9.1.1.0.25679895/2026-09-14T15-12-09Z"
    },
    {
      "point": "2026-09-14T15-21-40Z",
      "path": "sftp://svc-vcf-bck@10.167.173.134:22/media/backups/vcf/backups/f365cb61-
c1a4-42f1-b0c3-e77a21edabd6/9.1.1.0.25714471/salt-raas/edeac40e-76f2-44ab-83c8-
cb9dcddaed55/9.1.1.0.25679895/2026-09-14T15-21-40Z"
    },
    {
      "point": "2026-09-14T14-46-10Z",
      "path": "sftp://svc-vcf-bck@10.167.173.134:22/media/backups/vcf/
backups/f365cb61-c1a4-42f1-b0c3-e77a21edabd6/9.1.1.0.25714471/vidb/
b1408f7e-5fb3-47f9-8b4d-3461021bbea8/9.1.1.0.25679886/2026-09-14T14-46-10Z"
    },
    {
      "point": "2026-09-14T15-02-32Z",
      "path": "sftp://svc-vcf-bck@10.167.173.134:22/media/backups/vcf/backups/
f365cb61-c1a4-42f1-b0c3-e77a21edabd6/9.1.1.0.25714471/vcfms-metrics-store/1e58b436-
ce50-4f37-8f01-3678132a9a4b/9.1.1.0.25714471/2026-09-14T15-02-32Z"
    },
    {
      "point": "2026-09-14T14-51-59Z",
      "path": "sftp://svc-vcf-bck@10.167.173.134:22/media/backups/vcf/backups/
f365cb61-c1a4-42f1-b0c3-e77a21edabd6/9.1.1.0.25714471/vcf-obs-data-platform/1996e2f7-
f0a3-4ef9-8b20-2a23fa457db4/9.1.1.0.25679622/2026-09-14T14-51-59Z"
    },
    {
      "point": "2026-09-14T15-07-11Z",
      "path": "sftp://svc-vcf-bck@10.167.173.134:22/media/backups/vcf/backups/f365cb61-
c1a4-42f1-b0c3-e77a21edabd6/9.1.1.0.25714471/telemetry-acceptor/95dc43b7-749d-4014-
a479-96d51c67a6f0/9.1.1.0.25671600/2026-09-14T15-07-11Z"
```

```
    }
  ]
}'
```

Important:

After restoring the vcf-fleet-lcm component, there might be a failed backup task in the tasks view. This is a false negative and can be ignored.

15. The restore task returns a task id which can be used to monitor the restore

```
{"cancellable":true,"createTime":"2026-03-18T10:40:24Z","createdBy":"admin@vsp.local","endTime":null,"id":"selk5wjxb
{"components":[{"path":"sftp://svc-vcf-bck@10.167.173.126:22/media/backups/vcf/
backups/1f5c79fe-e3aa-41b1-a5cf-774a6497fa3d/9.1.0.0.25257932/vsp/1f5c79fe-e3aa-41b1-
a5cf-774a6497fa3d/9.1.0.0.25257932/2026-03-09T20-09-58Z","point":"2026-03-09T20-09-58Z"},
{"path":"sftp://svc-vcf-bck@10.167.173.126:22/media/backups/vcf/backups/1f5c79fe-
e3aa-41b1-a5cf-774a6497fa3d/9.1.0.0.25257932/vcf-fleet-lcm/3d7c01a5-55fb-4b2a-8f0e-
c4d24c6d2c03/9.1.0.0.25262128/2026-03-09T20-19-30Z","point":"2026-03-09T20-19-30Z"},{"path":"sftp://svc-
vcf-bck@10.167.173.126:22/media/backups/vcf/backups/1f5c79fe-e3aa-41b1-a5cf-774a6497fa3d/9.1.0.0.25257932/
vcf-fleet-depot/
a1a91288-7709-4ad6-978c-702a02b94195/9.1.0.0.25262075/2026-03-09T20-24-42Z","point":"2026-03-09T20-24-42Z"},
{"path":"sftp://svc-vcf-bck@10.167.173.126:22/media/backups/vcf/backups/1f5c79fe-e3aa-41b1-
a5cf-774a6497fa3d/9.1.0.0.25257932/vcf-sddc-lcm/
e8ad346d-7777-4b15-843b-2ec1d2964a5b/9.1.0.0.25262104/2026-03-09T20-13-46Z","point":"2026-03-09T20-13-46Z"}]},"resul
curl -X GET "https://$VCFMS_SR/api/v1/tasks/selk5wjxbvfhbksxiwvttwwxnu" \
  -k -H "Authorization: Bearer $VCFMS_SR_TOKEN" | jq
```

16. If you had to point SDDC Manager to an offline depot earlier in the process, then you need to point it back to the Fleet Depot Service once the redeployment is complete.
    a) To point SDDC Manager back to Fleet Depot Service, refer to Repoint SDDC Manager Depot to Fleet Depot Service for VCF Instance Backup and Restore

Apply Services Runtime Configuration of VCF Management Services for VCF Component Backup and Restore
To ensure the worker nodes are the correct size, you apply the kubernetes package deployment from the backup.

1. Discover the VCF management services control plane node.
   a) SSH to the Services Runtime FQDN e.g. sfo-sr01.sfo.rainpole.io.
   b) Find the EXTERNAL-IP address of the control plane node.

```
sudo su
export KUBECONFIG=/etc/kubernetes/admin.conf
kubectl get nodes -o wide
```

2. Copy all yaml files from Extract Services Runtime Configuration from the Services Runtime Backup for VCF Component Backup and Restore to the control plane node.
3. SSH to the control plane node and run the following.

```
sudo su
export KUBECONFIG=/etc/kubernetes/admin.conf
export YAML_PATH='./'
export WORKER_SIZE=$(awk '/worker:/{f=1} f && /size:/{print $2; exit}' $YAML_PATH/vmsp-platform.yaml)
export MACHINE_TYPE=$(awk '/worker:/{f=1} f && /machineType:/{print $2; exit}' $YAML_PATH/vmsp-
platform.yaml)
export PROFILE=$(awk '/profiles:/{f=1} f && /name:/{print $2; exit}' $YAML_PATH/vmsp-platform.yaml)

kubectl patch packagedeployment vmsp-platform -n vmsp-platform \
  --type=merge \
  -p "$(jq -n \
    --arg s "$WORKER_SIZE" \
    --arg m "$MACHINE_TYPE" \
    --arg p "$PROFILE" \
```

```
        '{
          "spec": {
            "values": {
              "cluster": {
                "worker": {
                  "size": $s,
                  "machineType": $m
                }
              },
              "profiles": {
                "name": $p
              }
            }
          }
        }')"
```

4. Monitor the task.

```
kubectl wait --for=jsonpath='{.status.phase}'=Successful packagedeployment/vmsp-platform -n vmsp-platform
 --timeout=2h

# Wait for this message
packagedeployment.releases.vmsp.vmware.com/vmsp-platform condition met

kubectl get packagedeployment vmsp-platform -n vmsp-platform -o jsonpath='{.status.phase}'

# Wait for this message
Successful
```

The worker nodes will be resized to match their size when the backup was taken. Do not proceed until this step has completed.

Configure Depot and Download Binaries

To enable the restore of the remaining components, you configure a software depot and download the required binaries.

The binaries show as downloaded in the Binary Management UI, however, the underlying binaries are not restored as part of the restore of the software depot component.

1. Log in to VCF Operations as a user with the Administrator role.
2. Configure the software depot.
   a) Navigate to **Build** > **Software Depot**.
   b) In the right pane, click Edit.
   c) Configure your depot type and click Finish.
3. Download the required binaries.
   a) Navigate to **Build** > **Lifecycle** > **VCF Instances** and click the VCF Instance name where you are performing the restore.
   b) In the right pane, click **Binary Management**.
   c) In the **Binary Management** pane, click **Install Binaries**.
   d) On the **Install Binaries** pane, click the checkbox for the following components, and click **Delete Download**.

Install Binaries to Download
Identity broker
Telemetry
License Server
Salt RaaS
Salt master

e) On the **Install Binaries** pane, click the checkbox for the following components, and click **Download**.

Install Binaries to Download
Identity broker
Telemetry
License Server

Wait for the binaries to download before proceeding.

Restore Log Management Component

After you restore the core VCF management services components, if you have the log management component, you can now restore it.

You must download the required binaries.

1. Retrieve a VCF management services fleet lifecycle access token.

```
export VCFMS_SR='sfo-sr01.sfo.rainpole.io'
export VCFMS_SR_PASSWORD='VMw@re1!VMw@re1!'

VCFMS_SR_TOKEN=$(curl -ks --request POST \
  --url "https://$VCFMS_SR/api/v1/identity/token" \
  --header 'Content-Type: application/x-www-form-urlencoded' \
  --data-urlencode grant_type=password \
  --data-urlencode "username=admin@vsp.local" \
  --data-urlencode "password=$VCFMS_SR_PASSWORD" \
  | jq -r '.access_token')
```

2. Retrieve a list of available backups for Log management.

```
# Retrieve list of backups for log management
  curl -k "https://$VCFMS_SR/api/v1/system/backups" \
  -H "Authorization: Bearer $VCFMS_SR_TOKEN" \
  -H 'accept: application/json' | \
  jq -r '
    ["ops-logs"] as $order
    | .backups as $all
    | $order[] as $current_type
    | $all[]
    | select(.component.type == $current_type)
    | (
        # Replace dashes with colons in the time portion to make it ISO-compliant
        (.name | sub("T(?<h>\\d{2})-(?<m>\\d{2})-(?<s>\\d{2})Z"; "T\(.h):\(.m):\(.s)Z") | fromdateiso8601)
      ) as $timestamp
    | ( (now - $timestamp) / 86400 | floor ) as $days_old
    | [ $current_type, .component.version, .name, ($days_old | tostring + " days ago"), .path ]
    | @tsv' | \
    column -t -s $'\t'
```

3. Perform the restore of Log management.

```
curl -k -X POST "https://$VCFMS_SR/api/v1/system/backups?action=restore" \
  -H "Authorization: Bearer $VCFMS_SR_TOKEN" \
  -H 'accept: application/json' \
  -H "Content-type: application/json" \
  -d '{
  "components": [
    {
      "path": "sftp://svc-vcf-bck@10.167.173.126:22/media/backups/vcf/
backups/1f5c79fe-e3aa-41b1-a5cf-774a6497fa3d/9.1.0.0.25257932/ops-logs/4e38afb4-
ac83-481b-876f-922497eaada7/9.1.0.0.25262140/2026-03-23T17-17-22Z",
```

```
        "point": "2026-03-23T17-17-22Z"
      }
    ]
  }'
```

4. Monitor the restore using the task id from the output of the above restore command.

```
curl -X GET "https://$VCFMS_SR/api/v1/tasks/<taskId>" \
  -k -H "Authorization: Bearer $VCFMS_SR_TOKEN" | jq
```

Important:

For VCF 9.1.0 only, after the Log Management restore completes, you must run a script to reconcile the log management state. Refer to this KB436904 for more information.

Update Log Management State

After the restore operation of log management is complete, you must perform steps to finalize the restore.

1. For VCF 9.1.1 do the following:
   a) Log in to VCF Operations as a user with the **Administrator** role.
   b) Navigate to **Build** > **Lifecycle** > **VCF Management** > **Components.**
   c) Click **Log management**.
   d) From the **Actions** menu, click **Stop Service**.
   e) Monitor the Stop Service task until it completes and then click **Start Service**.
   f) Monitor the Start Service task until it completes.
2. If the logs UI is not loading in the VCF Operations UI after DR failover, we need to follow these steps:
   a) Run following script to delete the Log Management component from Ops and re-registering it.

```
export KUBECONFIG=<path-to-kubeconfig-for VCF Services Runtime>
./update_registration_in_ops.sh --ops-fqdn "flt-ops01a.rainpole.io" --component-fqdn "flt-logs01.rainpole.io" --ops-username "admin"
```

   b) If step 2a fails, retry the task by visiting **https://<ops-fqdn>/vcf-operations/ui/operate/administration/control-panel/management-tasks/lcm/tasks** and clicking **Retry** on the failed task. Select the **Skip** option so that validations are skipped.
   c) After successful deletion of the Log Management component from Ops, navigate to **Build** > **Lifecycle** > **Tasks** to verify status of the **Refresh component Log management** task. If that task failed, click **Retry**.
3. For VCF 9.1.0 do the following:
   a) Follow the steps in Broadcom knowledge base article 436904.

File-Based Restore of VCF Automation

Depending on the failure scenario, the recovery process can vary. Use this table to guide you to the process that meets your scenario.

Table 1163: VCF Automation Failure Scenarios

Failure Scenario	Recovery Process
VCF Automation application corruption	In Place Restore of VCF Automation
Single node lost	• No action required • The failed node is automatically redeployed
Two nodes lost	VCF Automation Redeploy and Restore Tasks for Component Backup and Restore
Three nodes lost	VCF Automation Redeploy and Restore Tasks for Component Backup and Restore

In Place Restore of VCF Automation

The following procedure can be used to perform an in place restore of a VCF Automation instance if there is a failure at the application level.

- VCF Automation is in a failed state but the virtual machines are still running.
- You must have at least one recent backup of the component that you are restoring.
- Before proceeding with restoration of VCF Automation, the following components must be available:
  - vCenter
  - VCF Operations
  - VCF management services with Fleet lifecycle and SDDC lifecycle.

1. Log in to the VCF Operations interface at `https://<vcf_operations_fqdn>` as a user assigned the Administrator role.
2. Navigate to **Build** > **Lifecycle** > **VCF Management**.
3. On the VCF Management page, click **Backup & Restore** > **Restore**.
4. To restore your VCF component, select the VCF Instance.
5. On the Restore pane, select the VCF component from a specific backup or the latest backup.
   - Restore from the latest backup on the cluster.
   - Specify the date range to filter the backups available, select the backup and optionally select the restore point by selecting the incremental backup.
6. Optional: Enter the Encryption passphrase.
7. Click **Restore**.
8. Monitor the restore task from the **Tasks** pane.

Remediate Passwords for VMware Cloud Foundation Components After a VCF Automation Restore
After a successful restore of VCF Automation, some service accounts might become disconnected.

To manually remediate the VCF Automation service account in vCenter and NSX, use the VCF Automation UI.

1. Log in to the VCF Automation interface at https://<vcf_automation_fqdn> with a user assigned the **System Administrator** role.
2. In the left pane, navigate to **VCF Instances**.
3. On the **VCF Instances** page, click the vertical ellipsis next to the VCF Instance, and, from the drop-down menu, select **Rotate service account**.

   VCF Automation deletes and recreates the service account. This task might take a few minutes to complete.

VCF Automation Redeploy and Restore Tasks for Component Backup and Restore

Perform these tasks related to restoring VCF Automation, when VCF Automation is in a failed state and cannot be restored using the in-place restore process.

Restore VCF Automation

Note:

The following table contains the sequence of tasks along with PowerShell equivalents. To install the PowerShell module and dependencies, run the following commands:

```
Install-Module -Name VCF.PowerCLI -MinimumVersion 9.1.0 -Scope AllUsers -Force
Install-Module -Name Posh-SSH -RequiredVersion 3.0.8 -Scope AllUsers -Force
Install-Module -Name VMware.CloudFoundation.InstanceRecovery -Scope AllUsers -Force
```

If you have a password with a $ the recommendation is to use single quotes instead of double quotes.

Step	Manual Task	PowerShell Equivalent
1	Prepare to Restore VCF Automation for VCF Component Backup and Restore	```## Parameter Block ## $SddcManagerFqdn = "sfo-vcf01.sfo.rainpole.io" $VcfUserPassword = "**************" $RootPassword = "**************" $ClusterType = "CONSUMPTION" ## Command Block ## Remove-SddcManagerVspClusterEntry ` -SddcManagerFqdn $SddcManagerFqdn ` -VcfUserPassword $VcfUserPassword ` -RootPassword $RootPassword ` -ClusterType $ClusterType```
2	Redeploy the VCF Automation Services Runtime Cluster for Component Backup and Restore	```## Parameter Block ## $SddcManagerFqdn = "sfo-vcf01.sfo.rainpole.io" $SddcManagerUser = "administrator@vsphere.local" $SddcManagerPassword = "**************" $PlatformFqdn = "flt-vcfa-sr01.rainpole.io" $SystemUserPassword = "**************" $Ipv4Addresses = "10.11.99.29","10.11.99.30","10.11.99.31","10.11.99.32","10.11.99.33" $size = "medium" $NetworkMoId = "dvportgroup-28" $GatewayCidrIpv4 = "10.11.99.1/24" $ClusterId = "1f5c79fe-e3aa-41b1-a5cf-774a6497fa3d" $InternalClusterCidrIpv4 = "198.18.0.0/15" $type = "CONSUMPTION" ## Command Block ## New-ServicesRuntime ` -SddcManagerFqdn $SddcManagerFqdn ` -SddcManagerUser "administrator@vsphere.local" ` -SddcManagerPassword $SddcManagerPassword ` -PlatformFqdn $PlatformFqdn ` -SystemUserPassword $SystemUserPassword ` -Ipv4Addresses $Ipv4Addresses ` -Size $size ` -NetworkMoId $NetworkMoId ` -GatewayCidrIpv4 $GatewayCidrIpv4 ` -ClusterId $ClusterId ` -InternalClusterCidrIpv4 $InternalClusterCidrIpv4 ` -type $type```
3	Restore VCF Automation for VCF Component Backup and Restore	```## Parameter Block ## $ServicesRuntimeFqdn = "lax-sr01.lax.rainpole.io" $ServicesRuntimePassword = "**************" $VspId = "40e752c7-e940-4a9f-a49c-0f05efa9d915" $components = "vsp", "vcfa", "vcd-migrator" ## Command Block ## Get-ServicesRuntimeComponentBackups ` -ServicesRuntimeFqdn $ServicesRuntimeFqdn ` -ServicesRuntimePassword $ServicesRuntimePassword ` -VspId $VspId ` -components $components```

Step	Manual Task	PowerShell Equivalent
4	Restore VCF Automation for VCF Component Backup and Restore	```## Parameter Block ## $ServicesRuntimeFqdn = "flt-vcfa-sr01.rainpole.io" $ServicesRuntimePassword = "**************" ## Command Block ## Restore-ServicesRuntimeComponentBackup ` -ServicesRuntimeFqdn $ServicesRuntimeFqdn ` -ServicesRuntimePassword $ServicesRuntimePassword ` -RestoreJsonFile ".\restore-payload.json"```
5	Update Connections in VCF Automation for VCF Component Backup and Restore	Not Applicable. Follow the manual task.

Prepare to Restore VCF Automation for VCF Component Backup and Restore

Before you can deploy a new VCF Automation instance to restore to, you must delete the existing VCF Automation entry in the SDDC Manager database.

> **CAUTION:**
>
> This procedure should not be used to completely delete a deployed component. To fully delete VCF Automation refer to this KB441333

1. Open an SSH connnection to the SDDC Manger appliance as the **vcf** user and switch to the **root** user.
2. Connect to the Postgres database.

   ```
   psql -U postgres -h localhost -d platform;
   ```

3. Query the *vsp_clusters* table and record the id of the entry with type *CONSUMPTION*.

   ```
   select * from vsp_cluster;
   ```

4. Delete the *CONSUMPTION* entry.

   ```
   delete from vsp_cluster where type='CONSUMPTION';
   ```

5. Delete the corresponding credential.

   ```
   delete from credential where username='vsp/<CONSUMPTION-vsp_cluster_id>/svc-sddc-manager-admin';
   ```

   Example

   ```
   delete from credential where username='vsp/1f5c79fe-e3aa-41b1-a5cf-774a6497fa3d/svc-sddc-manager-admin';
   ```

6. Retrieve a token from the Fleet Lifecycle service.

   ```
   export VCFMS_FC='flt-fc01.rainpole.io'
   export VCFMS_FC_PASSWORD='VMw@re1!VMw@re1!'

   VCFMS_TOKEN=$(curl -ks --request POST \
     --url "https://$VCFMS_FC/api/v1/identity/token" \
     --header 'Content-Type: application/x-www-form-urlencoded' \
     --data-urlencode grant_type=password \
     --data-urlencode "username=admin@vsp.local" \
     --data-urlencode "password=$VCFMS_FC_PASSWORD" \
     | jq -r '.access_token')
   ```

7. Retrieve and record the component id for the VCF Automation.

   ```
   curl -k -s "https://$VCFMS_FC/fleet-lcm/v1/components?includeConsumptionVsp=true&includeVcdMigrator=true"
    \
     -H "Authorization: Bearer $VCFMS_TOKEN" | jq '.components[] |
     if .componentTypeDescription == "VCF services runtime"
     then {id, componentTypeDescription, fqdn}
   ```

```
    else {id, componentTypeDescription}
    end'
```

Redeploy the VCF Automation Services Runtime Cluster for Component Backup and Restore

You deploy a new VCF Automation services runtime instance using the SDDC Manager API.

- VCF Automation is in a failed state and cannot be restored using the in place procedure.
- VCF Automation VMs must no longer exist in vCenter.
- You must have at least one recent backup of VCF Automation.
- Before proceeding with restoration of VCF Automation, the following components must be available:
  - vCenter
  - VCF Operations
  - SDDC Manager
- The VCF Automation binary must be available in the fleet software depot.

1. Retrieve the *sddcLcmId* from VCF management services.
   a) Replace the values in the sample code with your values, these will be used in later steps.

```
export fleetLifecycleFqdn='flt-fc01.rainpole.io'
export fleetLifecycleUsername='admin@vsp.local'
export fleetLifecyclePassword='***************'
```

   b) Retrieve an access token from the VCF services runtime instance by running the following command.

```
fleetLifecycleToken=$(curl -k -s -X POST "https://${fleetLifecycleFqdn}/api/v1/identity/token" \
    --header "Content-Type: application/x-www-form-urlencoded" \
    --data-urlencode "grant_type=password" \
    --data-urlencode "username=$fleetLifecycleUsername" \
    --data-urlencode "password=$fleetLifecyclePassword" \
| jq -r '.access_token' | tee /dev/tty )
```

   c) Retrieve the *sddcLcmId* from VCF management services by running the following command.

```
sddcLcmId=$(curl -k -s -X GET "https://${fleetLifecycleFqdn}/fleet-lcm/v1/components?
includeConsumptionVsp=true&includeVcdMigrator=true" \
    --header "Authorization: Bearer ${fleetLifecycleToken}" \
    --header "Content-Type: application/json" \
    --header "Accept: application/json" \
| jq -r '.components[] | select(.componentType == "VCF_SDDC_LCM") | .id' | tee /dev/tty )
```

2. Retrieve token from SDDC Manager.

```
export sddcmFQDN='sfo-vcf01.sfo.rainpole.io'
export ssoAdminPassword='VMw@re1!VMw@re1!'
# generate token
token=$(curl -k https://"$sddcmFQDN"/v1/tokens \
  -X POST \
  -H "Content-type:application/json" \
  -d "{\"username\":\"administrator@vsphere.local\", \"password\":\"$ssoAdminPassword\"}" | jq -r
 '.accessToken')
```

3. Deploy a new VCF Automation services runtime instance.

```
# Domain id should be the management domain id from SDDC Manager
# FQDNs must match the original
# clusterId must match the original recorded in step 3 of Prepare to Restore VCF Automation for VCF
 Component Backup and Restore
# networkMoId must be from the portgroup you are deploying to
TASK_ID=$(curl -k -X POST -H 'accept: application/json' -H 'Content-Type: application/json' -H
 "Authorization: Bearer $token" https://"$sddcmFQDN"/v1/vsp-clusters \
-d '{
    "domainId": "a10b493e-0770-49ee-8179-d444ec5e8887",
    "deploymentType": "VspClusterSpec",
```

```
    "sddcLcmId": "3b9cdd99-9212-4e00-8bd6-53bf79c67fbf",
    "platformFqdn": "flt-vcfa-sr01.rainpole.io",
    "systemUserPassword": "**************",
    "type": "CONSUMPTION",
    "size": "medium",
    "ipv4Pool": {
        "addresses": [
            "10.11.10.46",
            "10.11.10.47",
            "10.11.10.48",
            "10.11.10.49",
            "10.11.10.50"
        ],
        "excludedAddresses": [
        ]
    },
    "clusterId": "6a079b0d-25e1-4032-9824-a7108d98a9b3",
    "networkMoId": "dvportgroup-22",
    "gatewayCidrIpv4": "10.11.10.1/24",
    "internalClusterCidrIpv4": "198.18.0.0/15"
}' | jq -r '.id')
```

4. Monitor the deployment until it completes.

```
curl -k -s -X GET "https://${sddcManagerFqdn}/v1/tasks/${TASK_ID}" \
    --header "Authorization: Bearer ${sddcManagerToken}" \
    --header "Content-Type: application/json" \
    --header "Accept: application/json" \
| jq
```

5. Update SFTP Settings on the new VCF Automation Instance.
   a) Retrieve the SFTP server fingerprint

```
ssh-keyscan -p 22 <sftp_server_ip> | ssh-keygen -lf -
# SSH fingerprints of type RSA and ED25519 are supported
# Enter the full fingerprint in the payload of step 9c. Example:
 SHA256:yA3FHIQMbiMnansI4+lONYBASLP2IQhGZkGJIMZ2Pi8
```

   b) Retrieve a VCF Automation services runtime token

```
export VCFA_SR='flt-vcfa-sr01.rainpole.io'
export VCFA_SR_PASSWORD='VMw@re1!VMw@re1!'

VCFA_SR_TOKEN=$(curl -ks --request POST \
  --url "https://$VCFA_SR/api/v1/identity/token" \
  --header 'Content-Type: application/x-www-form-urlencoded' \
  --data-urlencode grant_type=password \
  --data-urlencode "username=admin@vsp.local" \
  --data-urlencode "password=$VCFA_SR_PASSWORD" \
  | jq -r '.access_token')
```

   c) Apply the SFTP server settings

```
export COMPONENT_ID='765b4936-5fb5-49af-88bc-46ccc0a47a6b' # Should be the clusterId used in step 3

TASK_ID=$(curl -ks -X POST "https://$VCFA_SR/api/v1/components/${COMPONENT_ID}?action=apply" \
  -H "Authorization: Bearer $VCFA_SR_TOKEN" \
  -d  '{
    "spec": {
      "configuration": {
         "backups": {
            "destination": "sftp",
```

```
                "encryptionPassphrase": "**************",
                "storage": {
                  "sftp": {
                    "directory": "/media/backups/",
                    "host": "10.167.173.52",
                    "port": "22",
                    "username": "svc-vcf-bck",
                    "password": "**************",
                    "fingerprint": "SHA256:kxJ7dSHrVZpxcdbZD/s03B4J5UmMHAoW5iz4OxuBRkI"
                  }
                }
              }
            }
          },
          "options": {
          }
        }' | jq -r '.id')
```

d) Monitor the task to apply the SFTP server settings

```
curl -X GET "https://$VCFA_SR/api/v1/tasks/$TASK_ID"   -k -H "Authorization: Bearer $VCFA_SR_TOKEN" |
 jq

# Wait until the response shows "Succeeded" at the end of the task output
],
  "startTime": "2026-04-02T02:52:16Z",
  "status": "Succeeded",
  "type": "com.vmware.vcfms.task.ConfigureComponent"
}
```

Restore VCF Automation for VCF Component Backup and Restore

Restore VCF Automation from a file-based backup.

> **Note:**
>
> It is recommended to use backups taken by using a backup schedule. On-demand backups will not have a corresponding vsp backup. If you have a newer on-demand backup of VCF Automation, use the most recent available vsp backup along with it.

1. Retrieve a VCF Automation services runtime token

```
export VCFA_SR='flt-vcfa-sr01.rainpole.io'
export VCFA_SR_PASSWORD='VMw@re1!VMw@re1!'

VCFA_SR_TOKEN=$(curl -ks --request POST \
  --url "https://$VCFA_SR/api/v1/identity/token" \
  --header 'Content-Type: application/x-www-form-urlencoded' \
  --data-urlencode grant_type=password \
  --data-urlencode "username=admin@vsp.local" \
  --data-urlencode "password=$VCFA_SR_PASSWORD" \
  | jq -r '.access_token')
```

2. Retrieve list of backups for VCF Automation.

```
curl -k "https://$VCFA_SR/api/v1/system/backups" \
  -H "Authorization: Bearer $VCFA_SR_TOKEN" \
  -H 'accept: application/json' | \
  jq -r '
    ["vsp","vcfa","vcd-migrator"] as $order
    | .backups as $all
    | $order[] as $current_type
    | $all[]
```

```
    | select(.component.type == $current_type)
    | (
        (.name | sub("T(?<h>\\d{2})-(?<m>\\d{2})-(?<s>\\d{2})Z"; "T\(.h):\(.m):\(.s)Z") | fromdateiso8601)
      ) as $timestamp
    | ( (now - $timestamp) / 86400 | floor ) as $days_old
    | [ $current_type, .component.version, .name, ($days_old | tostring + " days ago"), .path ]
    | @tsv' | \
    column -t -s $'\t'
```

3. Choose the VCF Automation backup that you wish to restore and its corresponding VSP backup. Normally, this is the most recent backup of each component. The timestamps for the two entries will vary slightly as they are backed up in a sequential manner.,
4. Perform the Restore.

```
TASK_ID=$(curl -k -X POST "https://$VCFA_SR/api/v1/system/backups?action=restore" \
  -H "Authorization: Bearer $VCFA_SR_TOKEN" \
  -H 'accept: application/json' \
  -H "Content-type: application/json" \
  -d '{
  "components": [
    {
      "path": "sftp://svc-vcf-bck@10.167.173.126:22/media/backups/vcf/
backups/1f5c79fe-e3aa-41b1-a5cf-774a6497fa3d/9.1.1.0.25257932/vsp/1f5c79fe-e3aa-41b1-
a5cf-774a6497fa3d/9.1.1.0.25257932/2026-03-23T16-45-31Z",
      "point": "2026-03-23T16-45-31Z"
    },
    {
      "path": "sftp://svc-vcf-bck@10.167.173.126:22/media/backups/vcf/backups/1f5c79fe-
e3aa-41b1-a5cf-774a6497fa3d/9.1.1.0.25257932/vcfa/3d7c01a5-55fb-4b2a-8f0e-
c4d24c6d2c03/9.1.1.0.25262128/2026-03-23T16-54-12Z",
      "point": "2026-03-23T16-54-12Z"
    },
    {
      "path": "sftp://svc-vcf-bck@10.167.173.126:22/media/backups/vcf/backups/1f5c79fe-
e3aa-41b1-a5cf-774a6497fa3d/9.1.1.0.25257932/vcd-migrator/3d7c01a5-55fb-4b2a-8f0e-
c4d24c6d2c03/9.1.1.0.25262128/2026-03-23T16-54-12Z",
      "point": "2026-03-23T16-54-12Z"
    }
  ]
}' | jq -r '.id')
```

5. Monitor the restore task.

```
curl -X GET "https://$VCFA_SR/api/v1/tasks/$TASK_ID" \
  -k -H "Authorization: Bearer $VCFA_SR_TOKEN" | jq
```

Update Connections in VCF Automation for VCF Component Backup and Restore

After you restore VCF Automation as a part of the VCF Instance Backup and Restore operation, to ensure communication between VCF Automation and the VCF Instances it is connected to is active, you refresh the connections.

1. Log in to the VCF Automation provider portal as a user with the **System Administrator** role.
2. Under Administration, click **VCF Instances**.
3. Click the vertical ellipsis next to the first VCF instance, and click **Rotate Service Account**.
4. Repeat for each VCF Instance.
5. From the left menu, click **Connections**.
6. Verify each vCenter connection is enabled and connected.

Backup and Restore VCF Management Services in vSphere Foundation

vSphere Foundation deployments do not include the VCF Operations fleet management UI. Use the fleet lifecycle component API to perform all backup and restore operations for VCF management services components in a vSphere Foundation environment.

The VCF management services fleet lifecycle API provides programmatic access to backup configuration, on-demand and scheduled backup configuration, backup enumeration, and component restoration. All operations authenticate against the VCF Operations appliance and target the fleet lifecycle component discovered at runtime.

Prerequisites

- Verify that the workstation where you plan to run API calls has SSH access to VCF Operations.
- Verify that the workstation where you plan to run API calls has SSH access to the fleet lifecycle component.
- Verify that *curl*, *jq*, and *base64* are available on the workstation where you plan to run API calls.
- Verify that you have administrative access to VCF Operations.

Obtain VCF Management Services Bearer Token and Component IDs in VMware vSphere Foundation

Before configuring backup or performing backup and restore operations, obtain the unique component IDs for the VCF management services components. This procedure also establishes the session variables and bearer token required by all subsequent API calls in this workflow.

Acquire a fleet lifecycle bearer token.

1. Log in to the workstation where you are planning to run API calls.
2. Set the environment variables for the VCF Operations appliance.

```
export OPS_HOST="https://<vcf-operations-fqdn>"
export OPS_USER="admin"
export OPS_PASSWORD='password'
```

3. Encode the credentials as an HTTP Basic authorization value.

```
BASIC=$(printf '%s:%s' "$OPS_USER" "$OPS_PASSWORD" | base64 | tr -d ' \n')
```

4. To locate the fleet lifecycle service entry and record its host FQDN, query the `casa/services` endpoint.

   The `casa/services` endpoint uses HTTP Basic authentication. Locate the entry where `type == VCF_FLEET_LCM` and record both the key and the FQDN from `nodes[0].addresses[0].value`.

   Note:

   The response inlines PEM certificates with raw new lines, which breaks strict JSON parsers. Strip newlines using `tr -d '\n\r'` before passing output to *jq*.

```
curl -k -sS "${OPS_HOST}/casa/services" -H "Authorization: Basic ${BASIC}" | tr -d '\n\r' | jq -r
 '.services[] | select(.type == "VCF_FLEET_LCM") | [.key, .nodes[0].addresses[0].value] | @tsv'
# Example output:
# fleet-lcm    fleet-lcm.sfo.example.com
```

5. Set the `FLEET_HOST` variable from the FQDN.

```
export FLEET_HOST="https://<fleet-lcm-fqdn>"   # e.g. https://fleet-lcm.sfo.example.com
```

6. Set the `FLEET_KEY` variable from the key returned in the previous output.

   The service key identifies the fleet lifecycle service and is required to exchange the VCF Operations token for a fleet lifecycle JSON Web Token (JWT).

```
export FLEET_KEY="<fleet-lcm-key>"   # e.g. fleet-lcm
```

7. Acquire a bearer token from the VCF Operations appliance.

   The VCF Operations token authenticates against the suite-api endpoints on the VCF Operations appliance. This token cannot be used directly as a Bearer token against the fleet lifecycle API.

```
export OPS_TOKEN=$(curl -k -sS -X POST "${OPS_HOST}/suite-api/api/auth/token/acquire?_no_links=true" -H
 "Content-Type: application/json" -d "{\"username\": \"${OPS_USER}\", \"authSource\":\"LOCAL\", \"password
\": \"${OPS_PASSWORD}\"}" | tr -d '\n\r' | jq -r '.token')
```

8. Exchange the VCF Operations token for a fleet lifecycle JWT token.

   The fleet lifecycle API does not accept the VCF Operations token directly. Use the `/suite-api/api/auth/token/exchange` endpoint to exchange the VCF Operations token for a JWT token scoped to the fleet lifecycle service. Pass the VCF Operations token in the Authorization header using the `OpsToken` scheme and include the fleet lifecycle service key in the request body.

```
export FLEET_TOKEN=$(curl -k -sS -X POST "${OPS_HOST}/suite-api/api/auth/token/exchange" \
  -H "accept: application/json" \
  -H "Authorization: OpsToken ${OPS_TOKEN}" \
  -H "Content-Type: application/json" \
  -d "{\"serviceKeys\": [\"${FLEET_KEY}\"]}" | tr -d '\n\r' | jq -r '.jwtToken')
```

9. Define the authorization header array for reuse in subsequent API calls.

```
AUTH=(-H "Authorization: Bearer ${FLEET_TOKEN}" -H "Accept: application/json")
```

   List VCF management services components and obtain their IDs.

10. Query the fleet lifecycle endpoint to list all registered VCF management services components and their IDs.

```
curl -k -sS "${AUTH[@]}" "${FLEET_HOST}/fleet-lcm/v1/components" | tr -d '\n\r' | jq -r '.components[] |
 [.componentType, .id, .fqdn, .version] | @tsv' | column -t
```

    The output lists each component type, its UUID, its FQDN, and its current version. Example output:

```
OPS-LOGS    655d2c8b-2478-4057-80b2-60405cc45d96    ops-logs.sfo.example.com    9.1.0
FLEET-LCM   a1b2c3d4-0000-1111-2222-abcdef012345    fleet-lcm.sfo.example.com   9.1.0
SDDC-LCM    e5f6a7b8-3333-4444-5555-fedcba987654    sddc-lcm.sfo.example.com    9.1.0
```

11. Set the `COMPONENT_ID` variable to the UUID of the component to back up or restore.

```
export COMPONENT_ID="<component-uuid>"   # e.g. 655d2c8b-2478-4057-80b2-60405cc45d96
```

12. Retrieve the `sddcLcmId` for the selected component.

    This value is required to configure backup settings and initiate backup and restore operations.

```
export SDDC_LCM_ID=$(curl -k -sS "${AUTH[@]}" "${FLEET_HOST}/fleet-lcm/v1/components/${COMPONENT_ID}"
| tr -d '\n\r' | jq -r '.sddcLcmId')
```

The session variables `OPS_TOKEN`, `FLEET_TOKEN`, `FLEET_HOST`, `AUTH`, `COMPONENT_ID`, and `SDDC_LCM_ID` are set and ready for use in subsequent backup and restore procedures. The bearer token is valid for the duration of the session. The fleet lifecycle JWT token is valid for the duration of the session. Repeat Step 7 and Step 8 to renew the token if the token expires before the workflow is completed.

Configure the SFTP Backup Location for VCF Management Services in VMware vSphere Foundation

Configure an external SFTP server as the backup target for a VCF management services component. This configuration is set per component using the `SDDC_LCM_ID` obtained in the preceding procedure.

- Verify that you have the IP address or FQDN of the SFTP server.
- Verify that there is SSH access between the VCF management services component and the SFTP server.

1. Log in to the workstation where you are planning to run API calls.
2. Retrieve the SFTP server fingerprint.

   Supported SSH fingerprints are of type `RSA` or `ED25519`.

   ```
   ssh-keyscan -p 22 <sftp_server_ip> | ssh-keygen -lf -
   ```

3. Set the SFTP server credentials as environment variables.

   ```
   export SFTP_PASS='<sftp-password>'
   export SFTP_THUMBPRINT='<sftp-host-key-thumbprint>'
   ```

4. Send a `PATCH` request to update the backup configuration for the component.

   ```
   curl -k -sS -X PATCH "${AUTH[@]}" -H "Content-Type: application/json" "${FLEET_HOST}/fleet-lcm/v1/sddc-lcms/${SDDC_LCM_ID}" -d @- << JSON
   {
     "backupConfigSpec": {
       "storage": {
         "sftp": {
           "host": "<sftp-server-ip-or-fqdn>",
           "port": "22",
           "username": "root",
           "password": "${SFTP_PASS}",
           "directory": "/backup",
           "thumbprint": "${SFTP_THUMBPRINT}"
         }
       }
     }
   }
   JSON
   ```

   The API returns a `200 OK` response. A successful response body reflects the updated `backupConfigSpec` value.

The SFTP backup location is configured for the specified component. Subsequent on-demand and scheduled backup operations for this component write backup data to the configured SFTP server and directory.

Back up a VCF Management Services Component in VMware vSphere Foundation

Initiate a backup of a VCF management services component. Two approaches are available: trigger an immediate on-demand backup or configure a daily scheduled backup with a retention policy. Both approaches use the fleet lifecycle API and write backup data to the SFTP server configured in the preceding procedure.

- Verify that no backup of the same component is currently in progress. Do not submit a new backup request while another backup of the same component is running.
- Verify that the necessary environment variables are exported and you have completed the Obtain VCF Management Services Bearer Token and Component IDs in VMware vSphere Foundation and Configure the SFTP Backup Location for VCF Management Services in VMware vSphere Foundation procedures.

Trigger an immediate on-demand backup.

1. Log in to the workstation where you are planning to run API calls.
2. Submit a `POST` request to initiate an immediate backup of the specified component.

```
curl -k -sS -X POST "${AUTH[@]}" \
  -H "Content-Type: application/json" \
  "${FLEET_HOST}/fleet-lcm/v1/sddc-lcms/${SDDC_LCM_ID}/backups" \
  -d @- << 'JSON'
{"actionType": "ComponentsBackupSpec", "componentIds":["${COMPONENT_ID}"]}
JSON
```

   The response body contains a task object with a `taskId` field. Record this value to monitor the operation.

3. Poll the state of the task.

```
TASK_ID="<task-id-from-previous-response>"
while :; do
  S=$(curl -k -sS "${AUTH[@]}" \
    "${FLEET_HOST}/fleet-lcm/v1/tasks/${TASK_ID}" \
    | tr -d '\n\r' \
    | jq -r '.status')
  echo "$S"
  case "$S" in
    SUCCEEDED|FAILED|CANCELLED) break ;;
  esac
  sleep 20
done
```

   Configure a backup schedule.

4. Send a `PATCH` request to set the backup schedule and retention policy.

   The example configures a daily backup at 02:00 UTC with a maximum retention of seven backups. Adjust `startTime` to a time outside operational hours and `maxBackups` to align with your organization's retention policy. Remove days from the `days` array to exclude specific days of the week.

```
curl -k -sS -X PATCH "${AUTH[@]}" -H "Content-Type: application/json" "${FLEET_HOST}/fleet-lcm/v1/sddc-
lcms/${SDDC_LCM_ID}" -d @- << JSON
{
  "backupConfigSpec": {
    "storage": {
      "sftp": {
        "host": "<sftp-server-ip-or-fqdn>",
        "port": "22",
        "username": "root",
        "password": "${SFTP_PASS}",
        "directory": "/backup",
        "thumbprint": "${SFTP_THUMBPRINT}"
      }
    },
    "fullSchedule": {
      "enabled": true,
      "schedule": {
        "startTime": "02:00:00Z",
        "days": ["MON", "TUE", "WED", "THU", "FRI", "SAT", "SUN"]
      }
    },
    "retention": {
      "maxBackups": 7
    }
  }
}
```

```
JSON
```

5. Submit a task to create the backup schedule.

```
TASK_ID=$(curl -k -s -X POST "${AUTH[@]}" \
  -H "Content-Type: application/json" \
  "${FLEET_HOST}/fleet-lcm/v1/sddc-lcms/${SDDC_LCM_ID}?action=apply" \
  -d '{ "spec": { "configuration": { "backups": { "full": { "enable": true, "schedule": "0 2 * * 0" },
 "incremental": { "enable": true, "schedule": "0 2 * * 1-6" } } } }, "options": {} }' \
  | grep -o '"id":[^,]*' | grep -o '[^"]*$')
# Print the ID to verify it was captured
echo "Submitted Task ID: $TASK_ID"
```

The API returns a `200 OK` response. The response body reflects the updated schedule and retention values.

Restore VCF Management Services Components from Backup in VMware vSphere Foundation

Restore a VCF management services component from a previously created backup stored on the configured SFTP server. The restore operation is disruptive: the target component is taken offline and its data is replaced with the backup data. Plan and schedule this procedure during a maintenance window.

- Verify that the necessary environment variables are exported and you have completed the Obtain VCF Management Services Bearer Token and Component IDs in VMware vSphere Foundation and Configure the SFTP Backup Location for VCF Management Services in VMware vSphere Foundation procedures.

The restore operation brings the target component down and replaces all component data with the selected backup. This operation cannot be undone. Ensure that the correct backup path is specified before submitting the restore request.

1. Log in to the workstation where you are planning to run API calls.
2. Query the fleet lifecycle backup endpoint to list all backups available for the specified component.

```
curl -k -sS "${AUTH[@]}" "${FLEET_HOST}/fleet-lcm/v1/sddc-lcms/${SDDC_LCM_ID}/backups?componentId=
${COMPONENT_ID}" | tr -d '\n\r' | jq '.backups[] | {name, path, componentType, componentVersion, points}'
```

   The response lists each backup with its `name`, `path`, `componentType`, `componentVersion`, and the points array. Each entry in points corresponds to a point-in-time backup and includes the SFTP path required for restore. An example path has the following format:

```
sftp://root@<sftp-host>:22/backup/vcf/backups/<component-type>/<component-id>/<timestamp>
```

3. Export the full SFTP path of the backup to restore.

```
export BACKUP_PATH="sftp://root@<sftp-host>:22/backup/<path-to-backup>"
```

4. Submit a POST request to initiate the restore operation for the selected backup.

   Replace `<component-type>` with the `componentType` value returned in the backup list (for example, `OPS-LOGS`). Set `encryptionPassphrase` to the passphrase used when the backup was created, or leave it as an empty string if no passphrase was configured.

```
curl -k -sS -X POST "${AUTH[@]}" -H "Content-Type: application/json" "${FLEET_HOST}/fleet-lcm/v1/sddc-
lcms/${SDDC_LCM_ID}/backups" -d @- << JSON
{
  "actionType": "ComponentsRestoreSpec",
  "encryptionPassphrase": "",
  "components": [
    {
      "componentId": "${COMPONENT_ID}",
      "componentType": "<component-type>",
      "path": "${BACKUP_PATH}"
    }
  ]
}
JSON
```

The response body contains a task object with a `taskId` field. Record this value to monitor the operation.

5. Poll the state of the task.

```
TASK_ID="<task-id-from-previous-response>"
while :; do
  S=$(curl -k -sS "${AUTH[@]}" "${FLEET_HOST}/fleet-lcm/v1/tasks/${TASK_ID}" | tr -d '\n\r' | jq -r
 '.status')
  echo "$S"
  case "$S" in
    SUCCEEDED|FAILED|CANCELLED) break ;;
  esac
  sleep 20
done
```

The VCF management services component is restored from the selected backup. The component is restarted automatically as part of the restore process. Verify that the component is operational and that its services are accessible before resuming normal operations.

VCF Instance Backup and Restore

VCF Instance Backup and Restore provides guidance on restoring a VCF Instance by performing a complete reconstruction from a backup.

This guidance contains detailed instructions for both the management domain and workload domains where you must restore all components.

Example Failure Scenarios

You may need to restore all components in a VCF Instance if one of the following scenarios occurs:

- Complete VCF Instance failure
- Malware or ransomware attack
- Datastore corruption

Supported Topologies

Several topologies of VCF exist according to the number of availability zones and VCF Instances.

The following topologies are supported for restore using this guidance.

Supported Topology	Description
• Single VCF fleet • VCF domains with a single or multiple vSphere clusters • vSphere clusters with a single or multiple vSphere Distributed Switches • vSphere cluster models: — Single-Rack vSphere Cluster Model — Multi-Rack Layer 2 vSphere Cluster Model — Stretched vSphere Cluster Model • vSAN ESA and vSAN OSA	

VCF Components to Back Up

All important data in a VMware Cloud Foundation platform should be backed up and that backup data should survive the loss of any given site. Include all VMs in the management domain and VMs in workload domains that require data protection.

After initial deployment, the management domain contains a core set of VMs to manage the VMware Cloud Foundation platform. When you deploy a workload domain, VCF deploys additional VMs to manage the workload domain.

Note:

If you have other VMs in the management domain that require backup, for example, Microsoft SQL servers, Microsoft Active Directory servers, backup software VMs, and so on, you should plan to back them up too.

This guidance does not provide information about backing up VMs in workload domains, but your backup plan should also identify and back up critical VMs in workload domains.

Important:

- In order to capture all the required component data you must use scheduled backups for VCF Management Services and VCF Automation.
- From VMware Cloud Foundataion 9.1.1 and later, the SDDC Manager credentials database is no longer part of the SDDC Manager backup. You must use the SDDC Manager credentials API to regularly backup the credentials database and store it in a secure location. See On-Demand Backup of SDDC Manager Credential Store

Documenting the System Configuration of VCF

Keeping detailed as-built documentation on the system configuration eases the restore process if a failure in your VMware Cloud Foundation platform occurs.

Recording and subsequently providing the following information is mandatory when recovering a VCF Instance using the manual processes included in this guidance. while the automated recovery process recovers much of the following from the SDDC Manager backup, it is still a good idea to have an up-to-date record of the information. Save this information on a secure secondary storage.

- Topology diagrams of the VMware Cloud Foundation platform:
  - Physical networking
  - vSphere Distributed Switch networking
  - NSX networking
  - VCF domain configuration
  - Cluster configuration for each cluster in a VCF domain
  - ESX hosts assigned to each cluster
  - Custom VIBs installed on ESX hosts
- Networking information:
  - ESX host vmnic-to-switch port mappings
  - VM virtual NIC to distributed port group mappings
  - IP address information of the VMkernel interfaces on the ESX hosts
  - IP address information of VMs
  - DNS, NTP, AD, and other well-known services used by the VMware Cloud Foundation platform
- vCenter instance configuration — for each vCenter instance:
  - Fully qualified domain name (FQDN), and the IP address, prefix length, and default gateway for every configured network adapter
  - DNS servers and **DNS search domains**
  - **Static routes**, if any are configured
  - Number of virtual network adapters attached to the appliance
  - vCenter version and build number, from the **Summary** tab of the vCenter Management Interface
  - Deployment size and storage size
  - Single Sign-On domain name
  - Contents of **/root/.ssh/authorized_keys**, and any custom scripts stored on the appliance that are referenced by alarm actions or cron jobs

PowerShell Automation for This Guidance

You can perform the procedures in this guide manually or by running cmdlets in an associated PowerShell module.

The PowerShell cmdlets are available in an open-source module as code-based alternatives to completing certain procedures in each SDDC component's user interface. For more information on the PowerShell cmdlets for restore of VMware Cloud Foundation, see the VMware.CloudFoundation.InstanceRecovery open-source project in GitHub.

Table 1164: Prerequisites for CLI Implementation of VCF Instance Backup and Restore

Prerequisite

• Microsoft Windows 2022 (minimum)
• Microsoft PowerShell 7.5.x (minimum)
• VMware OVF Tool
• OpenSSL lite version 1.0.2g or higher installed and added to the Windows `PATH` system variable.
• PowerCLI **DefaultVIServerMode** setting must be configured as **Multiple** for the all **AllUsers** scope
• Install the supporting modules from the PowerShell Gallery by running the following commands:
`Install-Module -Name VCF.PowerCLI -MinimumVersion 9.1.0 -Scope AllUsers -Force`
`Install-Module -Name Posh-SSH -RequiredVersion 3.0.8 -Scope AllUsers -Force`
`Install-Module -Name VMware.CloudFoundation.InstanceRecovery -Scope AllUsers -Force`
• Open a new PowerShell console and run the following command to confirm all the prerequisites are installed.
`Confirm-VCFInstanceRecoveryPreReqs`

Sequence Optimization for Restore of a VCF Instance

When restoring an entire VCF Instance, you can run certain tasks in parallel to minimize the time that is required to restore it.

Example Restore Timeline for a VCF Instance

To minimize the overall restore time of a VCF instance, you can run multiple restore plans in parallel across VCF domains with the following guidelines:

1. If you do not use all of the components shown, then you can ignore the associated steps and proceed to the next step in the respective restore plan.
2. If you want to prioritize workload availability over management and operational components, you can do so.
3. You should not reorder the overall sequence of any given restore plan because that might compromise dependencies.

> **Important:**
>
> Where multiple items exist in a single cell, they may be run in parallel. Items across a single row may also be run in parallel.

The following timeline is for a sample setup with the following configuration:

- Two VCF domains.
  - One management domain (which is also running fleet components for the entire fleet).
  - Fleet components are not on an NSX overlay network.
  - One workload domain with an isolated vCenter Single Sign-On (SSO) domain.
- VCF 9.1.0 and 9.1.1 only.

> **Attention:**
>
> * If fleet components are on an NSX overlay network, then do not start the fleet component restore plan until the appropriate NSX Edge nodes are redeployed.

** Should there be more than one isolated workload domain, you can start the restore plan for all of them in parallel.

Task	Management Domain Restore Plan	Fleet Component Restore Plan	Workload Domain Restore Plan	Notes
1	Gather Information for VCF Instance Backup and Restore		Gather Information for VCF Instance Backup and Restore	
2	Deploy ESX Hosts		Deploy ESX Hosts	ESX hosts should be deployed with their original FQDNs and IP addresses.
3	Prepare Networking on Management ESX Hosts			
4	Add VMkernel Adapters to Management ESX Hosts			
5	Create a Single-Host vSAN Datastore			
6	• Deploy a vCenter Appliance from an OVA File • Deploy the First NSX Manager Cluster Node from an OVA File • Deploy the SDDC Manager Applinace from an OVA File		• Deploy a vCenter Appliance from an OVA File • Deploy the First NSX Manager Cluster Node from an OVA File	
7	• Restore vCenter from a File-Based Backup • Restore NSX Manager from a File-Based		• Restore vCenter from a File-Based Backup • Restore NSX Manager from a File-Based	
8	Verify the State of the First NSX Manager Cluster Node			
9	Update vCenter SSH Keys in the SDDC Manager Backup			
10	Restore SDDC Manager from a File-Based Backup			
11	Update Extracted SDDC Data **Note:** This task is only required when using the PowerShell Automation approach. If performing a fully manual restore, this information is gathered in the Documenting the System Configuration of VCF step.		Update Extracted SDDC Data **Note:** This task is only required when using the PowerShell Automation approach. If performing a fully manual restore, this information is gathered in the Documenting the System Configuration of VCF step.	
12	Rebuild Management vSphere Cluster (mulitple tasks)			

Task	Management Domain Restore Plan	Fleet Component Restore Plan	Workload Domain Restore Plan	Notes
13	Remediate the NSX Installation on the vSphere Cluster	Restore VCF Operations from an Image-Based Backup [1]	Rebuild the vSphere Cluster (multiple steps) [2]	
14	[Optional] Redeploy remaining nodes of the NSX Manager Cluster [3]	Prepare to Restore VCF Management Services	Remediate NSX Installation on the vSphere Cluster	You can start restoring workloads that *do not* use NSX Edge nodes into your workload domain.
15	Redeploy NSX Edge Nodes	Redeploy VCF Management Services	[Optional] Redeploy remaining nodes of the NSX Manager cluster [3]	
16	[Conditional] Redeploy remaining nodes of the NSX Manager cluster [4]	Retrieve a List of Backups for VCF Management Services Components	Redeploy an NSX Edge Cluster	
17		• RestoreVCF Management ServicesComponents • Redeploy a License Server	[Conditional] Redeploy remaining nodes of the NSX Manager cluster [4]	You can start restoring workloads that *do* use NSX Edge nodes into your workload domain.
18		Retrieve a List of Backups for Log Management		
19		Restore Log Management		
20		Recover the Remaining VCF Management Services Components		
21		Update Log Management State		
22		Prepare to Restore VCF Automation		
23		Redeploy the VCF Automation Services Runtime Cluster		
24		Retrieve a List of Backups for VCF Automation		
25		Prepare to Restore VCF Automation		
26		Update Connections in VCF Automation		
27		Install the Migration Service Engine Component		
28	Finalize Management Domain [5]		Finalize Workload Domain [5]	

[1] Dependent on the restore of the primary vSphere cluster in the management domain.

[2] Dependent on the restore of SDDC Manager being complete.

[3] If high availability of the NSX Manager nodes is considered less important than time to make workloads available, you can perform this task later.

[4] Perform this task if you skipped the deployment of the additional NSX Manager nodes earlier.

[5] Restore vSphere DRS, VM Location, VM Overrides, and VM Tags only after all workload VMs are restored to the cluster.

Post-Restore Cleanup Plan for VMware Cloud Foundation

- Delete the VM snapshots.

Preparing for Restore of the VCF Instance

Before you can perform a restore, you must perform some preparation tasks.

Table 1165: Preparation Tasks for VCF Restore

Task	Task Name	Additional Information
1	Locate the most recent known good file-based backups for the following components: • SDDC Manager • All vCenter instances • All NSX Manager instances • VCF Automation • VCF Management Services	**Note:** Ensure all backups are from the same backup window.
2	Locate the most recent SDDC Manager Credentials API backup	See On-Demand Backup of SDDC Manager Credential Store
3	Locate the most recent known good image-based backups for the following components: • VCF Operations • VCF Operations for Networks	
4	Gather Information for VCF Instance Backup and Restore	
5	Locate version and build number for vCenter, NSX Manager and SDDC Manager along with other relevant deployment details to ensure redeployment exactly as per failed environment: • IP address • Network information • Storage and appliance sizes	• You can see the vCenter build number and deployment size in the `backup-metadata.json` in the vCenter file-based backup. • You can see the NSX Manager version and build number from the file name of the backup file.
6	Download the OVA files of vCenter, SDDC Manager, and NSX Manager for the required version and build number. If the VCF Installer used to deploy the orginal environment is no longer available, then ensure you have local copies of the binaries used to deploy VCF, or an offline depot available that hosts these binaries.	
7	Download the Refresh SSH Key Script from Broadcom knowledge base article 313210.	The script is used to update SSH keys for recovered components that are stored in SDDC Manager.
8	Re-install ESX on the hosts with exactly the same personalities and network addresses as they had when the backup of the environment was taken.	See Deploying a New VCF or vSphere Foundation Platform.

Management Domain Restore Plan

Perform the following tasks to restore the components common to every VCF management domain.

For information about where tasks can be performed in parallel across VCF domains, see Sequence Optimization for Restore of a VCF Instance.

Management Domain Restore Plan Summary

- Gathering Required Information
- Restore Management vCenter, NSX Manager, and SDDC Manager
- Rebuild Management vSphere Cluster
- Restore Management NSX Edge Nodes
- Redeploy the Remaining Management NSX Manager Nodes
- Finalize the Restore of the Management Domain

Gathering Required Information

Step	Manual Task	PowerShell Equivalent
1	Gather Information for VCF Instance Backup and Restore **Note:** The *credentialsFilePath* is required from VCF 9.1.1 onwards. Older versions should be run without *credentialsFilePath*	```## Parameter Block ## $vcfBackupFilePath = ".\vcf-backup-sfo-vcf01-sfo-rainpole-io-2025-09-22-10-30-55.tar.gz" $encryptionPassword = "**********" $credentialsFilePath = "vcf-credentials-sfo-m01.json" ## Command Block ## New-ExtractDataFromSDDCBackup ` -vcfBackupFilePath $vcfBackupFilePath ` -encryptionPassword $encryptionPassword ` -credentialsFilePath $credentialsFilePath```

Restore Management vCenter, NSX Manager, and SDDC Manager

Step	Manual Task	PowerShell Equivalent
1	Prepare Networking on Management ESX Hosts for VCF Instance Backup and Restore	```## Parameter Block ## $extractedSDDCDataFile = ".\extracted-sddc-data.json" ## Command Block ## New-PrepareManagementHostNetworking ` -extractedSDDCDataFile $extractedSDDCDataFile```
2	Add VMkernel Adapters to Management ESX Hosts for VCF Instance Backup and Restore	```## Parameter Block ## $extractedSDDCDataFile = ".\extracted-sddc-data.json" ## Command Block ## Add-VMKernelsToManagementHosts ` -extractedSDDCDataFile $extractedSDDCDataFile```
3	Create a Single-Host vSAN Datastore for VCF Instance Backup and Restore	```## Parameter Block ## $extractedSDDCDataFile = ".\extracted-sddc-data.json" ## Command Block ## New-SingleHostVsanDatastore ` -extractedSDDCDataFile $extractedSDDCDataFile```
4	Deploy a vCenter Appliance from an OVA File for VCF Instance Backup and Restore	```## Parameter Block ## $targetType = "esx" $targetFqdn = "sfo01-m01-r01-esx01.sfo.rainpole.io" $targetAdmin = "root" $targetAdminPassword = "*********" $extractedSDDCDataFile = ".\extracted-sddc-data.json" $workloadDomain = "sfo-m01" $restoredvCenterDeploymentSize = "small-lstorage" $vCenterOvaFile = ".\VMware-vCenter-Server-Appliance-9.1.1.0.25712839_OVF10.ova" ## Command Block ## New-vCenterOvaDeployment ` -targetType $targetType ` -targetFqdn $targetFqdn ` -targetAdmin $targetAdmin ` -targetAdminPassword $targetAdminPassword ` -extractedSDDCDataFile $extractedSDDCDataFile ` -workloadDomain $workloadDomain ` -restoredvCenterDeploymentSize $restoredvCenterDeploymentSize ` -vCenterOvaFile $vCenterOvaFile```

Step	Manual Task	PowerShell Equivalent
5	Deploy the First NSX Manager Cluster Node from an OVA File for VCF Instance Backup and Restore	```## Parameter Block ## $targetType = "esx" $targetFqdn = "sfo01-m01-r01-esx01.sfo.rainpole.io" $targetAdmin = "root" $targetAdminPassword = "**********" $extractedSDDCDataFile = ".\extracted-sddc-data.json" $workloadDomain = "sfo-m01" $restoredNsxManagerDeploymentSize = "medium" $nsxManagerOvaFile = ".\nsx-unified-appliance-9.1.1.0.25691509.ova" ## Command Block ## New-NSXManagerOvaDeployment ` -targetType $targetType ` -targetFqdn $targetFqdn ` -targetAdmin $targetAdmin ` -targetAdminPassword $targetAdminPassword ` -extractedSDDCDataFile $extractedSDDCDataFile ` -workloadDomain $workloadDomain ` restoredNsxManagerDeploymentSize $restoredNsxManagerDeploymentSize ` -nsxManagerOvaFile $nsxManagerOvaFile```
6	Deploy the SDDC Manager Appliance from an OVA File for VCF Instance Backup and Restore	```## Parameter Block ## $targetType = "esx" $targetFqdn = "sfo01-m01-r01-esx01.sfo.rainpole.io" $targetAdmin = "root" $targetAdminPassword = "*********" $extractedSDDCDataFile = ".\extracted-sddc-data.json" $sddcManagerOvaFile = ".\VCF-SDDC-Manager- Appliance-9.1.1.0.25713928.ova" $rootUserPassword = "*********" $vcfUserPassword = "*********" $localUserPassword = "*********" $basicAuthUserPassword = "*********" ## Command Block ## New-SDDCManagerOvaDeployment ` -targetType $targetType ` -targetFqdn $targetFqdn ` -targetAdmin $targetAdmin ` -targetAdminPassword $targetAdminPassword ` -extractedSDDCDataFile $extractedSDDCDataFile ` -sddcManagerOvaFile $sddcManagerOvaFile ` -rootUserPassword $rootUserPassword ` -vcfUserPassword $vcfUserPassword ` -localUserPassword $localUserPassword ` -basicAuthUserPassword $basicAuthUserPassword```

Step	Manual Task	PowerShell Equivalent
7	Restore vCenter from a File-Based Backup for VCF Instance Backup and Restore	```## Parameter Block ## $extractedSDDCDataFile = ".\extracted-sddc-data.json" $workloadDomain = "sfo-m01" $vCenterBackupPath = "192.168.1.1/F$/ backups/sn_sfo-m01-vc01.sfo.rainpole.io/ M_9.0.2.0_20260406-084633_KBWGC5DGN5ZG2VDPN5WHGICCMFRWW5LQ" $locationtype = "SMB" $locationUser = "Administrator" $locationPassword = "*********" $vCenterBackupPassword = "*********" ## Command Block ## Invoke-vCenterRestore ` -extractedSDDCDataFile $extractedSDDCDataFile ` -workloadDomain $workloadDomain ` -vCenterBackupPath $vCenterBackupPath ` -locationtype $locationtype ` -locationUser $locationUser ` -locationPassword $locationPassword``` **Important:** If the vCenter backup was taken using a backup password, then append the following option to the command above. ```-backupPassword $vCenterBackupPassword```
8	Restore NSX Manager from a File-Based Backup for VCF Instance Backup and Restore	```## Parameter Block ## $extractedSDDCDataFile = ".\extracted-sddc-data.json" $workloadDomain = "sfo-m01" $sftpServer = "192.168.1.2" $sftpUser = "svc-vcf-bck" $sftpPassword = "*********" $sftpServerBackupPath = "/media/backups/sfo-m01- nsx01.sfo.rainpole.io" # FQDN of the NSX Manager cluster VIP $nsxBackupPassphrase = "*********"``` **Important:** For VCF 9.1.0 do not append the FQDN of the NSX Manager cluster VIP on the `$sftpServerBackupPath` variable. ```## Command Block ## Invoke-NSXManagerRestore ` -extractedSDDCDataFile $extractedSDDCDataFile ` -workloadDomain $workloadDomain ` -sftpServer $sftpServer ` -sftpUser $sftpUser ` -sftpPassword $sftpPassword ` -sftpServerBackupPath $sftpServerBackupPath ` -backupPassphrase $nsxBackupPassphrase```
9	Verify the State of the First NSX Manager Cluster Node for VCF Instance Backup and Restore	No PowerShell equivalent, follow the manual procedure.

Step	Manual Task	PowerShell Equivalent
10	Update vCenter SSH Keys in the SDDC Manager Backup for VCF Instance Backup and Restore	```## Parameter Block ## $targetType = "esx" $targetFqdn = "sfo01-m01-r01-esx01.sfo.rainpole.io" $targetAdmin = "root" $targetAdminPassword = "*********" $rootUserPassword = "*********" $vcfUserPassword = "*********" $vcfbackupFilePath = ".\vcf-backup-sfo-vcf01-sfo-rainpole-io-xxxx-xx-xx-xx-xx-xx.tar.gz" $encryptionPassword = "*********" $extractedSDDCDataFile = ".\extracted-sddc-data.json" ## Command Block ## New-UploadAndModifySDDCManagerBackup ` -targetType $targetType ` -targetFqdn $targetFqdn ` -targetAdmin $targetAdmin ` -targetAdminPassword $targetAdminPassword ` -rootUserPassword $rootUserPassword ` -vcfUserPassword $vcfUserPassword ` -backupFilePath $vcfbackupFilePath ` -encryptionPassword $encryptionPassword ` -extractedSDDCDataFile $extractedSDDCDataFile` ```
11	Restore SDDC Manager from a File-Based Backup for VCF Instance Backup and Restore	```## Parameter Block ## $extractedSDDCDataFile = ".\extracted-sddc-data.json" $vcfBackupFilePath = ".\vcf-backup-sfo-vcf01-sfo-rainpole-io-2023-11-21-10-42-38.tar.gz" $vcfUserPassword = "*********" $localUserPassword = "*********" $rootUserPassword = "*********" $encryptionPassword = "*********" ## Command Block ## Invoke-SDDCManagerRestore ` -extractedSDDCDataFile $extractedSDDCDataFile ` -backupFilePath $vcfBackupFilePath ` -vcfUserPassword $vcfUserPassword ` -localUserPassword $localUserPassword ` -rootUserPassword $rootUserPassword ` -encryptionPassword $encryptionPassword ```
12	Update Extracted SDDC Data for Instance Backup and Restore **Note:** This task is only required when using the PowerShell Automation approach. If performing a fully manual restore, this information is gathered in the Documenting the System Configuration of VCF step.	```## Parameter Block ## $extractedSDDCDataFile = ".\extracted-sddc-data.json" $sddcManagerFQDN = "sfo-vcf01.sfo.rainpole.io" $sddcManagerAdmin = "administrator@vsphere.local" $sddcManagerAdminPassword = "*********" $vCenterFqdn = "sfo-m01-vc01.sfo.rainpole.io" #fqdn of restored vCenter ## Command Block ## Update-ExtractedSDDCData ` -extractedSDDCDataFile $extractedSDDCDataFile ` -sddcManagerFQDN $sddcManagerFQDN ` -sddcManagerAdmin $sddcManagerAdmin ` -sddcManagerAdminPassword $sddcManagerAdminPassword ` -vCenterFQDN $vCenterFqdn ```

Rebuild Management vSphere Cluster

Step	Manual Task	PowerShell Equivalent
1	Record the Cluster Settings from the Restored vCenter for VCF Instance Backup and Restore	```## Parameter Block ## $vCenterFqdn = "sfo-m01-vc01.sfo.rainpole.io" #fqdn of restored vCenter $vCenterAdmin = "administrator@vsphere.local" $vCenterAdminPassword = "*********" $clusterName = "sfo-m01-cl01" ## Command Block ## Backup-ClusterVMOverrides ` -vCenterFQDN $vCenterFqdn ` -vCenterAdmin $vCenterAdmin ` -vCenterAdminPassword $vCenterAdminPassword ` -clusterName $clusterName Backup-ClusterVMLocations ` -vCenterFQDN $vCenterFqdn ` -vCenterAdmin $vCenterAdmin ` -vCenterAdminPassword $vCenterAdminPassword ` -clusterName $clusterName Backup-ClusterDRSGroupsAndRules ` -vCenterFQDN $vCenterFqdn ` -vCenterAdmin $vCenterAdmin ` -vCenterAdminPassword $vCenterAdminPassword ` -clusterName $clusterName Backup-ClusterVMTags ` -vCenterFQDN $vCenterFqdn ` -vCenterAdmin $vCenterAdmin ` -vCenterAdminPassword $vCenterAdminPassword ` -clusterName $clusterName```
2	Remove Non-Responsive ESX Hosts from the Inventory for VCF Instance Backup and Restore	```## Parameter Block ## $vCenterFQDN = "sfo-m01-vc01.sfo.rainpole.io" #fqdn of restored vCenter $vCenterAdmin = "administrator@vsphere.local" $vCenterAdminPassword = "*********" $clusterName = "sfo-m01-cl01" $nsxManagerFQDN = "sfo-m01-nsx01a.sfo.rainpole.io" #fqdn of restored NSX Manager node $nsxManagerAdmin = "admin" $nsxManagerAdminPassword = "*********" $nsxManagerRootPassword = "*********" ## Command Block ## Remove-NonResponsiveHosts ` -vCenterFQDN $vCenterFQDN ` -vCenterAdmin $vCenterAdmin ` -vCenterAdminPassword $vCenterAdminPassword ` -clusterName $clusterName ` -NsxManagerFQDN $nsxManagerFQDN ` -NsxManagerAdmin $nsxManagerAdmin ` -NsxManagerAdminPassword $nsxManagerAdminPassword ` -NsxManagerRootPassword $nsxManagerRootPassword```

Step	Manual Task	PowerShell Equivalent
3	Add New Hosts to the Cluster to the vCenter Instance for the Restored VCF Domain	```## Parameter Block ## $vCenterFQDN = "sfo-m01-vc01.sfo.rainpole.io" #fqdn of restored vCenter $vCenterAdmin = "administrator@vsphere.local" $vCenterAdminPassword = "*********" $clusterName = "sfo-m01-cl01" $sddcManagerFqdn = "sfo-vcf01.sfo.rainpole.io" $sddcManagerAdmin = "administrator@vsphere.local" $sddcManagerAdminPassword = "*********" $extractedSDDCDataFile = ".\extracted-sddc-data.json" ## Command Block ## Add-HostsToCluster ` -vCenterFQDN $vCenterFQDN ` -vCenterAdmin $vCenterAdmin ` -vCenterAdminPassword $vCenterAdminPassword ` -clusterName $clusterName ` -sddcManagerFqdn $sddcManagerFqdn ` -sddcManagerAdmin $sddcManagerAdmin ` -sddcManagerAdminPassword $sddcManagerAdminPassword ` -extractedSDDCDataFile $extractedSDDCDataFile```
4	Migrate the ESX Hosts and VMkernel Adapters to the vSphere Distributed Switch on the Restored vCenter for VCF Instance Backup and Restore	```## Parameter Block ## $vCenterFQDN = "sfo-m01-vc01.sfo.rainpole.io" #fqdn of restored vCenter $vCenterAdmin = "administrator@vsphere.local" $vCenterAdminPassword = "*********" $clusterName = "sfo-m01-cl01" $extractedSDDCDataFile = ".\extracted-sddc-data.json" ## Command Block ## New-RebuiltVdsConfiguration ` -vCenterFQDN $vCenterFQDN ` -vCenterAdmin $vCenterAdmin ` -vCenterAdminPassword $vCenterAdminPassword ` -clusterName $clusterName ` -extractedSDDCDataFile $extractedSDDCDataFile```
5	Wait for NSX Transport Self-Healing to Finish	```## Parameter Block ## $clusterName = "sfo-m01-cl01" $extractedSDDCDataFile = ".\extracted-sddc-data.json" ## Command Block ## Watch-NsxHostTransportNodeInstallation ` -clusterName $clusterName ` -extractedSDDCDataFile $extractedSDDCDataFile```

Step	Manual Task	PowerShell Equivalent
6	Add Disk Groups to Management Hosts for VCF Instance Backup and Restore **Important:** Perform this step only if the management cluster uses vSAN OSA.	```## Parameter Block ## $vCenterFQDN = "sfo-m01-vc01.sfo.rainpole.io" #fqdn of restored vCenter $vCenterAdmin = "administrator@vsphere.local" $vCenterAdminPassword = "*********" $clusterName = "sfo-m01-cl01" $extractedSDDCDataFile = ".\extracted-sddc-data.json" ## Command Block ## Add-DiskgroupsToManagementHosts ` -targetFqdn $vCenterFQDN ` -targetAdmin $vCenterAdmin ` -targetAdminPassword $vCenterAdminPassword ` -clusterName $clusterName ` -extractedSDDCDataFile $extractedSDDCDataFile```
7	Reconfigure vSAN Stretched Cluster **Important:** Perform this step only if the cluster was originally a stretched vSAN cluster	```## Parameter Block ## $vCenterFQDN = "sfo-m01-vc01.sfo.rainpole.io" #fqdn of restored vCenter $vCenterAdmin = "administrator@vsphere.local" $vCenterAdminPassword = "*********" $clusterName = "sfo-m01-cl01" $extractedSDDCDataFile = ".\extracted-sddc-data.json" New-ReconfiguredVsanStretchedCluster ` -vCenterFQDN $vCenterFQDN ` -vCenterAdmin $vCenterAdmin ` -vCenterAdminPassword $vCenterAdminPassword ` -clusterName $clusterName ` -extractedSDDCDataFile $extractedSDDCDataFile```
8	Set the Management Datastore Policy on the Rebuilt Datastore for VCF Instance Backup and Restore	```## Parameter Block ## $vCenterFQDN = "sfo-m01-vc01.sfo.rainpole.io" #fqdn of restored vCenter $vCenterAdmin = "administrator@vsphere.local" $vCenterAdminPassword = "*********" $clusterName = "sfo-m01-cl01" $extractedSDDCDataFile = ".\extracted-sddc-data.json" ## Command Block ## Set-ManagementDatastorePolicy ` -vCenterFQDN $vCenterFQDN ` -vCenterAdmin $vCenterAdmin ` -vCenterAdminPassword $vCenterAdminPassword ` -clusterName $clusterName ` -extractedSDDCDataFile $extractedSDDCDataFile```
9	Clear vCenter Alarms **Attention:** This step is optional, and may be executed if you want to clear all alarms in the recovered vCenter. You may also do this manually using the vCenter UI.	```## Parameter Block ## $vCenterFQDN = "sfo-m01-vc01.sfo.rainpole.io" #fqdn of restored vCenter $vCenterAdmin = "administrator@vsphere.local" $vCenterAdminPassword = "***************" ## Command Block ## Clear-vCenterAlarms ` -vCenterFQDN $vCenterFQDN ` -vCenterAdmin $vCenterAdmin ` -vCenterAdminPassword $vCenterAdminPassword```

Step	Manual Task	PowerShell Equivalent
10	Update the Datastore ID for Restored Cluster in SDDC Manager Inventory for VCF Instance Backup and Restore	```## Parameter Block ## $extractedSDDCDataFile = ".\extracted-sddc-data.json" $vCenterFqdn = "sfo-m01-vc01.sfo.rainpole.io" #fqdn of restored vCenter $clusterName = "sfo-m01-cl01" #name of cluster being recovered $vcfUserPassword = "*********" $rootPassword = "*********" ## Command Block ## Update-DomainDatastoreID ` -extractedSDDCDataFile $extractedSDDCDataFile ` -vCenterFqdn $vCenterFqdn ` -clusterName $clusterName ` -VcfUserPassword $vcfUserPassword ` -RootPassword $rootPassword```
11	Update the Cluster Host IDs for Restored Cluster in SDDC Manager Inventory for VCF Instance Backup and Restore	```## Parameter Block ## $extractedSDDCDataFile = ".\extracted-sddc-data.json" $vCenterFqdn = "sfo-m01-vc01.sfo.rainpole.io" #fqdn of restored vCenter $clusterName = "sfo-m01-cl01" #name of cluster being recovered $vcfUserPassword = "*********" $rootPassword = "*********" ## Command Block ## Update-ClusterHostSourceIDs ` -extractedSDDCDataFile $extractedSDDCDataFile ` -vCenterFqdn $vCenterFqdn ` -clusterName $clusterName ` -VcfUserPassword $vcfUserPassword ` -RootPassword $rootPassword```
12	Update ESX Host SSH Keys for VCF Instance Backup and Restore	```## Parameter Block ## $extractedSDDCDataFile = ".\extracted-sddc-data.json" $vCenterFQDN = "sfo-m01-vc01.sfo.rainpole.io" $vCenterAdmin = "administrator@vsphere.local" $vCenterAdminPassword = "*********" $clusterName = "sfo-m01-cl01" $workloadDomain = "sfo-m01" $vcfUserPassword = "*********" ## Command Block ## Invoke-SddcManagerSSHKeyRefresh ` -vCenterFQDN $vCenterFQDN ` -vCenterAdmin $vCenterAdmin ` -vCenterAdminPassword $vCenterAdminPassword ` -extractedSDDCDataFile $extractedSDDCDataFile ` -clusterName $clusterName ` -workloadDomain $workloadDomain ` -VcfUserPassword $vcfUserPassword```

Step	Manual Task	PowerShell Equivalent
13	Synchronize the SDDC Manager Service Account on the ESX Hosts for VCF Instance Backup and Restore	```## Parameter Block ## $vCenterFqdn = "sfo-m01-vc01.sfo.rainpole.io" #fqdn of restored vCenter $vCenterAdmin = "administrator@vsphere.local" $vCenterAdminPassword = "*********" $clusterName = "sfo-m01-cl01" $svcAccountPassword = "*********" $sddcManagerFqdn = "sfo-vcf01.sfo.rainpole.io" $sddcManagerAdmin = "administrator@vsphere.local" $sddcManagerAdminPassword = "*********" ## Command Block ## Resolve-PhysicalHostServiceAccounts ` -targetFQDN $vCenterFqdn ` -targetAdmin $vCenterAdmin ` -targetAdminPassword $vCenterAdminPassword ` -clusterName $clusterName ` -svcAccountPassword $svcAccountPassword ` -sddcManagerFqdn $sddcManagerFqdn ` -sddcManagerAdmin $sddcManagerAdmin ` -sddcManagerAdminPassword $sddcManagerAdminPassword```

Restore Management NSX Edge Nodes

Step	Manual Task	PowerShell Equivalent
1	Redeploy an NSX Edge Cluster for VCF Component Backup and Restore **Note:** You may also pass the `-monitor` flag on the end of the commmand block to track the status of the edge deployment through to completion.	```## Parameter Block ## $nsxManagerFqdn = "sfo-m01-nsx01.sfo.rainpole.io" #fqdn of restored NSX Manager $nsxManagerAdmin = "admin" $nsxManagerAdminPassword = "*********" $vCenterFqdn = "sfo-m01-vc01.sfo.rainpole.io" #fqdn of restored vCenter $vCenterAdmin = "administrator@vsphere.local" $vCenterAdminPassword = "********* $clusterName = "sfo-m01-cl01" $extractedSDDCDataFile = ".\extracted-sddc-data.json" ## Command Block ## Invoke-NSXEdgeClusterRecoverySelective ` -nsxManagerFqdn $nsxManagerFqdn ` -nsxManagerAdmin $nsxManagerAdmin ` -nsxManagerAdminPassword $nsxManagerAdminPassword ` -vCenterFQDN $vCenterFqdn ` -vCenterAdmin $vCenterAdmin ` -vCenterAdminPassword $vCenterAdminPassword ` -clusterName $clusterName ` -extractedSDDCDataFile $extractedSDDCDataFile```
2	Verify the State of the NSX Edge Cluster Nodes for VCF Instance Backup and Restore	No PowerShell equivalent, follow the manual procedure.

Redeploy the Remaining Management NSX Manager Nodes

Step	Manual Task	PowerShell Equivalent
1	Redeploy a Failed NSX Manager Node for VCF Instance Backup and Restore	```## Parameter Block ## $vCenterFqdn = "sfo-m01-vc01.sfo.rainpole.io" #fqdn of restored vCenter $vCenterAdmin = "administrator@vsphere.local" $vCenterAdminPassword = "*********" $extractedSDDCDataFile = ".\extracted-sddc-data.json" $workloadDomain = "sfo-m01" $restoredNsxManagerDeploymentSize = "medium" $nsxManagerOvaFile = ".\nsx-unified-appliance-9.0.0.0.24733065.ova" ## Command Block ## New-NSXManagerOvaDeployment ` -vCenterFqdn $vCenterFqdn ` -vCenterAdmin $vCenterAdmin ` -vCenterAdminPassword $vCenterAdminPassword ` -extractedSDDCDataFile $extractedSDDCDataFile ` -workloadDomain $workloadDomain ` -restoredNsxManagerDeploymentSize $restoredNsxManagerDeploymentSize ` -nsxManagerOvaFile $nsxManagerOvaFile```
2	Join NSX Manager Nodes to the NSX Manager Cluster for VCF Instance Backup and Restore	```## Parameter Block ## $workloadDomain = "sfo-m01" $extractedSDDCDataFile = ".\extracted-sddc-data.json" ## Command Block ## Add-AdditionalNSXManagers ` -workloadDomain $workloadDomain ` -extractedSDDCDataFile $extractedSDDCDataFile```

Finalize the Restore of the Management Domain

Important:

If this management domain is the one that runs the fleet management components, then ensure the VCF Fleet Component Restore Plan has been completed before proceeding. Refer to Sequence Optimization for Restore of a VCF Instance to understand the point at which you may start running the Fleet Component Restore Plan.

Step	Manual Task	PowerShell Equivalent
1	Update the Backup Configuration in SDDC Manager for VCF Instance Backup and Restore	No PowerShell equivalent, follow the manual procedure.
2	Update the Backup Configuration for vCenter for VCF Instance Backup and Restore	No PowerShell equivalent, follow the manual procedure.

Step	Manual Task	PowerShell Equivalent
3	Reapply Cluster Settings for the Restored vSphere Cluster for VCF Instance Backup and Restore	```## Parameter Block ## $vCenterFqdn = "sfo-m01-vc01.sfo.rainpole.io" #fqdn of restored vCenter $vCenterAdmin = "administrator@vsphere.local" $vCenterAdminPassword = "*********" $clusterName = "sfo-m01-cl01" $clusterVMOverridesJsonFile = ".\sfo-m01-cl01-vmOverrides.json" $clusterVMLocationsJsonFile = ".\sfo-m01-cl01-vmLocations.json" $clusterDRSConfigurationJsonFile = ".\sfo-m01-cl01-drsConfiguration.json" $clusterVMTagsJsonFile = ".\sfo-m01-cl01-vmTags.json" ## Command Block ## Restore-ClusterVMOverrides ` -vCenterFQDN $vCenterFqdn ` -vCenterAdmin $vCenterAdmin ` -vCenterAdminPassword $vCenterAdminPassword ` -clusterName $clusterName ` -jsonFile $clusterVMOverridesJsonFile Restore-ClusterVMLocations ` -vCenterFQDN $vCenterFqdn ` -vCenterAdmin $vCenterAdmin ` -vCenterAdminPassword $vCenterAdminPassword ` -clusterName $clusterName ` -jsonFile $clusterVMLocationsJsonFile Restore-ClusterDRSGroupsAndRules ` -vCenterFQDN $vCenterFqdn ` -vCenterAdmin $vCenterAdmin ` -vCenterAdminPassword $vCenterAdminPassword ` -clusterName $clusterName ` -jsonFile $clusterDRSConfigurationJsonFile Restore-ClusterVMTags ` -vCenterFQDN $vCenterFqdn ` -vCenterAdmin $vCenterAdmin ` -vCenterAdminPassword $vCenterAdminPassword ` -clusterName $clusterName ` -jsonFile $clusterVMTagsJsonFile```

VCF Fleet Component Restore Plan

Perform the following tasks to restore the fleet components in a management domain.

For information about where tasks can be performed in parallel across VCF domains, see Sequence Optimization for Restore of a VCF Instance .

Fleet Component Restore Plan Summary

- Restore VCF Operations
- Restore VCF Management Services
- Restore VCF Automation
- Restore Telemetry
- Redeploy VCF License Server

Restore VCF Operations

Step	Manual Task	PowerShell Equivalent
1	Restore VCF Operations from an Image-Based Backup for VCF Instance Backup and Restore	No PowerShell equivalent, follow the manual procedure.
2	Verify VCF Operations State for VCF Instance Backup and Restore	No PowerShell equivalent, follow the manual procedure.

Restore VCF Management Services

Step	Manual Task	PowerShell Equivalent
1	Retrieve Component Details from VCF Operations for VCF Instance Backup and Restore	```## Parameter Block ## $vcfOperationsFqdn = "flt-ops01a.rainpole.io" $username = "admin" $password = "***************" $authSource = "local" $outputDir = ".\" ## Command Block ## Get-VcfOperationsRegisteredComponents ` -VcfOperationsFqdn $vcfOperationsFqdn ` -Username $username ` -Password $password ` -AuthSource $authSource ` -OutputDir $outputDir``` **Attention:** Take note of the **vsp.id** from the resulting JSON. This will be used later as the **clusterId** when redeploying VCF Management Services and the **vspId** when retrieving backups of it.
2	Repoint SDDC Manager Depot to an Offline Depot for VCF Instance Backup and Restore	```## Parameter Block ## $sddcManagerFqdn = "sfo-vcf01.sfo.rainpole.io" $sddcManagerAdmin = "administrator@vsphere.local" $sddcManagerAdminPassword = "*************" $offlineDepotFqdn = "depot.rainpole.io" $offlineDepotPort = "443" $offlineDepotUsername = "depotUser" $offlineDepotPassword = "*******" ## Command Block ## Set-SDDCManagerOfflineDepot ` -sddcManagerFqdn $sddcManagerFqdn ` -sddcManagerUser $sddcManagerAdmin ` -sddcManagerPassword $sddcManagerAdminPassword ` -offlineDepotFqdn $offlineDepotFqdn ` -offlineDepotPort $offlineDepotPort ` -offlineDepotUsername $offlineDepotUsername ` -offlineDepotPassword $offlineDepotPassword```

Step	Manual Task	PowerShell Equivalent
3	Download Binaries for VCF Management Services Runtime for VCF Instance Backup and Restore	```## Parameter Block ## $sddcManagerFqdn = "sfo-vcf01.sfo.rainpole.io" $vcfUserPassword = "**********" $rootPassword = "**********" $adminPassword = "**********" $vcfVersion = "9.1.1.0" ## Command Block ## Invoke-SddcManagerBundleDownload ` -sddcManagerFqdn $sddcManagerFqdn ` -vcfUserPassword $vcfUserPassword ` -rootPassword $rootPassword ` -adminPassword $adminPassword ` -VcfVersion $vcfVersion ` -WaitForCompletion```
4	Extract Services Runtime Configuration from the Services Runtime Backup for VCF Instance Backup and Restore	```## Parameter Block ## $LocalArchivePath = "C:backups\2026-03-23T16-45-31Z.base.tgz" $EncryptionPassphrase = "**********" $OutputDir = ".\backup-yaml" ## Command Block ## New-ExtractVcfmsBackup ` -LocalArchivePath $LocalArchivePath ` -EncryptionPassphrase $EncryptionPassphrase ` -OutputDir $OutputDir```
5	Prepare to Restore VCF Management Services for VCF Instance Backup and Restore	```## Parameter Block ## $sddcManagerFqdn = "sfo-vcf01.sfo.rainpole.io" $vcfUserPassword = "**********" $rootPassword = "**********" $clusterType = "MANAGEMENT" ## Command Block ## Remove-SddcManagerVspClusterEntry ` -SddcManagerFqdn $sddcManagerFqdn ` -VcfUserPassword $vcfUserPassword ` -RootPassword $rootPassword ` -ClusterType $clusterType```

Step	Manual Task	PowerShell Equivalent
6	Redeploy VCF Management Services for VCF Instance Backup and Restore	```## Parameter Block ## $sddcManagerFqdn = "sfo-vcf01.sfo.rainpole.io" $sddcManagerUser = "administrator@vsphere.local" $sddcManagerPassword = "**************" $platformFqdn = "sfo-sr01.sfo.rainpole.io" #VCFMS services runtime FQDN $instanceFqdn = "sfo-ic01.sfo.rainpole.io" #VCFMS instance component FQDN $fleetFqdn = "flt-fc01.rainpole.io" $systemUserPassword = "**************" $ipv4Addresses = "10.11.10.31","10.11.10.32","10.11.10.33","10.11.10.34","10.11.10.35","10.11.1 $size = "medium" #Should match original VCFMS deployment size (small, medium or large) $networkMoId = "dvportgroup-28" $gatewayCidrIpv4 = "10.11.10.1/24" $clusterId = "1f5c79fe-e3aa-41b1-a5cf-774a6497fa3d" $internalClusterCidrIpv4 = "198.18.0.0/15" $type = "MANAGEMENT"``` **Attention:** The **$networkMoId** value must be obtained from vCenter using the following example command: (Get-VDPortgroup -Name "sfo-m01-cl01-vds01-pg-vm-mgmt").ExtensionData.MoRef.value The **$clusterId** value can be obtained from the output of Step 1. ```## Command Block ## New-ServicesRuntime ` -SddcManagerFqdn $sddcManagerFqdn ` -SddcManagerUser $sddcManagerUser ` -SddcManagerPassword $sddcManagerPassword ` -PlatformFqdn $platformFqdn ` -InstanceFqdn $instanceFqdn ` -FleetFqdn $fleetFqdn ` -SystemUserPassword $systemUserPassword ` -Ipv4Addresses $ipv4Addresses ` -Size $size ` -NetworkMoId $networkMoId ` -GatewayCidrIpv4 $gatewayCidrIpv4 ` -ClusterId $clusterId ` -InternalClusterCidrIpv4 $internalClusterCidrIpv4 ` -type $type```

Step	Manual Task	PowerShell Equivalent
7	Apply SFTP Settings on VCF Management Services for VCF Instance Backup and Restore	```## Parameter Block ## $servicesRuntimeFqdn = "sfo-sr01.sfo.rainpole.io" $servicesRuntimePassword = "**************" $sftpHost = "192.168.1.2" $sftpUsername = "svc-vcf-bck" $sftpPassword = "*******" $sftpDirectory = "/media/backups/" $encryptionPassphrase = "**************" $extractedSDDCDataFile = ".\extracted-sddc-data.json" ## Command Block ## Set-ServicesRuntimeSftpBackupSettings ` -ServicesRuntimeFqdn $servicesRuntimeFqdn ` -ServicesRuntimePassword $servicesRuntimePassword ` -SftpHost $sftpHost ` -SftpUsername $sftpUsername ` -SftpPassword $sftpPassword ` -SftpDirectory $sftpDirectory ` -EncryptionPassphrase $encryptionPassphrase ` -extractedSDDCDataFile $extractedSDDCDataFile```
8	Trust VCF Installer Certificate on VCF Management Services for VCF Instance Backup and Restore **Note:** This step is required if the binaries for the restore are hosted on the original VCF Installer that was used to deploy the environment.	```## Parameter Block ## $servicesRuntimeFqdn = "sfo-sr01.sfo.rainpole.io" $servicesRuntimePassword = "**************" $remoteHostFqdn = "sfo-ins01.sfo.rainpole.io" ## Command Block ## Add-VcfmsTrustedCertificate ` -ServicesRuntimeFqdn $servicesRuntimeFqdn ` -ServicesRuntimePassword $servicesRuntimePassword ` -RemoteHostFqdn $remoteHostFqdn```
9	Stage Component Binaries Prior to Restore **Note:** Use detail for VCFMS Services Runtime. This step is required if you do not have the original VCF Installer.	No PowerShell equivalent, follow the manual procedure.
10	Retrieve a List of Backups for VCF Management Services Components for VCF Instance Backup and Restore	```## Parameter Block ## $servicesRuntimeFqdn = "sfo-sr01.sfo.rainpole.io" $servicesRuntimePassword = "**************" $components = "vsp", "vcf-fleet-lcm", "vcf-fleet-depot", "vcf-sddc-lcm", "salt", "salt-raas", "vidb", "vcfms-metrics-store", "vcf-obs-data-platform" $extractedSDDCDataFile = ".\extracted-sddc-data.json" ## Command Block ## Get-ServicesRuntimeComponentBackups ` -ServicesRuntimeFqdn $servicesRuntimeFqdn ` -ServicesRuntimePassword $servicesRuntimePassword ` -Components $components ` -extractedSDDCDataFile $extractedSDDCDataFile```

Step	Manual Task	PowerShell Equivalent
11	Restore the VCF Management Services Components for VCF Instance Backup and Restore	```## Parameter Block ## $servicesRuntimeFqdn = "sfo-sr01.sfo.rainpole.io" $servicesRuntimePassword = "**************" ## Command Block ## Restore-ServicesRuntimeComponentBackup ` -ServicesRuntimeFqdn $servicesRuntimeFqdn ` -ServicesRuntimePassword $servicesRuntimePassword ` -RestoreJsonFile ".\restore-payload.json"```
12	Source the kubeconfig file for the new VCF Management Services Runtime **Note:** Only required if using PowerShell automation for recovery	```## Parameter Block ## $LocalArchivePath = "C:backups\2026-03-23T16-45-31Z.base.tgz" $EncryptionPassphrase = "**************" $OutputDir = ".\backup-yaml" $servicesRuntimeFqdn = "sfo-sr01.sfo.rainpole.io" $servicesRuntimePassword = "**************" ## Command Block ## New-ExtractVcfmsBackup ` -LocalArchivePath $LocalArchivePath ` -EncryptionPassphrase $EncryptionPassphrase ` -OutputDir $OutputDir ` -ServicesRuntimeFqdn $servicesRuntimeFqdn ` -ServicesRuntimePassword $servicesRuntimePassword```
13	Apply Services Runtime Configuration of VCF Management Services for VCF Instance Backup and Restore	```## Parameter Block ## $KubeconfigPath = ".\backup-yaml\sfo-sr01.kubeconfig" #Kubeconfig for the VCF Management Services cluster $BackupYamlDir = ".\backup-yaml" ## Command Block ## Set-ServicesRuntimeScale ` -KubeconfigPath $KubeconfigPath ` -BackupYamlDir $BackupYamlDir```
14	Retrieve a List of Backups for Log Management	```## Parameter Block ## $servicesRuntimeFqdn = "sfo-sr01.sfo.rainpole.io" $servicesRuntimePassword = "**************" $components = "ops-logs" $extractedSDDCDataFile = ".\extracted-sddc-data.json" ## Command Block ## Get-ServicesRuntimeComponentBackups ` -ServicesRuntimeFqdn $servicesRuntimeFqdn ` -ServicesRuntimePassword $servicesRuntimePassword ` -Components $components ` -extractedSDDCDataFile $extractedSDDCDataFile```
15	Restore Log Management for VCF Instance Backup and Restore	```## Parameter Block ## $servicesRuntimeFqdn = "sfo-sr01.sfo.rainpole.io" $servicesRuntimePassword = "**************" ## Command Block ## Restore-ServicesRuntimeComponentBackup ` -ServicesRuntimeFqdn $servicesRuntimeFqdn ` -ServicesRuntimePassword $servicesRuntimePassword ` -RestoreJsonFile ".\restore-payload.json"```
16	Update the Log Management State	No PowerShell equivalent, follow the manual procedure.

Step	Manual Task	PowerShell Equivalent
17	Repoint SDDC Manager Depot to Fleet Depot Service for VCF Instance Backup and Restore	```## Parameter Block ## $sddcManagerFqdn = "sfo-vcf01.sfo.rainpole.io" $sddcManagerAdmin = "administrator@vsphere.local" $sddcManagerAdminPassword = "**************" $originalConfigurationFile = "./originalDepotServicesconfig.json" ## Command Block ## Set-SDDCManagerFDSDepot ` -sddcManagerFqdn $sddcManagerFqdn ` -sddcManagerUser $sddcManagerAdmin ` -sddcManagerPassword $sddcManagerAdminPassword ` -originalConfigurationFile $originalConfigurationFile```

Restore VCF Automation

Step	Manual Task	PowerShell Equivalent
1	Retrieve Component Details from VCF Operations for VCF Instance Backup and Restore	```## Parameter Block ## $vcfOperationsFqdn = "flt-ops01a.rainpole.io" $username = "admin" $password = "***************" $authSource = "local" $outputDir = ".\" ## Command Block ## Get-VcfOperationsRegisteredComponents ` -VcfOperationsFqdn $vcfOperationsFqdn ` -Username $username ` -Password $password ` -AuthSource $authSource ` -OutputDir $outputDir``` **Attention:** Take note of the **vsp.id** from the resulting JSON. This will be used later as the **clusterId** when redeploying VCF Automation and the **vspId** when retrieving backups of it.
2	Prepare to Restore VCF Automation for VCF Instance Backup and Restore	```## Parameter Block ## $sddcManagerFqdn = "sfo-vcf01.sfo.rainpole.io" $vcfUserPassword = "**************" $rootPassword = "**************" $clusterType = "CONSUMPTION" ## Command Block ## Remove-SddcManagerVspClusterEntry ` -SddcManagerFqdn $sddcManagerFqdn ` -VcfUserPassword $vcfUserPassword ` -RootPassword $rootPassword ` -ClusterType $clusterType```

Step	Manual Task	PowerShell Equivalent
3	Redeploy the VCF Automation Services Runtime Cluster for VCF Instance Backup and Restore	```## Parameter Block ## $sddcManagerFqdn = "sfo-vcf01.sfo.rainpole.io" $sddcManagerUser = "administrator@vsphere.local" $sddcManagerPassword = "**************" $platformFqdn = "flt-vcfa-sr01.rainpole.io" #VCF Automation Services Runtime FQDN $systemUserPassword = "**************" #VCF Automation Services Runtime admin@vsp.local Password $ipv4Addresses = "10.11.10.46","10.11.10.47","10.11.10.48","10.11.10.49","10.11.10.50" $size = "medium" #Should match original VCFA deployment size (small, medium or large) $networkMoId = "dvportgroup-28" $gatewayCidrIpv4 = "10.11.10.1/24" $clusterId = "59cc7bb0-5436-4880-a710-b11933489dc7" $internalClusterCidrIpv4 = "198.18.0.0/15" $type = "CONSUMPTION"``` **Attention:** The **$networkMoId** value must be obtained from vCenter using the following example command: (Get-VDPortgroup -Name "sfo-m01-cl01-vds01-pg-vm-mgmt").ExtensionData.MoRef.value The **$clusterId** value can be obtained from the output of Step 1. ```## Command Block ## New-ServicesRuntime ` -SddcManagerFqdn $sddcManagerFqdn ` -SddcManagerUser $sddcManagerUser ` -SddcManagerPassword $sddcManagerPassword ` -PlatformFqdn $platformFqdn ` -SystemUserPassword $systemUserPassword ` -Ipv4Addresses $ipv4Addresses ` -Size $size ` -NetworkMoId $networkMoId ` -GatewayCidrIpv4 $gatewayCidrIpv4 ` -ClusterId $clusterId ` -InternalClusterCidrIpv4 $internalClusterCidrIpv4 ` -type $type```

Step	Manual Task	PowerShell Equivalent
4	Apply SFTP Settings on VCF Automation for VCF Instance Backup and Restore	```## Parameter Block ## $servicesRuntimeFqdn = "flt-vcfa-sr01.rainpole.io" #VCF Automation Services Runtime FQDN $servicesRuntimePassword = "**********" #VCF Automation Services Runtime admin@vsp.local Password $sftpHost = "192.168.1.2" $sftpUsername = "svc-vcf-bck" $sftpPassword = "*******" $sftpDirectory = "/media/backups/" $encryptionPassphrase = "**********" $extractedSDDCDataFile = ".\extracted-sddc-data.json" ## Command Block ## Set-ServicesRuntimeSftpBackupSettings ` -ServicesRuntimeFqdn $servicesRuntimeFqdn ` -ServicesRuntimePassword $servicesRuntimePassword ` -SftpHost $sftpHost ` -SftpUsername $sftpUsername ` -SftpPassword $sftpPassword ` -SftpDirectory $sftpDirectory ` -EncryptionPassphrase $encryptionPassphrase ` -extractedSDDCDataFile $extractedSDDCDataFile```
5	Trust VCF Installer Certificate on VCF Automation for VCF Instance Backup and Restore	```## Parameter Block ## $servicesRuntimeFqdn = "flt-vcfa-sr01.rainpole.io" #VCF Automation Services Runtime FQDN $servicesRuntimePassword = "**********" #VCF Automation Services Runtime admin@vsp.local Password $remoteHostFqdn = "sfo-ins01.sfo.rainpole.io" #VCF Installer FQDN ## Command Block ## Add-VcfmsTrustedCertificate ` -ServicesRuntimeFqdn $servicesRuntimeFqdn ` -ServicesRuntimePassword $servicesRuntimePassword ` -RemoteHostFqdn $remoteHostFqdn```
6	Stage Component Binaries Prior to Restore **Note:** Use detail for VCFA Services Runtime. This step is required if you do not have the original VCF Installer.	No PowerShell equivalent, follow the manual procedure.
7	Retrieve a List of Backups for VCF Automation for VCF Instance Backup and Restore	```## Parameter Block ## $servicesRuntimeFqdn = "flt-vcfa-sr01.rainpole.io" #VCF Automation Services Runtime FQDN $servicesRuntimePassword = "**********" #VCF Automation Services Runtime admin@vsp.local Password $components = "vsp", "vcfa", "vcd-migrator" $extractedSDDCDataFile = ".\extracted-sddc-data.json" ## Command Block ## Get-ServicesRuntimeComponentBackups ` -ServicesRuntimeFqdn $servicesRuntimeFqdn ` -ServicesRuntimePassword $servicesRuntimePassword ` -components $components ` -extractedSDDCDataFile $extractedSDDCDataFile```

Step	Manual Task	PowerShell Equivalent
8	Restore VCF Automation for VCF Instance Backup and Restore	```## Parameter Block ## $servicesRuntimeFqdn = "flt-vcfa-sr01.rainpole.io" #VCF Automation Services Runtime FQDN $servicesRuntimePassword = "**********" #VCF Automation Services Runtime admin@vsp.local Password ## Command Block ## Restore-ServicesRuntimeComponentBackup ` -ServicesRuntimeFqdn $servicesRuntimeFqdn ` -ServicesRuntimePassword $servicesRuntimePassword ` -RestoreJsonFile ".\restore-payload.json"```
9	Update Connections in VCF Automation for VCF Instance Backup and Restore	No PowerShell equivalent, follow the manual procedure.

Restore Telemetry

Step	Manual Task	PowerShell Equivalent
1	Retrieve a List of Backups for Telemetry Component	```## Parameter Block ## $servicesRuntimeFqdn = "sfo-sr01.sfo.rainpole.io" $servicesRuntimePassword = "**********" $components = "telemetry-acceptor" $extractedSDDCDataFile = ".\extracted-sddc-data.json" ## Command Block ## Get-ServicesRuntimeComponentBackups ` -ServicesRuntimeFqdn $servicesRuntimeFqdn ` -ServicesRuntimePassword $servicesRuntimePassword ` -components $components ` -extractedSDDCDataFile $extractedSDDCDataFile```
2	Restore Telemetry Component	```## Parameter Block ## $servicesRuntimeFqdn = "sfo-sr01.sfo.rainpole.io" $servicesRuntimePassword = "**********" #VCF Management Services Runtime admin@vsp.local Password ## Command Block ## Restore-ServicesRuntimeComponentBackup ` -ServicesRuntimeFqdn $servicesRuntimeFqdn ` -ServicesRuntimePassword $servicesRuntimePassword ` -RestoreJsonFile ".\restore-payload.json"```

Redeploy VCF License Server

Step	Manual Task	PowerShell Equivalent
1	Redeploy a License Server for VCF Instance Backup and Restore	No PowerShell equivalent, follow the manual procedure.

Workload Domain Restore Plan

Perform the following tasks to restore a workload domain.

For information about where tasks can be performed in parallel across domains, see Sequence Optimization for Restore of a VCF Instance.

Workload Domain Restore Plan Summary

- Restore Workload vCenter and NSX Manager
- Rebuild Workload vSphere Cluster
- Restore Workload NSX Edge Nodes
- Restore the Remaining Workload NSX Manager Nodes
- Finalize the Restore of the Workload Domain

Restore Workload vCenter and NSX Manager

Step	Manual Task	PowerShell Equivalent
1	Deploy a vCenter Appliance from an OVA File for VCF Instance Backup and Restore	```## Parameter Block ## $targetType = "esx" $targetFqdn = "sfo01-m01-r01-esx01.sfo.rainpole.io" $targetAdmin = "root" $targetAdminPassword = "*********" $extractedSDDCDataFile = ".\extracted-sddc-data.json" $workloadDomain = "sfo-w01" $restoredvCenterDeploymentSize = "medium" $vCenterOvaFile = ".\VMware-vCenter-Server-Appliance-9.1.1.0.25712839_OVF10.ova"``` **Note:** These parameters assume you plan to deploy OVA files for the workload domain before the management domain restore is complete. If you choose to fully restore the management domain before this process, then you can use the management vCenter details instead of the ESX details. ```## Command Block ## New-vCenterOvaDeployment ` -targetType $targetType ` -targetFqdn $targetFqdn ` -targetAdmin $targetAdmin ` -targetAdminPassword $targetAdminPassword ` -extractedSDDCDataFile $extractedSDDCDataFile ` -workloadDomain $workloadDomain ` -restoredvCenterDeploymentSize $restoredvCenterDeploymentSize ` -vCenterOvaFile $vCenterOvaFile```
2	Deploy the First NSX Manager Cluster Node from an OVA File for VCF Instance Backup and Restore	```## Parameter Block ## $targetType = "esx" $targetFqdn = "sfo01-m01-r01-esx01.sfo.rainpole.io" $targetAdmin = "root" $targetAdminPassword = "*********" $extractedSDDCDataFile = ".\extracted-sddc-data.json" $workloadDomain = "sfo-w01" $restoredNsxManagerDeploymentSize = "large" $nsxManagerOvaFile = ".\nsx-unified-appliance-9.1.1.0.25691509.ova"``` **Note:** These parameters assume you plan to deploy OVA files for the workload domain before the management domain restore is complete. If you choose to fully restore the management domain before this process, then you can use the management vCenter details instead of the ESX details. ```## Command Block ## New-NSXManagerOvaDeployment ` -targetType $targetType ` -targetFqdn $targetFqdn ` -targetAdmin $targetAdmin ` -targetAdminPassword $targetAdminPassword ` -extractedSDDCDataFile $extractedSDDCDataFile ` -workloadDomain $workloadDomain ` -restoredNsxManagerDeploymentSize $restoredNsxManagerDeploymentSize ` -nsxManagerOvaFile $nsxManagerOvaFile```

Step	Manual Task	PowerShell Equivalent
3	Restore vCenter from a File-Based Backup for VCF Instance Backup and Restore	```## Parameter Block ## $extractedSDDCDataFile = ".\extracted-sddc-data.json" $workloadDomain ="sfo-w01" $vCenterBackupPath = "192.168.1.1/F$/ backups/sn_sfo-w01-vc01.sfo.rainpole.io/ M_9.0.2.0_20260406-084633_KBWGC5DGN5ZG2VDPN5WHGICCMFRWW5LQ" $locationtype = "SMB" $locationUser = "Administrator" $locationPassword = "*********" $vCenterBackupPassword = "*********" ## Command Block ## Invoke-vCenterRestore ` -extractedSDDCDataFile $extractedSDDCDataFile ` -workloadDomain $workloadDomain ` -vCenterBackupPath $vCenterBackupPath ` -locationtype $locationtype ` -locationUser $locationUser ` -locationPassword $locationPassword``` **Important:** If the vCenter backup was taken using a backup password, then append the following option to the command above. ```-backupPassword $vCenterBackupPassword```
4	Restore NSX Manager from a File-Based Backup for VCF Instance Backup and Restore	```## Parameter Block ## $extractedSDDCDataFile = ".\extracted-sddc-data.json" $workloadDomain ="sfo-w01" $sftpServer ="192.168.1.2" $sftpUser ="svc-vcf-bck" $sftpPassword = "*********" $sftpServerBackupPath ="/media/backups" $nsxBackupPassphrase = "*********" ## Command Block ## Invoke-NSXManagerRestore ` -extractedSDDCDataFile $extractedSDDCDataFile ` -workloadDomain $workloadDomain ` -sftpServer $sftpServer ` -sftpUser $sftpUser ` -sftpPassword $sftpPassword ` -sftpServerBackupPath $sftpServerBackupPath ` -backupPassphrase $nsxBackupPassphrase```
5	Verify the State of the First NSX Manager Cluster Node for VCF Instance Backup and Restore	No PowerShell equivalent, follow the manual procedure.

Step	Manual Task	PowerShell Equivalent
6	Update Extracted SDDC Data for Instance Backup and Restore **Note:** This task is only required when using the PowerShell Automation approach. If performing a fully manual restore, this information is gathered in the Documenting the System Configuration of VCF step.	```## Parameter Block ## $extractedSDDCDataFile = ".\extracted-sddc-data.json" $sddcManagerFQDN = "sfo-vcf01.sfo.rainpole.io" $sddcManagerAdmin = "administrator@vsphere.local" $sddcManagerAdminPassword = "*********" $vCenterFQDN = "sfo-w01-vc01.sfo.rainpole.io" #fqdn of restored vCenter ## Command Block ## Update-ExtractedSDDCData ` -extractedSDDCDataFile $extractedSDDCDataFile ` -sddcManagerFQDN $sddcManagerFQDN ` -sddcManagerAdmin $sddcManagerAdmin ` -sddcManagerAdminPassword $sddcManagerAdminPassword ` -vCenterFQDN $vCenterFQDN```

Rebuild Workload vSphere Cluster

Step	Manual Task	PowerShell Equivalent
1	Record the Cluster Settings from the Restored vCenter for VCF Instance Backup and Restore	```## Parameter Block ## $vCenterFqdn = "sfo-w01-vc01.sfo.rainpole.io" #fqdn of restored vCenter $vCenterAdmin = administrator@sfo-w01.local" $vCenterAdminPassword = "*********" $clusterName = "sfo-w01-cl01" ## Command Block ## Backup-ClusterVMOverrides ` -vCenterFQDN $vCenterFqdn ` -vCenterAdmin $vCenterAdmin ` -vCenterAdminPassword $vCenterAdminPassword ` -clusterName $clusterName Backup-ClusterVMLocations ` -vCenterFQDN $vCenterFqdn ` -vCenterAdmin $vCenterAdmin ` -vCenterAdminPassword $vCenterAdminPassword ` -clusterName $clusterName Backup-ClusterDRSGroupsAndRules ` -vCenterFQDN $vCenterFqdn ` -vCenterAdmin $vCenterAdmin ` -vCenterAdminPassword $vCenterAdminPassword ` -clusterName $clusterName Backup-ClusterVMTags ` -vCenterFQDN $vCenterFqdn ` -vCenterAdmin $vCenterAdmin ` -vCenterAdminPassword $vCenterAdminPassword ` -clusterName $clusterName```
2	Remove Non-Responsive ESX Hosts from the Inventory for VCF Instance Backup and Restore	```## Parameter Block ## $vCenterFqdn = "sfo-w01-vc01.sfo.rainpole.io" #fqdn of restored vCenter $vCenterAdmin = "administrator@sfo-w01.local" $vCenterAdminPassword = "*********" $clusterName = "sfo-w01-cl01" $nsxManagerFqdn = "sfo-w01-nsx01.sfo.rainpole.io" #fqdn of restored NSX Manager $nsxManagerAdmin = "admin" $nsxManagerAdminPassword = "*********" $nsxManagerRootPassword = "*********" ## Command Block ## Remove-NonResponsiveHosts ` -vCenterFQDN $vCenterFqdn ` -vCenterAdmin $vCenterAdmin ` -vCenterAdminPassword $vCenterAdminPassword ` -clusterName $clusterName ` -NsxManagerFQDN $nsxManagerFqdn ` -NsxManagerAdmin $nsxManagerAdmin ` -NsxManagerAdminPassword $nsxManagerAdminPassword ` -NsxManagerRootPassword $nsxManagerRootPassword```

Step	Manual Task	PowerShell Equivalent
3	Add New Hosts to the Cluster to the vCenter Instance for the Restored VCF Domain	```## Parameter Block ## $vCenterFqdn = "sfo-w01-vc01.sfo.rainpole.io" $vCenterAdmin = "administrator@sfo-w01.local" $vCenterAdminPassword = "*********" $clusterName = "sfo-w01-cl01" $sddcManagerFqdn = "sfo-vcf01.sfo.rainpole.io" $sddcManagerAdmin = "administrator@vsphere.local" $sddcManagerAdminPassword = "*********" $extractedSDDCDataFile = ".\extracted-sddc-data.json" ## Command Block ## Add-HostsToCluster ` -vCenterFQDN $vCenterFqdn ` -vCenterAdmin $vCenterAdmin ` -vCenterAdminPassword $vCenterAdminPassword ` -clusterName $clusterName ` -sddcManagerFqdn $sddcManagerFqdn ` -sddcManagerAdmin $sddcManagerAdmin ` -sddcManagerAdminPassword $sddcManagerAdminPassword ` -extractedSDDCDataFile $extractedSDDCDataFile```
4	Migrate the ESX Hosts and VMkernel Adapters to the vSphere Distributed Switch on the Restored vCenter for VCF Instance Backup and Restore	```## Parameter Block ## $vCenterFqdn = "sfo-w01-vc01.sfo.rainpole.io" #fqdn of restored vCenter $vCenterAdmin = "administrator@sfo-w01.local" $vCenterAdminPassword = "*********" $clusterName = "sfo-w01-cl01" $extractedSDDCDataFile = ".\extracted-sddc-data.json" ## Command Block ## New-RebuiltVdsConfiguration ` -vCenterFQDN $vCenterFqdn ` -vCenterAdmin $vCenterAdmin ` -vCenterAdminPassword $vCenterAdminPassword ` -clusterName $clusterName ` -extractedSDDCDataFile $extractedSDDCDataFile```
5	Wait for NSX Transport Installation to finish	```## Parameter Block ## $clusterName = "sfo-w01-cl01" $extractedSDDCDataFile = ".\extracted-sddc-data.json" ## Command Block ## Watch-NsxHostTransportNodeInstallation ` -clusterName $clusterName ` -extractedSDDCDataFile $extractedSDDCDataFile```

Step	Manual Task	PowerShell Equivalent
6	Add VMkernel Adapters to Workload Domain ESX Hosts for VCF Instance Backup and Restore	```## Parameter Block ## $vCenterFqdn = "sfo-w01-vc01.sfo.rainpole.io" #fqdn of restored vCenter $vCenterAdmin = "administrator@sfo-w01.local" $vCenterAdminPassword = "*********" $clusterName = "sfo-w01-cl01" $sddcManagerFqdn = "sfo-vcf01.sfo.rainpole.io" $sddcManagerAdmin = "administrator@vsphere.local" $sddcManagerAdminPassword = "*********" $extractedSDDCDataFile = ".\extracted-sddc-data.json" ## Command Block ## Add-VMKernelsToHost ` -vCenterFQDN $vCenterFQDN ` -vCenterAdmin $vCenterAdmin ` -vCenterAdminPassword $vCenterAdminPassword ` -clusterName $clusterName ` -sddcManagerFqdn $sddcManagerFqdn ` -sddcManagerAdmin $sddcManagerAdmin ` -sddcManagerAdminPassword $sddcManagerAdminPassword ` -extractedSDDCDataFile $extractedSDDCDataFile```
7	Reconfigure vSAN Stretched Cluster **Important:** Perform this step only if the cluster was originally a stretched vSAN cluster	```## Parameter Block ## $vCenterFQDN = "sfo-w01-vc01.sfo.rainpole.io" #fqdn of restored vCenter $vCenterAdmin = "administrator@sfo-w01.local" $vCenterAdminPassword = "*********" $clusterName = "sfo-w01-cl01" $extractedSDDCDataFile = ".\extracted-sddc-data.json" New-ReconfiguredVsanStretchedCluster ` -vCenterFQDN $vCenterFQDN ` -vCenterAdmin $vCenterAdmin ` -vCenterAdminPassword $vCenterAdminPassword ` -clusterName $clusterName ` -extractedSDDCDataFile $extractedSDDCDataFile```
8	Recreate a vSAN Datastore for VCF Instance Backup and Restore	```## Parameter Block ## $vCenterFqdn = "sfo-w01-vc01.sfo.rainpole.io" #fqdn of restored vCenter $vCenterAdmin = "administrator@sfo-w01.local" $vCenterAdminPassword = "*********" $clusterName = "sfo-w01-cl01" $extractedSDDCDataFile = ".\extracted-sddc-data.json" ## Command Block ## New-RebuiltVsanDatastore ` -vCenterFqdn $vCenterFqdn ` -vCenterAdmin $vCenterAdmin ` -vCenterAdminPassword $vCenterAdminPassword ` -clusterName $clusterName ` -extractedSDDCDataFile $extractedSDDCDataFile```

Step	Manual Task	PowerShell Equivalent
9	Re-Enable vSphere HA on the Workload Domain Cluster for VCF Instance Backup and Restore	```## Parameter Block ## $vCenterFqdn = "sfo-w01-vc01.sfo.rainpole.io" #fqdn of restored vCenter $vCenterAdmin = "administrator@sfo-w01.local" $vCenterAdminPassword = "*********" $clusterName = "sfo-m01-cl01" ## Command Block ## Connect-VIServer ` -server $vCenterFqdn ` -user $vCenterAdmin ` -password $vCenterAdminPassword | Out-Null  Get-Cluster $clusterName | Set-Cluster ` -HAEnabled $false ` -confirm:$false | Out-Null  Get-Cluster $clusterName | Set-Cluster ` -HAEnabled $true ` -confirm:$false | Out-Null  Disconnect-VIServer * ` -confirm:$false```
10	Clear vCenter Alarms **Attention:** This step is optional, and may be executed if you want to clear all alarms in the recovered vCenter. You may also do this manually using the vCenter UI.	```## Parameter Block ## $vCenterFQDN = "sfo-w01-vc01.sfo.rainpole.io" #fqdn of restored vCenter $vCenterAdmin = "administrator@sfo-w01.local" $vCenterAdminPassword = "*********" $extractedSDDCDataFile = ".\extracted-sddc-data.json" ## Command Block ## Clear-vCenterAlarms ` -vCenterFQDN $vCenterFQDN ` -vCenterAdmin $vCenterAdmin ` -vCenterAdminPassword $vCenterAdminPassword```
11	Update ESX Host SSH Keys for VCF Instance Backup and Restore	No PowerShell equivalent, follow the manual procedure.
12	Update the Datastore ID for Restored Cluster in SDDC Manager Inventory for VCF Instance Backup and Restore	```## Parameter Block ## $extractedSDDCDataFile = ".\extracted-sddc-data.json" $vCenterFqdn = "sfo-w01-vc01.sfo.rainpole.io" #fqdn of restored vCenter $clusterName = "sfo-w01-cl01" #name of cluster being recovered $vcfUserPassword = "*********" $rootPassword = "*********" ## Command Block ## Update-DomainDatastoreID ` -extractedSDDCDataFile $extractedSDDCDataFile ` -vCenterFqdn $vCenterFqdn ` -clusterName $clusterName ` -VcfUserPassword $vcfUserPassword -RootPassword $rootPassword```

Step	Manual Task	PowerShell Equivalent
13	Update the Cluster Host IDs for Restored Cluster in SDDC Manager Inventory for VCF Instance Backup and Restore	```## Parameter Block ## $extractedSDDCDataFile = ".\extracted-sddc-data.json" $vCenterFqdn = "sfo-w01-vc01.sfo.rainpole.io" #fqdn of restored vCenter $clusterName = "sfo-w01-cl01" #name of cluster being recovered $vcfUserPassword = "*********" $rootPassword = "*********" ## Command Block ## Update-ClusterHostSourceIDs ` -extractedSDDCDataFile $extractedSDDCDataFile ` -vCenterFqdn $vCenterFqdn ` -clusterName $clusterName ` -VcfUserPassword $vcfUserPassword ` -RootPassword $rootPassword```
14	Update ESX Host SSH Keys for VCF Instance Backup and Restore	```## Parameter Block ## $extractedSDDCDataFile = ".\current\sddc-manager-backup\extracted-sddc-data.json" $vCenterFQDN = "sfo-w01-vc01.sfo.rainpole.io" $vCenterAdmin = "administrator@sfo-w01.local" $vCenterAdminPassword = "*********" $clusterName = "sfo-w01-cl01" $workloadDomain = "sfo-w01" $VcfUserPassword = "*********" Invoke-SddcManagerSSHKeyRefresh ` -vCenterFQDN $vCenterFQDN ` -vCenterAdmin $vCenterAdmin ` -vCenterAdminPassword $vCenterAdminPassword ` -extractedSDDCDataFile $extractedSDDCDataFile ` -clusterName $clusterName ` -workloadDomain $workloadDomain ` -VcfUserPassword $VcfUserPassword```

Restore Workload NSX Edge Nodes

Step	Manual Task	PowerShell Equivalent
1	Redeploy an NSX Edge Cluster for VCF Component Backup and Restore	```## Parameter Block ## $nsxManagerFqdn = "sfo-w01-nsx01.sfo.rainpole.io" #fqdn of restored NSX Manager $nsxManagerAdmin = "admin" $nsxManagerAdminPassword = "*********" $vCenterFqdn = "sfo-w01-vc01.sfo.rainpole.io" #fqdn of restored vCenter $vCenterAdmin = "administrator@sfo-w01.local" $vCenterAdminPassword = "*********" $clusterName = "sfo-w01-cl01" $extractedSDDCDataFile = ".\extracted-sddc-data.json" ## Command Block ## Invoke-NSXEdgeClusterRecoverySelective ` -nsxManagerFqdn $nsxManagerFqdn ` -nsxManagerAdmin $nsxManagerAdmin ` -nsxManagerAdminPassword $nsxManagerAdminPassword ` -vCenterFQDN $vCenterFqdn ` -vCenterAdmin $vCenterAdmin ` -vCenterAdminPassword $vCenterAdminPassword ` -clusterName $clusterName ` -extractedSDDCDataFile $extractedSDDCDataFile```
2	Verify the State of the NSX Edge Cluster Nodes for VCF Instance Backup and Restore	No PowerShell equivalent, follow the manual procedure.

Restore the Remaining Workload NSX Manager Nodes

Step	Manual Task	PowerShell Equivalent
1	Redeploy a Failed NSX Manager Node for VCF Instance Backup and Restore	```## Parameter Block ## $vCenterFqdn = "sfo-w01-vc01.sfo.rainpole.io" #fqdn of restored vCenter $vCenterAdmin = "administrator@sfo-w01.local" $vCenterAdminPassword = "*********" $extractedSDDCDataFile = ".\extracted-sddc-data.json" $workloadDomain = "sfo-w01" $restoredNsxManagerDeploymentSize = "large" $nsxManagerOvaFile = ".\nsx-unified-appliance-9.1.1.0.25691509.ova" ## Command Block ## New-NSXManagerOvaDeployment ` -vCenterFqdn $vCenterFqdn ` -vCenterAdmin $vCenterAdmin ` -vCenterAdminPassword $vCenterAdminPassword ` -extractedSDDCDataFile $extractedSDDCDataFile ` -workloadDomain $workloadDomain ` -restoredNsxManagerDeploymentSize $restoredNsxManagerDeploymentSize ` -nsxManagerOvaFile $nsxManagerOvaFile```
2	Join NSX Manager Nodes to the NSX Manager Cluster for VCF Instance Backup and Restore	```## Parameter Block ## $workloadDomain = "sfo-w01" $extractedSDDCDataFile = ".\extracted-sddc-data.json" ## Command Block ## Add-AdditionalNSXManagers ` -workloadDomain $workloadDomain ` -extractedSDDCDataFile $extractedSDDCDataFile```

Finalize the Restore of the Workload Domain

Step	Manual Task	PowerShell Equivalent
1	Synchronize the SDDC Manager Service Account on the ESX Hosts for VCF Instance Backup and Restore	```## Parameter Block ## $vCenterFqdn = "sfo-w01-vc01.sfo.rainpole.io" #fqdn of restored vCenter $vCenterAdmin = "administrator@sfo-w01.local" $vCenterAdminPassword = "*********" $clusterName = "sfo-w01-cl01" $svcAccountPassword = "*********" $sddcManagerFqdn = "sfo-vcf01.sfo.rainpole.io" $sddcManagerAdmin = "administrator@vsphere.local" $sddcManagerAdminPassword = "*********" ## Command Block ## Resolve-PhysicalHostServiceAccounts ` -vCenterFQDN $vCenterFqdn ` -vCenterAdmin $vCenterAdmin ` -vCenterAdminPassword $vCenterAdminPassword ` -clusterName $clusterName ` -svcAccountPassword $svcAccountPassword ` -sddcManagerFqdn $sddcManagerFqdn ` -sddcManagerAdmin $sddcManagerAdmin ` -sddcManagerAdminPassword $sddcManagerAdminPassword```
2	Update the Backup Configuration for vCenter for VCF Instance Backup and Restore	No PowerShell equivalent, follow the manual procedure.

Step	Manual Task	PowerShell Equivalent
3	Reapply Cluster Settings for the Restored vSphere Cluster for VCF Instance Backup and Restore	```## Parameter Block ## $vCenterFqdn = "sfo-w01-vc01.sfo.rainpole.io" #fqdn of restored vCenter $vCenterAdmin = "administrator@sfo-w01.local" $vCenterAdminPassword = "*********" $clusterName = "sfo-w01-cl01" $clusterVMOverridesJsonFile = ".\sfo-w01-cl01-vmOverrides.json" $clusterVMLocationsJsonFile = ".\sfo-w01-cl01-vmLocations.json" $clusterDRSConfigurationJsonFile = ".\sfo-w01-cl01-drsConfiguration.json" $clusterVMTagsJsonFile = ".\sfo-w01-cl01-vmTags.json" ## Command Block ## Restore-ClusterVMOverrides ` -vCenterFQDN $vCenterFqdn ` -vCenterAdmin $vCenterAdmin ` -vCenterAdminPassword $vCenterAdminPassword ` -clusterName $clusterName ` -jsonFile $clusterVMOverridesJsonFile Restore-ClusterVMLocations ` -vCenterFQDN $vCenterFqdn ` -vCenterAdmin $vCenterAdmin ` -vCenterAdminPassword $vCenterAdminPassword ` -clusterName $clusterName ` -jsonFile $clusterVMLocationsJsonFile Restore-ClusterDRSGroupsAndRules ` -vCenterFQDN $vCenterFqdn ` -vCenterAdmin $vCenterAdmin ` -vCenterAdminPassword $vCenterAdminPassword ` -clusterName $clusterName ` -jsonFile $clusterDRSConfigurationJsonFile Restore-ClusterVMTags ` -vCenterFQDN $vCenterFqdn ` -vCenterAdmin $vCenterAdmin ` -vCenterAdminPassword $vCenterAdminPassword ` -clusterName $clusterName ` -jsonFile $clusterVMTagsJsonFile```

vSphere Supervisor Restore Plan

To restore vSphere supervisor you perform the following steps.

Restore vSphere Supervisor

Step	Manual Task	PowerShell Equivalent
1	Mount the NFS Share for the VKr Content Library for VCF Instance Backup and Restore	Not Applicable. Follow the manual task,

2	Recreate the Content Library folder on the rebuilt vSAN Datatore	`## Parameter Block ##`
		`$restoredVcenterFqdn = "sfo-w01-vc01.sfo.rainpole.io"`
		`$restoredvCenterRootPassword = "*********"`
		`$contentLibraryName = "sv-cl01"`
		`## Command Block ##`
		`Confirm-ContentLibraryDatastoreFolder -vCenterFQDN $restoredVcenterFqdn -vCenterRootPassword $restoredvCenterRootPassword -contentLibraryName $contentLibraryName`

3	Update the VKr Content Library Datastore Mapping for VCF Instance Backup and Restore	`## Parameter Block ##`
		`$restoredVcenterFqdn = "sfo-w01-vc01.sfo.rainpole.io"`
		`$restoredvCenterRootPassword = "*********"`
		`$contentLibraryName = "sv-cl01"`
		`$datastoreName = "nfs01"`
		`## Command Block ##`
		`Set-ContentLibraryDatastoreMapping -vCenterFQDN $restoredVcenterFqdn -vCenterRootPassword $restoredvCenterRootPassword -contentLibraryName $contentLibraryName -datastoreName $datastoreName`

4	Monitor the Supervisor Cluster Self Healing Process	Not Applicable. Follow the manual task,
5	Restore the vSphere Supervisor Control Plane for VCF Instance Backup and Restore	Not Applicable. Follow the manual task,
6	Remediate vSphere Supervisor Nodes for VCF Instance Backup and Restore	Not Applicable. Follow the manual task,
7	Recreate the NSX NCP Pods for VCF Instance Backup and Restore	Not Applicable. Follow the manual task,

VCF Instance Backup and Restore Task Library

Use a hierarchy of tasks that are associated with the backup and restore of a VCF Instance, sorted by component or functional area. The order in which tasks are listed in the library is not necessarily the order in which you should perform them.

Data Gathering Tasks for VCF Instance Backup and Restore

Before you can restore a VCF Instance of your VMware Cloud Foundation platform its important to gather critical information that will be utilzed during the restore process.

Gather Information for VCF Instance Backup and Restore

You must gather critical information about your VCF Instance in order to prepare for the restore process. It is good practice to have this information captured, stored and regularly updated in preperation for such an event.

Procedure

1. Using a combination of as-built documentation and the most recent record of all required credentials, capture the following information for each component:

Component	Information
ESX hosts	• FQDN • IP address and network information • **root** user password • Service account user name
VMware NSX nodes	• FQDN • IP address and network information • **admin** user password • **root** user password • **audit** user password
NSX Edge nodes	• FQDN • IP address and network information • **root** user password • **admin** user password • **audit** user password
SDDC Manager	• FQDN • IP address and network information • **root** user password • Backup user password • FIPS configuration

vCenter Data Gathering

Item	Where to Find it	Why is it Needed
FQDN	vCenter Management Interface > **Summary**, or SDDC Manager inventory (**hostName**)	**Required.** Must match the backup exactly.
IP address, for every configured adapter and address family	vCenter Management Interface > **Networking**	**Required.** DNS must resolve the FQDN to this address before the restore starts.
Prefix length (subnet mask)	vCenter Management Interface > **Networking**	**Not validated.**
Default gateway	vCenter Management Interface > **Networking**	**Not validated.**
DNS servers	vCenter Management Interface > **Networking**	**Not validated**, and not restored from the backup.
DNS search domains	**appliancesh** > **networking.dns.domains.list**, or the Direct Console User Interface	**Not backed up and not restored.** Re-apply manually after the restore.
Static routes	**appliancesh** > **networking.routes.list**	**Not backed up and not restored.** Re-apply manually after the restore.
Number of virtual network adapters	vCenter Management Interface > **Networking**	The replacement appliance is deployed with one adapter. Additional adapters must be attached before the restore, or their addresses are silently dropped.
vCenter version **and build number**	vCenter Management Interface > **Summary**, or SDDC Manager inventory (**version**, recorded as **version_number - build_number**)	**Required.** The replacement appliance must be deployed from the image matching this version. Download that image and store it with the backup.

Item	Where to Find it	Why is it Needed
Deployment size and storage size	vCenter Management Interface > **Summary**	**Required.** The replacement appliance must be the same size or larger.
root user password	SDDC Manager inventory, or your credential store	**Required in practice.** This exact password must be set on the replacement appliance — see *Deploy a vCenter Appliance from an OVA File*.
SSO administrator password	SDDC Manager inventory, or your credential store	**Required** for Enhanced Linked Mode. After the restore this is the password in effect.
SSO domain name	vCenter Management Interface > **Summary**	Needed only if reconciliation has to be re-run manually.
Backup location URL, user name, and password	Backup schedule in the vCenter Management Interface	**Required.**
Backup encryption password	Your credential store	**Required if the backup is encrypted.** Not stored in the backup and not recoverable from it. A backup whose encryption password has been lost cannot be restored.

PowerShell Equivalent

To see the PowerShell equivalent of this process in context, with sample parameters, refer to the following plans this task is part of:

- Management Domain Restore Plan

ESX Configuration Tasks for VCF Instance Backup and Restore

Perform these tasks for the configuration of ESX when restoring a VCF Instance.

Prepare Networking on Management ESX Hosts for VCF Instance Backup and Restore

During the VCF Instance restore process, you map network adapters on the new management domain hosts to standard virtual switches

UI Procedure

1. Connect to the first ESX host in the management domain over SSH.
2. Based on the desired number of vSphere Distributed Switches that existed in the source system, and factoring in the VLANs that are available on each physical NICs in the new ESX hosts, run the following commands.

```
# Step 1: Inspect existing NICs and switches
esxcli network nic list
esxcli network vswitch standard list

# Step 2: Update MTU on the existing management vSS
esxcli network vswitch standard set --vswitch-name=vSwitch0 --mtu=9000

# Step 3: Add an additional NIC to the existing management vSS (if applicable)
esxcli network vswitch standard uplink add --vswitch-name=vSwitch0 --uplink-name=vmnic1

# Step 4: Create the vm_mgmt portgroup with the VM_MANAGEMENT VLAN
esxcli network vswitch standard portgroup add --vswitch-name=vSwitch0 --portgroup-name="vm_mgmt"
esxcli network vswitch standard portgroup set --portgroup-name="vm_mgmt" --vlan-id=<VM_MANAGEMENT_VLAN_ID>

# Step 5: Create a NEW vSS that maps to each non-management VDSes (repeat per additional vSS)
esxcli network vswitch standard add --vswitch-name=vSwitch1 --ports=128
esxcli network vswitch standard set --vswitch-name=vSwitch1 --mtu=9000
```

```
# Step 6: Add NICs to each new vSS (repeat per additional vSS)
esxcli network vswitch standard uplink add --vswitch-name=vSwitch1 --uplink-name=vmnic2
esxcli network vswitch standard uplink add --vswitch-name=vSwitch1 --uplink-name=vmnic3
```

3. Repeat on the remaining ESX hosts in the management domain.

PowerShell Equivalent

To see the PowerShell equivalent of this process in context, with sample parameters, refer to the following plans this task is part of:

- Management Domain Restore Plan

Add VMkernel Adapters to Management ESX Hosts for VCF Instance Backup and Restore

To enable vSphere vMotion and vSAN traffic, add VMkernel network adapters to the new management hosts.

UI Procedure

1. Connect to the first ESX host in the management domain over SSH.
2. Configure VMkernel network adapters on the ESX host, reinstating the documented network addresses for the vSphere vMotion and vSAN interfaces.

```
# ==============================
# vMotion VMkernel Configuration
# ==============================

# Step 1: Create the vMotion portgroup with the correct VLAN
esxcli network vswitch standard portgroup add --vswitch-name=<VMOTION_VSS> --portgroup-name="vmotion"
esxcli network vswitch standard portgroup set --portgroup-name="vmotion" --vlan-id=<VMOTION_VLAN_ID>

# Step 2: Create the vMotion netstack (if it doesn't already exist)
esxcli network ip netstack add --netstack=vmotion

# Step 3: Verify it exists
esxcli network ip netstack list

# Step 4: Add vmk1 on the vmotion netstack
esxcli network ip interface add --interface-name=vmk1 --portgroup-name="vmotion" --mtu=<VMOTION_MTU> --
netstack=vmotion
esxcli network ip interface ipv4 set --interface-name=vmk1 --type=static --ipv4=<HOST_VMOTION_IP> --
netmask=<VMOTION_MASK>

# Step 5: Set the default gateway on the vmotion netstack
esxcli network ip route ipv4 add --netstack=vmotion --network=default --gateway=<VMOTION_GATEWAY>

# ============================
# vSAN VMkernel Configuration
# ============================

#  Step 1: Create the vSAN portgroup with the correct VLAN
esxcli network vswitch standard portgroup add --vswitch-name=<VSAN_VSS> --portgroup-name="vsan"
esxcli network vswitch standard portgroup set --portgroup-name="vsan" --vlan-id=<VSAN_VLAN_ID>

#  Step 2: Add vmk2 (uses the default netstack)
esxcli network ip interface add --interface-name=vmk2 --portgroup-name="vsan" --mtu=<VSAN_MTU>
esxcli network ip interface ipv4 set --interface-name=vmk2 --type=static --ipv4=<HOST_VSAN_IP> --
netmask=<VSAN_MASK> --gateway=<VSAN_GATEWAY>
```

```
# Step 3: Enable vSAN traffic on vmk2
esxcli vsan network ipv4 add --interface-name=vmk2

# Step 4: Verify
esxcli network ip interface list
esxcli network ip route ipv4 list --netstack=vmotion
esxcli vsan network list
```

3. Repeat on the remaining ESX hosts in the management domain.

PowerShell Equivalent

To see the PowerShell equivalent of this process in context, with sample parameters, refer to the following plans this task is part of:

- Management Domain Restore Plan

Create a Single-Host vSAN Datastore for VCF Instance Backup and Restore

Create a single-host vSAN datastore to use as a target for the OVA deployments of vCenter, NSX Manager, and SDDC Manager.

UI Procedure

1. Connect to the first ESX host in the management domain over SSH.
2. If your rebuilt system will use vSAN ESA, run the following commands:

```
# Step 1: Initialize vSAN ESA
esxcli vsan cluster new --vsanesa

# Step 2: Verify vSAN is enabled
esxcli vsan cluster get

# Step 3: List eligible disks (Look for: IsLocal=true, IsBootDevice=false, deviceType=Direct-Access)
esxcli storage core device list | grep -E "^eui\.|^naa\.|^mpx\.|Size:|Is Boot Device:|Is Local:|Device
 Type:|Is SSD:"

# Step 4: Add each eligible disk to the ESA storage pool (repeat per disk)
esxcli vsan storagepool add --disk=<DISK_CANONICAL_NAME>
```

3. If your rebuilt system will use vSAN OSA, run the following commands:

```
# Step 1: Initialize vSAN OSA
esxcli vsan cluster new

# Step 2: Verify vSAN is enabled
esxcli vsan cluster get

# Step 3: List eligible disks (Look for: IsLocal=true, IsBootDevice=false, deviceType=Direct-Access)
esxcli storage core device list | grep -E "^eui\.|^naa\.|^mpx\.|Size:|Is Boot Device:|Is Local:|Device
 Type:|Is SSD:"

# Step 4: If any CAPACITY disk is an SSD, tag it as capacityFlash first
esxcli vsan storage tag add --disk=<CAPACITY_SSD_CANONICAL_NAME> --tag=capacityFlash

# Step 5: Add each disk group (append --disks parameter for each capacity disk to be added)
esxcli vsan storage add --ssd=<CACHE_DISK_CANONICAL_NAME> --disks=<CAPACITY_DISK_CANONICAL_NAME> --
disks=<CAPACITY_DISK_CANONICAL_NAME>
```

```
# Step 6: Set FTT to 0 to allow deployment of VMs
esxcli vsan policy setdefault -c vdisk -p '(("hostFailuresToTolerate" i0) ("forceProvisioning" i1))'
esxcli vsan policy setdefault -c vmnamespace -p '(("hostFailuresToTolerate" i0) ("forceProvisioning" i1))'
```

PowerShell Equivalent

To see the PowerShell equivalent of this process in context, with sample parameters, refer to the following plans this task is part of:

- Management Domain Restore Plan

vCenter Restore Tasks for VCF Instance Backup and Restore

Perform these tasks that are related to the restore of vCenter in a VCF Instance.

Deploy a vCenter Appliance from an OVA File for VCF Instance Backup and Restore

To restore a vCenter instance, you first deploy a new appliance from an OVA file. You then use this appliance to restore the management domain or workload domain.

Prerequesites

- Confirm the OVA is from the vCenter build **matching the version and build recorded for the failed instance**. The restore fails if the versions differ, and the mismatch is only detected after the appliance is deployed.
- Confirm you have the deployment size and storage size of the failed instance. The replacement appliance must be the same size or larger; a smaller one fails the restore.
- If the failed vCenter had more than one virtual network adapter, plan to attach the additional adapters to the replacement appliance before you run the restore.

UI Procedure

> **Important:**
>
> In the case of the management domain vCenter, the target for OVA deployment will always be the first management ESX host. For workload domains, the target can also be the first management ESX host (if doing in parallel with management domain recovery) or it can be the management vCenter if the workload domain recovery is being performed after management domain recovery.

1. Download the vCenter OVA and ensure its available to the workstation where you will be performing the recovery from.
2. Log in to the first management ESX host at `https://<esx_fqdn>/ui` as the `root` user.
3. Right-click the host and select **Deploy OVF Template**.
   a. Step through the wizard populating the fields with data gathered during Gather Information for VCF Instance Backup and Restore.

OVF Property	Source	If it is wrong
Host Network Identity (FQDN)	Gather Information — FQDN	Restore fails: **The appliance PNID does not match the backup PNID**.
Host Network IP Address	Gather Information — IP address	Restore fails: **Local machine IP(s) do not match to any of the PNID resolved IPs**.
Host Network Prefix	Gather Information — prefix length	**Not checked.** The restore succeeds with the wrong subnet mask.
Host Network Default Gateway	Gather Information — default gateway	**Not checked.** The restore succeeds with the wrong gateway.
Host Network DNS Servers	Gather Information — DNS servers	**Not checked, and not overwritten by the restore.** The value entered here is the value the restored vCenter keeps.

OVF Property	Source	If it is wrong
Root Password	Gather Information — root user password	**Not checked.** See the note below.
Deployment size / storage size	Gather Information — deployment size	Restore fails if smaller than the failed instance. Larger is accepted.
Network / port group	As-built documentation	The appliance is unreachable and the restore cannot start.

4. Submit and deploy the OVA.

Note:

This wizard does not read the backup. Unlike the vCenter installer used in *Restore a vCenter Instance from a File-Based Backup*, no field here is pre-filled from the backup metadata. Every value must come from Gather Information for VCF Instance Backup and Restore.

Important:

Set the root password to **the password the failed vCenter had when the backup was taken** — the value retrieved in *Retrieve the Credentials for Restoring a vCenter Instance*. The restore preserves the password you enter here; it does not restore the original password from the backup. SDDC Manager and VCF Operations store the original password and fail to connect to the restored vCenter if it differs. If you must change the root password, do so after the restore completes and update it in SDDC Manager and VCF Operations at the same time.

Note:

The Single Sign-On administrator password behaves differently: it **is** restored from the backup. After the restore, **administrator@<sso_domain>** has the password it had when the backup was taken, regardless of anything entered during recovery.

Note:

If the failed vCenter had more than one virtual network adapter, attach the additional adapters to the replacement appliance now, before you run the restore. The restore configures only adapters that already exist; addresses for missing adapters are dropped and the restore still reports success.

PowerShell Equivalent

To see the PowerShell equivalent of this process in context, with sample parameters, refer to the following plans this task is part:

- Management Domain Restore Plan
- Workload Domain Restore Plan

Restore vCenter from a File-Based Backup for VCF Instance Backup and Restore

Restore the file-based backup of a vCenter instance to the freshly deployed vCenter appliance you deployed from an OVA file.

UI Procedure

1. Log in to the appliance management interface of the vCenter instance at `https://<vcenter_fqdn>:5480` as **root**.
2. Select the **Restore** and click **Next**.
3. Provide the backup location, the user name, and password for the backup location and encryption password (if applicable), and click **Next**.
4. On the **Ready to complete** page, review the details and click **Finish**.

PowerShell Equivalent

To see the PowerShell equivalent of this process in context, with sample parameters, refer to the following plans this task is part of:

- Management Domain Restore Plan
- Workload Domain Restore Plan

Update the Backup Configuration for vCenter for VCF Instance Backup and Restore

If the backup target for vCenter has changed, you must update the backup configuration for each vCenter instance.

1. Log in to the vCenter appliance management interface at `https://<vcenter_fqdn>`:5480 with the root user.
2. Navigate to **Backup** and click **Edit**.
3. Enter the new backup target details and click **Save**.

vSphere Cluster Restore Tasks for VCF Instance Backup and Restore

Perform these tasks to recover a vSphere cluster in your VCF Instance.

Record the Cluster Settings from the Restored vCenter for VCF Instance Backup and Restore

Before you can recover the vSphere clusters in the restored vCenter instance, you must first record the settings so they can be reapplied to the cluster.

vCenter has been restored and powered on.

1. Log in to the restored vCenter instance at `https://<restored_vcenter_fqdn>/ui` with the `administrator@vsphere.local` user.
2. Record the following settings for every cluster in the restored vCenter.
   - Virtual machine overrides
   - Virtual machine locations
   - Virtual machine tags
   - vSphere DRS groups and rules

Remove Non-Responsive ESX Hosts from the Inventory for VCF Instance Backup and Restore

Before you can repair a failed vSphere cluster, you must first remove the non-responsive hosts from the cluster.

Prerequisites

Ensure you have exported all cluster settings before proceeding.

> **CAUTION:**
>
> Refer to SDDC Manager UI to determine if the cluster being recovered is a vSAN Compute-Only cluster. If so, ensure that the vSAN Storage cluster it depends on has already been recovered. If the cluster is the primary cluster in the management domain then by definition it cannot be a vSAN Compute-Only cluster.

UI Procedure

1. In the vSphere Client, browse to the cluster and identify every host whose connection state is **Not Responding** or **Disconnected**. Make a list of their exact display names — you will need them repeatedly in later steps.
2. Note the cluster's MoRef ID (visible in the cluster's URL/managed object browser, or via **Get-Cluster | select ExtensionData.MoRef**), and determine the NSX Manager's version.
   a. **GET https://<nsx-manager>/api/v1/node** and note **product_version**.

   b. If this call fails or returns no version, stop here — do not proceed until NSX Manager is confirmed healthy and reachable.
3. Identify the NSX transport nodes that correspond to the non-responsive hosts.
   a. **GET https://<nsx-manager>/api/v1/transport-nodes/** and filter results to entries where **resource_type** is **TransportNode** and **node_deployment_info.os_type** is **ESXI**.
   b. From that filtered list, match each entry's **display_name** against the non-responsive host names from step 1, and record their transport node **id** values.
4. Identify the cluster's NSX compute collection.
   a. **GET https://<nsx-manager>/api/v1/fabric/compute-collections**.
   b. Find the entry whose **cm_local_id** matches the cluster's MoRef ID from step 2, and record its **external_id** — this is the compute collection ID used in the next steps.
5. Remove the non-responsive hosts from vCenter Server inventory.
   a. For each host identified in step 1, right-click it in the vSphere Client and select **Remove from Inventory** (equivalent to **Remove-VMHost -Confirm:$false**). Confirm the removal.
   b. This step must be completed for *all* non-responsive hosts before continuing — later steps assume they are already gone from vCenter Server.
6. Detach the Transport Node Profile from the cluster's compute collection.
   a. **POST https://<nsx-manager>/api/v1/fabric/compute-collections/<compute-collection-id>?action=remove_nsx**.
   b. For each transport node ID recorded in step 3, poll **GET https://<nsx-manager>/api/v1/transport-nodes/<id>/state** repeatedly until its **state** reports **orphaned**. Do not proceed to step 7 until every affected transport node reaches this state.
7. Force-delete each stale transport node.
   a. For each transport node ID from step 3, **DELETE https://<nsx-manager>/api/v1/transport-nodes/<id>?force=true&unprepare_host=false**.
   b. **unprepare_host=false** is required here — the host is gone and unreachable, so NSX cannot attempt to uninstall its own agents from it; this only removes NSX's bookkeeping record for that node.
8. Wait for the deleted transport nodes to fully flush from NSX Manager's inventory before proceeding.
   a. Repeatedly **POST https://<nsx-manager>/policy/api/v1/search/aggregate?page_size=50** with body **{"primary": {"resource_type": "HostTransportNode"}}**.
   b. Keep polling until none of the returned results' **display_name** values match any of the removed host names from step 1. This can take some time — do not move on early.
9. Reattach the cluster to its Transport Node Profile so NSX resumes auto-preparing hosts in this cluster.
   a. **GET https://<nsx-manager>/policy/api/v1/infra/host-transport-node-profiles** and find the profile whose **display_name** contains the cluster's name.
   b. **POST https://<nsx-manager>/api/v1/transport-node-collections** with a body creating a **TransportNodeCollection** resource that links the compute collection ID (step 4) to that Transport Node Profile's ID, for example:

```
{
  "resource_type": "TransportNodeCollection",
  "display_name": "<clusterName>",
  "description": "<clusterName>",
  "compute_collection_id": "<compute-collection-id>",
  "transport_node_profile_id": "<transport-node-profile-id>"
}
```

Result

The non-responsive hosts are removed from both vCenter Server and NSX Manager, and the cluster's compute collection is once again associated with its Transport Node Profile.

PowerShell Equivalent

To see the PowerShell equivalent of this process in context, with sample parameters, refer to the plans this task is part of here.

- Management Domain Restore Plan
- Workload Domain Restore Plan

Add New Hosts to the Cluster to the vCenter Instance for the Restored VCF Domain

You add new ESX hosts to the cluster in the restored vCenter instance for the VCF domain.

UI Procedure

1. Log in to the restored vCenter instance at `https://<restored_vcenter_fqdn>/ui` with the `administrator@vsphere.local` user.
2. In the **Hosts and clusters** inventory, navigate to the cluster.
3. Right-click the cluster and select **Add Hosts**.
4. Enter the FQDN, user name, and password for each host, and click **Next**.
5. When prompted to accept the SSL certificate for the new hosts, click **Accept**.
6. Click **Finish**.

PowerShell Equivalent

To see the PowerShell equivalent of this process in context, with sample parameters, refer to the following plans this task is part of:

- Management Domain Restore Plan
- Workload Domain Restore Plan

Migrate the ESX Hosts and VMkernel Adapters to the vSphere Distributed Switch on the Restored vCenter for VCF Instance Backup and Restore

You connect the new cluster hosts to the vSphere Distributed Switch and migrate the VMkernel adapters during a VCF Instance restore operation.

UI Procedure

1. Log in to the restored vCenter instance at `https://<restored_vcenter_fqdn>/ui` with the `administrator@vsphere.local` user.
2. Add the hosts to the vSphere Distributed Switch.
   a. In **Networking** inventory, right-click the vSphere Distributed Switch and select **Add and Manage Hosts**.
   b. Select **Add Hosts**and click **Next**.
   c. On the **Select hosts** page, select all the ESX hosts, and click **Next**.
   d. On the **Manage physical adapters** page, select a free physical adapter, for example, vmnic0, and click **Assign Uplink**.
   e. Select uplink1, and click **Next**.
   f. On the **Manage VMkernel adapters** page, update the following VMkernel adapters to assign them to the appropriate port group on the new distributed switch.

Table 1166: VMkernel Migration by Domain Type

Management Domain	Workload Domain
vmk0 – Management Network port group	vmk0 – Management Network port group
vmk1 – vMotion Network port group	N/A - Not yet created
vmk2 – vSAN Network port group	N/A - Not yet created

   g. To assign the VMkernel adapters, select the adapter and under actions for the corresponding port group, click **Assign**.
   h. Click **Next**.

i. On the **Migrate VM networking** step, click **Next**.
j. Review the information on the **Ready to complete** page and click **Finish**.

3. For a management domain cluster, migrate the management VMs to the original management port group.
   a. Right-click the temporary management port group and select **Migrate VMs to Another Network**.
   b. For destination network, select the management port group on the vSphere Distributed Switch, for example sfo-m01-vc01-vds01-management, and click **Next**.
   c. On the **Select VMs to migrate** page, select all management VMs and click **Next**.
   d. On the **Ready to complete** page, click **Finish**.
4. Remove the temporary standard switch on each ESX Host.
   a. Select the first ESX host and, on the Configure tab, select **Networking** > **Virtual Switches**.
   b. Expand **vSwitch0** and click the horizontal ellipsis.
   c. Click **Remove** and click **Yes**.
5. Add more host uplinks to the vSphere Distributed Switch.
   a. Right-click the distributed switch and select **Add and Manager Hosts**.
   b. Select **Manage Host Networking** and click **Next**.
   c. On the **Select hosts** step, select all the ESX hosts and click **Next**.
   d. On the **Manage physical adapters** step, select one or more required free physical adapters, for example, vmnic1, and from **Assign Uplink** select the desired uplinks to corresponding physical adapters, and click**Next**.
   e. Click **Next** and click **Next**.
   f. Review the information on the **Ready to complete** page and click **Finish**.

PowerShell Equivalent

To see the PowerShell equivalent of this process in context, with sample parameters, refer to the following plans this task is part of:

- Management Domain Restore Plan
- Workload Domain Restore Plan

Add VMkernel Adapters to Workload Domain ESX Hosts for VCF Instance Backup and Restore

You add vSphere vMotion and vSAN VMkernel adapters to the new ESX hosts in the workload domain.

UI Procedure

1. Log in to the restored vCenter instance at `https://<restored_vcenter_fqdn>/ui` with the `administrator@vsphere.local` user.
2. In the **Hosts and clusters** inventory, select the first ESX host and on the Configure tab, select **Networking** > **VMkernel adapters**.
3. Click **Add networking**.
4. Select **VMkernel Network Adapter** and click **Next**.
5. Select **Select an existing network**, select the port group for the VMkernel type, and click **Next**.
6. On the **Port properties** page, configure the following settings, leaving the default values for all other settings, and click **Next**.

Setting	vMotion VMkernel	vSAN VMkernel
TCP/IP stack	vMotion	Default
Available services	N/A	vSAN

7. On the **IPv4 settings** page, select **Use static IPV4 settings** and enter the IP details for each VMkernel adapter from your system documentation, and click **Next**.
8. On the **Ready to complete** page, click **Finish**.
9. Repeat the procedure for each ESX host in the cluster.

PowerShell Equivalent

To see the PowerShell equivalent of this process in context, with sample parameters, refer to the following plans this task is part of:

- Workload Domain Restore Plan

Recreate a vSAN Datastore for VCF Instance Backup and Restore

To recreate the vSAN datastore, you claim disks and create a vSAN datastore.

> **CAUTION:**
>
> Recreate the vSAN datastore only on clusters that host vSAN datastores. Do not perform this operation on compute-only clusters.
>
> If the datastore is vSAN ESA, the datastore is recreated automatically. Only Step 5 and Step 6 are required.

UI Procedure

1. Log in to the restored vCenter instance at `https://<restored_vcenter_fqdn>/ui` with the `administrator@vsphere.local` user.
2. In the **Hosts and clusters** inventory, navigate to the cluster and click **Configure**.
3. Under **vSAN**, click **Disk Management**, and click **Claim Unused Disks**.
4. Select the relevant disks for each tier and click **Claim**.

   > **CAUTION:**
   >
   > The process to claim the disks and create the vSAN datastore might take some time. Wait for it to complete.
5. After the disk groups are created, right-click the datastore and select **Rename**.
6. Enter the original datastore name and click **OK**.

PowerShell Equivalent

To see the PowerShell equivalent of this process in context, with sample parameters, refer to the following plans this task is part of:

- Workload Domain Restore Plan

Add Disk Groups to Management Hosts for VCF Instance Backup and Restore

To include the remaining management domain hosts in a vSAN OSA configuration, you must create disk groups on each host.

> **CAUTION:**
>
> Recreate the vSAN datastore only on clusters that host vSAN datastores. Do not perform this operation on compute-only clusters.

UI Procedure

1. Log in to the restored vCenter instance at `https://<restored_vcenter_fqdn>/ui` with the `administrator@vsphere.local` user.
2. In the **Hosts and clusters** inventory, navigate to the cluster and click **Configure**.
3. Under **vSAN**, click **Disk Management**, and click **Claim Unused Disks**.

   > **Note:**
   >
   > Disk Groups should already exist on the first host in the cluster as a result of creating the single host vSAN earlier in the process, so you will only be claiming disks from the remaining hosts
4. Select the relevant disks for each tier and click **Claim**.

CAUTION:

The process to claim the disks and create the vSAN datastore might take some time. Wait for it to complete.

PowerShell Equivalent

To see the PowerShell equivalent of this process in context, with sample parameters, refer to the plans this task is part of here.

- Management Domain Restore Plan

Set the Management Datastore Policy on the Rebuilt Datastore for VCF Instance Backup and Restore

After you have rebuild the vSAN datastore, you re-apply the management datastore policy.

UI Procedure

1. Log in to the restored vCenter instance at `https://<restored_vcenter_fqdn>/ui` with the `administrator@vsphere.local` user.
2. In the **Datastore** inventory, navigate to the datastore and click **Configure**.
3. Under **Default Storage Policy** , click **Edit**.
4. In the **Change Default Storage Policy** dialog box, select the documented datastore policy and click **OK**.
5. On the **VMs** tab of the datastore, select all virtual machines and from the context menu select **VM Policies** and **Edit VM Storage Policies**
6. Select **Yes**, select the correct datastore policy ,and click **OK**.

PowerShell Equivalent

To see the PowerShell equivalent of this process in context, with sample parameters, refer to the following plans this task is part of:

- Management Domain Restore Plan

Re-Enable vSphere HA on the Workload Domain Cluster for VCF Instance Backup and Restore

After recreating a vSAN datastore, you turn off and turn on vSphere HA again on the cluster so that vCenter Cluster Service can self-heal.

UI Procedure

1. Log in to the restored vCenter instance at `https://<restored_vcenter_fqdn>/ui` with the `administrator@vsphere.local` user.
2. In the **Hosts and clusters** inventory, navigate to the cluster and click **Configure**.
3. Under **Services**, click **vSphere Availability**, and under **vSphere HA is Turned ON** click the **Edit** button.
4. Switch off the **vSphere HA** toggle button and Click **OK**.
5. Under **Services**, click **vSphere Availability**, and under **vSphere HA is Turned ON** click the **Edit** button.
6. Switch on the **vSphere HA** toggle button and Click **OK**.

PowerShell Equivalent

To see the PowerShell equivalent of this process in context, with sample parameters, refer to the following plans this task is part of:

- Workload Domain Restore Plan

Update the Datastore ID for Restored Cluster in SDDC Manager Inventory for VCF Instance Backup and Restore

After you have rebuild the vSAN datastore, the ID of that datastore is new and must be updated in SDDC Manager inventory.

UI Procedure

1. Log in to the restored vCenter instance at `https://<restored_vcenter_fqdn>/ui` with the `administrator@vsphere.local` user.
2. In the **Hosts & Clusters** inventory, select the cluster.
3. In the address bar of the browser, locate the ClusterComputeResource string e.g. *`ClusterComputeResource:domain-c9:c2e52e64-2d52-4ba0-8c4a-a5d5f4510ea0`*
4. Record the following detail:
   a. moRef of cluster, for example *`domain-c9`* in the above example.
   b. id of vCenter, for example *`c2e52e64-2d52-4ba0-8c4a-a5d5f4510ea0`* in the above example.
5. In the **Datastores** inventory, select the rebuilt datastore, and in the address bar locate the Datastore string e.g. *`Datastore:datastore-15`*
6. Record the following detail"
   a. moRef of Datastore, or example *`datastore-15`* in the above example.
7. Open an SSH connection to the SDDC Manager appliance as the **vcf** user.
8. Switch to the **root** user.
   ```
   su -
   ```
9. Connect to the Postgres database.
   ```
   psql -U postgres -h localhost -d platform;
   ```
10. Set wide view for readability.
    ```
    \x
    ```
11. Query the *vCenter* table.
    ```
    select * from vCenter;
    ```
12. Record *`id`* where *`vm_hostname`* matches the vCenter FQDN of the cluster being recovered.
13. Query the cluster table.
    ```
    select * from cluster;
    ```
14. Locate the record where the *`vcenter_id`* matches the *`id`* gathered in Step 11 and the *`source_id`* matches the moRef of the cluster that was captured in Step 4, and record the cluster *`id`*
15. Update the record using the *`moRef`* of the new datastore from Step 6 and the id from Step 13.
    ```
    update cluster set primary_datastore_source_id='<DATASTOREMOREF>' where id='CLUSTERID';
    ```

PowerShell Equivalent

To see the PowerShell equivalent of this process in context, with sample parameters, refer to the following plans this task is part of:

- Management Domain Restore Plan
- Workload Domain Restore Plan

Update the Cluster Host IDs for Restored Cluster in SDDC Manager Inventory for VCF Instance Backup and Restore

Procedure

1. Log in to the recovered vCenter instance at **https://<recovered_vcenter_fqdn>/ui** with an SSO administrator user (e.g. **administrator@vsphere.local**).
2. In the **Hosts & Clusters** inventory, expand the target cluster and select the first host to record.

3. In the address bar of the browser, locate the **HostSystem** string, for example: **HostSystem:host-87:c2e52e64-2d52-4ba0-8c4a-a5d5f4510ea0**
   a. Record the following detail:
      - moRef of the host, for example **host-87** in the above example.
      - FQDN (hostname) of the host, as shown in the inventory.
4. Repeat Step 3 for every host currently in the cluster.
5. Open an SSH connection to the SDDC Manager appliance as the **vcf** user.
6. Switch to the **root** user.
   ```
   su -
   ```
7. Connect to the Postgres database.
   ```
   psql -U postgres -h localhost -d platform;
   ```
8. Set wide view for readability.
   ```
   \x
   ```
9. Query the **host** table for the current **id** and **source_id** of the host being processed.
   ```
   SELECT id, source_id FROM host WHERE hostname='<HOST_FQDN>';
   ```
10. Record the **id** (the SDDC Manager host UUID) and compare the returned **source_id** against the moRef captured for this host in Step 3.
    – If **source_id** already matches the live moRef, no update is required for this host — continue to the next host.
    – If it differs (or is not set), continue to Step 11.
11. Update the record, using the **id** from Step 10 and the moRef captured in Step 3.
    ```
    UPDATE host SET source_id='<NEWMOREF>' WHERE id='<HOSTID>';
    ```
12. Verify the change.
    ```
    SELECT source_id FROM host WHERE id='<HOSTID>';
    ```
13. Repeat Steps 9 through 12 for every remaining host recorded in Step 3.

Reapply Cluster Settings for the Restored vSphere Cluster for VCF Instance Backup and Restore

After all workloads are restored on a vSphere cluster for the management or workload domain, you reapply the cluster settings for virtual machine overrides, locations, and vSphere DRS rules and groups.

1. Log in to the restored vCenter instance at `https://<restored_vcenter_fqdn>/ui` with the `administrator@vsphere.local` user.
2. Re-apply the following settings for every cluster in the restored vCenter.
   - Virtual machine overrides
   - Virtual machine locations
   - Virtual machine tags
   - vSphere DRS groups and rules

NSX Restore Tasks for VCF Instance Backup and Restore

Perform these tasks that are related to restoring NSX.

Deploy the First NSX Manager Cluster Node from an OVA File for VCF Instance Backup and Restore

When the NSX Manager cluster nodes (single node or three node cluster) are in a failed state, you begin the restore process by restoring the first NSX Manager cluster node from an OVA file.

> **Note:**
>
> The first NSX Manager node is defined as the node that took the backup of the NSX Manager cluster. To identify this node, examine the name of the folder that is created during backup. The folder name includes the node's IP address.

UI Procedure

Important:

For the management domain NSX Manager the target for OVA deployment is always the first management ESX host. For workload domains, the target can also be the first management ESX host (if doing in parallel with management domain recovery) or it can be the management vCenter if the workload domain recovery is being performed after management domain recovery.

1. Download the NSX Manager OVA and ensure its available to the workstation where you will be performing the recovery from.
2. Log in to the first management ESX host at `https://<esx_fqdn>/ui` as the `root` user.
3. Right-click the host and select **Deploy OVF Template**.
4. Follow the wizard, filling the information you gathered during Gather Information for VCF Instance Backup and Restore.
5. Submit and deploy the OVA file.

PowerShell Equivalent

To see the PowerShell equivalent of this process in context, with sample parameters, refer to the following plans this task is part of:

- Management Domain Restore Plan
- Workload Domain Restore Plan

Restore NSX Manager from a File-Based Backup for VCF Instance Backup and Restore

You restore the file-based backup of the first NSX Manager cluster node to the newly-deployed NSX Manager instance.

Before you Begin

- Verify that you have the login credentials (username and password or the SSH private key matching the public key stored on the backup file server) for the backup file server.
- Verify that you have the SSH fingerprint of the backup file server. Support includes key size 256-bit, 384-bit, and 521-bit. Ensure whatever key size is used at time of backup is used at time of restore.
- Verify that you have the passphrase of the backup file.
- Identify which backup you want to restore. Take note of the IP or FQDN of the NSX appliance that took the backup.
- Ensure the network setup where you are performing the restore has the same set of network connectivity as the system on which you performed the backup. For example, the same VIPs, DNS, NTP communication,and so on. If network connectivity is not same, fix the inconsistencies before adding a second or third node to the restored system.
- Familiarize yourself with the Management Plane upgrade process as part of restoring a backup during an upgrade.

UI Procedure

1. Log in to the NSX Manager user interface at `https://<nsx_manager-fqdn>` with the **admin** user.
2. On the main navigation bar, click **System**.
3. In the left navigation pane, under **Lifecycle Management**, click **Backup & Restore**.
4. In the **Backup & Restore** pane, under **SFTP server**, click **Edit**.
5. In the **Backup configuration** dialog box, enter the details for your backup server, and click **Save**.
6. Under **Backup history**, select the source backup, and click **Restore**.
7. During the restore, when prompted, reject adding NSX Manager nodes by clicking **I understand** and **Resume**.
8. If a **Confirm CM/VC Connectivity** dialog box appears, click **I understand the message mentioned above and wish to proceed**, and click **Resume**.
9. If a fabric node discovery timeout dialog box appears, click **Resume** to continue the restore operation.

PowerShell Equivalent

To see the PowerShell equivalent of this process in context, with sample parameters, refer to the following plans this task is part:

- Management Domain Restore Plan
- Workload Domain Restore Plan

Verify the State of the First NSX Manager Cluster Node for VCF Instance Backup and Restore

After you restore the first NSX Manager cluster node, you verify the services state from the Web console of the restored node VM.

1. Log in to the first management ESX host at `https://<esx_fqdn>/ui` as the `root` user.
2. Click the VM name of the newly-deployed first NSX Manager cluster node, click **Launch Web Console**, and log in by using administrator credentials.
3. Run the command to view the cluster status.

   ```
   get cluster status
   ```

   The services on the single-node NSX Manager cluster appear as `UP`.

Redeploy a Failed NSX Manager Node for VCF Instance Backup and Restore

You deploy a new NSX Manager instance by using the configuration of the failed node.

UI Procedure

1. Log in to the management domain vCenter instance at `https://<vcenter_fqdn>/ui` with the `administrator@vsphere.local` user.
2. In the **Hosts and clusters** inventory, right-click the management cluster and select **Deploy OVF Template**.
3. On the **Select an OVF template** page, select **Local file**, click **Upload files**, navigate to the location of the NSX Manager OVA file, click **Open**, and click **Next**.
4. On the **Select a name and folder** page, enter the VM name and click **Next**.
5. On the **Select a compute resource** page, select the cluster and click **Next**.
6. On the **Review details** page, click **Next**.
7. **Note:**

   For the management domain, select **Medium**, and for workload domains, select **Large** unless you changed these defaults during deployment.
8. On the **Select storage** page, select the management vSAN datastore, and click **Next**.
9. On the **Select networks** page, from the **Destination network** drop-down menu, select management distributed port group, and click **Next**.
10. On the **Customize template** page, enter these values and click **Next**.

Setting	Value
System root user password	<failed_nsx_cluster_node_root_password>
CLI admin user password	<failed_nsx_cluster_node_admin_password>
CLI audit user password	<failed_nsx_cluster_node_audit_password>
Hostname	<failed_nsx_cluster_node_fqdn>
Default IPv4 gateway	<failed_nsx_cluster_node_gw>
Management network IPv4 address	<failed_nsx_cluster_node_ip>
Management network netmask	<failed_nsx_cluster_node_mask>
DNS server list	<dns_server_list>
NTP server list	<ntp_server_list>

Setting	Value
Enable SSH	Selected
Allow root SSH logins	Deselected

11. On the **Ready to complete** page, review the deployment details and click **Finish**.
12. Repeat for the remaining failed node.

PowerShell Equivalent

To see the PowerShell equivalent of this process in context, with sample parameters, refer to the plans this task is part of here.

- Management Domain Restore Plan
- Workload Domain Restore Plan

Join NSX Manager Nodes to the NSX Manager Cluster for VCF Instance Backup and Restore

You retrieve the ID and API thumbprint of the NSX Manager cluster, and use them to join the newly-deployed NSX Manager instance to the cluster.

UI Procedure

1. Log in to the management domain vCenter instance at `https://<vcenter_fqdn>/ui` with the `administrator@vsphere.local` user.
2. In the **VMs and templates** inventory, click the VM of an operational NSX Manager node in the cluster.
3. Click **Launch web console**, and log in by using **admin** credentials.
4. Retrieve the ID of the NSX Manager cluster.
   a. Run the command to view the cluster ID.
      ```
      get cluster config | find Id:
      ```
   b. Write down the cluster ID.
5. Retrieve the API thumbprint of the NSX Manager API certificate.
   a. Run the command to view the certificate API thumbprint.
      ```
      get certificate api thumbprint
      ```
   b. Write down the certificate API thumbprint.
6. Close the VM Web console.
7. In the vSphere Client, click the VM of the newly deployed NSX Manager node.
8. Click **Launch Web console**, and log in by using **admin** credentials.
9. Run the command to join the new NSX Manager node to the cluster.
   ```
   join newly_deployed_node_ip_address cluster-id cluster_id thumbprint api_thumbprint username admin
   ```
10. Repeat for the remaining failed node.

PowerShell Equivalent

To see the PowerShell equivalent of this process in context, with sample parameters, refer to the following plans this task is part of:

- Management Domain Restore Plan
- Workload Domain Restore Plan

Redeploy Specific NSX Edges in an NSX Edge Cluster

To restore only the NSX Edge nodes that are missing from the vCenter inventory (for example, after a partial cluster failure), identify which NSX Edge nodes are missing, recreate a placeholder VM shell for each one in vCenter, then use the NSXNSX API to point NSX Manager at the new VM and redeploy the NSX Edge node.

- Administrator credentials for the NSX Manager that owns the NSX Edge cluster.

- Administrator credentials for the vCenter that hosts the cluster.
- The **extracted-sddc-data.json** file produced during backup extraction, in case the original management portgroup or vSphere Distributed Switch can no longer be resolved directly in vCenter.
- Sufficient compute and storage capacity in the target cluster to place new NSX Edge VM shells.

1. Identify all NSX Edge transport nodes associated with the cluster.
   a) In NSX Manager, send a **GET** request to **https://<nsx-manager>/api/v1/transport-nodes/** and note every result whose **node_deployment_info.resource_type** is **EdgeNode**.
   b) In vCenter note the MoRef IDs of the cluster's resource pools (excluding the default **Resources** pool) and of the cluster object itself.
   c) Match each Edge's **node_deployment_info.deployment_config.vm_deployment_config.compute_id** against those MoRef IDs to determine which NSX Edge belong to this cluster, and whether each one belongs to a specific resource pool or directly to the cluster.
2. Determine which of those NSX Edge VMs are missing from vCenter.
   a) In the vSphere Client, browse to the cluster and check whether a VM with the same name as each NSX Edge node**display_name** exists in inventory.
   b) Make a list of the NSX Edge nodes that have no matching VM — these are the ones that need to be redeployed.
3. For each missing NSX Edge node, retrieve its current placement configuration.
   a) Send a **GET** request to **https://<nsx-manager>/api/v1/transport-nodes/<edgenode-id>** and note the **vm_deployment_config** section — specifically **resource_allocation.cpu_count**, **resource_allocation.memory_allocation_in_mb**, **reservation_info.cpu_reservation.reservation_in_shares**, **data_network_ids**, and **management_network_id**.
   b) In vCenter Server, resolve **management_network_id** to a distributed portgroup name and VDS name (match it against the portgroup's MoRef in the Networking inventory view). If it can no longer be resolved (for example, the

VDS was rebuilt during recovery), instead use the management portgroup and VDS name recorded for this cluster in **extracted-sddc-data.json**.

c) Note the cluster's primary datastore.

4. Create a placeholder VM shell for the missing NSX Edge node.
   a) In the vSphere Client, right-click the resource pool identified in step 1 (or the cluster, if the NSX Edge node was not in a resource pool) and select **New Virtual Machine**.
   b) Name the VM exactly the same as the NSX Edge node's **display_name** from step 3.
   c) Place it on any host in the cluster, using the datastore identified in step 3.3.
   d) Set the guest OS to a generic Linux 64-bit guest, a 200 GB thin-provisioned disk, and connect its first network adapter to the management portgroup identified in step 3.2.
   e) Do not power on the VM.
5. Set the new VM's resource allocation to match the original NSX Edge node.
   a) Edit the VM's resource settings and set the memory reservation to the value from **resource_allocation.memory_allocation_in_mb** (step 3.1).
   b) Set the CPU shares level (Low/Normal/High) to match **reservation_info.cpu_reservation.reservation_in_shares** (step 3.1).
6. Add the NSX Edge node's remaining network adapters.
   a) For every network ID in **data_network_ids** (step 3.1) other than the management network, add a VMXNET3 network adapter to the VM, connected to the matching distributed portgroup, set to connect at power on.
7. Update the NSX Edge node's placement references in NSX Manager .
   a) Note the new VM's MoRef ID in vCenter.
   b) In the **vm_deployment_config** JSON retrieved in step 3.1, set **storage_id** to the datastore's MoRef ID.
   c) Build a JSON payload containing **vm_id** (the new VM's MoRef), the updated **vm_deployment_config**, and a **node_user_settings** object with **cli_username** set to **admin** and **audit_username** set to **audit**.
   d) **POST** that payload to **https://<nsx-manager>/api/v1/transport-nodes/<edgenode-id>?action=addOrUpdatePlacementReferences**.
8. Check the NSX Edge node's deployment state.
   a) **GET https://<nsx-manager>/api/v1/transport-nodes/<edgenode-id>/state** and check **node_deployment_state.state**.
   b) If the state is already **success**, no further action is required for this Edge.
   c) If the state is **MPA_DISCONNECTED**, **VM_PLACEMENT_REFRESH_FAILED**, or **NODE_READY**, continue to step 9.
   d) If the state is any other value, do not proceed — review the Edge in NSX Manager before retrying.
9. Redeploy the NSX Edge node.
   a) **GET https://<nsx-manager>/api/v1/transport-nodes/<edgenode-id>** and copy the full JSON response body.
   b) **POST** that same JSON body to **https://<nsx-manager>/api/v1/transport-nodes/<edgenode-id>?action=redeploy**.
10. Repeat steps 3 – 9 for each remaining missing Edge.

NSX Manager provisions each redeployed NSX Edge node onto the new VM shell you created and rejoins it to the cluster.

In the NSX Manager UI, monitor the configuration status of each redeployed NSX Edge node until it shows **Success**.

If an NSX Edge node instead reports that its NSX Edge VM is present in NSX inventory but missing in vCenter go to **Alarms** in NSX Manager, select the alarm with Event Type **Edge VM Present In NSX Inventory Not Present In vCenter**, choose **Resolve**, and retry the redeploy step (step 9) for that NSX Edge node.

PowerShell Equivalent

To see the PowerShell equivalent of this process in context, with sample parameters, refer to the following plans this task is part of:

- Management Domain Restore Plan

- Workload Domain Restore Plan

Verify the State of the NSX Edge Cluster Nodes for VCF Instance Backup and Restore

After completing all NSX Edge node redeployments, you must verify the state of the NSX Edge cluster nodes.

1. Log in to the NSX Manager user interface at `https://<nsx_manager-fqdn>` with the **admin** user.
2. On the main navigation bar, click **System**.
3. In the left pane, under **Configuration**, click **Fabric** > **Edges**.
4. Click the **Edge clusters** tab.
5. Verify that all edge transport nodes show these values.

Setting	Value
Configuration state	Success
Node status	Up
Tunnels	Up

SDDC Manager Restore Tasks or VCF Instance Backup and Restore

Perform these tasks that are related to restoring SDDC Manager.

Deploy the SDDC Manager Appliance from an OVA File for VCF Instance Backup and Restore

You deploy a new SDDC Manager appliance by using the OVA file that you downloaded during the preparation for restore.

UI Procedure

1. Mount the SDDC Manager ISO in your file manager.
2. Browse to the first management ESX host login as an administrator.
3. Right-click the host and select **Deploy OVF Template**.
4. Step through the wizard populating the fields with data gathered during Gather Information for VCF Instance Backup and Restore.
5. Submit and deploy the OVA.

PowerShell Equivalent

To see the PowerShell equivalent of this process in context, with sample parameters, refer to the plans this task is part of here.

- Management Domain Restore Plan

Update vCenter SSH Keys in the SDDC Manager Backup for VCF Instance Backup and Restore

To be able to restore of SDDC Manager in VCF versions earlier than VCF 9.1.1, you must update the SSH keys for the new management vCenter appliance in the backup archive.

Prerequisites

Verify that the vCenter restore operation is complete.

Manual Procedure

1. Copy the encrypted backup file of SDDC Manager to the `/tmp` folder on the newly-deployed SDDC Manager appliance.

2. Open an SSH connection to the SDDC Manager appliance, connect using the **vcf** user.
3. Switch to the **root** user and change to the `/tmp` folder.
   ```
   su -
   cd /tmp
   ```
4. Replace the values in the sample code with your values, these will be used in later steps.
   ```
   export tmpFolderPath='/tmp/'
   export backupFile='vcf-backup-sfo-vcf01-sfo-rainpole-io-xxxx-xx-xx-xx-xx-xx.tar.gz'
   export vcenterFqdn='sfo-m01-vc01.sfo.rainpole.io'
   export encryptionPassword='VMw@re1!VMw@re1!'
   backupFolder="${backupFile%.tar.gz}"
   ```
5. In the appliance console, run the following command to extract the backup archive.
   ```
   tail -c +5 $tmpFolderPath$backupFile | OPENSSL_FIPS=1 openssl enc -d -aes-256-cbc -pbkdf2 -iter 600000 -md
    sha256 | tar -xz
   ```
   When prompted, enter the `encryption_password`.
6. Retrieve the new **ecdsa-sha2-nistp256** and **ssh-rsa** SSH keys for the management vCenter.
   ```
   ecdsaKey=$(ssh-keyscan -t ecdsa $vcenterFqdn 2>/dev/null | awk '/ecdsa-sha2-nistp256/ {print $3}')
   rsaKey=$(ssh-keyscan -t rsa $vcenterFqdn 2>/dev/null | awk '/ssh-rsa/ {print $3}')
   ```
7. Replace the existing SSH keys with the new SSH keys for the management vCenter returned in the previous step.
   ```
   sed -i '/"host": "'"${vcenterFqdn}"'"/ { N; /"keyType": "ssh-rsa"/ { N; s|"key": "[^"]*"|"key":
    "'"${rsaKey}"'"|; } }' $backupFolder/appliancemanager_ssh_knownHosts.json
   sed -i '/"host": "'"${vcenterFqdn}"'"/ { N; /"keyType": "ecdsa-sha2-nistp256"/ { N; s|"key":
    "[^"]*"|"key": "'"${ecdsaKey}"'"|; } }' $backupFolder/appliancemanager_ssh_knownHosts.json
   ```
8. Rename the original backup archive.
   ```
   mv $tmpFolderPath$backupFile $tmpFolder$backupFile.original
   ```
9. Re-encrypt the backup folder as an archive using the same archive name.
   ```
   tar -cz $tmpFolderPath$backupFolder | OPENSSL_FIPS=1 openssl enc -aes-256-cbc -md sha256 -out
    $tmpFolderPath$backupFile -pass env:encryptionPassword
   ```

PowerShell Equivalent

To see the PowerShell equivalent of this process in context, with sample parameters, refer to the following plans this task is part of:

- Management Domain Restore Plan

Restore SDDC Manager from a File-Based Backup for VCF Instance Backup and Restore

You restore the file-based backup on the newly-deployed SDDC Manager appliance.

Manual Procedure

1. If you have not already uploaded the SDDC Manager backup to the `/tmp` folder on the newly-deployed SDDC Manager appliance then do so now
2. Open an SSH connection to the SDDC Manager appliance, connect using the **vcf** user.
3. Switch to the **root** user.
   ```
   su -
   ```
4. Backup the `restore_status.json` file.
   ```
   cp /opt/vmware/sddc-support/backup/restore_status.json /opt/vmware/sddc-support/backup/
   restore_status.json.bak
   ```
5. Replace the values in the sample code with your values, these will be used in later steps.
   ```
   export sddcManagerFqdn='sfo-vcf01.sfo.rainpole.io'
   export sddcManagerUsername='admin@local'
   export sddcManagerPassword='***************'

   export tmpFolderPath='/tmp/'
   ```

```
export backupFile='/tmp/vcf-backup-sfo-vcf01-sfo-rainpole-io-xxxx-xx-xx-xx-xx-xx.tar.gz'
export encryptionPassword='***************'
```

6. Retrieve an access token from the SDDC Manager instance by running the following command.

```
sddcManagerToken=$(curl -k -s -X POST "https://${sddcManagerFqdn}/v1/tokens" \
    --header "Content-Type: application/json" \
    -d "{\"username\":\"${sddcManagerUsername}\", \"password\":\"${sddcManagerPassword}\"}" \
| jq -r '.accessToken' | tee /dev/tty )
```

7. Create the JSON payload for the restore by running the following command.

```
sddcRestorePayload=$(jq -n \
  --arg file "${tmpFolderPath}${backupFile}" \
  --arg pass "${encryptionPassword}" \
  '{
    "elements": [ { "resourceType": "SDDC_MANAGER" } ],
    "backupFile": $file,
    "encryption": { "passphrase": $pass }
  }' | tee /dev/tty )
```

8. Perform a restore of the SDDC Manager instance by running the following command.

```
restoreId=$(curl -k -s -X POST "https://${sddcManagerFqdn}/v1/restores/tasks" \
    --header "Authorization: Bearer ${sddcManagerToken}" \
    --header "Content-Type: application/json" \
    --header "Accept: application/json" \
    --data "$sddcRestorePayload" \
| jq -r '.id' | tee /dev/tty )
```

9. Monitor the restore task by running the following command.

```
curl -k -s -X GET "https://${sddcManagerFqdn}/v1/restores/tasks/${restoreId}" \
    --header "Authorization: Bearer ${sddcManagerToken}" \
    --header "Content-Type: application/json" \
    --header "Accept: application/json" \
| jq
```

PowerShell Equivalent

To see the PowerShell equivalent of this process in context, with sample parameters, refer to the following plans this task is part of:

- Management Domain Restore Plan

Update ESX Host SSH Keys for VCF Instance Backup and Restore

For each ESX host that you rebuilt as part of restoring the management cluster or a cluster in a workload domain, you must update the ESX host SSH keys in the `known_hosts` file on SDDC Manager.

Download `refreshsshkeys.py` from Broadcom knowledge base article 313210.

1. Enable SSH on all hosts in the cluster.
2. Copy the `refreshsshkeys.py` file to the `/tmp` directory on the SDDC Manager appliance by using a secure file copy utility as the **vcf** user.
3. Open an SSH connection to the SDDC Manager appliance, connect using the **vcf** user.
4. Switch to the **root** user and change to the `/tmp` folder.

```
su -
cd /tmp
```

5. Update the ESX host SSH keys in the `known_hosts` file by running the following command.`refreshsshkeys.py`.

```
"yes" | python refreshsshkeys.py
```

Synchronize the SDDC Manager Service Account on the ESX Hosts for VCF Instance Backup and Restore

The service account of SDDC Manager on the ESX hosts must be in sync with the SDDC Manager inventory. You must manually set a new password on each ESX host and perform a password remediate operation in SDDC Manager.

UI Procedure

1. If the service account does not exist, create a service account on the ESX host.
   a. Log in to the first ESX host in the cluster using the host client as **root**.
   b. Navigate to **Manage** > **Security & users** > **Users**, and click **Add user**.
   c. Enter the SDDC Manager service user name and password, and click **Add**.

      Note:

      The service account user name must be in the **svc-vcf-*ESX_hostname*** format.
2. If the service account exists, set a new password on the ESX host.
   a. Log in to the first ESX host in the cluster using the host client as **root**.
   b. In the host client on the first ESX host, navigate to **Manage** > **Security & users** > **Users**.
   c. Select the SDDC Manager service account **svc-vcf-*ESX_hostname*** and click **Edit user**.
   d. Set a new password and click **Save**.
3. Perform password remediation in SDDC Manager.
   a. Log in to SDDC Manager
   b. Navigate to **Developer Center** > **Credentials** > **GET**.
   c. Enter the ESX FQDN in the resourceName field and click Execute.
   d. Locate the service account from Step 1 or Step 2 and expand the output and record the output of the account with **accountType** of **SERVICE**.
   e. Construct the following JSON payload using the password you set in step 1 or 2.

```
{
    "elements": [ {
        "credentials": [ {
            "accountType": "SERVICE",
            "credentialType": "SSH",
            "password": "************",
            "username": "svc-vcf-sfo01-m01-r01-esx02"
        } ],
        "resourceId": "0f11d65d-1b08-4943-b920-5418c7e43c40",
        "resourceName": "sfo01-m01-r01-esx02.sfo.rainpole.io",
        "resourceType": "ESXI"
    } ],
    "operationType": "REMEDIATE"
}
```

4. Expand /v1/credentials/PATCH and paste the JSON payload into the body field, and click Execute. Wait for the password remediation to complete.
5. Repeat the steps for all hosts in the workload domain.

PowerShell Equivalent

To see the PowerShell equivalent of this process in context, with sample parameters, refer to the following plans this task is part of:

- Management Domain Restore Plan
- Workload Domain Restore Plan

Update the Backup Configuration in SDDC Manager for VCF Instance Backup and Restore

If the SFTP backup target for SDDC Manager and NSX has changed, you must update the SSH key for the backup configuration in SDDC Manager.

1. Log in to the SDDC Manager console at `https://<sddc_manager_fqdn>/` with the **administrator@vsphere.local** user.
2. Navigate to **Administration** > **Backup**.
3. Click the **Site Settings** tab.
4. On the **Site Settings** tab, click **Edit**.
5. Enter the new backup target details and click **Save**.

Repoint SDDC Manager Depot to an Offline Depot for VCF Instance Backup and Restore

To deploy a new VCF management services runtime cluster, you must point SDDC Manager to an offline depot.

UI Procedure

1. Replace the values in the sample code with your values, these will be used in later steps.
```
export sddcManagerFqdn='sfo-vcf01.sfo.rainpole.io'
export sddcManagerUsername='admin@local'
export sddcManagerPassword='***************'

export offlineDepotFqdn="depot.rainpole.io"
export offlineDepotPort="443"
export offlineDepotUsername="depotuser"
export offlineDepotPassword="***************"
```
2. Retrieve an access token from the SDDC Manager instance by running the following command.
```
sddcManagerToken=$(curl -k -s -X POST "https://${sddcManagerFqdn}/v1/tokens" \
    --header "Content-Type: application/json" \
    --data "{\"username\":\"${sddcManagerUsername}\", \"password\":\"${sddcManagerPassword}\"}" \
| jq -r '.accessToken' | tee /dev/tty )
```
3. Retrive the depot service key and export the current configuration details to `original-depotServices-config.json` by running the following command.
```
serviceConfigKey=$(curl -k -s -X GET "https://${sddcManagerFqdn}/v1/services-config" \
    --header "Authorization: Bearer ${sddcManagerToken}" \
    --header "Content-Type: application/json" \
    --header "Accept: application/json" \
| jq . | tee original-depotServices-config.json | jq -r '.services[0].key')
```
4. Delete the current depot service configuration by running the following command.
```
curl -k -s -X DELETE "https://${sddcManagerFqdn}/v1/services-config/${serviceConfigKey}" \
    --header "Authorization: Bearer ${sddcManagerToken}" \
    --header "Content-Type: application/json" \
    --header "Accept: application/json"
```
5. Retrieve the offline depot certificate to `trusted-cert-spec.json` by running the following command.
```
echo "{ \"certificate\" : $(openssl s_client \
    -connect ${offlineDepotFqdn}:${offlineDepotPort} \
    2>/dev/null </dev/null \
    | sed -ne '/-BEGIN CERTIFICATE-/,/-END CERTIFICATE-/p' \
    | jq -sR), \"certificateUsageType\" : \"TRUSTED_FOR_OUTBOUND\"}" \
  > trusted-cert-spec.json
```
6. Trust the offline depot certificate on SDDC Manager by running the following command.
```
curl -k -s -X POST "https://${sddcManagerFqdn}/v1/sddc-manager/trusted-certificates" \
    --header "Authorization: Bearer ${sddcManagerToken}" \
    --header "Content-Type: application/json" \
```

```
        --header "Accept: application/json" \
        --data @trusted-cert-spec.json
```

7. Configure the offline depot by running the following command.

```
curl -k -s -X PUT "https://${sddcManagerFqdn}/v1/system/settings/depot" \
    --header "Authorization: Bearer ${sddcManagerToken}" \
    --header "Content-type: application/json" \
    --header "Accept: application/json" \
    --data "{
        \"offlineAccount\": {
        \"username\": \"${offlineDepotUsername}\",
        \"password\": \"${offlineDepotPassword}\"
        },
        \"depotConfiguration\": {
        \"isOfflineDepot\": true,
        \"hostname\": \"${offlineDepotFqdn}\",
        \"port\": ${offlineDepotPort}
        }
    }"
```

PowerShell Equivalent

To see the PowerShell equivalent of this process in context, with sample parameters, refer to the plans this task is part of here.

- VCF Fleet Component Restore Plan

Download Binaries for VCF Management Services Runtime for VCF Instance Backup and Restore

To redeploy VCF management services runtime, you must download the binary on SDDC Manager.

UI Procedure

1. Log in to the SDDC Manager UI.
   a. Navigate to **Lifecycle Management** > **Binary Management** > **Install Binaries**
2. Select **VCF services runtime** and select **Download**.
3. Wait until binary has been downloaded and validated.

PowerShell Equivalent

To see the PowerShell equivalent of this process in context, with sample parameters, refer to the plans this task is part of here.

- VCF Fleet Component Restore Plan

Repoint SDDC Manager Depot to Fleet Depot Service for VCF Instance Backup and Restore

Once VCF management services and fleet management components have been restored, you reinstate the original SDDC Manager depot configuration.

> **Important:**
>
> You will need the `original-depotServices-config.json` created as part of the Repoint SDDC Manager Depot to an Offline Depot for VCF Instance Backup and Restore procedure.

UI Procedure

1. Open an SSH connection to the SDDC Manager appliance, connect using the **vcf** user.
2. Switch to the **root** user.

```
su -
```

3. Set variables

```
export ADMIN_PASSWORD='VMw@re1!VMw@re1!'
export ORIGINAL_CONFIG_FILE="originalDepotServicesconfig.json"
```

4. Replace the values in the sample code with your values, these will be used in later steps.

```
export sddcManagerFqdn='sfo-vcf01.sfo.rainpole.io'
export sddcManagerUsername='admin@local'
export sddcManagerPassword='***************'
```

5. Retrieve an access token from the SDDC Manager instance by running the following command.

```
sddcManagerToken=$(curl -k -s -X POST "https://${sddcManagerFqdn}/v1/tokens" \
    --header "Content-Type: application/json" \
    --data "{\"username\":\"${sddcManagerUsername}\", \"password\":\"${sddcManagerPassword}\"}" \
| jq -r '.accessToken' | tee /dev/tty )
```

6. Delete the existing offline depot configuration by running the following command.

```
curl -k -s -X DELETE "https://${sddcManagerFqdn}/v1/system/settings/depot" \
    --header "Authorization: Bearer ${sddcManagerToken}" \
    --header "Content-Type: application/json" \
    --header "Accept: application/json" \
| jq
```

7. Reinstate original fleet depot services-config from the saved `original-depotServices-config.json` file by running the following command.

```
curl -k -s -X PUT "https://${sddcManagerFqdn}/v1/services-config" \
    --header "Authorization: Bearer ${sddcManagerToken}" \
    --header "Content-type: application/json" \
    --header "Accept: application/json" \
    --data @original-depotServices-config.json \
| jq
```

PowerShell Equivalent

To see the PowerShell equivalent of this process in context, with sample parameters, refer to the plans this task is part of here.

- VCF Fleet Component Restore Plan

VCF Operations Restore Tasks for VCF Instance Backup and Restore

Perform these tasks that are related to restoring the VCF Operations nodes.

Restore VCF Operations from an Image-Based Backup for VCF Instance Backup and Restore

Restore a simple or high availability deployment of VCF Operations by using an image-based backup tool.

- **Backup Data Available**: Ensure that you have a recent backup of your VCF Operations cluster.
- **VCF Operations Cluster Configuration**: Ensure you know the details of your VCF Operations cluster configuration, such as cluster nodes, user credentials.
- **Backup Software**: Ensure that the backup solution you are using is compatible with your VCF Operations version.

Table 1167: VCF Operations Failure Scenarios

Component	Deployment Model	Failure Scenario	Restore Process
VCF Operations	Simple	Primary node lost	Restore primary node from image-based backup.
		Cloud proxy lost	Restore cloud proxy from image-based backup.
	High Availability	Primary node lost	Restore primary node from image-based backup.

Component	Deployment Model	Failure Scenario	Restore Process
		Replica node lost	Restore replica node from image-based backup.
		Data node lost	Restore data node from image-based backup.
		Cloud proxy lost	Restore cloud proxy from image-based backup.

1. If the VCF Operations virtual machines are running in the management domain, power them off and rename or delete them.
2. Access your backup software interface.
3. Locate the backup of the VCF Operations nodes.
4. Start the restore process.
5. After all virtual machines are restored, power them on.

Verify VCF Operations State for VCF Instance Backup and Restore

After a successful restore of VCF Operations, you verify the state of the analytics cluster and each integration adapter.

The cluster must be online. Not all integration adapters may be in a collecting state as they may not yet be recovered.

1. Verify the VCF Operations cluster state.
   a) Log in to the VCF Operations admin console at `https://<vcf-operations-primary-node-fqdn>/admin` with the **admin** user.
   b) On the **System Status** page, verify that all nodes are in a **Running** state and their status is **Online**.
   c) Navigate to the **Cloud Proxies** page and verify that the **Health Status** for each cloud proxy is **Healthy**.
2. Verify the VCF Operations integration adapters state.
   a) Log in to the VCF Operations interface at `https://<vcf-operations-primary-node-fqdn>/` as a user assigned the **Administrator** role.
   b) Navigate to **Administration** > **Integrations**.
   c) Expand each integration and ensure that its status is **OK**.

VCF Management Services Restore Tasks for VCF Instance Backup and Restore

Perform the tasks that are related to restoring the nodes running VCF management services.

Retrieve Component Details from VCF Operations for VCF Instance Backup and Restore

Before you can redeploy a VCF services runtime cluster, you must find the current version from the VCF Operations inventory.

You use the VCF Operations API to retrieve the component IDs and versions.

1. Replace the values in the sample code with your values, these will be used in later steps.

```
# $vcfOperationsFqdn should be the VCF Operations master node FQDN
# $vcfOperationsAdminPassword should be the VCF Operations local admin password

export vcfOperationsFqdn='flt-ops01a.rainpole.io'
export vcfOperationsUsername='admin'
export vcfOperationsPassword='***************'
```

2. Retrieve an access token from the VCF Operations instance by running the following command.

```
vcfOperationsToken=$(curl -k -s -X POST "https://${vcfOperationsFqdn}/suite-api/api/auth/token/acquire" \
    --header "Content-Type: application/json" \
    --header "Accept: application/json" \
    --data "{
      \"authSource\": \"local\",
```

```
      \"username\": \"${vcfOperationsUsername}\",
      \"password\": \"${vcfOperationsPassword}\"
    }" \
| jq -r '.token' | tee /dev/tty )
```

3. Retrieve a list of component IDs and versions from VCF Operations and save them to a JSON file named `component-ids-versions.json` for later use.

```
curl -k -s -X GET "https://${vcfOperationsFqdn}/suite-api/internal/components?_no_links=true" \
    --header "Authorization: OpsToken ${vcfOperationsToken}" \
    --header "Content-Type: application/json" \
    --header "Accept: application/json" \
    --header "X-Ops-API-use-unsupported: true" \
| jq '
    .components as $all
    | ( $all
        | map(
            select(
            (.componentType | ascii_upcase) as $t
            | $t == "FLEET_LCM" or $t == "SALT_RAAS" or $t == "VIDB" or $t == "LI" or $t == "FDS" or $t ==
 "VODAP" or $t == "SALT_MASTER" or $t == "SDDC_LCM"
            )
        )
    ) as $filtered
    | ( $all
        | map(select((.componentType | ascii_upcase) == "VSP"))
        | map({
            id: .componentUuid,
            componentVersion: .componentVersion,
            fqdn: .properties.fqdn,
            fleetFqdn: .properties.fleetFqdn,
            ip: (.properties.ip // .vcfInstance.ip),
            instanceFqdn: .properties.instanceFqdn,
            instanceName: .vcfInstance.instanceName,
            vcfInstanceFqdn: .vcfInstance.fqdn
        })
        | unique_by(.id)
    ) as $vsp
    | ( $all
        | map(select((.componentType | ascii_upcase) == "VCFA"))[0]
        | if . then {
            fqdn: .properties.fqdn,
            componentVersion: .componentVersion,
            platformFqdn: .properties.platformFqdn,
            vspComponentUuid: .properties.vspComponentUuid
        } else null end
    ) as $vcfa
    | {
        components: (
            $filtered
            | map({
                componentType,
                id: .componentUuid,
                componentVersion: .componentVersion,
                fqdn: .properties.fqdn
            })
        ),
        vsp: $vsp,
```

```
            vcfa: $vcfa
        }' \
    | tee component-ids-versions.json
```

Sample Output


```
{
  "components": [
    {
      "componentType": "SALT_MASTER",
      "id": "12571e84-16fe-4daa-94c1-2385b6ceebf6",
      "componentVersion": "9.1.1.0.25679895",
      "fqdn": "sfo-ic01.sfo.rainpole.io"
    },
    {
      "componentType": "SALT_RAAS",
      "id": "edeac40e-76f2-44ab-83c8-cb9dcddaed55",
      "componentVersion": "9.1.1.0.25679895",
      "fqdn": "flt-fc01.rainpole.io"
    },
    {
      "componentType": "VIDB",
      "id": "b1408f7e-5fb3-47f9-8b4d-3461021bbea8",
      "componentVersion": "9.1.1.0.25679886",
      "fqdn": "flt-idb01.rainpole.io"
    },
    {
      "componentType": "FDS",
      "id": "6bc3a1a1-c045-431d-bc12-193ae61a08cc",
      "componentVersion": "9.1.1.0.25713941",
      "fqdn": "flt-fc01.rainpole.io"
    },
    {
      "componentType": "VODAP",
      "id": "1996e2f7-f0a3-4ef9-8b20-2a23fa457db4",
      "componentVersion": "9.1.1.0.25679622",
      "fqdn": "sfo-ic01.sfo.rainpole.io"
    },
    {
      "componentType": "SDDC_LCM",
      "id": "68161b94-63b5-4843-be76-a99dcc7b07ff",
      "componentVersion": "9.1.1.0.25713940",
      "fqdn": "sfo-ic01.sfo.rainpole.io"
    },
    {
      "componentType": "FLEET_LCM",
      "id": "aaf5baa7-0dde-43f4-8dc3-d3ac5243c63f",
      "componentVersion": "9.1.1.0.25713934",
      "fqdn": "flt-fc01.rainpole.io"
    },
    {
      "componentType": "LI",
      "id": "5a43748c-59e0-4333-9a8b-de957bc2dee0",
      "componentVersion": "9.1.1.0.25679624",
      "fqdn": "flt-logs01.rainpole.io"
    }
  ],
  "vsp": [
```

```
    {
      "id": "f365cb61-c1a4-42f1-b0c3-e77a21edabd6",
      "componentVersion": "9.1.1.0.25714471",
      "fqdn": "sfo-sr01.sfo.rainpole.io",
      "fleetFqdn": "flt-fc01.rainpole.io",
      "ip": "10.11.10.13",
      "instanceFqdn": "sfo-ic01.sfo.rainpole.io",
      "instanceName": "San Francisco",
      "vcfInstanceFqdn": "sfo-vcf01.sfo.rainpole.io"
    }
  ],
  "vcfa": {
    "fqdn": "flt-auto01.rainpole.io",
    "componentVersion": "9.1.1.0.25714559",
    "platformFqdn": "flt-vcfa-sr01.rainpole.io",
    "vspComponentUuid": "0eabd66d-003a-4322-831d-d72aeec4d637"
  }
}
```

Extract Services Runtime Configuration from the Services Runtime Backup for VCF Instance Backup and Restore

Using the VCF services runtime backup, you extract the package deployment configuration.

- OpenSSL 1.1.1 or later must be available on your Linux machine that you use to access the environment.
- Download and extract the scripts from Broadcom knowledge base article 443506 to the machine you use to access your VMware Cloud Foundation platform.
- A linux workstation with access to the SFTP server.

The VCF services runtime package deployment configuration is captured as part of the VCF management services backup. You use this extracted data in a later step.

1. Browse the SFTP server and locate the VCF management services backup that you plan to use for the restore operation.
2. Replace the values in the sample code with your values, these will be used in later steps.

```
export sftpHost='192.168.1.2'
export sftpPort='22'
export sftpDirectory='/media/backups/'
export sftpUsername='svc-vcf-bck'
export sftpPassword='***************'
export sftpEncryptionPassphrase='***************'
export remoteArchive='/media/backups/vcf/backups/5d29166a-780d-4481-81df-59cc3cefb697/9.1.1.0.25714471/
vsp/5d29166a-780d-4481-81df-59cc3cefb697/9.1.1.0.25714471/2026-09-01T14-53-53Z/2026-09-01T14-53-53Z.base.tgz'
 # remote-archive should be the path to the vsp backup for the VCF Management Services instance
```

3. Run the following command to execute the script to extract the configuration from the backup.

```
./vmsp-backup-sftp-to-yaml.sh \
    --sftp-host ${sftpHost} \
    --sftp-port ${sftpPort} \
    --sftp-user ${sftpUsername} \
    --sftp-password ${sftpPassword} \
    --sftp-dir ${sftpDirectory} \
    --remote-archive ${remoteArchive} \
    --passphrase ${sftpEncryptionPassphrase} \
    --output-dir "./tmp/backup-yaml"
```

Sample Output

```
vmsp-backup-sftp-to-yaml.sh: using local file: /media/backups/vcf/
backups/5d29166a-780d-4481-81df-59cc3cefb697/9.1.1.0.25714471/
vsp/5d29166a-780d-4481-81df-59cc3cefb697/9.1.1.0.25714471/2026-09-01T14-53-53Z/2026-09-01T14-53-53Z.base.tgz
tmp/backup-yaml/vmsp-platform.yaml
tmp/backup-yaml/ingress-fleet-tls-ndc.yaml
tmp/backup-yaml/ingress-instance-tls-ndc.yaml
tmp/backup-yaml/ingress-platform-tls-ndc.yaml
tmp/backup-yaml/ingress-fleet-tls.yaml
tmp/backup-yaml/ingress-instance-tls.yaml
tmp/backup-yaml/ingress-platform-tls.yaml

vmsp-backup-sftp-to-yaml.sh: YAML written under: ./tmp/backup-yaml
total 148
drwxr-xr-x 2 root root  4096 Sep  2 14:23 .
drwxr-xr-x 3 root root  4096 Sep  2 14:23 ..
-rw-r--r-- 1 root root 19019 Sep  2 14:23 ingress-fleet-tls-ndc.yaml
-rw-r--r-- 1 root root 19232 Sep  2 14:23 ingress-fleet-tls.yaml
-rw-r--r-- 1 root root 19046 Sep  2 14:23 ingress-instance-tls-ndc.yaml
-rw-r--r-- 1 root root 19262 Sep  2 14:23 ingress-instance-tls.yaml
-rw-r--r-- 1 root root 19054 Sep  2 14:23 ingress-platform-tls-ndc.yaml
-rw-r--r-- 1 root root 19356 Sep  2 14:23 ingress-platform-tls.yaml
-rw-r--r-- 1 root root 17409 Sep  2 14:23 vmsp-platform.yaml

# The file you require is backup-yaml/vmsp-platform.yaml
```

Prepare to Restore VCF Management Services for VCF Instance Backup and Restore

Before you can deploy a new VCF management services instance to restore to, you must delete the existing VCF management services entry in the SDDC Manager database.

> **CAUTION:**
>
> This procedure should not be used to completely delete a deployed component. To fully delete day 2 deployed components on VCF management services refer to Broadcom knowledge base article 441333. Only follow either Step 1 or Step 2 depending on the version of VCF you are recovering.

1. For **VCF 9.1.1** and later, follow these steps:
   a) Replace the values in the sample code with your values, these will be used in later steps.

   ```
   export sddcManagerFqdn='sfo-vcf01.sfo.rainpole.io'
   export sddcManagerUsername='admin@local'
   export sddcManagerPassword='***************'
   ```

   b) Retrieve an access token from the SDDC Manager instance by running the following command.

   ```
   sddcManagerToken=$(curl -k -s -X POST "https://${sddcManagerFqdn}/v1/tokens" \
       --header "Content-Type: application/json" \
       --data "{\"username\":\"${sddcManagerUsername}\", \"password\":\"${sddcManagerPassword}\"}" \
   | jq -r '.accessToken' | tee /dev/tty )
   ```

   c) Export the VCF services runtime cluster of type *MANAGEMENT* details to the `vcfms-vsp-cluster-details.json` file for later use by running the following command.

   ```
   curl -k -s -X GET "https://${sddcManagerFqdn}/v1/vsp-clusters" \
       --header "Authorization: Bearer ${sddcManagerToken}" \
       --header "Content-type: application/json" \
       --header "Accept: application/json" \
   | jq '.elements[] | select(.type == "MANAGEMENT")' \
   | tee vcfms-vsp-cluster-details.json
   ```

   d) Retrieve the ID of the VCF services runtime cluster of type *MANAGEMENT* by running the following command.

   ```
   vspManagementId=$(curl -k -s -X GET "https://${sddcManagerFqdn}/v1/vsp-clusters" \
       --header "Authorization: Bearer ${sddcManagerToken}" \
   ```

```
    --header "Content-type: application/json" \
    --header "Accept: application/json" \
| jq -r '.elements[] | select(.type == "MANAGEMENT") | .id' | tee /dev/tty )
```

e) Delete the VCF services runtime cluster of type *MANAGEMENT*, using the **id** obtained in the previous step by running the following command.

```
curl -k -s -X DELETE "https://${sddcManagerFqdn}/v1/vsp-clusters/${vspManagementId}" \
    --header "Authorization: Bearer ${sddcManagerToken}" \
    --header "Content-Type: application/json" \
    --header "Accept: application/json" \
| jq
```

2. For **VCF 9.1.0**, follow these steps:
   a) Open an SSH connection to the SDDC Manager appliance, connect using the **vcf** user.
   b) Switch to the **root** user.

```
su -
```

   c) Connect to the Postgres database by running the following command.

```
psql -U postgres -h localhost -d platform;
```

   d) Query the *vsp_cluster* table and record the **vsp_cluster_id** of the entry with type *MANAGEMENT* by running the following command.

```
select * from vsp_cluster;
```

   e) Delete the *MANAGEMENT* entry by running the following command.

```
delete from vsp_cluster where type='MANAGEMENT';
```

   f) Delete the corresponding credential by running the following command.

```
delete from credential where username='vsp/<MANAGEMENT-vsp_cluster_id>/svc-sddc-manager-admin';
```

   For Example

```
delete from credential where username='vsp/1f5c79fe-e3aa-41b1-a5cf-774a6497fa3d/svc-sddc-manager-
admin';
```

Redeploy VCF Management Services for VCF Instance Backup and Restore

You deploy a new VCF management services instance using the SDDC Manager API.

- VCF management services is in a failed state and cannot be restored using the in place procedure.
- VCF management services VMs must no longer exist in vCenter.
- You must have at least one recent backup of the component that you are restoring.
- Before proceeding with restoration of VCF management services, the following components must be available:
  - vCenter
  - VCF Operations
  - SDDC Manager
- VCF Installer with the correct binary versions for the release you are running for VCF services runtime, Fleet lifecycle, Software depot, SDDC lifecycle, Salt master, Salt RaaS, and Identity broker.

Restore the specific components according to your scenario.

VCF Management Services in the First VCF Instance	VCF Management Services in an Additional VCF Instance
• VCF services runtime • Fleet lifecycle • SDDC lifecycle • Software depot • Salt RaaS • Salt master • Identity broker	• VCF services runtime • SDDC lifecycle • Software depot * • Salt master • Identity broker *

* Might not be deployed in every instance.

1. Replace the values in the sample code with your values, these will be used in later steps.

```
export sddcManagerFqdn='sfo-vcf01.sfo.rainpole.io'
export sddcManagerUsername='admin@local'
export sddcManagerPassword='***************'

export systemUserPassword='***************'
export size='medium'
export networkMoId='dvportgroup-25'
export
 ipv4Pool='"10.11.10.31","10.11.10.32","10.11.10.33","10.11.10.34","10.11.10.35","10.11.10.36",,"10.11.10.37","10.11
export gatewayCidrIpv4='10.11.10.1/24'
export internalClusterCidrIpv4='198.18.0.0/15'
```

   Attention:

   The **$networkMoId** value must be obtained from vCenter using the following example command: (Get-VDPortgroup -Name "sfo-m01-cl01-vds01-pg-vm-mgmt").ExtensionData.MoRef.value

2. Retrieve an access token from the SDDC Manager instance by running the following command.

```
sddcManagerToken=$(curl -k -s -X POST "https://${sddcManagerFqdn}/v1/tokens" \
    --header "Content-Type: application/json" \
    --data "{\"username\":\"${sddcManagerUsername}\", \"password\":\"${sddcManagerPassword}\"}" \
| jq -r '.accessToken' | tee /dev/tty )
```

3. Generate the `vcfms-deployment-payload.json` file by running the following command.

```
cat << EOF | tee vcfms-deployment-payload.json
{
    "domainId": "$(jq -r '.domainId' vcfms-vsp-cluster-details.json)",
    "platformFqdn": "$(jq -r '.platformFqdn' vcfms-vsp-cluster-details.json)",
    "instanceFqdn": "$(jq -r '.vsp[].instanceFqdn' component-ids-versions.json
)",
    "fleetFqdn": "$(jq -r '.vsp[].fleetFqdn' component-ids-versions.json
)",
    "systemUserPassword": "${systemUserPassword}",
    "type": "MANAGEMENT",
    "ipv4Pool": {
      "addresses": [
        ${ipv4Pool}
      ]
    },
    "size": "${size}",
    "version": "$(jq -r '.vsp[].componentVersion' component-ids-versions.json
)",
    "networkMoId": "${networkMoId}",
    "gatewayCidrIpv4": "${gatewayCidrIpv4}",
    "clusterId": "$(jq -r '.vspClusterId' vcfms-vsp-cluster-details.json)",
    "internalClusterCidrIpv4": "${internalClusterCidrIpv4}"
}
EOF
```

   Important:

   The `clusterId`, `domainId` and `platformFqdn` are obtained from the `vcfms-vsp-cluster-details.json` that is created during the Prepare to Restore VCF Management Services for VCF Instance Backup and Restore procedure.

Important:

The `instanceFqdn`, `fleetFqdn`, and `version` and are obtained from the `component -ids-versions.json` that is created during the Retrieve Component Details from VCF Operations for VCF Instance Backup and Restore procedure.

4. Deploy a new VCF management services runtime instance using the SDDC Manager API by running the following command.

```
deploymentId=$(curl -k -s -X POST "https://${sddcManagerFqdn}/v1/vsp-clusters" \
    --header "Authorization: Bearer ${sddcManagerToken}" \
    --header "Content-Type: application/json" \
    --header "Accept: application/json" \
    --data @vcfms-deployment-payload.json \
| jq -r '.id' | tee /dev/tty )
```

5. Monitor the task by running the following command.

```
curl -k -s -X GET "https://${sddcManagerFqdn}/v1/tasks/${deploymentId}" \
    --header "Authorization: Bearer ${sddcManagerToken}" \
    --header "Content-Type: application/json" \
    --header "Accept: application/json" \
| jq
```

Apply SFTP Settings on VCF Management Services for VCF Instance Backup and Restore

To enable the restore from SFTP backup, you apply the SFTP server settings on the newly deployed VCF management services instance.

1. Replace the values in the sample code with your values, these will be used in later steps. The `vspId` can be obtained from the output of the Retrieve Component Details from VCF Operations for VCF Instance Backup and Restore procedure.

```
export servicesRuntimeFqdn='sfo-sr01.sfo.rainpole.io'
export servicesRuntimeUsername='admin@vsp.local'
export servicesRuntimePassword='***************'

export sftpHost='192.168.1.2'
export sftpPort='22'
export sftpDirectory='/media/backups/'
export sftpUsername='svc-vcf-bck'
export sftpPassword='***************'
export sftpEncryptionPassphrase='***************'
```

2. Extract the `vspId` from the `components-ids-versions.json` file generated from the Retrieve Component Details from VCF Operations for VCF Instance Backup and Restore procedure.

```
vspId=$(jq -r '.vsp[].id' component-ids-versions.json | tee /dev/tty )
```

3. Retrieve an access token from the VCF services runtime by running the following command.

```
servicesRuntimeToken=$(curl -k -s -X POST "https://${servicesRuntimeFqdn}/api/v1/identity/token" \
    --header "Content-Type: application/x-www-form-urlencoded" \
    --data-urlencode "grant_type=password" \
    --data-urlencode "username=${servicesRuntimeUsername}" \
    --data-urlencode "password=${servicesRuntimePassword}" \
| jq -r '.access_token' | tee /dev/tty )
```

4. Retrieve the fingerprint of the SFTP server by running the following command.

```
sftpFingerprint=$(ssh-keyscan -t ed25519 -p $sftpPort $sftpHost 2>/dev/null | ssh-keygen -lf - | awk
 '{print $2}' | tee /dev/tty )
```

5. Generate the `sftp-config-payload.json` file by running the following command.

```
cat << EOF | tee sftp-config-payload.json
{
    "spec": {
        "configuration": {
            "backups": {
                "destination": "sftp",
                "encryptionPassphrase": "${sftpEncryptionPassphrase}",
                "storage": {
                    "sftp": {
                        "directory": "${sftpDirectory}",
                        "host": "${sftpHost}",
                        "port": "${sftpPort}",
                        "username": "${sftpUsername}",
                        "password": "${sftpPassword}",
                        "fingerprint": "${sftpFingerprint}"
                    }
                }
            }
        }
    },
    "options": {}
}
EOF
```

6. Apply the SFTP server settings but running the following command.

```
taskId=$(curl -k -s -X POST "https://${servicesRuntimeFqdn}/api/v1/components/${vspId}?action=apply" \
    --header "Authorization: Bearer ${servicesRuntimeToken}" \
    --header "Content-Type: application/json" \
    --header "Accept: application/json" \
    --data @sftp-config-payload.json \
| jq -r '.id' | tee /dev/tty )
```

7. Monitor the task by running the following command.

```
while true; do
    # Capture the filtered JSON output
    output=$(curl -k -s -X GET "https://${servicesRuntimeFqdn}/api/v1/tasks/${taskId}" \
        --header "Authorization: Bearer ${servicesRuntimeToken}" \
        --header "Content-Type: application/json" \
        --header "Accept: application/json" \
        | jq '{id: .id, stages: .stages, status: .status}')

    # Extract just the raw status string for evaluation
    status=$(echo "$output" | jq -r '.status')

    # Check if the status matches our looping conditions
    if [[ "$status" == "Pending" || "$status" == "Running" ]]; then
        echo "Current status is '$status'. Checking again in 60 seconds..."
        sleep 60
    else
        # Stop the loop and print the final JSON payload
        echo "Task completed with status: '$status'."
        echo "$output"
        break
    fi
done
```

Trust VCF Installer Certificate on VCF Management Services for VCF Instance Backup and Restore

To enable the restore from backup, you trust the VCF Installer certificate on the newly deployed VCF management services instance. This enables VCF management services to download binaries from the VCF Installer during the restore process.

1. Replace the values in the sample code with your values, these will be used in later steps.

```
export servicesRuntimeFqdn='sfo-sr01.sfo.rainpole.io'
export servicesRuntimeUsername='admin@vsp.local'
export servicesRuntimePassword='***************'

export vcfInstallerFqdn='sfo-ins01.sfo.rainpole.io'
```

2. Retrieve the VCF Installer certificate details by running the following command.

```
vcfInstallerCertData=$(echo | openssl s_client -connect "${vcfInstallerFqdn}:443" 2>/dev/null | openssl
 x509 -outform pem | tee /dev/tty )
```

3. Retrieve an access token from the VCF services runtime by running the following command.

```
servicesRuntimeToken=$(curl -k -s -X POST "https://${servicesRuntimeFqdn}/api/v1/identity/token" \
    --header "Content-Type: application/x-www-form-urlencoded" \
    --data-urlencode "grant_type=password" \
    --data-urlencode "username=${servicesRuntimeUsername}" \
    --data-urlencode "password=${servicesRuntimePassword}" \
| jq -r '.access_token' | tee /dev/tty )
```

4. Trust the VCF Installer certificate by running the following command.

```
taskId=$(jq -n --arg cert "$vcfInstallerCertData" '{cert: $cert}' | \
curl -k -s -X POST "https://${servicesRuntimeFqdn}/api/v1/system/trusted-certificates?action=add" \
    --header "Authorization: Bearer ${servicesRuntimeToken}" \
    --header "Content-Type: application/json" \
    --header "Accept: application/json" \
    --data @- | jq -r '.id' | tee /dev/tty )
```

5. Monitor the task by running the following command.

```
curl -k -s -X GET "https://${servicesRuntimeFqdn}/api/v1/tasks/${taskId}" \
    --header "Authorization: Bearer ${servicesRuntimeToken}" \
    --header "Content-Type: application/json" \
    --header "Accept: application/json" \
| jq
```

Stage Component Binaries Prior to Restore

If you do not have the original VCF Installer, you must pre-stage the component binaries onto the new services runtime cluster.

1. Replace the values in the sample code with your values, these will be used in later steps.

```
export servicesRuntimeFqdn='sfo-sr01.sfo.rainpole.io' #fqdn of VCFMS services runtime or VCFA Services
 Runtime as appropriate
export servicesRuntimeUsername='admin@vsp.local'
export servicesRuntimePassword='***************'

export offlineDepotFqdn="depot.rainpole.io"
```

   Important:

   When using offline depot this must not require authentication.

2. Retrieve an access token from the VCF services runtime by running the following command.

```
servicesRuntimeToken=$(curl -k -s -X POST "https://${servicesRuntimeFqdn}/api/v1/identity/token" \
```

```
    --header "Content-Type: application/x-www-form-urlencoded" \
    --data-urlencode "grant_type=password" \
    --data-urlencode "username=${servicesRuntimeUsername}" \
    --data-urlencode "password=${servicesRuntimePassword}" \
| jq -r '.access_token' | tee /dev/tty )
```

3. Replace the values in the sample code with your build numbers.

```
## for VCFMS Services Runtime use this ##
export vcfFleetLcm='9.1.1.0.25713934'
export vcfFleetDepot='9.1.1.0.25713941'
export vcfSddcLcm='9.1.1.0.25713940'
export salt='9.1.1.0.25679895'
export saltRaas='9.1.1.0.25679895'
export vidb='9.1.1.0.25679886'
export vcfmsMetricsStore='9.1.1.0.25714471'
export vcfObsDataPlatform='9.1.1.0.25679622'
export telemetryAcceptor='9.1.1.0.25671600'
## for VCFA Services Runtime use this ##
export vcfa='9.1.1.0.25714559'
export vcdMigrator='9.1.1.0.25714559'
```

4. Create an array of JSON strings for each component by running the following command.

```
## for VCFMS Services Runtime use this ##
components=(
    '{"type": "vcf-fleet-lcm", "version": "$vcfFleetLcm", "url": "https://$offlineDepotFqdn/PROD/release/
depot-manifest-vcf-fleet-lcm-$vcfFleetLcm.yaml"}'
    '{"type": "vcf-fleet-depot", "version": "$vcfFleetDepot", "url": "https://$offlineDepotFqdn/PROD/
release/depot-manifest-vcf-fleet-lcm-$vcfFleetDepot.yaml"}'
    '{"type": "vcf-sddc-lcm", "version": "$vcfSddcLcm", "url": "https://$offlineDepotFqdn/PROD/release/
depot-manifest-vcf-sddc-lcm-$vcfSddcLcm.yaml"}'
    '{"type": "salt", "version": "$salt", "url": "https://$offlineDepotFqdn/PROD/release/depot-manifest-
salt-$salt.yaml"}'
    '{"type": "salt-raas", "version": "$saltRaas", "url": "https://$offlineDepotFqdn/PROD/release/depot-
manifest-salt-raas-$saltRaas.yaml"}'
    '{"type": "vidb", "version": "$vidb", "url": "https://$offlineDepotFqdn/PROD/release/depot-manifest-
vidb-$vidb.yaml"}'
    '{"type": "vcfms-metrics-store", "version": "$vcfmsMetricsStore", "url": "https://$offlineDepotFqdn/
PROD/release/depot-manifest-vcfms-metrics-store-$vcfmsMetricsStore.yaml"}'
    '{"type": "vcf-obs-data-platform", "version": "$vcfObsDataPlatform", "url": "https://
$offlineDepotFqdn/PROD/release/depot-manifest-vcf-obs-data-platform-$vcfObsDataPlatform.yaml"}'
    '{"type": "telemetry-acceptor", "version": "$telemetryAcceptor", "url": "https://$offlineDepotFqdn/
PROD/release/depot-manifest-telemetry-acceptor-$telemetryAcceptor.yaml"}'
)
## for VCFA Services Runtime use this ##
components=(
    '{"type": "vcfa", "version": "$vcfa", "url": "https://$offlineDepotFqdn/PROD/release/depot-manifest-
vcf-fleet-lcm-$vcfa"}'
    '{"type": "vcd-migrator", "version": "$vcdMigrator", "url": "https://$offlineDepotFqdn/PROD/release/
depot-manifest-vcf-fleet-lcm-$vcdMigrator.yaml"}'
)
```

5. Stage the binaries for the *vcf-fleet-lcm*, *vcf-fleet-depot*, *vcf-sddc-lcm*, *salt*, *salt-raas*, and *vidb* componets outputting the API request results to the `staging-results.log` file by running the following command.

```
{
    for ROW in "${components[@]}"; do
        TYPE=$(echo "$ROW" | jq -r '.type')
        VERSION=$(echo "$ROW" | jq -r '.version')
```

```
        URL=$(echo "$ROW" | jq -r '.url')

        echo "Staging $TYPE version $VERSION..."

        curl -k -s -X POST "https://${servicesRuntimeFqdn}/api/v1/components?action=stage" \
            --header "Authorization: Bearer ${servicesRuntimeToken}" \
            --header "Content-Type: application/json" \
            --header "Accept: application/json" \
            --data '{
            "type": "'"$TYPE"'",
            "version": "'"$VERSION"'",
            "repository": {
                "url": "'"$URL"'"
            },
            "options": {
                "timeout": "30m"
            }
            }' | jq
    done
} 2>&1 | tee staging-results.log
```

6. Using the `staging-results.log` you can retrieve the task ids and monitor a staging task by running the following command.

```
export taskId='ynepfibqjfawhex6ioyfr2ixri'
curl -k -s -X GET "https://${servicesRuntimeFqdn}/api/v1/tasks/${taskId}" \
    --header "Authorization: Bearer ${servicesRuntimeToken}" \
    --header "Content-Type: application/json" \
    --header "Accept: application/json" \
| jq
```

Retrieve a List of Backups for VCF Management Services Components for VCF Instance Backup and Restore

Before you restore VCF management services components, which include VCF services runtime, Software depot, Fleet lifecycle, SDDC lifecycle, Salt master, Salt RaaS, and Identity broker you retrieve a list of available backups.

1. Replace the values in the sample code with your values, these will be used in later steps.

```
export servicesRuntimeFqdn='sfo-sr01.sfo.rainpole.io'
export servicesRuntimeUsername='admin@vsp.local'
export servicesRuntimePassword='***************'
```

2. Retrieve an access token from the VCF services runtime by running the following command.

```
servicesRuntimeToken=$(curl -k -s -X POST "https://${servicesRuntimeFqdn}/api/v1/identity/token" \
    --header "Content-Type: application/x-www-form-urlencoded" \
    --data-urlencode "grant_type=password" \
    --data-urlencode "username=${servicesRuntimeUsername}" \
    --data-urlencode "password=${servicesRuntimePassword}" \
| jq -r '.access_token' | tee /dev/tty )
```

3. Retrieve a list of backups for VCF services runtime, Software depot, Fleet lifecycle, SDDC lifecycle, Salt master, Salt RaaS, and Identity broker by running the following command.

```
curl -k -s -X GET "https://${servicesRuntimeFqdn}/api/v1/system/backups" \
    --header "Authorization: Bearer ${servicesRuntimeToken}" \
    --header "Content-Type: application/json" \
    --header "Accept: application/json" \
| jq -r '
    ["TYPE", "VERSION", "NAME", "AGE"],
    (
```

```
        ["vsp", "vcf-fleet-lcm", "vcf-fleet-depot", "vcf-sddc-lcm", "salt", "salt-raas", "vidb", "vcfms-
metrics-store", "vcf-obs-data-platform"] as $order
        | .backups as $all
        | $order[] as $current_type
        | $all[]
        | select(.component.type == $current_type)
        | (
            (.name | sub("T(?<h>\\d{2})-(?<m>\\d{2})-(?<s>\\d{2})Z"; "T\(.h):\(.m):\(.s)Z") |
 fromdateiso8601)
          ) as $timestamp
        | ( (now - $timestamp) / 86400 | floor ) as $days_old
        | [ $current_type, .component.version, .name, ($days_old | tostring + " days ago") ]
    )
    | @tsv' | \
    column -t -s $'\t'
```

4. Identity the `NAME` of the backup job and replace the values in the sample code.

```
export vspBackupName='2026-09-01T14-53-53Z'
export vcfFleetLcmBackupName='2026-09-01T15-03-05Z'
export vcfFleetDepotBackupName='2026-09-01T15-08-32Z'
export vcfSddcLcmBackupName='2026-09-01T14-57-39Z'
export saltBackupName='2026-09-01T15-23-53Z'
export saltRaasBackupName='2026-09-01T15-18-19Z'
export vidb='2026-09-01T15-12-57Z'
```

5. Generate the `vcfms-restore-payload.json` file by running the following command.

```
curl -k -s -X GET "https://${servicesRuntimeFqdn}/api/v1/system/backups" \
    --header "Authorization: Bearer ${servicesRuntimeToken}" \
    --header "Content-Type: application/json" \
    --header "Accept: application/json" \
| jq \
    --arg vsp "${vspBackupName}" \
    --arg vcfFleetLcm "${vcfFleetLcmBackupName}" \
    --arg vcfFleetDepot "${vcfFleetDepotBackupName}" \
    --arg vcfSddcLcm "${vcfSddcLcmBackupName}" \
    --arg salt "${saltBackupName}" \
    --arg saltRaas "${saltRaasBackupName}" \
    --arg vidb "${vidb}" \
    '{
       "components": [
         .backups[]
         | select(
             (.component.type == "vsp" and .name == $vsp) or
             (.component.type == "vcf-fleet-lcm" and .name == $vcfFleetLcm) or
             (.component.type == "vcf-fleet-depot" and .name == $vcfFleetDepot) or
             (.component.type == "vcf-sddc-lcm" and .name == $vcfSddcLcm) or
             (.component.type == "salt" and .name == $salt) or
             (.component.type == "salt-raas" and .name == $saltRaas) or
             (.component.type == "vidb" and .name == $vidb)
           )
         | {
             "point": .name,
             "path": .path
           }
       ]
     }' | tee vcfms-restore-payload.json
```

Restore the VCF Management Services Components for VCF Instance Backup and Restore

Restore VCF management services components, which include VCF services runtime, Software depot, Fleet lifecycle, SDDC lifecycle, Salt master, Salt RaaS, and Identity broker from a file based backup.

1. Replace the values in the sample code with your values, these will be used in later steps.

```
export servicesRuntimeFqdn='sfo-sr01.sfo.rainpole.io'
export servicesRuntimeUsername='admin@vsp.local'
export servicesRuntimePassword='***************'
```

2. Retrieve an access token from the VCF services runtime by running the following command.

```
servicesRuntimeToken=$(curl -k -s -X POST "https://${servicesRuntimeFqdn}/api/v1/identity/token" \
    --header "Content-Type: application/x-www-form-urlencoded" \
    --data-urlencode "grant_type=password" \
    --data-urlencode "username=${servicesRuntimeUsername}" \
    --data-urlencode "password=${servicesRuntimePassword}" \
| jq -r '.access_token' | tee /dev/tty )
```

3. Perform the restore operation of VCF management services using the `vcfms-restore-payload.json` file generated in the Retrieve a List of Backups for VCF Management Services Components for VCF Instance Backup and Restore procedure by running the following command.

```
restoreTaskId=$(curl -k -X POST "https://${servicesRuntimeFqdn}/api/v1/system/backups?action=restore" \
    --header "Authorization: Bearer ${servicesRuntimeToken}" \
    --header "Content-Type: application/json" \
    --header "Accept: application/json" \
    --data @vcfms-restore-payload.json \
| jq -r '.id' | tee /dev/tty )
```

   Important:

   After restoring the vcf-fleet-lcm component, there might be a failed backup task in the tasks view. This is a false negative and can be ignored.

4. Monitor the restore task of VCF management services by running the following command.

```
while true; do
    # Capture the filtered JSON output
    output=$(curl -k -s -X GET "https://${servicesRuntimeFqdn}/api/v1/tasks/${restoreTaskId}" \
        --header "Authorization: Bearer ${servicesRuntimeToken}" \
        --header "Content-Type: application/json" \
        --header "Accept: application/json" \
        | jq '{id: .id, stages: .stages, status: .status, resource: .resource}')

    # Extract just the raw status string for evaluation
    status=$(echo "$output" | jq -r '.status')

    # Check if the status matches the looping conditions
    if [[ "$status" == "Pending" || "$status" == "Running" ]]; then
        echo "Current status is '$status'. Checking again in 120 seconds..."
        sleep 120
    else
        # Stop the loop and print the final JSON payload
        echo "Task completed with status: '$status'."
        echo "$output"
        break
    fi
done
```

When the restore completes, verify the Software Depot connection is active in VCF Operations. If using an online connection, the activation token may need to be refreshed.

Apply Services Runtime Configuration of VCF Management Services for VCF Instance Backup and Restore

To ensure the worker nodes are the correct size, you apply the kubernetes package deployment from the backup.

1. Discover the VCF management services control plane node.
   a) Open an SSH connection to the VCF services runtime, connect using the **vmware-system-user** user.
   b) Switch the user to administrative access.

      ```
      sudo su
      ```

   a) Find the `EXTERNAL-IP` address of the control plane node.

      ```
      export KUBECONFIG=/etc/kubernetes/node-agent.conf
      kubectl get nodes -o wide
      ```

2. Copy all yaml files from Extract Services Runtime Configuration from the Services Runtime Backup for VCF Instance Backup and Restore to the control plane node.
3. Open an SSH connection to control plane node, connect using the **vmware-system-user** user.
4. Switch the user to administrative access.

   ```
   sudo su
   ```

5. Replace the values in the sample code with your values, these will be used in later steps.

   ```
   export KUBECONFIG=/etc/kubernetes/admin.conf
   export YAML_PATH='./'
   export WORKER_SIZE=$(awk '/worker:/{f=1} f && /size:/{print $2; exit}' $YAML_PATH/vmsp-platform.yaml)
   export MACHINE_TYPE=$(awk '/worker:/{f=1} f && /machineType:/{print $2; exit}' $YAML_PATH/vmsp-
   platform.yaml)
   export PROFILE=$(awk '/profiles:/{f=1} f && /name:/{print $2; exit}' $YAML_PATH/vmsp-platform.yaml)
   ```

6. Apply the services runtime configuration by running the following command.

   ```
   kubectl patch packagedeployment vmsp-platform -n vmsp-platform \
     --type=merge \
     -p "$(jq -n \
       --arg s "$WORKER_SIZE" \
       --arg m "$MACHINE_TYPE" \
       --arg p "$PROFILE" \
       '{
         "spec": {
           "values": {
             "cluster": {
               "worker": {
                 "size": $s,
                 "machineType": $m
               }
             },
             "profiles": {
               "name": $p
             }
           }
         }
       }')"
   ```

7. Monitor the task.

   ```
   kubectl wait --for=jsonpath='{.status.phase}'=Successful packagedeployment/vmsp-platform -n vmsp-platform
    --timeout=2h

   # Wait for this message
   packagedeployment.releases.vmsp.vmware.com/vmsp-platform condition met
   ```

```
kubectl get packagedeployment vmsp-platform -n vmsp-platform -o jsonpath='{.status.phase}'

# Wait for this message
Successful
```

The worker nodes will be resized to match their size when the backup was taken. Do not proceed until this step has completed.

Retrieve a List of Backups for Log Management

If your VMware Cloud Foundation platform included log management as an additional deployed component retrieve a list of backups for log management to perform the restore.

1. Replace the values in the sample code with your values, these will be used in later steps. The `vspId` can be obtained from the output of the Retrieve Component Details from VCF Operations for VCF Instance Backup and Restore procedure.

   ```
   export servicesRuntimeFqdn='sfo-sr01.sfo.rainpole.io'
   export servicesRuntimeUsername='admin@vsp.local'
   export servicesRuntimePassword='***************'
   ```

2. Extract the *`vspId`* from the `components-ids-versions.json` file generated from the Retrieve Component Details from VCF Operations for VCF Instance Backup and Restore procedure.

   ```
   vspId=$(jq -r '.vsp[0].id' component-ids-versions.json | tee /dev/tty )
   ```

3. Retrieve an access token from the VCF services runtime by running the following command.

   ```
   servicesRuntimeToken=$(curl -k -s -X POST "https://${servicesRuntimeFqdn}/api/v1/identity/token" \
       --header "Content-Type: application/x-www-form-urlencoded" \
       --data-urlencode "grant_type=password" \
       --data-urlencode "username=${servicesRuntimeUsername}" \
       --data-urlencode "password=${servicesRuntimePassword}" \
   | jq -r '.access_token' | tee /dev/tty )
   ```

4. Retrieve a list of available log management backups from the VCF services runtime by running the following command.

   ```
   curl -k -s -X GET "https://${servicesRuntimeFqdn}/api/v1/system/backups" \
       --header "Authorization: Bearer ${servicesRuntimeToken}" \
       --header "Content-Type: application/json" \
       --header "Accept: application/json" \
   | jq -r --arg VSP_ID "$vspId" '
       ["TYPE", "VERSION", "NAME", "AGE"],
       (
           ["ops-logs"] as $order
           | .backups as $all
           | $order[] as $current_type
           | $all[]
           | select(.component.type == $current_type)
           # Filter based on the VSP_ID variable passed from the shell
           | select(.path | contains("/vcf/backups/" + $VSP_ID))
           | (
               (.name | sub("T(?<h>\\d{2})-(?<m>\\d{2})-(?<s>\\d{2})Z"; "T\(.h):\(.m):\(.s)Z") |
    fromdateiso8601)
           ) as $timestamp
           | ( (now - $timestamp) / 86400 | floor ) as $days_old
           | [ $current_type, .component.version, .name, ($days_old | tostring + " days ago") ]
       )
       | @tsv' | \
       column -t -s $'\t'
   ```

5. Identity the `NAME` of the backup job and replace the values in the sample code.

```
export backupName='2026-09-01T15-28-34Z'
```

6. Generate the `logs-restore-payload.json` file by running the following command.

```
curl -k -s -X GET "https://${servicesRuntimeFqdn}/api/v1/system/backups" \
    --header "Authorization: Bearer ${servicesRuntimeToken}" \
    --header "Content-Type: application/json" \
    --header "Accept: application/json" \
| jq --arg target "${backupName}" '{
    "components": [
      .backups[]
      | select(.component.type == "ops-logs" and .name == $target)
      | {
          "point": .name,
          "path": .path
        }
    ]
  }' | tee logs-restore-payload.json
```

Restore Log Management for VCF Instance Backup and Restore

After you restore the core VCF management services components, you can restore the log management component if you have it in your environment.

Download the required binaries.

1. Replace the values in the sample code with your values, these will be used in later steps.

```
export servicesRuntimeFqdn='sfo-sr01.sfo.rainpole.io'
export servicesRuntimeUsername='admin@vsp.local'
export servicesRuntimePassword='***************'
```

2. Retrieve an access token from the VCF services runtime instance by running the following command.

```
servicesRuntimeToken=$(curl -k -s -X POST "https://${servicesRuntimeFqdn}/api/v1/identity/token" \
    --header "Content-Type: application/x-www-form-urlencoded" \
    --data-urlencode "grant_type=password" \
    --data-urlencode "username=${servicesRuntimeUsername}" \
    --data-urlencode "password=${servicesRuntimePassword}" \
| jq -r '.access_token' | tee /dev/tty )
```

3. Perform the restore operation of log management using the `logs-restore-payload.json` file generated in the Retrieve a List of Backups for Log Management procedure by running the following command.

```
restoreTaskId=$(curl -k -s -X POST "https://${servicesRuntimeFqdn}/api/v1/system/backups?action=restore" \
    --header "Authorization: Bearer ${servicesRuntimeToken}" \
    --header "Content-Type: application/json" \
    --header "Accept: application/json" \
    --data @logs-restore-payload.json \
| jq -r '.id' | tee /dev/tty )
```

4. Monitor the restore task of log management by running the following command.

```
curl -k -s -X GET "https://${servicesRuntimeFqdn}/api/v1/tasks/${restoreTaskId}" \
    --header "Authorization: Bearer ${servicesRuntimeToken}" \
    --header "Content-Type: application/json" \
    --header "Accept: application/json" \
| jq
```

Important:

For VCF 9.1.0 only, after the log management restore operation completes, you must run a script to reconcile the log management database state. See Broadcom knowledge base article 436904.

Update Log Management State

After the restore operation of log management is complete, you must perform steps to finalize the restore.

1. For VCF 9.1.1 do the following:
   a) Log in to VCF Operations as a user with the **Administrator** role.
   b) Navigate to **Build** > **Lifecycle** > **VCF Management** > **Components.**
   c) Click **Log management**.
   d) From the **Actions** menu, click **Stop Service**.
   e) Monitor the Stop Service task until it completes and then click **Start Service**.
   f) Monitor the Start Service task until it completes.
2. If the logs UI is not loading in the VCF Operations UI after DR failover, we need to follow these steps:
   a) Run following script to delete the Log Management component from Ops and re-registering it.

      ```
      export KUBECONFIG=<path-to-kubeconfig-for VCF Services Runtime>
      ./update_registration_in_ops.sh --ops-fqdn "flt-ops01a.rainpole.io" --component-fqdn "flt-logs01.rainpole.io" --ops-username "admin"
      ```

   b) If step 2a fails, retry the task by visiting **https://<ops-fqdn>/vcf-operations/ui/operate/administration/control-panel/management-tasks/lcm/tasks** and clicking **Retry** on the failed task. Select the **Skip** option so that validations are skipped.
   c) After successful deletion of the Log Management component from Ops, navigate to **Build** > **Lifecycle** > **Tasks** to verify status of the **Refresh component Log management** task. If that task failed, click **Retry**.
3. For VCF 9.1.0 do the following:
   a) Follow the steps in Broadcom knowledge base article 436904.

Retrieve a List of Backups for Telemetry Component

Before you restore VCF management services telemetry, you retrieve a list of available backups.

1. Replace the values in the sample code with your values, these will be used in later steps. The `vspId` can be obtained from the output of the Retrieve Component Details from VCF Operations for VCF Instance Backup and Restore procedure.

   ```
   export servicesRuntimeFqdn='sfo-sr01.sfo.rainpole.io'
   export servicesRuntimeUsername='admin@vsp.local'
   export servicesRuntimePassword='***************'
   ```

2. Extract the *`vspId`* from the `components-ids-versions.json` file generated from the Retrieve Component Details from VCF Operations for VCF Instance Backup and Restore procedure.

   ```
   vspId=$(jq -r '.vsp[0].id' component-ids-versions.json | tee /dev/tty )
   ```

3. Retrieve an access token from the VCF services runtime by running the following command.

   ```
   servicesRuntimeToken=$(curl -k -s -X POST "https://${servicesRuntimeFqdn}/api/v1/identity/token" \
       --header "Content-Type: application/x-www-form-urlencoded" \
       --data-urlencode "grant_type=password" \
       --data-urlencode "username=${servicesRuntimeUsername}" \
       --data-urlencode "password=${servicesRuntimePassword}" \
   | jq -r '.access_token' | tee /dev/tty )
   ```

4. Retrieve a list of available log management backups from the VCF services runtime by running the following command.

   ```
   curl -k -s -X GET "https://${servicesRuntimeFqdn}/api/v1/system/backups" \
       --header "Authorization: Bearer ${servicesRuntimeToken}" \
       --header "Content-Type: application/json" \
       --header "Accept: application/json" \
   | jq -r --arg VSP_ID "$vspId" '
       ["TYPE", "VERSION", "NAME", "AGE"],
   ```

```
    (
        ["telemetry-acceptor"] as $order
        | .backups as $all
        | $order[] as $current_type
        | $all[]
        | select(.component.type == $current_type)
        # Filter based on the VSP_ID variable passed from the shell
        | select(.path | contains("/vcf/backups/" + $VSP_ID))
        | (
            (.name | sub("T(?<h>\\d{2})-(?<m>\\d{2})-(?<s>\\d{2})Z"; "T\(.h):\(.m):\(.s)Z") |
 fromdateiso8601)
        ) as $timestamp
        | ( (now - $timestamp) / 86400 | floor ) as $days_old
        | [ $current_type, .component.version, .name, ($days_old | tostring + " days ago") ]
    )
    | @tsv' | \
    column -t -s $'\t'
```

5. Identity the `NAME` of the backup job and replace the values in the sample code.

```
export backupName='2026-09-01T15-28-34Z'
```

6. Generate the `component-restore-payload.json` file by running the following command.

```
curl -k -s -X GET "https://${servicesRuntimeFqdn}/api/v1/system/backups" \
    --header "Authorization: Bearer ${servicesRuntimeToken}" \
    --header "Content-Type: application/json" \
    --header "Accept: application/json" \
| jq --arg target "${backupName}" '{
    "components": [
      .backups[]
      | select(.component.type == "telemetry-acceptor" and .name == $target)
      | {
          "point": .name,
          "path": .path
        }
    ]
  }' | tee component-restore-payload.json
```

Restore Telemetry Component

Download the required binaries.

1. Replace the values in the sample code with your values, these will be used in later steps.

```
export servicesRuntimeFqdn='sfo-sr01.sfo.rainpole.io'
export servicesRuntimeUsername='admin@vsp.local'
export servicesRuntimePassword='***************'
```

2. Retrieve an access token from the VCF services runtime instance by running the following command.

```
servicesRuntimeToken=$(curl -k -s -X POST "https://${servicesRuntimeFqdn}/api/v1/identity/token" \
    --header "Content-Type: application/x-www-form-urlencoded" \
    --data-urlencode "grant_type=password" \
    --data-urlencode "username=${servicesRuntimeUsername}" \
    --data-urlencode "password=${servicesRuntimePassword}" \
| jq -r '.access_token' | tee /dev/tty )
```

3. Perform the restore operation of telemetry using the `component-restore-payload.json` file generated in the Retrieve a List of Backups for Telemetry Component procedure by running the following command.

```
restoreTaskId=$(curl -k -s -X POST "https://${servicesRuntimeFqdn}/api/v1/system/backups?action=restore" \
    --header "Authorization: Bearer ${servicesRuntimeToken}" \
```

```
    --header "Content-Type: application/json" \
    --header "Accept: application/json" \
    --data @component-restore-payload.json \
| jq -r '.id' | tee /dev/tty )
```

4. Monitor the restore task of telemetry by running the following command.

```
curl -k -s -X GET "https://${servicesRuntimeFqdn}/api/v1/tasks/${restoreTaskId}" \
    --header "Authorization: Bearer ${servicesRuntimeToken}" \
    --header "Content-Type: application/json" \
    --header "Accept: application/json" \
| jq
```

VCF Automation Restore Tasks for VCF Instance Backup and Restore

Perform these tasks related to restoring VCF Automation.

Prepare to Restore VCF Automation for VCF Instance Backup and Restore

Before you can deploy a new VCF Automation instance to restore to, you must delete the existing VCF Automation entry in the SDDC Manager database.

> **CAUTION:**
>
> This procedure should not be used to completely delete a deployed component. To fully delete day 2 deployed components on VCF management services refer to Broadcom knowledge base article 441333.

1. For **VCF 9.1.1** and newer, follow these steps:
   a) Replace the values in the sample code with your values, these will be used in later steps.

```
export sddcManagerFqdn='sfo-vcf01.sfo.rainpole.io'
export sddcManagerUsername='admin@local'
export sddcManagerPassword='***************'
```

   b) Retrieve an access token from the SDDC Manager instance by running the following command.

```
sddcManagerToken=$(curl -k -s -X POST "https://${sddcManagerFqdn}/v1/tokens" \
    --header "Content-Type: application/json" \
    --data "{\"username\":\"${sddcManagerUsername}\", \"password\":\"${sddcManagerPassword}\"}" \
| jq -r '.accessToken' | tee /dev/tty )
```

   c) Export the VCF services runtime cluster of type *CONSUMPTION* details to the `vcfa-vsp-cluster-details.json` file for later use by running the following command.

```
curl -k -s -X GET "https://${sddcManagerFqdn}/v1/vsp-clusters" \
    --header "Authorization: Bearer ${sddcManagerToken}" \
    --header "Content-type: application/json" \
    --header "Accept: application/json" \
| jq '.elements[] | select(.type == "CONSUMPTION")' \
| tee vcfa-vsp-cluster-details.json
```

   d) Retrieve the ID of the VCF services runtime cluster of type *CONSUMPTION* by running the following command.

```
vcfaVspManagementId=$(curl -k -s -X GET "https://${sddcManagerFqdn}/v1/vsp-clusters" \
    --header "Authorization: Bearer ${sddcManagerToken}" \
    --header "Content-type: application/json" \
    --header "Accept: application/json" \
| jq -r '.elements[] | select(.type == "CONSUMPTION") | .id' | tee /dev/tty )
```

   e) Delete the VCF services runtime cluster of type *CONSUMPTION*, using the **id** obtained in the previous step by running the following command.

```
curl -k -s -X DELETE "https://${sddcManagerFqdn}/v1/vsp-clusters/${vcfaVspManagementId}" \
    --header "Authorization: Bearer ${sddcManagerToken}" \
```

```
    --header "Content-type: application/json" \
    --header "Accept: application/json" \
| jq
```

2. For **VCF 9.1.0**, follow these steps:
   a) Open an SSH connection to the SDDC Manger appliance as the **vcf** user and switch to the **root** user.
   b) Connect to the Postgres database.

```
psql -U postgres -h localhost -d platform;
```

   a) Query the *vsp_clusters* table and record the **vsp_cluster_id** of the entry with type *CONSUMPTION*.

```
select * from vsp_cluster;
```

   a) Delete the *CONSUMPTION* entry.

```
delete from vsp_cluster where type='CONSUMPTION';
```

   a) Delete the corresponding credential.

```
delete from credential where username='vsp/<CONSUMPTION-vsp_cluster_id>/svc-sddc-manager-admin';
```

   For Example

```
delete from credential where username='vsp/1f5c79fe-e3aa-41b1-a5cf-774a6497fa3d/svc-sddc-manager-
admin';
```

Redeploy the VCF Automation Services Runtime Cluster for VCF Instance Backup and Restore

You deploy a new VCF Automation services runtime instance using the SDDC Manager API.

- VCF Automation is in a failed state and cannot be restored using the in place procedure.
- VCF Automation VMs must no longer exist in vCenter.
- You must have at least one recent backup of VCF Automation.
- Before proceeding with restoration of VCF Automation, the following components must be available:
  – vCenter
  – VCF Operations
  – SDDC Manager
- The VCF Automation binary must be available in the fleet software depot.

1. Retrieve the *sddcLcmId* from VCF management services.
   a) Replace the values in the sample code with your values, these will be used in later steps.

```
export fleetLifecycleFqdn='flt-fc01.rainpole.io'
export fleetLifecycleUsername='admin@vsp.local'
export fleetLifecyclePassword='***************'
```

   b) Retrieve an access token from the VCF services runtime instance by running the following command.

```
fleetLifecycleToken=$(curl -k -s -X POST "https://${fleetLifecycleFqdn}/api/v1/identity/token" \
    --header "Content-Type: application/x-www-form-urlencoded" \
    --data-urlencode "grant_type=password" \
    --data-urlencode "username=$fleetLifecycleUsername" \
    --data-urlencode "password=$fleetLifecyclePassword" \
| jq -r '.access_token' | tee /dev/tty )
```

   c) Retrieve the *sddcLcmId* from VCF management services by running the following command.

```
sddcLcmId=$(curl -k -s -X GET "https://${fleetLifecycleFqdn}/fleet-lcm/v1/components?
includeConsumptionVsp=true&includeVcdMigrator=true" \
    --header "Authorization: Bearer ${fleetLifecycleToken}" \
    --header "Content-Type: application/json" \
    --header "Accept: application/json" \
| jq -r '.components[] | select(.componentType == "VCF_SDDC_LCM") | .id' | tee /dev/tty )
```

2. Retrieve an access token from the SDDC Manager instance
   a) Replace the values in the sample code with your values, these will be used in later steps.

```
export sddcManagerFqdn='sfo-vcf01.sfo.rainpole.io'
export sddcManagerUsername='admin@local'
export sddcManagerPassword='***************'
```

   b) Retrieve an access token from the SDDC Manager instance by running the following command.

```
sddcManagerToken=$(curl -k -s -X POST "https://${sddcManagerFqdn}/v1/tokens" \
    --header "Content-Type: application/json" \
    --data "{\"username\":\"${sddcManagerUsername}\", \"password\":\"${sddcManagerPassword}\"}" \
| jq -r '.accessToken' | tee /dev/tty )
```

3. Build the JSON payload file `vcfa-deployment-payload.json` to redeploy VCF Automation.
   a) Replace the values in the sample code with your values, these will be used in later steps.

```
export systemUserPassword='***************'
export size='medium'
export networkMoId='dvportgroup-25'
export ipv4Pool='"10.11.99.46","10.11.99.47","10.11.99.48","10.11.99.49","10.11.99.50"'
export ipv4Vips='"10.11.99.51","10.11.99.52","10.11.99.53"'
export gatewayCidrIpv4='10.11.99.1/24'
export internalClusterCidrIpv4='198.18.0.0/15'
```

   Attention:

   The **$networkMoId** value must be obtained from vCenter using the following example command: (Get-VDPortgroup -Name "sfo-m01-cl01-vds01-pg-vm-mgmt").ExtensionData.MoRef.value

   b) Generate the `vcfa-deployment-payload.json` file by running the following command. The `clusterId`, `domainId` and `platformFqdn` are obtained from the `vcfa-vsp-cluster-details.json` that is created during the Prepare to Restore VCF Automation for VCF Instance Backup and Restore procedure.

```
cat << EOF | tee vcfa-deployment-payload.json
{
    "domainId": "$(jq -r '.domainId' vcfa-vsp-cluster-details.json)",
    "deploymentType": "VspClusterSpec",
    "sddcLcmId": "${sddcLcmId}",
    "platformFqdn": "$(jq -r '.platformFqdn' vcfa-vsp-cluster-details.json)",
    "systemUserPassword": "${systemUserPassword}",
    "type": "CONSUMPTION",
    "size": "${size}",
    "ipv4Pool": {
        "addresses": [
            ${ipv4Pool}
        ],
      "excludedAddresses": [
        ]
    },
    "ingress": {
        "vcfa": {
            "fqdn": "$(jq -r '.vcfa.fqdn' component-ids-versions.json
)",
            "vips": {
                "ipv4": [${ipv4Vips}]
          }
        }
      },
    "version": "$(jq -r '.vcfa.componentVersion' component-ids-versions.json
)",
    "networkMoId": "${networkMoId}",
    "gatewayCidrIpv4": "${gatewayCidrIpv4}",
```

```
    "clusterId": "$(jq -r '.vspClusterId' vcfa-vsp-cluster-details.json)",
    "internalClusterCidrIpv4": "${internalClusterCidrIpv4}"
}
EOF
```

Important:

The `clusterId`, `domainId` and `platformFqdn` are obtained from the `vcfa-vsp-cluster-details.json` that is created during the Prepare to Restore VCF Automation for VCF Instance Backup and Restoreprocedure.

Important:

The `fqdn`, and `version` and are obtained from the `component -ids-versions.json` that is created during the Retrieve Component Details from VCF Operations for VCF Instance Backup and Restore procedure.

4. Deploy a new VCF Automation services runtime instance either with or without an external load balancer.
   a) Deploy a new VCF Automation services runtime instance with an external load balancer.

```
vcfaDeploymentId=$(curl -k -s -X POST "https://${sddcManagerFqdn}/v1/vsp-clusters" \
    --header "Authorization: Bearer ${sddcManagerToken}" \
    --header "Content-Type: application/json" \
    --header "Accept: application/json" \
    --data @vcfa-deployment-payload.json \
| jq -r '.id' | tee /dev/tty )
```

   b) Deploy a new VCF Automation services runtime instance without an external load balancer.

```
curl -k -s -X GET "https://${sddcManagerFqdn}/v1/tasks/${vcfaDeploymentId}" \
    --header "Authorization: Bearer ${sddcManagerToken}" \
    --header "Content-Type: application/json" \
    --header "Accept: application/json" \
| jq '{id: .id, name: .name, status: .status}'
```

Apply SFTP Settings on VCF Automation for VCF Instance Backup and Restore

To enable the restore from SFTP backup, you apply the SFTP server settings on the newly deployed VCF Automation instance.

1. Replace the values in the sample code with your values, these will be used in later steps. The `vspId` can be obtained from the output of the Retrieve Component Details from VCF Operations for VCF Instance Backup and Restore procedure.

```
export vcfaServicesRuntimeFqdn='flt-vcfa-sr01.rainpole.io'
export vcfaServicesRuntimeUsername='admin@vsp.local'
export vcfaSservicesRuntimePassword='***************'

export sftpHost='192.168.1.2'
export sftpPort='22'
export sftpDirectory='/media/backups/'
export sftpUsername='svc-vcf-bck'
export sftpPassword='***************'
export sftpEncryptionPassphrase='***************'
```

2. Extract the vspId from the `components-ids-versions.json` file generated from the Retrieve Component Details from VCF Operations for VCF Instance Backup and Restore procedure.

```
vspId=$(jq -r '.vcfa.vspComponentUuid' component-ids-versions.json | tee /dev/tty )
```

3. Retrieve an access token from the VCF services runtime by running the following command.

```
vcfaServicesRuntimeToken=$(curl -k -s -X POST "https://${vcfaServicesRuntimeFqdn}/api/v1/identity/token" \
    --header "Content-Type: application/x-www-form-urlencoded" \
    --data-urlencode "grant_type=password" \
    --data-urlencode "username=${vcfaServicesRuntimeUsername}" \
    --data-urlencode "password=${vcfaSservicesRuntimePassword}" \
```

```
| jq -r '.access_token' | tee /dev/tty )
```

4. Retrieve the fingerprint of the SFTP server by running the following command.

```
sftpFingerprint=$(ssh-keyscan -t ed25519 -p $sftpPort $sftpHost 2>/dev/null | ssh-keygen -lf - | awk
 '{print $2}' | tee /dev/tty )
```

5. Generate the `sftp-config-payload.json` file by running the following command.

```
cat << EOF | tee sftp-config-payload.json
{
    "spec": {
        "configuration": {
            "backups": {
                "destination": "sftp",
                "encryptionPassphrase": "${sftpEncryptionPassphrase}",
                "storage": {
                    "sftp": {
                        "directory": "${sftpDirectory}",
                        "host": "${sftpHost}",
                        "port": "${sftpPort}",
                        "username": "${sftpUsername}",
                        "password": "${sftpPassword}",
                        "fingerprint": "${sftpFingerprint}"
                    }
                }
            }
        }
    },
    "options": {}
}
EOF
```

6. Apply the SFTP server settings but running the following command.

```
taskId=$(curl -k -s -X POST "https://${vcfaServicesRuntimeFqdn}/api/v1/components/${vspId}?action=apply" \
    --header "Authorization: Bearer ${vcfaServicesRuntimeToken}" \
    --header "Content-Type: application/json" \
    --header "Accept: application/json" \
    --data @sftp-config-payload.json \
| jq -r '.id' | tee /dev/tty )
```

7. Monitor the task by running the following command.

```
while true; do
    # Capture the filtered JSON output
    output=$(curl -k -s -X GET "https://${vcfaServicesRuntimeFqdn}/api/v1/tasks/${taskId}" \
        --header "Authorization: Bearer ${vcfaServicesRuntimeToken}" \
        --header "Content-Type: application/json" \
        --header "Accept: application/json" \
        | jq '{id: .id, stages: .stages, status: .status}')

    # Extract just the raw status string for evaluation
    status=$(echo "$output" | jq -r '.status')

    # Check if the status matches our looping conditions
    if [[ "$status" == "Pending" || "$status" == "Running" ]]; then
        echo "Current status is '$status'. Checking again in 60 seconds..."
        sleep 60
    else
        # Stop the loop and print the final JSON payload
        echo "Task completed with status: '$status'."
```

```
            echo "$output"
            break
        fi
    done
```

Trust VCF Installer Certificate on VCF Automation for VCF Instance Backup and Restore

To enable the restore from backup, you trust the VCF Installer certificate on the newly deployed VCF Automation instance. This enables VCF Automation to download binaries from the VCF Installer during the restore process.

1. Replace the values in the sample code with your values, these will be used in later steps.

```
export vcfaServicesRuntimeFqdn='flt-vcfa-sr01.rainpole.io'
export vcfaServicesRuntimeUsername='admin@vsp.local'
export vcfaServicesRuntimePassword='***************'

export vcfInstallerFqdn='sfo-ins01.sfo.rainpole.io'
```

2. Retrieve the VCF Installer certificate details by running the following command.

```
vcfInstallerCertData=$(echo | openssl s_client -connect "${vcfInstallerFqdn}:443" 2>/dev/null | openssl
 x509 -outform pem | tee /dev/tty )
```

3. Retrieve an access token from the VCF Automation by running the following command.

```
vcfaServicesRuntimeToken=$(curl -k -s -X POST "https://${vcfaServicesRuntimeFqdn}/api/v1/identity/token" \
    --header "Content-Type: application/x-www-form-urlencoded" \
    --data-urlencode "grant_type=password" \
    --data-urlencode "username=${vcfaServicesRuntimeUsername}" \
    --data-urlencode "password=${vcfaServicesRuntimePassword}" \
| jq -r '.access_token' | tee /dev/tty )
```

4. Trust the VCF Installer certificate by running the following command.

```
taskId=$(jq -n --arg cert "$vcfInstallerCertData" '{cert: $cert}' | \
curl -k -s -X POST "https://${vcfaServicesRuntimeFqdn}/api/v1/system/trusted-certificates?action=add" \
    --header "Authorization: Bearer ${vcfaServicesRuntimeToken}" \
    --header "Content-Type: application/json" \
    --header "Accept: application/json" \
    --data @- | jq -r '.id' | tee /dev/tty )
```

5. Monitor the task by running the following command.

```
curl -k -s -X GET "https://${vcfaServicesRuntimeFqdn}/api/v1/tasks/${taskId}" \
    --header "Authorization: Bearer ${vcfaServicesRuntimeToken}" \
    --header "Content-Type: application/json" \
    --header "Accept: application/json" \
| jq
```

Retrieve a List of Backups for VCF Automation for VCF Instance Backup and Restore

Before you restore VCF Automation, you retrieve a list of available backups.

> **Note:**
>
> It is recommended to use backups taken by using a backup schedule. On-demand backups will not have a corresponding vsp backup. If you have a newer on-demand backup of VCF Automation, use the most recent available vsp backup along with it.

1. Replace the values in the sample code with your values, these will be used in later steps.

```
export vcfaServicesRuntimeFqdn='flt-vcfa-sr01.rainpole.io'
export vcfaServicesRuntimeUsername='admin@vsp.local'
export vcfaServicesRuntimePassword='***************'
```

2. Retrieve an access token from the VCF Automation by running the following command.

```
vcfaServicesRuntimeToken=$(curl -k -s -X POST "https://${vcfaServicesRuntimeFqdn}/api/v1/identity/token" \
    --header "Content-Type: application/x-www-form-urlencoded" \
    --data-urlencode "grant_type=password" \
    --data-urlencode "username=${vcfaServicesRuntimeUsername}" \
    --data-urlencode "password=${vcfaServicesRuntimePassword}" \
| jq -r '.access_token' | tee /dev/tty )
```

3. Retrieve list of backups for VCF Automation by running the following command.

```
curl -k -s -X GET "https://${vcfaServicesRuntimeFqdn}/api/v1/system/backups" \
    --header "Authorization: Bearer ${vcfaServicesRuntimeToken}" \
    --header "Content-Type: application/json" \
    --header "Accept: application/json" \
| jq -r '
    ["TYPE", "VERSION", "NAME", "AGE"],
    (
        ["vsp", "vcfa", "vcd-migrator"] as $order
        | .backups as $all
        | $order[] as $current_type
        | $all[]
        | select(.component.type == $current_type)
        | (
            (.name | sub("T(?<h>\\d{2})-(?<m>\\d{2})-(?<s>\\d{2})Z"; "T\(.h):\(.m):\(.s)Z") |
 fromdateiso8601)
        ) as $timestamp
        | ( (now - $timestamp) / 86400 | floor ) as $days_old
        | [ $current_type, .component.version, .name, ($days_old | tostring + " days ago") ]
    )
    | @tsv' | \
    column -t -s $'\t'
```

4. Identity the `NAME` of the backup job and replace the values in the sample code.

```
export vspBackupName='2026-09-01T14-54-30Z'
export vcfaBackupName='2026-09-01T14-54-40Z'
```

5. Generate the `vcfa-restore-payload.json` file by running the following command.

```
curl -k -s -X GET "https://${vcfaServicesRuntimeFqdn}/api/v1/system/backups" \
    --header "Authorization: Bearer ${vcfaServicesRuntimeToken}" \
    --header "Content-Type: application/json" \
    --header "Accept: application/json" \
| jq \
    --arg vsp "${vspBackupName}" \
    --arg vcfa "${vcfaBackupName}" \
    '{
       "components": [
         .backups[]
         | select(
             (.component.type == "vsp" and .name == $vsp) or
             (.component.type == "vcfa" and .name == $vcfa)
           )
         | {
             "point": .name,
             "path": .path
           }
       ]
     }' | tee vcfa-restore-payload.json
```

Restore VCF Automation for VCF Instance Backup and Restore

Restore VCF Automation from a file-based backup.

1. Replace the values in the sample code with your values, these will be used in later steps.

```
export vcfaServicesRuntimeFqdn='flt-vcfa-sr01.rainpole.io'
export vcfaServicesRuntimeUsername='admin@vsp.local'
export vcfaServicesRuntimePassword='***************'
```

2. Retrieve an access token from the VCF Automation by running the following command.

```
vcfaServicesRuntimeToken=$(curl -k -s -X POST "https://${vcfaServicesRuntimeFqdn}/api/v1/identity/token" \
    --header "Content-Type: application/x-www-form-urlencoded" \
    --data-urlencode "grant_type=password" \
    --data-urlencode "username=${vcfaServicesRuntimeUsername}" \
    --data-urlencode "password=${vcfaServicesRuntimePassword}" \
| jq -r '.access_token' | tee /dev/tty )
```

3. Perform the restore operation of VCF Automation using the `vcfa-restore-payload.json` file generated in the Retrieve a List of Backups for VCF Automation for VCF Instance Backup and Restoreprocedure by running the following command.

```
restoreTaskId=$(curl -k -X POST "https://${vcfaServicesRuntimeFqdn}/api/v1/system/backups?action=restore"
 \
    --header "Authorization: Bearer ${vcfaServicesRuntimeToken}" \
    --header "Content-Type: application/json" \
    --header "Accept: application/json" \
    --data @vcfa-restore-payload.json \
| jq -r '.id' | tee /dev/tty )
```

4. Monitor the restore task of VCF Automation by running the following command.

```
while true; do
    # Capture the filtered JSON output
    output=$(curl -k -s -X GET "https://${vcfaServicesRuntimeFqdn}/api/v1/tasks/${restoreTaskId}" \
        --header "Authorization: Bearer ${vcfaServicesRuntimeToken}" \
        --header "Content-Type: application/json" \
        --header "Accept: application/json" \
        | jq '{id: .id, stages: .stages, status: .status, resource: .resource}')

    # Extract just the raw status string for evaluation
    status=$(echo "$output" | jq -r '.status')

    # Check if the status matches the looping conditions
    if [[ "$status" == "Pending" || "$status" == "Running" ]]; then
        echo "Current status is '$status'. Checking again in 120 seconds..."
        sleep 120
    else
        # Stop the loop and print the final JSON payload
        echo "Task completed with status: '$status'."
        echo "$output"
        break
    fi
done
```

Update Connections in VCF Automation for VCF Instance Backup and Restore

After you restore VCF Automation as a part of the VCF Instance Backup and Restore operation, to ensure communication between VCF Automation and the VCF Instances it is connected to, you refresh the connections.

Procedure

1. Log in to the VCF Automation provider portal as a user with the **System Administrator** role.
2. Under Administration, click **VCF Instances**.
3. Click the vertical ellipsis next to the first VCF instance, and click **Rotate Service Account**.
4. Repeat for each VCF Instance.
5. From the left menu, click **Connections**.
6. Verify each vCenter connection is enabled and connected.

vSphere Supervisor Restore Tasks

Tasks related to the restore of vSphere Supervisor.

Mount the NFS Share for the VKr Content Library for VCF Instance Backup and Restore

If you are using an NFS datastore for the content library with the VMware vSphere Kubernetes releases (VKrs), mount the relevant NFS share.

Procedure

1. Log into vCenter for the workload domain as a user with the **administrator** role.
2. In the **Hosts and Clusters** inventory, right- click the vSphere cluster and select **Storage** > **New Datastore**.
3. On the **Type** page, select NFS and click **Next**.
4. On the **NFS version** page, select your NFS version and click **Next**.
5. On the **Name and configuration** page, enter the datastore details and click **Next**.
6. On the **Hosts accessibility** page, select all hosts in the cluster and click **Next**.
7. On the **Ready to complete** page, review the settings and click **Finish**.

Recreate the Content Library folder on the rebuilt vSAN Datatore

After rebuilding a vSAN Datastore, you must create the folder for any dependent Content Libraries

Create vSAN Folder

1. Log into vCenter for the workload domain as a user with the **administrator** role
2. In the Content Libraries inventory, select the VKr Content Library and note the moRef of the library in your browser address bar using he following example for reference

   ```
   https://<vcenter_fqdn>/ui/app/content-libraries/cl/
   urn:vapi:com.vmware.content.Library:<content_library_moref:<vcenter_moref>/summary
   ```
3. In the **Datastores** inventory, select the rebuilt datastore.
4. Select the **Files** tab.
5. Click **New Folder**.
6. Enter the name of the folder in the format **contentlib-<content_library_moref>**

Update the VKr Content Library Datastore Mapping for VCF Instance Backup and Restore

The NFS datastore backing the vSphere content library has a new moref-id. You update the vCenter Content Library Datastore Mapping in the vCenter database to reflect the change.

Procedure

1. Log in to the vCenter appliance as **root** over SSH or directly to appliance console.
2. Type **shell** to switch to the BASH Shell.
3. Connect to the VCDB by running this command.

```
/opt/vmware/vpostgres/current/bin/psql -d VCDB -U postgres
```

4. Find the id of your content library.
   ```
   select id from cl_library where name='content-library-name';
   ```
5. Find the id of the old datastore that corresponds to the content library id from the previous step.
   ```
   select * from cl_library_storage;
   ```
6. Find the storageuri to be updated with the new datastore details.
   ```
   select * from cl_storage;
   ```
7. Get the storageuri for the new NFS datastore.
   a. In a browser, log into the vCenter server where the new NFS datastore is located.
   b. Navigate to Inventory > Storage and click on the datastore.
   c. In the URL select and copy the following string.
      ```
      Datastore:datastore-4682698:c7481195-b0a1-4c74-a67f-c5bce8b2145d
      ```
8. Update the datastore mapping in the cl_storage table.
   ```
   update from cl_storage SET storageuri='storageuri-from-step-7' where id='storage-id-from-step-5';
   ```
9. Reboot the vCenter server.

Monitor the Supervisor Cluster Self Healing Process

After altering the content library datastore reference and rebooting the vCenterm you monitor the supervisor self-healing process.

Monitor the Self-Healing Process

1. In the vSphere Client, go to the **Supervisor Management** inventory and select the relevant supervisor cluster.
2. The supervisor installation will be self-healing after the reboot.
3. On the **Summary** tab monitor the installation until both the status in the **Host Configuration** and **Config Status** columen is **Running**.

Restore the vSphere Supervisor Control Plane for VCF Instance Backup and Restore

You can restore the Supervisors control pane in vCenter from the backup files of the vCenter system itself.

Configure recording the Supervisor state from the vCenter management interface file-based backups.

Restoring the Supervisor control plane in vCenter and restoring the state of vCenter are two different workflows. Restoring vCenter does not lead to restoring the Supervisor control plane.

1. In the vSphere Client, go to **Supervisor Management**.
2. Select **Supervisors**, select the Supervisor to be restored, and click **Restore**.
3. Enter the backup details.

Option	Description
vCenter	Select the vCenter system that manages the Supervisor.
Backup selection	• Select **Browse backup server folder** to upload the files that are stored in the root folder of the backup file server configured with this vCenter system. • Select **Single backup location** to upload a particular backup file, and then enter the URL to that backup file. • Select **Use backup location and user name from vCenter backup schedule** to populate the root folder location and user name for the backup location that are configured with vCenter.
Location	Enter the location of the backup root folder.
Username	Enter the user name to access the backups
Password	Enter the password for that user name.

4. Click **Next**.
5. Select a backup file to restore from and click **Next** to start the download of the backup file.
6. Select the Supervisor to restore and click **Next**.
7. Review the settings and click **Finish**.

   The Supervisor goes back to configuring state and all the control plane VMs are deleted and re-deployed with the data from the backup file. You can monitor the process by clicking **view** in the **Config Status** column.

What to do next:

- Restore vSphere Pods. See Backup and Restore vSphere Pods Using the Velero Plugin for vSphere.
- Restore Stateless and Stateful Workloads on a vSphere Supervisor Cluster. See Backing Up and Restoring VKS Cluster Workloads.
- Restore VM Service VMs on a vSphere Supervisor cluster. See Backing Up and Restoring VM Service VMs in vSphere Supervisor.

Remediate vSphere Supervisor Nodes for VCF Instance Backup and Restore

To ensure the vSphere Supervisor spherelet configuration and certificate are pushed to each ESX host, you must remove and add again each host as a Supervisor node.

1. Retrieve the Supervisor **root** password from vCenter.
   a) Open an SSH connection to vCenter as the **root** user.
   b) Run the following command.

      ```
      /usr/lib/vmware-wcp/decryptK8Pwd.py
      ```

2. Using the **root** password from the previous step, open an SSH connection to the Supervisor.
3. List the Supervisor nodes that are in a **Not Ready** state by running the following command.

   ```
   kubectl get nodes
   ```

4. Check to see if the supervisor-admin has the necessary priviledges.

   ```
   kubectl auth can-i delete nodes
   ```

5. If the result is No, then temporarily grant permissions

   ```
   kubectl --kubeconfig=/etc/kubernetes/admin.conf create clusterrolebinding supervisor-admin-cluster-admin \
     --clusterrole=cluster-admin \
     --user=supervisor-admin
   ```

6. For each ESX host that is in **Not Ready** state
   a) In the vSphere Client, place the host in the list into maintenance mode.
   b) In the SSH session to the Supervisor, run the following command.

      ```
      kubectl delete node <esx-fqdn>
      ```

   c) Return to the vSphere Client and exit the host from maintenance mode.
7. If you had to grant temporary permissions to delete the nodes, then revoke the permission

   ```
   kubectl --kubeconfig=/etc/kubernetes/admin.conf delete clusterrolebinding supervisor-admin-cluster-admin
   ```

Recreate the NSX NCP Pods for VCF Instance Backup and Restore

If the NSX NCP Pods are in a crashed state after the Supervisor restore, you must restore them back to running state.

Procedure

1. Retrieve the Supervisor **root** password from vCenter.
   a. Open an SSH connection to vCenter as the **root** user.
   b. Run the following command.

      ```
      /usr/lib/vmware-wcp/decryptK8Pwd.py
      ```

2. Using the **root** password from the previous step, open an SSH connection to the Supervisor.
3. List the NSX-NCP Pod by running the following command.
   ```
   kubectl get pods -n vmware-system-nsx | grep nsx-ncp
   ```
4. If a pod is in a **Not Ready** state, then restart it NSX-NCP Pod by deleting it.
   When you delete an NSX-NCP Pod, it is automatically recreated.
   ```
   kubectl delete pod <nsx-ncp pod name> -n vmware-system-nsx
   ```
5. After the NSX-NCP Pod is recreated, confirm it is now in a `Running` state:
   ```
   kubectl get pods -n vmware-system-nsx | grep nsx-ncp
   ```
6. Repeat these steps for all NSX-NCP Pods.

License Server Restore Tasks for VCF Instance Backup and Restore

Perform these tasks for restoring a license server when you restore a VCF Instance.

Redeploy a License Server for VCF Instance Backup and Restore

You redeploy the license server to ensure continuity of licensing after you restore a VCF Instance.

Delete a License Server

If you want to replace your License server with a new one, and you want to delete the first License server you must delete it from the VCF Business Services Console and the VCF Operations. See Delete a License Server.

Add and Register an Additional License Server

You can download and install a License server manually. For VCF 9.1.0 see Deploy a License Server Version 9.1.0, for VCF 9.1.1 see Deploy a License Server Version 9.1.1.

Verification and Remediation Tasks for VCF Instance Backup and Restore

Perform these tasks that are related to the verification and remediation after you restore a VCF Instance.

To verify a restored vCenter appliance — network configuration, identity sources, certificates, and inventory reconciliation — complete *Validate the vCenter Instance State* in *Restoring a vCenter Instance*.

A restored vCenter reflects the environment as it was when its backup was taken. For the complete list of configuration that reverts, that may change, or that requires manual reconciliation — including storage policy providers, Distributed Power Management, vSphere HA cluster state, orphaned virtual machines, resource pool and DRS settings, and vSAN — see Considerations and Limitations for File-Based Backup and Restore in the vSphere documentation.

Check SDDC Manager Health for VCF Instance Backup and Restore

After rebuilding a cluster, you should perform a health check using the SoS utility on SDDC Manager.

1. Open an SSH connection to the SDDC Manager appliance, connect using the vcf user.
2. Switch to the root user and change to the `/opt/vmware/sddc-support` folder.
   ```
   su -
   cd /opt/vmware/sddc-support
   ```
3. Run the `sos` health check utility by running the following command.
   ```
   ./sos --health-check --domain-name ALL --skip-cert-check
   ```

4. Remediate any issues reported by the health check.

Perform SDDC Manager Prechecks for VCF Instance Backup and Restore

Perform a pre-check on each workload domain to ensure that the system is in a healthy state.

1. Log in to the VCF Operations interface at `https://<vcf-operations-fqdn>/` as a user assigned the **Administrator** role.
2. Select the **Build** tab.
3. In the navigation pane, click **Lifecycle**.
4. In the **Lifecycle** navigation pane, expand **VCF Instances** > **Instance Name**.
5. Cick the VCF domain where you want to run the pre-check operation on.
6. On the **Summary** pane, click the **Upgrades** tab.
7. Click **Run Precheck**.
8. On the **Run Precheck** dialog, select the **Upgrade Readiness** radio button.
9. On the **Precheck Scope**, select **Entire Workload Domain**.
10. Click **Run Prechecks**.
11. To see details for a task, click the expand arrow.

    If a precheck task failed, fix the issue, and click **Retry** to run the task again. You can also click **Retry all failed resources** to retry all failed tasks.

VCF Fleet Disaster Recovery

This *VCF Fleet Disaster Recovery* guide provides the ability to handle a failure of the VCF Instance or a site and recover VCF fleet components in another location within the same VCF fleet, where that location is typically outside metro radius.

> **Important:**
>
> Use this VCF fleet disaster recovery process together with the design elements of the VCF Fleet Disaster Recovery blueprint. The VCF fleet disaster recovery process outlines the workflow to recover critical VCF fleet components on another VCF Instance within the same fleet, covering both failover and failback of these components.

Table 1168: Plans in this Guide

The *VCF Fleet Disaster Recovery* guide includes structured, runbook-style plans featuring PowerShell cmdlets, manual task links, and supporting documentation for key steps.

Plan	Description
Protection and Recovery Implementation Plan for VCF Fleet Disaster Recovery	Deployment and configuration guidance for protection and recovery, including setting up recovery plans and configuring failover for VCF Operations and VCF Operations for networks.
Failover Plan for VCF Fleet Disaster Recovery	Step-by-step guidance to failover VCF fleet level components to another VCF Instance or site.
Failback Plan for VCF Fleet Disaster Recovery	Step-by-step guidance to fail back VCF fleet level components to another VCF Instance or site.

PowerShell Automation for This Guidance

You can perform the procedures in the plans above manually or by running cmdlets in an associated PowerShell module where available.

The PowerShell cmdlets are available in an open-source module as code-based alternatives to completing certain procedures in each SDDC component's user interface. For more information on the PowerShell cmdlets, see the project on GitHub.

Table 1169: Prerequisites for CLI Implementation of VCF Fleet Disaster Recovery

Prerequisite

• Microsoft Windows 2022 (minimum)
• Microsoft PowerShell 7.5.x (minimum)
• OpenSSL lite version 1.0.2g or higher installed and added to the Windows `PATH` system variable.
• Kubectl latest version.
• PowerCLI **DefaultVIServerMode** setting must be configured as **Multiple** for the all **AllUsers** scope.
• Install the supporting modules from the PowerShell Gallery by running the following commands:
`Install-Module -Name VCF.PowerCLI -MinimumVersion 9.1.0 -Scope AllUsers -Force`
`Install-Module -Name Posh-SSH -RequiredVersion 3.0.8 -Scope AllUsers -Force`
`Install-Module -Name VMware.CloudFoundation.InstanceRecovery -Scope AllUsers -Force`
• Open a new PowerShell console and run the following command to confirm all the prerequisites are installed.
`Confirm-VCFInstanceRecoveryPreReqs`

Note:

For steps in the plan that have PowerShell guidance, the variables in the `## Parameter Block ##` contain sample values. You must update these values to match your environment.

Protection and Recovery Implementation Plan for VCF Fleet Disaster Recovery

You use protection and recovery to protect VCF Operations and VCF Operations for networks. across VCF Instances.

Step	Task	Description
1	Ensure DNS servers from both sites are configured on all components	Managing Infrastructure Services for VMware Cloud Foundation
2	Prerequisites for VCF Fleet Disaster Recovery	Fulfill the prerequisites for installing protection and recovery in workload domains.
3	Deploy Protection and Recovery for Fleet Disaster Recovery for VMware Cloud Foundation	Deploy the protection and recovery appliance and place the Protection and Recovery appliances in the workload domain vCenter where you register them.
4	Replace the Certificate of Protection and Recovery for Fleet Disaster Recovery for VMware Cloud Foundation	Replace the certificate of protection and recovery for your workload domain instance of protection and recovery .
5	Register Protection and Recovery with vCenter Single Sign-On for Fleet Disaster Recovery for VMware Cloud Foundation	Register protection and recovery with vCenter Single Sign-On for your workload domain.
6	Isolate Network Traffic of Protection and Recovery for Fleet Disaster Recovery for VMware Cloud Foundation	Isolate network traffic of protection and recovery for the ESX hosts for the workload domain.
7	Deploy Protection and Recovery for Fleet Disaster Recovery for VMware Cloud Foundation	Deploy protection and recovery for the VCF domain of your VMware Cloud Foundation instance and place the protection and recovery appliances in the management domain.
8	Replace the Certificate of Protection and Recovery for Fleet Disaster Recovery for VMware Cloud Foundation	Replace the certificate of protection and recovery for your workload domain instance of protection and recovery .

Step	Task	Description
9	Register Protection and Recovery with vCenter Single Sign-On for Fleet Disaster Recovery for VMware Cloud Foundation	Register protection and recovery with vCenter Single Sign-On for your workload domain instance of protection and recovery .
10	Assign Licenses to Protection and Recovery using Offline Mode for Fleet Disaster Recovery for VMware Cloud Foundation	Assign a license to the protection and recovery instance for your workload domain.
11	Create a Site Pair Between the Protected and Recovery VMware Cloud Foundation Instances	Create a site pair between the protected and recovery VMware Cloud Foundation instances for your workload domain.
12	Configure Mappings between the Protected and the Recovery VMware Cloud Foundation Instances	Configure mappings between the protected and the recovery VMware Cloud Foundation instances for your workload domain.
13	Configure Replication, Create a Protection Group and a Recovery Plan for VCF Operations for Fleet Disaster Recovery for VMware Cloud Foundation	Configure the replication for VCF Operations.
14	Customize the Recovery Plan for VCF Operations for Fleet Disaster Recovery for VMware Cloud Foundation	Customize the recovery plan for VCF Operations.
15	Configure Replication, Create a Protection Group and a Recovery Plan for VCF Operations for Networks for Fleet Disaster Recovery for VMware Cloud Foundation	Configure the replication for VCF Operations for networks
16	Customize the Recovery Plan for VCF Operations for Networks for Fleet Disaster Recovery for VMware Cloud Foundation	Customize the recovery plan for VCF Operations for networks.
17	Create Anti-Affinity Rules for the Placeholder Virtual Machines in the Recovery VMware Cloud Foundation Instance	Create anti-affinity rules for the placeholder virtual machines.
18	Create Virtual Machine Groups and Restart Order in the Recovery VMware Cloud Foundation Instance	Create virtual machine groups to define the startup order in the recovery site.

Failover Plan for VCF Fleet Disaster Recovery

To perform the failover of fleet components to a recovery VCF Instance after a disaster, you perform the following steps.

Summary of Failover Plan for VCF Fleet Disaster Recovery

- NSX Federation Tasks in the Recovery VCF Instance
- Recover the VCF Operations Appliances
- Recover the VCF Management Services for the VCF Fleet
- Recover VCF Automation
- NSX Federation Post-Failover Configuration After the Original Protected VMware Cloud Foundation Instance Is Available Again
- Reprotect the VCF Operations Appliances

Note:

The steps in this plan are for a full disaster recovery of the VCF fleet components. It should not be used to recover individual components.

NSX Federation Tasks in the Recovery VCF Instance

The following tasks are applicable if you are using NSX Federation between VCF Instances.

Step	Manual Task	PowerShell Equivalent
1	Assign the Active Role to the NSX Global Manager Instance in the Recovery VMware Cloud Foundation Instance	No PowerShell equivalent, follow the manual procedure.
2	Reconfigure the NSX Tier-1 Gateway for Dynamic Routing in the Recovery VMware Cloud Foundation Instance	No PowerShell equivalent, follow the manual procedure.
3	Connect the NSX Load Balancer in the Recovery VMware Cloud Foundation Instance to the Standalone NSX Tier-1 Gateway	No PowerShell equivalent, follow the manual procedure.

Recover the VCF Operations Appliances

Step	Manual Task	PowerShell Equivalent
1	Initiate a Disaster Recovery of VCF Operations for Fleet Disaster Recovery for VMware Cloud Foundation	No PowerShell equivalent, follow the manual procedure.
2	Initiate a Disaster Recovery of VCF Operations for networks for Fleet Disaster Recovery for VMware Cloud Foundation	No PowerShell equivalent, follow the manual procedure.

Recover the VCF Management Services for the VCF Fleet

Step	Manual Task	PowerShell Equivalent
1	Modify DNS Entries for Fleet Component VIPs Before Recovery	No PowerShell equivalent, follow the manual procedure.
2	Retrieve Component Details from VCF Operations	```## Parameter Block ## $VcfOperationsFqdn = "flt-ops01a.rainpole.io" $Username = "admin" $Password = "**************" $AuthSource = "local" $outputDir = ".\" ## Command Block ## Get-VcfOperationsRegisteredComponents ` -VcfOperationsFqdn $VcfOperationsFqdn ` -Username $Username ` -Password $Password ` -AuthSource $AuthSource ` -OutputDir $outputDir```

Step	Manual Task	PowerShell Equivalent
3	Extract Fleet VIP Address Configuration from the Source Services Runtime Backup	```## Parameter Block ## $LocalArchivePath = ".\2026-03-23T16-45-31Z.base.tgz" $EncryptionPassphrase = "**********" $OutputDir = ".\backup-yaml" $ServicesRuntimeFqdn = "lax-sr01.lax.rainpole.io" $ServicesRuntimePassword = "**********" ## Command Block ## New-ExtractVcfmsBackup ` -LocalArchivePath $LocalArchivePath ` -EncryptionPassphrase $EncryptionPassphrase ` -OutputDir $OutputDir ` -ServicesRuntimeFqdn $ServicesRuntimeFqdn ` -ServicesRuntimePassword $ServicesRuntimePassword```
4	Scale the Recovery Site VCF Management Services Instance for VCF Fleet Disaster Recovery	```## Parameter Block ## $KubeconfigPath = ".\backup-yaml\lax-sr01.kubeconfig" # Kubeconfig for the recovery VCF Management Service cluster $BackupYamlDir = ".\backup-yaml" ## Command Block ## Set-VcfmsServicesRuntimeScale ` -KubeconfigPath $KubeconfigPath ` -BackupYamlDir $BackupYamlDir```
5	Disable Log Management Integration on Recovery VCF Management Services Instance **Important:** Only required on VCF 9.1.0.x	```## Parameter Block ## $ServicesRuntimeFqdn = "lax-sr01.lax.rainpole.io" $ServicesRuntimePassword = "**********" $KubeconfigPath = ".\backup-yaml\lax-sr01.kubeconfig" ## Command Block ## Disable-VcfmsClusterLogging ` -ServicesRuntimeFqdn $ServicesRuntimeFqdn ` -ServicesRuntimePassword $ServicesRuntimePassword ` -KubeconfigPath $KubeconfigPath```
6	Apply Fleet VIP on Recovery VCF Management Services Instance	```## Parameter Block ## $BackupYamlDir = ".\backup-yaml" $ServicesRuntimeFqdn = "lax-sr01.lax.rainpole.io" $ServicesRuntimePassword = "**********" $KubeconfigPath = ".\backup-yaml\lax-sr01.kubeconfig" $FleetFqdn = "flt-fc01.rainpole.io" $FleetVip = "10.21.99.20" ## Command Block ## Set-VcfmsFleetIdentity ` -BackupYamlDir $BackupYamlDir ` -KubeconfigPath $KubeconfigPath ` -ServicesRuntimeFqdn $ServicesRuntimeFqdn ` -ServicesRuntimePassword $ServicesRuntimePassword ` -FleetFqdn $FleetFqdn ` -FleetVip $FleetVip```

Step	Manual Task	PowerShell Equivalent
7	Retrieve a List of Backups for VCF Management Services Fleet Components	```## Parameter Block ## $sftpServer = "10.167.173.130" $sftpUser = "svc-vcf-bck" $sftpPassword = "**************" $sftpServerBackupPath = "/media/backups" $vspId = "40e752c7-e940-4a9f-a49c-0f05efa9d915" # ID of the VCF Management Services instance from step 2 $componentNames = "vcf-fleet-lcm","vcf-fleet-depot","salt-raas","vidb" ## Command Block ## Get-BackupsFromSFTPServer ` -sftpServer $sftpServer ` -sftpUser $sftpUser ` -sftpPassword $sftpPassword ` -sftpServerBackupPath $sftpServerBackupPath ` -vspId $vspId ` -componentNames $componentNames```
8	Stage Binaries on the Recovery VCF Management Services Instance	No PowerShell equivalent, follow the manual procedure.
9	Restore the Fleet Components on the Recovery VCF Management Services Instance	```## Parameter Block ## $ServicesRuntimeFqdn = "lax-sr01.lax.rainpole.io" $ServicesRuntimePassword = "**************" ## Command Block ## Restore-ServicesRuntimeComponentBackup ` -ServicesRuntimeFqdn $ServicesRuntimeFqdn ` -ServicesRuntimePassword $ServicesRuntimePassword ` -RestoreJsonFile ".\component-restore-payload.json"```
10	Update SDDC LCM in the Target Site with Credentials from the Source Site	No PowerShell equivalent, follow the manual procedure.
11	Update the Identity Broker VIP Address	```## Parameter Block ## $ServicesRuntimeFqdn = "lax-sr01.lax.rainpole.io" $ServicesRuntimePassword = "**************" $ComponentType = "vidb" $vips = "10.21.99.23" ## Command Block ## Set-ServicesRuntimeComponentVips ` -ServicesRuntimeFqdn $ServicesRuntimeFqdn ` -ServicesRuntimePassword $ServicesRuntimePassword ` -ComponentType $ComponentType ` -Vips $vips```
12	Update the Identity Broker VCF Instance Association	```## Parameter Block ## $VcfOpsFqdn = "flt-ops01a.rainpole.io" $VcfOpsRootPassword = "**************" $VcfOpsAdminPassword = "**************" $VidbFqdn = "flt-idb01.rainpole.io" ## Command Block ## Invoke-VcfOpsVidbVcfInstanceUpdate ` -VcfOpsFqdn $VcfOpsFqdn ` -VcfOpsRootPassword $VcfOpsRootPassword ` -VcfOpsAdminPassword $VcfOpsAdminPassword ` -VidbFqdn $VidbFqdn```

Step	Manual Task	PowerShell Equivalent
13	Update Fleet Component Registration	```## Parameter Block ## $ServicesRuntimeFqdn = "lax-sr01.lax.rainpole.io" $ServicesRuntimePassword = "**************" $TargetVcfInstance = "lax-ic01.lax.rainpole.io" ## Command Block ## Invoke-VcfmsFleetComponentRegistration ` -ServicesRuntimeFqdn $ServicesRuntimeFqdn ` -ServicesRuntimePassword $ServicesRuntimePassword ` -TargetVcfInstance $TargetVcfInstance ` -ConfirmOpsMove ` -ConfirmOpsNetworksMove``` **Note:** Service registration might take up to 15 minutes to complete.
14	Update VCF Operations Inventory	No PowerShell equivalent, follow the manual procedure.
15	Retrieve a List of Backups for Log Management	```## Parameter Block ## $sftpServer = "10.167.173.130" $sftpUser = "svc-vcf-bck" $sftpPassword = "**************" $sftpServerBackupPath = "/media/backups" $vspId = "40e752c7-e940-4a9f-a49c-0f05efa9d915" # ID of the VCF Management Services instance from step 2 $componentNames = "ops-logs" ## Command Block ## Get-BackupsFromSFTPServer ` -sftpServer $sftpServer ` -sftpUser $sftpUser ` -sftpPassword $sftpPassword ` -sftpServerBackupPath $sftpServerBackupPath ` -vspId $vspId ` -componentNames $componentNames```
16	Restore Log Management	```## Parameter Block ## $ServicesRuntimeFqdn = "lax-sr01.lax.rainpole.io" $ServicesRuntimePassword = "**************" ## Command Block ## Restore-ServicesRuntimeComponentBackup ` -ServicesRuntimeFqdn $ServicesRuntimeFqdn ` -ServicesRuntimePassword $ServicesRuntimePassword ` -RestoreJsonFile ".\component-restore-payload.json"```
17	Update the Log Management VIP Address	```## Parameter Block ## $ServicesRuntimeFqdn = "lax-sr01.lax.rainpole.io" $ServicesRuntimePassword = "**************" $ComponentType = "ops-logs" $vips = "10.21.99.26" ## Command Block ## Set-ServicesRuntimeComponentVips ` -ServicesRuntimeFqdn $ServicesRuntimeFqdn ` -ServicesRuntimePassword $ServicesRuntimePassword ` -ComponentType $ComponentType ` -Vips $vips```
18	Update the Log Management State	No PowerShell equivalent, follow the manual procedure.

Step	Manual Task	PowerShell Equivalent
19	Enable Logging on the Recovery Site VCF Management Services Instance **Important:** Only required on VCF 9.1.0.x	```## Parameter Block ## $ServicesRuntimeFqdn = "lax-sr01.lax.rainpole.io" $ServicesRuntimePassword = "**************" $LogManagementVip = "flt-logs01.rainpole.io" ## Command Block ## Enable-VcfmsClusterLogging ` -ServicesRuntimeFqdn $ServicesRuntimeFqdn ` -ServicesRuntimePassword $ServicesRuntimePassword ` -LogManagementVip $LogManagementVip```
20	Redeploy a License Server for VCF Instance Backup and Restore **Important:** Only required if the failover will last for more than 90 days.	No PowerShell equivalent, follow the manual procedure.

Recover VCF Automation

Step	Manual Task	PowerShell Equivalent
1	Deploy a VCF Automation Services Runtime Cluster	```## Parameter Block ## $SddcManagerFqdn = "lax-vcf01.lax.rainpole.io" $SddcManagerUser = "administrator@vsphere.local" $SddcManagerPassword = "**************" $PlatformFqdn = "flt-vcfa-sr01.rainpole.io" $SystemUserPassword = "**************" $Ipv4Addresses = "10.11.99.29","10.11.99.30","10.11.99.31","10.11.99.32","10.11.99.33" $size = "small" $NetworkMoId = "dvportgroup-1001" $GatewayCidrIpv4 = "10.11.99.1/24" $ClusterId = "afbdb184-25de-4025-9cc2-8569a54d093d" $InternalClusterCidrIpv4 = "198.18.0.0/15" $Type = "CONSUMPTION" ## Command Block ## New-ServicesRuntime ` -SddcManagerFqdn $SddcManagerFqdn ` -SddcManagerUser "administrator@vsphere.local" ` -SddcManagerPassword $SddcManagerPassword ` -PlatformFqdn $PlatformFqdn ` -SystemUserPassword $SystemUserPassword ` -Ipv4Addresses $Ipv4Addresses ` -Size $size ` -NetworkMoId $NetworkMoId ` -GatewayCidrIpv4 $GatewayCidrIpv4 ` -ClusterId $ClusterId ` -InternalClusterCidrIpv4 $InternalClusterCidrIpv4 ` -Type $Type```
2	Update SFTP Settings on a new Services Runtime Instance	```## Parameter Block ## $ServicesRuntimeFqdn = "flt-vcfa-sr01.rainpole.io" $ServicesRuntimePassword = "**************" $ComponentId = "1f5c79fe-e3aa-41b1-a5cf-774a6497fa3d" $SftpHost = "10.167.173.130" $SftpUsername = "svc-vcf-bck" $SftpPassword = "********" $SftpDirectory = "/media/backups/" $EncryptionPassphrase = "**************" ## Command Block ## Set-ServicesRuntimeSftpBackupSettings ` -ServicesRuntimeFqdn $ServicesRuntimeFqdn ` -ServicesRuntimePassword $ServicesRuntimePassword ` -ComponentId $ComponentId ` -SftpHost $SftpHost ` -SftpUsername $SftpUsername ` -SftpPassword $SftpPassword ` -SftpDirectory $SftpDirectory ` -EncryptionPassphrase $EncryptionPassphrase```

Step	Manual Task	PowerShell Equivalent
3	Retrieve a List of Backups for VCF Automation	```## Parameter Block ## $sftpServer = "10.167.173.130" $sftpUser = "svc-vcf-bck" $sftpPassword = "********" $sftpServerBackupPath = "/media/backups" $vspId = "afbdb184-25de-4025-9cc2-8569a54d093d" # clusterId from step 1 $componentNames = "vsp", "vcfa", "vcd-migrator" ## Command Block ## Get-BackupsFromSFTPServer ` -sftpServer $sftpServer ` -sftpUser $sftpUser ` -sftpPassword $sftpPassword ` -sftpServerBackupPath $sftpServerBackupPath ` -vspId $vspId ` -componentNames $componentNames```
4	Stage Binaries for VCF Automation	No PowerShell equivalent, follow the manual procedure.
5	Restore VCF Automation for VCF Fleet Disaster Recovery	```## Parameter Block ## $ServicesRuntimeFqdn = "flt-vcfa-sr01.rainpole.io" $ServicesRuntimePassword = "**************" ## Command Block ## Restore-ServicesRuntimeComponentBackup ` -ServicesRuntimeFqdn $ServicesRuntimeFqdn ` -ServicesRuntimePassword $ServicesRuntimePassword ` -RestoreJsonFile ".\restore-payload.json"```
6	Update the VCF Automation VIP Address	```## Parameter Block ## $ServicesRuntimeFqdn = "flt-vcfa-sr01.rainpole.io" $ServicesRuntimePassword = "**************" $ComponentType = "vcfa" $vips = "10.21.99.25" ## Command Block ## Set-ServicesRuntimeComponentVips ` -ServicesRuntimeFqdn $ServicesRuntimeFqdn ` -ServicesRuntimePassword $ServicesRuntimePassword ` -ComponentType $ComponentType ` -Vips $vips```

Step	Manual Task	PowerShell Equivalent
8	Update VCF Automation Registration to VCF Operations	```## Parameter Block ## $ServicesRuntimeFqdn = "lax-sr01.lax.rainpole.io" $ServicesRuntimePassword = "**************" $VcfaRuntimeFqdn = "flt-vcfa-sr01.rainpole.io" $OpsFqdn = "flt-ops01a.rainpole.io" $VspPassword = "**************" $VcfaSystemPassword = "**************" $OpsPassword = "**************" ## Command Block ## Update-VcfmsVcfaRegistration ` -ServicesRuntimeFqdn $ServicesRuntimeFqdn ` -ServicesRuntimePassword $ServicesRuntimePassword ` -VcfaRuntimeFqdn $VcfaRuntimeFqdn ` -OpsFqdn $OpsFqdn ` -VspPassword $VspPassword ` -VcfaSystemPassword $VcfaSystemPassword ` -OpsPassword $OpsPassword```

Update Backup Configuration

Step	Manual Task	PowerShell Equivalent
1	Update Backup Schedule Configuration for Recovered Components	Not applicable. Follow the manual task.

NSX Federation Post-Failover Configuration After the Original Protected VMware Cloud Foundation Instance Is Available Again

Note:

The following tasks are applicable if you are using NSX Federation between VCF Instances.

Step	Manual Task	PowerShell Equivalent
1	Disconnect the NSX Load Balancer in the Original Protected VMware Cloud Foundation Instance from the Standalone Tier-1 Gateway	Not applicable. Follow the manual task.
2	Obtain the Certificate Thumbprint for the NSX Global Manager Virtual IP for the Original Protected VMware Cloud Foundation Instance	Not applicable. Follow the manual task.
3	Reconfigure the NSX Global Manager in the Original Protected VMware Cloud Foundation Instance	Not applicable. Follow the manual task.
4	Reconfigure the Tier-1 Gateway for Dynamic Routing in the Original Protected VMware Cloud Foundation Instance	Not applicable. Follow the manual task.

Reprotect the VCF Operations Appliances

Step	Manual Task	PowerShell Equivalent
1	Prerequisites for Reprotection for VCF Fleet Disaster Recovery	Not applicable. Follow the manual task.
2	Reprotect VCF Operations for Fleet Disaster Recovery for VMware Cloud Foundation	Not applicable. Follow the manual task.
3	Reprotect VCF Operations for networks for Fleet Disaster Recovery for	Not applicable. Follow the manual task.
4	Back up the Recovered Components to a Highly Available SFTP Server.	Not applicable. Follow the manual task.

Important:

Do not leave the reprotection operations until after the failback operation is complete. By reprotecting your VCF Operations and VCF Operations for networks immediately, you can fail back to the original site at any time, even in an unplanned manner.

Failback Plan for VCF Fleet Disaster Recovery

To perform the failback of fleet components to the original primary VCF Instance after it is back online, you perform the following steps.

NSX Federation Tasks in the Recovery VCF Instance

Note:

The following tasks are applicable if you are using NSX Federation between VCF Instances.

Step	Manual Task	PowerShell Equivalent
1	Assign the Active Role to the NSX Global Manager Instance in the Recovery VMware Cloud Foundation Instance	Not applicable. Follow the manual task.
2	Reconfigure the NSX Tier-1 Gateway for Dynamic Routing in the Recovery VMware Cloud Foundation Instance	Not applicable. Follow the manual task.
3	Connect the NSX Load Balancer in the Recovery VMware Cloud Foundation Instance to the Standalone NSX Tier-1 Gateway	Not applicable. Follow the manual task.

Reprotect VCF Operations Appliances

Step	Manual Task	PowerShell Equivalent
1	Reprotect VCF Operations for Fleet Disaster Recovery for VMware Cloud Foundation	Not applicable. Follow the manual task.
2	Reprotect VCF Operations for networks for Fleet Disaster Recovery for	Not applicable. Follow the manual task.

Fail Back the VCF Operations Appliances

Step	Manual Task	PowerShell Equivalent
1	Initiate a Planned Migration of VCF Operations for Fleet Disaster Recovery for VMware Cloud Foundation	Not applicable. Follow the manual task.
2	Initiate a Planned Migration of VCF Operations for networks for Fleet Disaster Recovery for VMware Cloud Foundation	Not applicable. Follow the manual task.

Revert the VCF Management Services for the VCF Fleet

Step	Manual Task	PowerShell Equivalent
1	Back up the Recovered Components to a Highly-Available SFTP Server.	Not applicable. Follow the manual task.
2	Retrieve a List of Backups for VCF Management Services Fleet Components	`## Parameter Block ##` `$sftpServer = "10.167.173.130"` `$sftpUser = "svc-vcf-bck"` `$sftpPassword = "**************"` `$sftpServerBackupPath = "/media/backups"` `$vspId = "40e752c7-e940-4a9f-a49c-0f05efa9d915" # ID of the VCF Management Services instance` `$componentNames = "vcf-fleet-lcm","vcf-fleet-depot","salt-raas","vidb"` `## Command Block ##` `Get-BackupsFromSFTPServer `` `-sftpServer $sftpServer `` `-sftpUser $sftpUser `` `-sftpPassword $sftpPassword `` `-sftpServerBackupPath $sftpServerBackupPath `` `-vspId $vspId `` `-componentNames $componentNames`
3	Cleanup the recovered components from the recovery Site VCF Management Services Instance	`## Parameter Block ##` `$ServicesRuntimeFqdn = "lax-sr01.lax.rainpole.io"` `$ServicesRuntimePassword = "**************"` `## Command Block ##` `Remove-VcfmsRecoveredComponents `` `-ServicesRuntimeFqdn $ServicesRuntimeFqdn `` `-ServicesRuntimePassword $ServicesRuntimePassword`
4	Cleanup Fleet VIP Prior to Failback	`## Parameter Block ##` `$ServicesRuntimeFqdn = "lax-sr01.lax.rainpole.io"` `$ServicesRuntimePassword = "**************"` `## Command Block ##` `Clear-VcfmsFleetIdentity `` `-ServicesRuntimeFqdn $ServicesRuntimeFqdn `` `-ServicesRuntimePassword $ServicesRuntimePassword`
5	Modify DNS Entries for Fleet Component VIPs Before Recovery	Not Applicable. Follow the manual task.
6	Power on the VCF Management Services Instance virtual machines on the original protected Site	Not Applicable. Perform this task in vCenter.

Step	Manual Task	PowerShell Equivalent
7	Restore the Fleet Components on the Recovery VCF Management Services Instance **Note:** Restoring backups taken while in a failed over state is only required if changes were made to a component while in a failed over state.	```## Parameter Block ## $ServicesRuntimeFqdn = "sfo-sr01.sfo.rainpole.io" $ServicesRuntimePassword = "**************" ## Command Block ## Restore-ServicesRuntimeComponentBackup ` -ServicesRuntimeFqdn $ServicesRuntimeFqdn ` -ServicesRuntimePassword $ServicesRuntimePassword ` -RestoreJsonFile ".\restore-payload.json"```
8	Update SDDC LCM in the Target Site with Credentials from the Source Site	No PowerShell equivalent, follow the manual procedure.
9	Update the Identity Broker VCF Instance Association	```## Parameter Block ## $VcfOpsFqdn = "flt-ops01a.rainpole.io" $VcfOpsRootPassword = "**************" $VcfOpsAdminPassword = "**************" $VidbFqdn = "flt-idb01.rainpole.io" ## Command Block ## Invoke-VcfOpsVidbVcfInstanceUpdate ` -VcfOpsFqdn $VcfOpsFqdn ` -VcfOpsRootPassword $VcfOpsRootPassword ` -VcfOpsAdminPassword $VcfOpsAdminPassword ` -VidbFqdn $VidbFqdn``` **Note:** Service registration might take up to 15 minutes to complete.
10	Update Fleet Component Registration	```## Parameter Block ## $ServicesRuntimeFqdn = "sfo-sr01.lax.rainpole.io" $ServicesRuntimePassword = "**************" $TargetVcfInstance = "sfo-ic01.sfo.rainpole.io" ## Command Block ## Invoke-VcfmsFleetComponentRegistration ` -ServicesRuntimeFqdn $ServicesRuntimeFqdn ` -ServicesRuntimePassword $ServicesRuntimePassword ` -TargetVcfInstance $TargetVcfInstance```

Revert VCF Automation

Step	Manual Task	PowerShell Equivalent
1	Perform an on-demand backup of VCF Automation in the recovery site	Not applicable.
2	Delete VCF Automation component from the Recovery Site	Not applicable. Follow step 3 in Broadcom knowledge base article 441333.
3	Prepare to Fail Back VCF Automation	
4	Power on the VCF Automation Instance on the Original Protected Site	Not applicable. Perform this task on vCenter.

Step	Manual Task	PowerShell Equivalent
3	Retrieve a List of Backups for VCF Automation **Note:** Restoring backups taken while in a failed over state is only required if changes were made to a component while in a failed over state.	```## Parameter Block ## $ServicesRuntimeFqdn = "sfo-sr01.sfo.rainpole.io" $ServicesRuntimePassword = "**************" $VspId = "40e752c7-e940-4a9f-a49c-0f05efa9d915" $components = "vcfa" ## Command Block ## Get-VcfmsBackups ` -ServicesRuntimeFqdn $ServicesRuntimeFqdn ` -ServicesRuntimePassword $ServicesRuntimePassword ` -VspId $VspId ` -components $components```
4	Restore VCF Automation for VCF Fleet Disaster Recovery **Note:** Restoring backups taken while in a failed over state is only required if changes were made to a component while in a failed over state.	```## Parameter Block ## $ServicesRuntimeFqdn = "sfo-sr01.sfo.rainpole.io" $ServicesRuntimePassword = "**************" ## Command Block ## Restore-VcfmsBackup ` -ServicesRuntimeFqdn $ServicesRuntimeFqdn ` -ServicesRuntimePassword $ServicesRuntimePassword ` -RestoreJsonFile ".\restore-payload.json"```

Update Backup Configuration

Step	Manual Task	PowerShell Equivalent
1	Update Backup Schedule Configuration for Recovered Components	Not applicable. Follow the manual task.

Disaster Recovery Task Library

A hierarchy of tasks that are associated with fleet disaster recovery, sorted by component, or functional area.

CAUTION:

The order in which tasks are listed in the library is not necessarily the order in which they should be performed. Refer to the relevant restore plan to see the sequence of tasks appropriate to the elements you are restoring.

Prerequisites for VCF Fleet Disaster Recovery

Before you implement the *VCF Fleet Disaster Recovery* guidance, your environment must meet certain requirements for networking and component deployments.

When you prepare your environment for disaster recovery, you must determine which of your two VCF Instances functions as the protected site and which one as the recovery site.

- The protected site hosts the business-critical SDDC services.
- The recovery site is an alternative location to which these applications migrate to if a disaster occurs.

In this guide, the protected site is referred to as the protected VCF Instance and the recovery site is referred to as the recovery VCF Instance.

Prerequisites

To complete the implementation of *Fleet Disaster Recovery for VMware Cloud Foundation*, verify that your system fulfills the following prerequisites.

Table 1170: Prerequisites for Implementation of Fleet Disaster Recovery for VMware Cloud Foundation

Category	Prerequisite
Environment	• Verify that your VCF version is 9.1.x on the protected and recovery sites. • Verify that you have sufficient resources on the recovery site to run the recovered management components. • Verify that the chosen recovery site does not have a running instance of any of the components you plan to recover, such as Identity broker or software depot.
Domain Name Service	• Verify that the required DNS entries are created in the DNS server for the associated forward and reverse zones. • Verify that all components to be failed over have the DNS servers from the protected and recovery sites configured. • Reserve IP addresses in the DNS server for the fleet components that will be failed over.
Network	• Verify that you have satisfied the IP mobility requirements between VCF Instances. • Verify that your VCF Instances have connectivity between each other and that the connection supports jumbo frames and Layer 3 routing. • Verify that all uplinks, port channels, and VLANs that carry overlay and Protection and Recovery traffic are configured for jumbo frames. • Verify that your environment supports the maximum supported latency between VCF instances. • Verify that sufficient bandwidth is available for replication traffic. See the Protection and Recovery Calculator.
Software	• Download the Protection and Recovery `.ova` image and mount it on the machine that you use to access the vSphere Client.
License	• Verify that you have obtained a Protection and Recovery license with a quantity that fulfills the requirements of your design.

Implementation Tasks for VCF Fleet Disaster Recovery

Library of tasks used to implement VCF fleet disaster recovery.

Deploy Protection and Recovery for Fleet Disaster Recovery for VMware Cloud Foundation

To manage virtual machine replication, protection, and recovery, deploy the protection and recovery appliance in the protected and recovery instances.

For the protected VMware Cloud Foundation instance procedure, use the VMware Cloud Foundation Planning and Preparation Workbook.

Procedure

1. Log in to the management domain vCenter at `https://<management_vcenter_fqdn>/ui` in the protected instance as **administrator@vsphere.local**.
2. In the **Hosts and clusters** inventory, expand the management domain vCenter tree and expand the management domain data center.
3. Deploy the new protection and recovery appliance.
   a. Right-click the default management cluster and select **Deploy OVF template**.
   b. On the **Select an OVF template** page, select **Local file**, click **Upload files**, and click **Next**.

Protection-and-recovery-9.1.x.0.XXXXXXXX.ova

   c. On the **Select a name and folder** page, in the **Virtual machine name** text box, enter a virtual machine name according to your VMware Cloud Foundation Planning and Preparation Workbook, and click **Next**.
   d. On the **Select a compute resource** page, select the default management cluster and click **Next**.
   e. On the **Review details** page, click **Next**.

f. On the **License agreements** page, accept the license agreement and click **Next**.
g. On the **Select storage** page, select the management domain vSAN datastore and click **Next**.
h. On the **Select networks** page, select the management network and click **Next**.
i. On the **Customize template** page, configure the following settings and configure the remaining settings according to your VMware Cloud Foundation Planning and Preparation Workbook, and click **Next**.

Setting	Value
Enable SSHD	Selected
Host network IP address family	ipv4
Host network mode	static

j. On the **Ready to complete** page, click **Finish** and wait for the completion of the process.
k. Once the deployment is completed successfully, right-click the *Protection and Recovery* appliance and select **Power** > **Power on**.

4. Repeat this procedure to deploy the *Protection and Recovery* appliance in the recovery VMware Cloud Foundation instance by using the corresponding values in your VMware Cloud Foundation Planning and Preparation Workbook for the Recovery Instance.

Replace the Certificate of Protection and Recovery for Fleet Disaster Recovery for VMware Cloud Foundation

To ensure secure communication with the connected management components, replace the certificates on the *Protection and Recovery* appliance in the protected and the recovery instances.

For the protected instance procedure, use the VMware Cloud Foundation Planning and Preparation Workbook.

Procedure

1. Log in to the VMware Live Recovery appliance configuration interface at `https://<vlr_fqdn>:5480` in the protected instance as **admin**.
2. In the left pane, click **Certificates**.
3. Generate the CSR and download the file.
   a. In the **Appliance certificate** section, click **Generate CSR**.
   b. In the **New Private Key and CSR** dialog box, configure the settings according to your VMware Cloud Foundation Planning and Preparation Workbook, and click **Generate and Download**.
4. Use the generated CSR file to submit a certificate request to the Microsoft Active Directory Certificate Services and get a signed certificate file.
5. Replace the SSL certificate.
   a. In the **Appliance certificate** section, click **Change**.
   b. In the **Change certificate** dialog box, select certificate type to **Use a CA-signed certificate generated from CSR**, upload the signed certificate file, and click **Change**.
6. Repeat this procedure to replace the certificate on the protection and recovery appliance in the recovery VMware Cloud Foundation instance by using the corresponding values in your VMware Cloud Foundation Planning and Preparation Workbook for the Recovery Instance.

Create vCenter Single Sign-On User for Registering *Protection and Recovery* for Fleet Disaster Recovery for VMware Cloud Foundation

You create vCenter Single Sign-On local service account to register *Protection and Recovery* with vCenter.

Procedure

1. Log in to the management domain vCenter at `https://<management_vcenter_fqdn>/ui` in the protected instance as **administrator@vsphere.local**.
2. From the vSphere Client Menu, select **Administration**.
3. In the **Single-Sign On** section, click **Users and groups**.
4. Click the **Users** tab.
5. From the **Domain** drop-down menu, select the vCenter Single Sign-On domain and click **Add**.
6. In the **Add user** dialog box, create the *Protection and Recovery* service account according to your values in the VMware Cloud Foundation Planning and Preparation Workbook, and click **Add**.
7. Repeat this procedure to create a new vCenter Single Sign-On local service account on the isolated workload domain of the recovery instance by using the corresponding values in your VMware Cloud Foundation Planning and Preparation Workbook for the Recovery Instance.

Assign vCenter Single Sign-On User with Group Role for Fleet Disaster Recovery for VMware Cloud Foundation

You assign vCenter Single Sign-On local service account with **Administrators** group access in vCenter.

Procedure

1. Log in to the management domain vCenter at `https://<management_vcenter_fqdn>/ui` in the protected instance as **administrator@vsphere.local**.
2. From the vSphere Client Menu, select **Administration**.
3. In the **Single-Sign On** section, click **Users and groups**.
4. Click the **Groups** tab.
5. Select the **Administrators** group and click **Edit**.
6. In the **Edit group** dialog box, from the **Add a member** drop-down menu, select the vCenter Single Sign-On domain.
7. In the **Search** box, enter the vCenter Single Sign-On local service account according to your VMware Cloud Foundation Planning and Preparation Workbook, and press **Enter**.
8. Click **Save**.
9. Repeat this procedure to assign roles to the vCenter Single Sign-On local service account on the isolated workload domain of the recovery instance by using the corresponding values in your VMware Cloud Foundation Planning and Preparation Workbook for the Recovery Instance.

Register Protection and Recovery with vCenter Single Sign-On for Fleet Disaster Recovery for VMware Cloud Foundation

After you deploy the *Protection and Recovery* appliances in the protected and recovery instances, to start protecting virtual machines, you register the *Protection and Recovery* appliances with the protected and recovery vCenter instances.

For the protected VMware Cloud Foundation instance procedure, use the VMware Cloud Foundation Planning and Preparation Workbook.

Procedure

1. Log in to the Protection and Recovery appliance configuration interface at `https://<vlr_fqdn>:5480` as **admin**.
2. In the left pane, click **Summary** and click **Configure appliance**.
   The **Configure Protection and Recovery** wizard opens.
3. On the **vCenter Connection** page, configure the following settings, configure the remaining settings according to your VMware Cloud Foundation Planning and Preparation Workbook, and click **Next**.

Setting	Value
vCenter port	443

4. In the **Security alert** dialog box, click **Connect**.
5. On the **vCenter** page, select the management domain vCenter for the protected VMware Cloud Foundation instance, and click **Next**.
6. In the **Security alert** dialog box, click **Connect**.
7. On the **Name and extension** page, configure the following settings, configure the remaining settings according to your VMware Cloud Foundation Planning and Preparation Workbook, and click **Next**.

Setting	Value
Live Site Recovery Extension ID	Default extension ID (com.vmware.vcDr)

   On the **Ready to complete** page, click **Finish**.
8. In the left pane, click **Services** and verify that the status of the *hms* service is `Started`.
9. Repeat this procedure to register the *Protection and Recovery* appliance in the recovery VMware Cloud Foundation instance with vCenter Single Sign-On by using the corresponding values in your VMware Cloud Foundation Planning and Preparation Workbook for the Recovery Instance.

Isolate Network Traffic of Protection and Recovery for Fleet Disaster Recovery for VMware Cloud Foundation

Protection and Recovery can consume a lot of bandwidth during initial replication, and when virtual machines are added or destroyed. To avoid network problems and enhance network performance in the data center, isolate replication traffic from other network traffic.

Create a Port Group for Protection and Recovery Traffic for Fleet Disaster Recovery for VMware Cloud Foundation

You isolate the network traffic to the *Protection and Recovery* server by creating a dedicated port group for *Protection and Recovery* traffic.

For the protected VMware Cloud Foundation instance procedure, use the VMware Cloud Foundation Planning and Preparation Workbook.

Procedure

1. Log in to the management domain vCenter at `https://<management_vcenter_fqdn>/ui` in the protected instance as **administrator@vsphere.local**.
2. Create a port group in the distributed switch for the *Protection and Recovery* traffic.
   a. In the **Networking** inventory, navigate to the management virtual distributed switch.
   b. Right-click the management virtual distributed switch and select **Distributed port group** > **New distributed port group**.
   c. On the **Name and location** page, enter a name and click **Next**.
   d. On the **Configure settings** page, configure the following settings, configure the remaining settings according to your VMware Cloud Foundation Planning and Preparation Workbook, and click **Next**.

Setting	Value
Port binding	Static
Port allocation	Elastic
Number of ports	8
VLAN type	VLAN
Customize default policies configuration	Selected

   e. On the **Security** page, click **Next**.
   f. On the **Traffic shaping** page, click **Next**.
   g. On the **Teaming and failover** page, from the **Load balancing** drop-down menu, select **Route based on physical NIC load**, and click **Next**.

h. On the **Monitoring** page, click **Next**.
i. On the **Miscellaneous** page, click **Next**.
j. On the **Ready to complete** page, click **Finish**.

3. Repeat the procedure to create the dedicated replication traffic port group on the distributed switch in the recovery VMware Cloud Foundation instance by using the corresponding values from your Planning and Preparation Workbook.

Create a VMkernel Adapter on the ESX Hosts for Protection and Recovery Traffic for Fleet Disaster Recovery for VMware Cloud Foundation

You isolate the network traffic to the *Protection and Recovery* server by dedicating a VMkernel network adapter on each management ESX host in both the protected and the recovery VMware Cloud Foundation instances.

For the protected VMware Cloud Foundation instance procedure, use the VMware Cloud Foundation Planning and Preparation Workbook.

Procedure

1. Log in to the management domain vCenter at `https://<management_vcenter_fqdn>/ui` in the protected instance as **administrator@vsphere.local**.
2. Create a VMkernel adapter for Protection and Recovery traffic on the management domain ESX hosts.
   a. In the **Hosts and clusters** inventory, expand the management domain vCenter tree and expand the management domain data center.
   b. Expand the default management cluster and select the first ESX host.
   c. Click the **Configure** tab.
   d. In the left pane, navigate to **Networking** > **VMkernel adapters**.
   e. On the **VMkernel adapters** page, click **Add networking**.
   The **Add networking** wizard opens.
   f. On the **Select connection type** page, select **VMkernel network adapter** and click **Next**.
   g. On the **Select target device** page, select **Select an existing network**, click **Browse**, select the Protection and Recovery storage replication port group, and click **Next**.
   h. On the **Port properties** page, in the **Available services** section, select *vSphere Replication NFC*, and click **Next**.
   i. On the **IPv4 setting** page, select **Use static IPv4 settings**, select the **Override default gateway for this adapter** check box, configure the settings according to your VMware Cloud Foundation Planning and Preparation Workbook, and click **Next**.
   j. On the **Ready to complete** page, click **Finish**.
3. Repeat this step to create VMkernel network adapters on the remaining management ESX hosts in the protected VMware Cloud Foundation instance.
4. Repeat this procedure to create VMkernel network adapters on the ESX hosts in the recovery VMware Cloud Foundation instance by using the corresponding values in your VMware Cloud Foundation Planning and Preparation Workbook for the Recovery Instance.
5. If you add new ESX hosts to the management domain by using SDDC Manager, you must follow this procedure to add and configure a new Protection and Recovery traffic VMkernel adapter on each new ESX host.

Configure Failover of the VCF Management Components for Fleet Disaster Recovery for VMware Cloud Foundation

After you deployed and configured *Protection and Recovery* in the protected and recovery VMware Cloud Foundation instances, prepare the management components in the SDDC for failover or planned migration. Replicate application-specific virtual machines and create recovery plans for them by using *Protection and Recovery*. To fully recreate the SDDC configuration of the protected VMware Cloud Foundation instance in the recovery VMware Cloud Foundation instance if failover occurs, configure vSphere DRS rules for high availability of the management virtual machines.

Protected Site Configuration Tasks for Fleet Disaster Recovery for VMware Cloud Foundation

The following set of procedures must be performed in the protected VMware Cloud Foundation instance using the relevant VMware Cloud Foundation Planning and Preparation Workbook.

Create a Site Pair Between the Protected and Recovery VMware Cloud Foundation Instances

To use *Protection and Recovery* between the protected and recovery VMware Cloud Foundation instances, you create a new site pair.

For the following procedure, use the VMware Cloud Foundation Planning and Preparation Workbook.

Procedure

1. Log in to the management domain vCenter at `https://<management_vcenter_fqdn>/ui` in the protected instance as **administrator@vsphere.local**.
2. From the vSphere Client menu, select **VMware Live Cyber Recovery**.
3. On the *Protection and Recovery* page, click **Open Protection and Recovery.**
4. Pair the protected and the recovery *Protection and Recovery* instances.
   a. On the *Protection and Recovery* page, click **New site pair**.
      The **New pair** wizard opens.
   b. On the **Pair type** page, select the vCenter for the protected instance, select the **pair type** to **Pair with a peer vCenter Server located in a different SSO domain**, and click **Next**.
   c. On the **Peer vCenter Server** page, configure the vCenter port to 443, configure the remaining settings according to your VMware Cloud Foundation Planning and Preparation Workbook, and click **Find vCenter Server Instances**.
   d. On the **Services** page, **Data Protection**, and click **Next**.
   e. In the **Security alert** dialog box, click **Connect**.
   f. On the **Ready to complete** page, click **Finish**.
   g. Click **View Details** and click **Site pair**.

Assign Licenses to *Protection and Recovery* using Offline Mode for Fleet Disaster Recovery for VMware Cloud Foundation

Before you configure the replication for the management components in the SDDC, you must create and apply an offline license key.

Procedure

1. 1. Log in to the management domain vCenter at `https://<management_vcenter_fqdn>/ui` in the protected instance as **administrator@vsphere.local**.
2. From the vSphere Client menu, select **VMware Site Recovery Manager**.
3. On the VMware Site Recovery Manager page, click **Open VMware Site Recovery Manager.**
4. Click **Connect to Protection and Recovery**.
5. In the **Connect to Protection and Recovery** dialog box, select the vCenter according to your VMware Cloud Foundation Planning and Preparation Workbook, and click **Next**.
6. Select the **VMware Site Recovery Manager** server to use to connect to *Protection and Recovery* cloud services and click **Enter an offline license key**.
7. Copy or download the **Change request key** .
8. Keep the the **VMware Live Cyber Recovery** UI open. You will return to it to enter the offline license key. Closing the window invalidates the change request key.
9. Log in to the VMware Site Recovery Manager global console and select the **VMware Site Recovery Manager** tile.
10. From the **Set Up Deployment** drop-down menu, select **Create offline license key**.
11. Paste or upload the **Change request key** you just generated in the VMware Live Cyber Recovery UI, and click **Next**.
12. Under **New license information**, define the license by entering the number of VMs you want to protect and set an expiration date. Confirm that you understand that after clicking Next, the license will be created and the number of protected VMs in use will immediately increase, and click **Next**.

    Attention:

    The offline license key is unique for the site pair for which it was generated. You cannot use the key for a different VMware Live Cyber Recovery site pair.
13. Copy or download the license key and click **Finish**.
14. Switch back to the **VMware Live Cyber Recovery** UI, paste or upload the license key, and select the checkbox to confirm that offline sites have limited functionality, and click **Activate**.
15. In the site pair pane, click **View details**.

16. Verify the information about your offline license in the **Summary** tab under **VMware Live Cyber Recovery.> Entitlement**.

    Note:

    You can also use connected mode to activate **VMware Live Cyber Recovery.**. See Connect Site Recovery Manager to Protection and Recovery and Activate Protection and Recovery for the Site Pair.

Configure Mappings between the Protected and the Recovery VMware Cloud Foundation Instances

Configure resource mappings between the protected and the recovery VMware Cloud Foundation instances to provide failover of the VCF management components.

Procedure

1. Log in to the management domain vCenter at `https://<management_vcenter_fqdn>/ui` in the protected instance as **administrator@vsphere.local**.
2. From the vSphere Client menu, select
3. On the VMware Site Recovery Manager page, click **Open VMware Site Recovery Manager.**
4. In the site pair pane, click **View details**.
5. Configure network mappings.
   a. On the main navigation bar, click **Site pair**.
   b. In the left pane, navigate to **Configure** > **Network mappings**.
   c. Click **New**.
      The **New network mappings** wizard opens.
   d. On the **Creation mode** page, select **Prepare mappings manually** and click **Next**.
   e. On the **Recovery networks** page, expand the object trees, select the port groups to map from the protected to the recovery instance, click **Add mappings**, and click **Next**.
   f. On the **Reverse mappings** page, select the configured mappings and click **Next**.
   g. On the **Test networks** page, click **Next**.
   h. On the **Ready to complete** page, click **Finish**.
6. Configure folder mappings.
   a. In the left pane, navigate to **Configure** > **Folder mappings**.
   b. Click **New**.
      The **New folder mappings** wizard opens.
   c. On the **Creation mode** page, select **Prepare mappings manually** and click **Next**.
   d. On the **Recovery folders** page, expand the object trees, select the virtual machine folders to map from the protected to the recovery instance, click **Add mappings**, and click **Next**.
   e. Repeat the previous step to create additional folder mappings according to your VMware Cloud Foundation Planning and Preparation Workbook.
   f. On the **Reverse mappings** page, select the configured mappings and click **Next**.
   g. On the **Ready to complete** page, click **Finish**.
7. Configure resource mappings.
   a. In the left pane, navigate to **Configure** > **Resource mappings**.
   b. Click **New**.
      The **New resource mappings** wizard opens.
   c. On the **Recovery resources** page, expand the object trees, select the clusters to map from the protected to the recovery VMware Cloud Foundation instances, click **Add mappings**, and click **Next**.
   d. On the **Reverse mappings** page, select the configured mappings and click **Next**.
   e. On the **Ready to complete** page, click **Finish**.
8. Repeat this procedure for the recovery VMware Cloud Foundation instance.

Configure Replication, Create a Protection Group and a Recovery Plan for VCF Operations for Fleet Disaster Recovery for VMware Cloud Foundation

Configure the replication of the virtual machines for VCF Operations. Create a protection group and a recovery plan to support failover to the recovery instance. Replica virtual machines become active upon failover. You create a protection group and recovery plan along with the replication.

For the following procedure, use the VMware Cloud Foundation Planning and Preparation Workbook.

Procedure

1. Log in to Protection and Recovery at `https://<vlr_fqdn/dr>` in the protected VMware Cloud Foundation instance as **administrator@vsphere.local**.
2. In the site pair pane, click **View details**.
   If the **Log In Site** dialog box appears, enter the login details for the remote vCenter.
3. On the main navigation bar, click **Replications**.
4. In the left pane, click **Outgoing** and click **New**.
   The **Configure replication** wizard opens.
5. On the **Target site** page, review **Target site** and click **Next**.
6. On the **Virtual machines** page, select the virtual machines according to your VMware Cloud Foundation Planning and Preparation Workbook, and click **Next**.
7. On the **Target datastore** page, configure the following settings, configure the remaining settings according to your VMware Cloud Foundation Planning and Preparation Workbook, and click **Next**.

Setting	Value
Disk format	Same as source
VM storage policy	vSAN Default Storage Policy

8. On the Replication mapping tests page click **Next**.
9. On the **Replication settings** page, configure the settings, configure the remaining settings according to your VMware Cloud Foundation Planning and Preparation Workbook, and click **Next**.

Setting	Value
Enable point in time instances	Selected • Instances per day
Enable network compression for VR data	Selected
Enable network encryption for VR data	Selected (required for Enhanced vSphere Replication)
Enable DataSets replication	Selected

Note:

Do not activate guest OS quiescing because VCF Operations databases do not support quiescing. Quiescing might result in a cluster failure because virtual disks remain in a frozen state.

Compression requires additional host CPU resources. Do not activate it if the hosts are over-utilized.

10. On the **Protection group** page, select **Add to new protection group**, enter a protection group name, and click **Next**.
11. On the **Recovery plan** page, select **Add to new recovery plan**, enter a recovery plan name, and click **Next**.
12. On the **Ready to complete** page, click **Finish**.

Customize the Recovery Plan for VCF Operations for Fleet Disaster Recovery for VMware Cloud Foundation

After you create the recovery plan, configure the startup priority, and the startup and shutdown options for the VCF Operations.

For the following procedure, use the VMware Cloud Foundation Planning and Preparation Workbook.

Procedure

1. Log in to Protection and Recovery at `https://<vlr_fqdn/dr>` in the protected VMware Cloud Foundation instance as **administrator@vsphere.local**.

2. In the site pair pane, click **View details**.
3. On the main navigation bar, click **Recovery plans**.
4. In the left pane, navigate to the recovery plan and click the **Recovery steps** tab.
5. Change the startup priority of the VCF Operations virtual machines.
   a. Expand the **Power on priority 3 VMs** section.
   b. Right-click the virtual machine of the VCF Operations primary node and select **Priority group** > **1 (Highest)**.
   c. In the **Change priority** dialog box, click **Yes**.
6. Repeat this step to configure the startup priority for the remaining analytics cluster nodes.

Virtual machine	Power on Priority Order
VCF Operations primary node	1
VCF Operations replica node	2
VCF Operations data node(s)	3

7. If using a load balancer, add a user prompt to verify the availability of the load balancer for VCF Operations.
   a. Right-click **Power on priority 1 VMs** and select **Add step before**.
   b. In the **Add step** dialog box, configure the settings and click **Add**.

Setting	Value
Type	Prompt
Name	Verify the availability of the load balancer for VCF Operations.
Content	Verify the availability of the load balancer for VCF Operations.

8. Add a user prompt to monitor the status of the VCF Operations primary node.
   a. Right-click **Power on priority 1 VMs** and select **Add step before**.
   b. In the **Add step** dialog box, configure the settings and click **Add**.

Setting	Value
Type	Prompt
Name	Monitor the status of the VCF Operations primary node
Content	For planned migration: • State is `Not Running` • Status is `Offline` For disaster recovery: • State is `Running` • Status is `Waiting For Analytics`

9. Add a user prompt to monitor the status of the VCF Operations replica node.
   a. Right-click **Power on priority 2 VMs** and select **Add step before**.
   b. In the **Add step** dialog box, configure the settings and click **Add**.

Setting	Value
Type	Prompt
Name	Monitor the status of the VCF Operations replica node
Content	For planned migration: • State is `Not Running` • Status is `Offline` For disaster recovery: • State is `Running` • Status is `Waiting For Analytics`

10. Add a user prompt to monitor the status of the VCF Operations data node.
    a. Right-click **Power on priority 3 VMs** and select **Add step before**.

b. In the **Add step** dialog box, configure the settings and click **Add**.

Setting	Value
Type	Prompt
Name	Monitor the status of the VCF Operations data node
Content	For planned migration: • State is `Not Running` • Status is `Offline` For disaster recovery: • State is `Running` • Status is `Online`

Configure Replication, Create a Protection Group and a Recovery Plan for VCF Operations for Networks for Fleet Disaster Recovery for VMware Cloud Foundation
Configure the replication of the virtual machines for VCF Operations for Networks. Create a protection group and a recovery plan to support failover to the recovery instance. Replica virtual machines become active upon failover. You create a protection group and recovery plan along with the replication.

For the following procedure, use the VMware Cloud Foundation Planning and Preparation Workbook.

Procedure

- Log in to Protection and Recovery at `https://<vlr_fqdn/dr>` in the protected VMware Cloud Foundation instance as **administrator@vsphere.local**.
- In the site pair pane, click **View details**.
  If the **Log In Site** dialog box appears, enter the login details for the remote vCenter.
- On the main navigation bar, click **Replications**.
- In the left pane, click **Outgoing** and click **New**.
  The **Configure replication** wizard opens.
- On the **Target site** page, review **Target site** and click **Next**.
- On the **Virtual machines** page, select the virtual machines according to your VMware Cloud Foundation Planning and Preparation Workbook, and click **Next**.
- On the **Target datastore** page, configure the following settings, configure the remaining settings according to your VMware Cloud Foundation Planning and Preparation Workbook, and click **Next**.

Setting	Value
Disk format	Same as source
VM storage policy	vSAN Default Storage Policy

- On the Replication mapping tests page click **Next**.
- On the **Replication settings** page, configure the settings, configure the remaining settings according to your VMware Cloud Foundation Planning and Preparation Workbook, and click **Next**.

Setting	Value
Enable point in time instances	Selected • Instances per day
Enable network compression for VR data	Selected
Enable encryption for VR data	Deselected

Note:

Do not activate guest OS quiescing because VCF Operations databases do not support quiescing. Quiescing might result in a cluster failure because virtual disks remain in a frozen state.

Compression requires additional host CPU resources. Do not activate it if the hosts are over-utilized.

- On the **Protection group** page, select **Add to new protection group**, enter a protection group name, and click **Next**.
- On the **Recovery plan** page, select **Add to new recovery plan**, enter a recovery plan name, and click **Next**.
- On the **Ready to complete** page, click **Finish**.

Customize the Recovery Plan for VCF Operations for Networks for Fleet Disaster Recovery for VMware Cloud Foundation
After you create the recovery plan, configure the startup priority, the startup and shutdown options for the VCF Operations for networks virtual machines. After failover or planned migration, the platform node1 of VCF Operations for networks must start first.

Procedure

1. Log in to Protection and Recovery at `https://<vlr_fqdn/dr>` in the protected VMware Cloud Foundation instance as **administrator@vsphere.local**.
2. In the site pair pane, click **View details**.
3. On the main navigation bar, click **Recovery plans**.
4. In the left pane, navigate to the recovery plan and click the **Recovery steps** tab.
5. Change the startup priority of the VCF Operations for networks platform node.
   a. Expand the **Power on priority 3 VMs** section.
   b. Right-click the virtual machine of the VCF Operations for networks platform node 1 and select **Priority group** > **1 (Highest)**.
   c. In the **Change priority** dialog box, click **Yes**.
6. Repeat this step to configure the startup priority for the remaining nodes.

Virtual machine	Power on Priority Order
VCF Operations for networks platform node 2	2
VCF Operations for networks platform node 3	3
VCF Operations for networks collector nodes **Note:** Applicable if you are replicating the colector nodes	4

7. If using a load balancer, add a user prompt to verify the availability of the load balancer for VCF Operations for networks.
   a. Right-click **Power on priority 1 VMs** and select **Add step before**.
   b. In the **Add step** dialog box, configure the settings and click **Add**.

Setting	Value
Type	Prompt
Name	Verify the availability of the load balancer for VCF Operations for networks.
Content	Verify the availability of the load balancer for VCF Operations for networks.

Recovery Site Configuration Tasks for Fleet Disaster Recovery for VMware Cloud Foundation
The following set of procedures must be performed in the recovery VMware Cloud Foundation instance using the relevant VMware Cloud Foundation Planning and Preparation Workbook.

Create Anti-Affinity Rules for the Placeholder Virtual Machines in the Recovery VMware Cloud Foundation Instance
Anti-affinity rules do not persist across VMware Cloud Foundation instances during recovery using Protection and Recovery. You must create the anti-affinity rules for the placeholder virtual machines in the recovery VMware Cloud Foundation instance so that the rules apply after a failover of the protected virtual machines.

You create separate anti-affinity rules to keep the virtual machines of VCF Operations and VCF Operations for networks on separate hosts.

For the following procedure, use the VMware Cloud Foundation Planning and Preparation Workbook.

Procedure

1. Log in to the management domain vCenter at `https://<management_vcenter_fqdn>/ui` in the recovery instance as **administrator@vsphere.local**.
2. In the **Hosts and clusters** inventory, expand the management domain vCenter tree and expand the management domain data center.
3. Select the default management cluster and click the **Configure** tab.
4. In the left pane, navigate to **Configuration** > **VM/Host rules** and click **Add**.
5. On the **Create VM/Host rule** dialog box, configure the settings, configure the remaining settings according to your VMware Cloud Foundation Planning and Preparation Workbook, and click **OK**.

Setting	Value
Enable rule	Selected
Type	Separate virtual machines

Create Virtual Machine Groups and Restart Order in the Recovery VMware Cloud Foundation Instance

To define the startup order and activate continued power on and power off operations of protected virtual machines in the recovery VMware Cloud Foundation instance, you create virtual machine groups.

You create separate VM/Host groups and VM/Host rules for the virtual machines of VCF Operations and VCF Operations for networks.

For the following procedure, use the VMware Cloud Foundation Planning and Preparation Workbook.

Procedure

1. Log in to the management domain vCenter at `https://<management_vcenter_fqdn>/ui` in the recovery instance as **administrator@vsphere.local**.
2. In the **Hosts and clusters** inventory, expand the management domain vCenter tree and expand the management domain data center.
3. Select the default management cluster and click the **Configure** tab.
4. In the left pane, navigate to **Configuration** > **VM/Host groups** and click **Add**.
5. On the **Create VM/Host group** dialog box, configure the settings, configure the remaining settings according to your VMware Cloud Foundation Planning and Preparation Workbook, and click **OK**.

Setting	Value
Type	VM group

6. In the left pane, navigate to **Configuration** > **VM/Host rules** and click **Add**.
7. On the **Create VM/Host rule** dialog box, configure the settings, configure the remaining settings according to your VMware Cloud Foundation Planning and Preparation Workbook, and click **OK**.

Setting	Value
Enable rule	Selected
Type	Virtual machines to virtual machines

8. Repeat steps for remaining VM group dependency restart condition for VCF Automation.

Fleet Component Recovery Tasks

Modify DNS Entries for Fleet Component VIPs Before Recovery

Before you can recover VCF Management Services fleet components to a new instance you must update the DNS entry for each fleet component VIP to use an IP from the recovery site VCF Management Services VLAN.

To recover the VCF Management Services fleet components, you must maintain their VIP FQDNs. This process is required when the source and target VCF Management Services instances are using different VLAN/subnets to allow the recovery of the fleet components using different IPs.

Modify DNS for the following FQDNs to point to a free IP in the same VLAN/subnet that the VCF Management Services instance on the recovery site is using.

- Fleet Components FQDN
- Identity Broker FQDN
- Log Management FQDN
- VCF Automation FQDN
- VCF Automation services runtime FQDN

Retrieve Component Details from VCF Operations

Before you can restore the fleet components across sites, you must retrieve the component IDs from VCF Operations.

You use the VCF Operations API to retrieve the component IDs and versions.

1. Retrieve a VCF Operations access token.

```
# VCF_OPS_FQDN should be the VCF Operations master node FQDN
# VCF_OPS_ADMIN_PASSWORD should be the VCF Operations local admin password

export VCF_OPS_FQDN='flt-ops01a.rainpole.io'
export VCF_OPS_ADMIN_PASSWORD='*****************'

OPS_TOKEN=$(curl -k -s -X POST "https://$VCF_OPS_FQDN/suite-api/api/auth/token/acquire" \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d "{
    \"username\": \"admin\",
    \"password\": \"$VCF_OPS_ADMIN_PASSWORD\",
    \"authSource\": \"local\"
  }" | jq -r .token)
```

2. Retrieve a list of component IDs and versions from VCF Operations and save them for later use.

```
curl -k -s -X GET \
  "https://${VCF_OPS_FQDN}/suite-api/internal/components?_no_links=true" \
  -H 'accept: application/json' \
  -H 'X-Ops-API-use-unsupported: true' \
  -H "Authorization: OpsToken ${OPS_TOKEN}" \
| jq '
  .components as $all
  | ( $all
    | map(
        select(
          (.componentType | ascii_upcase) as $t
          | $t == "FLEET_LCM" or $t == "SALT_RAAS" or $t == "VIDB" or $t == "LI"
        )
      )
    ) as $filtered
  | ( $all
    | map(select((.componentType | ascii_upcase) == "VSP"))
    | map({
```

```
            id: .componentUuid,
            componentVersion: .componentVersion,
            fqdn: .properties.fqdn,
            fleetFqdn: .properties.fleetFqdn,
            ip: (.properties.ip // .vcfInstance.ip),
            instanceName: .vcfInstance.instanceName,
            vcfInstanceFqdn: .vcfInstance.fqdn
          })
        | unique_by(.id)
        ) as $vsp
      | ( $all
        | map(select((.componentType | ascii_upcase) == "VCFA"))[0]
        | if . then {
            fqdn: .properties.fqdn,
            componentVersion: .componentVersion,
            platformFqdn: .properties.platformFqdn,
            vspComponentUuid: .properties.vspComponentUuid
          } else null end
        ) as $vcfa
      | {
          components: (
            $filtered
            | map({
                componentType,
                id: .componentUuid,
                componentVersion: .componentVersion,
                fqdn: .properties.fqdn
              })
          ),
          vsp: $vsp,
          vcfa: $vcfa
        }
    '
```

Sample output

```
{
  "components": [
    {
      "componentType": "FLEET_LCM",
      "id": "7bee057f-a164-4bce-8eb6-dea66f916008",
      "componentVersion": "9.1.1.0.25626340",
      "fqdn": "flt-fc01.rainpole.io"
    },
    {
      "componentType": "SALT_RAAS",
      "id": "1786c678-8565-43a0-a6e7-42405a24c3d8",
      "componentVersion": "9.1.1.0.25628391",
      "fqdn": "flt-fc01.rainpole.io"
    },
    {
      "componentType": "LI",
      "id": "7555d644-5d8c-4695-8b54-6b741ac9bafb",
      "componentVersion": "9.1.0.0400.25544947",
      "fqdn": "flt-logs01.rainpole.io"
    },
    {
      "componentType": "VIDB",
```

```
      "id": "b90d023a-b33a-469a-a4b6-096776bb732e",
      "componentVersion": "9.1.1.0.25629297",
      "fqdn": "flt-idb01.rainpole.io"
    }
  ],
  "vsp": [
    {
      "id": "6dfa787b-f697-44b4-8b13-7cc4b35bd8b7",
      "componentVersion": "9.1.1.0.25628383",
      "fqdn": "lax-sr01.lax.rainpole.io",
      "fleetFqdn": null,
      "ip": "10.21.10.13",
      "instanceName": "Los Angeles",
      "vcfInstanceFqdn": "lax-vcf01.lax.rainpole.io"
    },
    {
      "id": "c722868a-3db6-4991-9c22-d6610c2bed5d",
      "componentVersion": "9.1.1.0.25628383",
      "fqdn": "sfo-sr01.sfo.rainpole.io",
      "fleetFqdn": "flt-fc01.rainpole.io",
      "ip": "10.11.10.13",
      "instanceName": "San Francisco",
      "vcfInstanceFqdn": "sfo-vcf01.sfo.rainpole.io"
    }
  ],
  "vcfa": {
    "fqdn": "flt-auto01.rainpole.io",
    "componentVersion": "9.1.1.0.25629010",
    "platformFqdn": "flt-vcfa-sr01.rainpole.io",
    "vspComponentUuid": "c7862cc3-2857-4df8-a985-50467545d9cd"
  }
}
```


Extract Fleet VIP Address Configuration from the Source Services Runtime Backup

Using the services runtime backup from the source site, you extract the fleet component configuration.

- OpenSSL 1.1.1 or later must be available on your Linux machine that you use to access the environment.
- Download and extract the scripts from Broadcom knowledge base article 443506 to the machine you use to access your VCF environment.
- A linux workstation with access to the SFTP server

The fleet component configuration is captured as part of the VCF management services runtime backup. You use this extracted data in a later step.

1. Browse the SFTP server and locate the VCF management services runtime backup that you plan to use for the restore operation.
2. Run the following command to execute the script to extract the configuration from the backup.

```
# remote-archive should be the path to the vsp backup for the source VCF Management Services instance
export VMSP_SFTP_PASSWORD='************'
export VMSP_ENCRYPTION_PASSPHRASE='**************'
 ./vmsp-backup-sftp-to-yaml.sh \
  --sftp-host "10.167.173.130" \
  --sftp-user "svc-vcf-bck" \
  --sftp-dir "/media/backups/" \
```

```
  --remote-archive "/media/backups/vcf/backups/e6b2ad0a-b76f-4080-b9db-aa338bacdc64/9.1.1.0.25662438/vsp/
e6b2ad0a-b76f-4080-b9db-aa338bacdc64/9.1.1.0.25662438/2026-08-13T14-23-26Z/2026-08-13T14-23-26Z.base.tgz"
 \
  --output-dir ./backup-yaml
```

3. Save the resulting YAML files for later use.

```
./vmsp-backup-sftp-to-yaml.sh   --sftp-host "10.167.173.130"   --sftp-user "svc-vcf-
bck"   --sftp-dir "/media/backups/"   --remote-archive "/media/backups/vcf/backups/
e6b2ad0a-b76f-4080-b9db-aa338bacdc64/9.1.1.0.25662438/vsp/e6b2ad0a-b76f-4080-b9db-
aa338bacdc64/9.1.1.0.25662438/2026-08-13T14-23-26Z/2026-08-13T14-23-26Z.base.tgz"   --output-dir ./backup-
yaml

vmsp-backup-sftp-to-yaml.sh: downloading sftp:/media/backups/vcf/backups/
e6b2ad0a-b76f-4080-b9db-aa338bacdc64/9.1.1.0.25662438/vsp/e6b2ad0a-b76f-4080-b9db-
aa338bacdc64/9.1.1.0.25662438/2026-08-13T14-23-26Z/2026-08-13T14-23-26Z.base.tgz -> /tmp/vmsp-
backup.EX1XSg/2026-08-13T14-23-26Z.base.tgz

backup-yaml/vmsp-platform.yaml
backup-yaml/ingress-fleet-tls-ndc.yaml
backup-yaml/ingress-instance-tls-ndc.yaml
backup-yaml/ingress-platform-tls-ndc.yaml
backup-yaml/ingress-fleet-tls.yaml
backup-yaml/ingress-instance-tls.yaml
backup-yaml/ingress-platform-tls.yaml

vmsp-backup-sftp-to-yaml.sh: YAML written under: ./backup-yaml
total 140
drwxr-xr-x  2 root root   180 Aug 13 15:33 .
drwxrwxrwt 39 root root   860 Aug 13 15:33 ..
-rw-r--r--  1 root root 19019 Aug 13 15:33 ingress-fleet-tls-ndc.yaml
-rw-r--r--  1 root root 19232 Aug 13 15:33 ingress-fleet-tls.yaml
-rw-r--r--  1 root root 19046 Aug 13 15:33 ingress-instance-tls-ndc.yaml
-rw-r--r--  1 root root 19262 Aug 13 15:33 ingress-instance-tls.yaml
-rw-r--r--  1 root root 19046 Aug 13 15:33 ingress-platform-tls-ndc.yaml
-rw-r--r--  1 root root 19348 Aug 13 15:33 ingress-platform-tls.yaml
-rw-r--r--  1 root root 17404 Aug 13 15:33 vmsp-platform.yaml
```

Scale the Recovery Site VCF Management Services Instance for VCF Fleet Disaster Recovery

To support restoring components to the recovery site, the target VCF Management Services instance must be the same size profile. You use this process to scale it up if necessary.

This task is only required if the target VCF Management Services instance is smaller than the protected site instance. You use the YAML files that are generated in Extract Fleet VIP Address Configuration from the Source Services Runtime Backup

1. Scale the target cluster.
   a) On a workstation with access to the KUBECONFIG for the recovery VCF Management Services instance and the extracted backup YAML files, run the following commands

```
export KUBECONFIG=/path/to/kubeconfig
export WORKER_SIZE=$(awk '/worker:/{f=1} f && /size:/{print $2; exit}' ./<path-to-yaml-files>/vmsp-
platform.yaml)
export PROFILE=$(awk '/profiles:/{f=1} f && /name:/{print $2; exit}' ./<path-to-yaml-files>/vmsp-
platform.yaml)

kubectl patch packagedeployment vmsp-platform -n vmsp-platform \
  --type=merge \
  -p "$(jq -n --arg s "$WORKER_SIZE" --arg p "$PROFILE" '
```

```
          {
            "spec": {
              "values": {
                "cluster": {
                  "worker": {
                    "size": $s
                  }
                },
                "profiles": {
                  "name": $p
                }
              }
            }
          }')"
```

2. Monitor the task.

```
kubectl wait --for=jsonpath='{.status.phase}'=Successful packagedeployment/vmsp-platform -n vmsp-platform
 --timeout=2h

# Wait for this message
packagedeployment.releases.vmsp.vmware.com/vmsp-platform condition met

kubectl get packagedeployment vmsp-platform -n vmsp-platform -o jsonpath='{.status.phase}'

# Wait for this message
Successful
```

The worker nodes will be resized to match the size from the protected site when the backup was taken. Do not proceed until this step has completed.

Disable Log Management Integration on Recovery VCF Management Services Instance

You temporarily disable log management integration on the recovery site VCF Management Services Instance as log management is unavailable until it is restored.

Disable logging on the target VCF Management Services instance

Flush log buffer on fluentd on the target VCF Management Services instance

Scale down & up fluentd pod

1. Disable logging on the target VCF Management Services instance.
   a) Retrieve a token from the VCF Management Services cluster on the recovery site.

```
export VCFMS_SR='lax-sr01.lax.rainpole.io'
export VCFMS_SR_PASSWORD='****************'

VCFMS_SR_TOKEN=$(curl -ks --request POST \
  --url "https://$VCFMS_SR/api/v1/identity/token" \
  --header 'Content-Type: application/x-www-form-urlencoded' \
  --data-urlencode grant_type=password \
  --data-urlencode "username=admin@vsp.local" \
  --data-urlencode "password=$VCFMS_SR_PASSWORD" \
  | jq -r '.access_token')
```

   b) Query the VCF Management Services cluster on the recovery site for a list of existing components.

```
curl -ks "https://${VCFMS_SR}/api/v1/components" \
  -H "Authorization: Bearer ${VCFMS_SR_TOKEN}" | jq '.components[] | {type, id, status}'
```

   c) Disable logging on the target VCF Management Services instance.

```
# VSP_COMP_ID should be the id for the VSP component from the previous step
```

```
export VSP_COMP_ID='e319236e-867e-4779-9270-4921bddf4f1f'
export VCFMS_SR='lax-sr01.lax.rainpole.io'

TASK_ID=$(curl -ks -X POST "https://${VCFMS_SR}/api/v1/components/${VSP_COMP_ID}?action=apply" \
-H "Authorization: Bearer ${VCFMS_SR_TOKEN}" \
-H "Content-Type: application/json" \
-d '{
  "spec": {
    "configuration": {
      "logs": {
        "type": "none"
      }
    }
  },
  "options": {
    "precheckOnly": false,
    "precheck": true,
    "timeout": "1h"
  }
}' | jq -r '.id')

# Poll task until complete
curl -ks "https://${VCFMS_SR}/api/v1/tasks/${TASK_ID}" \
  -H "Authorization: Bearer ${VCFMS_SR_TOKEN}" | jq
```

2. Verify that logging is disabled before proceeding.
   a) SSH to the control plane node of the target VCF Management Services instance and run the following:

```
sudo su
export KUBECONFIG=/etc/kubernetes/admin.conf
kubectl -n vmsp-platform rollout status sts/logging-operator-fluentd
kubectl -n vmsp-platform get sts logging-operator-fluentd
```

   Important:

   Sample Healthy Response

```
kubectl -n vmsp-platform rollout status sts/logging-operator-fluentd
kubectl -n vmsp-platform get sts logging-operator-fluentd
partitioned roll out complete: 1 new pods have been updated...
NAME                        READY   AGE
logging-operator-fluentd   1/1     29d
```

3. If the command does not return a response similar to the sample healthy response, perform the following steps to flush the log buffer, as not doing so will block the restore.
   a) Flush the fluentd log buffer on the target VCF Management Services instance.

```
kubectl exec logging-operator-fluentd-0 -n vmsp-platform -- find /buffers -name "*.buffer" -type f -
delete
```

   b) Scale down & up the fluentd pod.

```
#Scale down
kubectl scale sts logging-operator-fluentd -n vmsp-platform --replicas=0

# Scale up
kubectl scale sts logging-operator-fluentd -n vmsp-platform --replicas=1
```

Apply Fleet VIP on Recovery VCF Management Services Instance

You apply the fleet component VIP configuration you extracted form the backup in an earlier step.

You must have access to the *KUBECONFIG* for the recovery VCF Management Services instance. The *KUBECONFIG* is located on the control plane node at **/etc/kubernetes/admin.conf**.

Applying the configuration from the extracted backup configures the fleet FQDN and associated certificates and secrets on the recovery VCF Management Services instance.

1. Apply the configuration from the backup.
   a) On a workstation with access to the KUBECONFIG for the recovery VCF Management Services instance and the extracted backup YAML files, run the following commands

```
export KUBECONFIG=/path/to/kubeconfig
kubectl apply -f ./<path-to-yaml-files>/ingress-fleet-tls.yaml
kubectl apply -f ./<path-to-yaml-files>/ingress-fleet-tls-ndc.yaml
```

2. Retrieve a token from the VCF Management Services cluster on the recovery site.

```
export VCFMS_SR='lax-sr01.lax.rainpole.io'
export VCFMS_SR_PASSWORD='*****************'

VCFMS_SR_TOKEN=$(curl -ks --request POST \
  --url "https://$VCFMS_SR/api/v1/identity/token" \
  --header 'Content-Type: application/x-www-form-urlencoded' \
  --data-urlencode grant_type=password \
  --data-urlencode "username=admin@vsp.local" \
  --data-urlencode "password=$VCFMS_SR_PASSWORD" \
  | jq -r '.access_token')
```

3. Apply Fleet FQDN and TLS Certificates to the VCF Management Services cluster on the recovery site.

```
# VSP_COMP_ID should be the id for the recovery site VSP component from the previous step
# FLEET_FQDN should be the existing fleet components FQDN from the primary site
# FLEET_VIP should be the new IP that was assigned in DNS to the existing fleet FQDN

export VSP_COMP_ID='e319236e-867e-4779-9270-4921bddf4f1f'
export FLEET_FQDN='flt-fc01.rainpole.io'
export FLEET_VIP='10.21.99.20'

TASK_ID=$(curl -ks -X POST \
  "https://${VCFMS_SR}/api/v1/components/${VSP_COMP_ID}?action=apply" \
  -H "Authorization: Bearer ${VCFMS_SR_TOKEN}" \
  -H "Content-Type: application/json" \
  -d '{
  "spec": {
    "configuration": {
      "ingress": {
        "fleet": {
          "fqdn": "'$FLEET_FQDN'",
          "vips": {
            "ipv4": ["'${FLEET_VIP}'"]
          }
        }
      }
    }
  },
  "options": {
    "timeout": "30m"
  }
}' | jq -r '.id')
```

4. Monitor the task until it reports as *SUCCEEDED*.

```
curl -ks "https://${VCFMS_SR}/api/v1/tasks/${TASK_ID}" \
```

```
  -H "Authorization: Bearer ${VCFMS_SR_TOKEN}" | jq
```

Retrieve a List of Backups for VCF Management Services Fleet Components

Retrieve list of backups for the VCF Management Services Fleet Components to perform the restore.

1. Connect to the SFTP server and retrieve a list of backups for the source VCF management services cluster.

```
SFTP_HOST="192.168.0.2"
SFTP_USER="svc-vcf-bck"
SFTP_PASS="***************"
BASE_DIR="/home/fleet_sftp_user/backup-inst2/vcf/backups/"    # parent directory to scan
VSP_ID="9facfd64-c8a0-446e-81fd-ab048c3ad113"  # the ID of the source VSP cluster
COMPONENTS=("vcf-fleet-lcm" "vcf-fleet-depot" "salt-raas" "vidb")

curl_ls() {
  # Lists a remote dir, returns "<type-char> <name>" per entry, skipping . and ..
  local path="$1"
  curl -sk --user "${SFTP_USER}:${SFTP_PASS}" "sftp://${SFTP_HOST}${path}/" \
    | awk '{print substr($1,1,1), $NF}' \
    | grep -Ev ' \.\.?$'
}

for COMPONENT in "${COMPONENTS[@]}"; do
  echo "== ${COMPONENT} =="
  FOUND=""

  # Level 1: VSP_ID/VERSION
  while read -r vtype VERSION; do
    [[ "$vtype" == d ]] || continue
    COMP_PATH="${BASE_DIR}/${VSP_ID}/${VERSION}/${COMPONENT}"

    # Level 2: does this COMPONENT exist under this VERSION?
    if ! curl -sk --head --user "${SFTP_USER}:${SFTP_PASS}" \
         "sftp://${SFTP_HOST}${COMP_PATH}/" >/dev/null 2>&1; then
      continue
    fi

    # Level 3: COMPONENT/SUB_ID
    while read -r stype SUB_ID; do
      [[ "$stype" == d ]] || continue
      SUB_PATH="${COMP_PATH}/${SUB_ID}"

      # Level 4: SUB_ID/VERSION2
      while read -r v2type VERSION2; do
        [[ "$v2type" == d ]] || continue
        V2_PATH="${SUB_PATH}/${VERSION2}"

        # Level 5: dated backup folders
        while read -r btype BACKUP; do
          [[ "$btype" == d ]] || continue
          # Prefix with timestamp for chronological sorting
          FOUND+="${BACKUP}|${V2_PATH}/${BACKUP}"$'\n'
        done < <(curl_ls "$V2_PATH")

      done < <(curl_ls "$SUB_PATH")

    done < <(curl_ls "$COMP_PATH")
```

```
  done < <(curl_ls "${BASE_DIR}/${VSP_ID}")

  # Sort numerically/lexicographically by timestamp (field 1) and grab latest
  LATEST_ENTRY=$(printf '%s' "$FOUND" | grep -v '^$' | sort -t'|' -k1,1 | tail -n1)
  LATEST="${LATEST_ENTRY#*|}"

  if [[ -n "$LATEST" ]]; then
    echo "Most recent backup: $(basename "$LATEST")"
    echo "Full path: ${LATEST}"
  else
    echo "No backups found for ${COMPONENT}"
  fi
done
```

2. Create a restore JSON payload using the backup paths from step 1.

```
{
  "components": [
    {
      "path": "sftp://svc-vcf-bck@10.167.173.130:22/media/backups/lax/vcf/backups/40e752c7-
e940-4a9f-a49c-0f05efa9d915/9.1.1.0.25370367/vcf-fleet-lcm/1769b822-e28e-455a-bc28-
e2711a78d13c/9.1.1.0.25371109/2026-04-27T07-57-25Z/",
      "point": "2026-04-27T07-57-25Z"
    },
    {
      "path": "sftp://svc-vcf-bck@10.167.173.130:22/media/backups/lax/vcf/
backups/40e752c7-e940-4a9f-a49c-0f05efa9d915/9.1.1.0.25370367/vcf-fleet-depot/
cb06b1c3-15ca-42f3-9512-2527804a55cf/9.1..10.25371105/2026-04-27T08-03-07Z/",
      "point": "2026-04-27T08-03-07Z"
    },
    {
      "path": "sftp://svc-vcf-bck@10.167.173.130:22/media/backups/lax/vcf/
backups/40e752c7-e940-4a9f-a49c-0f05efa9d915/9.1.1.0.25370367/vidb/4038c5bf-
f77d-4c54-9739-12bc6e7825e1/9.1.1.0.25368698/2026-04-27T08-07-44Z/",
      "point": "2026-04-27T08-07-44Z"
    },
    {
      "path": "sftp://svc-vcf-bck@10.167.173.130:22/media/backups/lax/vcf/
backups/40e752c7-e940-4a9f-a49c-0f05efa9d915/9.1..10.25370367/salt-raas/31ec7008-
a780-4776-9078-31e7086bae6e/9.1.1.0.25346036/2026-04-27T08-13-22Z/",
      "point": "2026-04-27T08-13-22Z"
    }
  ]
}
```

Stage Binaries on the Recovery VCF Management Services Instance

To enable the restore of the fleet level components you pre-stage the binaries on the recovery VCF Management Services instance.

1. Replace the values in the sample code with your values, these will be used in later steps.

```
export servicesRuntimeFqdn='lax-sr01.lax.rainpole.io'
export servicesRuntimeUsername='admin@vsp.local'
export servicesRuntimePassword='***************'
```

2. Retrieve an access token from the VCF services runtime by running the following command.

```
servicesRuntimeToken=$(curl -k -s -X POST "https://${servicesRuntimeFqdn}/api/v1/identity/token" \
    --header "Content-Type: application/x-www-form-urlencoded" \
    --data-urlencode "grant_type=password" \
```

```
    --data-urlencode "username=${servicesRuntimeUsername}" \
    --data-urlencode "password=${servicesRuntimePassword}" \
| jq -r '.access_token')
```

3. Stage the binary for the fleet lifecycle component by running the following command.

```
curl -k -s -X POST "https://${servicesRuntimeFqdn}/api/v1/components?action=stage" \
    --header "Authorization: Bearer ${servicesRuntimeToken}" \
    --header "Content-type: application/json" \
    --header "Accept: application/json" \
    --data-urlencode -d '{
        "type": "vcf-fleet-lcm",
        "version": "9.1.1.0.25371109",
        "repository": {
            "url": "https://offline-depot.rainpole.io/PROD/release/depot-manifest-vcf-fleet-
lcm-9.1.1.0.25371109.yaml"
        },
        "options": {
            "timeout": "30m"
        }
    }' \
| jq
```

4. Repeat step 3 for vcf-fleet-depot, salt-raas, and vidb.

Restore the Fleet Components on the Recovery VCF Management Services Instance

Restoring the fleet components onto the VCF management services instance on the recovery site including restoring the fleet lifecycle, software depot, Salt RaaS, and identity broker.

Stage the binaries prior to restore by following Stage Binaries on the Recovery VCF Management Services Instance.

1. Retrieve a token from the VCF management services runtime on the recovery site.

```
export VCFMS_SR='lax-sr01.lax.rainpole.io'
export VCFMS_SR_PASSWORD='****************'

VCFMS_SR_TOKEN=$(curl -ks --request POST \
  --url "https://$VCFMS_SR/api/v1/identity/token" \
  --header 'Content-Type: application/x-www-form-urlencoded' \
  --data-urlencode grant_type=password \
  --data-urlencode "username=admin@vsp.local" \
  --data-urlencode "password=$VCFMS_SR_PASSWORD" \
  | jq -r '.access_token')
```

2. Restore the components by using a JSON payload.

```
TASK_ID=$(curl -k -X POST "https://$VCFMS_SR/api/v1/system/backups?action=restore" \
  -H "Authorization: Bearer $VCFMS_SR_TOKEN" \
  -H 'accept: application/json' \
  -H "Content-type: application/json" \
  -d '{
  "components": [
    {
      "path": "/vcf/backups/40e752c7-e940-4a9f-a49c-0f05efa9d915/9.1.0.0.25370367/vcf-fleet-lcm/1769b822-
e28e-455a-bc28-e2711a78d13c/9.1.0.0.25371109/2026-04-27T07-57-25Z/",
      "point": "2026-04-27T07-57-25Z"
    },
    {
      "path": "/vcf/backups/40e752c7-e940-4a9f-a49c-0f05efa9d915/9.1.0.0.25370367/vcf-fleet-depot/
cb06b1c3-15ca-42f3-9512-2527804a55cf/9.1.0.0.25371105/2026-04-27T08-03-07Z/",
      "point": "2026-04-27T08-03-07Z"
```

```
    },
    {
      "path": "/vcf/backups/40e752c7-e940-4a9f-a49c-0f05efa9d915/9.1.0.0.25370367/vidb/4038c5bf-
f77d-4c54-9739-12bc6e7825e1/9.1.0.0.25368698/2026-04-27T08-07-44Z/",
      "point": "2026-04-27T08-07-44Z"
    },
    {
      "path": "/vcf/backups/40e752c7-e940-4a9f-a49c-0f05efa9d915/9.1.0.0.25370367/salt-raas/31ec7008-
a780-4776-9078-31e7086bae6e/9.1.0.0.25346036/2026-04-27T08-13-22Z/",
      "point": "2026-04-27T08-13-22Z"
    }
  ],
  "storage": {
    "sftp": {
      "directory": "/sftp-directory-for-source-cluster",
      "host": "<source cluster SFTP server>",
      "port": "22",
      "username": "<sftp username>",
      "password": "<sftp password>"
    }
  }
}' | jq -r '.id')
```

Important:

After restoring the vcf-fleet-lcm component, there might be a failed backup task in the tasks view. This is a false negative and can be ignored.

3. Monitor the task until it becomes *SUCCEEDED* for all components.

```
curl -ks "https://${VCFMS_SR}/api/v1/tasks/${TASK_ID}" \
  -H "Authorization: Bearer ${VCFMS_SR_TOKEN}" | jq
```

Update SDDC LCM in the Target Site with Credentials from the Source Site

To successfully recover the fleet components you restore the credentials from SDDC LCM in the source site to the SDDC LCM in the target site. This applies to failover and failback processes.

- Download and extract the scripts from Broadcom knowledge base article 443506 to the machine you use to access your VCF environment.
- Kubectl is required

1. Extract the credentials to access fleet component from the SDDC LCM backup of the source cluster.

```
 export SDDCLCM_ENCRYPTION_PASSPHRASE='*************'

./sddclcm-backup-secrets-to-yaml.sh --sftp-host 10.167.173.130  --sftp-user svc-vcf-bck --sftp-
remote-dir "/media/backups/vcf/backups/0685bace-2881-4a73-98a3-cdb83743b303/9.1.1.0.25714471/vcf-sddc-
lcm/23d24425-3dd6-48aa-a8eb-253e3f46b328/9.1.1.0.25713940/2026-09-08T10-21-17Z" --sftp-password 'VMw@re1!'
 --output-dir ./backup-yaml
```

2. Apply the credentials.

```
export KUBECONFIG=<path to target services runtime kubeconfig>

cd ./backup-yaml
for f in vcf-iam-ops-admin.json vcf-iam-ops-networks-admin.json vcf-iam-salt-raas-admin.json vcf-iam-vcfa-
admin.json vcf-iam-vsp-admin.json;
do
kubectl replace --force -f $f
done
```

3. Update VCF Operations node discovery.

```
export KUBECONFIG=<path to target services runtime kubeconfig>

cd ./backup-yaml
for f in vcf-iam-ops-admin.json vcf-iam-ops-networks-admin.json vcf-iam-salt-raas-admin.json vcf-iam-vcfa-admin.json vcf-iam-vsp-admin.json;
do
kubectl replace --force -f $f
done
```

Update the Identity Broker VIP Address

After restoring the backup of identity broker, you must update it to use the new IP address that is mapped to the identity broker FQDN on the DNS server.

Download and extract the scripts from Broadcom knowledge base article 443506 to the machine you use to access your VCF environment.

On your Linux machine, run the following commands.

```
export PLATFORM_URI='https://lax-sr01.lax.rainpole.io'
export PASSWORD='****************'
export VIDB_VIP='10.11.99.20'
export COMPONENT_TYPE='vidb'
update-component-vip.sh --platform-url $PLATFORM_URI --password $PASSWORD --vips $VIDB_VIP --component-type
 $COMPONENT_TYPE
```

Update the Identity Broker VCF Instance Association

After identity broker is restored on the recovery site, you must update the VCF Operations inventory with the new location of the identity broker.

- Enable SSH on the VCF Operations primary node.
- Download and extract the scripts from Broadcom knowledge base article 443506 to the machine you use to access your VCF environment.

1. Confirm that identity broker is reachable on the new VCF instance by running the following commands.

```
curl -sk https://flt-idb01.rainpole.io/acs/t/CUSTOMER/.well-known/openid-configuration | python3 -m
 json.tool | head -5

#Expected response: a valid JSON response with issuer, authorization_endpoint
```

2. Copy the scripts `vidb-sso-info.sh` and `migrate-vidb-vcf-instance.sh` to the primary VCF Operations appliance and run the following commands.

```
chmod 777 vidb-sso-info.sh
chmod 777 migrate-vidb-vcf-instance.sh
```

3. Run the following command to retrieve the current identity broker registration information, entering the VCF Operations admin password when prompted.

```
./vidb-sso-info.sh --ops-fqdn flt-ops01a.rainpole.io
```

Sample output

```
./vidb-sso-info.sh --ops-fqdn flt-ops01a.rainpole.io
Enter Ops Admin Password:
[INFO]  Fetching VIDB <-> VCF <-> SSO Realm mapping from flt-ops01a.rainpole.io...
[OK]    Token acquired

Table 1: VIDB -> VCF Instance -> SSO Realm  (GET /suite-api/internal/vidb/vidbs)
┌──────────────────────────────┬──────────────────────────────────────┬──────────────────────────────────────┐
```

```
| VIDB FQDN                    | VCF Instance ID                      | SSO Realm ID
   |
├──────────────────────────────┼──────────────────────────────────────┼──────────────────────────────────────┤
| sfo-m01-vc01.sfo.rainpole.io | c783cb34-4b4d-4d95-827e-30f9ebee54da | d8757ff0-d3e0-4e5c-
b08e-0f5855cbfd05 |
| flt-idb01.rainpole.io        | c783cb34-4b4d-4d95-827e-30f9ebee54da | d8757ff0-d3e0-4e5c-
b08e-0f5855cbfd05 |
| lax-m01-vc01.lax.rainpole.io | f7bba463-8dad-45ce-9ec1-6fc0c583a3b3 | —
   |
└──────────────────────────────┴──────────────────────────────────────┴──────────────────────────────────────┘
```

Table 2: VIDB -> VCF -> SSO Realm (GET /suite-api/api/fleet-management/iam/ssorealms)

```
┌──────────────────────────────────────┬──────────────────────────────────────┬──────────────────────────────────────
| VIDB ID                              | VCF ID                               | SSO Realm ID
          | SSO Realm Name        |
├──────────────────────────────────────┼──────────────────────────────────────┼──────────────────────────────────────
| 2d5d0503-dda1-44b8-936f-f3978efcb6ff | c783cb34-4b4d-4d95-827e-30f9ebee54da | d8757ff0-d3e0-4e5c-
b08e-0f5855cbfd05 | flt-idb01.rainpole.io |
└──────────────────────────────────────┴──────────────────────────────────────┴──────────────────────────────────────
```

4. Update the identity broker component registration.
   a) Set the following variables:

```
export OPS_FQDN="flt-ops01a.rainpole.io"
export VIDB_FQDN="<VIDB_FQDN_FROM_TABLE_1>"
export TARGET_VCF="<RECOVERY_VCF_INSTANCE_ID_FROM_TABLE_1>"
export SSO_REALM="<SSO_REALM_ID_FROM_TABLE_2>"
```

   b) Run the following command to update the identity broker component registration, entering the VCF Operations admin password when prompted.

```
./migrate-vidb-vcf-instance.sh \
  --ops-fqdn   $OPS_FQDN \
  --vidb-host  $VIDB_FQDN \
  --target-vcf $TARGET_VCF \
  --sso-realm  $SSO_REALM
```

   Sample output

```
╔═══════════════════════════════════════════════════╗
║        VIDB VCF Instance Migration (auto flow)    ║
╚═══════════════════════════════════════════════════╝
[INFO]  OPS Host       : flt-ops01a.rainpole.io
[INFO]  VIDB Host      : flt-idb01.rainpole.io
[OK]    Token acquired
[INFO]  Detected VCF Operations version: 9.1.1

==> Feature flag resolved: VIDB_MIGRATION_FLOW=api
[INFO]  Target VCF      : f7bba463-8dad-45ce-9ec1-6fc0c583a3b3
[INFO]  SSO Realm       : d8757ff0-d3e0-4e5c-b08e-0f5855cbfd05

==> Step 1/7  Acquiring auth token
[OK]    Token acquired

==> Step 2/7  Fetching VIDB record and discovering current configuration
[OK]    External VIDB found: 82d855f4-e208-4e4e-870c-9b64a326866f
[INFO]    VIDB Host              : flt-idb01.rainpole.io
[INFO]    Current VCF Instance   : c783cb34-4b4d-4d95-827e-30f9ebee54da
```

```
[INFO]    Target VCF Instance    : f7bba463-8dad-45ce-9ec1-6fc0c583a3b3
[INFO]    Current MGMT VC GUID   : 5a734067-41b3-41a9-a80f-ce5dc714de3a
[INFO]    Current Collector ID   :

==> Step 3/7  Resolving management VC GUID for target VCF
[OK]    Target Management VC GUID: 1b6e7e17-8f13-4df7-b248-f6f33f38627c

==> Step 4/7  Verifying pre-conditions
[INFO]  VCF PATCH needed: true
[INFO]  Collector update will be checked in Step 5.5 regardless of PATCH status
[INFO]  Pre-conditions verified — proceeding with PATCH

==> Step 5/7  Updating external VIDB (PATCH)
[INFO]  PATCH payload: {"id":"82d855f4-e208-4e4e-870c-9b64a326866f","vcGUID":"1b6e7e17-8f13-4df7-b248-
f6f33f38627c"}
[OK]    External VIDB updated (HTTP 200)
[INFO]    VIDB ID                : 82d855f4-e208-4e4e-870c-9b64a326866f
[INFO]    New Management VC GUID : 1b6e7e17-8f13-4df7-b248-f6f33f38627c

==> Step 5.5/7  Checking collector consistency (mandatory)
[INFO]    [5.5a] Fetching current VIDB adapter via public API
[INFO]    Current adapter collector ID: 3
[INFO]    Target MGMT VC GUID        : 1b6e7e17-8f13-4df7-b248-f6f33f38627c
[INFO]    [5.5b] Fetching VMWARE adapter list to resolve target collector
[INFO]    Target adapter collector ID: 3
[INFO]    Comparing collectors:
[INFO]      Current: 3
[INFO]      Target:  3
[OK]    Adapters collectors match — no update needed

==> Step 6/7  Waiting for VIDB to become eligible (timeout: 1200s)
[INFO]    Polling every 10s for VIDB ID=82d855f4-e208-4e4e-870c-9b64a326866f
[OK]    VIDB is associated with target VCF instance (elapsed: 0s)
[INFO]    VIDB ID          : 82d855f4-e208-4e4e-870c-9b64a326866f
[INFO]    VCF Instance ID  : f7bba463-8dad-45ce-9ec1-6fc0c583a3b3

==> Step 7/7  Updating SSO Realm VCF instance ID
[INFO]    [7a] Verifying SSO Realm exists
[OK]    SSO Realm verified to exist
[INFO]    [7b] Patching SSO Realm with target VCF instance ID
[OK]    SSO Realm updated (HTTP 200)
[INFO]    SSO Realm ID    : d8757ff0-d3e0-4e5c-b08e-0f5855cbfd05
[INFO]    New VCF Instance : f7bba463-8dad-45ce-9ec1-6fc0c583a3b3

[OK]    All 7 steps completed successfully

Summary:
  VIDB ID              : 82d855f4-e208-4e4e-870c-9b64a326866f
  Current VCF Instance : c783cb34-4b4d-4d95-827e-30f9ebee54da
  Target VCF Instance  : f7bba463-8dad-45ce-9ec1-6fc0c583a3b3
  VIDB Host            : flt-idb01.rainpole.io
  SSO Realm ID         : d8757ff0-d3e0-4e5c-b08e-0f5855cbfd05
```

Update Fleet Component Registration

To enable day 2 operations on the fleet components, you update their registration in VCF Operations.

Download and extract the scripts from Broadcom knowledge base article 443506 to the machine you use to access your VCF environment.

1. Confirm that identity broker is reachable on the new VCF instance.

```
curl -sk https://flt-idb01.rainpole.io/acs/t/CUSTOMER/.well-known/openid-configuration | python3 -m
 json.tool | head -5

#Expected response: a valid JSON response with issuer, authorization_endpoint
```

2. Open an SSH connection to the control plane node of the target VCF management services instance and run the following commands.

```
sudo su
export KUBECONFIG=/etc/kubernetes/admin.conf
```

3. Copy the `update_fleet_component_registration.sh` script to the control plane node and run the following command.

```
./update_fleet_component_registration.sh --target-fqdn <fleet-lcm-fqdn>
```

Update VCF Operations Inventory

- Download and extract the scripts from Broadcom knowledge base article 443506 to the machine you use to access your VCF environment.
- Kubectl is required

Update VCF Operations node discovery.

```
export KUBECONFIG=<path to recovery services runtime kubeconfig>

./trigger_ops_node_discovery.sh
```

Sample output

```
./trigger_ops_node_discovery.sh
========================================================================
 DR Fix: Seed OPS Node + Trigger SDDC LCM Refresh (V1.0)
========================================================================

Step 1: Locating primary SDDC LCM DB pod...
  SDDC LCM DB: vcf-sddc-lcm-db-0

Step 2: Reading OPS component from SDDC LCM DB...
  id (internal PK)  : 01a06d97-cb70-75a9-9fb7-59c372891272
  component_id      : 365f1e32-7984-48b6-b3cf-ee719265a53b
  fqdn               : flt-ops01a.rainpole.io
  version            : 9.1.1.0.25679751

Step 3: Seeding MASTER node row (SDDC LCM DB)...
    OK   INSERT MASTER node row for OPS component (365f1e32-7984-48b6-b3cf-ee719265a53b)

Step 4: Acquiring admin@vsp.local token...
  identity endpoint  : https://lax-sr01.lax.rainpole.io
admin@vsp.local password for lax-sr01.lax.rainpole.io:
  OK   token acquired

Step 5: Triggering SDDC LCM refresh for component 365f1e32-7984-48b6-b3cf-ee719265a53b...
  sddc-build-service pod: vcf-sddc-build-service-sddcbuild-86975f88d4-vbxqz
  HTTP 202
```

```
  {"id":"01a0a551-739f-7082-9fb6-059cd659fe07","name":"REFRESH_COMPONENT_WORKFLOW","description":
{"id":"com.broadcom.vcf.lcm.build.sbs.workflow.RefreshComponentWorkflowSddc","defaultMessage":"Refresh
 component OPS flt-ops01a.rainpole.io","localizedMessage":"Refresh component
 OPS flt-ops01a.rainpole.io","args":{"componentType":"OPS","componentFqdn":"flt-
ops01a.rainpole.io"}},"status":"RUNNING","createdBy":"system","createTime":"2026-09-15T13:46:13.791Z","startTime":"2026
{"descriptionArgs":{"componentFqdn":"flt-ops01a.rainpole.io","componentType":"OPS"}}}
  OK   refresh triggered

=====================================================================
 Done.
 SDDC LCM will now (re)discover OPS component 365f1e32-7984-48b6-b3cf-ee719265a53b's real
 node topology live from flt-ops01a.rainpole.io. Verify with:
   kubectl exec -n vcf-sddc-lcm vcf-sddc-lcm-db-0 -c postgres -- \
     psql -U postgres -d vcfsddclcmdb -c "SELECT * FROM node WHERE component_id = '01a06d97-
cb70-75a9-9fb7-59c372891272';"
=====================================================================
```

Retrieve a List of Backups for Log Management

If your VMware Cloud Foundation platform included log management as an additional deployed component retrieve a list of backups for log management to perform the restore.

1. Replace the values in the sample code with your values, these will be used in later steps. The `vspId` can be obtained from the output of the Retrieve Component Details from VCF Operations for VCF Instance Backup and Restore procedure.

   ```
   export servicesRuntimeFqdn='sfo-sr01.sfo.rainpole.io'
   export servicesRuntimeUsername='admin@vsp.local'
   export servicesRuntimePassword='***************'
   ```

2. Extract the *`vspId`* from the `components-ids-versions.json` file generated from the Retrieve Component Details from VCF Operations for VCF Instance Backup and Restore procedure.

   ```
   vspId=$(jq -r '.vsp[0].id' component-ids-versions.json | tee /dev/tty )
   ```

3. Retrieve an access token from the VCF services runtime by running the following command.

   ```
   servicesRuntimeToken=$(curl -k -s -X POST "https://${servicesRuntimeFqdn}/api/v1/identity/token" \
       --header "Content-Type: application/x-www-form-urlencoded" \
       --data-urlencode "grant_type=password" \
       --data-urlencode "username=${servicesRuntimeUsername}" \
       --data-urlencode "password=${servicesRuntimePassword}" \
   | jq -r '.access_token' | tee /dev/tty )
   ```

4. Retrieve a list of available log management backups from the VCF services runtime by running the following command.

   ```
   curl -k -s -X GET "https://${servicesRuntimeFqdn}/api/v1/system/backups" \
       --header "Authorization: Bearer ${servicesRuntimeToken}" \
       --header "Content-Type: application/json" \
       --header "Accept: application/json" \
   | jq -r --arg VSP_ID "$vspId" '
       ["TYPE", "VERSION", "NAME", "AGE"],
       (
           ["ops-logs"] as $order
           | .backups as $all
           | $order[] as $current_type
           | $all[]
           | select(.component.type == $current_type)
           # Filter based on the VSP_ID variable passed from the shell
           | select(.path | contains("/vcf/backups/" + $VSP_ID))
           | (
   ```

```
            (.name | sub("T(?<h>\\d{2})-(?<m>\\d{2})-(?<s>\\d{2})Z"; "T\(.h):\(.m):\(.s)Z") |
 fromdateiso8601)
       ) as $timestamp
       | ( (now - $timestamp) / 86400 | floor ) as $days_old
       | [ $current_type, .component.version, .name, ($days_old | tostring + " days ago") ]
    )
    | @tsv' | \
    column -t -s $'\t'
```

5. Identity the `NAME` of the backup job and replace the values in the sample code.

```
export backupName='2026-09-01T15-28-34Z'
```

6. Generate the `logs-restore-payload.json` file by running the following command.

```
curl -k -s -X GET "https://${servicesRuntimeFqdn}/api/v1/system/backups" \
    --header "Authorization: Bearer ${servicesRuntimeToken}" \
    --header "Content-Type: application/json" \
    --header "Accept: application/json" \
| jq --arg target "${backupName}" '{
    "components": [
      .backups[]
      | select(.component.type == "ops-logs" and .name == $target)
      | {
          "point": .name,
          "path": .path
        }
    ]
  }' | tee logs-restore-payload.json
```

Restore Log Management

You restore Log Management into the VCF Management Services instance on the recovery site.

The VCF Management Services instance on the recovery site will be scaled out as appropriate to accommodate the required number of log management replicas.

1. Retrieve a token from the VCF Management Services cluster on the recovery site.

```
export VCFMS_SR='lax-sr01.lax.rainpole.io'
export VCFMS_SR_PASSWORD='****************'

VCFMS_SR_TOKEN=$(curl -ks --request POST \
  --url "https://$VCFMS_SR/api/v1/identity/token" \
  --header 'Content-Type: application/x-www-form-urlencoded' \
  --data-urlencode grant_type=password \
  --data-urlencode "username=admin@vsp.local" \
  --data-urlencode "password=$VCFMS_SR_PASSWORD" \
  | jq -r '.access_token')
```

2. Perform the restore using the JSON payload.

```
TASK_ID=$(curl -k -X POST "https://$VCFMS_SR/api/v1/system/backups?action=restore" \
  -H "Authorization: Bearer $VCFMS_SR_TOKEN" \
  -H 'accept: application/json' \
  -H "Content-type: application/json" \
  -d '{
  "components": [
    {
      "path": "sftp://svc-vcf-bck@10.167.173.130:22/media/backups/lax/vcf/backups/40e752c7-
e940-4a9f-a49c-0f05efa9d915/9.1.1.0.25370367/ops-logs/1769b822-e28e-455a-bc28-
e2711a78d13c/9.1.1.0.25371109/2026-04-27T07-57-25Z/",
```

```
        "point": "2026-04-27T07-57-25Z"
      }
    ],
    "storage": {
      "sftp": {
        "directory": "/sftp-directory-for-source-cluster",
        "host": "<source cluster SFTP server>",
        "port": "22",
        "username": "<sftp username>",
        "password": "<sftp password>"
      }
    }
  }' | jq -r '.id')
```

3. Monitor the task until it reports as *SUCCEEDED*.

```
curl -ks "https://${VCFMS_SR}/api/v1/tasks/${TASK_ID}" \
  -H "Authorization: Bearer ${VCFMS_SR_TOKEN}" | jq
```

Update the Log Management VIP Address

After restoring the backup of log management, you must update it to use the new IP address that is mapped to the log management FQDN on the DNS server.

Download and extract the scripts from Broadcom knowledge base article 443506 to the machine you use to access your VCF environment.

On your Linux machine, run the following commands.

```
export PLATFORM_URI='https://lax-sr01.lax.rainpole.io'
export PASSWORD='****************'
export LOG_VIP='10.11.99.25'
update-component-vip.sh --platform-url $PLATFORM_URI --password $PASSWORD --log-vips $LOG_VIP
```

Update the Log Management State

After the restore operation of log management is complete, you must perform steps to finalize the restore.

1. For VCF 9.1.1 do the following:
   a) Log in to VCF Operations as a user with the **Administrator** role.
   b) Navigate to **Build** > **Lifecycle** > **VCF Management** > **Components.**
   c) Click **Log management**.
   d) From the **Actions** menu, click **Stop Service**.
   e) Monitor the Stop Service task until it completes and then click **Start Service**.
   f) Monitor the Start Service task until it completes.
2. If the logs UI is not loading in the VCF Operations UI after DR failover, we need to follow these steps:
   a) Run following script to delete the Log Management component from Ops and re-registering it.

```
export KUBECONFIG=<path-to-kubeconfig-for VCF Services Runtime>
./update_registration_in_ops.sh --ops-fqdn "flt-ops01a.rainpole.io" --component-url "flt-
logs01.rainpole.io" --ops-username "admin"
```

b) If step 2a fails, retry the task by visiting **https://<ops-fqdn>/vcf-operations/ui/operate/administration/control-panel/management-tasks/lcm/tasks** and clicking **Retry** on the failed task. Select the **Skip** option so that validations are skipped.
c) After successful deletion of the Log Management component from Ops, navigate to **Build** > **Lifecycle** > **Tasks** to verify status of the **Refresh component Log management** task. If that task failed, click **Retry**.

3. For VCF 9.1.0 do the following:
   a) Follow the steps in Broadcom knowledge base article 436904.

      Attention:

      Log management integration to the VCF Services Runtime in the primary site will fail as that instance is currently down. This is expexted behaviour.

Enable Logging on the Recovery Site VCF Management Services Instance

After a successful restore of log management you enable logging on the VCF Management Services instance on the recovery site.

1. Enable logging on the target VCF Management Services instance.
   a) Retrieve a token from the VCF Management Services instance on the recovery site.

```
export VCFMS_SR='lax-sr01.lax.rainpole.io'
export VCFMS_SR_PASSWORD='****************'

VCFMS_SR_TOKEN=$(curl -ks --request POST \
  --url "https://$VCFMS_SR/api/v1/identity/token" \
  --header 'Content-Type: application/x-www-form-urlencoded' \
  --data-urlencode grant_type=password \
  --data-urlencode "username=admin@vsp.local" \
  --data-urlencode "password=$VCFMS_SR_PASSWORD" \
  | jq -r '.access_token')
```

   b) Query the VCF Management Services cluster on the recovery site for a list of existing components.

```
curl -ks "https://${VCFMS_SR}/api/v1/components" \
  -H "Authorization: Bearer ${VCFMS_SR_TOKEN}" | jq '.components[] | {type, id, status}'
```

   c) Enable logging on the target VCF Management Services instance.

```
# VSP_COMP_ID should be the id for the VSP component from the previous step

export VSP_COMP_ID='e319236e-867e-4779-9270-4921bddf4f1f'
export VCFMS_SR='lax-sr01.lax.rainpole.io'
export LOG_MANAGEMENT_VIP='flt-logs01.rainpole.io'

TASK_ID=$(curl -ks -X POST "https://${VCFMS_SR}/api/v1/components/${VSP_COMP_ID}?action=apply" \
-H "Authorization: Bearer ${VCFMS_SR_TOKEN}" \
-H "Content-Type: application/json" \
-d '{
  "spec": {
    "configuration": {
      "logs": {
          "type": "ops",
          "ops": {
            "host": "'${LOG_MANAGEMENT_VIP}'",
            "scheme": "https",
            "port": "9543"
          }
    }
   }
  },
```

```
    "options": {
      "precheckOnly": false,
      "precheck": true,
      "timeout": "1h"
    }
  }' | jq -r '.id')

  # Poll task until complete
  curl -ks "https://${VCFMS_SR}/api/v1/tasks/${TASK_ID}" \
    -H "Authorization: Bearer ${VCFMS_SR_TOKEN}" | jq
```

2. Verify that logging is enabled.
   a) SSH to the control plane node of the target VCF Management Services instance and run the following:

```
sudo su
export KUBECONFIG=/etc/kubernetes/admin.conf
kubectl -n vmsp-platform rollout status sts/logging-operator-fluentd
kubectl -n vmsp-platform get sts logging-operator-fluentd
```

Note:

Example response

```
kubectl -n vmsp-platform rollout status sts/logging-operator-fluentd
kubectl -n vmsp-platform get sts logging-operator-fluentd
partitioned roll out complete: 1 new pods have been updated...
NAME                        READY   AGE
logging-operator-fluentd    1/1     29d
```

Deploy a VCF Automation Services Runtime Cluster

You deploy a new VCF Automation services runtime instance using the SDDC Manager API.

- VCF Automation should not already exist in the recovery VCF Instance.
- You must have at least one recent backup of VCF Automation.
- Before proceeding with restoration of VCF Automation, the following components must be available:
  - VCF Operations
  - vCenter
  - SDDC Manager
- The VCF Automation binary must be available in the software depot.

1. Retrieve token from SDDC Manager.

```
# sddcmFQDN should be the recovery site SDDC Manager
export sddcmFQDN='lax-vcf01.lax.rainpole.io'
export ssoAdminPassword='****************'
# generate token
token=$(curl -k https://"$sddcmFQDN"/v1/tokens \
  -X POST \
  -H "Content-type:application/json" \
  -d "{\"username\":\"administrator@vsphere.local\", \"password\":\"$ssoAdminPassword\"}" | jq -r
 '.accessToken')
```

2. Deploy a new VCF Automation services runtime instance.

```
# Domain id should be the management domain id from SDDC Manager
# FQDNs must match the original
# clusterId must match the original
# networkMoId must be from the portgroup you are deploying to
TASK_ID=$(curl -k -X POST -H 'accept: application/json' -H 'Content-Type: application/json' -H
 "Authorization: Bearer $token" https://"$sddcmFQDN"/v1/vsp-clusters \
-d '{
```

```
        "domainId": "a10b493e-0770-49ee-8179-d444ec5e8887",
        "deploymentType": "VspClusterSpec",
        "sddcLcmId": "3b9cdd99-9212-4e00-8bd6-53bf79c67fbf",
        "platformFqdn": "flt-vcfa-sr01.rainpole.io",
        "systemUserPassword": "***************",
        "type": "CONSUMPTION",
        "size": "medium",
        "ipv4Pool": {
            "addresses": [
                "10.21.99.46",
                "10.21.99.47",
                "10.21.99.48",
                "10.21.99.49",
                "10.21.99.50"
            ],
            "excludedAddresses": [
            ]
        },
        "clusterId": "6a079b0d-25e1-4032-9824-a7108d98a9b3",
        "networkMoId": "dvportgroup-22",
        "gatewayCidrIpv4": "10.11.10.1/24",
        "internalClusterCidrIpv4": "198.18.0.0/15"
    }' | jq -r '.id')
```

Trust VCF Installer Certificate on VCF Automation

To enable the restore from backup, you trust the VCF installer certificate on the newly deployed VCF Automation instance. This enables VCF Automation to download binaries from the VCF installer during the restore process.

Trust the VCF Installer certificate on the new VCF Automation instance.

a) Retrieve the VCF Installer certificate.

```
CERT_DATA=$($echo | openssl s_client -connect sfo-ins01.sfo.rainpole.io:443 2>/dev/null | \
openssl x509 -outform pem)
```

b) Retrieve a VCF Automation runtime token.

```
export VCFA_SR='flt-vcfa-sr01.rainpole.io'
export VCFA_SR_PASSWORD='****************'

VCFA_SR_TOKEN=$(curl -ks --request POST \
  --url "https://$VCFA_SR/api/v1/identity/token" \
  --header 'Content-Type: application/x-www-form-urlencoded' \
  --data-urlencode grant_type=password \
  --data-urlencode "username=admin@vsp.local" \
  --data-urlencode "password=$VCFA_SR_PASSWORD" \
  | jq -r '.access_token')
```

c) Trust the VCF installer certificate.

```
jq -n --arg cert "$CERT_DATA" '{cert: $cert}' | \
curl -k -X POST "https://$VCFA_SR/api/v1/system/trusted-certificates?action=add" \
     -H "Content-Type: application/json" \
     -H "Authorization: Bearer $VCFA_SR_TOKEN" \
     -d @-
```

Update SFTP Settings on a new Services Runtime Instance

Before you can perform a restore, you must update the SFTP settings on the newly deployed VCF Automation services runtime cluster.

Update SFTP Settings on the new VCF Automation Instance.

a) Retrieve the SFTP server fingerprint

```
ssh-keyscan -p 22 <sftp_server_ip> | ssh-keygen -lf -
# SSH fingerprints of type RSA and ED25519 are supported
# Enter the full fingerprint in the payload of step 1c. Example:
 SHA256:yA3FHIQMbiMnansI4+lONYBASLP2IQhGZkGJIMZ2Pi8
```

b) Retrieve a VCF Automation services runtime token

```
export VCFA_SR='flt-vcfa-sr01.rainpole.io'
export VCFA_SR_PASSWORD='****************'

VCFA_SR_TOKEN=$(curl -ks --request POST \
  --url "https://$VCFA_SR/api/v1/identity/token" \
  --header 'Content-Type: application/x-www-form-urlencoded' \
  --data-urlencode grant_type=password \
  --data-urlencode "username=admin@vsp.local" \
  --data-urlencode "password=$VCFA_SR_PASSWORD" \
  | jq -r '.access_token')
```

c) Apply the SFTP server settings

```
TASK_ID=$(curl -ks -X POST "https://$VCFA_SR/api/v1/components/765b4936-5fb5-49af-88bc-46ccc0a47a6b?
action=apply" \
  -H "Authorization: Bearer $VCFA_SR_TOKEN" \
  -d  '{
    "spec": {
      "configuration": {
         "backups": {
            "destination": "sftp",
            "encryptionPassphrase": "***************",
            "storage": {
              "sftp": {
                "directory": "/media/backups/",
                "host": "10.167.173.52",
                "port": "22",
                "username": "svc-vcf-bck",
                "password": "*********",
                "fingerprint": "SHA256:kxJ7dSHrVZpxcdbZD/s03B4J5UmMHAoW5iz4OxuBRkI"
              }
            }
          }
        }
    },
    "options": {
    }
  }' | jq -r '.id')
```

d) Monitor the task to apply the SFTP server settings

```
curl -X GET "https://$VCFA_SR/api/v1/tasks/$TASK_ID"   -k -H "Authorization: Bearer $VCFMS_SR_TOKEN" | jq

# Wait until the response shows "Succeeded" at the end of the task output
],
  "startTime": "2026-04-02T02:52:16Z",
  "status": "Succeeded",
  "type": "com.vmware.vcfms.task.ConfigureComponent"
}
```

Retrieve a List of Backups for VCF Automation

Retrieve list of backups for VCF Automation to perform the restore.

> **Note:**
>
> It is recommended to use backups taken by using a backup schedule. On-demand backups will not have a corresponding vsp or vcd-migrator backup. If you have a newer on-demand backup of VCF Automation, use the most recent available vsp and vcd-migrator backup along with it.

1. Retrieve a VCF Automation services runtime token.

```
export VCFA_SR='flt-vcfa-sr01.rainpole.io'
export VCFA_SR_PASSWORD='****************'

VCFA_SR_TOKEN=$(curl -ks --request POST \
  --url "https://$VCFA_SR/api/v1/identity/token" \
  --header 'Content-Type: application/x-www-form-urlencoded' \
  --data-urlencode grant_type=password \
  --data-urlencode "username=admin@vsp.local" \
  --data-urlencode "password=$VCFA_SR_PASSWORD" \
  | jq -r '.access_token')
```

2. Retrieve list of backups for VCF Automation.

```
curl -k "https://$VCFA_SR/api/v1/system/backups" \
  -H "Authorization: Bearer $VCFA_SR_TOKEN" \
  -H 'accept: application/json' | \
  jq -r '
    ["vsp","vcfa","vcd-migrator"] as $order
    | .backups as $all
    | $order[] as $current_type
    | $all[]
    | select(.component.type == $current_type)
    | (
        (.name | sub("T(?<h>\\d{2})-(?<m>\\d{2})-(?<s>\\d{2})Z"; "T\(.h):\(.m):\(.s)Z") | fromdateiso8601)
      ) as $timestamp
    | ( (now - $timestamp) / 86400 | floor ) as $days_old
    | [ $current_type, .component.version, .name, ($days_old | tostring + " days ago"), .path ]
    | @tsv' | \
    column -t -s $'\t'
```

3. Create a restore JSON payload using the backup paths from step 2.

```
{
  "components": [
    {
      "path": "sftp://svc-vcf-bck@10.167.173.126:22/media/backups/vcf/
backups/1f5c79fe-e3aa-41b1-a5cf-774a6497fa3d/9.1.0.0.25257932/vsp/1f5c79fe-e3aa-41b1-
a5cf-774a6497fa3d/9.1.0.0.25257932/2026-03-23T16-45-31Z",
      "point": "2026-03-23T16-45-31Z"
    },
    {
      "path": "sftp://svc-vcf-bck@10.167.173.126:22/media/backups/vcf/backups/1f5c79fe-
e3aa-41b1-a5cf-774a6497fa3d/9.1.0.0.25257932/vcfa/3d7c01a5-55fb-4b2a-8f0e-
c4d24c6d2c03/9.1.0.0.25262128/2026-03-23T16-54-12Z",
      "point": "2026-03-23T16-54-12Z"
    },
    {
```

```
      "path": "sftp://svc-vcf-bck@10.167.173.126:22/media/backups/vcf/backups/1f5c79fe-
e3aa-41b1-a5cf-774a6497fa3d/9.1.0.0.25257932/vcd-migrator/3d7c01a5-55fb-4b2a-8f0e-
c4d24c6d2c03/9.1.0.0.25262128/2026-03-23T16-54-12Z",
      "point": "2026-03-23T16-54-12Z"
    }
  ]
}
```

Stage Binaries for VCF Automation

To enable the restore of VCF Automation you pre-stage the binaries on the recovery services runtime instance.

1. Replace the values in the sample code with your values, these will be used in later steps.

```
export servicesRuntimeFqdn='flt-vcfa-sr01.rainpole.io'
export servicesRuntimeUsername='admin@vsp.local'
export servicesRuntimePassword='***************'
```

2. Retrieve an access token from the VCF services runtime by running the following command.

```
servicesRuntimeToken=$(curl -k -s -X POST "https://${servicesRuntimeFqdn}/api/v1/identity/token" \
    --header "Content-Type: application/x-www-form-urlencoded" \
    --data-urlencode "grant_type=password" \
    --data-urlencode "username=${servicesRuntimeUsername}" \
    --data-urlencode "password=${servicesRuntimePassword}" \
| jq -r '.access_token')
```

3. Stage the binary for the vcfa component by running the following command.

```
curl -k -s -X POST "https://${servicesRuntimeFqdn}/api/v1/components?action=stage" \
    --header "Authorization: Bearer ${servicesRuntimeToken}" \
    --header "Content-type: application/json" \
    --header "Accept: application/json" \
    --data-urlencode '{
        "type": "vcfa",
        "version": "9.1.1.0.25714559",
        "repository": {
            "url": "https://offline-depot.rainpole.io/PROD/release/depot-manifest-
vca-9.1.1.0.25714559.yaml"
        },
        "options": {
            "timeout": "30m"
        }
    }' \
| jq
```

Restore VCF Automation for VCF Fleet Disaster Recovery

Restore VCF Automation from a file-based backup.

1. Retrieve a VCF Automation services runtime token.

```
export VCFA_SR='flt-vcfa-sr01.rainpole.io'
export VCFA_SR_PASSWORD='***************'

VCFA_SR_TOKEN=$(curl -ks --request POST \
  --url "https://$VCFA_SR/api/v1/identity/token" \
  --header 'Content-Type: application/x-www-form-urlencoded' \
  --data-urlencode grant_type=password \
  --data-urlencode "username=admin@vsp.local" \
  --data-urlencode "password=$VCFA_SR_PASSWORD" \
  | jq -r '.access_token')
```

2. Perform the Restore.

```
TASK_ID=$(curl -k -X POST "https://$VCFA_SR/api/v1/system/backups?action=restore" \
  -H "Authorization: Bearer $VCFA_SR_TOKEN" \
  -H 'accept: application/json' \
  -H "Content-type: application/json" \
  -d '{
  "components": [
    {
      "path": "sftp://svc-vcf-bck@10.167.173.126:22/media/backups/vcf/
backups/1f5c79fe-e3aa-41b1-a5cf-774a6497fa3d/9.1.1.0.25257932/vsp/1f5c79fe-e3aa-41b1-
a5cf-774a6497fa3d/9.1.1.0.25257932/2026-03-23T16-45-31Z",
      "point": "2026-03-23T16-45-31Z"
    },
    {
      "path": "sftp://svc-vcf-bck@10.167.173.126:22/media/backups/vcf/backups/1f5c79fe-
e3aa-41b1-a5cf-774a6497fa3d/9.1.1.0.25257932/vcfa/3d7c01a5-55fb-4b2a-8f0e-
c4d24c6d2c03/9.1.1.0.25262128/2026-03-23T16-54-12Z",
      "point": "2026-03-23T16-54-12Z"
    }
  ]
}' | jq -r '.id')
```

3. Monitor the restore task.

```
curl -X GET "https://$VCFA_SR/api/v1/tasks/$TASK_ID" \
  -k -H "Authorization: Bearer $VCFA_SR_TOKEN" | jq
```

Update the VCF Automation VIP Address

After restoring the backup of VCF Automation, you must update it to use the new IP address that is mapped to the VCF Automation FQDN on the DNS server.

Download and extract the scripts from Broadcom knowledge base article 443506 to the machine you use to access your VCF environment.

This step is only required if the IP address for the VCF Automation VIP address has been changed on the recovery site.

On your linux workstation, run the following command.

```
export PLATFORM_URI='https://flt-vcfa-sr01.rainpole.io'
export PASSWORD='****************'
export VCFA_VIP='10.21.99.25'
echo COMPONENT_TYPE='vcfa'
update-component-vip.sh --platform-url $PLATFORM_URI --password $PASSWORD --vips $VCFA_VIP  --component-type
 $COMPONENT_TYPE
```

Update VCF Automation Registration to VCF Operations

After a successful recovery of VCF Automation, you update the registration in VCF Operations.

Download and extract the scripts from Broadcom knowledge base article 443506 to the machine you use to access your VCF environment.

On your linux workstation, run the following command.

```
export VCFA_RUNTIME_FQDN='flt-vcfa-sr01.rainpole.io'

./update_vcfa_registration.sh --vcfa-runtime-fqdn $VCFA_RUNTIME_FQDN --password <admin@vsp.local passwd>
```

Update Backup Schedule Configuration for Recovered Components

To ensure the recovered components are backed up according to the schedule that is assigned by the VCF Instance they currently reside in, you update the backup schedule.

1. Log in to VCF Operations as a user with the ADMIN role.
2. Navigate to **Build** > **Lifecycle** > **VCF Management** > **Backup and Restore**.
3. Click the vertical elipsis next to the schedule for the VCF Instance where the recovered components currently reside, and click **Edit**.
4. On the **Edit Schedule** page, click **Update**.
5. Monitor the task on the **Tasks** pane.

Cleanup Fleet VIP Prior to Failback

Before you can revert to the fleet VIP to the original site, you cleanup the VIP that was created on the recovery site.

1. Retrieve a token from the VCF Management Services cluster on the recovery site.

```
export VCFMS_SR='lax-sr01.lax.rainpole.io'
export VCFMS_SR_PASSWORD='****************'

VCFMS_SR_TOKEN=$(curl -ks --request POST \
  --url "https://$VCFMS_SR/api/v1/identity/token" \
  --header 'Content-Type: application/x-www-form-urlencoded' \
  --data-urlencode grant_type=password \
  --data-urlencode "username=admin@vsp.local" \
  --data-urlencode "password=$VCFMS_SR_PASSWORD" \
  | jq -r '.access_token')
```

2. Cleanup the fleet VIP.

```
export VCFMS_SR='lax-sr01.lax.rainpole.io'
export VSP_COMP_ID='e319236e-867e-4779-9270-4921bddf4f1f'

TASK_ID=$(curl -ks -X POST \
  "https://${VCFMS_SR}/api/v1/components/${VSP_COMP_ID}?action=apply" \
  -H "Authorization: Bearer ${VCFMS_SR_TOKEN}" \
  -H "Content-Type: application/json" \
  -d '{
    "spec": {
      "configuration": {
        "ingress": {
          "fleet": {
            "fqdn": "",
            "vips": { "ipv4": [] }
          }
        }
      }
    },
    "options": {"timeout": "30m"}
  }' | jq -r '.id')
```

3. Monitor the task.

```
curl -X GET "https://$VCFMS_SR/api/v1/tasks/$TASK_ID" \
  -k -H "Authorization: Bearer $VCFMS_SR_TOKEN" | jq
```

Cleanup the recovered components from the recovery Site VCF Management Services Instance

Before you can failback to the original site, you must cleanup the recovered components from the recovery site.

> **CAUTION:**
>
> This procedure is specific to failing back from a disaster recovery failover and should not be used to completely delete a deployed component outside of this scenario. To fully delete day 2 deployed components on VCF Management Services refer to this KB441333

SSH to the control plane node of the target VCF Management Services instance and run the following:

a) Cleanup the recovered components

```
sudo su
export KUBECONFIG=/etc/kubernetes/admin.conf
kubectl delete component ops-logs salt-raas vidb vcf-fleet-depot vcf-fleet-lcm
```

b) Remove the recovered components.

```
# Set VCFA Service runtime fqdn
export vcfaServicesRuntimeFqdn='flt-vcfa-sr01.rainpole.io'

# Find leader
export DB_LEADER=`kubectl exec vcf-sddc-lcm-db-0 -n vcf-sddc-lcm -c postgres -- patronictl list | awk -
F'|' '/Leader/ {print $2}' | xargs`

# Cleanup nodes of VCF Operations and VCF Operations for networks
kubectl exec $DB_LEADER -n vcf-sddc-lcm -it -- psql -U postgres -d vcfsddclcmdb -c "DELETE FROM node
 WHERE component_id IN ( SELECT id FROM component WHERE component_type in ('OPS', 'OPS_NETWORKS') OR
 fqdn='$vcfaServicesRuntimeFqdn');"

# Cleanup component entries in SDDC LCM
kubectl exec $DB_LEADER -n vcf-sddc-lcm -it -- psql -U postgres -d vcfsddclcmdb -c "DELETE FROM component
 WHERE component_type IN ('OPS', 'VIDB', 'OPS_LOGS', 'SALT_RAAS', 'VCF_FLEET_LCM', 'VCF_FLEET_DEPOT',
 'OPS_NETWORKS', 'VCFA') OR fqdn='$vcfaServicesRuntimeFqdn');"
```

Prepare to Fail Back VCF Automation

Before you fail back VCF Automation to the original site, you clean up the SDDC Manager inventory on the recovery site.

1. Open an SSH to the SDDC Manger appliance in the recovery site as the **vcf** user and elevate to the `root` user.
2. Connect to the Postgres database.

   ```
   psql -U postgres -h localhost -d platform;
   ```

3. Query the *vsp_clusters* table and record the ID of the entry with type *CONSUMPTION*.

   ```
   select * from vsp_cluster;
   ```

4. Delete the *CONSUMPTION* entry.

   ```
   delete from vsp_cluster where type='CONSUMPTION';
   ```

5. Delete the corresponding credential.

   ```
   delete from credential where username='vsp/<CONSUMPTION-vsp_cluster_id>/svc-sddc-manager-admin';
   ```

   For example:

   ```
   delete from credential where username='vsp/1f5c79fe-e3aa-41b1-a5cf-774a6497fa3d/svc-sddc-manager-admin';
   ```

Failover Tasks

Configure and perform a failover of the management applications in the SDDC from the protected to the recovery VMware Cloud Foundation instance. Failing over these applications maintains the operational state of the SDDC. You follow

the step-by-step instructions to perform disaster recovery of management components running within VMware Cloud Foundation.

You fail over the following management applications:

Table 1171: Support for Failover of the SDDC Management Applications

Management Application	Supports Failover
VCF Identity Broker	Yes
VCF Automation	Yes
VCF Operations	Yes

Prerequisites

Prerequisite	Value
Compute	The compute infrastructure in the recovery VMware Cloud Foundation instance must mirror the compute infrastructure in the protected VMware Cloud Foundation instance.
Storage	• The storage configuration and capacity in the recovery VMware Cloud Foundation instance must mirror the storage configuration and capacity in the protected VMware Cloud Foundation instance. • Datastore space on the management cluster with sufficient capacity must be available for all the virtual machines of VCF Automation and VCF Operations.
External services	Provide the following services in the recovery site. • Active Directory domain controller for each domain in the forest. Required for VCF Identity Broker to synchronize users when a domain controller in the protected site is unavailable. • DNS • NTP • SMTP • Syslog
Virtual infrastructure	• Verify that the VMware Cloud Foundation versions are the same in both sites. • Verify that Protection and Recovery are deployed in both VMware Cloud Foundation instances and paired. • Verify that the NSX load balancer is deployed and configured in both VMware Cloud Foundation instances. • Verify the NSX Edge cluster for North-South routing is deployed and configured in both VMware Cloud Foundation instances.

Failover Checklist for the VCF Management Applications for VMware Cloud Foundation

Use the checklist to verify that you have fulfilled all the requirements to initiate disaster recovery or planned migration of the SDDC management applications and to complete the configuration of these applications.

Table 1172: Checklist for Failover and Failback in an SDDC

Checklist	Tasks
Activation and assessment	• Verify that disaster failover or failback is required: For example, an application failure might not be a cause for a failover or failback, while an extended VMware Cloud Foundation instance outage is a valid cause. • Plan for business continuity events such as scheduled building maintenance or the probability of a natural disaster.
Approval	• Submit the required documentation for approval to the following roles: – IT management staff – CTO – Business users – Other stakeholders
Activation logistics	• Verify that all the required facilities and personnel are available for the complete duration of the disaster recovery process. • Verify that Protection and Recovery is available in the recovery VMware Cloud Foundation instance. • Verify the replication status of the SDDC management components. • Verify the state of NSX Data Center in the recovery VMware Cloud Foundation instance: – Verify that the NSX Edge nodes are available. – Verify that the IP addresses for the overlay-backed networks are correct. – Verify that the NSX load balancer is correctly configured according to the design.
Communication, initiation, and failover or failback validation	• In case of a planned migration: – Notify all stakeholders for the planned outage and the expected duration of the maintenance window. – At the scheduled time, initiate the failover or failback process. • In case of a failover or failback for disaster recovery: – Before initiating a failover or a failback, notify all stakeholders for the event. • After completing a failover or a failback: – Test applications availability. – Notify all stakeholders for the completed event.
Multiple Availability Zones	If your environment consists of multiple availability zones, perform additional configuration for failback for disaster recovery: • In case of failback for disaster recovery in which the recovery VMware Cloud Foundation instance remains unavailable, the vSAN witness appliance might not available too. As a result, you might be unable to provision the VCF Automation and VCF Operations virtual machines in the protected VMware Cloud Foundation instance according to the active vSAN storage policy. To ensure the recovery of the VCF Automation and VCF Operations virtual machines, turn on the force-provisioning option in the storage policy. • In case of a planned migration in which both the protected and recovery VMware Cloud Foundation instances are still operational, the vSAN witness appliance is available and the active storage policy is satisfied.

Checklist	Tasks
Configuration after failover	In case of disaster recovery failover, perform additional configuration: • Redirect the log data from the failed over or failed back applications to Log Management in the recovery VMware Cloud Foundation instance. • Complete a post-recovery assessment: – Note which items worked and which did not work, and identify improvements that you can include in the recovery plan.

Perform a Disaster Recovery of the VCF Management Applications for Fleet Disaster Recovery for VMware Cloud Foundation

If the protected VMware Cloud Foundation instance becomes unavailable, prepare networking in the recovery VMware Cloud Foundation instance and perform a failover of the protected components to the recovery VMware Cloud Foundation instance.

NSX Federation Tasks in the Recovery VMware Cloud Foundation Instance
Perform the following steps if you are using NSX Federation to provide IP mobility between VMware Cloud Foundation instances.

Assign the Active Role to the NSX Global Manager Instance in the Recovery VMware Cloud Foundation Instance

If the protected VMware Cloud Foundation instance becomes unavailable, you must prepare the network layer in the recovery VMware Cloud Foundation instance for a failover of the management applications. To provide full management of the network infrastructure in the recovery VMware Cloud Foundation instance, first change the role of the NSX Global Manager instance in the recovery VMware Cloud Foundation instance.

Procedure

You set the NSX Global Manager instance in the recovery VMware Cloud Foundation instance as active.

1. Log in to NSX Global Manager for the management domain in the recovery VMware Cloud Foundation instance at `https://<standby_nsx_gm_fqdn>/login.jsp?local=true` as **admin**.
2. On the main navigation bar, click **System**.
3. In the left navigation pane, under **Configuration**, click **Location manager**.
4. In the card for the NSX Global Manager in the recovery VMware Cloud Foundation instance, click **Actions** and select **Make active**.
5. In the **Make active** dialog box, click **Make active**.
6. In the **Do you still want to continue?** dialog box, click **Force to active**.
7. In the **Do you want to force this standby Global Manager to be active?** dialog box, click **Yes**.
8. Wait until the NSX Global Manager instance changes from `Standby` to `Active`.
   This process takes several minutes to complete.

Reconfigure the NSX Tier-1 Gateway for Dynamic Routing in the Recovery VMware Cloud Foundation Instance

To support dynamic routing in the recovery VMware Cloud Foundation instance, before you initiate a disaster recovery from the protected VMware Cloud Foundation instance, configure the NSX Tier-1 gateway to set the recovery VMware Cloud Foundation instance as primary. This configuration ensures that the management components of the SDDC continue to communicate using optimal routes in a fault-tolerant network.

1. Log in to NSX Global Manager for the management domain in the recovery VMware Cloud Foundation instance at `https://<active_nsx_gm_fqdn>/login.jsp?local=true` as **admin**.
2. On the main navigation bar, click **Networking**.
3. In the left navigation pane, click **Tier-1 gateways**.

4. Click the vertical ellipsis for the cross-instance Tier-1 gateway to which the protected components are connected and select **Edit**.
5. In the **Locations** panel, configure the settings.

Location	Mode
Protected VMware Cloud Foundation instance	Secondary
Recovery VMware Cloud Foundation instance	Primary

6. In the **Change primary location** dialog box, click **Yes**.
7. Click **Save** and click **Close editing**.

Connect the NSX Load Balancer in the Recovery VMware Cloud Foundation Instance to the Standalone NSX Tier-1 Gateway

Activate the network connectivity on the load balancer in the recovery VMware Cloud Foundation instance to support high availability and distribute the network traffic load for VCF Operations after disaster recovery to the recovery VMware Cloud Foundation instance.

1. Log in to NSX Global Manager for the management domain in the recovery VMware Cloud Foundation instance at `https://<active_nsx_gm_fqdn>/login.jsp?local=true` as **admin**.
2. From the drop-down menu at the top, select the **recovery site NSX location**.
3. On the main navigation bar, click **Networking**.
4. In the left navigation pane, click **Load balancing**.
5. On the **Load balancers** tab, click the vertical ellipsis for the standby load balancer and select **Edit**.
6. In the **Attachment** column, from the drop-down menu, select the standalone Tier-1 gateway for the recovery VMware Cloud Foundation instance, and click **Save**.

Initiate a Disaster Recovery of VCF Operations for Fleet Disaster Recovery for VMware Cloud Foundation

If a site failure occurs in the protected VMware Cloud Foundation instance, initiate a disaster recovery of VCF Operations to ensure continued operations.

Procedure

1. Log in to Protection and Recovery at `https://<vlr_fqdn/dr>` in the recovery VMware Cloud Foundation instance as **administrator@vsphere.local**.
2. In the existing site pair, click **View details**.
3. Click the **Recovery plans** tab and click the recovery plan for the VCF Operations according to your VMware Cloud Foundation Planning and Preparation Workbook.
4. Click the **Recovery steps** tab and click **Run**.
   The **Recovery** wizard appears.
5. On the **Confirmation options** page, configure these settings and click **Next**.

Setting	Value
I understand that this process will permanently alter the virtual machines and infrastructure of both the protected and recovery datacenters.	Selected
Recovery type	Disaster recovery

6. On the **Ready to complete** page, click **Finish**.
7. If using a load balancer, when prompted, verify that the load balancer virtual server for VCF Operations is available and responds to ping requests, and click **Dismiss**.
8. When prompted, power off all cloud proxiess.
   a. Log in to the management domain vCenter at `https://<management_vcenter_fqdn>/ui` in the recovery instance as **administrator@vsphere.local**.
   b. In the **VMs and Templates** inventory, navigate to the cloud proxies virtual machine folder.
   c. Right-click the cloud proxies virtual machine and select **Power** > **Shut down guest OS**.
   d. Repeat these steps for all cloud proxies in the recovery VMware Cloud Foundation instance.

e. On the recovery plan prompt, click **Dismiss**.

9. When prompted, monitor the status of each analytics cluster node.
   a. Log in to the primary VCF Operations node by using the administrator interface at `https://<vcf_operations_primary_node_fqdn>/admin` with a user assigned the **Administrator** role.
   b. On the **System status** page, under **Nodes in the VCF Operations cluster**, locate the node.
   c. Monitor the node until the **State** and the **Status** match the expected results in the recovery plan prompt.
   d. On the recovery plan prompt, click **Dismiss**.
10. When prompted, power on all available cloud proxies.
    a. Log in to the management domain vCenter at `https://<management_vcenter_fqdn>/ui` in the recovery instance as **administrator@vsphere.local**.
    b. In the **VMs and Templates** inventory, navigate to the cloud proxies virtual machine folder.
    c. Right-click each cloud proxies virtual machine and select **Power** > **Power on**.
    d. Repeat these steps for all cloud proxies in the recovery VMware Cloud Foundation instance.
    e. On the recovery plan prompt, click **Dismiss**.

What to do next:

1. Prepare VCF Operations for failback by reprotecting the virtual machines of the analytics cluster in Protection and Recovery. See Reprotect VCF Operations for Fleet Disaster Recovery for VMware Cloud Foundation.

Initiate a Disaster Recovery of VCF Operations for networks for Fleet Disaster Recovery for VMware Cloud Foundation

If a site failure occurs in the protected VMware Cloud Foundation instance, initiate a disaster recovery of VCF Operations for networks to ensure continued operations.

Procedure

1. Log in to Protection and Recovery in the protected VMware Cloud Foundation instance at `https://<vlr_fqdn>/dr` as **administrator@vsphere.local**.
2. In the existing site pair, click **View details**.
3. Click the **Recovery plans** tab and click the recovery plan for VCF Operations for networks according to your VMware Cloud Foundation Planning and Preparation Workbook.
4. Click the **Recovery steps** tab and click **Run**.
   The **Recovery** wizard appears.
5. On the **Confirmation options** page, configure these settings and click **Next**.

Setting	Value
I understand that this process will permanently alter the virtual machines and infrastructure of both the protected and recovery datacenters.	Selected
Recovery type	Disaster recovery

6. To initiate the disaster recovery of VCF Operations for networks, on the **Ready to complete** page, click **Finish**.
   This operations takes several minutes to complete.
7. On the **Recovery steps** tab, monitor the progress of the recovery plan until all recovery steps complete successfully.

What to do next

1. Prepare VCF Operations for networks for failback by reprotecting the VCF Operations for networks virtual machines in Protection and Recovery. See Reprotect VCF Operations for Fleet Disaster Recovery for VMware Cloud Foundation.

NSX Federation Post-Failover Configuration after the Original Protected VMware Cloud Foundation Instance is Available Again

When the original protected VMware Cloud Foundation instance is back online, you can fully transfer all features of the SDDC configuration to the recovery VMware Cloud Foundation instance. Perform the following steps if you are using NSX Federation to provide IP mobility between VMware Cloud Foundation instances.

Disconnect the NSX Load Balancer in the Original Protected VMware Cloud Foundation Instance from the Standalone Tier-1 Gateway

To avoid IP conflicts when the original protected VMware Cloud Foundation instance comes back online, disconnect the NSX load balancer for the SDDC management applications in the original protected VMware Cloud Foundation instance.

Procedure

1. Log in to NSX Global Manager for the management domain in the recovery VMware Cloud Foundation instance at `https://<active_nsx_gm_fqdn>/login.jsp?local=true` as **admin**.
2. From the drop-down menu at the top, select the original **protected site NSX location**.
3. On the main navigation bar, click **Networking**.
4. In the left navigation pane, click **Load balancing**.
5. On the **Load balancers** tab, click the vertical ellipsis for the standby load balancer and select **Edit**.
6. In the **Attachment** column, remove the standalone Tier-1 gateway, and click **Save**.

Obtain the Certificate Thumbprint for the NSX Global Manager Virtual IP for the Original Protected VMware Cloud Foundation Instance

Before you configure the NSX Global Manager from the original protected VMware Cloud Foundation instance as Standby to provide high availability, you must obtain the certificate thumbprint for the virtual IP of the NSX Global Manager cluster in the original protected VMware Cloud Foundation instance.

Procedure

1. Log in to the management domain vCenter in the recovery VMware Cloud Foundation instance `<recovery_vcenter_server_fqdn>:22` by using a Secure Shell (SSH) client.
2. To switch to the bash shell, run the `shell` command.
3. To retrieve the SHA-256 thumbprint of the NSX Global Manager certificate, run the command.
   ```
   echo -n | openssl s_client -connect <original_protected_NSX_Global_Manager_FQDN>:443 2>/dev/null | openssl
    x509 -noout -fingerprint -sha256
   ```
4. Save the fingerprint to later add the NSX Global Manager from the original protected VMware Cloud Foundation instance as standby for the NSX Global Manager cluster in the recovery VMware Cloud Foundation instance.

Reconfigure the NSX Global Manager in the Original Protected VMware Cloud Foundation Instance

When the original protected VMware Cloud Foundation instance is back online after a disaster recovery to the recovery VMware Cloud Foundation instance, to avoid conflicts in the management and control planes of NSX, modify the roles of the NSX Global Manager instances.

When the NSX Global Manager for the management cluster in the original protected VMware Cloud Foundation instance is back online, this NSX Global Manager instance is still partially active. To avoid any conflicts you set it to standby.

Procedure

1. Log in to NSX Global Manager for the management domain in the recovery VMware Cloud Foundation instance at `https://<active_nsx_gm_fqdn>/login.jsp?local=true` as **admin**.
2. Log in to NSX Global Manager for the management domain in the recovery VMware Cloud Foundation instance at `https://<active_nsx_gm_fqdn>/login.jsp?local=true` as **admin**.
3. On the main navigation bar, click **System**.
4. In the left navigation pane, under **Configuration**, click **Location manager**.
5. Remove the NSX Global Manager from the original protected VMware Cloud Foundation instance.
   a. Locate the original protected VMware Cloud Foundation instance NSX Global Manager.
   b. From the **Actions** menu, select **Remove**.

c. In the **Confirm remove** dialog box, click **Remove**.

6. Add the NSX Global Manager from the original protected VMware Cloud Foundation instance as standby.
   a. On the **Location manager** page, click **Add standby**, and configure the settings.

Setting	Value
Location name	Enter a location name.
FQDN/IP	Enter the FQDN for the original protected VMware Cloud Foundation instance NSX Global Manager.
Username	Enter the admin user for the original protected VMware Cloud Foundation instance NSX Global Manager.
Password	Enter the admin password for the original protected VMware Cloud Foundation instance NSX Global Manager.
SHA-256 thumbprint	Enter the NSX Global Manager virtual IP thumbprint value that you obtained earlier.

   b. Click **Check compatibility** and click **Save**.

Reconfigure the Tier-1 Gateway for Dynamic Routing in the Original Protected VMware Cloud Foundation Instance

When the original protected VMware Cloud Foundation instance is back online after a disaster recovery, you must update the routing configuration to restore routing between the protected to the recovery VMware Cloud Foundation instances and to optimize route selection.

Procedure

1. Log in to NSX Global Manager for the management domain in the recovery VMware Cloud Foundation instance at `https://<active_nsx_gm_fqdn>/login.jsp?local=true` as **admin**.
2. On the dashboard, click **Networking**.
3. In the left pane, click **Tier-1 Gateways**.
4. Click the vertical ellipsis for the cross-instance Tier-1 gateway to which the protected components are connected and select **Edit**.
5. In the **Locations** panel, configure these settings.

Location	Mode
Original Protected VMware Cloud Foundation instance	Primary
Recovery VMware Cloud Foundation instance	Secondary

6. In the **Change primary location** dialog box, click **Yes**.
7. Click **Save** and click **Close editing**.

Synchronize the VCF Operations Fleet Management Inventory for Fleet Disaster Recovery for VMware Cloud Foundation

If the virtual machines reside in the recovery VMware Cloud Foundation instance for an extended period of time, to activate day two operations, such as power on and power off of virtual machines managed by VCF Operations fleet management, you must update the VCF Operations fleet management inventory to ensure that it synchronizes the new location of the virtual machines. To update the VCF Operations fleet management inventory with the new location of the failed over virtual machines, you perform an inventory sync for each component.

Prerequisites

Verify that the virtual machines for each component are powered on.

Procedure

1. Log in to the VCF Operations interface at *https://<vcf_operations_fqdn>* with a user assigned the Administrator role.

2. Navigate to **Fleet Management** > **Lifecycle** > **VCF Management** > **Components**.
3. On the **Components** page, select the first component to synchronize.
4. On the component tab, click **Trigger inventory sync**.
5. Monitor the inventory sync request in the Tasks page.
6. If the inventory sync fails when validating the vCenter details, click **Retry**, enter the FQDN of the vCenter instance where the virtual machines are currently running, and click **Submit**.
7. Repeat the procedure for each component.

What to do next

If SDDC Manager in the original protected VMware Cloud Foundation instance is unavailable, when SDDC Manager in the original protected VMware Cloud Foundation instance becomes available again, repeat the inventory sync procedure.

Reprotection Tasks

After a disaster recovery or planned migration, the recovery VMware Cloud Foundation instance becomes the protected instance, but the virtual machines are not yet protected. If the original protected VMware Cloud Foundation instance becomes operational, you can reverse the direction of protection.

Prerequisites for Reprotection for VCF Fleet Disaster Recovery

During the reprotect operation, after Protection and Recovery reverses the direction of protection, it forces a synchronization of the storage from the new protected VMware Cloud Foundation instance to the new recovery VMware Cloud Foundation instance. Forcing data synchronization ensures that the recovery VMware Cloud Foundation instance has a current copy of the protected virtual machines running at the protected VMware Cloud Foundation instance. Recovery is possible immediately after the reprotect operation finishes.

Prerequisites

To reprotect the virtual machines of the SDDC management applications, your environment must meet certain requirements for the availability of the original protected region and state of the recovery plans.

- The new recovery VMware Cloud Foundation instance must be available. The vCenter instance, ESX hosts, Protection and Recovery Server instances, Protection and Recovery, and corresponding databases must all be recovered.
  If you cannot restore the original protected region, you must reinstall Protection and Recovery on the protected and recovery regions.
- Verify that all steps of the failover recovery plan finish successfully.

Reprotect VCF Operations for Fleet Disaster Recovery for VMware Cloud Foundation

After the inaccessible VMware Cloud Foundation instance is back online, prepare VCF Operations for failover by reprotecting the virtual machines in VMware Site Recovery Manager.

Procedure

1. Log in to Protection and Recovery in the protected VMware Cloud Foundation instance at `https://<vlr_fqdn>/dr` as **administrator@vsphere.local**.
2. In the existing site pair, click **View details**.
3. Click the **Recovery plans** tab.
4. Right-click the recovery plan for VCF Operations and select **Reprotect**.
5. On the **Confirmation options** page, select the check box confirming that you understand the reprotect operation is irreversible, and click **Next**.
6. On the **Ready to complete** page, review the reprotect information and click **Finish**.
7. To monitor the progress of the reprotect operation, click the recovery plan for VCF Operations and click the **Recovery steps** tab.
8. If the status of the recovery plan changes to `Reprotect interrupted`, repeat steps Step 4 to Step6 by selecting the **Force cleanup** check box on the **Confirmation options** page.

9. After the status of the recovery plan changes to `Ready`, click the **History** tab and click **Export all**.
10. Verify that the history report contains no errors.

    The recovery plan can return to the `Ready` state even if errors occurred during the reprotect operation.
11. If errors occurred during the reprotect operation, resolve them and run a test recovery to verify that the errors are fixed.

    If you skip resolving the errors and attempt to run a planned migration or disaster recovery later, the recovery of some virtual machines might fail.

Results

After reprotect, Protection and Recovery performs the following actions:

- Reverses the recovery site and the protected site.
- Creates placeholder copies of the virtual machines of VCF Operations from the new protected site to the new recovery site.

Reprotect VCF Operations for networks for Fleet Disaster Recovery for

After the successful migration to the recovery site, prepare VCF Operations for networks for failover by reprotecting the virtual machines in Protection and Recovery.

Procedure

1. Log in to Protection and Recovery in the protected VMware Cloud Foundation instance at `https://<vlr_fqdn>/dr` as **administrator@vsphere.local**.
2. In the existing site pair, click **View details**.
3. Click the **Recovery plans** tab.
4. Right-click the VCF Operations for networks recovery plan and select **Reprotect**.
5. On the **Confirmation options** page, select the check box for confirming that you understand the reprotect operation is irreversible, and click **Next**.
6. On the **Ready to complete** page, review the reprotect information and click **Finish**.
7. To monitor the progress of the reprotect operation, click the VCF Operations for networks recovery plan and click the **Recovery steps** tab.
8. If the status of the recovery plan changes to `Reprotect interrupted`, repeat steps Step 4 to Step 6 by selecting the **Force cleanup** check box on the **Confirmation options** page.
9. After the status of the recovery plan changes to `Ready`, click the **History** tab and click **Export all**.
10. Verify that the history report contains no errors.

    The recovery plan can return to the `Ready` state even if errors occurred during the reprotect operation.
11. If errors occurred during the reprotect operation, resolve them and run a test recovery to verify that the errors are fixed.

    If you skip resolving the errors and attempt to run a planned migration or disaster recovery later, the recovery of some virtual machines might fail.

Results

After successful reprotect, Protection and Recovery performs the following actions:

- Reverses the recovery site and the protected site.
- Creates placeholder copies of the VCF Operations for networks virtual machines from the new protected site to the new recovery site.

Failback Tasks

Initiate a Planned Migration of VCF Operations for Fleet Disaster Recovery for VMware Cloud Foundation

To migrate the virtual machines of the VCF Operations from the protected to the recovery VMware Cloud Foundation instance under planned circumstances, run the recovery plan for VCF Operations.

Procedure

1. Log in to Protection and Recovery in the protected VMware Cloud Foundation instance at `https://<vlr_fqdn>/dr` as **administrator@vsphere.local**.
2. In the existing site pair, click **View details**.
3. Click the **Recovery plans** tab and click the recovery plan for VCF Operations according to your VMware Cloud Foundation Planning and Preparation Workbook.
4. Click the **Recovery steps** tab and click **Run**.
   The **Recovery** wizard appears.
5. On the **Confirmation options** page, configure these settings and click **Next**.

Setting	Value
I understand that this process will permanently alter the virtual machines and infrastructure of both the protected and recovery datacenters.	Selected
Recovery type	Planned migration

6. To initiate the planned migration of VCF Operations, on the **Ready to complete** page, click **Finish**.
   This operations takes several minutes to complete.
7. If you are using a load balancer for VCF Operations, when prompted, verify that the load balancer virtual server for VCF Operations is available and responds to ping requests, and click **Dismiss**.

   Note:

   If you failed over the NSX load balancer, the NSX load balancer is running in the recovery VMware Cloud Foundation instance. Otherwise, the NSX load balancer is still running in the protected VMware Cloud Foundation instance.
8. When prompted, power off all running cloud proxies nodes.
   a. Log in to the management domain vCenter in the protected VMware Cloud Foundation instance at `https://<protected_instance_vcenter_server_fqdn>/ui` as **administrator@vsphere.local**.
   b. In the **VMs and Templates** inventory, navigate to the cloud proxies virtual machine folder.
   c. Right-click each cloud proxies virtual machine and select **Power** > **Shut down guest OS**.
   d. On the **Confirm guest shut down** dialog box, click **Yes**.
   e. Repeat these steps for all cloud proxies nodes in the recovery VMware Cloud Foundation instance.
   f. On the recovery plan prompt, click **Dismiss**.
9. When prompted, monitor the status of each analytics cluster node.
   a. Log in to the primary VCF Operations node by using the administrator interface at `https://<vcf_operations_primary_node_fqdn>/admin` with a user assigned the **Administrator** role.
   b. On the **System status** page, under **Nodes in the VCF Operations cluster**, locate the node.
   c. Monitor the node until the **State** becomes `Not Running` and the **Status** becomes `Offline`.
   d. On the recovery plan prompt, click **Dismiss**.
   e. Repeat these steps for all analytics cluster nodes.
10. When prompted, power on all VCF Operations cloud proxies.
    a. Log in to the management domain vCenter in the protected VMware Cloud Foundation instance at `https://<protected_instance_vcenter_server_fqdn>/ui` as **administrator@vsphere.local**.
    b. In the **VMs and Templates** inventory, navigate to the cloud proxies virtual machine folder.
    c. Right-click each cloud proxy virtual machine and select **Power** > **Power On**.
    d. Repeat these steps for all remote collector nodes in the recovery VMware Cloud Foundation instance.
    e. On the recovery plan prompt, click **Dismiss**.

11. Monitor the status of each cloud proxy node.
    a. Log in to the primary VCF Operation node by using the administrator interface at `https://<vcf_operations_primary_node_fqdn>/admin` with a user assigned the **Administrator** role.
    b. On the **System status** page, under **Nodes in the VCF Operations cluster**, locate the node.
    c. Monitor all cloud proxies until their **State** becomes `Not Running` and the **Status** becomes `Offline`.
12. In the administrator interface of the primary VCF Operations node, on the **System status** page, under **Cluster status**, click **Bring cluster online**.
    This operations takes several minutes to complete.
13. On the **Recovery steps** tab, monitor the progress of the recovery plan until all recovery steps complete successfully.

What to do next

1. Prepare VCF Operations for failback by reprotecting the VCF Operations virtual machines in Protection and Recovery. See Reprotect VCF Operations for Fleet Disaster Recovery for VMware Cloud Foundation.

Initiate a Planned Migration of VCF Operations for networks for Fleet Disaster Recovery for VMware Cloud Foundation

To migrate the virtual machines of the VCF Operations for networks from the protected to the recovery VMware Cloud Foundation instance under planned circumstances, run the recovery plan for VCF Operations for networks.

Procedure

1. Log in to Protection and Recovery in the protected VMware Cloud Foundation instance at `https://<vlr_fqdn>/dr` as **administrator@vsphere.local**.
2. In the existing site pair, click **View details**.
3. Click the **Recovery plans** tab and click the recovery plan for VCF Operations for networks according to your VMware Cloud Foundation Planning and Preparation Workbook.
4. Click the **Recovery steps** tab and click **Run**.
   The **Recovery** wizard appears.
5. On the **Confirmation options** page, configure these settings and click **Next**.

Setting	Value
I understand that this process will permanently alter the virtual machines and infrastructure of both the protected and recovery datacenters.	Selected
Recovery type	Planned migration

6. To initiate the planned migration of VCF Operations for networks, on the **Ready to complete** page, click **Finish**.
   This operations takes several minutes to complete.
7. On the **Recovery steps** tab, monitor the progress of the recovery plan until all recovery steps complete successfully.

What to do next

1. Prepare VCF Operations for networks for failback by reprotecting the Log Management virtual machines in Protection and Recovery. See Reprotect VCF Operations for Fleet Disaster Recovery for VMware Cloud Foundation.

FIPS Configuration for VCF Components

VCF supports the Federal Information Processing Standards (FIPS) 140-3, which specify security requirements for cryptographic modules.

Many VCF components support the FIPS standard inherently, and cannot be configured in ways that do not meet the standard. For example, ESX and VMware NSX. Other components might require enablement for FIPS compliance or have varying behavior depending on how they were deployed.

The following components have FIPS enabled by default:

- ESX
- vCenter
- NSX
- VCF Operations HCX
- Protection and Recovery
- Data Services Manager
- VCF management services

The table below summarizes FIPS enablement for components where FIPS is either not enabled by default or can be disabled in certain circumstances.

Component	FIPS Configuration
SDDC Manager	In new VCF 9.x deployments, FIPS compliance in SDDC Manager is on by default and cannot be turned off. For deployments upgraded from VCF 5.x, the FIPS setting remains the same as it was in the original deployment. This setting cannot be altered after upgrade.
VCF Operations	FIPS compliance is off by default. You can activate FIPS compliance during deployment by selecting the **Activate FIPS Mode** option during deployment.
Log management	FIPS compliance is off by default. You can enable FIPS compliance during deployment by selecting the the **On** option for **FIPS Compliance**. See Deploy Log Management.
VCF Operations for networks	VCF Operations for networks uses FIPS compliant cryptographic algorithms for internal connections. FIPS compliance for external connections is off by default. You can enable FIPS compliance for external connections by navigating to **Settings** > **System Configuration** > **FIPS Mode for External Connections** and toggling the setting to **On**.
Cloud Proxy for VCF Operations	FIPS compliance is not enabled by default. You can enable FIPS compliance for cloud proxies after first enabling FIPS compliance for VCF Operations. For instructions on enabling FIPS compliance, see Using the Cloud Proxy Command-Line Interface.
VCF Automation	In new VCF 9.x deployments, FIPS compliance in VCF Automation is enabled by default and cannot be turned off. For deployments upgraded from VMware Aria Operations 18.x, the FIPS setting remains the same as it was in the original deployment. This setting cannot be altered after upgrade.
VMware vSphere Kubernetes Service	VKS 3.3 and later provide the ability to enable OS FIPS mode by configuring the osConfiguration cluster class variable. See osConfiguration for more details.
VMware Virtual Disk Developer Kit (VDDK)	By default, VDDK does not enable FIPS. You can configure it by setting the option `vixDiskLib.ssl.enableSslFIPS=1` in the VDDK configuration file. See the instructions provided with the development kit at https://developer.broadcom.com/sdks/vmware-virtual-disk-development-kit-vddk/latest for more information.

FIPS Configuration for VMware vSphere Foundation Components

You can activate Federal Information Processing Standards (FIPS) for the VCF Operations component of VMware vSphere Foundation to make your environment FIPS compliant. You can also activate firewall hardening.

Activate FIPS

You can activate FIPS in the VCF Operations cluster at the time of installation or after VCF Operations is up and running. Adding FIPS at installation is less intrusive because the cluster has not yet started.

If the cluster is running, to activate FIPS, you must take the cluster offline.

FIPS mode is supported in Cloud Proxy. You can continue using your cloud proxy after enabling FIPS for the VCF Operations cluster.

1. In a Web browser, navigate to the master node administration interface. `https://master-node-name-or-ip-address/admin`.
2. Enter the VCF Operations administrator credentials and click **Log In**.
3. Click **Administrator Settings**.

   Note:
   The **Activate FIPS** button is deactivated when the cluster is running.
4. Click **Activate FIPS** after you take your cluster offline.

   Note:
   After you activate FIPS, you cannot deactivate the FIPS mode in the current setup. To revert to a FIPS deactivated setup, you must re-deploy VCF Operations.
5. In the **Are you sure you want to activate FIPS** dialog box, read the note and provide your consent for enabling FIPS and then click **Yes**.

   Note:
   After you activate FIPS, the cluster restarts and is not be available during this time. The cluster nodes are rebooted and after the cluster is online, all the nodes are FIPS activated.

Activate Firewall Hardening

Activating firewall hardening restricts network access to internal services in VCF Operations.

1. In a Web browser, navigate to the master node administration interface. `https://master-node-name-or-ip-address/admin`.
2. Enter the VCF Operations administrator credentials and click **Log In**.
3. Click **Administrator Settings**, and then click **Security Settings** from the **Administrator Settings** page.
4. Click **Activate Firewall Hardening**.

Using the FIPS API to Query FIPS Status of Components

VCF 9.1 includes an API that allows you to retrieve information about the FIPS cryptographic modules used by each component, including version number, CMVP certificate, and other information.

Note:

The FIPS API is available as a Tech Preview only.

The FIPS API is a read-only API that can be used to retrieve FIPS cryptographic module information about the following components:

- vCenter
- ESX
- NSX
- HCX
- SDDC Manager
- Data Services Manager
- VCF Automation
- VCF Operations
- VCF Operations for networks
- VMware Live Recovery

Components aggregate information about subcomponents and return that information in the API response. For example, a query to vCenter returns component information for ESX.

The API returns the following fields.

Field	Description
`name`	The official title of the cryptographic module.
`version`	Specifies the version that has undergone FIPS 140 validation.
`cmvp_cert`	The CMVP certificate number for the cryptographic module. If the CMVP certificate was not issued at the time of release, this field returns nothing.
`cavp_cert`	The CAVP certificate number for the cryptographic module. The Cryptographic Algorithm Validation Program (CAVP) is a prerequisite for CMVP validation. A CAVP certificate number indicates progress in the validation process. If the CAVP certificate was not issued at the time of release, this field returns nothing.
`component_name`	Name of the component containing the cryptographic module.
`component_id`	Component id.
`component_version`	Component version.

FIPS API Limitations

- If FIPS has been disabled for a component, the API might return an empty list.
- Parent components silently exclude subcomponents that are at versions below 9.1. For example, consider a vCenter running 9.1 that is managing a mixture of ESX hosts running versions 9.0 and 9.1. The API query returns information for the ESX 9.1 hosts, but omits the ESX 9.0 hosts.

FIPS API Endpoints

Component	Endpoint
ESX	`/api/esx/crypto/fips/modules`
vCenter	`/api/vcenter/crypto/fips/modules`
SDDC Manager	`/v1/system/security/fips/modules`
NSX	`/api/v1/node/fips/module`
HCX Manager	`/hybridity/api/admin/fips/modules`
HCX Photon-based data plane	`/fips/modules`
Data Services Manager	`/provider/fips/modules`
VCF Operations	`/suite-api/internal/fips/modules`
VCF Automation	`/vcfa/fips/modules`
Log Management	`/api/v2/fips/modules`
VCF Operations for networks	`/api/ni/settings/fips/modules`

Querying vCenter

1. Obtain an API token.

   ```
   curl -k -X POST -u 'userName:password' https://<host>/api/session
   ```

2. Use the token obtained in the following command.

   ```
   curl -k -X GET https://<host>/api/vcenter/crypto/fips/modules -H "vmware-api-session-id: <token>" -H
    'Content-Type: application/json'
   ```

Example output:

```
[
  {
    "cmvp_cert": "4743",
    "component_name": "vCenter",
    "name": "VMware's BC-FJA (Bouncy Castle FIPS Java API)",
    "version": "2.0.0",
    "component_version": "9.1.0"
  },
   {
    "cmvp_cert": "4865",
    "component_name": "vCenter",
    "name": "VMware’s Linux Kernel Cryptographic Module",
    "version": "5.0.0",
    "component_version": "9.1.0"
  },
  {
    "cmvp_cert": "4442",
    "component_name": "ESXi",
    "name": "VMware's ESXboot Cryptographic Module",
    "cavp_cert": "A1357",
    "version": "1.0",
    "component_version": "9.1.0"
  },
  {
    "cmvp_cert": "4815",
    "component_name": "ESXi",
    "name": "VMware VMkernel Cryptographic Module",
    "cavp_cert": "A2792",
    "version": "2.0",
    "component_version": "9.1.0"
  }
]
```


Querying ESX

1. Obtain an API token.

   ```
   curl -k -X POST -u 'userName:password' https://<host>/api/session
   ```

2. Use the token obtained in the following command.

   ```
   curl -k -X GET https://<host>/api/esx/crypto/fips/modules -H "vmware-api-session-id: <token>" -H
    'Content-Type: application/json’
   ```

Querying SDDC Manager

1. Obtain an API token.

   ```
   curl https://<host>/v1/tokens -k -X POST -H 'Content-Type: application/json' -H 'Accept: application/json'
    -d '{"username" : "username","password" : "password"}'
   ```

2. Use the token obtained in the following command.

   ```
   curl -H 'Authorization: <token>' https://<api_host>/v1/system/security/fips/modules
   ```

Querying NSX

Query the NSX Manager host as follows.

```
curl -k -X GET https://<nsx-manager-host>/api/v1/node/fips/modules -u 'username:password'
```

Querying VCF Automation

1. Obtain an API token.

```
curl https://<host>/api/v1/tokens -k -X POST -H 'Content-Type: application/json' -H 'Accept: application/
json' -d '{"username" : "username","password" : "password"}'
```

2. Use the token obtained in the following command.

```
curl -vk https://{$HOST_IP_ADDRESS}/vcfa/fips/modules
```

Configuring Management Components

After deploying VCF Operations, VCF Automation, or VCF management services components, you can update a component configuration or address issues in a component deployment.

Performing Actions on Components

You can perform post-deployment actions on VCF Operations, VCF Automation, VCF Operations for networks, and VCF management services components. VCF management services components include log management, identity broker, software depot, real-time metrics, fleet lifecycle, SDDC lifecycle, Salt RaaS, and Salt master. You can also perform most post-deployment actions on VCF services runtime.

Table 1173: Post-deployment actions and applicable components

Post-deployment actions are grouped into categories such as, Scaling operations, Node Management, Maintenance, Network, and State Management. The following table lists some of the post-deployment actions that you can perform.

Category: Post-deployment Action	Operates on:
Scaling operations: Scale Up Increase the resources such as CPU and memory that is allocated to the management components by increasing the size from small to medium, or large. See Scaling VCF Management Components.	VCF Operations , VCF Automation, VCF Operations for networks, VCF services runtime, VCF management services
Node Management: Add Replica or Data Node, Add platform node, Add collector node. See Add Nodes in VCF Operations and VCF Operations for networks.	VCF Operations, VCF Operations for networks
Node Management: Add Cloud Proxy See Add a Cloud Proxy.	VCF Operations
Scaling operations: Increase Storage Size Extend the capacity of existing storage volumes for a component. See Increase Storage Size for VCF Automation and VCF Management Services Components.	VCF Automation, VCF services runtime, VCF management services and VCF management services components
Maintenance: Generate Support Bundle Create a support bundle for the component to use for troubleshooting. See Support Bundle Actions for VCF Automation and VCF Services Runtime. **Note:** Support bundle actions are being deprecated. In its place, use Log Assist to provide log bundles for troubleshooting. See Use Log Assist to Accelerate Your Support Cases.	VCF Automation, VCF services runtime, VCF management services and VCF management services components
Maintenance: Delete Support Bundle See Support Bundle Actions for VCF Automation and VCF Services Runtime.	VCF Automation, VCF services runtime, VCF management services and VCF management services components

Category: Post-deployment Action	Operates on:
Maintenance: Download Support Bundle Download a support bundle for the component to use for troubleshooting. See Support Bundle Actions for VCF Automation and VCF Services Runtime. **Note:** Support bundle actions are being deprecated. In its place, use Log Assist to provide log bundles for troubleshooting. See Use Log Assist to Accelerate Your Support Cases.	VCF Automation, VCF services runtime, VCF management services and VCF management services components
Network: Add Cluster VIPs Add virtual IP addresses to the component FQDNs to improve resiliency and performance of components. See Network Actions for VCF Services Runtime.	VCF management services
Network: Expand Node IP Pool Add IP addresses to the VCF services runtime nodes when scaling components. See Network Actions for VCF Services Runtime.	VCF management services

Viewing and Accessing Post-Deployment Actions

To view the post-deployment actions that are available for your component or cluster:

1. Log in to the VCF Operations interface at `https://<vcf_operations_fqdn>` as a user assigned the Administrator role.
2. Navigate to **Build** > **Lifecycle**.
3. In the Lifecycle navigation, select **VCF Management**.
4. On the Components tab, select the component listed that you want to operate on.
5. Click **Actions** to view the list of actions for the component selected.

Figure 441: Example: Actions for VCF Operations

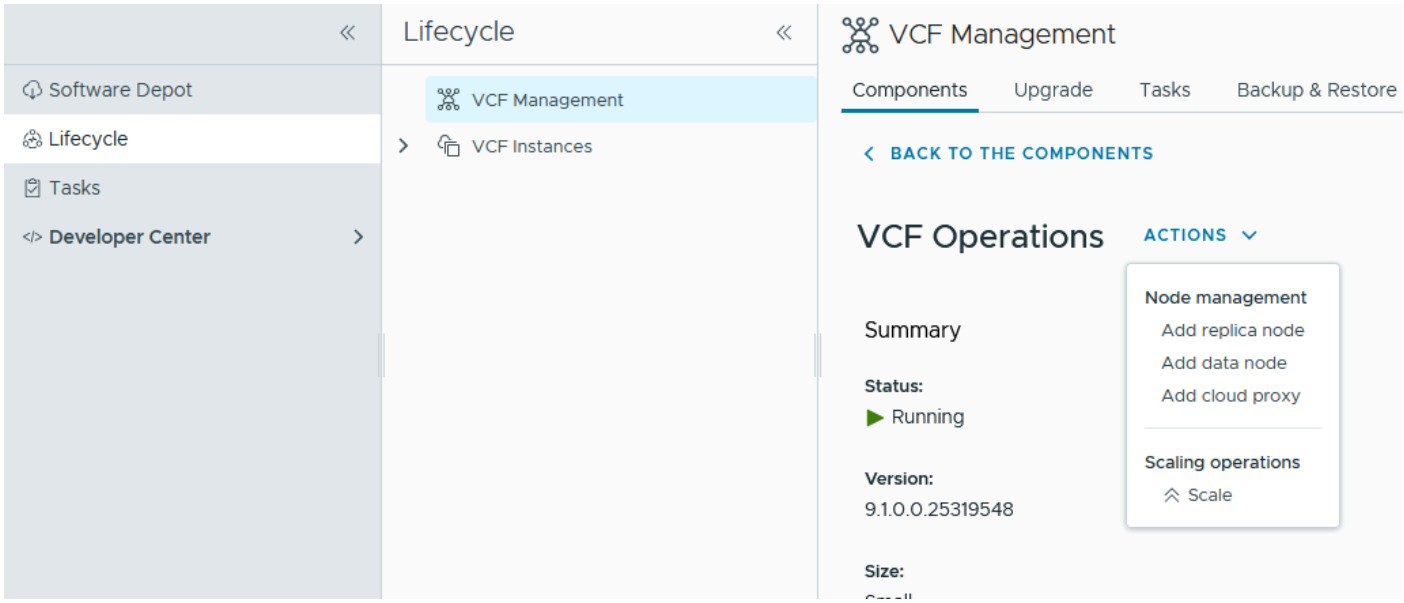
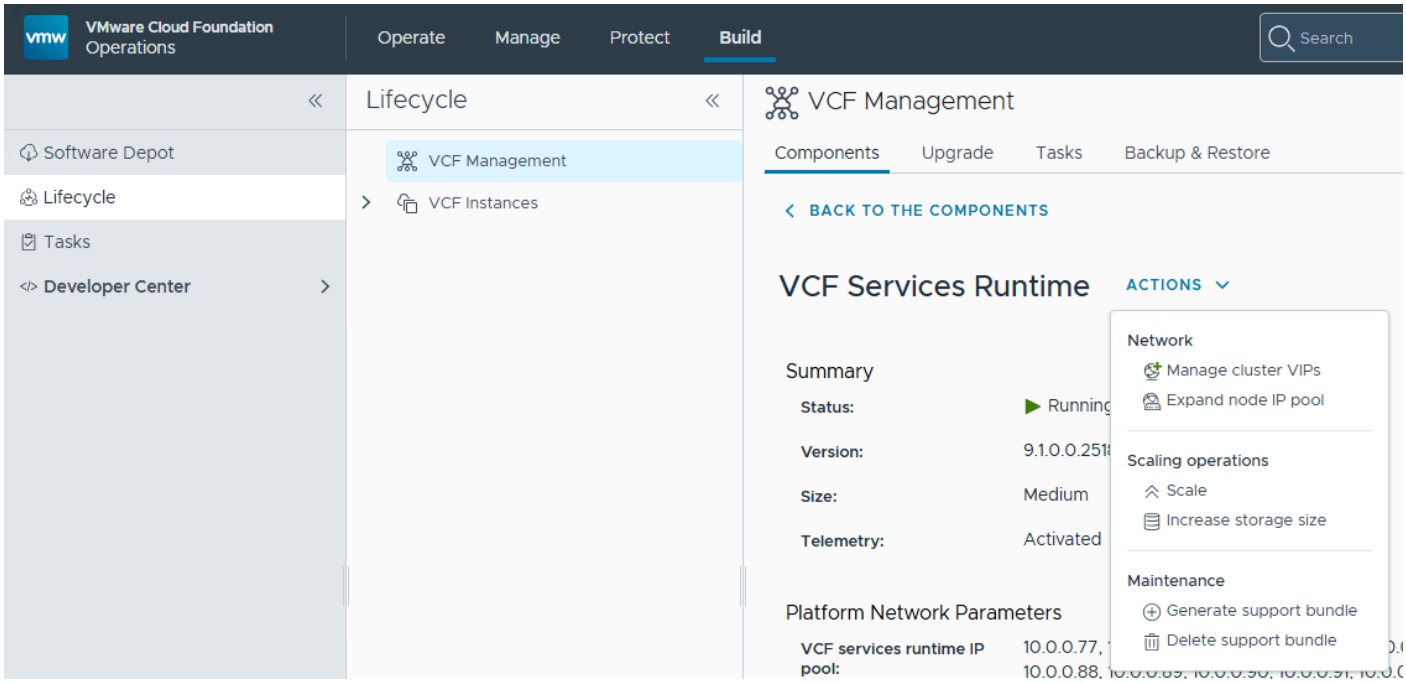


Figure 442: Example: Actions for VCF Services Runtime Cluster



Scaling VCF Management Components

You can scale up and can scale out your VCF Operations, VCF Operations for networks, VCF Automation, VCF services runtime, log management, identity broker, Salt RaaS, and Salt master components.

For all components, the scale up procedure begins in the same way. Some components include additional selections:

- For VCF Operations, you increase the size of nodes in your VCF Operations cluster. You also can add disk size and select a datastore where the new disk is added.

  Important:

  To scale a VCF Operations cluster for continuous availability, you must:

  - Deploy a VCF Operations cluster for continuous availability. See Configure VCF Operations Cluster for Continuous Availability.
  - Log in to the VCF Operations administrator interface in the primary fault domain to specify options and start the scale up process for VCF Operations.
- For VCF Operations for networks, you select the platform or collector nodes to scale up.
- For log management, you choose the size of your log management deployment and select the number of replica nodes.

- For VCF services runtime, you scale up capacity by increasing size and adding IP addresses to the Node IP pool.

1. Log in to the VCF Operations interface at `https://<vcf_operations_fqdn>` as a user assigned the Administrator role.
2. Under **Build** > **Lifecycle** > **VCF Management**, click the component that you want to change.
3. Specify the options for your component and start the scale up process.
   - For VCF Operations:
     1. Under Actions, click **Scale**.
     2. Select the node size: XS, Small, Medium, Large, XL
     3. Optionally provide additional disk size in GB to be added to every node in the VCF Operations cluster.
     4. Select **Advanced Datastore Configuration** to select a datastore where the disk is to be added on each node.
     5. Review the summary of your request and click **Scale**.
   - For VCF Operations for networks:
     1. Under Actions, click **Scale Up**.
     2. Select the platform or collector nodes to scale up.
     3. Specify the size, datastore, and additional disk size for the node, and click **Confirm**.
     4. Repeat the process for the collector nodes, as needed.
     5. Review the summary of your request and click **Scale Up**.
   - For VCF services runtime:
     1. Under Actions, click **Scale**.
     2. For size, the current size is pre-selcted. Specify the new size of the deployment: Small, Medium, Large.
     3. (optional) Add IP addresses to the Node IP pool.
     4. Review the summary of your request and click **Finish**.
   - For log management:
     1. Under Actions, click **Scale**.
     2. For size, the current size is pre-selected. Specify the new size of the deployment: Small, Medium, Large.
     3. For number of replicas, the current number is pre-selected. Specify the new number of nodes.
     4. Click **Scale**.
   - For VCF Automation, identity broker , and real-time metrics select the size of deployment:
     1. Under Actions, click **Scale**.
     2. For size, the current size is pre-selcted. Specify the new size of the deployment: Small, Medium, Large.
     3. Acknowledge the possible service disruption and click **Scale**.

Scale up begins. Click the **Tasks** tab to check the progress of the scale up task. During the process, the nodes are restarted and services are not available during the downtime.

> **Important:**
>
> Rolling back to a backup created before scale up will revert the cluster to its earlier size and permanently delete data from the node added during scale up. After the scale up process completes, take a new backup to avoid data loss.

Add Nodes in VCF Operations and VCF Operations for networks

For VCF Operations, you add replica or data nodes. For VCF Operations for networks, you add platform nodes or collector nodes.

Update the VCF Operations certificate to include the FQDN or IP address of the nodes to be added. See Managing Certificates in VMware Cloud Foundation.

Note:

If you do not update the certificate prior to performing the scale-out operation, the lifecycle management process will fail. To remediate the failure, see Broadcom knowledge base article 430384.

1. Log in to the VCF Operations interface at `https://<vcf_operations_fqdn>` as a user assigned the Administrator role.
2. Under **Build** > **Lifecycle** > **VCF Management**, select the component that you want to update.
3. Under Actions, select the Node Management action that you want to perform.
   - To add a replica or data node in VCF Operations:
     1. Select the type of node to add.
     2. Enter the Fully Qualified Domain Name.

        Important:

        The VCF Operations certificate must be updated to include the FQDN of the new node.
     3. Enter the password for the VCF Operations admin.
     4. Enter the password for the new node.
     5. After verifying that any necessary changes to the VCF Operations certificate have been made, click **ADD** to start the node deployment.
   - To add a platform node in VCF Operations for networks:
     1. Select **Add platform node**.
        a. Enter the IP address of the new platform node.

        Important:

        The VCF Operations certificate must be updated to include the IP address of the new node.
        b. (optional) To add multiple platform nodes, click **Add Another Platform Node** and enter additional IP addresses.
        c. Enter the credentials of the platform node or regenerate and consume a new password. The password is used for the console user and support.
     2. To add a node or multiple nodes to the Operations-Network cluster, click **ADD**.
   - To add a collector node in VCF Operations for networks:
     1. Select **Add collector node**.
     2. Enter the IP address of the new collector node.

        Important:

        The VCF Operations certificate must be updated to include the IP address of the new node.
     3. Enter the credentials of the collector node or regenerate and consume a new password. The password is used for the console user and support.
     4. Select the VCF Instance where Operations-Collector node should be deployed. The node is deployed on the Management Domain vCenter of the VCF Instance selected.
     5. To add the node to the Operations-Network cluster, click **ADD**.

Add a Cloud Proxy

Add a cloud proxy to your VCF Operations environment.

- Verify that you have an IP address and an FQDN for the cloud proxy.

- Verify that you have the VCF Operations admin account password.
- Use the following information to verify that you have enough compute resources for the cloud proxy.

Cloud Proxy Type	VM Scale Requirements	CPU Requirements	Memory Requirements
Unified Small Cloud Proxy	Up to 16,000 VMs	4 vCPUs	16GB
Unified Standard Cloud Proxy	Between 16,000 to 80,000 VMs	8 vCPUs	48 GB

To add a cloud proxy as a post-deployment action, perform the following steps:

1. Log in to the VCF Operations interface at `https://<vcf_operations_fqdn>` as a user assigned the Administrator role.
2. Under **Build** > **Lifecycle** > **VCF Management**, select VCF Operations.
3. Under Actions, select **Add cloud proxy**.
   a) Provide input for the cloud proxy.
      - FQDN: Enter the FQDN for the cloud proxy.
      - Size: Select a small or standard size for the new cloud proxy. See Broadcom knowledge base article 397782.
      - VCF Operations admin password: Enter the admin account password for the VCF Operations instance where you are adding the cloud proxy.
      - Password: Create a password for the cloud proxy.

      Note:

      Save the password to a secure place for future use.
      - VCF Instance: Select the VCF Instance where you want to add the cloud proxy.

   b) Click **Add**.

   If you want to start collecting data using the newly added cloud proxy, you must update the integration settings for your VCF Instance or vCenter.
4. (Optional) After the deployment completes, update the VCF Instance integration to use the new collector for new workload domains.
   a) Select **Operate** > **Administration** > **Integrations** and click **Accounts**.
   b) Expand **VMware Cloud Foundation** and expand the integration that you want to configure.
   c) At the VCF Instance level, click the **vertical ellipsis** and click **Edit**.
   d) On the **Cloud Account Information** page, from the **Cloud Proxy / Group** drop-down menu, select the newly deployed collector, click **Validate Connection**, and click **Save**.
5. (Optional) For the existing domains of the same VCF Instance, update the integrations for all vCenter instances to use the new collector/collector group.
   a) Select **Operate** > **Administration** > **Integrations** and click **Accounts**.
   b) Expand **VMware Cloud Foundation** and expand the integration that you want to configure.
   c) At the domain level, click the **vertical ellipsis** and click **Edit**.
   d) On the **Connect vCenter** page, from the **Cloud Proxy / Group** drop-down menu, select the newly deployed collector, click **Validate Connection**, and click **Save**.
   e) Repeat the step for all remaining workload domains in that VCF Instance.

After deploying the cloud proxy, you can monitor the appliance health status and activate data persistence and log assist. For more information, see Monitoring the Health of Cloud Proxies.

Increase Storage Size for VCF Automation and VCF Management Services Components

Increase storage size increases the storage capacity for different volume groups in your VCF Automation, VCF services runtime, or VCF Management Services Component deployment.

1. Log in to the VCF Operations interface at `https://<vcf_operations_fqdn>` as a user assigned the Administrator role.
2. Under **Build** > **Lifecycle** > **VCF Management**, select VCF services runtime or the VCF Management Services component that you want to update.
3. Under Actions, select **Increase Storage Size**.
4. Update Volume Size values for the various Volume Groups.

   Important:

   Once increased, storage size cannot be reduced.

   - Volume Group Name: Identifies the volume groups that are mapped to specific components.
   - Volumes in group: Indicates the number of volumes that are currently assigned to each volume group.
   - Volume Size: Displays the current size of each volume in Gigabytes (GB). Use the - and + buttons or manually enter a value to specify the new, increased size per volume.
   - Used/Total Storage Capacity: Displays the current storage usage versus the total available capacity for the group. The total capacity is calculated by multiplying the number of "Existing Volumes" by the specified "Volume Size".
5. After ensuring that you are ready to irreversibly increase storage size, click **Resize**.

Support Bundle Actions for VCF Automation and VCF Services Runtime

For VCF Automation and VCF services runtime, you use the following actions to manage the support bundles that collect logs. The VCF Automation support bundle also includes the information for the available VCF Services.

- Generate Support Bundle
- Delete Support Bundle
- Download Support Bundle

1. Log in to the VCF Operations interface at `https://<vcf_operations_fqdn>` as a user assigned the Administrator role.
2. Under **Build** > **Lifecycle** > **VCF Management**, select the component that you want to generate a support bundle for.
3. Under Actions, select the Maintenance action that you want to perform.
   - To generate a support bundle:
     1. Select **Generate Support Bundle**.
     2. For Period, click the -/+ buttons or enter the number of days that you would like to collect logs and click **Generate**.
   - To download a support Bundle:
     1. Select **Download Support Bundle**.
     2. Review the support bundle names and dates generated, then select the bundle would like to download and click **Download**.
   - To delete a support Bundle:
     1. Select **Delete Support Bundle**.
     2. Review the support bundle names and dates generated, then select the bundle would like to delete and click **Delete**.

Network Actions for VCF Services Runtime

Use the following network actions to manage the IP addresses in your deployment.

- Manage Cluster VIPs
- Expand Node IP Pool

New IP address ranges do not need to be continuous.

1. Log in to the VCF Operations interface at `https://<vcf_operations_fqdn>` as a user assigned the Administrator role.
2. Under **Build** > **Lifecycle** > **VCF Management**, select the component that you want to update.
3. Under Actions, select the Network action that you want to perform.
   - To manage cluster VIPs:
     1. Select **Manage Cluster VIPs**.
     2. Assign IP addreses or click **Add IP Address** for the three FQDNs listed as needed and click **Save**.
        – VCF services runtime FQDN
        – VCF Instance Services FQDN
        – VCF Fleet Services FQDN

     Note:

     Any added IP addresses must resolve to the displayed FQDNs and must be in the same subnet as existing IPs. Ensure that both forward and reverse lookups are configured correctly on your DNS server before adding them.
   - To expand a node IP Pool:
     1. Select **Expand Node IP Pool**.
        The Expand Node IP Pool displays existing IP addresses for VCF services runtime nodes that are being used by VCF management services components.
     2. Click **Add New IP Pool** to add IP addresses and click **Save**.

Configure a Proxy Server for VCF Management Services Components and VCF Automation

If your VCF Management Services components do not have direct Internet access, you can configure a proxy server. You configure proxy settings for a VCF services runtime instance. The configuration is then valid for all the components hosted on that instance. Components hosted on VCF services runtime instances that might connect to the Internet, are software depot, telemetry, identity broker, and VCF Automation.

- Verify that you have administrative credentials for the VCF Operations primary node.
- Verify that you have configured a proxy server to access the Internet.
- Verify that you have `curl` and `jq` utilities installed on a machine with management access to your VCF environment.

Note:

In environments where SSL/TLS interception is enabled, software depot configured in connected mode might not establish a trusted connection to the Broadcom authorization server. For more information on adding your proxy's CA certificate to the truststore of software depot, see Broadcom knowledge base article 442978.

The following procedure allows you to route component traffic through a proxy server for a specific VCF services runtime instance. The configuration includes TLS settings, authentication credentials, and certificate information.

This procedure includes the following steps.

- Retrieving the service key for the fleet lifecycle component which stores proxy configuration settings for all VCF services runtime instances in the fleet.
- Obtaining a VCF Operations token that you use to authenticate and authorize API calls to the fleet lifecycle component.
- Exchanging the VCF Operations token for a bearer token.
- Configuring the VCF services runtime instance configuration with proxy settings.

You must perform this procedure on each VCF services runtime instance, where the component requires the traffic to be routed through a configured proxy server.

Note:

Alternatively, instead of manually performing the tasks in the procedure below, you can use the `vmsp-utility.py` script. The script automates the API calls and includes a menu with options to show, configure,

and remove a proxy configuration for a VCF services runtime instance. See Broadcom knowledge base article 442978.

1. On a machine with management access to your VCF environment, set the VCF Operations host environment variable.

   OPS_FQDN is the VCF Operations primary node FQDN and *password* is the VCF Operations admin user password.

   ```
   export OPS_HOST=https://OPS_FQDN
   export OPS_PASSWORD="password"
   ```

2. Retrieve the service key for the fleet lifecycle component.

   The Basic authorization header contains the base64-encoded *username:password*. To encode your credentials in base64, use the `printf 'username:password' | base64` command. For example, `admin:vAU^MJ43#I0opiDRrYH4` equals `YWRtaW46dkFVXk1KNDMjSTBvcGlEUnJZSDQ=`.

   ```
   curl -k --location --request GET ${OPS_HOST}/casa/services \
     --header 'Authorization: Basic YWRtaW46dkFVXk1KNDMjSTBvcGlEUnJZSDQ=' | jq | less
   ```

   The response contains service information. Locate and record the *key* value. For example:

   ```
   {
     "name": "VCF Fleet LCM",
     "type": "VCF_FLEET_LCM",
     "key": "key",
     "nodes": [
       {
         "name": "VCF Fleet LCM",
         "addresses": [
           {
             "type": "Fqdn",
             "value": "vsp-fleet.vcf.rainpole.io"
           }
         ]
       }
     ]
   }
   ```

3. Use the authentication endpoint to obtain a VCF Operations token.

   ```
   export OPS_TOKEN=$(curl -k --location --request POST ${OPS_HOST}/suite-api/api/auth/token/acquire \
     --header 'Content-Type: application/json' \
     --data-raw "$(jq -n --arg pass "$OPS_PASSWORD" '{
     "username": "admin",
     "authSource": "LOCAL",
     "password": $pass
     }')" | jq -r '.token')
   echo $OPS_TOKEN
   ```

4. Exchange the VCF Operations token for a fleet lifecycle bearer token.

   key is your service key.

   ```
   export TOKEN=$(curl -k -X 'POST' \
     ${OPS_HOST}/suite-api/api/auth/token/exchange \
     -H 'accept: application/json' \
     -H "Authorization: OpsToken $OPS_TOKEN" \
     -H 'Content-Type: application/json' \
     -d '{
     "serviceKeys": [
       "key"
       ]
     }' -s | jq -r '.jwtToken')
   ```

```
echo $TOKEN
```

You have successfully obtained the fleet lifecycle bearer token. The `TOKEN` environment variable contains the fleet lifecycle bearer token and you can use it to authenticate API request to the fleet lifecycle component.

5. Set the required fleet lifecycle environment variables.

   `FLEET_FQDN` is the fleet components FQDN and `TOKEN` is your fleet lifecycle bearer token.

   ```
   export FLEET_HOST=https://FLEET_FQDN
   export VSP_COMP_ID=($(curl -ks -XGET -H "Authorization: Bearer ${TOKEN}" "${FLEET_HOST}/fleet-lcm/v1/
   components" | jq -r '.components[] | select(.componentType == "VSP") | .id'))
   ```

   VSP_COMP_ID returns the IDs of all VCF services runtime instances in the VCF fleet.

6. View the current proxy configuration for a VCF services runtime instance.

   `TOKEN` is your fleet lifecycle bearer token and *`VSP_COMP_ID`* is the ID of the VCF services runtime instance.

   ```
   curl -ks -XGET -H "Authorization: Bearer ${TOKEN}" \
     ${FLEET_HOST}/fleet-lcm/v1/components/${VSP_COMP_ID}/config | jq | less
   ```

   The response returns the `peerProxy` object with the currently applied proxy settings.

7. Edit the VCF services runtime configuration with proxy settings.

   The proxy configuration supports the following parameters. The encodedCertificate value must be a base64-encoded SSL/TLS certificate for the proxy server. Do not use asterisks in the `excludeDomains` and `excludeIpAddresses` properties.

   ```
   "peerProxy": {
     "tlsEnabled": true | false,
     "credentialsEnabled": true | false,
     "encodedCertificate": "certificate",
     "excludeDomains": "<domains to exclude>",
     "excludeIpAddresses": "<IP addresses to exclude>",
     "host": "<proxy host name>",
     "port": <proxy port number>,
     "password": "<proxy password>",
     "username": "<proxy user name>"
   }
  }
   ```

   If necessary, to remove all proxy configurations, use the following configuration.

   ```
   "peerProxy":
      {  "host": "",
         "port": 0
     }
   ```

   Perform the following command on each VCF services runtime instance, where the component requires the traffic to be routed through a configured proxy server. In the command, `TOKEN` is your fleet lifecycle bearer token and *`VSP_COMP_ID`* is the ID of the VCF services runtime instance.

   ```
   export TASK_ID=$(curl -k -s -X PATCH -H "Authorization: Bearer ${TOKEN}" \
   "${FLEET_HOST}/fleet-lcm/v1/components/${VSP_COMP_ID}/config" \
     -H "Content-Type: application/json" \
     -d '{
     "type": "VspClusterConfigSpec",
     "peerProxy": {
         "tlsEnabled": true,
         "credentialsEnabled": true,
         "encodedCertificate": "LS0tLS1CRUdJTiBDRVJUSUZJQ0FURS0t…",
   ```

```
        "excludeDomains": "internal.example.com,*.local",
        "excludeIpAddresses": "10.0.0.0/24,192.168.1.10",
        "host": "proxy.example.com",
        "port": 3128,
        "username": "proxyuser",
        "password": "proxypassword"
      }
    }' | jq -r '.id')

  echo "Started Task ID: $TASK_ID"
```

8. (Optional) Check the status of the editing task.

```
curl -ks -XGET -H "Authorization: Bearer ${TOKEN}" ${FLEET_HOST}/fleet-lcm/v1/tasks/$TASK_ID | jq | less
```

9. (Optional) Verify the applied proxy configuration values.

   `TOKEN` is your fleet lifecycle bearer token and *`VSP_COMP_ID`* is the ID of the VCF services runtime instance.

```
curl -ks -XGET -H "Authorization: Bearer ${TOKEN}" \
  ${FLEET_HOST}/fleet-lcm/v1/components/${VSP_COMP_ID}/config | jq | less
```

The proxy configuration is successfully applied for the VCF services runtime instances that you configured. The traffic for the components that this instances hosts are routed through the configured proxy server.

Establish an SSH Connection to a VCF Services Runtime Instance

As a VCF Operations administrator, you can open an SSH connection to a VCF Services Runtime control plane node and directly access its underlying Kubernetes cluster.

Verify that you have the VCF services runtime SSH credentials used during the deployment of the component.

Perform the following steps to obtain access to the Kubernetes cluster.

1. Log in to the VCF Operations interface at `https://<vcf_operations_fqdn>` as a user assigned the Administrator role.
2. Navigate to **Build** > **Lifecycle** > **VCF Management**.
3. On the Components tab, select the **VCF services runtime.**
   a) Scroll to the **Nodes** section.
   b) Identify a control plane node and note the IP address.
4. Using the IP address, open an SSH connection to the control plane node as the `vmware-system-user` using the `APPLIANCE_ROOT_PASSWORD`.
   a) To obtain administrative access to the control plane, enter the following commands.

```
sudo -i
export KUBECONFIG=/etc/kubernetes/admin.conf
```

   b) To confirm that you have logged in to a control plane node, enter the following command.

```
kubectl get nodes -o wide
```

   The response gets nodes with IP addresses listed.

Renaming VCF 9.1 Components After Deployment

After you deploy VMware Cloud Foundation or vSphere Foundation 9.1 or later, you can rename certain vCenter and NSX components without breaking your SDDC Manager inventory.

VCF 9.1 reduces the number of component names that SDDC Manager persists in its inventory. As a result, renaming many vCenter objects no longer causes inconsistencies between vCenter, NSX, and SDDC Manager. However, some components are still subject to restrictions, and renaming outside the supported paths can produce unexpected behavior.

vCenter and NSX Components You Can Rename

- Verify that you have the required privileges in vCenter, NSX Manager, or SDDC Manager, depending on the component.
- Verify that you have a current backup of SDDC Manager, vCenter, and NSX Manager.

Component	Rename Location	Synchronization Behavior
ESX host name	vCenter	SDDC Manager automatically synchronizes the updated host name. No manual intervention is required. **Note:** Do not edit the SDDC Manager inventory directly. Rename ESX hosts in your vCenter inventory.
Cluster	vCenter	SDDC Manager does not have a dependency on the cluster name and renaming the cluster in vCenter does not break SDDC Manager.
Port group	vCenter	SDDC Manager does not have a dependency on the port group name and renaming does not break SDDC Manager.
Datastore	vCenter	SDDC Manager does not have a dependency on the datastore name and renaming does not break SDDC Manager.
Datacenter	vCenter	SDDC Manager does not have a dependency on the datacenter name and renaming does not break SDDC Manager.
vSphere Distributed Switch (VDS)	vCenter	SDDC Manager does not have a dependency on the VDS name and renaming does not break SDDC Manager.
SDDC Manager VM	vCenter	The VM display name in vCenter is a label and does not affect the FQDN, certificates, or SDDC Manager registration. **Note:** The FQDN of the SDDC Manager VM cannot be renamed. The FQDN is embedded in certificates, DNS, and SDDC Manager registration records. Changing the FQDN requires certificate regeneration and re-registration.
vCenter VM	vCenter	The VM display name in vCenter is a label and does not affect the FQDN, certificates, or SDDC Manager registration. **Note:** The FQDN of the vCenter VM cannot be renamed. The FQDN is embedded in certificates, DNS, and SDDC Manager registration records. Changing the FQDN requires certificate regeneration and re-registration.
NSX Manager VM	vCenter	The VM display name in vCenter is a label and does not affect the FQDN, certificates, or SDDC Manager registration. **Note:** The FQDN of the NSX Manager VM cannot be renamed. The FQDN is embedded in certificates, DNS, and SDDC Manager registration records. Changing the FQDN requires certificate regeneration and re-registration.
Domain name	SDDC Manager	A domain name is an SDDC Manager construct. You manage and rename it by using the SDDC Manager UI.
Network pool	SDDC Manager	Network pool is an SDDC Manager construct. You manage and rename it by using the SDDC Manager UI.
Uplink profile	NSX Manager	SDDC Manager does not persist the uplink profile name in its inventory and renaming the profile does not affect SDDC Manager.
NSX transport zone	NSX Manager	SDDC Manager does not persist the transport zone name in its inventory and renaming the transport zone does not affect SDDC Manager.

Component	Rename Location	Synchronization Behavior
NSX Tier-0 gateway	NSX Manager	SDDC Manager does not persist the Tier-0 gateway name in its inventory and renaming the gateway does not affect SDDC Manager.
NSX Tier-1 gateway	NSX Manager	SDDC Manager does not persist the Tier-1 gateway name in its inventory and renaming the gateway does not affect SDDC Manager.

vCenter and NSX Components You Cannot Rename

CAUTION:

The following components are known to have naming dependencies. Do not rename these components.

Component	Reason
NSX segment	Segment names are referenced in NSX policy objects.
VCF services runtime VMs	Name modifications in vCenter cause synchronization failures.
VCF services runtime datastores	Name modifications in vCenter cause synchronization failures.
VCF services runtime network objects	Name modifications in vCenter cause synchronization failures.
VCF services runtime folders	Name modifications in vCenter cause synchronization failures.